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The Book of Daniel the Prophet:
How the Ancient Hebrew Lunisolar Calendar Unseals the Hebrew Bible as well as Precisely Synchronizes and Recalibrates Ancient Near East Chronologies and Histories

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For Jeremiah W. (*Z''L* – ז"ל):[1]

May GOD show you the proof of Absolute Truth as you rest in Peace,
and may He bless your soul at the Resurrection
with Eternal Life, Compassion, Mercy, Grace, Kindness, and Love.

1 Zichrono/Zichro Livrachah – זכרונו / זכרו לברכה – [May] his memory [be] for a blessing.

"To the conductor, a Psalm of David: The skies tell the honor of GOD, and the deed of His hands the firmament says. Day to day will express a saying, and night to night will make experience knowledge."

– (Psalms 19:1-2)

"To count our days: Yes, inform [us], and we will bring a heart of wisdom."

– (Psalms 90:12)

"And you, Daniel, close [up] these things and seal the book until the time of the End; many will wander, and knowledge will increase."

– (Daniel 12:4)

"And he said, 'Go, Daniel, because closed [up] and sealed are these things until the time of the End. [They] will become clear, and will be clarified, and many will be refined, and [they] will convict evil [ones], and all evil [ones] will not understand, and [those who] contemplate will understand.'"

– (Daniel 12:9-10)

"And you run and run to catch up with the Sun /
But it's sinking /
Racing around to come up behind you again. /
The Sun is the same in a relative way /
But you're older /
Shorter of breath /
And one day closer to death. /"

– ("Time", *Dark Side of the Moon*, Pink Floyd)

"Emancipate yourselves from mental slavery /
None but ourselves can free our minds. /"

– ("Redemption Song", Bob Marley)

"Thus said YHVH of Armies: 'Because it will be wondrous in the eyes of the remnant of this people in those days, also in My eyes [it will be wondrous]' – Speech of YHVH of Armies".

– (Zechariah 8:6)

0.0. AUTHOR'S FOREWORD:

Long ago, like the prophet Isaiah (cf. Isaiah 6), I offered myself and my life to God and asked Him to use me as a vessel for His use and for His glory to sanctify His Name here on earth in these End-of-Days before the coming of the Promised Messiah.

I began by translating the Book of Daniel to English, and writing a supplementary commentary that included citations from ancient Jewish sages, rabbis, the Talmud, as well as other holy books while attempting to reconcile it to secular academic history. Eventually, the book became too large, and I had to narrow down the focus.

I decided to evaluate Sir Robert Anderson's claims that the Book of Daniel 9:24-27 accurately predicted the coming of the Messiah, purportedly fulfilled by Jesus Christ, but I had no way to confirm this scientifically. In order to do this, I would to need to calculate precise ancient proleptic Julian and Gregorian dates of events recorded in the Hebrew Bible (that uses the Hebrew calendar) including the exact date of the destruction of Jerusalem in 586 BCE (Julian) as well as the reign of Artaxerxes I (464 – 424 BCE (Julian)) – most especially the exact date of the event recorded in Nehemiah 2:1 where Artaxerxes I orders the reconstruction of the city itself with emphasis on the wall of Jerusalem.

Ancient biblical chronology had always fascinated me, and I wanted to know exact dates of ancient events recorded in the Hebrew Bible. I was not satisfied to only know the year or its approximation (since ancient dates are infamously difficult to date precisely) – and I questioned how scholars were so certain of their proposed chronologies. I also wanted to know not only the years, but also the exact months, as well as exact days (if at all possible), especially the critical dates of the destruction of Jerusalem in 586 BCE (Julian), and the Exodus – as these events seemed to be possible to date precisely because exact years, months, and days were given in the Hebrew Bible.

I searched for a calendar converter calculator that could convert ancient Hebrew dates that was precise to the very day all the way back through ancient periods, but it did not exist at the time (2006 – 2013). Therefore, finding the algorithms in early 2013 by Dershowitz and Reingold (1997) for these calendrical calculations was a Godsend. However, the only way that I could be certain that it was providing accurate calculations was by developing the custom program myself and testing it against known dates to see if they were correct and/or feasible – but I didn't know how to program or code well, so I had to learn by doing, and teach myself.

The majority of the research for this book was conducted between 2006 and 2016. In Summer 2013, after approximately six months of intensive research and work, I completed the development of the necessary software in Python for the critical calendrical calculations that produced the data which are detailed, discussed, and unfolded in this book. This required me to learn the crucial mathematics – including the calculus of infinite sequences and series – as well as the essential algorithms by

Dershowitz and Reingold (1997, 2008, 2018), and how to program and code.

I began with a working theory that contained several parts:

1.) The Hebrew calendar as we have it today is essentially the same as it was in ancient times, as well as identical to the Babylonian calendar, and could be used interchangeably and seamlessly;

2.) The capture of Jerusalem by Nebuchadnezzar II in 597 BCE (Julian) is recorded in both the Hebrew Bible and the Babylonian record, but only the Babylonian record gives the day: day two (2) of the month of Addaru / Adar in the 7th Regnal Year of this Babylonian king. The Hebrew Bible confirms the 7th Regnal Year of Nebuchadnezzar by dating the deportation that followed the capture to "the seventh year"[2] in agreement with the Babylonian count, whereas elsewhere the Bible records the exile of Yehoyachin (Jehoiachin) as having occurred in the 8th Regnal Year of Nebuchadnezzar;[3]

3.) By synchronizing Babylonian dates with the Hebrew calendar, it would be possible to determine the Julian and Gregorian equivalents of both the ancient Hebrew and Babylonian dates;

4.) Jeremiah 25:1 is the key to unlocking the biblical chronologies recorded in the Hebrew Bible using the Hebrew calendar, and thus is the key to synchronizing Hebrew and Babylonian chronologies;

Subsequently, I began to calculate and plot ancient dates one by one in an attempt to synchronize ancient chronologies. I confirmed the Jewish dates as recorded in *Seder Olam Rabbah* for the Judean king Yehoyakim (Jehoiakim) as well as the Babylonian king Nebuchadnezzar II were not possible.

According to this timeline, the 1st Regnal Year of Yehoyakim (Jehoiakim) – one of the last kings of Judah – was Hebrew Year 3317 – 3318 AM (i.e. 443 – 442 BCE (Julian)). However, this would be impossible as it would require that the 1st Regnal Year of Nebuchadnezzar (i.e. the 4th Regnal Year of Yehoyakim (Jehoiakim))[4] to be Hebrew Year 3320 – 3321 AM (i.e. 440 – 439 BCE (Julian)) – when in fact, according to established academic consensus, the 1st Regnal Year of Nebuchadnezzar began Spring 604 – 603 BCE (i.e. Hebrew Year 3157 – 3158 AM).

I recalibrated using the secular academic date for the 1st Regnal Year of Nebuchadnezzar as the given, and continued to plot dates and construct a timeline of relative chronology of this period for the last four (4) kings of Judah that synchronized perfectly with the Babylonian dates. This then unfolded into working out the precise chronology of all the kings of Judah and Israel back through king David who ruled from 1061 – 1022 BCE (Julian).

2 cf. Jeremiah 52:28
3 cf. 2 Kings 24:12
4 cf. Jeremiah 25:1

This chronology of the Hebrew kings corrects the work of E.R. Thiele which has been the standard, conventional chronology used in establishment Academia (despite being extremely problematic) for over 70 years only because of lack of a better alternative.

Having the precise years for the reign of king Solomon (1021 – 982 BCE (Julian)), I was able to calculate back from his 4th Regnal Year, 1018 – 1017 BCE (Julian) and determine the precise date of the Exodus to be 1497 BCE (Julian).

I moved forward in time and determined precise dates for the reigns of all kings of the Neo-Babylonian as well as Persian Achaemenid empires through the critical Artaxerxes I (464 – 424 BCE (Julian)). In January 2014, I made the most incredible discovery of this work: The Book of Daniel 9:24-27 does indeed mathematically confirm the exact date of the coming of a Promised Messiah in 33 CE (Julian) in not only one (1) way, but in two (2) ways that overlap each other with precise mathematics.

Therefore, this book – for the first time in history – proves both mathematically and scientifically that the Hebrew Bible is absolute truth, and thus that the God of Abraham, Isaac, and Jacob – the Creator of the universe – exists. I am sharing this preprint free with the world because everyone needs to have free access to this information in order to see and know the absolute truth of the undeniable, irrevocable, mathematical, scientifically verifiable, and confirmable proof of God's existence for themselves.

Until its publication here, I was the only person who knew of these discoveries, insights, and implications. The pressure has been enormous because I knew that if something happened to this research or to me, or if I died, this information (which is more important than my life) would tragically die with me and be lost forever – God forbid. Thank God this work has been completed after 20 years of research and struggle that has cost me my own time, my own sacrifice, and years of economic opportunity – but it has been worth it because this is my life's mission and the purpose for my existence.

Completing this book has been the most difficult thing I've ever set out to accomplish. Life circumstances beyond my control kept throwing obstacles in my path (like what happened to Job) and delaying the completion of the research and its publication. They forced me into an extended, long-term, WWW-shaped midlife crisis (i.e. trying to recover and then falling again), personal trials and tribulations (that were more than most people could believe and/or endure, and so severe and traumatic that I prefer not to detail them here), medical issues, as well as financial hardships.

This was nothing less than a spiritual battle to complete a spiritual mission for the sake of heaven and for the glory of the God of Israel: the Creator of the universe Who graciously agreed to use me as a vessel for His use.

בָּרוּךְ אַתָּה ה' אֱלֹהֵינוּ מֶלֶךְ הָעוֹלָם, שֶׁהֶחֱיָנוּ וְקִיְּמָנוּ וְהִגִּיעָנוּ לַזְּמַן הַזֶּה

Blessed are You, HASHEM, our God – King of the Universe – Who has made us live, and established us, and made us arrive to this time.

Amen.

Note on citations made in this book:

Text followed by a footnote reproduces or closely follows the wording of the cited source. Each footnote covers the text back to the previous footnote or the start of the paragraph. For sake of readability, quotation marks are mostly omitted. Words in square brackets [...] are the author's own additions or clarifications. Text that is not followed by a footnote is the author's own.

Claude (Anthropic) was used to verify data, confirm calculated dates, check citations, and assist in drafting parts of the concluding sections; a full statement appears in Section 78.8.

For the reader of this book, you can help in two (2) ways:

1.) Share this preprint book and its supplementary appendices, as well as data files, and the websites with links to these resources freely with as many people as possible.

2.) If this research and information are valuable to you, please consider to help financially support what comes next: the physical and digital editions of this book, websites, applications (apps), social media channels, and of course further research (including translations of books of the Bible) and publications, God willing.

You can find links to support this work via the following websites:

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0.0.1. ABSTRACT:

XX. Mathematics never lies: The ancient Babylonian / Hebrew calendar is arguably the most accurate and predictable lunisolar calendar system that exists, and can be used to proleptically calculate both its Julian and Gregorian equivalents that is precise to the very day. It can then be utilized to determine the precise, absolute, astronomical, mathematical, historical, ancient chronology of the Bible (to the very day) that is anchored to this ancient Babylonian / Hebrew mathematical measurement of astronomy. This subsequently can be used to recalibrate (and thus precisely synchronize) both Assyrian / Babylonian (i.e. Mesopotamian) as well as Egyptian chronologies and timelines to the Hebrew Bible's precise, definitive chronology (and not vice-versa).

XX. This recalibration provides precise confirmations and/or corrections (as well as fresh insights) to the Babylonian vs. Julian dates specified in Parker and Dubberstein's lunar tables[5] (that according to Walker (1997) have never been seriously questioned) thus making it possible to gain further insights into ancient Babylonian intercalation practices, namely whether their intercalation month for any particular leap year used an extra month Ululu / Elul or Addaru / Adar as "arbitrarily" decreed by the king upon the advice of his priest-astronomers.

XX. This in turn can lead Assyriologists to ascertain a precise year for the advent of the method of predictive, mathematical calculation of – and subsequent standardization of – the correct fixed 19-year "Metonic cycle" intercalary system of Babylonian / Hebrew years (which evidently seems to have been established as standard practice during the exact time of the events recorded in the Book of Daniel) during the Persian Achaemenid Era, i.e. from the 10th Regnal Year of Xerxes I (476 BCE (Julian)), becoming permanent by 381 BCE (Julian).[6]

XX. Moreover, this corrected, synchronized, and recalibrated ancient chronology can be used to unlock and mathematically determine the precise dates of many other events across recorded ancient history both within as well as outside of the Bible including (but not limited to):

5 Parker and Dubberstein (1956, 1971): *Babylonian Chronology 626 B.C. – A.D. 75*

6 This raises the possibility that the historical Daniel was among those who first pioneered this phenomenon in "Babylonian mathematical astronomy", even though its consistent application as standard practice is attested only from the 10th Regnal Year of Xerxes I (476 BCE (Julian)), becoming permanent by 381 BCE (Julian)

1. The Exodus (Saturday 18 April 1497 BCE (Julian));[7] [8] [9] [10] [11]

2. The receiving of the Torah at Mount Sinai (Sunday 07 June 1497 BCE (Julian)),[12] [13]

3. The wandering in the desert for 40 years (1497 – 1457 BCE (Julian));[14] [15]

4. The deaths of Miriam (March – April 1458 BCE (Julian)),[16] [17] [18] and Aharon (Monday 21 July 1458 BCE (Julian));[19] [20] [21]

5. The final address of Moses to the Nation of Israel (Wednesday 14 January 1457 BCE (Julian));[22] [23] [24] [25]

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7 Saturday 18 April 1497 BCE (Julian); Saturday 05 April 1497 BCE (Gregorian); 15 Nisannu / Nisan 2264 AM; Astronomical Julian Day JD_X: 1174751.5; RD_Y: 826755; RD_X: -546673; Sabbatical Cycle: 3; Metonic Cycle: 3;

8 cf. Appendix Table X.110

9 As there is no year zero (0) in the Julian calendar, for most dates of ancient history, the Proleptic Gregorian (Calibrated) year is (for all practical purposes) equivalent to the Julian year except for potentially the dates close to 01 January in December / January when the Julian and Gregorian calendars may be in different Julian Years, e.g. a Gregorian Astronomical Year in December, and a Julian Year in January or vice-versa. Therefore, for purposes of dating ancient events – for most of the year – the Julian Year BCE will be used also for the Gregorian Astronomical Year, and thus year zero (0) of the Gregorian Astronomical Year will be replaced with the Gregorian Calibrated (to the Julian) year unless there are dates during this December / January period that require precise specificity to differentiate the Julian vs. Gregorian years.

10 Naturally, this will eventually lead Egyptologists to definitively determine (finally) who was the actual Pharaoh of the Exodus.

11 Since the Exodus itself out of Egypt occurred on a Sabbath, it is now possible to understand one of the several reasons mentioned for remembering the commandment to keep the Sabbath is to remember that we were slaves in Egypt; cf. Deuteronomy 5:12-15

12 Sunday 07 June 1497 BCE (Julian); Saturday 25 May 1497 BCE (Gregorian); 06 Simannu / Sivan 2264 AM; Astronomical Julian Day JD_X: 1174801.5; RD_Y: 826805; RD_X: -546623; Sabbatical Cycle: 3; Metonic Cycle: 3;

13 cf. Appendix Table X.112

14 15 Nisan 2264 – 10 Nisan 2304 AM

15 cf. Appendix Table X.109-X.149

16 March – April 1458 BCE (Julian); March – April 1458 BCE (Gregorian); Month of Nisannu / Nisan 2303 AM; Sabbatical Cycle: 7; Metonic Cycle: 4;

17 cf. Appendix Table X.136

18 cf. Numbers 20:1

19 Monday 21 July 1458 BCE (Julian); Monday 08 July 1458 BCE (Gregorian); 01 Abu / Av 2303 AM; Astronomical Julian Day JD_X: 1189089.5; RD_Y: 841093; RD_X: -532335; Sabbatical Cycle: 7; Metonic Cycle: 4;

20 cf. Appendix Table X.140

21 cf. Numbers 20:22-29; 33:38

22 Wednesday 14 January 1457 BCE (Julian); Wednesday 01 January 1457 BCE (Gregorian); 01 Shabatu / Shvat 2304 AM; Astronomical Julian Day JD_X: 1189266.5; RD_Y: 841270; RD_X: -532158; Sabbatical Cycle: 1; Metonic Cycle: 5;

23 cf. Appendix Table X.146

24 cf. Footnote #9

25 cf. Deuteronomy 1:1-3

6. The death of Moses (Thursday 19 February 1457 BCE (Julian));[26] [27] [28]

7. The Entering of the Promised Land (Monday 22 March 1457 BCE (Julian));[29] [30] [31] [32]

8. The period of the Judges;

9. The reigns of King David (1061 – 1022 BCE (Julian))[33] and King Solomon (1021 – 982 BCE (Julian)),[34] as well as all of the Hebrew kings of the Divided Kingdom (982 – 586 BCE (Julian))[35] [36] thus definitively correcting Thiele's proposed chronology of the Bible which (despite it being extremely problematic with its clear errors in logic, reasoning, and consistency) has been the standard, most-accepted consensus view among the majority of scholars within established academia including the majority of Assyriologists and Egyptologists (through today and until now) simply because of lack of a better alternative;

10. The 31st (FINAL) Regnal Year of Yoshiyahu (Josiah)[37] is hereby to be corrected from the 17th Regnal Year of Nabopolassar (609 – 608 BCE (Julian)) to precisely the 18th Regnal Year of Nabopolassar (608 – 607 BCE (Julian));[38] == Year 0 Accession Year of Yeho-ahaz (Jeho-ahaz / Shalum) (i.e. 3 months) == Year 0 Accession Year of Yehoyakim (Jehoiakim / Elyakim) == YEAR ONE (1) OF SERVITUDE;[39]

11. The Exile of Daniel, Hananyah, Mishael, and Azaryah in the 3rd Regnal Year of Yehoyakim (Jehoikim) == the Year 0 Accession Year of Nebuchadnezzar (605 – 604 BCE (Julian));[40]

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26 Thursday 19 February 1457 BCE (Julian); Thursday 06 February 1457 BCE (Gregorian); 07 Addaru / Adar 2304 AM; Astronomical Julian Day JD_X: 1189302.5; RD_Y: 841306; RD_X: -532122; Sabbatical Cycle: 1; Metonic Cycle: 5;

27 cf. Appendix Table X.147

28 cf. Deuteronomy 34:1-12; Babylonian Talmud, *Kiddushin* 38a

29 Monday 22 March 1457 BCE (Julian); Monday 09 March 1457 BCE (Gregorian); 10 Nisannu / Nisan 2304 AM; Astronomical Julian Day JD_X: 1189334.5; RD_Y: 841338; RD_X: -532090; Sabbatical Cycle: 1; Metonic Cycle: 5;

30 cf. Appendix Table X.149

31 cf. Joshua 4:19

32 The date of Entering the Promised Land in the Hebrew year 2304 AM (1457 BCE) marks the beginning of exactly 860 Babylonian / Hebrew years (to the very day) until the Exile of Yehoyachin (Jehoiachin) on 10 Nisannu / Nisan, i.e. Saturday 23 March 597 BCE (Julian) / Saturday 17 March 597 BCE (Gregorian) (3164 AM). In other words, Israel sojourned exactly 860 Babylonian / Hebrew years (i.e. exactly 314116 days) in the land of Israel from 1457 BCE (2304 AM) until being exiled to Babylon in 597 BCE (3164 AM).

33 cf. Appendix Table X.1, X.194

34 cf. Appendix Table X.1, X.2, X.194

35 cf. Appendix Table X.2, X.3, X.194

36 i.e. 396 years of the Divided Kingdom from 982 – 586 BCE (Julian); 3175 – 2779 AM (Hebrew) == 396 years.

37 cf. Appendix Table X.2, X.3, X.4, X.194

38 3153 – 3154 AM

39 i.e. There is no Year zero (0) Accession Year for COUNTING YEARS OF 70-YEARS OF SERVITUDE: YEAR ONE (1) OF SERVITUDE BEGINS ON DAY ONE (1) AND IS COMPLETE AT THE END OF YEAR.

40 cf. Appendix Table X.2, X.4, X.14, X.194

12. The 1st Capture of Jerusalem by Nebuchadnezzar II (Saturday 16 March 597 BCE (Julian)));[41] [42]

13. The Exile of Yehoyachin (Jehoiachin) (Saturday 23 March 597 BCE (Julian)) == YEAR ONE (1) OF EXILE OF YEHOYACHIN (JEHOIACHIN);[43] [44]

14. Any and all events recorded in the biblical chronologies of the Torah, the Prophets, and the Writings in the Hebrew Scriptures (i.e. the Hebrew Bible: the "Old Testament") that include precise dates such as the arrival of Nebuzaradan to Jerusalem (Sunday 15 July 586 BCE (Julian)),[45] [46] [47] and his subsequent destroying of Solomon's First Temple (Wednesday 18 July 586 BCE (Julian));[48] [49] [50]

15. The final week and days of Jesus Christ including the date of Palm Sunday (Sunday 29 March 33 CE (Julian))[51] [52] and the Last Supper (the evening / night of Sunday 29 March 33 CE (Julian));[53] [54] [55] [56] [57]

16. The arrest of Jesus Christ after nightfall following the Last Supper (the night of Sunday 29 March 33 CE (Julian) through early morning hours of Monday 30 March 33 CE (Julian) after

41 Saturday 16 March 597 BCE (Julian); Saturday 10 March 597 BCE (Gregorian); 03 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503443.5; RD_Y: 1155447; RD_X: -217981; Sabbatical Cycle: 7; Metonic Cycle: 10;

42 cf. Appendix Table X.13, X.32, X.33, X.34

43 Saturday 23 March 597 BCE (Julian); Saturday 17 March 597 BCE (Gregorian); 10 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503450.5; RD_Y: 1155454; RD_X: -217974; Sabbatical Cycle: 7; Metonic Cycle: 10;

44 cf. Appendix Table X.34

45 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;

46 cf. Appendix Table X.59

47 cf. 2 Kings 25: 8-11

48 Wednesday 18 July 586 BCE (Julian); Wednesday 12 July 586 BCE (Gregorian); 10 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507584.5; RD_Y: 1159588; RD_X: -213840; Sabbatical Cycle: 4; Metonic Cycle: 2;

49 cf. Appendix Table X.59

50 cf. Jeremiah 52: 12-14

51 Sunday 29 March 33 CE (Julian); Sunday 27 March 33 CE (Gregorian); 09 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733198.5; RD_Y: 1385202; RD_X: 11774; Sabbatical Cycle: 6; Metonic Cycle: 12;

52 cf. Appendix Table X.185

53 Sunday 29 March 33 CE (Julian); Sunday 27 March 33 CE (Gregorian); 09 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733198.5; RD_Y: 1385202; RD_X: 11774; Sabbatical Cycle: 6; Metonic Cycle: 12;

54 cf. Appendix Table X.185

55 Monday 30 March 33 CE (Julian); Monday 28 March 33 CE (Gregorian); 10 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733199.5; RD_Y: 1385203; RD_X: 11775; Sabbatical Cycle: 6; Metonic Cycle: 12;

56 cf. Appendix Table X.185

57 Since the Babylonian / Hebrew day begins at Sunset (at approximately 18:00), the evening/night of Sunday 29 March 33 CE (Julian) technically is the beginning of the new day 10 Nisannu / Nisan 3793 AM which overlaps the Julian / Gregorian day which begins at Midnight (at exactly 00:00), and thus the precise dates for both of these days (that overlap) are included here.

midnight);[58] [59] [60] [61] [62] [63]

17. The Crucifixion of Jesus Christ (Friday 03 April 33 CE (Julian));[64] [65]

18. The Resurrection of Jesus Christ (Sunday 05 April 33 CE (Julian));[66] [67]

19. The seamless integration of this same ancient Babylonian / Hebrew (i.e. the Standard Mesopotamian calendar by this time) calendar system that continued as the standard calendar system during the Era of Contracts (*Era Contractuum*) also known as the Era of Alexander (311 – 63 BCE (Julian)) which began in the Greek / Seleucid period and remained the mandatory, standard calendar for the Jewish community throughout Talmudic and Middle Ages (e.g. Maimonides / Rambam), that subsequently seamlessly transitioned into today's modern Hebrew calendar system as *Anno Mundi* (AM)[68] [69] up to the present day.

XX. For example, the date of the death of Alexander the Great, 29 Ayyaru / Iyar in the Babylonian record and previously dated to 10 / 11 June 323 BCE (Julian), is confirmed by the proleptic Hebrew calendar as Monday 11 June 323 BCE (Julian), i.e. Hebrew 30 Sivan 3438 AM, because in that year the Hebrew calendar ran one (1) month ahead of the Babylonian.[70]

XX. As a direct result of observation of the mathematics of this ancient chronology, it becomes clearly evident without a doubt that the mysterious, elusive identity of the heretofore unknown historical figure of "Darius the Mede" can now be confirmed as none other than Darius the Great (r. 521 – 486 BCE (Julian)) himself (i.e. hidden in plain sight). Thus the Book of Daniel must be read and correctly

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58 Sunday 29 March 33 CE (Julian); Sunday 27 March 33 CE (Gregorian); 09 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733198.5; RD_Y: 1385202; RD_X: 11774; Sabbatical Cycle: 6; Metonic Cycle: 12;

59 cf. Appendix Table X.185

60 Monday 30 March 33 CE (Julian); Monday 28 March 33 CE (Gregorian); 10 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733199.5; RD_Y: 1385203; RD_X: 11775; Sabbatical Cycle: 6; Metonic Cycle: 12;

61 cf. Appendix Table X.185

62 Since the Babylonian / Hebrew day begins at Sunset (at approximately 18:00), the evening/night of Sunday 29 March 33 CE (Julian) technically is the beginning of the new day 10 Nisannu / Nisan 3793 AM which overlaps the Julian / Gregorian day which begins at Midnight (at exactly 00:00), and thus the precise dates for both of these days (that overlap) are included here.

63 cf. Exodus 12:3-6

64 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;

65 cf. Appendix Table X.185

66 Sunday 05 April 33 CE (Julian); Sunday 03 April 33 CE (Gregorian); 16 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733205.5; RD_Y: 1385209; RD_X: 11781; Sabbatical Cycle: 6; Metonic Cycle: 12;

67 cf. Appendix Table X.185

68 i.e. *Anno Mundi* (AM): Year of the World, i.e. the Hebrew Year counted from the first day of Adam.

69 The *Anno Mundi* (AM) years here indicate the Hebrew Year civil year count each year that begins in the 7th month, and ends in the 6th month: 7, 8, 9, 10, 11, 12 (or 13 in leap years), 1, 2, 3, 4, 5, 6.

70 Saturday 12 May 323 BCE (Julian); Saturday 07 May 323 BCE (Gregorian); 29 Ayyaru / Iyar 3438 AM; Astronomical Julian Day JD_X: 1603578.5; RD_Y: 1255582; RD_X: -117846; Sabbatical Cycle: 1; Metonic Cycle: 18;

understood in this context.

XX. Consequently, this synchronization of ancient historical chronologies will allow scholars of Jewish studies to recalibrate the events recorded throughout Jewish history in order to find and account for the so-called "missing ~165 years", and thus determine and update precise dates.

XX. Moreover, this mathematical, lunisolar Babylonian / Hebrew calendar can then be used to fix precisely the day of every recorded ancient lunar and solar eclipse catalogued by NASA via Espenak and Meeus,[71] [72] (and others before them)[73] as well as to systematically determine more precise ancient values for the critical astronomical variable, ΔT (Delta T), which reflects the irregularities of the Earth's rate of rotation and its measurement throughout history. It should be stressed that this variable, ΔT (Delta T), has particular relevance, significance, and critical importance for the study and measurement of times and positions of ancient eclipses.

XX. Furthermore, this same Babylonian / Hebrew calendar system is the key to unsealing the Book of Daniel, and specifically chapter 9 with its Messianic prophecy[74] that Sir Robert Anderson[75] (as well as both ancient Jewish and Christian scholars before him) attempted to interpret and decipher as a sealed prediction of the exact day of the coming of the Messiah. Applying this same Hebrew calendar (together with a correct literal reading of the cryptic Hebrew text), this passage in the Book of Daniel can be unsealed to correctly interpret, reveal, and confirm the precise mathematical prediction and fulfillment of the exact year of Jesus' baptism by John the Baptist (28 CE (Julian)),[76] [77] as well as both the exact year and month of Jesus' Crucifixion (33 CE (Julian)).[78] [79] [80]

XX. Therefore – for the first time in history – all of this correctly, thoroughly, and precisely proves both mathematically and scientifically not only that the Hebrew Bible is the most accurate historical book of ancient history (including the Book of Daniel), and not only that Jesus (Yeshua) was the Promised Messiah, but also by direct implication: It literally, undeniably, thoroughly, precisely, and irrevocably proves the existence of the Creator, GOD, for all to see with their own eyes, and scientifically (and mathematically) verify for themselves.

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71 NASA/TP–2009–214172: Espenak and Meeus (2009): *Five Millennium Canon of Lunar Eclipses: –1999 to +3000 (2000 BCE to 3000 CE)*

72 NASA/TP–2006–214141: Espenak and Meeus (2006): *Five Millennium Canon of Solar Eclipses: –1999 to +3000 (2000 BCE to 3000 CE)*

73 e.g. Huber and De Meis (2004); Beaulieu and Britton (1994); as well as others previously.

74 cf. Daniel 9:24-27

75 Anderson 1894: *The Coming Prince*

76 cf. Appendix Table X.174

77 cf. Daniel 9:24-25 – "to anoint the Holy of Holies".

78 cf. Appendix Table X.184, X.185

79 cf. Daniel 9:24-26 – "a Messiah will be cut off".

80 Crucifixion or Cruci-Fiction?

0.0.2. HOW THIS WORK IS STRUCTURED:

Section 0 provides the technical and historical background in the context of calendar systems, time measurement, and ancient Babylonian astronomy in order to understand how measurements of time will be measured and presented. It can be skipped and referred back to as necessary.

Section 1 introduces some of the academic issues to be discussed.

Sections 2 through 63 discuss the research findings and analysis of ancient history, and reconciles the ancient biblical chronology to the Babylonian, Assyrian, and Egyptian chronologies.

Sections 64 through 73 discuss applying this system of time measurement to the Book of Daniel.

Section 74 discusses the issue of Darius the Mede from the Book of Daniel.

Section 75 discusses the timeline of the week prior (and up) to the Crucifixion of Jesus Christ (Yeshua the Messiah).

Section 76 discusses the issues (and problems) of chronology of the ancient Near East.

Section 77 discusses the measurement of time, ΔT (Delta T), and eclipses

Section 78 provides the conclusions of this research.

Section 79 provides the references and bibliography.

0.1. Introduction

XX. Is it possible to prove GOD exists? If so, how would it be done?

XX. According to Isaac Newton (1642 – 1727 CE (Gregorian)), if GOD spoke, He spoke to be understood. The prophecies therefore must contain an ascertainable significance, straightforward and comprehensible to the intellect, appropriate for all men without requiring the meditation of the learned.

XX. However, these prophecies were so obscure that Newton himself held that not even the most learned men could ever have understood them.[81] Indeed, from review of Newton's writings on theological manuscripts of Daniel and John, it is evident that there is much that Newton himself did not – and could not – understand in his day, having died in 1727 CE (Julian in the UK), approximately 100 years before the great discoveries of Assyriology of the 18th / 19th centuries CE (Gregorian), as well as all other modern knowledge since then.

XX. Newton reasoned that the obscurity itself was relevant to God's plans. The obscurity was God's way of prolonging the revelation in time and of – according to Newton – distinguishing true Christians from apostates.[82]

XX. As observed by Newton: amongst the old Prophets, Daniel is most distinct in order of time, and – according to Newton – easiest to be understood. Therefore in those things which relate to the last times, he must be made the key to the rest.[83]

XX. Establishing the precise date of any ancient event is well known to be extremely difficult. It is probably not an exaggeration to state that only when ancient chronology is based upon calculable astronomical phenomena can we have certainty and precision.[84]

XX. Chronology is based on calendars, and ancient calendars are based upon astronomy. As we will see: knowledge of ancient calendars turns out to be critical in reconstructing [not only] the events of Jesus' last days, [but indeed all of ancient history].[85]

XX. However, as we look back in time, the Julian calendar has no meaning for modern society that uses the Gregorian calendar. If we are to understand (i.e. experience) ancient dates at the same time of the solar year as we experience today, we need to use the proleptic Gregorian calendar – which of course did not exist before the Gregorian calendar reform of the previous Julian calendar in October 1582 CE (Julian / Gregorian).

81 Mamiani 2002: 392
82 Mamiani 2002: 392
83 Newton 1733: 15
84 Humphreys and Waddington 1992: 350
85 Humphreys 2011: 11

0.2. Julian Years

XX. The Julian calendar, which dates from 01 January 709 A.U.C. (i.e. 01 January 45 BCE (Julian), was introduced by Julius Caesar in 45 BCE (Julian). It is a solar calendar with months of fixed lengths, and was a modification of the Roman Republican and ancient Egyptian calendars.[86] [In the Julian calendar in] every 4th year an intercalary day was added to maintain synchrony between the [Julian] calendar year [of (365.25 mean days) per Julian year] and the tropical year [of (365.242177 mean days) per tropical solar year]. It served as a standard for European civilization until the Gregorian Reform of 1582 CE (Julian / Gregorian).[87]

XX. As an average Julian year is slightly longer than a tropical year (365.25 days compared with 365.2422 days), the start of each year on the Julian calendar became progressively delayed relative to the seasons. The difference amounted to one day every 128 years. Over the centuries, this defect produced a marked change in the vernal equinox (and thus in the date of Easter).[88]

XX. The year 45 BCE (Julian) has been called the "year of confusion" because in that year Julius Caesar inserted 90 days to bring the months of the Roman calendar back to their traditional place with respect to the seasons. This was Caesar's first step in replacing a calendar that had gone badly awry.[89]

XX. Although the pre-Julian calendar was lunisolar in inspiration, its months no longer followed the lunar phases and its year had lost step with the cycle of the seasons. Following the advice of Sosigenes, an Alexandrian astronomer, Caesar created a solar calendar with twelve (12) months of fixed lengths and a provision for an intercalary day to be added every 4th year.[90]

XX. As a result, the average length of the Julian calendar year was 365.25 days. This is consistent with the length of the tropical year as it was known at the time.[91]

XX. [Julian] years are classified as normal years of 365 days and leap years of 366 days. Leap years occur in years that are evenly divisible by 4. For this purpose, year zero (0) (i.e. 1 BCE (Julian)) is considered evenly divisible by 4. The year is divided into twelve (12) formalized months that were eventually adopted for the Gregorian calendar.[92]

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86 Seidelmann 1992: 600; Dershowitz and Reingold 2008: 46
87 Seidelmann 1992: 600
88 Stephenson 1997: 31
89 Seidelmann 1992: 601
90 Seidelmann 1992: 601; Dershowitz and Reingold 2008: 46
91 Seidelmann 1992: 601
92 Seidelmann 1992: 601

XX. Following Caesar's death [i.e. the ides of March, i.e. 15 March 44 BCE (Julian)], the Roman calendrical authorities misapplied the leap-year rule with the result that every third – rather than every fourth – year was intercalary. Although detailed evidence is lacking, it is generally believed that Emperor Augustus corrected the situation by omitting intercalation from the Julian years 8 BCE (Julian) through 4 CE (Julian). After this the Julian calendar finally began to function as planned.[93]

XX. Through the Middle Ages the use of the Julian calendar evolved and acquired local peculiarities that continue to snare the unwary historian: There were variations in the initial epoch for counting years, the date for beginning the year, and the method of specifying the day of the month.[94]

XX. Not only did these vary with time and place, but also with purpose. Different conventions were sometimes used for dating ecclesiastical records, fiscal transactions, and personal correspondence.[95]

XX. Caesar designated 01 January as the beginning of the year. However, other conventions flourished at different times and places. The most popular alternatives were 01 March, 25 March, and 25 December. This continues to cause problems for historians since – for example – 28 February 998 CE (Julian) as recorded in a city that began its year on 01 March would be the same as 28 February 999 CE of a city that began the year 01 January.[96]

XX. Days within the month were originally counted from designated division points within the month: Kalends, Nones, Ides.[97]

XX. The Kalend is the 1st day of the month. The Ides is the 13th of the month except in March, May, July, and October when it was the 15th day. The Nones is always eight (8) days before the Ides [i.e. the Nones can be the 5th or 7th day depending on whether the Ides of month is the 13th or 15th day respectively].[98]

XX. Dates falling between these division points are designated by counting inclusively backward from the upcoming division point [i.e. Kalends, Nones, Ides, and the 1st day (i.e. Kalend) of the next month].[99]

XX. Intercalation was performed by repeating the day VI Kalends March, i.e. inserting a day between VI Kalends March (24 February) and VII Kalends March (23 February).[100]

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93 Seidelmann 1992: 601
94 Seidelmann 1992: 601
95 Seidelmann 1992: 601
96 Seidelmann 1992: 601
97 Seidelmann 1992: 602
98 Seidelmann 1992: 602
99 Seidelmann 1992: 602
100 Seidelmann 1992: 602-603

XX. Confusing? Yes! Absolutely! Remember these days are designated by counting inclusively backward from the upcoming division point – in this case: counting backward from the upcoming Kalend division point, Kalend (01) March. See Seidelmann 1992 for more details with table as well as Dershowitz and Reingold (1997; 2008; 2018) for algorithms.

XX. By the 11th century CE (Julian) consecutive counting of days from the beginning of the month came into use. However, local variations continued including counts of days from dates that commemorated local saints. The inauguration and spread of the [later] Gregorian calendar [from October 1582 CE (Julian)] resulted in the adoption of a uniform standard for recording dates.[101]

XX. Today the principles of the Julian calendar continue to be used by chronologists.[102] The Julian proleptic calendar is formed by applying the rules of the Julian calendar to times before Caesar's reform.[103] This provides a simple chronological system for correlating other calendars and serves as the basis for the Julian day [JD or JD_X] numbers.[104]

0.3. Gregorian Years

XX. The Gregorian calendar – designed at the end of the 16th century CE (Julian) and used today throughout the world – is purely solar in nature.[105] The most popular [civil] calendar currently in use today in most of the world is the Gregorian (or "new-style") calendar designed by a commission assembled by Pope Gregory XIII in the 16th century CE (Julian).[106] The main author of the new system was the Naples astronomer Aloysius Lilius.[107]

XX. The previous Julian calendar continued in force until Thursday 04 October 1582 CE (Julian), when – in a small correction of accumulated error – Pope Gregory XIII dropped ten (10) days so that the day following Thursday 04 October 1582 CE (Julian) became Friday 05 October 1582 CE (Gregorian).[108]

XX. This strictly solar calendar is based on a 365-day common year divided into 12 months of lengths [of 31, 30, and 28 days (with one (1) leap day according to either Julian and/or Gregorian intercalation rules which mostly overlap, but differ slightly)].[109] The Julian calendar is so close in substance to the Gregorian [as the Gregorian calendar is simply] a modification to the Julian calendar [that occurred] in

101 Seidelmann 1992: 603
102 Seidelmann 1992: 600
103 Seidelmann 1992: 600-601
104 Seidelmann 1992: 601
105 Dershowitz and Reingold 2013: 22
106 Dershowitz and Reingold 1997: 5; Dershowitz and Reingold 2008: 3, 45
107 Dershowitz and Reingold 2008: 45
108 Finegan 1998: 406
109 Dershowitz and Reingold 2008: 45

the 16th century CE (Julian).[110]

XX. The Gregorian calendar:[111]

1.) January – 31 days

2.) February – 28 or 29 days

3.) March – 31 days

4.) April – 30 days

5.) May – 31 days

6.) June – 30 days

7.) July – 31 days

8.) August – 31 days

9.) September – 30 days

10.) October – 31 days

11.) November – 30 days

12.) December – 31 days.

XX. Over the centuries the [Julian] calendar year (of 365.25 mean solar days) slipped with respect to the seasons [that are measured relative to the tropical/solar year (with current value that is 365.242177 mean solar days)]. By the 16th century CE (Julian), the true date of the vernal (spring) equinox had shifted from around 21 March in the 4th century CE (Julian) – when the date of Easter was fixed – to around 11 March.[112] If this error were not corrected, gradually Easter (whose date depends on the ecclesiastical approximation of 21 March for the vernal equinox) would migrate through the seasons, eventually to become a summer holiday.[113]

110 Dershowitz and Reingold 1997: 5; Dershowitz and Reingold 2008: 3
111 Dershowitz and Reingold 1997: 33; Dershowitz and Reingold 2008: 45
112 Dershowitz and Reingold 2008: 46
113 Dershowitz and Reingold 2008: 46-47

XX. In most societies a calendar reform is an extraordinary event. Adoption of a calendar depends on the forcefulness with which it is introduced and on the willingness of society to accept it. For example, the acceptance of the Gregorian calendar as a worldwide standard spanned more than three (3) centuries.[114]

XX. Pope Gregory XIII instituted only a minor change in the [Gregorian] calendar: century years not divisible by 400 would no longer be leap years [e.g. 100, 200, 300, 500, 600, 700, 900, 1000, 1100, 1300, 1400, 1500, 1700, 1800, 1900, 2100, 2200, 2300, etc.]. The Pope also modified the rules for Easter.[115] Thus 3 out of 4 century years are common years [i.e. not leap years], giving a cycle of 400 years containing (400 x 365 + 97) [i.e. the number of leap years] == (146,097 days) and an average year length of (146097/400) == (365.2425 days) [i.e. the Gregorian year (365.2425 mean solar days) is slightly more accurate than the Julian year (365.25 mean solar days) when compared to the length of the Tropical year (365.242177 mean solar days)].[116]

XX. Pope Gregory XIII also corrected the accumulated 10-day error in the date of the [vernal] equinox by proclaiming that Thursday 04 October 1582 CE (Julian) (according to the "old-style" Julian calendar then in use) would be followed by Friday 15 October 1582 CE (Gregorian), i.e. the first day of the "new-style" (Gregorian) calendar. By universal current custom, the new Gregorian year number begins on 01 January [of every year].[117]

XX. In [Gregorian] leap years, [the year is] 366 days [and] the extra day is added to make the 2nd month [i.e. February] 29 days long.[118] A [Gregorian] year is a leap year if: 1.) it is divisible by 4 and is [also] not a century year ([i.e. it is both divisible by 4 and also not a] multiple of 100 [with both conditions together]); or 2.) [it is a Gregorian leap year] if it is divisible by 400.[119]

XX. The Gregorian calendar differs from its predecessor (i.e. the old-style, or Julian calendar) only in that the Julian calendar did not include the century rule for leap years, i.e. all century years were leap years [in the Julian calendar system].[120] The months of the Julian calendar are the same as on the Gregorian calendar.[121]

XX. Because every 4th year was a leap year [in the Julian calendar], a cycle of 4 years contained (4 x 365 + 1) == (1461 days), giving an average length of year of (365.25 days). This [as well as the Gregorian year of (365.2425 mean solar days)] is somewhat more than the mean length of the tropical

114 Seidelmann 1992: 578
115 Dershowitz and Reingold 2008: 47
116 Dershowitz and Reingold 2008: 47
117 Dershowitz and Reingold 2008: 47; Dershowitz and Reingold 2013: 22
118 Dershowitz and Reingold 2008: 45
119 Dershowitz and Reingold 2008: 45
120 Dershowitz and Reingold 2008: 45
121 Dershowitz and Reingold 2008: 45, 63

year [of (365.242177 mean solar days)].[122]

XX. The Gregorian calendar today serves as an international standard for civil use. In addition, it regulates the ceremonial cycle of the Roman Catholic and Protestant churches. In fact, its original purpose was ecclesiastical. Although a variety of other calendars are in use today, they are restricted to particular religions or cultures.[123]

0.4. The Tropical Year

XX. The tropical year is the time it takes for the mean Sun to travel from one mean vernal equinox to the next. As the speed of the mean Sun on the ecliptic is slowly increasing, the tropical year [as well as Length of Day (LOD)] as determined from the instantaneous speed of the mean Sun for an arbitrary instant is slowly decreasing.[124]

XX. The tropical year is defined as the mean interval between vernal equinoxes, and it corresponds to the cycle of the seasons [throughout one (1) tropical year].[125] In the following formulas, *T* is measured in Julian centuries of Terrestrial Dynamical Time (TDT) which is independent of the variable rotation of [i.e. it is not a measure of the true, variable rotation of] the Earth. Thus the lengths of the tropical year and synodic [lunar] month are here defined in days of 86400 seconds of International Atomic Time (TAI).[126]

XX. The following expression, based on the orbital elements of Laskar (1986), is used for calculating the length of the tropical year:[127]

$$(365.2421896698\ ^{days}) - (0.00000615359\ T) - (7.29) \times (10^{-10})\ (T^2) + (2.64) \times (10^{-10})\ (T^3)$$

where (T) == $((JD_X - 2451545.0) / 36525)$ and (JD_X) is the Julian Day number. However, the interval from a particular vernal equinox [i.e. 0° Longitude] to the next may vary from this mean by several minutes.[128]

XX. The synodic [lunar] month, the mean interval between conjunctions of the Moon and Sun, corresponds to the cycle of lunar phases [i.e. a lunation]. The following expression for the synodic [lunar] month is based on the lunar theory of Chapront-Touzé and Chapront (1988):[129]

122 Dershowitz and Reingold 2008: 46
123 Seidelmann 1992: 580
124 Dershowitz and Reingold 2008: 187
125 Seidelmann 1992: 576
126 Seidelmann 1992: 577
127 Seidelmann 1992: 576
128 Seidelmann 1992: 576
129 Seidelmann 1992: 576

$(29.5305888531^{days}) + (0.00000021621T) – (3.64) \times (10^{-10}) (T^2)$

where again (T) == (JD_X – 2451545.0) / 36525) and (JD_X) is the Julian day number. Any particular [lunar] phase cycle may vary from the mean by up to seven (7) hours.[130]

XX. From these formulas we see that the cycles change slowly with time. Furthermore, the formulas should not be considered to be absolute facts, [but rather] are the best approximations possible today. Therefore a calendar year of an integral [whole] number of days cannot be perfectly synchronized to the tropical year.[131]

XX. Approximate synchronization of calendar months with the lunar phases requires a complex sequence of months of 29 and 30 days. For convenience, it is common to speak of a lunar year of twelve (12) synodic [lunar] months, or (354.36707 days).[132]

XX. The length of the tropical year is today defined with respect to a "dynamical" equinox. Its current value is (365.242177 mean solar days), and is decreasing by about (1.3×10^{-5} solar days per century).[133]

XX. For the formulas, calculations, and algorithms provided by Dershowitz and Reingold, the defined value used for Mean Tropical Year is (365.242189 [mean solar days) and not the value (365.242177 mean solar days) mentioned above]. [It should be noted that] the time intervals between successive vernal equinoxes differ from those between successive autumnal equinoxes, as well as also from the tropical year. [Indeed,] equinox-to-equinox and solstice-to-solstice year lengths are fluctuating.[134]

XX. A sidereal year is the time it takes for the Earth to revolve once around the Sun, or for the (mean) Sun to return to the same position relative to the background of the fixed stars. The sidereal year is about 20 minutes more than the tropical [year], with the Mean Sidereal Year being 365.25636 [mean solar days].[135]

XX. Three (3) distinct types of calendars have resulted from this situation:[136]

1.) A solar calendar – of which the Gregorian [(365.2425 days per mean year), or Julian (365.25 days per mean year)] calendar in its civil usage is an example – is designed to maintain synchrony with the tropical year [(365.242177 days per mean year)]. To do so, days are intercalated (forming leap

130 Seidelmann 1992: 576
131 Seidelmann 1992: 577
132 Seidelmann 1992: 577
133 Dershowitz and Reingold 2008: 187
134 Dershowitz and Reingold 2008: 187
135 Dershowitz and Reingold 2008: 187
136 Seidelmann 1992: 577

years) to increase the average length of the calendar year [in order to stay calibrated and synchronized with the shorter tropical year – otherwise the longer (365.25-day Julian year) and (365.2425-day Gregorian year) – which are numbered as an integral 365.0 whole days – will slip/migrate in relation to the longer tropical year of (365.242177 mean days per tropical solar year)].[137]

The inaccuracy of the Julian calendar causes the Summer Solstice to recede by about one (1) month in 4000 years rather than [recede in] two (2) months [in 4000 years] as expected for an accurate solar (Gregorian) calendar.[138]

2.) A lunar calendar such as the Islamic calendar [and some scholars (cf. Bloch (2010, 2012a CE, 2012b) and Jeffers (2017)) include the Middle Assyrian Calendar] follows the lunar phase cycle without regard for the tropical year. Thus the months of the Islamic calendar [and according to Bloch, Jeffers, and also Frahm (2017) perhaps likely also the Ancient Middle Assyrian Calendar prior to the 11th century BCE (Julian)] systematically shift [i.e. wander (or slip) 11 days per year] with respect to the months of the Gregorian calendar.[139]

3.) The third type of calendar, the lunisolar calendar, has a sequence of [twelve (12)] months based on the lunar phase cycle, but every few years a whole [13th] month is intercalated to bring the calendar back in phase with the tropical year [of 365.242177 mean days per tropical solar year]. The Hebrew, [Babylonian,] and Chinese calendars are examples of this type of calendar [of a lunar calendar that is synchronized with the seasons in a solar year through intercalation].[140]

0.5. Biblical Chronology and Biblical Recording of Dates

XX. The period before us is – in some respects – the most interesting and fruitful of all for biblical chronological study because for no other period is there such a wealth of detailed chronological information – both biblical and extra-biblical. There are frequent cross-references between Hebrew and Babylonian kings that facilitate the precise and certain dating of events. Dates are now expressed not only in terms of years, but at times in terms of months or days.[141]

XX. Cohen cites Vanderkam who notes “No part of the Bible or even the Bible as a whole presents a full calendar; information about these matters must be gleaned from occasional, often incidental references to dates, days, months, seasons, and years.” To this, Cohen voices his skepticism about the value of biblical material for reconstructing the pre-exilic calendar for the Hebrews with any degree of

137 Seidelmann 1992: 577
138 De Jong 2006: 438
139 Seidelmann 1992: 577
140 Seidelmann 1992: 577
141 Thiele 1983: 179

certainty.[142]

XX. Cohen is correct when he states that the only extant copies are the product of varying degrees of editing [e.g. the various biblical texts/manuscripts extant] over the centuries. Though he admits that the Bible contains information on the early Hebrew cultic calendar, Cohen argues that without contemporaneous source material, the study of the pre-exilic Hebrew calendar cannot progress beyond scholarly speculation.[143]

XX. It will be proven here otherwise.

0.6. Mathematics Never Lies

XX. Mathematics never lies. If we are to scientifically study the dates and math involved with the chronologies that are recorded in the Hebrew Scriptures and the Babylonian Chronicles (including the Astronomical Diaries), then we must be willing to go with math, logic, and reason no matter where the math, logic, and reason lead us. We must draw logical conclusions based upon what can be proven with (and inferred from) the precise and exact mathematics.

XX. When something conflicts with the mathematics (that never lies), then we must find a way to reconcile our new understandings to the pure mathematics, and we must learn to accept our newly-understood reality more precisely according to the mathematics. Then others can scientifically repeat to test and challenge the data of mathematics to either confirm or disprove with the repeatable (and testable) scientific method.

XX. Mathematics allows us to reconstruct and calibrate a single accurate, ancient historical chronology not only to the exact year, and not only to the exact month, but even unto the exact day(!) in ancient history. Amazingly, it is precisely the exact mathematics (exclusively) that allow us to reconstruct and unfold in more detail the circumstances of the final days of Yehoyakim (Jehoiakim / Elyakim) king of Judah, as well as the mathematically evident co-regency of his final months co-reigning with his son Yehoyachin (Jehoiachin) who ruled for exactly 3 months and 10 days.[144]

XX. Proper, exact, and precise calibration of the Hebrew Biblical Chronologies with the Akkadian Mesopotamian (i.e. Babylonian and Assyian) Chronologies, as well as with the Egyptian Chronology, will be unfolded throughout this study. It will become evident through mathematics, logic, and reason that it was Yehoyakim (Jehoiakim / Elyakim) who is the king that is mentioned in the Babylonian Chronicle as having been captured on the 2nd day of the (12th and final) month of Addaru / Adar in

142 Cohen 1993: 20
143 Cohen 1993: 20
144 cf. 2 Kings 24: 8-9; 2 Chronicles 36: 9-10

Nebuchadnezzar's 7th Regnal Year (598 – 597 BCE (Julian)).

0.7. Astronomical Methodologies: Ancient Greek and Modern Geometric (Arcus Visionis) vs. Ancient Babylonian Algebraic

XX. Traditional modern astronomical methods are (and have been) geometric in that they attempt to calculate the lunar crescent geometrically by angle and position of the Moon in the sky. Yet ancient Babylonian astronomical methodologies were arithmetic and algebraic. As we will begin to see detailed below, there are a complexity of variables that most certainly – individually and collectively – affect the accuracy of the overall results in data from these modern attempts with geometrical methodologies attempting to study ancient Babylonian mathematical astronomy that used ancient arithemetic and algebraic methodologies.

XX. The following is a recount of historical attempts to apply geometrical astronomical, calendrical dating methods including a discussion of the critical variable of arcus visionis. In this section, as well as all sections that follow, many of these astronomical variables will be noted with square brackets, e.g. [i.e. variable]. This is in order to bring to the awareness of the reader the complexity of the calculations that require these numerous astronomical variables within this traditional geometrical method of arcus visionis. Most (but not all) variables will be noted. Therefore, the existence of these variables will affect the accuracy of calculations, and should be kept in the back of the reader's mind throughout. We will see that the more ancient Babylonian / Hebrew algebraic method is both simpler and more elegant, and precludes the need for many of these variables while providing more accurate astronomical, calendrical calculations.

XX. The so-called *arcus visionis* [i.e. arc of vision] is the distance in degrees between the true position of the star [or Moon] above – and that of the Sun below – the horizon, measured perpendicular to the horizon at the moment of first appearance.[145] The concept of the arcus visionis – probably introduced into Greek Astronomy at the time of Eudoxos in the 4th century BCE (Julian) – was first properly treated by Ptolemy in about 150 CE (Julian).[146]

XX. Values of the arcus visionis in Alexandria for the brightest stars may be derived from the dates of first/last appearance given in the remaining part of another work of Ptolemy "Phases of the fixed stars and collection of weather changes".[147] [148] From the dates given in the "Phases" Heinrich Vogt (1920) showed that Ptolemy – in his theoretical treatment – takes account of the dependence of the arcus visionis on stellar magnitude, and apparently also on Elongation of the star from the sun (measured

145 De Jong 2007: 112
146 Almagest VIII,6 and XIII,7; Toomer 1984 – cited by De Jong 2007: 116
147 cf. De Jong 2007: 116 for the Greek
148 De Jong 2007: 116

along the horizon in Azimuth).[149] The dependence on Elongation is expected to be most pronounced for stars far above or below the ecliptic.[150]

XX. Carl Schoch (1924b) used Babylonian observations of stars and planets and the theoretical formulation of Vogt (1920) to come up with the following algorithms for the arcus visionis at the first appearance of stars in Babylon as a function of stellar magnitude and Elongation (E):[151]

Table 0.1. Proposed Values for Arcus Visionis from "Astronomical dating of the rising star list in MUL.APIN", De Jong 2007: 117

Magnitude	**Arcus visionis (h)**
-1.0	6.9° + 2.5° cos (E)
0.0	7.8° + 2.6° cos (E)
+1.0	8.9° + 2.7° cos (E)
+2.0	10.6° + 2.9° cos (E)
+3.0	12.8° + 3.3° cos (E)

XX. These algorithms were used by Van der Waerden (1949) in his analysis of the MUL.APIN star list.[152]

XX. In the words of C. Schoch: "The computation of the [lunar] crescent is very complicated and is the most difficult exercise in ancient astronomy." [Johannes] Kepler [1571 – 1630 CE (Julian – Gregorian)] regarded the computation of the [lunar] crescent as impossible.[153] De Meis notes that [Isaac] Newton said that the Moon caused his head to ache, which is something for such a genius![154] The reason for this is the complexity of the Moon's orbit together with the requirement of extreme accuracy.[155]

XX. The tables of R.G. Schram (1908) and the early tables of P.V. Neugebauer (1914) were not sufficiently accurate for this purpose. Besides the difficulty of calculating the coordinates [i.e. variables] of the Sun [i.e. variable] and Moon [i.e. variable], it is also necessary to develop empirical criteria [i.e. variables] for the minimum conditions [i.e. variables] under which the [lunar] crescent [i.e. variable] would be visible to a sharp-sighted [i.e. variable] observer [i.e. variable] at the location of interest [i.e. variable].[156]

149 De Jong 2007: 116-117
150 De Jong 2007: 117
151 De Jong 2007: 117
152 De Jong 2007: 117
153 Casperson 1986: 142
154 De Meis 2004: 121
155 Casperson 1986: 142
156 Casperson 1986: 142

XX. J.K. Fotheringham analyzed a series of [lunar] crescent observations made primarily at Athens in the latter half of the last 19th century CE (Gregorian). From this data he obtained an approximate relationship [i.e. variable] between the minimum altitude of the Moon [i.e. variable] at Sunrise [i.e. variable] or Sunset [i.e. variable] and the Sun-Moon Azimuth difference [i.e. Δ (Delta): variable]. That relationship was employed by Neugebauer in setting up an explicit tabular procedure for computing [lunar] crescent visibility [i.e. variable].[157]

XX. Unfortunately, Neugebauer's inclusion of refraction [i.e. variable] and the finite size of the solar disk [i.e. variable] was inconsistent with the manner in which Fotheringham had established his visibility criteria [i.e. variables].[158]

XX. Thus, in spite of their complexity [i.e. the complexity of variables, calculations, and thus room for accumulated error in this complicated geometrical methodology], Neugebauer's results were inaccurate. A series of tables for lunar month calculations was then developed by Schoch (1927). He also derived a table of [lunar] crescent visibility criteria [i.e. variables] based on an analysis of over 400 ancient Babylonian observations.[159]

XX. Then Neugebauer (1929) published a new set of tables claiming still better accuracy and incorporating Schoch's visibility criteria [i.e. variables] for Babylon [i.e. variable]. Parker's lunar month calculations have been based on Schoch's tables with all doubtful cases resolved by reference to Neugebauer's tables. It is probable that most other lunar month calculations have been based on Neugebauer's tables as well.[160]

XX. Since the time of Schoch and Neugebauer, there have been only modest improvements in knowledge about lunar and planetary orbits, but in recent years there have been very substantial improvements in techniques for carrying out orbit calculations quickly and accurately.[161]

XX. According to Casperson, Neugebauer's tables for calculating lunar height [i.e. variable] are fundamentally incorrect when applied to the problem of lunar visibility [i.e. variable] at Sunrise [i.e. variable].[162]

XX. Casperson asserts that the difficulty stems from Neugebauer's erroneous computation of the angle [i.e. variable] between the horizon [i.e. variable] and the parallel of declination [i.e. variable] through

157 Casperson 1986: 142
158 Casperson 1986: 142
159 Casperson 1986: 142
160 Casperson 1986: 142
161 Casperson 1986: 142
162 Casperson 1986: 142

the Sun [i.e. variable] at Sunrise [i.e. variable].[163] Casperson notes that this error had already been noted many years ago in a somewhat inaccessible article by B.L. van der Waerden (1942).[164]

XX. Already implicit is another type of ambiguity [i.e. variable] that can affect the accuracy of a lunar month calculation [i.e. variable]: It is not known with certainty where [i.e. variable] the Egyptians made their various astronomical observations during the Eighteenth Dynasty [i.e. variable]. The most common opinions are that these observations were made either at Thebes or at Memphis/Heliopolis. Parker did not indicate the location that was assumed in his calculations.[165]

XX. Another limitation of Neugebauer's tabular procedure is that it assumes a constant value of 23.7° for the obliquity of the ecliptic (the angle [i.e. variable] by which the rotational axis of the Earth is tilted from the axis of the Earth's orbit) [i.e. variable]. In fact, the obliquity [of the Earth's ecliptic, i.e. variable] has been gradually decreasing for the last few thousand years.[166]

XX. According to Casperson, the obliquity of the Earth's ecliptic [i.e. variable] was about 23.7° in 1 CE (Julian), but about 23.9° in 1500 BCE (Julian). While this difference is small compared to other uncertainties [i.e. variables], it could on very rare occasions affect the starting date [i.e. variable] of the lunar months [i.e. variable].[167]

XX. Casperson emphasizes the errors that are inherent in applying Neugebauer's procedure for calculating the astronomical coordinates of the Moon at the beginning of the month [i.e. trying to geometrically calculate the lunar crescent visibility of the New Moon]. Equally important is the problem of estimating the visibility of the Moon [i.e. variable] after its coordinates [i.e. variable] are known.[168]

XX. The most detailed data, mentioned above, are those derived by Schoch from hundreds of ancient crescent observations made in Babylon. To a first approximation, the minimum height [i.e. variable] for visibility [i.e. variable] should be a quadratic function of the Sun-Moon Azimuth difference [i.e. Δ (Delta): variable], and a formula that agrees very well with the data for Babylon – according to Casperson – is:[169]

$$(h) == (10.4) - (0.0106\Delta^2) \text{ (Babylon)}$$

where the result is expressed in degrees.[170]

163 Casperson 1986: 142-143
164 Casperson 1986: 143
165 Casperson 1986: 143
166 Casperson 1986: 143
167 Casperson 1986: 143
168 Casperson 1986: 143
169 Casperson 1986: 143
170 Casperson 1986: 144

XX. Maimonides developed a simple procedure for estimating the visibility of the Moon in Jerusalem, and his procedure was transformed into modern notation by Fotheringham (1910).[171] It is noteworthy that Fotheringham cites a German translation of Maimonides' Sanctification of the New Moon (i.e. *Kiddush HaChodesh*), with astronomical comments, by von Littrow (1872),[172] and therefore elements of the original Hebrew may or may not have been lost in translation into Fotheringham's English through the German translation that he read.

XX. Fotheringham comments that Maimonides, who though silent as to the observations on which his conclusions rest, gives a detailed rule for determining the date of the first visibility of the Moon in Palestine. Fotheringham notes that there is no reason for doubting that the rule given by Maimonides is the result of trustworthy observations.[173]

XX. Fotheringham reasons that the result – according to Maimonides' theory – depends partly on the true elongation [i.e. variable] or difference in true longitude [i.e. variable] between Sun and Moon [i.e. variables], and partly on the apparent angle of vision [i.e. arcus visionis: variable] at the moment [i.e. variable] when the Moon might be expected to appear. Fotheringham asserts that this moment is – according to Maimonides – on an average [approximately] twenty (20) minutes after Sunset. By the angle of vision – Fotheringham reckons – Maimonides appears to mean the apparent difference in zenith distance [i.e. variable] between Sun and Moon.[174]

XX. Fotheringham summarizes Maimonides' conclusion as follows:

1.) If the angle of vision [i.e. arcus visionis: variable] exceeds 11°, the Moon is visible;

2.) If the angle of vision is between 10° and 11°, the Moon is only visible if the elongation exceeds 12°;

3.) If the angle of vision is between 9° and 10°, the Moon is only visible if the elongation exceeds 13°;

4.) If the angle of vision is less than 9°, the Moon is only visible if the elongation exceeds 24°.[175]

XX. Fotheringham notes that the conditions for observation [of the lunar crescent visibility] are slightly more favorable at Jerusalem than at Athens [whose data Fotheringham studied]. Maimonides also gives

171 Casperson 1986: 144
172 Fotheringham 1910: 121
173 Fotheringham 1910: 121
174 Fotheringham 1910: 121
175 Fotheringham 1910: 121

rules for computing the Moon's elongation [i.e. variable] and angle of vision [i.e. variable], but Fotheringham asserts that they are very inaccurate compared with the methods of modern astronomy.[176]

XX. Fotheringham's data for Jerusalem can be represented approximately by the formula:[177]

(h) == (11.5) – ($0.008\Delta^2$) (Jerusalem)

XX. Fotheringham's data for Athens can be represented approximately by the formula:[178]

(h) == (12.0) – ($0.008\Delta^2$) (Athens)

XX. Modern observations from many parts of the world have also yielded good agreement with the Athens data.[179]

XX. The principal conclusion to be drawn from the previously described data is that the Babylonians were able to see the lunar crescent under conditions of height [i.e. variable] and azimuth difference [i.e. variable] where it might not have been visible to observers elsewhere. This fact has been attributed to the relatively clear air of Babylon [i.e. variable], and to the possibility that the Babylonians – unique among ancient astronomers – may have been able to calculate the position at which the [lunar] crescent should be found.[180]

XX. It is also possible, however, that calculations may sometimes have been used in place of observations. For other locations and for Egypt in particular, the Athens or Jerusalem data probably provide our best estimate of the conditions under which an ordinary sharp-sighted observer might be able to see the lunar crescent. In commenting on observations made in Egypt, Schoch has remarked that "the Egyptian sky is far worse than the Babylonian".[181]

XX. In setting up his tables of [lunar] crescent visibility, Neugebauer incorporated Schoch's visibility criteria [i.e. variables] for Babylon. Parker used those tables to calculate the visibility of the lunar crescent in ancient Egypt. Casperson believes that this procedure may exaggerate the observational capabilities of the Egyptians.[182]

0.8. Traditional Greek / Modern Geometric Astronomical Methodology for Chronological

176 Fotheringham 1910: 121
177 Casperson 1986: 144
178 Casperson 1986: 144
179 Casperson 1986: 144
180 Casperson 1986: 144
181 Casperson 1986: 144
182 Casperson 1986: 144

Calendrical Calculations is Problematic because of the Multitude and Complexity of Variables

XX. The first modern attempts to formulate a [geometric] crescent visibility criterion [i.e. variable] incorporated two (2) parameters [i.e. variables]:

1.) True Lunar Altitude [i.e. variable];

2.) Azimuthal difference between Sun and Moon at Sunset [i.e. variable].[183]

XX. The most commonly used criterion [i.e. variable] of this kind is the one of Schoch. Bruin chose a different ansatz [i.e. variable] in 1977, which was later modified [i.e. variable] by Schaefer (1988). Yallop (1997) tried to merge the approach of Bruin [i.e. variable] with the criterion of Schoch [i.e. variable], from whom he adopted the values [i.e. variables] for the minimal altitude of the Moon [i.e. variable] at a certain difference in Azimuth [i.e. variable]. At the same time he resorted to the moment of the best possible sighting ("best time") [i.e. variable] and the width of the lunar crescent [i.e. variable] defined by Bruin.[184]

XX. Yallop accounted for the lunar parallax [i.e. variable] and the topocentric width of the lunar crescent [i.e. variable] and introduced a parameter q [i.e. variable] that describes the threshold [i.e. variable] for a possible successful sighting.[185]

XX. John K. Fotheringham was the first to calculate old and new crescent in general (for earlier attempts see Casperson (1986); and Krauss (2006)). According to Fotheringham, the old or new crescent is to be considered visible if it has a minimal altitude h [i.e. arcus visionis: variable] which is dependent on the distance [i.e. difference; delta] Δ in Azimuth [i.e. variable] of the Sun and Moon at the moment [i.e. variable] when the center of the Sun is in the mathematical horizon [i.e. variable] – the position of the crescent [Moon] is computed geocentrically [i.e. geometrically from the perspective of the Earth at the center, thus geocentric], the position of the Sun [i.e. variable] is computed without refraction [i.e. variable]. Paul Viktor Neugebauer and Schoch improved upon Fotheringham's [criteria of] minimal altitudes [i.e. variables].[186]

XX. The visibility criteria [i.e. variables] of Schoch, Neugebauer, and more recently Krauss (2012) are based on [geometric] geocentric calculations for the instance of Sunset [i.e. variable]. However, it is impossible to observe the crescent at the moment of Sunset [i.e. variable], and observers are not situated [i.e. variable] at the Earth's center. Yallop (1997) introduced the concept of "best time" [i.e. variable] for observing the crescent Moon and this concept [i.e. variable] has also been adopted [by

183 Gautschy 2014: 79
184 Gautschy 2014: 79
185 Gautschy 2014: 79-80
186 Krauss 2009: 138

Gautschy (2014)].[187]

XX. It has been customary to discuss the first and last appearance [i.e. variables] of stars and planets [i.e. variables] in terms of the arcus visionis [i.e. variable]. One assumes that a star or planet becomes visible when the arcus visionis exceeds a limiting value [i.e. variable] that depends on its brightness (visual magnitude) [i.e. variable]. The arcus visionis of Venus [i.e. variable] at first/last appearance [i.e. variables] is defined as the distance [i.e. variable] between Venus and the Sun measured perpendicular to the horizon [i.e. variable].[188]

XX. As pointed out by De Jong, it is important to notice the large variations [i.e. variables] in ecliptic latitude [i.e. variable] of Venus at EL [i.e. variable] and MF [i.e. variable] when Venus is close [i.e. variable] to the Earth [i.e. variable] near to its inferior conjunction [i.e. variable] with the Sun [i.e. variable].[189]

XX. This [arcus visionis] method was already used – and may in fact have been introduced – by Claudius Ptolemy in his Almagest (~130 CE (Julian)). However in reality the visibility of a star or planet is not determined only by the relative geometry [i.e. variable] of the star or planet [i.e. variables] and the Sun with respect to the horizon [i.e. variable], but depends additionally on a number of [other variables of] physical parameters: the brightness of the object [i.e. variable], the brightness of the sky [i.e. variable], atmospheric refraction [i.e. variable] and [atmospheric] extinction [i.e. variable], and the sensitivity of the human eye [i.e. variable].[190]

XX. Early models attempting to take these physical factors [i.e. variables] into account were published by Bruin (1979) and Schaeffer (1985). De Jong uses the approach followed in the more detailed model [i.e. more variables] in the program developed by his former student, Frank Inklaar (1989) to include not only the aforementioned variables, but also allowing the computations of the visibility time (NA-SU / *na-su*) variable of Venus on the dates of first appearance [i.e. variable].[191]

XX. For De Jong's methodology, Frank Inklaar (1989) had just finished a master thesis in which he had developed a new method to compute dates of first and last appearance of stars and planets based on a physical [i.e. geometrical] model of stellar visibility. In his thesis he applied this method to an interpretation of the observations of rising stars in MUL.APIN inspired by the earlier work of Bartel van der Waerden (1949).[192]

XX. According to Hunger and Pingree (1989) MUL.APIN is a composite text with many different

187 Gautschy 2014: 87
188 De Jong 2012: 395
189 De Jong 2012: 395
190 De Jong 2012: 395
191 De Jong 2012: 395, 406, 407
192 De Jong 2007: 107

sources.[193] Their edition of MUL.APIN is based on about 40 (fragments of) copies of the text. The oldest copy dates from 687 BCE (Julian) which provides a firm "terminus ante quem" [i.e. last/newest possible date].[194] Hunger and Pingree favor a date around 1000 BCE (Julian) for the composition of the text of MUL.APIN but allow for the possibility that certain parts of the text may be several centuries older.[195]

XX. The most quantitative attempt to date one particular part of the text of MUL.APIN is by van der Waerden (1949). He based his dating on an analysis of the calendar dates of the first appearance of 34 stars and/or constellations in MUL.APIN. By comparing with computed dates of first appearances he concluded that these observations must have been carried out somewhere between 1400 and 900 BCE (Julian). In his paper (2007), De Jong follows van der Waerden's approach in a new attempt to date the observations underlying the rising star list in MUL.APIN.[196]

XX. Hunger and Pingree (1989) in the Astronomical Commentary to MUL.APIN identify six (6) lists of stars in the text, indicated as lists I – VI. Following van der Waerden (1949), De Jong (2007) discusses only those two (2) star lists that contain sufficiently quantitative information to allow an attempt to astronomically date the observations on which they are based. Star list II contains dates of the first appearances of 34 stars and/or constellations in an ideal lunar calendar containing 12 months of 30 days each (Tablet I, column ii, line 36 – column iii, line 12). Star list IV contains intervals in days between dates of first appearance of 16 stars or constellations (Tablet I, column iii, lines 34 – 48). The stars in list IV are a subset of those in list II and the intervals given are consistent with the dates in list II.[197]

XX. Since the orientation of the sky with respect to the horizon slowly varies due to precession[! – i.e. variable] a comparison between observed and computed dates in principle allows a determination of the epoch of observation.[198] Van der Waerden (1949) concluded that the observations [found in MUL.APIN] were most probably carried out between 1400 and 900 BCE (Julian), and in Babylon rather than in Nineveh.[199]

XX. In De Jong's version of this [astronomical computer program] model the magnitude threshold of human vision [i.e. variable] as a function of sky background brightness [i.e. variable] is taken from Tousey and Koomen, the sky brightness in Babylon [i.e. variable] as a function of solar depression [i.e. variable] below the horizon [i.e. variable] is adapted from measurements of the sky brightness [i.e. variable] in Maryland (USA) [i.e. variable], and atmospheric refraction [i.e. variable] as a function of

193 Hunger and Pingree 1989: 10 – cited by De Jong 2007: 107
194 De Jong 2007: 107
195 Hunger and Pingree 1989; see also Hunger and Pingree 1999 – cited by De Jong 2007: 107-108
196 De Jong 2007: 108
197 De Jong 2007: 109
198 De Jong 2007: 110
199 De Jong 2007: 110

altitude [i.e. variable] is computed according to an algorithm [i.e. variable] taken from *Astrophysical quantities*.[200]

XX. De Jong calculates variable positions [i.e. variable] of the ecliptic longitude of the Sun [i.e. variable] as well as the ecliptic longitude and latitude of Venus [i.e. variables] at the moment of its first/last appearance [i.e. variables]. De Jong computes these positions using algorithms to compute the positions of the Sun and the planets back to 4000 BCE (Julian).[201]

XX. In his computations, De Jong applies a time correction ΔT [i.e. variable] to convert Universal Time [UT, i.e. variable] into Ephemeris Time [ET, i.e. variable], i.e. $\Delta T = E.T. – U.T. = 32.5\ t^2$ seconds, where t is the time measured [i.e. variable] in centuries counted from 1800 CE (Gregorian).[202] According to De Jong, this correction [i.e. variable] must be applied because of the slow gradual deceleration of the Earth rotation due to tidal friction [i.e. variable] and other geophysical effects [i.e. variable], and according to De Jong it amounts to 3–5 hours over the time-span covered by the Venus observations.[203]

XX. De Jong concludes that he has shown that the arcus visionis values [i.e. variables] of the planet Venus [i.e. variable] derived from Babylonian observations show two (2) asymmetries [i.e. two (2) variable inequalities]: most pronounced between first and last appearances [i.e. variables] but also between appearances near inferior (EL and MF observations [i.e. variables]) and near superior conjunction (ML and EF observations [i.e. variables]). It turns out that these asymmetries [i.e. inequalities] are due to the interplay of a number of different causes [i.e. even more variables!] (arranged in order of decreasing scale):[204]

1.) Solar system dynamics; the different relative velocity of Venus and the Sun at the near side and far side of its orbit as seen from Earth.[205]

2.) Terrestrial weather; the day-to-day variations in weather (clouds) and atmospheric extinction, including the effect of the 1–5 day correlation in the extinction.[206]

3.) The human factor; the observational selection effect that last appearance observations are easier to make than first appearance observations.[207]

XX. Like others before him, De Jong attempts to calculate Julian dates of ancient observations. These

200 De Jong 2012: 395
201 De Jong 2012: 394
202 De Jong 2012: 394
203 De Jong 2012: 394-395
204 De Jong 2012: 406
205 De Jong 2012: 406
206 De Jong 2012: 406
207 De Jong 2012:406

dates were determined by generating a theoretical Babylonian calendar based on the first visibility of the lunar crescent computed using the geometric crescent visibility criterion [i.e. variable] of P. V. Neugebauer (1929). According to Huber et al. (1982), for the astronomical texts this criterion [i.e. variable] is slightly superior to that [crescent visibility criterion (i.e. variable)] of Langdon et al. (1928) used by Parker and Dubberstein (1956) for their Babylonian chronological tables.[208]

XX. Therefore, as is clearly evident, De Jong's methodology (as well as all those before him) of attempting to calculate chronology using the geometric arcus visionis depends upon too many variables (i.e. too many "moving parts") of astronomical and atmospheric parameters that each most certainly affects the accuracy of calculations.

XX. All of the above variables that are necessary for the computation of celestial positions of celestial objects via the traditional Greek geometrical method (since Ptolemy) of attempting to measure the angle of the arcus visionis (or arcus apparentus) can produce small errors that accumulate the farther we go back in time, and thus provide a high level of uncertainty for calculations of ancient dates, eclipse times, and eclipse locations. After a careful and thorough examination, this brings one to the reasonable and logical conclusion that there are too many variables (i.e. too many "moving parts") and too much room for uncertainty with this traditional Greek / Modern method and methodology.

XX. Another better arithmetic / algebraic methodology with its more logical and reasonable methods will be unfolded throughout. It seems plausible to test the theory that such a methodology inherently precludes many of the variables of uncertainty that are necessary for Ptolemy's traditional geometrical method of computing planetary positions (i.e. more specifically the Moon and ancient dates), but also it is quite plausible that this new (ancient Babylonian / Hebrew) methodology provides a better method to be able to recalibrate more accurate values for ΔT for ancient dates– i.e. this methodology therefore simplifies and provides more accurate calculations.

XX. As should already be evident – and will be unfolded further – there are so many variables of uncertainty in such precise astronomical calculations (each of which require precise measurements), it will be recognized that the current geometric methodology is riddled with problems when considering ancient dates. After a review of just some of these variable uncertainties which will affect the accuracy of astronomical calculations, a better methodology will be presented to be repeated, tested, and proven (or disproven) scientifically.

XX. A simpler, better, and more elegant methodology must be found – and indeed already exists.

0.9. The Ancient Babylonian (Simpler) Algebraic Methodology: Babylonian Astronomy and Calendars

208 De Jong 2012: 394

XX. Like no other cultural achievement, astronomy stands out as a legacy of Babylon.[209] Babylonian astronomy is a strongly interdisciplinary area of research.[210] Indeed, in the course of the last few decades it has become increasingly clear that all Western efforts in the exact sciences are descendants in direct line from the work of the Late Babylonian astronomers.[211] Our present real knowledge of Babylonian astronomical science is based chiefly on the work of three (3) Jesuit scholars:[212]

1.) Assyriologist J. N. Strassmaier;

2.) Astronomer J. Epping;

3.) Astronomer F. X. Kugler;

XX. In no domain has the influence of ancient Mesopotamia on Western civilization been more profound and decisive than in theoretical astronomy and – principally through it – mathematics.[213] Babylonian mathematical astronomy is, in its core, perceived as a form of pure mathematics.[214] A concept of negative numbers exists neither in Babylonian mathematical astronomy nor in Babylonian mathematics.[215]

XX. The deciphering of the cuneiform script and the languages written therein was one of the great achievements of the 19th century CE (Gregorian) which brought to light an entirely lost period of history and culture. Previous knowledge – mostly from Greek sources (mutilated and anecdotal) – did not go back beyond about 700 BCE (Julian).[216] Before the cuneiform script was deciphered in the middle of the 19th century CE (Gregorian) the only available sources of information concerning Babylonian astronomy were the Bible and the works of several Greek and Roman scholars.[217]

XX. The distinction customarily made between the two (2) chief astronomical traditions of antiquity is that Greek astronomy was geometrical, whereas Babylonian astronomy was arithmetical.[218] Two (2) features of Babylonian theoretical astronomy seem particularly peculiar to a modern eye:

1.) first is the total absence of any underlying geometrical models or “orbits”;

209 Ossendrijver 2011: 213
210 De Jong 2007: 107
211 Aaboe 1991: 276
212 Pannekoek 1961: 30
213 Aaboe 1991: 276
214 Ossendrijver 2016b: 336
215 Høyrup 1993 – cited by Ossendrijver 2016b: 338
216 Pannekoek 1961: 29
217 Ossendrijver 2012: 3
218 Jones 1991: 441

2.) secondly is its choice of independent variable.[219]

XX. In ancient Greece and indeed up to the time of Johannes Kepler (1571 – 1630 CE (Julian)), all astronomers assumed the orbital motions of Sun, Moon, planets, and stars either to be circular or a combination of circular motions. Modern astronomy has removed the stars from orbiting Earth and has them orbit the galactic center, which itself moves with respect to other galaxies.[220]

XX. In Babylonian astronomy, the planetary and lunar models (as we still call them), are entirely arithmetical in character. The mathematical skills required for the computation of a [Babylonian] ephemeris[221] [i.e. a table showing the positions of a heavenly body on a number of dates in a regular sequence] are limited to the four (4) basic operations of arithmetic:[222]

1.) addition;

2.) subtraction;

3.) multiplication;

4.) division;

XX. That is to say, the Babylonian astronomers of the last five (5) centuries BCE (Julian) devised elaborate combinations of arithmetical sequences to predict the apparent motions of the heavenly bodies, while the Greeks persistently tried to explain the same phenomena by hypothesizing kinematic models compounded out of circular motions.[223]

XX. The anonymous creators of Babylonian theoretical astronomy, probably of the 5th or 4th century BCE (Julian), drew their essential ingredients from several branches of learning and literature – chief among them mathematics – and for observations: the Astronomical Diaries, closely linked to the celestial omen texts.[224]

XX. Ossendrijver cites Neugebauer (1938) who wrote, “Here [in Babylonia], for the first time in the history of mankind, one has succeeded in controlling (i.e. German: *beherrschen*) the laws of a very complicated natural phenomenon through purely mathematical methods.”[225] Hence Babylonian mathematical astronomy is – in its core – perceived as a form of pure mathematics.[226]

219 Aaboe 1991: 285
220 Kelley and Milone 2011: 30
221 Plural: ephemerides
222 Aaboe 1991: 285
223 Jones 1991: 441
224 Aaboe 1991: 276
225 Ossendrijver 2016b: 336
226 Ossendrijver 2016b: 336

XX. Elsewhere [Neugebauer (1946)] expressed the primacy of mathematics in ancient astronomy using even more drastic terms, “For methodological reasons it is obvious that a drastic restriction in terminology must be made. We shall here call ‘astronomy’ only those parts of human interest in celestial phenomena which are amenable to mathematical treatment.”[227]

XX. During the last seven (7) and half centuries BCE (Julian), Babylonian astronomers kept systematic records of night-by-night observations of a range of astronomical phenomena. From these observations, lunar and planetary cycles were identified which were used to make predictions of future astronomical phenomena from previous observations.[228]

XX. During the 1st millennium BCE (Julian), Babylonian astronomy encompassed the careful and systematic observation of astronomical phenomena, the identification of planetary and lunar cycles that allowed future phenomena to be predicted from past observations, and the development of a theoretical astronomy which used mathematical methods to calculate astronomical phenomena.[229]

XX. These different aspects of Babylonian astronomy coexisted throughout the 2nd half of the 1st millennium BCE (Julian).[230] Mathematical astronomy did not replace the prediction using [arithmetic] cycles, nor did[231] prediction and calculation replace observation. Citing Rochberg, Steele notes that all three (3) types of astronomy (observational, mathematical, and predictive) were practiced by the same scribes.[232]

XX. Babylonian mathematical astronomy comprises about 440 cuneiform tablets and fragments from Babylon and Uruk, all written within the period 450 – 50 BCE (Julian). This makes Babylonian mathematical astronomy the earliest known form of mathematical astronomy of the ancient world.[233] A distinctive feature of these texts, which represent only a small fraction of the Babylonian astronomical corpus, is that astronomical quantities are computed with mathematical algorithms.[234]

XX. The corpus can be divided into tabular texts, currently numbering about 330, and procedure texts numbering about 110.[235] The former contain computed data for the Moon and the planets, e.g. positions and times, arranged into columns and rows [i.e. ephemerides]; the latter contain detailed instructions for computing and verifying these tables [i.e. procedure texts].[236]

227 Ossendrijver 2016b: 336
228 Steele 2015: 1855
229 Steele 2015: 1855
230 Steele 2015: 1855
231 Steele 2015: 1855
232 Steele 2015: 1856
233 Ossendrijver 2016b: 333
234 Ossendrijver 2016b: 333
235 Ossendrijver 2016b: 333-334
236 Ossendrijver 2016b:334

XX. Babylonian mathematical astronomy is attested in the form of tables with computed quantities and procedure texts with the corresponding computational instructions.[237] The partly overlapping skills required for mathematics suggests that astral science and mathematics were often pursued by the same individuals.[238]

XX. In ancient Mesopotamia, the experts who observed and studied the sky and compiled astronomical texts ("astronomers") were one-and-the-same as those who observed the sky in search of omens ("astrologers").[239] Babylonian astronomy and astrology were [then] transmitted to the Greco-Roman world, where the word "Chaldaean" – which originally designated the inhabitants of Babylonia – became synonymous with "astrologer".[240]

XX. Two (2) further major innovations of the astral sciences occurred in the late Achaemenid era.[241] Sometime toward the end of the 5th century BCE (Julian),[242] the [Babylonian mathematical] zodiac was invented as a 2nd system for recording positions in the sky.[243]

XX. All computations are based on the sexagesimal place-value system which, along with some of the mathematical terminology, were handed down from Old Babylonian mathematics,[244] which had flourished more than a 1000 years earlier.[245] Babylonian astronomers used the sexagesimal place-value notation, which means that numbers were represented as sequences of digits, each assuming a value between 0 and 59 [i.e. base 60]. Successive digits denote successively decreasing powers of 60, entirely analogous to our decimal system [which is base 10].[246]

XX. The Babylonians had successfully constructed mathematical models to predict a number of phenomena beginning in the Old Babylonian period.[247] The [Astronomical] Diaries, as well as the tradition that they initiated that culminated in the mathematical astronomy of the Astronomical Cuneiform Texts (ACT) material [studied and published by Neugebauer 1955], attest to the vigor of the Babylonian belief in an ordered universe.[248]

XX. Further developments in Babylonian astral science are connected to the introduction of the zodiac

237 Ossendrijver 2015a: 9
238 Ossendrijver 2013b: 10
239 Horowitz 1996: 40
240 Ossendrijver 2013a: 6072
241 Ossendrijver 2013b: 8; Ossendrijver 2015a: 9
242 Hunger and Pingree 1999: 17; Steele 2015: 1856
243 Steele 2015: 1856
244 ca. 1894 – 1595 BCE (Julian) Middle Chronology; 1830 – 1531 BCE (Julian) Low Chronology
245 Ossendrijver 2016b:334
246 Ossendrijver 2015b: 1865
247 ca. 1894 – 1595 BCE (Julian) Middle Chronology; 1830 – 1531 BCE (Julian) Low Chronology
248 Hunger and Pingree 1999: 140

near 400 BCE (Julian).[249] Within a few decades, mathematical astronomy emerged as a second method for predicting astronomical phenomena alongside the [more ancient] Goal-Year method.[250]

XX. In 2018, Ossendriver revises not only the number of tablets and fragments to be approximately 450, but also the dates to be between about 400 BCE and 50 BCE (Julian).[251]

XX. In these texts, essentially the same phenomena known from the Astronomical Diaries and the Goal-Year method are computed by means of mathematical algorithms that are based on the sexagesimal place-value notation known from Old Babylonian mathematics.[252] [253]

XX. Astronomical phenomena could be predicted either with the existing Goal-Year method or with the newly developed algorithms of mathematical astronomy.[254] [That is to say: Within a very short period] mathematical astronomy emerged as a new method for predicting essentially the same astronomical phenomena as possible with the Goal-Year method.[255]

XX. The corpus of mathematical astronomy stands out in that it is strictly dedicated to astronomical prediction, with no reference to observations or astrological interpretation. Furthermore, while the diaries do indirectly reveal potential astrological applications, as they include reports of terrestrial phenomena, the tablets with mathematical astronomy only contain computed astronomical phenomena.[256]

XX. The separation between computed tables and verbal instructions within this corpus reflects a differentiation between concrete numerical predictions that were produced for specific dates and the underlying procedural knowledge which is generally applicable and not connected to any particular date.[257]

XX. It is easy to imagine the pragmatic advantages of such an arrangement, because the procedures were meant to be applied again and again with different initial conditions when ever a table had to be computed for a new range of dates.[258]

XX. Hence it made perfect sense to preserve these rules on separate tablets and, presumably, store them at a different location in the astronomer's library. The distinct status of the two (2) genres is confirmed

249 Ossendrijver 2018b: 76
250 Ossendrijver 2018b: 76
251 Ossendrijver 2018b: 76
252 ca. 1894 – 1595 BCE (Julian) Middle Chronology; 1830 – 1531 BCE (Julian) Low Chronology
253 Ossendrijver 2018b: 76
254 Ossendrijver 2013b: 8
255 Ossendrijver 2013b: 8
256 Ossendrijver 2018b: 76
257 Ossendrijver 2018b: 76
258 Ossendrijver 2018b: 76-77

by the fact that relatively few tabular texts can be shown to be copies, while the opposite is true for the procedure texts.[259]

XX. A remarkable feature of Babylonian mathematical astronomy is the simultaneous existence of different computational systems (sets of algorithms) for predicting the same phenomena of a given planet or the Moon. These different systems form two (2) main families known as type A and type B.[260]

XX. By about 330 BCE (Julian), the complex lunar algorithms known as Systems A and B had reached their final state of development, with little evidence of further change. Most of the available tablets were written after 250 BCE (Julian) and reflect that final stage. About 300 tablets of this corpus are numerical tables with computed predictions.[261]

XX. The most common class are the so-called synodic tables, which contain data for the synodic phenomena of the Moon (Full Moon and New Moon) or a planet (first appearances, last appearances, stations and oppositions). Another class is formed by about forty (40) daily motion tables, in which the position of the Moon, the sun or a planet is tabulated from day to day.[262]

XX. The remaining 110 tablets are procedure texts, which consist of verbal instructions mainly concerned with computing or verifying the aforementioned tables. Except for a few planetary tables that include the corresponding procedures at the end of the tablet, the procedure texts usually occupy different tablets.[263]

0.10. Ancient Babylonian Astronomers

XX. Little is known about the astronomers who wrote the [Astronomical] Diaries during the [Late Neo-Assyrian //] Neo-Babylonian era (ca. 750 – 538 BCE (Julian)).[264] Next to nothing is known about the astronomers of the Neo-Babylonian era, since we have only anonymous observational texts from that period.[265] We do not know – for instance – whether they were primarily associated with the court like the astrologers of the [earlier and immediately preceding] Neo-Assyrian era, or [primarily associated] with the temples as is evident in the later periods.[266]

XX. In the following Late Babylonian period, when Babylonia was successively ruled by the

259 Ossendrijver 2018b: 77
260 Ossendrijver 2015b: 1865
261 Ossendrijver 2018b: 76
262 Ossendrijver 2018b: 76
263 Ossendrijver 2018b: 76
264 Ossendrijver 2016a: 128
265 Ossendrijver 2011: 213
266 Ossendrijver 2011: 213

Achaemenids [i.e. Persians],[267] [Macedonian] Alexander the Great and his dynasty,[268] the [Greek] Seleucids[269] and the Parthians [i.e. Persians],[270] most astronomers are known to have been employed as temple priests [and this would include the time of Daniel during the early years of the Achaemenid Persian period which would have been the final years of his life]. In Babylon they were associated with the Esagila, temple of the supreme god Bēl (Marduk), [and] in Uruk with the Rēš, temple of the skygod Anu.[271]

XX. Observation of some celestial phenomena was necessary for the calendar. To fix the exact first day of the month, the New Moon had to be caught at its first appearance. This was not so very difficult: In that marvelously radiant climate [of Babylon] (some winter months and occasional sand-storms excepted) with its clear visibility across the wide plains from horizon to horizon, the [Babylonian] priest-astronomers, from their terraced towers, could easily detect the first slender lunar arc in the [western] evening sky.[272]

XX. Babylonian priest-astronomers also had to give attention to the Full Moon for ceremonial purposes, and to observe (for the sake of prediction) the last appearance of the morning crescent [of the Old Moon] toward the end of the [lunar] month. It is highly probable – and this is supported by some texts – that this regular observation of the Moon must gradually have led to an increasing practical concern with the stars.[273]

XX. Can one imagine that in so doing they would not have noticed the stars silently tracing their courses, progressing onward every successive month [westward]? Or that they were not struck by the brilliant planets appearing occasionally among them, and most of all by the peerless evening star?[274]

XX. The first phenomenon manifesting itself in the observation of the young Moon is the regular progression and shifting of the constellations [westward] in the course of a year. The stars visible in the western evening sky are characteristic of the season, hence they can verify the calendar and the intercalations (this also applies to the first appearance of the stars in the eastern morning sky). Intercalation by means of the regular heliacal rising and setting of stars must therefore have gradually superseded the irregular empirical method of observing the crops.[275]

XX. This knowledge of the stars persisted [beginning with the Sumerian period (or perhaps earlier), followed by the Old Babylonian period] through the succeeding centuries of political decay when

267 538 – 331 BCE (Julian)
268 330 – 312 BCE (Julian)
269 311 – 146 BCE (Julian)
270 145 – 100 CE (Julian)
271 Ossendrijver 2016a: 128
272 Pannekoek 1961:31
273 Pannekoek 1961:31
274 Pannekoek 1961:31
275 Pannekoek 1961:31

Babylon was dominated by eastern conquerors, the Cassites [Kassites], and the western countries were a battlefield of Egyptian and Hittite expansion.[276]

XX. Babylonian astronomers were primarily interested in the Moon, the [visible] planets (Mercury, Venus, Mars, Jupiter, Saturn), and the Sun.[277] The Moon was studied more intensely than any other celestial body.[278] According to Ossendrijver, the [ancient Babylonian] algorithms for the Moon are – by far – the most complicated part of Babylonian mathematical astronomy.[279]

XX. The stars mainly served as [background] reference objects in the [Astronomical] Diary-type texts for reporting the motion of the wandering bodies. A more or less fixed group of 32 stars, known by the modern term "Normal Stars", was used for that [purpose serving as quasi-fixed (albeit slowly rotating westward) background reference points]. In mathematical astronomy, stars were not considered at all.[280]

XX. Apart from lunar and planetary positions, a lot of effort was spent on synodic phenomena which are conspicuous events during the apparent motion of a planet or the Moon around the Earth. They form a periodically repeating sequence [i.e. repeating pattern of motion as viewed from the Earth] which is distinct for inner planets (Mercury, Venus) and outer planets (Mars, Jupiter, Saturn).[281]

XX. In the Astronomical Diaries [the Moon's] position was reported from day to day. Around the synodic phenomena of New Moon (when Moon and Sun are in conjunction), and Full Moon (when they are in opposition), eclipses were observed and so-called Lunar Six intervals were measured.[282]

XX. At the end of each monthly paragraph [in the Astronomical Diaries] we find a summary of the positions and phenomena of the planets.[283] If a planet moved from one (1) zodiacal sign into another in the course of a month, this is expressed by: "planet X was in sign A, at the end of the month in sign B". Later in the Seleucid period, this is replaced by "on day *n*, planet X reached sign B".[284] Planetary observations – some of them from the 6th century BCE (Julian) – are also found in the LBAT material.[285]

XX. The evidence from Babylon therefore suggests that the various astral disciplines were pursued by a single community of collaborating scholars[286] [which evidently would have included

276 Pannekoek 1961:32
277 Ossendrijver 2016a: 128
278 Ossendrijver 2016a: 130
279 Ossendrijver 2012: p. vii
280 Ossendrijver 2016a: 128
281 Ossendrijver 2016a: 128
282 Ossendrijver 2016a: 130
283 Sachs and Hunger 1988: 25
284 Sachs and Hunger 1988: 25-26
285 Sachs and Hunger 1988: 13
286 Ossendrijver 2011: 214

Daniel]. [Even in ancient times,] Babylonian astronomy [was] a strongly interdisciplinary area of research [and occupation].[287]

0.11. The Concept of Time in Ancient Mesopotamia

XX. Time is one of the most basic categories – if not the most basic one – in the rational matrix by which human beings create meaning in the world.[288] From remote antiquity the celestial bodies – the Sun, Moon, and stars – have been the fundamental reckoners of time. The rising and setting of the Sun and the stars determine the day and night; the phases of the Moon determine the month; and the positions of the Sun and stars along the horizon determine the seasons.[289]

XX. This apparent motion of the Sun [observed against the apparent motion of a westwardly-spinning celestial sphere] as viewed by an earthbound observer provided the earliest timekeeping standard for human kind. The Sun moves from East to West, and night follows day with predictable regularity. Accordingly, the day is the basic unit of time underlying all calendars. However, different calendars use different conventions to structure days into larger units of weeks, months, years, and cycles of years.[290]

0.12. The UŠ ("USH")

XX. [In ancient Babylon,] time intervals shorter than one (1) day were usually expressed in UŠ[291] [292] The Babylonian unit of time was the [Akkadian from Sumerian:] UŠ and was precisely equal to 4 minutes. This is the time for the [360-degree] celestial sphere to rotate through one (1) degree in right ascension, and so it has become customary to translate UŠ as "degree".[293] 1/60 of one (1) UŠ is called one (1) NINDA [which is equivalent to modern-day division of the degree into 60 arcminutes per degree (which can be further divided into 60 arcseconds per arcminute)]. These units are well known from other astronomical texts (see ACT pp.39f).[294]

XX. 360 UŠ correspond to one (1) day (i.e. one (1) UŠ == four (4) minutes). This unit must usually be inferred from the context because the sign UŠ was rarely written after the number. [Though this Sumerian logogram adopted by the Akkadian was pronounced (and is read today) "ush" or "oosh" or "ouche" in the more ancient Sumerian,] its Akkadian reading and literal meaning are unknown.[295]

287 De Jong 2007: 107
288 Ben-Dov, Horowitz, Steele 2012: 1
289 Nelson et al. 2001: 510
290 Dershowitz and Reingold 1997: 6; Dershowitz and Reingold 2008: 4
291 Pronounced: "ush" or "oosh" or "ouche"
292 Sachs and Hunger 1988: 16; Ossendrijver 2016a: 132
293 Sachs and Hunger 1988: 16; Steele, Stephenson, and Morrison 1997: 338
294 Sachs and Hunger 1988: 16
295 Ossendrijver 2016a: 132

XX. In the [current Assyriology] literature the UŠ is also referred to as the "time degree", but [in Ossendriver's translations (including of VAT 4956) published in 2017] it is left untranslated. The next larger unit of time is the [Akkadian:] *bēru* ([Sumerian:] DANNA), equivalent to 30 UŠ (2 hours). It is sometimes translated as "double hour", but left untranslated [by Ossendrijver (2017)].[296]

XX. Both the *bēru* and the UŠ were originally units of geographical distance, i.e. ancient units of length that were transferred to astronomy from daily life. As a unit of time, the UŠ is not attested before about 1200 BCE (Julian), but as a unit of geographical distance, corresponding to about 360 meters, it can be traced back at least to the Ur III period (2100 – 2000 BCE (Julian)), as is [also] true for the *bēru* (DANNA), "mile".[297]

XX. For early records (before about 560 BCE (Julian)) – with one (1) exception – it seems that the times given are rounded to the nearest 5 (or possibly 10 in the earliest cases) degrees / UŠ. However, Steele notes that after 562 BCE (Julian) [i.e. Nebuchadnezzar's 43rd (FINAL) Regnal Year which began Spring of that year],[298] times are consistently recorded to the nearest degree / UŠ. It is not known exactly how the Babylonian astronomers made their measurements of time, but it seems that some form of clepsydra (water clock) was used.[299]

XX. Although the Babylonian astronomers rounded the time-intervals that they determined for these phenomena to the nearest four (4) minutes [i.e. to the nearest degree / UŠ], the real accuracy they achieved – according to Stephenson and Fatoohi – was far less than this. Typical errors of at least ½ hour in measuring intervals of no more than six (6) hours represents a poor performance by any reasonable standards – even allowing for the occasional scribal error. Not until medieval [Islamic] Arab astronomers introduced altitude measurements as an alternative technique to direct timing did the measurement of time in any part of the world significantly improve.[300]

XX. It has been learned from LBAT 1494 and LBAT 1495 as well as elsewhere that the Babylonians employed gnomons [i.e. sundials] to tell the time of day. From Mesopotamia the use of the gnomon spread to the Greeks, among whom either Anaximander, or his pupil Anaximenes, is alleged to have been the first to set up a gnomon. The Greek indebtedness to Mesopotamia is acknowledged by Herodotus ([*Histories*] II 109, 3).[301]

XX. It is of course also possible to measure time at night by means of fixed stars. Lists of *ziqpu* stars which give the time in UŠ between successive culminations of stars could be used for exactly this

296 Ossendrijver 2016a: 132
297 Ossendrijver 2016a: 132
298 i.e. the time that Daniel was purportedly serving in Nebuchadnezzar's royal court
299 Steele, Stephenson, and Morrison 1997: 338
300 Stephenson and Fatoohi 1993: 267
301 Hunger and Pingree 1999: 80

purpose. Unfortunately, most time intervals recorded in the [Astronomical] Diaries – especially those concerning the Moon – fall around Sunrise or Sunset when only few stars are visible. On the other hand, the inaccuracy of time measurement seems to have been of little importance for the development of Babylonian astronomy.[302]

0.13. Artificial-Day Units

XX. In [ancient Babylonian] mathematical astronomy, dates and time intervals of planetary phenomena were usually computed in an ["ideal"] artificial unit corresponding to 1/30 of the mean synodic month (29.53 days) [i.e. artificial-day units of less than 24 hours per unit]. In the [Assyriology] literature this unit is referred to as the (mean) *tithi* – a term borrowed from Indian astronomy – but the Babylonian astronomers simply called it "day" (Akkadian: *ūmu* [cf. Hebrew: *yom*; Arabic: *yawm, yom*]).[303]

XX. By using mean tithis instead of [actual 24-hour] days the [ancient Babylonian] astronomers circumvented the computation of the varying lengths of future months, which would be required if predicted dates were to be expressed in the actual calendar [using actual days of actual 24-hour day-lengths within the mean synodic month of 29.53 days)].[304]

0.14. The Day

XX. Different calendars begin their day at different times. The Islamic, Hebrew, and ancient Babylonian days begin at sunset.[305] Dershowitz and Reingold note that day names for calendars are generally distinct, but stress that this is not always the case. For example: the day of the week is a calendar (in a trivial sense) with infinitely many days having the same day name[306] [every seven (7) days: Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday].

XX. The Babylonian calendar uses days, months, and years.[307] No weeks or units longer than a year are attested.[308] The number of days in a year is of no interest in the Babylonian calendar.[309]

0.15. The Week

302 Sachs and Hunger 1988: 16
303 Ossendrijver 2016a: 132
304 Ossendrijver 2016a: 132
305 Dershowitz and Reingold 1997: 6; Dershowitz and Reingold 2008: 4; Sachs and Hunger 1988: 15
306 Dershowitz and Reingold 1997: 6; Dershowitz and Reingold 2008: 5
307 Sachs and Hunger 1988: 13
308 Sachs and Hunger 1988: 13
309 Sachs and Hunger 1988: 14

XX. Given the thoroughly astronomical character of the Babylonian calendar, the continuous 7-day cycle known as the "week" – which is entirely disassociated from the period of a lunation [of the Moon] and that is in use throughout much of the world today – was unknown in [ancient] Mesopotamia.[310]

XX. The origin of the seven-(7)-day week is uncertain. As a continuously running, uninterrupted cycle, it comes to us as part of Jewish tradition.[311] The week is traceable to the [Hebrew] Bible and is thought to be a Hebraic innovation. The Jewish weekdays were counted, but planetary names for the days (based on an astrological week of Hellenistic [Greek] invention) subsequently entered the western European calendar.[312]

XX. A seven-(7)-day week is[313] almost universal today. [Though unknown in ancient Babylon,] in many cultures, the day of the week were named after the seven (7) "wandering stars" (or after the gods associated with those heavenly bodies), namely: the Sun, the Moon, and the five (5) planets visible to the naked eye (Mercury, Venus, Mars, Jupiter, and Saturn).[314] All seven (7) are manifestations of major deities of the [Babylonian] pantheon.[315]

XX. The number seven (7) had mystical and cosmological significance throughout the [ancient] Semitic cultures. It was used in the Babylonian and Assyrian calendars, though not as a continuous cycle. Although the seven-(7)-day week may have its origin in number mysticism, astronomy may have contributed to the mysticism.[316]

XX. The Sun, Moon, and [five (5)] naked-eye planets comprise the seven (7) "wandering stars"[317] Seven is a useful – if inexact – count of days between the four (4) Moon phases, and four (4) times seven (7) is a useful – if inexact – estimate of days from first to last visibility of the Moon in its phase cycle.[318]

XX. In the Greek world of the 3rd century BCE (Julian) there were seven (7) planetary gods:[319]

1.) Helios (the Sun);

2.) Hermes (Mercury);

310 Rochberg 1995: 1938
311 Seidelmann 1992: 577
312 Rochberg 1995: 1938
313 Dershowitz and Reingold 1997: 6; Dershowitz and Reingold 2008: 5
314 Dershowitz and Reingold 1997: 7; Dershowitz and Reingold 2008: 5
315 Ossendrijver 2015a: 3
316 Seidelmann 1992: 577
317 Seidelmann 1992: 577
318 Seidelmann 1992: 577-578
319 Kelley and Milone 2011: 36

3.) Aphrodite (Venus);

4.) Selene (the Moon);

5.) Ares (Mars);

6.) Zeus (Jupiter);

7.) Kronos (Saturn);

XX. In some languages – e.g. Hebrew, Arabic, and Portuguese – days of the week are just numbered [1, 2, 3, 4, 5, 6, with day 7 named *Shabbat*, *Sabt*, *Sábado* respectively].[320] In Jewish practice, the days of the week were designated by numbers rather than by names, except that day 7 was known as the Sabbath. Recognition of the [seven-(7)-day] week and the observance of the Sabbath as a rest day gradually spread to the Roman world.[321]

XX. The use of names to designate the days of the week developed in the Roman culture of the 2nd and 1st centuries BCE (Julian). These names come from the astrological practice of naming each day after the planet that governed the day. At that time the Sun and Moon were included among the planets. "Saturn's day" coincided with the Jewish Sabbath. The planetary names [for the days] were gradually – if reluctantly – accepted by the Jewish and Christian cultures.[322]

XX. It is a reasonable – but ultimately unprovable [according to Seidelmann] – assumption that the cyclic continuity of the [seven-(7)-day] week was maintained without interruption by religious authorities from its origin in Biblical times to the present day. Although ten (10) days were dropped from the Christian [Julian] calendar in the Gregorian Reform of 1582 CE (Julian / Gregorian), the cycle of weekdays was not disturbed. Thus on Thursday 04 October 1582 CE (Julian) of the Julian calendar was followed by Friday 15 October 1582 CE (Gregorian) of the Gregorian calendar.[323]

0.16. The Month

XX. Ancient man also needed to measure time periods shorter than the seasonally-based [solar] year, yet greater than several days, and [since the seven-(7)-day week was unknown in Mesopotamia,] for this he [ancient man] looked to the Moon which he observed endlessly repeating its cycle of waxing

320 Dershowitz and Reingold 1997: 7; Dershowitz and Reingold 2008: 5
321 Seidelmann 1992: 578
322 Seidelmann 1992: 578
323 Seidelmann 1992: 578

and waning.[324] Like the day, the notions of "month" and "year" were originally based on observations of heavenly phenomena, namely the waxing and waning of the Moon, and the cycle of seasons respectively.[325]

XX. The month in ancient Babylon was observed as follows:

XX. The month begins after Sunset on the evening when the lunar crescent becomes visible for the 1st time after [its] Conjunction [with the Sun respective to our position on the Earth, i.e. No Moon]. Each [lunar] month contains either 29 or 30 days.[326]

XX. The day was determined by marking each day of the month by the Moon's specific:

A.) shape;

B.) location;

C.) time of rising;

D.) [time of] setting;[327]

1.) The crescent New Moon on the western horizon immediately after Sunset marks the 1st day of the month;[328]

2.) From the 1st of the month through the 15th, the duration of the visibility of the Moon increases;[329]

3.) The middle of the month is marked by the Full Moon on the eastern horizon across from the setting Sun [that is] on the western horizon (i.e. the rising Moon is in opposition to the setting Sun);[330]

4.) From the 15th day through the end of the month, the duration of the visibility of the Moon decreases;[331]

5.) At the end of the lunar month towards dawn, the crescent Old Moon on the eastern horizon

324 Cohen 1993: 3
325 Dershowitz and Reingold 1997: 7
326 Sachs and Hunger 1988: 13
327 Horowitz 2012: 9
328 Horowitz 2012: 9
329 Horowitz 1996: 42
330 Horowitz 1996: 42
331 Horowitz 1996: 42

(and subsequent No Moon when it disappears from view) marks the waning days of the month.[332]

6.) The "Day of Disappearance" [i.e. No Moon] when no portion of the Moon is visible, occurs at the end of the lunar month.[333]

XX. Thus any given date of the year, e.g. the 15th day of the 1st month Nisannu / Nisan (i.e. Passover), could be either counted from the New Moon of the month, or established by observation of the Moon's phases in the sky.[334]

XX. The Moon orbits the Earth once [approximately] every 27.3 days relative to the background stars (i.e. the "sidereal period") at an average distance of 384,400 km (238,900 miles) [from the Earth]. The orbit is rather eccentric [with] the Moon going from a "perigee" (closest to Earth) distance of 363,300 km to an "apogee" [farthest from Earth] distance of 405,400 km.[335]

XX. [However,] the Moon takes [approximately] 29.5 days to go through its phases, which is the orbital period [of the Moon around the Earth] relative to the Sun (i.e. longer than the 27.3-day period relative to the stars, since the Sun is moving continuously eastward along the ecliptic[336] [thus making the celestial sphere of the stars appears to spin annually westward] i.e. the Sun's [own] orbit[337] [in space] and the Moon must catch up with it[338][and its movement eastward along the ecliptic]).

XX. Like the Sun, the Moon moves to the East against the starry background [i.e. the celestial sphere]. It closely follows the ecliptic, and therefore moves (at a rate of thirteen (13) degrees per day, its own angular diameter [of ½ degree] in an hour) through the constellations of the Zodiac. However, a five-(5)-degree tilt to the orbit sends the Moon the same angle to the North and South of the ecliptic.[339]

XX. The Moon does not rotate relative to Earth. [That is to say] the Moon [is tidally-locked with the Earth and] keeps one (1) face pointed to us at all times (the result of tides raised by [sic: in] the Earth in [sic: by] the solid body of the Moon). Only the side facing us [on Earth], the "near side", is visible. The "far side" is perpetually invisible [to us from the face of Earth looking at the Moon].[340]

XX. The term "dark side of the Moon" for the "far side" is a misnomer. The far side [of the Moon] receives just as much daylight per month as the near side.[341]

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332 Horowitz 1996: 36; Horowitz 2012: 9
333 Horowitz 1996: 36; Horowitz 2012: 9
334 Horowitz 1996: 36; Horowitz 2012: 9
335 Kaler 2003: 32
336 Kaler 2003: 32
337 Ossendrijver 2013a: 6071
338 Kaler 2003: 32
339 Kaler 2003: 32
340 Kaler 2003: 34
341 Kaler 2003: 34

XX. [Though it does not rotate relative to the Earth], the Moon does rotate relative to the Sun and stars. If you were on the Moon you would see the Sun rise and set [approximately] every 29.5 [Earth-]days, and the stars rising and setting [approximately] every 27.3 [Earth-]days. The Earth, however, would hang nearly motionless in the black lunar sky as it went through its phases.[342]

XX. The orbit of the Moon is inclined against the ecliptic by roughly 5°, so that the Moon crosses it twice every month at points called the nodes. Like the apogee, these nodes are not fixed, but they revolve around the Earth in a cycle of roughly 18 years. The position of the Moon near a node is crucial for the occurrence of an eclipse.[343]

0.17. The Month and Moon in Ancient Mesopotamia

XX. The main focus of attention for the Babylonian astronomers was the Moon (Sumerian: NANNA; Akkadian: *Sîn*). Indeed, the Moon god NANNA/*Sîn* even regularly played the key role in Babylonian religion.[344]

XX. [The ancient Sumerians[345] were a people in southern Mesopotamia [who were] considered ancient even at the time of the ancient Babylonians themselves, and this applies for both Old Babylonian,[346] as well as Neo-Babylonian[347] periods.]

XX. The term for "month" in Sumerian was ITI (Akkadian: *arḫu*; Ugaritic: *yrḫ*; Hebrew: *yeraḥ*,[348]and also: *ḥōdeš*[349]). Cohen cites Rochberg-Halton who observes that the monthly cycle in Mesopotamia was lunar, i.e. each month began at the moment when – following the period of invisibility due to the nearness of the Sun [i.e. the "Day of Disappearance" / No Moon] – the lunar crescent [of the Moon] first appears again briefly on the western horizon in the evening just after Sunset.[350]

XX. Huber[351] has calculated the length of every month in Babylon for the period from 2457 BCE to 212 CE (Julian) and found that only three (3) types of months occur: 29-day (46.9%), 30-day (53.1%), and 31-day months (<0.1%).[352]

342 Kaler 2003: 34
343 Locher 2006: 394
344 O'Maly 2011: 80
345 ca. 4500-1750 BCE (Julian)
346 ca. 1894 – 1595 BCE (Julian) Middle Chronology; 1830 – 1531 BCE (Julian) Low Chronology
347 625 – 539 BCE (Julian)
348 Literally: Moon
349 Literally: New-Month
350 Cohen 1993: 4; Rochberg 1995: 1931
351 Huber 1982: 24-25 – cited by Steele 2011a: 479
352 Steele 2011a: 479

XX. No examples of 31-day months are known from Mesopotamia.[353] From letters and reports by Assyrian and Babylonian scholars from the 7th century BCE (Julian), we can understand how the calendar operated in practice.[354]

XX. So it seems that the beginning of the month was not strictly governed by the visibility of the Moon, but [rather] through a rule that can be expressed as follows:[355]

XX. Beginning of the 30th day of the month a watch was kept for the *New Moon* crescent. If the lunar crescent is seen [in the western horizon] on the evening (an hour or so after Sunset which is the beginning of the day in Mesopotamia) at the beginning of the 30th day, that day would be "turned back" or "rejected" (Akkadian: *turru*) and become instead the 1st day of the new month; otherwise if the Moon was not seen that night of the 30th day, then that day would be "completed" or "confirmed" (Akkadian: *kunnu*) as the 30th day and the next evening would mark the beginning of the 1st [day] of the new month.[356]

XX. In other words, the beginning of the month is determined purely by whether the New Moon crescent is visible on the 30th day.[357] [Maimonides (who lived and wrote during the Middle Ages in Europe, Africa, and the Ancient Near East and known to Jews as RaMBaM),[358] in his exposition of the Hebrew calendar echoes this point: "And thus if they [i.e. witnesses] saw it [i.e. the Moon] in its time [i.e. the presumed 30th day gets turned back to a 29 day month → 1st day of new month], and in the night of its intercalation [i.e. adding a 30th day as confirmed following the 29 previous days] it is not seen, then these [witnesses] are faithful because we are not responsible except for sighting [the Moon during] the 30th day [of the Month] only."][359]

XX. In the Seleucid Era (311 – 146 BCE (Julian)), the beginning of the month was no longer established by observing the first [appearance of the] lunar crescent, but [rather] by predicting it [using mathematical calculations]. However, according to Steele (2007), it appears that this was not done with the algorithms of mathematical astronomy.[360]

XX. Thus each month was 29 or 30 days [making a non-leap] year of approximately 354 days. The length of a particular month was based upon sightings of the Moon and not upon a predetermined pattern of alternating 29- and 30-day months. Cohen stresses that the existence of archives which include tablets dated to the 30th of the month for all or most of the months of the year is not proof that

353 Steele 2011a: 472, 479
354 Steele 2011a: 472
355 Steele 2011a: 472, 479
356 Beaulieu 1993; Britton 2007; Steele 2007 – cited by Steele 2011a: 472, 479
357 Steele 2011a: 479
358 Rabbi Moshe Ben Maimon
359 Maimonides / Rambam, *Mishneh Torah Sefer Z'manim Kiddush HaChodesh*, Chapter 2, Halacha 6
360 Ossendrijver 2015b: 1870

some sites had a 12-month year consisting of 30 days every month, i.e. a year of 360 days.[361]

XX. This is because the 360-day "ideal calendar" is an approximation of what was observed in the heavens between the Sun, the Moon, the constellations of stars, and the seasons. It likely was the original primitive calendar used which was refined over many centuries as the ancient Sumerians and Babylonians observed this calendar migrate through the constellations, the Moon, and the seasons – and thus Babylonian / Hebrew calendar as it exists today is evidence of their refinement and precision of their understanding.

XX. Though ancient man could not intuitively sense the Moon's effect upon him as he could the Sun's, the Mesopotamians were aware of the difference between the lengths of the lunar and solar years. Eventually the Moon (Sumerian: NANNA; Akkadian: *Sîn*) achieved dominance over the Sun (Sumerian: UTU; Akkadian: *Šamaš, Shamash*) as determiner of the year.[362]

XX. The importance of the Moon for determining the length of the year was noted by Rim-Sîn,[363] the ruler of [ancient] Larsa,[364] who praised the Moon as "Nanna, who establishes the months, who completes the year." The Sumerians' perceptions of the superiority of the Moon was reflected also in their mythology [theology] wherein the Moon was considered the father of the Sun.[365]

XX. When the sky was clear, each month began with the 1st sighting of the crescent [New] Moon in the [western] evening sky. If cloud prevailed, the Babylonian astronomers used instead a fairly accurate rule to predict whether the crescent should have been visible or not.[366]

XX. There are several examples of the month beginning after a 29-day month even though the Moon had not been seen because of the weather. The earliest example of this comes from the Astronomical Diary for 374 BCE (Julian).[367]

XX. This implies that the decision to begin the new month must have been made based on a prediction of the visibility of the Moon. In other words: sometimes the date of the beginning of a month was determined purely on the basis of advanced [mathematical] prediction – not on observation.[368]

XX. As with intercalation, notice of the beginning of the month seems to have been sent out from the major cities and cult centers to smaller [surrounding] towns.[369] The time taken for communication

361 Cohen 1993: 4
362 Cohen 1993: 3
363 r. c. 1822 – 1763 BCE (Julian) Middle Chronology; 1758 – 1699 BCE (Julian) Low Chronology
364 c. 2025 – 1763 BCE (Julian) Middle Chronology; 1961 – 1699 BCE (Julian) Low Chronology
365 Cohen 1993: 3
366 Stephenson and Fatoohi 1993: 257
367 Steele 2011a: 480
368 Steele 2011a: 481
369 Steele 2011a: 481

between cities and the importance of practicing ritual activities on the correct day simultaneously throughout the empire may have contributed to the development and acceptance of a predicted [standard] calendar that could be produced a year in advance.[370]

XX. These methods [of New Month prediction] mainly utilized six (6) time intervals recorded every month, named by Sachs (1948) as the "Lunar Six (6)". The most important of these intervals is the time between Sunset and Moonset on the 1st day of the month (called NA). Four (4) intervals between Sunrise/Sunset and Moonrise/Moonset on days around Full Moon (called ŠÚ, NA, ME, and GE_6) and the time between Moonrise and Sunrise on the night when the Moon is seen for the last time during the month (called KUR) complete the Lunar Six (6).[371] Regular observations of the lunar six (6) began by at least 643 BCE (Julian) in Babylon and continued to be made until the end of cuneiform astronomy.[372]

XX. This was the definition of the month for most of the cultures of the ancient Near East and Mediterranean. Only the Egyptians (and later the Romans) did not conform, disregarding the natural, yet slightly irregular time indicators in favor of regular, arbitrary [man-made] measures, such as the fixed 30-day month, and the 365-day year.[373]

0.18. The Year

XX. The year [in ancient Babylonia was] not defined by some independent observation of natural phenomena. It is intended to have the same seasonal events occur at approximately the same point in the year. On the other hand, the year has to contain a whole number of months.[374]

XX. The solar year, also known as the tropical year, defines the seasons. It is the time between the return of the Sun to its same place in the ecliptic, i.e. averaging little less than 365.25 days, specifically 365.24219878 days.[375]

XX. The Sun's annual journey involves visits to successive areas of the sky. Twelve (12) constellations follow one another in a band around the sky. They straddle the ecliptic which is the path of the Sun during its annual journey among the stars [as observed by an observer on the Earth].[376] The distance of the Sun from the Earth does indeed vary throughout the year, but at the present time the Earth's passage through perihelion – or nearest point to the Sun – occurs during the Northern Hemisphere winter[377] in

370 Steele 2011a: 481
371 Steele 2011a: 479
372 Huber and Steele 2007 – cited by Steele 2011a: 480
373 Rochberg 1995: 1931
374 Sachs and Hunger 1988: 14
375 Steele 2011a: 471; Kelley and Milone 2011: 100
376 Kelley and Milone 2011: 26
377 Kelley and Milone 2011: 25

January.[378]

XX. The Sun's eastward motion, combined with its apparent northern motion from winter to summer (and [apparent] southern motion from summer to winter), appears to spiral through the sky. Some cultures saw the weaving of a pattern.[379] A camera recording the Noon (12:00) position of the Sun a regular number of days apart over the course of a year will produce a figure-eight (8) pattern called an *analemma*.[380]

XX. Because the Sun must cover 360° in the course of a year, it must move eastward at slightly less than 1° per day on average. As a consequence, the groups of stars that can be seen during the night change slowly from night to night, amounting to an angular displacement of about 1/12 of the sky's circumference – or 30° – in a month.[381]

XX. The use of celestial longitudes spread over 360° is a relatively modern development. The Babylonians and Greeks used degrees of the zodical sign, measuring from the western edge.[382] Ptolemy's values differ from current values because of precession [of the equinoxes], the variation of the obliquity of the ecliptic, and possibly other factors [i.e. variables].[383]

XX. The Sun and Moon always move eastward relative to the stars so that the angular rates of their westward diurnal motions are always less than that of the stars.[384] The Sun's motion among the stars means that successive constellations fade as the Sun nears [i.e. passes in front of] them and [they] become visible again as [the Sun] passes East of them.[385]

XX. The obvious diurnal westward movement of the Sun is shared by the Moon, planets, and stars. However, the diurnal westward motions of the Sun, Moon, and planets are different from those of the stars, as well as from each other.[386]

XX. The planets' apparent motions are more complex, sometimes halting their eastward motions and briefly moving westward before resuming eastward motion. Thus their diurnal motions are usually slower but sometimes slightly faster than are those of the stars.[387]

XX. However, the supremacy of the Moon [over the Sun for the ancient Mesopotamians] was evident

378 Kelley and Milone 2011: 24
379 Kelley and Milone 2011: 25
380 Kelley and Milone 2011: 25
381 Kelley and Milone 2011: 25
382 Kelley and Milone 2011: 29
383 Kelley and Milone 2011: 29
384 Kelley and Milone 2011: 21
385 Kelley and Milone 2011: 22
386 Kelley and Milone 2011: 21
387 Kelley and Milone 2011: 21

in the year being measured by complete lunar cycles. As an acknowledgement of the unbreakable bond between the solar year and its religious festivals, the Mesopotamians intercalated the year to ensure that festivals observed in a particular month coincided with the seasonal phenomenon being celebrated.[388]

XX. Even this intercalation was defined in terms of the Moon by adding a complete lunar cycle to the year. Thus – as Cohen phrases it – the Mesopotamian year was, in effect, a solar year squeezed into a lunar straightjacket.[389]

XX. Ancient man's concept of "year" – recurring cycles by which he could count the length of his life and measure the span between major events – was an inevitable development, one based upon the cycle of seasons.[390] The Babylonian year commenced around the time of the vernal equinox.[391]

XX. The year was primarily measured in Ancient Mesopotamia by means of the Moon. A cycle of 12 New Moons marked the passage of regular years with the occasional intercalary leap year of 13 New Moons.[392]

XX. The Sumerian term for "year" – usually twelve (12) lunar cycles (sometimes 13) – was MU (Akkadian: *šattu*; Ugaritic: *šnt*; Hebrew: *šānâ*; Arabic: *sana*). The inhabitants of Mesopotamia referred to these years by designating the number of the regnal year of a leader or king, or by assigning each year a name based upon an event of the past year or at the beginning of the current year.[393]

XX. The months were assigned names, usually a reference to that month's distinguishing ritual observance or to an agriculturally significant event. Intercalated months were named with the same name as the preceding month – i.e. the scribe appending the number "2" after the month name or adding the word "extra" (*diri* / DIRI) to distinguish it from the previous month.[394]

XX. In leap-years an extra lunar month was added which allowed for the 1st [day] of the 1st month (Akkadian: *Nissanu*; Hebrew: *Nisan*) and the 1st [day] of the 7th month (Akkadian: *Tashritu*; Hebrew: *Tishrei/Tishri*) to return to the eve of the 1st New Moon after the [Vernal and Autumnal] Equinoxes (according to the Babylonian system).[395]

0.19. The Cycle between the Equinoxes

388 Cohen 1993: 3
389 Cohen 1993: 4
390 Cohen 1993: 3
391 Stephenson and Fatoohi 1993: 257
392 Horowitz 2012: 9
393 Cohen 1993: 4
394 Cohen 1993: 5
395 Horowitz 1996: 37

XX. In addition to the seasonal cycle of the lunar cycle, the Mesopotamians were affected by a 3rd cycle: the cycle between the equinoxes, i.e. a period when the Sun and the Moon vied with each other for time in the sky. The ancient Hebrews recognized the significance of this cycle, referring to the equinoxes (the times when the year [re]turns) in Exodus 34:22 ["And the Holiday of Weeks [Shavuot] you will do for yourself, the firstfruits of the harvest of wheat, and the Holiday of the Gathering [is in] the Period of the Year"[396] – with the "Period of the Year" identified as the Autumn] and 2 Samuel 11:1 ["And it was at the return of the year at the time of the leaving of the messengers..."[397] – with the "Return of the Year" identified as the Spring].[398] The concept of a 6-month equinox year appears to have been a major factor in the establishment of the cultic calendar throughout the [ancient] Near East.[399]

XX. In many locations there were parallel major festivals[400] in the 1st month (Nisannu / Nisan) and 7th month (Tashritu / Tishri) – suggesting that rather than considering one of these festivals as marking the beginning and the other the half-way point of the year, the ancients viewed each as a beginning, the onset of this 6-month "equinox year".[401] The Israelite incorporation of this 6-month "year" into its ritual can further [still] be detected in the duration and timing of the festival of the 1st month (i.e. Passover, the Feast of Unleavened Bread), and the festival of the 7th month (i.e. Sukkot, the Feast of the Gathering [also known as the Feast of the Tabernacles]).[402]

XX. The concept of two (2) 6-month periods defined by the equinoxes was further ingrained in the Mesopotamian psyche by the natural weather pattern of the seasonal year. The Babylonians divided the seasonal year into two (2) periods, each beginning in the month of an equinox:[403] The summer (Sumerian: EMEŠ; Akkadian: *ummātum*), the hot season (Akkadian: *umšum*), or the harvest season (Akkadian: *ebūrum*) began around March; the winter (Sumerian: ENTEN) or the cold season (Akkadian: *kuṣṣum*) began around September. The impact of the solstices was also understood, though they did not define the year to the extent that the equinoxes did.[404]

XX. The rainy season [in ancient Mesopotamia] began around September, resulting in the sporadic flooding of the Tigris and Euphrates rivers from November through March.[405] The main flooding occurred in April and May, the two (2) rivers being swollen by winter rains and snows in the mountains of [now modern-day] Turkey.[406] The winter months preceding the major spring flooding were spent

396 cf. Exodus 34:22
397 cf. 2 Samuel 11:1
398 Cohen 1993: 6
399 Cohen 1993: 6
400 i.e. the *Akitu* Festival
401 Cohen 1993: 7
402 Cohen 1993: 6
403 Cohen 1993: 7
404 Cohen 1993: 7
405 Cohen 1993: 7
406 Cohen 1993: 7-8

shoring up the dikes and cleaning out the canals in preparation for the flooding. This was the time of festivals of the storm-gods – such as Ninurta – hurling lighting and vanquishing the *anzû*-bird and the *asakku*-demon of the stormy mountains.[407]

XX. After the winter rains subsided, the dry season set in culminating with the very hot and dry months of July and August when the parched and merciless steppe seemed to be a place of death and demons. These hot summer months were the time throughout Mesopotamia for the cult of the dead, Dumuzi, who – as embodiment of the power in the grain – had disappeared, i.e. the grain had already been harvested and the new seeds not yet sown.[408]

0.20. Intercalations

XX. Intercalary months were called *atru* in Akkadian, but usually written with the [Sumerian] logogram DIRI.[409] The insertion of an intercalary [i.e. a leap] month was usually a reaction to a natural event (e.g. harvest, rains, flooding) being out of phase with the cultic calendar.[410]

XX. This custom of adding a month to a year can be found in the oldest documents which give any calendric information (from the 3rd millennium BCE (Julian)) and continues to the end of the attested use of cuneiform writing.[411] According to Hunger (2009), intercalary months are attested since the 24th century BCE (Julian) with the decision of when a month had to be intercalated laying with the king, who will have had his advisers. In the 2nd millennium BCE (Julian), intercalation was irregular, becoming more regular in the 8th and 7th centuries, which may be due to the use of [the discovery and adoption of] intercalation rules.[412]

XX. Although many [ancient Mesopotamian] cities waited until the end of the year (February / March) to correct this problem, there are instances of cities inserting an intercalary month at various points throughout the year. The Assyrian lunar calendar of the 2nd millennium BCE (Julian) was the exception: it did not use intercalation and thereby wandered throughout the solar year[413] [exactly as the Islamic calendar wanders today in relation to the fixed solar, Tropical, Gregorian civic year].

XX. Parker and Dubberstein (1956) suggest that it may have been in the reign of Nabonassar (747 BCE (Julian)) that Babylonian astronomers began to recognize – as the result of centuries of observations of the heavens – that 235 lunar months have almost exactly the same number of days as 19 solar years.

407 Cohen 1993: 8
408 Cohen 1993: 8
409 Steele 2011a: 471
410 Cohen 1993: 5
411 Sachs and Hunger 1988: 14
412 Hunger 2009: 145
413 Cohen 1993: 5

They note, however, that this is not to be interpreted as acceptance of that date as the point at which Babylonian astronomers consciously recognized the principle.[414]

XX. Parker and Dubberstein (1956) suggest that during the Babylonian period the directives for intercalation came from the king, whereas during the subsequent [Persian] Achaemenid period priestly officials in Babylon gave the orders.[415] A mathematical scheme producing a regular intercalary cycle was finally introduced into the Babylonian calendar – according to Rochberg – during the [Persian] Achaemenid period. Precisely when this regular intercalary cycle was introduced is a matter of some debate, but it would seem to be either shortly after 500 BCE (Julian) or – Rochberg believes – possibly closer to 380 BCE (Julian).[416]

XX. According to Sachs and Hunger (1988), from 381 BCE (Julian) onwards, a fixed pattern of intercalation was followed which had seven (7) intercalary years in a 19-year period.[417] However, writing in 2009, Hunger revises his position by stating an intercalation cycle of 19 years is evident from around 500 BCE (Julian).[418] Even before this time one [in ancient Babylonia] had evidently tried to establish such a pattern as can be seen from the almost (but not quite) regular distribution of intercalary months during the preceding centuries.[419]

XX. Aaboe (1991) reckons that it is likely that Babylonian theoretical [mathematical] astronomy was created sometime in the 4th century BCE (Julian).[420] According to Cohen (1993), by 367 BCE (Julian) but perhaps by as early as 383 BCE (Julian), the intercalations became standardized, and the 19-year cycle thus came into being.[421]

XX. However, Steele (2011) observes that from the beginning of the reign of Xerxes I [r. 486 – 465 BCE], a regular 19-year cycle of intercalations continues uninterrupted until the end of the cuneiform record [with the general decline by the 1st century BCE (Julian) with the absolute last known tablet in 79/80 CE (Julian)]. This 19-year cycle distributes seven (7) intercalations[422] within a 19-year period, all but one (1) of which are intercalary months XII [Akkadian: *Addaru*; Hebrew: *Adar*].[423] According to Hunger (2009), this Metonic of intercalary years cycle contained seven (7) additional months in a fixed pattern: 1st, 3rd, 6th, 9th, 12th, 14th, and 17th. [Some scholars say that it occurred in the 1st, 3rd, 6th, 8th, 11th, 14th, 17th years]. This 19-year cycle remained in use in the Seleucid and Arsacid periods.[424]

414 Cohen 1993: 6
415 Cohen 1993: 6
416 Rochberg 1995: 1938
417 Sachs and Hunger 1988: 14
418 Hunger 2009: 145
419 Sachs and Hunger 1988: 14
420 Aaboe 1991: 285
421 Cohen 1993: 6
422 Steele 2011a: 476
423 Steele 2011a: 477
424 Hunger 2009: 145

XX. Intercalations were effected by royal decree.[425] The intercalation was performed by an additional month after the 6th month (Akkadian: *Ulūlu*, Hebrew: *Elul*) or after the 12th month (Akkadian: *Addaru*, Hebrew: *Adar*).[426] Documents from the reigns of Hammurabi,[427] Nabonidus (555 – 539 BCE (Julian)), Cyrus II (538 – 530 BCE (Julian)), and Cambyses II (529 – 522 BCE (Julian)) – among others – attest to the procedure. This *ad hoc* intercalation of months represented by the royal letters was the standard procedure for controlling the calendar throughout the ancient Near East from approximately the 3rd millennium BCE (Julian) until about the middle of the 1st millennium BCE (Julian) certainly until 526 BCE (Julian).[428] [429]

XX. In the 1st millennium BCE (Julian), there are two (2) points in the year where an intercalary month can be added: either after the 6th [i.e. Akkadian: *Ulūlu*; Hebrew: *Elul*] or after the 12th [i.e. Akkadian: *Addaru*; Hebrew: *Adar*] month. Until about the middle of the 1st millennium BCE (Julian), this intercalation was done when one [i.e. the king in consultation with his priest-astronomers] felt that it was needed. We know this from royal letters commanding intercalation as late as the reign of Nabonidus (555 – 539 BCE (Julian)) and from other official letters on the same topic from the time of Cyrus II (538 – 530 BCE (Julian)) or even Cambyses II (529 – 522 BCE (Julian)).[430]

XX. Pioneer work in collecting and tabulating intercalary months was done by Kugler and Sidersky.[431] According to Parker and Dubberstein, reconstruction [of the intercalary months] before 626 BCE (Julian) is much too hazardous at present and must await further additions to our knowledge.[432]

XX. In the earlier periods [from 626 BCE (Julian)] – according to Parker and Dubberstein – there is a marked preference for a 2nd [intercalary 7th month] Ululu / Elul.[433] A gradual shift from Ululu / Elul to [a 2nd intercalary 12th month] Addaru / Adar in later periods can be seen.[434]

XX. When an [intercalary 2nd] Addaru / Adar is missed for any reason – as in 538 BCE (Julian)[435] – the year after Cyrus conquered Babylon, an Ululu / Elul is often inserted in the following year [i.e. in six (6) months from the 1st month of Nisannu / Nisan] to bring the calendar up to date more quickly.[436]

XX. The extra sixth month (Ululu / Elul II), added in late summer about 26 times by the Babylonians

425 Rochberg 1995: 1932
426 Cohen 1993: 6
427 ca. 1792 – 1750 BCE (Julian) Middle Chronology; ca. 1728 – 1686 BCE (Julian) Low Chronology
428 i.e. during the exact time when Daniel purportedly served as a high official in the Babylonian and Persian empires.
429 Rochberg 1995: 1932
430 Sachs and Hunger 1988: 14
431 Parker and Dubberstein 1956: 2; Parker and Dubberstein 1971: 2
432 Parker and Dubberstein 1956: 2; Parker and Dubberstein 1971: 2
433 Parker and Dubberstein 1956: 2-3; Parker and Dubberstein 1971: 2-3
434 Parker and Dubberstein 1956: 3; Parker and Dubberstein 1971: 3
435 i.e. during the exact time when Daniel purportedly served as a high official in the Babylonian and Persian empires.
436 Parker and Dubberstein 1956: 3; Parker and Dubberstein 1971: 3

and Persians (Nabopolassar to Darius III),[437] dislodges Tishri from its midpoint in the year, providing another reason why this means of regulating the year is never seen in ancient Judaism.[438]

XX. Parker and Dubberstein (1956, 1971) list the intercalary months (i.e. either Ululu / Elul II or Addaru / Adar II) together with the year of intercalation mentioned in then-unpublished or recently published (1956, 1971) texts including from Nabopolassar through Artaxerxes II.[439]

XX. In making the transition from an observation-based calendar to one governed by astronomical theory, the king and his government placed their trust in the expert knowledge of a small group of specialists.[440] [441]

XX. In this regard it is interesting to note how the evidence for intercalation in the royal archives from Persepolis complements the evidence from Babylonia. The Persepolis Fortification Tablets date to the Regnal Years 13 – 28[442] [443] of Darius I (521 – 486 BCE (Julian)),[444] while the Persepolis Treasury Tablets date from Regnal Year 30[445] of Darius I to Regnal Year 7[446] of Artaxerxes I.[447] [448]

XX. Most interestingly, the pattern of intercalations is the same at Persepolis as in Babylonia. One wonders how this was achieved. Did Babylonian astronomers advise or instruct the Persian court?[449] If the "Metonic Cycle" really had its genesis in this period, did Darius himself [or Daniel as one of his priest-astronomer advisers] have a hand in its origin?[450]

XX. Walker postulates that it is not difficult to imagine the man who [had] won an empire for himself and ordered the invention of a new cuneiform script for the Persian language [also to have] instructed his scholars, "Give me a calendar". However – as Walker concedes – such speculation may be too fanciful, and the astronomical argument probably stands on firmer ground.[451]

XX. However, it will put forward here that if Daniel was indeed an historical figure who would have previously served in the Neo-Babylonian court, and subsequently lived during the time of Darius as

437 Parker and Dubberstein 1956: 27-36 – cited by Derstine 2022: 14
438 Derstine 2022: 14
439 Parker and Dubberstein 1956: 4-9; 1971: 4-9
440 i.e. Daniel would have been among this small group of specialists.
441 Steele 2011a: 482
442 i.e. 13th Regnal Year of Darius I from 509 – 508 BCE (Julian)
443 i.e. 28th Regnal Year of Darius I from 494 – 493 BCE (Julian)
444 cf. Appendix Table X.4
445 i.e. 30th Regnal Year of Darius I from 492 – 491 BCE (Julian)
446 i.e. 7th Regnal Year of Artaxerxes I from 458 – 457 BCE (Julian)
447 cf. Appendix Table X.4
448 Walker 1997: 23
449 i.e. e.g. Daniel
450 Walker 1997: 24
451 Walker 1997: 24

adviser to the king, then it would at least be feasible that it was this very same Daniel who discovered the Metonic cycle.

0.21. Lunar Calendars vs. Luni-Solar Calendars

XX. Most calendars that use lunar months, such as the Islamic [and Hebrew] calendars [today], begin the month on the evening when the thin lunar crescent [of the New Moon] can be seen for the first time.[452]

XX. In the Ancient Near East, the Moon-god was the most important deity of the three (3) astronomical siblings.[453] The Moon-god NANNA-Sîn was the big brother, and the Sun-god (UTU-Šamaš) and Venus (INANNA-Ištar) were his little brother and little sister respectively. According to Horowitz, this reflects the paramount role of the Moon in keeping time where the Moon has a role in determining not only the month by means of its phases, but also the day and the year.[454]

XX. Most lunar calendars incorporate the notion of a year. Pure lunar calendars may approximate the solar year with twelve (12) lunar months (as does the Islamic), though this is about eleven (11) days short of the astronomical year.[455] True lunar months do not permit the beginning of the year to remain at a fixed distance from the equinox.[456]

XX. Lunisolar calendars invariably alternate 12- and 13-month years, either according to some fixed [mathematical] rule (as in the Hebrew calendar), or [according to an] astronomically-determined pattern (e.g. Chinese and modern Hindu calendars).[457]

XX. It is generally the case that in Ancient Mesopotamian civilization, the Sun[-god] comes in 3rd as timekeeper after the Moon and the stars. One can mark the parts of the day by observing the Sun's progressions across the sky from east to west, and even keep a closer record of solar time by means of a sundial.[458] For the Sun there are numerous examples of astronomical reports relating to the length of the day and night around the time of the Spring [Vernal] Equinox.[459]

XX. True lunar calendars [like the Islamic calendar] which base the beginning of the month on the observation of a particular phase of the Moon (generally the 1st sighting of the crescent of the New Moon) are subject to inherent uncertainty in the length of each month caused by unpredictable

452 Steele 2011a: 478
453 Elsewhere the Moon-god is mentioned as the father of the Sun-god.
454 Horowitz 2012: 9
455 Dershowitz and Reingold 1997: 9
456 Sachs and Hunger 1988: 14
457 Dershowitz and Reingold 1997: 9
458 Horowitz 2012: 11
459 Horowitz 2012: 9

variations in atmospheric conditions, cloud cover, and the eyesight of the observer [i.e. variables]. These factors prevent members of a community who use a lunar calendar from knowing in advance the precise number of days between an event in one month and a planned event in the next month.[460]

XX. For a true lunar calendar to operate effectively within a community, it is necessary for there to be an agreed consensus of "observation" of the New Moon crescent in order to proclaim the beginning of a new month. However, although this process leads to an "official" calendar used by all members of a particular community, it still does not solve the problem of lack of foreknowledge of the length of any particular month.[461]

XX. According to Schiffman (2012), the establishment of the current calculated [luni-solar] "Jewish Calendar" as we have it today involved a complex series of developments that took place during Late Antiquity and in the early Middle Ages.[462] It is clear that much of the astronomical and calendrical knowledge of Jews was not reduced to writing until the Middle Ages [by Maimonides/Rambam], and therefore its origins – according to Schiffman – will not be recovered from any written source. Schiffman proposes an historical analysis that can be evaluated as the most likely reconstruction.[463]

XX. According to Schiffman, it is evident that Israel and the ancient Canaanites shared a common calendar. Citing Stieglitz (1998), Schiffman asserts that we can reconstruct the [ancient] Canaanite calendar as a 12-month lunar calendar, with the periodic interpolation of a 13th month to maintain synchronization with the seasons.[464]

XX. The last seven (7) centuries of the 1st Millennium BCE (Julian) therefore witnessed a transformation from a luni-solar calendar[465] based primarily on observation – and regulated at the discretion of the king – to a calculated luni-solar calendar that was regulated by astronomical scholars. This transition brought advantages to Babylonian society in the form of increased certainty for administration, the preparation of rituals, and the harmonization of cultic and civil life throughout the empire.[466]

XX. In Mesopotamia the transition to a calculated calendar was driven mostly by practical needs (albeit some were ideological), in particular through the simplification of administration and bureaucracy, the desire for certainty in the preparation for rituals, and in ensuring that all subjects of the empire[467] maintained the same calendar.[468]

460 Steele 2012: 373
461 Steele 2012: 373
462 Schiffman 2012: 231
463 Schiffman 2012: 231
464 Schiffman 2012: 232
465 Steele 2012: 379
466 Steele 2012: 379-380
467 Steele 2012: 383
468 Steele 2012: 384

XX. A centrally organized calendar was needed to ensure that the local population knew which day the public parts of the *akitu*-festival were to take place. On a practical level, the use of a common calendar no doubt simplified business transactions and the collection of taxes throughout the countryside.[469]

XX. Before the articulation of a solar theory in the mathematical astronomy of the Late Babylonian period, no value for the length of a year in days can be cited. As a consequence of the lunisolar character of the calendar, the length of the Babylonian civil year varied, depending upon whether an extra lunar month was intercalated.[470]

0.22. The Calendars of Ancient Mesopotamia

XX. The purpose of a calendar is to give a name to each day. The most simple naming convention – mathematically – is to assign a [number] integer to each day. Fixing day one (1) would determine the whole calendar. The Babylonians had such a day count (in base 60). Such *diurnal* calendars are used by astronomers [today] and calendarists.[471]

XX. It is impossible to date the invention of the lunar calendar in ancient Mesopotamia and the Near East.[472] The earliest direct evidence for both the Sumerian and Semitic calendars, those of the middle 3rd millennium BCE (Julian), demonstrates that these calendars already included intercalary months.[473] Earlier indirect evidence for the lunar calendar dates to late 4th millennium BCE (Julian) southern Mesopotamia. At approximately the same time, the earliest surviving evidence for the 360-day [so-called] "ideal calendar" emerges as well.[474]

XX. Both the 360-day year and the three-(3)-year (37-month) cycle are present in the Ur III period a millennium later,[475] and this same 360-day year length finds explicit expression in the 1st half of the 2nd millennium BCE (Julian) in two (2) Old Babylonian period[476] documents where a year from one 20th [day] of Nisannu / Nisan to the next is 360 days and 360 nights long.[477]

XX. This ideal 360-day astronomical year corresponded to an ancient Mesopotamian astronomical theory known from 1st millennium BCE (Julian) astronomical texts which held that the stars, Sun, and Moon moved along 360-degree circuits. According to this model, each day of the ideal astronomical

469 Steele 2007: 140
470 Rochberg 1995: 1938
471 Dershowitz and Reingold 1997: 6
472 Horowitz 1996: 38
473 Horowitz 1996: 38-39
474 Horowitz 1996: 39
475 ca. 2112 – 2004 BCE (Julian) Middle Chronology; 2048 – 1940 BCE (Julian) Low Chronology
476 ca. 1894 – 1595 BCE (Julian) Middle Chronology; 1830 – 1531 BCE (Julian) Low Chronology
477 Horowitz 1996: 39

year of 360 days corresponded to one (1) degree UŠ ("USH") of stellar or solar movement.[478]

XX. In the Persian [Achaemenid] and Hellenistic [Greek] periods the 360-day calendar appears in astrological texts such as the so-called *Kalendertexte*, which relate dates, positions in the zodiac, and (for example) ingredients used in medicine.[479]

XX. The calendar used by the people of Mesopotamia from at least the late 4th millennium BCE (Julian) to the beginning of the 1st millennium CE (Julian) was a luni-solar calendar in which the beginning of the month was dependent upon the visibility of the crescent of the New Moon, and intercalary (leap) months were added in certain years.[480] The calendar played an important role in many aspects of daily life for the people of ancient Mesopotamia.[481]

XX. The ancient Mesopotamian civil and religious calendar was built, in principle at least, around an ideal three-(3)-year cycle of 37 lunar months consisting of two (2) regular years of 12 lunar-months (approximately 354 days) and a leap year of 13 lunar-months (approximately 384 days).[482]

XX. Given the importance of the calendar in civic, cultic, and daily life, it is not surprising that as soon as we have evidence for real interest in predictive astronomy in [ancient] Mesopotamia (around the 8th century BCE (Julian)), we find attempts to use astronomy to provide ways of removing these uncertainties from the calendar.[483] During the Neo-Babylonian and Persian [Achaemenid] periods, an increasing interest in predicting the length of the month in advance began to develop.[484]

XX. The transition from irregular intercalation ordered by the king to an astronomically regulated system for intercalation raises interesting questions about the importance of the calendar and a possible change in the power relationship between the king and his astronomical advisors.[485] Many of the issues surrounding the various calendars of Mesopotamia are discussed in greater detail in Cohen (1993), Brown (2000), Britton (2007), and Steele (2007).[486]

0.23. The Earliest Calendars

XX. During the middle of the 3rd millennium BCE (Julian), an early Semitic calendar was in use across much of northern and central Mesopotamia, being attested on tablets from across the region. It may be

478 Horowitz 1996: 40
479 Steele 2011a: 474
480 Steele 2012: 374
481 Steele 2012: 378
482 Horowitz 1996: 37
483 Steele 2012: 378
484 Steele 2012: 379
485 Steele 2011a: 477
486 Steele 2011a: 482

significant that the Early Semitic calendar month names do not contain theophoric elements, but rather are agricultural in nature (with month names meaning “Sheep”, “Plowing”, etc.) – a feature which enabled this early Semitic calendar to be somewhat universal in its application.[487]

XX. Adherence to this calendar was widespread, reaching from northwestern [now modern-day] Syria down to south-central [now modern-day] Iraq.[488] However, no texts from before the 1st millennium BCE (Julian) describe the technical details of the calendar.[489]

XX. Most Assyriologists remain convinced that the luni-solar calendar formed the basis of calendrical practice throughout most of Mesopotamian history.[490] Concurring with Cohen, Steele believes that the luni-solar calendar operated from at least the latter part of the 4th millennium BCE (Julian), and quite probably earlier, down to the end of cuneiform writing [with the general decline by the 1st century BCE (Julian) with the absolute last known tablet in 79/80 CE (Julian)]. However, Steele notes that it must be admitted that the evidence in support of this assumption is largely circumstantial before the 1st millennium BCE (Julian).[491]

XX. In southern Mesopotamia ca. 2400 – 2200 BCE (Julian), local [non-Semitic] Sumerian calendars were in use, documented in tablet archives mainly from the cities Adab, Lagash, Nippur, and Ur.[492]

XX. The Sumerian Nippur calendar was adopted throughout much of southern Mesopotamia after the fall of Ibbi-Sîn of Ur (ca. 2000 BCE (Julian)). This Nippur calendar was most likely an innovation of the 1st monarch of the new Isin dynasty, Išbi-Erra.[493] Beginning with his reign (ca. 2017 – 1985 BCE (Julian)[494] a true *Reichskalendar* was established based on the [Sumerian] Nippur Calendar, and was used almost exclusively throughout southern Mesopotamia.[495]

XX. This calendar [i.e. the Southern-Mesopotamian Sumerian calendar of Nippur] became a source of unification for the Isin empire, both politically and economically. The symbolism of Nippur was not lost on the Isin monarchs, who made a special effort to participate in the Nippur rites. As merchants and scribes from the Isin Empire conducted business in the Diyala region, the official calendar probably followed so that when the First Dynasty of Babylon arose,[496] it too was already using this Southern-Mesopotamian Sumerian calendar [of Nippur].[497]

487 Cohen 1993: 8
488 Cohen 1993: 15
489 Steele 2011a: 472
490 Steele 2011a: 473
491 Steele 2011a: 472
492 Cohen 1993: 9
493 Cohen 1993: 11
494 Date provided by Cohen 1993: 10
495 Cohen 1993: 10
496 ca. 1894 – 1595 BCE (Julian) Middle Chronology; ca. 1830 – 1531 BCE (Julian) Low Chronology
497 Cohen 1993: 11

XX. Variations of a basically Semitic calendar were employed by many of the cities stretching across the middle of the Tigris and Euphrates valley during the last century of the 3rd millennium and the 1st three (3) centuries of the 2nd millennium BCE (Julian) – the last attestations coinciding with the reign of Samsu-iluna (i.e. son of Hammurabi) of Babylon.[498] The [previous] centuries [had] witnessed the influx of Semitic tribes referred to in Mesopotamian sources as Amorites "westerners" (Akkadian: *amurru*; Sumerian: MAR.TU) into the central regions of Mesopotamia, followed by the establishment of the Amorite kingdoms throughout the area, including at such important centers as Eshnunna and Babylon. Although there is no conclusive evidence that the new calendars were introduced by the immigrating Amorite tribes, the appearance of a new calendar at the same period as the introduction of a new, large, dominating population suggests just such a relationship.[499]

XX. Furthermore, the middle of the 2nd millennium BCE (Julian) witnessed the arrival throughout north-central Mesopotamia of peoples speaking Hurrian.[500] Therefore – according to Cohen – it is likely that the Standard Mesopotamian calendar was an artificial creation commissioned by Samsu-iluna of Babylon as a means to unify a heterogeneous and rebellious empire.[501] The economic and political advantages of a single, standard calendar were as obvious in the 2nd millennium BCE (Julian) as it had been on a smaller scale hundreds of years earlier to Išbi-Erra of Isin.[502]

0.24. The Standard Mesopotamian Calendar

XX. Based upon the peculiarities of the hybrid nature of this calendar, the Standard Mesopotamian calendar may have been an artificial creation, a means to unify a divergent empire.[503]

XX. Rather than select one particular city's non-Sumerian calendar as the new *Reichskalendar* – a move which might have alienated those cities on whom another's city's calendar would have been imposed – the Babylonian administration invented a hybrid *Reichskalendar*, culling months from various calendars throughout the realm and beyond, thereby hoping to gain international acceptance. The use of Southern-Mesopotamian Sumerian month names as logograms for this new calendar is a clear signal that there was something highly unusual and "unnatural" about the development and imposition of this new calendar. The retention of the Southern-Mesopotamian Sumerian month names [from Nippur] on written documents may have been a negotiating point to gain the acceptance of the former Sumerian cities and of their proud scribes.[504]

498 ca. 1749 – 1712 BCE (Julian) Middle Chronology; ca. 1686 – 1648 BCE (Julian) Low Chronology
499 Cohen 1993: 12
500 Cohen 1993: 12
501 Cohen 1993: 13
502 Cohen 1993: 13
503 Cohen 1993: 303
504 Cohen 1993: 13; 303-304

XX. The Standard Mesopotamian calendar may have been introduced by Samsu-iluna[505] who sensed the urgency of fostering a sense of nationhood among the cities of his empire which had recently rebelled against him. Use of this new calendar spread without the use of military conquest. It was probably international commerce and contracting that were the catalysts for the Standard Mesopotamian calendar's wide-spread acceptance in places that were not under Babylonian control, such as Alalakh and Assyria.[506]

XX. At some period during or after the Judean exile in Babylonia [or perhaps long before], the Judeans adopted the Standard Mesopotamian calendar, as did the Nabateans, the Palmyrans, and other Aramaic-speaking peoples.[507] This calendrical system was used down through the close of the 1st millennium BCE (Julian).[508] During the 1st century BCE (Julian) it was the most widely used calendar in Mesopotamia and in the Near East because the great Near Eastern empires of the Assyrians, Babylonians, Persians, and eventually Seleucids (Hellenistic Greek) adopted and disseminated it as their official, imperial calendar.[509]

XX. Considering who controlled the Jewish calendar in Judea in the centuries leading up to the institution of the Jewish calendar as we now have it in the rabbinic period, under the great Near Eastern empires of the 1st millennium BCE (Julian) – especially the Achaemenid (6th – 4th centuries BCE (Julian)) and Seleucid dynasties (3rd – 2nd centuries BCE (Julian)) – the Jews appear to have simply adopted and used the official, imperial Babylonian calendar for their own purposes.[510] According to Morgenstern (1935), the [Hebrew] calendar was borrowed during the Babylonian exile, between about 450 and 419 BCE (Julian), and first came into regular use – in existing materials – in the writings of Nehemiah.[511]

XX. The Babylonian exile also affected the Israelite calendar. Exposure to the Babylonians allowed for a rich infusion of astronomical and calendrical knowledge into the Jewish elite [in Babylon as well as] even in Judea. The Israelites also adopted the Babylonian month names. Initially, the [Hebrew] Bible referred to months in numerical, ordinal procession. However, the post-exilic books of *[Daniel,] Zechariah, [Haggai,] Ezra, Nehemiah, [Kings,] Chronicles, Esther*, and the [non-canonical] books of the *Maccabees* often identify numbered months with Babylonian names. Eventually the Babylonian month names became normative among the Jewish people.[512]

XX. Cited by Stern, Wacholder and Weisberg (1971) argue that comparing the calendar and "New

505 ca. 1749 – 1712 BCE (Julian) Middle Chronology; ca. 1686 – 1648 BCE (Julian) Low Chronology
506 Cohen 1993: 305
507 Cohen 1993: 299
508 Cohen 1993: 9
509 Stern 2012: 213
510 Stern 2012: 220
511 Kelley and Milone 2011: 219
512 Schiffman 2012: 232

Moon" procedures of the Jewish *Mishnah* with those attested in ancient Mesopotamia reveals that procedures apparent in Mesopotamian cuneiform sources for sighting the "New Moon" and declaring the beginning of the month were so similar to those of the Jewish Rabbinic calendar that the latter must have been in direct, historical continuity with the much earlier calendars of the ancient Near East.[513]

XX. Following the Talmudic dating of the adoption of the Babylonian month names to the exilic period,[514] Schiffman argues that much of the astronomic and calendrical knowledge came into the Jewish community during the Babylonian exile, more or less at the very same time that the mathematically-calculated lunar, 19-year calendar cycle became known in Babylonia.[515]

XX. However, this does not necessarily prove or confirm that the Judeans learned these innovations from the Babylonians. Indeed logically this only suggests a *Terminus Ad Quem*. Indeed it is the converse that is logically implied and more reasonable since it was only after the Jews were in Babylon that the Neo-Babylonian court began to use this 19-year Metonic cycle, and it was the Jews (via Daniel) who plausibly brought (or discovered) these innovations in Babylon when they were exiled.

XX. Thus it is possible that it was precisely the person of Daniel who discovered and standardized these innovations who had had the highest access in the Babylonian and Persian courts and "in whom there is not any defect, and good-looking, and contemplating in all wisdom, and knowers of knowledge, and understanders of science, and in whom there is strength to stand in the Hall of [the palace of] the king,"[516] and trained in "book and tongue of the Chaldeans." – thus Daniel would have had the opportunity to influence such astronomical-calendrical matters as adviser to several Babylonian and Persian kings.[517]

XX. This [19-year, mathematically-calculated lunar intercalation] cycle began to be consistently applied [later] in [Persian] Achaemenid times. Because of this advanced knowledge of lunar calendation and astronomy, the claim of the *Tannaim* (i.e. Mishnaic rabbis of the 10 – 220 CE (Julian)) to have had the ability to determine the lunation and position of the nascent Moon (such as was done in ancient Babylonia) is completely plausible.[518]

XX. By the 1st millennium BCE (Julian), we have unambiguous evidence for the luni-solar calendar[519] [in the form of this Standard Mesopotamian calendar] which was a calendar governed by [mathematical] astronomical calculation.[520] From the middle of the 8th century BCE [i.e. 747 BCE (Julian) year one (1) of the Era of Nabonassar] onwards we have an almost complete record of

513 Stern 2012: 212
514 Babylonian Talmud *Rosh HaShanah* 1:2 (56d)
515 Schiffman 2012: 233-234
516 cf. Daniel 1:4
517 cf. Daniel 2; 5; 6
518 Schiffman 2012: 233-234
519 Steele 2011a: 476
520 Steele 2011a: 482

intercalary months preserved in astronomical texts.[521]

XX. Additionally, during this period Babylonian astronomers recorded observations of eclipses of the Sun and Moon. Surviving datable observational texts range from around 750 BCE to 50 BCE and contain many descriptions of lunar and solar eclipses. Sadly, only a small proportion of the original material is known to be extant.[522]

XX. From 7th-century BCE (Julian) Assyria we have a large body of correspondence sent by Assyrian and Babylonian scholars to the [then overlords of Babylon] Assyrian kings Esarhaddon and Aššurbanipal, a significant number of which are concerned with the watch for the New Moon at the end of a month, intercalation, and other calendrical matters.[523]

0.25. New Year

XX. In Mesopotamia the New Year began on the 1st day of the 1st month (Akkadian: *Nisannu*; Hebrew: *Nisan*), the day of the 1st New Moon after the Spring [Vernal] Equinox – according to the Babylonian convention – which was ideally meant to occur [i.e. the Vernal Equinox] on the 15th day of the 12th (normal year) month or [on the 15th day of the] 13th (leap year) month of the year, [Akkadian: *Addaru 1*; Hebrew: *Adar 1*] or [Akkadian: *Addaru 2*; Hebrew: *Adar 2*] (i.e. months 12 and 13 respectively).[524]

XX. The 1st day of the 7th month (Akkadian: *Tashritu*; Hebrew: *Tishrei/Tishri*), should[525] therefore have fallen one (1) half-year later, ideally on the 1st day of the 7th month (Akkadian: *Tashritu*; Hebrew: *Tishrei/Tishri*), the day of the 1st New Moon after the Autumn Equinox (ideally the 15th day of the 6th month, Akkadian: *Ulūlu*; Hebrew: *Elul*). However, because of the difference of 11¼ days between the 354-day lunar year and the true solar year [of 365.2422 days], the 1st of *Nisannu / Nisan* and the 1st of *Tashritu / Tishri* slipped backwards in respect to the equinoxes and solstices at the rate of just over 1/3 month each year, i.e. approximately one (1) lunar month every three (3) years.[526]

XX. In lunar calendars without intercalation, such as the modern Islamic calendar, this shortfall of approximately 11¼ days per year is ignored so the lunar months fall back through the seasons of the solar year over a 32½-year cycle. As a result, in Islam for instance, holidays [during the Islamic month of] Ramadan occur in different seasons in different years.[527]

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521 Steele 2011a: 476
522 Steele, Stephenson, and Morrison 1997: 337
523 Steele 2011a: 472
524 Horowitz 1996: 36
525 Horowitz 1996: 36
526 Horowitz 1996: 37
527 Horowitz 1996: 37

XX. One period[528] of the Moon is – on the average – 29.53059 days; a [mean] solar year is 365.24220 days, i.e. [approximately] eleven (11) days more than twelve (12) lunar periods which amount to 354.3671 days [in a mean lunar year]. After three (3) [lunar] years, the Moon calendar is [approximately] 33 days behind the Sun's progress [and thus requires an intercalary lunar month to resynchronize the lunar calendar with the solar year].[529]

XX. Al-Biruni, a Muslim scholar, writes:

"*In pagan [i.e.* Jahiliyyah *pre-Islamic] times the Arabs dealt with their months as the Moslems do now, and their pilgrimage moved through all the four (4) seasons of the year. Then, however, they resolved to fix their pilgrimage at a time when their wares, hides and fruits, were ready for marketing; so they tried to make it immovable, to have it always in the most abundant season.*[530]

"*So they learned from the Jews the system of intercalation, 200 years before the* Hegira*. And they applied it in the same way as did the Jews, by adding the difference between their year and the solar year, when it had increased to a month, to the months of the year.*[531]

"*Then the* Kalammas *(i.e. the Sheiks of a certain tribe, in charge of this task) at the end of the pilgrimage ceremonies came forward, spoke to the people, and intercalated a month by giving to the next month the name of the present one. People expressed their approbation of the* Kalammas*' decision by their applause.*[532]

"*This procedure they called* Nasi*, i.e. shifting, because every 2nd or 3rd year the beginning of the year was shifted... They could judge it after the risings and settings of the* menazil*. Thus it remained up to the flight of the Prophet from Mecca to Medina.*"[533]

XX. In the 9th year after the Hegira Mohammed forbade the shiftings, perhaps to break the spiritual power of the Kalammas by taking away their function, and, in addition, to separate himself more sharply from the Jews. So now the Mohammedan calendar [i.e. the Islamic pure lunar calendar] is based upon [pure] lunar years of 354 days, each consisting of 12 lunar months, which in 33 years run through all the seasons.[534]

XX. Islamic astronomers developed an arithmetic approximation that is used for estimation [of exact (varying) dates to be confirmed strictly by observation in each region of the world]. In it there are

528 i.e. One (1) Lunation
529 Pannekoek 1961: 25
530 Pannekoek 1961: 27
531 Pannekoek 1961: 27
532 Pannekoek 1961: 27
533 Pannekoek 1961: 27
534 Pannekoek 1961: 27

twelve (12) months which contain alternately 30 or 29 days. The 12th month contains 29 days in an ordinary year and 30 days in a leap year, with 11 out every 30 years comprising 355 days, rather than 354.[535] An important classic source for understanding the visibility of the crescent was *The Handbook of Astronomy* by al-Battani (850 – 920 CE (Julian)) who cited "ancient opinion" as well as methods for determining visibility.[536]

XX. Here [with the purely lunar Islamic calendar] we have a calendar which – as Pannekoek puts it is – divorced from social practice by formal prescript, has turned into a petrified religious tradition of a primitive Bedouin [and perhaps also Ancient Middle Assyrian] way of life.[537]

XX. [The ancient Babylonians understood that] in order to remain in accordance with the Sun, an extra month had to be added every 3rd year (sometimes more often) so that the [lunar] year had thirteen (13) instead of twelve (12) months. The [mathematical] calendar problem consists in finding a larger period which is a common multiple of the [lunar] month and the [solar] year [in which] – after this period – the Sun and Moon will return to the same mutual position.[538]

XX. Of course an exact common multiple does not exist, but more or less satisfactory approximations may be found. With our precise modern knowledge of the solar and the lunar period, we are able to deduce [and mathematically calculate] such approximate calendar periods theoretically.[539]

XX. Of course in earliest times peoples did not have this exact knowledge of periods, [and thus] the finding of a good calendar period for them was a difficult practical [observational] problem – only to be solved by a laborious process of adapting solar and lunar reckonings [though centuries of refinement]. Therefore this problem acted as a driving force toward more careful observation of the celestial phenomena.[540]

XX. Ancient Mesopotamians, like the [ancient] Jews, celebrated seasonal-agricultural holidays, and so could not allow the dates of their holidays to wander over the seasons of the year. Thus ancient Mesopotamians declared a leap-year approximately every 3rd year to keep the lunar months in their proper seasons.[541]

XX. We should bear in mind that in those times agricultural activities were accompanied by religious ceremonies and festivities. The agricultural festivities, like all great and important social happenings, were at the same time religious ceremonies.[542]

535 Dershowitz and Reingold 2013:27
536 Kelley and Milone 2011: 75
537 Pannekoek 1961: 27
538 Pannekoek 1961: 25
539 Pannekoek 1961: 25
540 Pannekoek 1961: 25
541 Horowitz 1996: 37
542 Pannekoek 1961: 25

XX. In leap-years an extra lunar month was added – almost always as intercalary months XII2 [Akkadian: *Addaru 2*; Hebrew: *Adar 2*] or VI2 [Akkadian: *Ululu 2*; Hebrew: *Elul 2*] – which allowed for the 1st day of the 1st month (Akkadian: *Nisannu*; Hebrew: *Nisan*) and the 1st day of the 7th month (Akkadian: *Tashritu*; Hebrew: *Tishrei/Tishri*) to return to the eve of the 1st *New Moon* after the equinoxes (according to the Babylonian system).[543]

XX. The decision as to whether to intercalate was important for the ritual calendar, and in particular for the preparations for the major festivals. The uncertainty in whether a festival should be performed caused by irregular intercalation was clearly a serious matter.[544] As the empire grew, the potential for confusion would increase: orders sent by the kings to cities throughout the empire would take different lengths of time to arrive, perhaps arriving too late in some cases to prevent the festival from being delayed [sic: celebrated] until the next month.[545]

XX. Avoiding this problem may have been one of the reasons why intercalation cycles of 8 and 19 years were developed in the Neo-Babylonian and early Persian [Achaemenid] periods. Nevertheless, during the reign of the last Neo-Babylonian king Nabonidus (555 – 539 BCE (Julian), only the king continued to have the right to declare an intercalation as [evidenced in] notices sent out in the king's name that proclaimed when intercalations should occur.[546]

XX. However, it is interesting to note that similar letters from the reigns of either Cyrus II or Cambyses II were written by officials in the *Esangila* temple, and do not mention the king's authority. The absence of a reference to the king seems to suggest a shift in responsibility from the king to the temple, although it may be that the temple was merely a conduit through which the king chose to communicate at this time.[547]

XX. Nevertheless, combined with the increased adherence to intercalation governed by the 19-year cycle, Steele notes that it is suggestive of a decline in royal control of the calendar, as well as possibly also a decline in royal interest in the calendar.[548]

0.26. Nisannu / Nisan: The 1st Month of the New Year

XX. Until the closing centuries of the 1st millennium BCE (Julian), there were only two (2) instances which reasonably present the possibility of a New Year other than during the Spring: at the Summer

543 Horowitz 1996: 37
544 Steele 2011a: 477
545 Steele 2011a: 477-478
546 Steele 2011a: 478
547 Steele 2011a: 478
548 Steele 2011a: 478

solstice at Sargonic[549] Umma, and at the Autumnal Equinox at pre-Old Babylonian[550] Sippar. Aside from these two (2), the evidence overwhelmingly indicates a "universal" Spring New Year.[551] An analysis of each city's calendar, beginning with sites from the 3rd millennium BCE (Julian), provides mounting evidence of an almost total adherence to a Spring New Year throughout Mesopotamia.[552]

XX. The concept of a beginning to the year is manifested in three (3) ways: 1.) as a starting point for counting and naming years; 2.) as the start of the fiscal year (i.e. performing year-end accounting and starting the ledgers for the coming year); and 3.) as the start of the religious/agricultural year's cycle. Throughout almost all of Mesopotamia the beginning of the year occurred in the Spring.[553]

XX. The Sumerians called the beginning of the year ZAG-MU, literally "border of the year", from which derived the Akkadian designation for New year, *zag-mukku*. The beginning of the seasonal year, ZAG-MU, took on an important economic role within society, becoming the occasion for reviewing the previous year's financial accounts and administrative affairs in the temple and royal residences, as documented by the large number of year-end account tablets from the Ur III period[554] onward.[555]

XX. The Early Semitic calendar began in the month *Za-'a-tum*, which can be assigned to the Spring on the basis of the agricultural meanings of several month names. Thus the Semitic-speaking peoples of middle 3rd millennium BCE (Julian) Mesopotamia observed a Spring New Year.[556] A Spring New year existed in the calendars of the 3rd-millennium BCE (Julian) Sumerian cities, Lagash, Ur, and Nippur.[557]

XX. [Babylonian] *Nisannu* (Hebrew: *Nisan*) was the 1st month of the Standard Mesopotamian calendar as noted in lexical lists, economics documents, and in other texts such as an inscription of [Assyrian king] Sennacherib: "The month Nisannu, the first month".[558] The name of this 1st month probably derives from Sumerian "NESAG", first.[559] According to Assyrian Astrolabe B, the month of Nisannu was the month of the Moon-god, Sîn.[560]

XX. Astrolabe B states that during this month of Nisannu / Nisan there was an installation of the king. The 1st month was the occasion of one (1) of the two (2) yearly [i.e. semi-annual] *akitu*-festivals, at which time the chief god grandly marched into his city from the *akitu*-house, [thereby] reenacting the

549 ca. 2334 – 2279 BCE (Julian) Middle Chronology; ca. 2270 – 2215 BCE (Julian) Low Chronology
550 ca. 1894 – 1595 BCE (Julian) Middle Chronology; ca. 1830 – 1531 BCE (Julian) Low Chronology
551 Cohen 1993: 14
552 Cohen 1993: 15
553 Cohen 1993: 14
554 ca. 2112 – 2004 BCE (Julian) Middle Chronology; ca. 2048 – 1940 BCE (Julian) Low Chronology
555 Cohen 1993: 14
556 Cohen 1993: 15
557 Cohen 1993: 15
558 Cohen 1993: 19
559 Cohen 1993: 305
560 Cohen 1993: 306

god's original entry into the city and his assumption of suzerainty there.[561] The *akitu*-festival – the major religious observance of the year – occurred during the first eleven (11) days of the month in Babylon, Uruk, and in Assyria, as well as during the first days of the 7th month Tashritu / Tishri.[562]

XX. According to Steele, in the 1st millennium BCE (Julian), the *akitu*-festival began on the 1st day of the 1st month Nisannu / Nisan, and lasted for twelve (12) days.[563]

XX. The schedule of events at Babylon for the *akitu*-festival in Nisannu / Nisan can be partially reconstructed based upon ritual texts, commentaries, and historical inscriptions. However large gaps in our knowledge remain. This festival – which spanned eleven (11) days – may have evolved from separate *akitu*-festivals for [Babylonian gods] Nabu and Marduk[564] [According to evidence that is available to us], there seems to have been an *akitu*-observance [also] in the 7th month Tashritu / Tishri at Babylon.[565]

XX. In the 1st millennium BCE (Julian), with the ever-increasing importance of the cult of Nabu, there may have been a syncretism of two (2) separate *akitu*-festivals: the *akitu* of Nabu and the *akitu* of Marduk. At some period, the *akitu*-festival of Nabu in Borsippa and that of Marduk in Babylon were combined into one (1) Babylonian *akitu*-festival. The appearance of the king before Marduk was one of Marduk's acts in organizing the city he had just entered for the first time.[566]

0.27. Akitu Festival

XX. The *akitu*-festival probably originated in [the city of] Ur as a celebratory event at the onset of the semi-annual equinox-year. The major theme of the festival was the coming of the [Sumerian] moon-god Nanna [Akkadian: Sîn], symbolized by the waxing of the Moon in the sky and reenacted by the entry of his statue by barge into Ur from outside the city, where it had temporarily resided in a building called and named *á-ki-ti*.[567] The full name of the festival of the 1st month [Akkadian: *Nisannu*; Hebrew: *Nisan*] was "the á-ki-ti festival of the harvesting (season)", which is differentiated in documents from the festival of the 7th month [Akkadian: *Tashritu*; Hebrew: *Tishrei/Tishri*] as "the *á-ki-ti* festival of the seeding (season)".[568]

XX. [Though many of us today may sometimes use the term "crescent" to describe the shape of a waxing or waning Moon, for the ancient Mesopotamians,] the form of the New Moon resembled a boat

561 Cohen 1993: 306
562 Cohen 1993: 307
563 Steele 2007: 140
564 Cohen 1993: 308
565 Cohen 1993: 329
566 Cohen 1993: 441
567 Cohen 1993: 403, 406
568 Cohen 1993: 406

(the image frequently cited by the Sumerian poets to describe the Moon) who was identified as a ship: "Father Nanna, when you sail (across the sky) like a ship on flood waters." In fact, in Ur on the 3rd day of the *á-ki-ti* of the 1st month special offerings were made to the Boat of Nanna.[569]

XX. The [Akkadian] term *elip nuri*, "the boat of light", referring to the Moon, occurs in a text from the 1st millennium BCE (Julian). Thus the *á-ki-ti* was "the festival of the boat of light" or "the festival of the boat of the Moon". The length of the festival – at least in the 7th month – of eleven (11) days may have been to enable the Moon to nearly complete its waxing, i.e. its full arrival into Ur. Perhaps the festival lasted as long as the Moon retained the shape of a barge – once the Moon became almost full, the illusion was done. Cohen concludes that the *á-ki-ti* festival [at Ur] was intricately tied to the Moon.[570]

XX. The festival was adopted at Nippur as part of its function as a religious center representing all|of Sumeria, but was adapted to Nippur's own [Southern-Mesopotamian Sumerian] calendar, losing much of the significance of the Ur cult. The festival had great appeal to the other cities of Sumeria and eventually the rest of Mesopotamia because each city saw the occasion as a reenactment of the original entry of its own chief god into the city. It was a spectacular opportunity to welcome its god and show him the respect he deserved, for which – in turn – the god would administer his city justly and decree a good fate for it.[571]

XX. The basic ritual format of the festival was rather straight-forward: the statue of the god left the city in fitting procession for temporary residence in the *akitu*-house, where it received the standard offerings and prayers during its stay. The statue returned to the city in a grand-procession, after which the god set in order the administration of his city, including the determination of the city's fate.[572]

XX. In addition to this apparent prerequisite of the *akitu*-house being placed outside the city walls, the other unexpected feature is that nothing unusually significant occurred at the *akitu*-house. At Babylon, the reading of the Enuma Elish to Marduk and the humbling of the king before Marduk – as well as the determining of destiny by the gods – occurred back in the *Esagil* [temple].[573]

XX. The [essence of the] *akitu*-house's main function was to serve as temporary residence for the chief god of the city until the moment arrived for his glorious reentry into his city. It was a holding station from which [the Sumerian Moon-god] Nanna returned to Ur by barge, just as the Boat of the Moon was approaching in the sky.[574]

569 Cohen 1993: 402
570 Cohen 1993: 402
571 Cohen 1993: 406
572 Cohen 1993: 406
573 Cohen 1993: 403
574 Cohen 1993: 404

XX. Although the location of the *akitu*-house at Babylon was not specified, it was far removed from the *Esagil* [temple], necessitating a long procession and barges to reach it of great pomp and circumstance.[575] There had to be two (2) processions associated with the *akitu*-house: 1.) one going, and 2.) the other returning. In Babylon, the opulence of the procession way between the *Esagil* [temple] and the *akitu*-house, as well as the richness and decoration of Marduk's barge, is described by Nebuchadnezzar. This luxuriance obviously was applicable to the return trip as well as to the procession [out of the city] to the *akitu*, as it is hard to imagine the [statue of the] god being given a lesser reception upon his return than upon his departure.[576]

XX. The most important ritual aspect of the entire celebration was the entrance of the [statue of the] god into his city, i.e. the procession in which the king escorted Marduk from the *akitu*-house into the *Esagil* [temple].[577] As at Uruk, Babylon celebrated an *akitu*-festival in the 7th month as well [as in the 1st month]. Although the tablets do not mention the festival by name, the ritual information contained clearly refers to an *akitu*-celebration.[578]

XX. Uruk, Babylon, and probably Assyria held *akitu*-festivals in both the 1st and 7th months.[579] The goddess Nana[580] had a major cult at Uruk and throughout 1st millennium BCE Mesopotamia [she] was identified with Tashmetum (Nabu's spouse), and in liturgy with Ishtar, for whom special *akitu*-festivals were held in Uruk and in Nineveh.[581]

XX. The Babylonian *akitu*-festival included a wedding on the 11th day, although the preserved texts provide no further information about it. The wedding was presumably between Marduk and Sarpanitum, just as the following month began with a wedding between the king (or his substitute) and the goddess Nana[582] – the first clear attestation of a sacred marriage rite since Iddin-Dagan of Isin at the beginning of 2nd millennium BCE (Julian).[583] The celebration of cultic festivals, in particular the *Akitu*-festival, acted as a link between the king, the people, and the gods, [thus] legitimizing the king's rule.[584]

0.28. Intercalation's Effects on Ritual and Civic Activity

XX. The insertion of an extra month into the calendar for a particular year is not only an astronomical

575 Cohen 1993: 403-404
576 Cohen 1993: 405
577 Cohen 1993: 451
578 Cohen 1993: 451
579 Cohen 1993: 452
580 i.e. Cohen means here INANNA (Sumerian) / Ishtar (Akkadian), i.e. Venus; not to be confused with NANNA (Sumerian) / Sîn (Akkadian), i.e. the Moon.
581 Cohen 1993: 452-453
582 i.e. INANNA / Ishtar
583 Cohen 1993: 453
584 Steele 2007: 140; Steele 2011a: 475

issue but had direct consequences for civic, ritual, and everyday life. Tributes and other payments scheduled to be paid on a day in a month following the insertion of an intercalary months may have to be rescheduled to the intercalary month or accepted 29 or 30 days later than previously planned.[585]

XX. Ritual activities could also be disrupted by the practice of intercalation. For example, the *akītu* festival held at the beginning of the new year, which acted as a crucial link between the king, the people, and the gods, legitimizing the king's rule might have to be effectively put off for a month if an intercalary month was added at the end of the previous year. Therefore it is unsurprising that decisions concerning intercalation were of concern to the king and his advisors.[586]

XX. To ensure stability in the correspondence between specific months and times of the year – whether defined agriculturally, fiscally, or religiously – an intercalary 13th month was added to the year whenever necessary to maintain the proper "place" of a month[587] within a given season within the solar year. Either an extra 6th month [Akkadian: Ululu 2; Hebrew: Elul 2] or an extra 12th month [Akkadian: Addaru 2; Hebrew: Adar 2] would normally be intercalated. An intercalary 1st month is occasionally attested, albeit rarely, in ancient economic texts.[588]

XX. We continue to hear of such practice [of calendrical intercalation] throughout ancient Babylonian history. For example, the Babylonian king Hammurabi[589] also intercalated the calendar. By the time the Sun had set on the Neo-Babylonian Empire (625 – 539 BCE (Julian)), this calendar was well rooted. This tradition passed on to the Persian Empire. According to Schiffman, by 503 BCE (Julian), under Darius I the Great, the Persian Empire was already using a calendar calculated fully in advance with an intercalary cycle of 7 out of 19 leap years with an added month.[590]

XX. From the middle of the 8th century BCE (Julian) onwards we have an almost complete record of intercalary months preserved in astronomical texts and attested in contracts and letters. These show a generally increasing regularity, with the pattern of years containing intercalary months repeating in either 8- or 19-year cycles, interspersed with occasional unexpected or delayed intercalations, some of which at least, can be explained by periods of political instability.[591]

XX. Intercalations previous to the implementation and establishment of the 19-year cycle were effected by royal decree. Documents from the reigns of Hammurabi,[592] Nabonidus,[593] Cyrus II,[594] and Cambyses

585 Steele 2011a: 475
586 Steele 2011a: 475
587 Rochberg 1995: 1931
588 Rochberg 1995: 1931-1932
589 ca. 1792 – 1750 BCE (Julian) Middle Chronology; ca. 1728 – 1686 BCE (Julian) Low Chronology
590 Schiffman 2012: 234
591 Steele 2011a: 475
592 ca. 1792 – 1750 BCE (Julian) Middle Chronology; ca. 1728 – 1686 BCE (Julian) Low Chronology
593 r. 555 – 539 BCE (Julian)
594 r. 538 – 530 BCE (Julian), i.e. years of rule over Babylon

II[595] – among others – attest to the procedure [of intercalation effected by royal decree]. This *ad hoc* intercalation of months represented by the royal letters was the standard procedure for controlling the calendar throughout the ancient Near East from approximately the 3rd millennium BCE (Julian) until about the middle of the 1st millennium BCE – certainly until 526 BCE – (Julian).[596]

XX. In the 6th century BCE (Julian), Babylonian astronomers introduced a highly effective 19-year intercalation cycle, whereby one (1) extra month, either a [leap 6th month, Akkadian: *Ulūlu arkû*; Hebrew: *Elul 2*] VI2 or a [leap 12th month, Akkadian: *Addāru arkû*; Hebrew: *Adar 2*] XII2, was inserted in 7 out of 19 years according to a fixed pattern that could be readily continued to arbitrary future dates.[597]

XX. According to Rochberg, a mathematical scheme producing a regular intercalary cycle was finally introduced into the Babylonian calendar during the [Persian] Achaemenid period. Precisely when this regular intercalary cycle was introduced is a matter of some debate, but Rochberg suggests that it would seem to be either shortly after 500 BCE (Julian) or closer to 380 BCE (Julian).[598]

XX. According to Stephenson and Fatoohi, although intercalation was rather irregular before about 400 BCE, the dates of many intercalary months (always following the 6th or 12th month) are still preserved in Babylonian records. Hence even at this early period, conversion of dates to the Julian Calendar presents [theoretically] few difficulties. After about 400 BCE (Julian), a systematic scheme based on the 19-year Metonic cycle was used for intercalation, the details of which are well known.[599]

0.29. Astral Texts: Astronomical Diaries and Other Related Texts

XX. The 1st millennium BCE (Julian) was a period of profound change in the astral sciences [in Babylonia]. Between 750 BCE and 250 BCE (Julian), several new textual genres concerning the observation, prediction, and interpretation of celestial phenomena emerged in Babylonia. These developments occurred against the background of a longstanding textual tradition of celestial omens and simple quantitative schemes for lunar and planetary phenomena that can be traced back to the Old Babylonian period.[600] [601]

XX. With more than one thousand (1000) tablets and fragments, Astronomical Diaries and related texts form the largest coherent group among the new genres. They are the fruits of systematic observations

595 r. 529 – 522 BCE (Julian)
596 Rochberg 1995: 1932
597 Ossendrijver 2015b: 1867
598 Rochberg 1995: 1938
599 Stephenson and Fatoohi 1993: 257
600 ca. 1894 – 1595 BCE (Julian) Middle Chronology; 1830 – 1531 BCE (Julian) Low Chronology
601 Ossendrijver 2016a: 126

that were probably initiated in Babylon near 750 BCE (Julian) and continued without major interruptions until about 100 CE (Julian).[602]

XX. The Akkadian term for [these Astronomical Diaries] is *naṣāru ša ginê* "regular watching" [or "regular guarding"][603] which is written at the end and on the edges of the tablets. That a regular watch was kept by observers specifically trained and employed for this purpose is shown by two (2) documents dealing with such employment by the assembly of the temple Esangila [temple] in Babylon.[604] Scribes are only infrequently mentioned on the [Astronomical] Diary tablets.[605]

XX. Since the Babylonian day began at Sunset, the [Astronomical] Diaries record first the events of the nighttime, and then the daytime segment of the a day. Nighttime is identified with the [Sumerian] word GE_6 "night" before the day number; daytime has no specification. For a given night, the weather situation is usually reported first; then follow [this] lunar and finally planetary observations. During daytime, [the report] of weather phenomena naturally dominate.[606]

XX. Most of these texts are [Astronomical] Diaries, six-(6)-monthly [i.e. semi-annual, every six (6) months] compilations of astronomical, meteorological, economic and historical data.[607] They are end products of an elaborate data management operation in which short-term reports written by different scholars were[608] collected and compiled into the desired format [of Astronomical Diaries]. By about 600 BCE (Julian),[609] a functioning predictive method had been discovered for most of the lunar and planetary phenomena reported in the diaries: the so-called "Goal-Year" method [which will be discussed below].[610]

XX. A typical [Astronomical] Diary contains information on the following topics:[611]

1.) Moon;

2.) Planets;

3.) Solstices and Equinoxes, Sirius Phenomena;

4.) Meteors, Comets, etc.;

602 Ossendrijver 2016a: 126
603 cf. Hebrew: *Lintzor* (לנצור) to guard; to save; to keep
604 Sachs and Hunger 1988: 11
605 Sachs and Hunger 1988: 12
606 Sachs and Hunger 1988: 13
607 Ossendrijver 2016a: 126
608 Ossendrijver 2016a: 126-127
609 Ossendrijver 2016a: 126-127
610 i.e. the precise time of Daniel in Babylon
611 Sachs and Hunger 1988: 13

5.) Weather;

6.) Prices of Commodities;

7.) River Level;

8.) Historical Events;

XX. From at least the 8th century BCE (Julian), Babylonian astronomers recorded observations of eclipses of the Sun and Moon [in these Astronomical Diaries and other related texts]. Surviving datable observational texts range from around 750 BCE to 50 BCE (Julian) and contain many descriptions of lunar and solar eclipses.[612] Sadly, only a small portion of the original material is known to be extant.[613]

XX. Tablets from Babylon and Uruk contain the earliest known form of mathematical astronomy.[614] Uruk is the only other site apart from Babylon where all major categories of Babylonian astral science are represented in cuneiform tablets.[615] Due to the technical nature of the texts, they cannot be approached without some background knowledge about Babylonian astral science, its concepts, and cultural context.[616]

XX. The [Babylonian] Astronomical Diaries or more simply, the Diaries, are records of observations and computations made during each period of half a year (six (6) or seven (7) months).[617] Since most astronomical diaries contain observations for six (6) months, they must have been compiled from individual reports of different scholars (or astronomers) to be understood in the broad sense of specialist of the astral sciences, i.e. astronomer / astrologer [/ priest / mathematician].[618]

XX. Also the fact that observational texts [like the Astronomical Diaries] (unlike other scholarly texts) are anonymous (with very few exceptions) [means that they] can be viewed as a consequence of the collaborative effort by which they were produced. Astronomical diaries and related texts are therefore testimonies of a systematic, collaborative program of observation. This program is believed to have existed continuously from the Neo-Babylonian era (i.e. 625 – 539 BCE (Julian)) – perhaps as early as the middle of the 8th century BCE (Julian) – until the very end of cuneiform writing in the 1st century CE (Julian).[619]

612 Steele, Stephenson, and Morrison 1997: 337
613 Steele, Stephenson, and Morrison 1997: 337
614 Ossendrijver 2013a: 6072
615 Ossendrijver 2013b: 1
616 Ossendrijver 2016a: 125-126
617 Hunger and Pingree 1999: 139; Ossendrijver 2013b: 2
618 Ossendrijver 2011: 213
619 Ossendrijver 2011: 213

XX. The oldest [Astronomical Diary] so far dated was inscribed in 652 BCE (Julian),[620] but the series probably began in Babylon in the 1st year of Nabû-nasir (Nabonassar), 747 BCE (Julian), a date proposed for reasons given below, while the latest securely dated [Astronomical] Diary is from 61 BCE (Julian). This means that the tradition of keeping the [Astronomical] Diaries persisted through seven (7) centuries, or even eight (8) if the [Astronomical] Diaries continued to be kept until the end of cuneiform writing the late 1st century CE (Julian).[621]

XX. However, we know that observation of the sky with the intention to control the observed phenomena is older than this date [of 652 BCE (Julian)]. Eclipse reports preserved on tablets go back to the 2nd half of the 8th century BCE (Julian), thus confirming the well-known claim of Ptolemy (*Almagest* III, 7) that he had at his disposal more or less continuous eclipse records from the time of Nabonassar (747 – 734 BCE (Julian)) onwards.[622]

XX. The earliest [Astronomical] Diaries that we have (652 BCE and 568 BCE (Julian)) are for a whole year, as well as some later ones (383 BCE; 371 BCE; 287 BCE; 257 BCE; and 183 BCE (Julian)).[623] The remains of the [Astronomical] Diary for 568 BCE (Julian), preserved as VAT 4956, were used by Weidner (1912a) to prove that Babylonian "scientific" astronomy is older than the famous report of observations made in year seven (7) of Cambyses II [that began in Spring] 523 BCE (Julian).[624]

XX. The [Astronomical] Diary for 568 BCE (Julian) was published as a transcription [of the cuneiform] with German translation, as well as a commentary by Neugebauer-Weidner (1915).[625] The [Astronomical] Diaries for 164 BCE (Julian) and 87 BCE (Julian) are transliterated and translated in Stephenson-Walker (1985: 21-40).[626]

XX. Our recorded knowledge of astronomical events in Mesopotamia begins with records of lunar eclipses. We know of seven (7) references to lunar eclipses and two (2) to solar eclipses prior to the 8th century BCE (Julian). These eclipses are important both to our understanding of Mesopotamian astrology and for determining the chronology of the area. Most of these eclipses are considered extensively by Stephenson and Clark (1978), by Huber (1982), and by Mitchell (1990).[627]

XX. The altitude of the Moon above the horizon at any given time depends on:[628]

1.) the ecliptical position of the Moon;

620 Sachs and Hunger 1988: 12; Hunger and Pingree 1999: 139; Ossendrijver 2013b: 2
621 Hunger and Pingree 1999: 139
622 Sachs and Hunger 1988: 12
623 Hunger and Pingree 1999: 143
624 Hunger and Pingree 1999: 141
625 Hunger and Pingree 1999: 141
626 Hunger and Pingree 1999: 141
627 Kelley and Milone 2011: 223
628 Dershowitz and Reingold 2008: 203

2.) the time;

3.) the viewing location;

XX. Babylon lies in a very flat plain, and thus in clear weather Sunset or Sunrise [as well as eclipses] would be accurately defined. Customarily, the Babylonian astronomers recorded the time interval between an eclipse and Sunset or Sunrise.[629]

XX. Sachs and Schaumberger classified the texts into nine (9) groups:[630]

1.) those devoted to mathematical astronomy;

2.) Astronomical Diaries;

3.) Normal-Star almanacs;

4.) almanacs;

5.) "Goal-Year texts";

6.) tables of planetary and lunar observations;

7.) miscellaneous astronomical texts;

8.) astrological texts;

9.) tablets devoted to aspects of mathematics;

XX. Eclipse observations are recorded on texts in three (3) of the above categories:[631]

1.) [Astronomical] Diaries;

2.) Goal-Year texts;

3.) Eclipse Tables;

629 Steele, Stephenson, and Morrison 1997: 338
630 Stephenson and Fatoohi 1993: 256
631 Stephenson and Fatoohi 1993: 256

XX. The appearance of planets is an item of interest in the eclipse reports.[632]

XX. Ossendrijver (2011) classifies the large and diverse body of literature created by Babylonian astronomers as belonging to four (4) groups consisting of, and composed, as follows:[633]

1.) Observational astronomy: astronomical diaries, planetary reports, eclipse reports, Lunar-Six reports, Goal-Year texts, almanacs, normal-star almanacs;

2.) Mathematical astronomy: procedure texts, auxiliary tables, synodic tables, template tables and daily motion tables;

3.) Zodiacal astrology: horoscopes, calendar texts, and various other texts;

4.) Omen astrology: the series "When Anu and Enlil" [i.e. *Enuma Anu Enlil*] and its commentaries;

XX. Although both solar and lunar eclipses are noted on the various tablets, the latter are much more numerous and cover a considerably longer time-span. This is possibly the result of chance [because] it is known that only a small proportion of the original archive has come to light.[634]

XX. Citing Sachs and Hunger, Steele notes that the Babylonian Astronomical Diaries and related texts provide extensive evidence for the operation of the luni-solar calendar from the mid-7th century BCE (Julian) onwards.[635] The term "[Astronomical] Diaries" is the term now universally used for the regular series of Babylonian observational records.[636] They contain observations of celestial phenomena recorded on a daily basis by the astronomers.[637]

XX. The large number of astronomical observations preserved in the Babylonian Astronomical Diaries and related texts demonstrate that the Babylonian astronomers undertook regular and systematic observations of astronomical phenomena. Citing Sachs, Steele notes that these observations were made on a night-by-night basis from the mid-8th century BCE (Julian) down to sometime in the 1st century CE.[638]

XX. The preserved observations indicate that the Babylonians were primarily interested in observing cyclical phenomena such as the 1st and last visibilities of the planets, stellar passages by the Moon and

632 Hunger and Pingree 1999: 17
633 Ossendrijver 2011: 213
634 Stephenson and Fatoohi 1993: 256
635 Steele 2011a: 472
636 Walker 1997: 18
637 Steele, Stephenson, and Morrison 1997: 337
638 Steele 2015: 1856

the planets, eclipses, as well as the duration of visibility of the Moon [both] after Sunset and before Sunrise.[639] Observations of irregular astronomical phenomena such as comets, meteors, or aurorae are only recorded very occasionally. On the whole, the Babylonian astronomers observed astronomical phenomena that were capable of being predicted in advance.[640]

XX. Celestial bodies most suitable for presages were those that presented great diversity and irregularity in their phenomena. Therefore the most important were not the Sun and the fixed stars, but [rather] the Moon and the planets [with] the fixed stars [that appeared to spin westward in the celestial sphere around the Earth once per sidereal year] forming the constant background for the phenomena.[641]

XX. Although less than about 10% of the content of the original [Astronomical] Diaries has survived (largely in the period from about 380 to 70 BCE (Julian)), many additional observations – particularly of lunar eclipses – are to be found in surviving Goal-Year texts and Eclipse Tables. Reports of lunar and solar eclipses appear with comparable frequency in the extant [Astronomical] Diaries and Goal-Year texts. However, whereas many lengthy lunar eclipse tables are still preserved, their solar counterparts are virtually non-existent.[642]

XX. Hunger and Pingree note that the astronomical data recorded in the [Astronomical] Diaries seem to have been particularly stable. This is remarkable in several respects, [perhaps most notable of which for our study] is that only phenomena which are periodic – and therefore capable of being described mathematically – are included.[643]

XX. The importance of the [Astronomical] Diaries lies not only in the kind of information they contain – astronomical, meteorological, economic, and historical (as well as the long time interval they span) – but also in the fact that, because of the ever-presence of the swiftly moving Moon, we can date them to the very day[!], if we can date them at all.[644]

XX. The records of observations of both lunar and solar eclipses made by these Late Babylonian astronomers contain many measurements of the time interval of an eclipse from Sunrise or Sunset and of the duration of the eclipse phases.[645] The locations and times of the ancient eclipses provided checks on theories of the motion of the Moon and Sun in antiquity for Hipparchus and Ptolemy.[646] Comparison of these time intervals with modern computation provides a method of investigating the accuracy of the clocks used at this early period.[647]

639 Steele 2015: 1856
640 Steele 2015: 1856
641 Pannekoek 1961: 38
642 Stephenson and Fatoohi 1993: 256
643 Hunger and Pingree 1999: 144
644 Aaboe 1991: 283
645 Steele, Stephenson, and Morrison 1997: 337
646 Kelley and Milone 2011: 121
647 Steele, Stephenson, and Morrison 1997: 337

XX. Two (2) types of planetary phenomena are reported in Late Babylonian astronomical texts: synodic phenomena (also known as "Greek Letter Phenomena" – see Neugebauer (1955: 280)), which depend upon the spatial relationship between the planet and the sun as seen from the earth, and sidereal phenomena, which depend upon the position of the planet within the fixed stars.[648]

XX. [Furthermore, some of these] ancient Babylonian clay tablets [e.g. Astronomical Diaries] buried for centuries beneath the sands of the desert are part of an extensive historical archive which contains vital information about the rotation of the Earth.[649] Many eclipse records are preserved from a variety of cultures, and using these seemingly crude ancient and medieval observations, variations in the Earth's rotation[650] can be traced back over the past 2500 years.[651]

XX. [A comparison of these ancient time intervals provides us a means to check, test, and confirm current accepted modern calculations for ancient values of ΔT (Delta T)], (i.e. the accumulated clock error ΔT in the Earth's rotation).[652] These data today provide information about the rate of change Δ [i.e. delta; difference] of the [time of the] Earth's rotation.[653] [It will be proposed and argued here that a comparison of these precise ancient time intervals for ancient eclipses as precisely measured by the Babylonian / Hebrew calendar also allows for the possibility to calibrate and calculate more accurate values for ancient instances of ΔT.]

0.30. Astral Texts: Goal-Year Texts

XX. Among the related [astral] texts, the Goal-Year texts ought to be mentioned.[654] It is generally assumed that by about 600 BCE (Julian),[655] the astronomers who produced the [Astronomical] Diaries had discovered accurate cycles for nearly all of the reported astronomical phenomena.[656] By the beginning of the 6th century BCE (Julian) in Babylonia, very effective techniques had been developed for predicting the beginning of the month.[657]

XX. By [about] 600 BCE [i.e. the 5th Regnal Year of Nebuchadnezzar II beginning Spring 600 – 599] BCE (Julian),[658] accurate periods for nearly all of the reported lunar and planetary phenomena had been

648 Steele 2011b: 104
649 ΔT (Delta T)
650 ΔT (Delta T)
651 Morrison and Stephenson 2001: 247
652 Kelley and Milone 2011: 122
653 Kelley and Milone 2011: 121
654 Aaboe 1991: 283
655 i.e. the time of Daniel in Babylon
656 Ossendrijver 2018b: 74-75
657 Steele 2011a: 479
658 i.e. the time of Daniel in Babylon

discovered.[659] These so-called Goal-Year periods are expressed as a whole number of years: [e.g. Saros cycle of approximately] eighteen (18) [Earth-] years for the Moon;[660] eight (8) [Earth-] years for Venus;[661] and fifty-nine (59) [Earth-] years for Saturn, etc.).[662]

XX. After one such period, the phenomena repeat near the same position in the sky and the same Babylonian calendar date. Henceforth nearly all of the reported lunar and planetary phenomena were routinely predicted by projecting reported past phenomena into the future by adding the appropriate period.[663]

XX. After one (1) Goal-Year period, the phenomena repeat near the same celestial position and the same Babylonian calendar date. With this method, planetary and lunar phenomena were predicted by projecting reported past phenomena into the future year by adding the appropriate period.[664] This so-called Goal-Year method exploits that lunar and planetary phenomena repeat near the same celestial position and calendar date after a certain (characteristic) period which is different for each planet (e.g. 8 years for Venus, 59 years for Saturn, or 18 years [of the so-called Saros cycle] for the Moon).[665]

XX. Thus astronomical predictions, presumably obtained in this manner, were incorporated in the [Astronomical] Diaries whenever an expected phenomenon had not been observed due to bad weather. We know this because these phenomena are recorded but marked as "not observed" (in Akkadian: *ul attaṣar*, usually rendered logographically as [Sumerian:] NU PAP.[666]

XX. Henceforth, their phenomena could be predicted by adding the appropriate period to a recorded date of the same phenomenon.[667] Such a text predicts the behavior of Moon and planets for a given year – the Goal Year – by presenting information extracted from Diaries that antedate the Goal Year by astronomically significant periods (e.g. 46 years for Mercury; 8 years for Venus; 79 years (synodic phenomena) or 47 years (passages by Normal Stars) for Mars; 71 years (synodic phenomena) or 83 years (passages by Normal Stars); 59 years for Saturn, and so on.[668] [669]

XX. The prediction of planetary phenomena using periods is based upon the assumption that after a certain number of years, a particular phenomenon will occur again on about the same day in the Babylonian calendar. For example, roughly every thirteen (13) months Jupiter will be seen in the night

659 Ossendrijver 2015a: 9
660 Ossendrijver 2015a: 9; Ossendrijver 2018b: 75
661 Ossendrijver 2015a: 9; Ossendrijver 2018b: 75
662 Ossendrijver 2015a: 9; Ossendrijver 2018b: 75
663 Ossendrijver 2015a: 9
664 Ossendrijver 2018b: 75
665 Ossendrijver 2013b: 3; Ossendrijver 2016a: 127
666 Sachs and Hunger 1988: 21 – cited by Ossendrijver 2018b: 75
667 Ossendrijver 2016a: 127
668 Aaboe 1991: 283
669 Steele 2011b: 105

sky shortly before dawn for the first time. After 71 years, however, Jupiter will make this first appearance on very nearly the same day in the Babylonian calendar.[670]

XX. In other words, if Jupiter makes its first appearance on the 10th of Month IV (i.e. Du'uzu / Tammuz) of a certain year, then 71 years later Jupiter will again make its first appearance on about the 10th of Month IV.[671] The same principal applies to the other planets with different numbers of years.[672]

XX. Goal-Year Texts contain collections of planetary and lunar phenomena to be used in making predictions of the same kind of phenomena for a coming 'Goal Year'. The texts are divided into several sections; sections are normally separated by horizontal rulings, and the lunar data is subdivided into three (3) columns separated by vertical rulings:[673]

(i) Jupiter's synodic phenomena taken from 71 years before the Goal Year;

(ii) Jupiter's passages by Normal Stars taken from 83 years before the Goal Year;

(iii) Venus's synodic phenomena and passages by Normal Stars for 8 years before the Goal Year;

(iv) Mercury's synodic phenomena and passages by Normal Stars for 46 years before the Goal Year;

(v) Saturn's synodic phenomena and passages by Normal Stars for 59 years before the Goal Year;

(vi) Mars's synodic phenomena for 79 years before the Goal Year;

(vii) Mars's passages by Normal Stars for 47 years before the Goal Year;

(viii-x) Lunar data for 18 and 19 years before the Goal Year;

XX. The principal underlying the Goal-Year Texts is that by copying out observational material (presumably taken from the Astronomical Diaries) for one Goal-Year period earlier than the year for which the predictions are intended, the phenomena recorded will correspond to the phenomena that can be expected in the coming year. The periods chosen for each planet were those that give the most exact repetition of phenomena on dates in the Babylonian calendar, whilst still being short enough to ensure

670 Steele 2011b: 101
671 Steele 2011b: 101-102
672 Steele 2011b: 102
673 Steele 2011b: 106

that records of observations from the earlier years were readily available. Two (2) separate periods were used for each of Jupiter and Mars to allow more precise predictions of synodic phenomena and passages of the planet by Normal Stars.[674]

XX. Several texts are known that contain details of the planetary periods and the corrections in days, and sometimes also in degrees along the zodiac, that are to be applied when using the periods to predict future phenomena. It is clear that the periods chosen to be used in the Goal-Year Texts are generally those that required the smallest corrections in either date or position in the zodiac. Knowledge of the number of 29- and 30-day months would have been useful in estimating the number of days correction that must be made to the period.[675]

XX. It is interesting to note the frequent small discrepancies as to the size of the corrections that should be applied to the Goal Year periods that are specified in the different texts. Steele believes that the values given in later texts represent intended improvements over earlier values, but rather illustrate the plurality of astronomy in Babylonia, in much the same way as the co-existence of different versions of the ACT schemes for each planet.[676]

XX. The corrections are broadly in line with what we find from modern computations of planetary data.[677] For example, according to modern computations, after 59 Babylonian calendar years synodic phenomena of Saturn should take place about 6 days back in the Babylonian calendar. After 46 Babylonian calendar years Mercury phenomena should take place about 1 day earlier.[678]

XX. The utility of the method of predicting planetary phenomena using periodic repetitions was exploited in a group of astronomical texts called “First days, appearances, passings and eclipses which are established for year x”, generally known to modern scholars as “Goal-Year Texts”.[679] [680]

XX. The development of this technique for predicting planetary phenomena through the identification of periodic repetitions of observed events probably antedates the so-called ACT methods of mathematical astronomy (Neugebauer 1955), but continued to be used alongside the ACT methods until the end of Babylonian astronomy.[681]

XX. Unlike the ACT methods, which are directed towards the calculation of the date and longitude of the synodic phenomena of a planet (e.g., first and last visibilities), the application of Goal-Year periods

674 Steele 2011b: 106
675 Steele 2011b: 104
676 Steele 2011b: 105
677 Gray and Steele 2008 – cited by Steele 2011b: 105
678 Steele 2011b: 105
679 Hunger 2006: ix – cited by Steele 2011b: 105-106
680 Steele 2011b: 105-106
681 Steele 2011b: 101

to observed records also allowed conjunctions of the planets with stars to be predicted.[682]

XX. The predictions resulting from the Goal-Year Texts are almost certainly found in the Almanacs and Normal Star Almanacs. These texts contain predictions of astronomical phenomena for a coming year, arranged month-by-month. Common to both Almanacs and Normal Star Almanacs are predictions of the synodic phenomena of the planets, the length of the month, solar and lunar eclipses, and the dates of solstices, equinoxes and Sirius phenomena.[683]

XX. In addition, the Normal Star Almanacs contain predicted passages of the planets by the Normal Stars and the set of time intervals known as the Lunar Six, whereas the Almanacs contain instead the dates when the planets pass from one sign of the zodiac to the next and the dates of the so-called Lunar Three. The Goal-Year Texts contain the requisite material to predict all of the data in the Normal Star Almanacs except for the dates of solstices, equinoxes and Sirius phenomena.[684] [685]

XX. However, this latter data is all given by a simple scheme known today as the "Uruk Scheme".[686] [687]

XX. The dates of entrances by planets in zodiacal signs found in the Almanacs were derived from the dates of passages by certain Normal Star.[688] Thus everything in the Almanacs and Normal Star Almanacs could be predicted using the Goal-Year Texts or the Uruk scheme.[689]

XX. Comparison of planetary data in Almanacs and Normal Star Almanacs which are preserved for years where the Goal-Year Text for that Goal Year is also preserved indicates that the predicted data in these two (2) types of text were indeed based upon the material in the Goal-Year Texts.[690] [691]

XX. Eventually, this [Goal-Year] method led to three (3) new textual genres,[692] i.e. three (3) types of compilations with astronomical predictions for a future calendar year were also derived from the [Astronomical] Diaries by this [Goal-Year] method:

1.) Goal-Year texts;[693]

2.) Almanacs;

682 Steele 2011b: 101
683 Steele 2011b: 106
684 Sachs 1948 – cited by Steele 2011b: 106
685 Steele 2011b: 106
686 Neugebauer 1948; Sachs 1952 – cited by Steele 2011b: 106-107
687 Steele 2011b: 106-107
688 Huber 1958 – cited by Steele 2011b: 106-107
689 Steele 2011b: 107
690 Hunger 1999; Gray and Steele 2008 – cited by Steele 2011b: 107
691 Steele 2011b: 107
692 Ossendrijver 2018b: 75
693 Hunger 2006 – cited by Ossendrijver 2013b: 3; Ossendrijver 2018b: 75

3.) Normal-Star Almanacs[694] which are all attested from the Seleucid Era (311 – 63 BCE (Julian)) onwards;[695]

XX. It appears that none of these [three (3) new textual genres] are attested before the 3rd century BCE (Julian), but the underlying Goal-Year method is believed to have been available since about 600 BCE (Julian), as mentioned.[696] That the Goal-Year method was fully developed by about 600 BCE is suggested by the fact that – from then on – in so-called Lunar Six tablets ([i.e.] observational reports with Lunar Six data) weather related gaps in the observations were completed with predictions that were presumably obtained with that method.[697] [698]

XX. Like the [Astronomical] Diaries, all three (3) [new textual genres] employ a highly standardized format. With very few exceptions, these texts are anonymous, perhaps because the data were produced by a community of astronomers. A Goal-Year text always ends with the following standard label near the bottom of the reverse: "First days (of the month), (first and last) appearances, passings and eclipses which were established for year Y, king NN."[699]

XX. Since the Goal-Year method operates with recorded phenomena, it partly explains why [Astronomical] Diaries continued to be written in a roughly identical fashion for many centuries.[700] This happens to be a quite efficient way of predicting phenomena of the sort considered, and it is based on a recognition of the periodic character of planetary and lunar behavior – a notion that is fundamental in Babylonian theoretical astronomy.[701]

XX. Before producing the almanac for a selected year, the observations that would assist in making the necessary predictions were assembled in the so-called Goal-Year texts, the planned year being the Goal-Year. Such texts might cite [lunar] eclipses from 18 years previously, Venus data from 8 years beforehand, and so on.[702]

XX. As we would expect, the dates of the predicted phenomena are not exactly the same as those of the observations recorded in the Goal-Year Texts but have been corrected by a number of days forward or backward. By and large the corrections applied are in rough accord with both astronomical reality and with the corrections given in procedure texts,[703] although there is a considerable variation in the

694 Hunger 2014 – cited by Ossendrijver 2013b: 3; Ossendrijver 2018b: 75
695 Ossendrijver 2013b: 3
696 i.e. the precise time of Daniel in Babylon
697 Huber and Britton 2007 – cited by Ossendrijver 2018b: 75
698 Ossendrijver 2018b: 75
699 Ossendrijver 2018b: 75
700 Ossendrijver 2016a: 127
701 Aaboe 1991: 283
702 Stephenson and Fatoohi 1993: 256
703 See Steele 2011b: 107, Table 1

corrections applied for certain planets.[704]

XX. Because the Babylonian calendar inserts intercalary months roughly every three (3) years, a further correction of one (1) month to the date of predicted phenomena was required in certain years.[705]

XX. Steele (2011b) shows an example of the number of months between a given month in an observation year and the same month in a Goal Year 8 years later (the Goal-Year period for Venus), for each year of a 19-year intercalation cycle beginning in SE 1.[706]

XX. Since at least the middle of the 5th century BCE (Julian) continuing throughout the whole of the Seleucid period, the placement of intercalary months was strictly governed by a 19-year cycle;[707] thus the pattern repeats every 19 years.[708]

XX. For Venus we see that the number of months in most cases is 99. However, where the observation date is in months VI2 to XII of year SE 18, or Goal date is in months I to VI of year SE 18 (implying the observation is in year SE 8), the number of months is only 98. Thus, in these two circumstances, a one-month correction to the predicted date is required, and so an observed Venus phenomena in Month III of year SE 18 would result in a predicted Venus phenomena in Month IV of year SE 26. Similar tables can be constructed for the different planets, each with a different pattern of years in which corrections are necessary.[709] Comparison of Goal-Year Texts with Almanacs and Normal Star Almanacs indicates that these corrections were indeed made when required.[710]

XX. An interesting situation surrounds the one-(1)-month correction for the prediction of lunar phenomena. For the lunar period of 18 years, a correction is needed on those occasions where the Goal Year contains an intercalary month. In some Goal-Year Texts for years where this correction would be necessary, the Lunar Six data was re-labelled one (1) month later than it was observed.[711] [712]

XX. For example, an observed Lunar Six measurement in Month III would be recorded as being in Month IV. Lunar Six data would be recorded for a Month XII2, when that particular year did not contain an intercalary month.[713] In other words, the one-(1)-month correction was in these cases made when the lunar data was compiled and copied onto the Goal-Year Text, rather than when the predictions

704 Steele 2011b: 107
705 Steele 2011b: 107
706 Steele 2011b: 107-108
707 Parker and Dubberstein 1956; Britton 2007 – cited by Steele 2011b: 107-108
708 Steele 2011b: 108
709 Gray and Steele in preparation – cited by Steele 2011b: 108
710 Steele 2011b: 108
711 Brack-Bernsen 1999 – cited by Steele 2011b: 109
712 Steele 2011b: 109
713 Kugler 1909-24: 459-460 – cited by Steele 2011b: 109

were made.[714]

XX. It appears that this practice only took place in Goal-Year Texts for SE 107 and earlier. In later Goal-Year Texts the Lunar Six data are always recorded without any relabeling, and the correction would have to be made when making the predictions of the coming Lunar Six. Interestingly, this practice is restricted to the lunar data. Nothing equivalent occurs with the recording of planetary data on the Goal-Year Texts.[715]

XX. The Goal-Year Texts represent a remarkably simple yet powerful aspect of Babylonian astronomy. They reduced the prediction of planetary phenomena to little more than record-keeping, something well within the scope of any scribe. At the same time, the results of Babylonian Goal-Year astronomy were surprisingly accurate, producing predictions of visibility phenomena that were frequently as good as those that were predicted using the more complicated methods of mathematical astronomy.[716]

XX. The main motive behind these activities was astrological. Eclipse Tables (also produced from the [Astronomical] Diaries during the Hellenistic period) contain lists of eclipses, usually at 18-year intervals. Sometimes these extend back several centuries. It seems clear from the content of these subsidiary texts that (in general) the Late Babylonian astronomers had access to [many] series of [Astronomical] Diaries covering vast periods of time.[717]

XX. Virtually all the evidence for horoscopy is from Hellenistic and Roman times when personal astrology gained great popularity. In fact, until 1952 CE (Julian) it was possible to build a persuasive case for a Hellenistic origin of personal astrology. At the moment it seems likely, but not certain, that Hellenistic and Roman personal astrology was derived from its Babylonian predecessor.[718]

XX. [Therefore, it is possible to] see in Babylonian theoretical astronomy the first successful attempt at addressing the problem that ever since has remained central in the exact sciences: to give a mathematical description of a well-defined class of natural phenomena, a description capable of yielding numerical predictions that can be tested against observations.[719]

0.31. Eclipses

XX. Among the many celestial events observed in Ancient Mesopotamia, eclipses – particularly eclipses of the Moon – were considered to be among the most astrologically significant. More than

714 Steele 2011b: 109
715 Steele 2011b: 109
716 Steele 2011b: 109
717 Stephenson and Fatoohi 1993: 256
718 Aaboe 1991: 281
719 Aaboe 1991: 290

eight (8) of the seventy (70) or so tablets of the great astronomical omen series *Enuma Anu Enlil* are devoted to their interpretation, and a number of rituals to be performed during an eclipse are known from the Neo-Assyrian, Achaemenid, and Hellenistic periods. It is perhaps not surprising, therefore, that attempts were made to predict eclipses.[720]

XX. Eclipses were viewed as important astrological events from an early time in Mesopotamia,[721] and were viewed as ominous signs of concern to the king.[722] The earliest observations of eclipses come from Babylon and China.[723] Eclipse records and observations of the Moon are also among the scholarly correspondence in the royal archives of Nineveh.[724] Many [ancient] eclipse records made [in both Babylon and China] are now proving invaluable in the determination of the rates of change for the Earth's rotation[!][725] and the Moon's mean motion.[726]

XX. Numerous observations of lunar eclipses are recorded on the Late Babylonian Texts that were recovered from the site of Babylon during the 1870s and 1880s CE (Gregorian). These astronomical texts were for the most part discovered accidentally, and hence there is little archaeological context. Only a few tens of these cuneiform tablets were ever excavated officially. The rest, numbering about 2000 texts, seem to have been dug up by inhabitants of nearby communities, and were eventually sold to antique dealers in Baghdad.[727]

XX. Eclipse observations are recorded on texts in three (3) of the categories [of Astronomical Cuneiform Texts]:[728]

1.) [Astronomical] Diaries;

2.) Goal-Year texts;

3.) Eclipse Tables [i.e. Ephemerides].

XX. Although both solar and lunar eclipses are noted on the various tablets, the latter [i.e. lunar eclipses] are much more numerous and cover a considerably longer time-span. As Stephenson and Fatoohi note, this is possibly the result of chance: it is known that only a small proportion of the original archive has come to light.[729]

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720 Steele 2000a: 421
721 Steele 2000b: 94
722 Ossendrijver 2016a: 127
723 Banjević 2006:252
724 Sachs and Hunger 1988: 12-13
725 ΔT (Delta T)
726 Kelley and Milone 2011: 121
727 Stephenson and Fatoohi 1993: 255
728 Stephenson and Fatoohi 1993: 256
729 Stephenson and Fatoohi 1993: 256

XX. Our earliest contemporary records of eclipse observations from Mesopotamia come from the first half of the 7th century BCE (Julian), although records stretching back to the middle of the 8th century are preserved in later compilations [of copies of more ancient previous texts].[730] It is evident from these accounts that primitive attempts were being made to predict the eclipses before they were observed.[731]

XX. In Babylon, by at least the middle of the 7th century BCE (Julian), and we have good reason to believe stretching back to as early as the middle of the 8th century BCE (Julian), astronomical observations were being systematically conducted and recorded in a group of texts which we have come to call the Astronomical Diaries.[732]

XX. These [Astronomical] Diaries, and other [related] texts which are related to them [i.e. ADRT], contain many observations and predictions of eclipses. The predictions generally include the expected time of the eclipse, apparently calculated quite precisely. By the last three (3) centuries BCE (Julian), the Babylonian astronomers had developed highly advanced mathematical theories of the Moon and planets.[733]

XX. This lunar theory could be used to calculate the times and magnitudes of [both] lunar and solar eclipses.[734] Before the development of such a theory, early astronomers had to rely on simpler schemes [e.g. Goal-Year Texts] which made use of the periodicities in the Moon's motion.[735] Various methods appear to have been formulated by the Mesopotamian astronomers to predict eclipses of the Sun and Moon. This leads into the question of which of these methods were actually used – and why?[736]

XX. [What follows is the] case for lunar eclipses, and solar eclipse possibilities can be treated analogously. Once an eclipse has occurred, it is clear from the rules [known to Babylonian astronomers] that another eclipse will take place when (a) the Moon is in the same phase again, and (b) the Moon is at its same position in its orbit with respect to the node. In other words, an eclipse will occur after there has been both a whole number of synodic months and a whole number of draconic months.[737]

XX. Although there is no reasonably small integral common multiple for these two (2) intervals, a number of short periods are close. For example, 47 synodic months is only one tenth (1/10) of a day different from 51 draconic months, and 135 synodic months is about half (½) a day more than 146

730 Steele 2000a: 421
731 Steele 2000a: 421
732 Steele 2000a: 421-422
733 Steele 2000a: 422
734 Steele 2000a: 422
735 Steele 2000a: 424
736 Steele 2000a: 422
737 Steele 2000a: 424

draconic months.[738]

XX. Because the Moon's orbit is not exactly circular, its distance from the Earth varies. The average interval between successive closest approaches to the Earth is known as an anomalistic month and is equal to 27.5545 days.[739]

XX. The most useful of these periods, however, is 223 synodic months which is very close to 242 draconic months. This period is useful because it is also very close to 239 anomalistic months, which means that the recurring eclipses will have similar magnitudes and durations. This period, which is equal to about 6585 1/3 days or slightly more than 18 years, has become known as the "Saros".[740]

XX. With respect to the fixed background of stars, the Moon moves around the Earth in an approximately circular orbit with an average period of 27.3216 days, known as the sidereal month. However, from the [perspective of the] Earth, the Sun also appears to circle us returning to the same location relative to the fixed stars in a period of 365.2564 days, known as the sidereal year.[741]

XX. Therefore, over the course of a sidereal month the Sun has moved slightly ahead of the fixed stars, and so it takes a little more than another two (2) days for the Moon and Sun to reach conjunction [i.e. No Moon]. The average time interval between two (2) conjunctions [i.e. No Moon] or oppositions [i.e. Full Moon] of the Moon and Sun is equal to 29.5306 days and is known as the synodic month.[742]

XX. There are two (2) types of eclipses: 1.) lunar, and 2.) solar. Lunar eclipses occur when the Moon at opposition [i.e. Full Moon] passes through the Earth's shadow, whereas solar eclipses may occur whenever the Moon at conjunction [i.e. No Moon] covers some part of the Sun's disc.[743]

XX. If the two (2) planes in which the Moon and Sun move were the same then one (1) luminary would be eclipsed every conjunction or opposition [i.e. approximately every 14.7653 days, about twice per month]. However, these two (2) planes are in fact inclined at an angle of about 5° to each another, intersecting at points called nodes.[744] The average interval between successive passages of the Moon by a given node [i.e. from node to node], known as a draconic month, is equal to 27.2122 days.[745]

XX. Only when the Earth's shadow at opposition (for a lunar eclipse) – or when the Sun at conjunction (for a solar eclipse) – is near to a node will an eclipse be possible. This is equivalent to saying that eclipses only occur when the latitude of the Moon is sufficiently close to zero at the moment of

738 Steele 2000a: 424
739 Steele 2000a: 424
740 Steele 2000a: 424
741 Steele 2000a: 422
742 Steele 2000a: 422
743 Steele 2000a: 422
744 Steele 2000a: 422
745 Steele 2000a: 422-423

conjunction [i.e. No Moon] or opposition [i.e. Full Moon].[746]

XX. Due to the different lengths of the synodic and draconic months, the lunar node recedes in longitude by about 1;34° per month. During this same month, the Sun on average travels about 29;6° forward in longitude. Therefore, the difference in longitude between the node and the Sun (or the Earth's shadow) at syzygy increases by roughly 30;40° per month.[747]

XX. Eclipse possibilities are defined as being the syzygy at which the lunar latitude is closest to zero.[748] This moment of syzygy occurs when this difference in longitude is either 0° or 180°.[749] If we assume that eclipses do not occur in consecutive months – as it is apparent that the Babylonian astronomers did – it is possible to define an "eclipse possibility" as the syzygy at which the Earth's shadow or the Sun is closest to the node every time it passes by that node.[750]

XX. The average interval between successive eclipse possibilities is equal to about 5;52,7,44 months. Of course, this does not imply that eclipses possibilities occur every 5;52,7,44 months, for then Moon and Sun would not be at syzygy, but rather that eclipses occur every six (6) months, with a five (5) month interval every now and again.[751]

XX. This rule that eclipses can be predicted by simply moving on by six (6) or occasionally five (5) lunar months from the preceding eclipse possibility is the most basic scheme for calculating eclipses that can be identified. Its use is complicated by the uncertainty as to when the five (5) month interval is needed. However, once the months of eclipse possibilities have been identified it is even possible to make a rough estimate of the time of the expected eclipses by measuring the time interval during which the Moon and Sun had been seen together on the days running up to syzygy.[752]

XX. Among the many cuneiform tablets recovered from the [Assyrian] site of Nineveh were a large number of letters and reports sent by Assyrian and Babylonian scholars to the [Assyrian] kings Esarhaddon and Assurbanipal. These probably all date from within the period 674 – 645 BCE (Julian) and contain the earliest series of celestial observations from Mesopotamia preserved in a contemporary source. Earlier astronomical records are preserved from Babylon, but – with the exception of a single diary from 652 BCE (Julian) – they are all contained in compilations which were probably made at a later date.[753]

XX. A number of the letters and reports describe eclipses of the Sun or Moon. Sometimes these have

746 Steele 2000a: 423
747 Steele 2000a: 423
748 Steele 2000a: 446
749 Steele 2000a: 423
750 Steele 2000a: 423
751 Steele 2000a: 423
752 Steele 2000a: 423
753 Steele 2000a: 425

been observed, but on a significant number of occasions they relate to eclipses that were predicted and then not seen.[754]

XX. Only a small amount of information is given in the letters and reports themselves about how the eclipse predictions were made. The report ARAK 502 and others indicate that lunar and solar eclipses were generally expected to take place within a month of one another:[755]

...An eclipse of the Moon and Sun in Sivan (III) will take place. These signs are of bad fortune for Akkad, for the kings of Westland and of Akkad; and now, in this month of Kislev (IX), an eclipse will take place...[756]

XX. This report appears to have been written in the 9th month of a year. During this month an eclipse was expected to take place. Six (6) months later, in the 3rd month of the following year, both a lunar and a solar eclipse were expected. In addition to indicating that lunar and solar eclipses were expected to occur in the same month, this report also implies that the six (6) month interval between eclipse possibilities had been identified.[757]

XX. Other texts, for example LABS 45 indicate that occasionally two (2) successive months were considered as possible dates for an eclipse:[758]

...Concerning the watch of the sun about which the king, my lord, wrote to me, it is (indeed) the month for a watch of the Sun. We will keep the watch twice, on the 28th of Marchesvan (VIII) and the 28th of Kislev (IX). Thus we will keep the watch of the Sun for 2 months...[759]

XX. This suggests that the scholars had realized that not only are eclipses separated by 6, 12, 18 etc. months possible, but also those at 5, 11, 17 etc. months. In other words, eclipse possibilities are separated not only by (6*n*) months (where *n* is an integer less than about 9), but also by (6*n* – 1) months.[760]

XX. Only one (1) letter (LABS 71 which discusses lunar eclipses) makes any reference to this rule and, unfortunately, it is somewhat damaged. The realization that eclipse possibilities can occur at a five-(5)-month interval is also implied by the report ARAK 4 which reports an observation of an eclipse at an "unappointed time" (*ina la mi-na-ti-šú*).[761]

754 Steele 2000a: 426
755 Steele 2000a: 426
756 Steele 2000a: 426
757 Steele 2000a: 426
758 Steele 2000a: 426
759 Steele 2000a: 427
760 Steele 2000a: 427
761 Steele 2000a: 427

XX. It would appear from the [previous] examples that, in general, the scholars were unsure as to when a five-(5)-month interval might occur. One way to eliminate this problem would be to use eclipse periods such as the Saros.[762]

XX. Once the months of the eclipse possibilities have been established, the next question facing the scholars was to decide the day of the eclipse and whether it would be seen, or would "pass by" (or be "omitted") when the luminary was below the horizon. In some texts the scholars were obviously uncertain as to the answer to this question, but in others it would seem that – at least for lunar eclipses – they were even able to predict the watch during which it would occur.[763]

XX. The approximate time of the expected eclipse was apparently determined by observing whether the Moon and Sun were visible together on the previous few days. In the Babylonian Astronomical Diaries measurements of this time interval and five (5) other similar intervals were systematically recorded each month. These are the so called "Lunar Six" described by A. Sachs, "A Classification of the Babylonian Astronomical Tablets of the Seleucid Period," *Journal of Cuneiform Studies 2* (1948), 271-290.[764]

XX. The large number of astronomical tablets recovered from Babylon may be split into two (2) main categories: 1.) texts of mathematical astronomy which have become known as *Astronomical Cuneiform Texts* (ACT), and 2.) texts of non-mathematical astronomy, known as NMAT texts.[765]

XX. The fundamental observational text of the Babylonian astronomers was the Astronomical Diary. These texts typically contain a day by day account of the observations made over a six (6) or seven (7) month period. They include data on the length of the month, measurements of the Lunar Six, the passing by of the Normal Stars by the Moon, eclipses, planetary phases, solstices and equinoxes, Sirius phenomena, and occasionally events such as comets and meteors.[766]

XX. Most of the contents of the [Astronomical] Diaries represent observations; however, where observations were unavailable, for example because of bad weather or because an event was expected to occur at a moment when the heavenly body was below the horizon, then predictions were entered in their place. In addition, some data recorded in the Diaries, such as solstices and equinoxes, were always predicted. Surviving Diaries range in date from 652 to 61 BCE (Julian), although we have reason to believe that the Diary tradition began at least one hundred (100) years earlier [i.e. approximately 747 BCE (Julian)].[767]

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762 Steele 2000a: 427
763 Steele 2000a: 428
764 Steele 2000a: 428
765 Steele 2000a: 428-429
766 Steele 2000a: 429
767 Steele 2000a: 429

XX. From the Diaries, the Babylonians appear to have abstracted records to compile two (2) further types of texts: Goal-Year texts, and texts devoted to individual phenomena.[768] It will suffice here to state that the Goal-Year texts contain, among other things, descriptions of eclipse observations and predictions from eighteen years (18) earlier that were to be used in making predictions for a specific "Goal-Year", and that the texts of individual phenomena include many texts devoted to eclipses that Steele refers to as "Eclipse Texts".[769]

XX. These Eclipse Texts may take one of three (3) forms:[770]

1.) simple lists of successive eclipse observations and predictions;

2.) lists arranged in Saros cycles;

3.) individual reports of particular eclipse observations;

XX. The source of the predictions found in the [Astronomical] Diaries may have been the Almanacs and the Normal Star Almanacs. These texts contain predictions for a coming year of many of the phenomena that are to be found in the [Astronomical] Diaries. Together with the [Astronomical] Diaries they may also have been the source of the astronomical data recorded in a group of texts known as Horoscopes. Finally, we have a group of texts that Steele terms "Theoretical Texts." These texts contain, for example, the dates of specific phenomena such as eclipse possibilities.[771]

XX. If we ignore – for the moment – the theoretical texts, then all of the NMAT described above include references to eclipses of the Sun and Moon. As we would expect, the [Astronomical] Diaries, Goal-Year texts, and the Eclipse Texts contain references to both observed and predicted eclipses, whereas the Almanacs and Normal Star Almanacs contain only predictions. It is not always clear whether the Horoscopes refer to observations or predictions, but at least in some cases they must be predictions. Often accompanying the prediction of the eclipse is a [Lunar Six] time given relative to Sunrise or Sunset.[772]

XX. The Assyrians [before the Babylonians] seem to have realized that eclipse possibilities were separated by six (6) or occasionally five (5) month intervals. To reliably predict eclipses, the Babylonian astronomers needed to formulate a scheme to determine when these five (5) month intervals were required. Commencing in the middle of the 8th century BCE (Julian) it seems that a more or less complete record of observed lunar eclipses was available to the Babylonian astronomers.[773]

768 Steele 2000a: 429
769 Steele 2000a: 429-430
770 Steele 2000a: 430
771 Steele 2000a: 430
772 Steele 2000a: 430
773 Steele 2000a: 431

XX. Britton (1989) has shown that by a fairly basic analysis of this observational record, simple schemes for the arrangement of the five (5) and six (6) month intervals could be identified. The most important of these is the Saros [cycle] of 223 [synodic] months. Within each Saros cycle there are 38 eclipse possibilities: 33 of which are separated by six-(6)-month intervals, and the remaining five (5) by five-(5)-month intervals.[774]

XX. Using the simple rule that these should be distributed as evenly as possible we get the following arrangement: If the first eclipse in a Saros cycle comes five (5) months after the preceding eclipse possibility, then it will be followed by:[775]

– seven (7) eclipses (nos. 2-8) – each of which is six (6) months after the preceding eclipse;

– then an eclipse (no. 9) at five (5) months;

– six (6) more (nos. 10-15) at six (6) months;

– another (no. 16) at five (5) months;

– seven (7) (nos. 17-23) at six (6) months;

– another (no. 24) at five (5) months;

– six (6) at six (6) months (nos. 25-30);

– one (1) more at five (5) months (no. 31);

– and finally seven (7) (nos. 32-38) at six (6) months;[776]

XX. [An alternating pattern of the numbers can be seen: 1, 7, 1, 6, 1, 7, 1, 6, 1, 7].

XX. Thus the 38 eclipse possibilities are divided into five (5) groups, each of which begins with an eclipse possibility five (5) months after the preceding eclipse, containing eight (8), seven (7), eight (8), seven (7), and eight (8) eclipses respectively. This is often written as (8-7-8-7-8).[777]

XX. Of course, the definition of the beginning of the Saros period is arbitrary, and the distribution

774 Steele 2000a: 431
775 Steele 2000a: 431-432
776 Steele 2000a: 431-432
777 Steele 2000a: 432

could equally well be (7-8-7-8-8); (8-7-8-8-7); (7-8-8-7-8); or (8-8-7-8-7). The last distribution is that found by Aaboe (1972) from a theoretical analysis of eclipse possibilities equally spaced in longitude.[778]

XX. That such arrangements were recognized by the Babylonian astronomers is proven by a number of tablets such as the "Saros Canon" (LBAT 1428) and the Eclipse Texts LBAT *1414, LBAT 1415 + 1416 + 1417, and LBAT *1419 which are all laid out in a format based upon this distribution of eclipse possibilities. These texts were all written sometime after the middle of the 4th century BCE (Julian) but refer to dates stretching back to 731 BCE (Julian).[779]

XX. However, the preserved part of the text LBAT *1420 contains eclipse observations and predictions from 604 to 576 BCE (Julian) which also follow the (8-7-8-7-8) arrangement of eclipse possibilities, and this text was probably compiled not long after its final entry in 576 BCE (Julian), thus indicating that the Saros was in use [and discovered] by this period.[780] Temple documents describing the ritual performed in anticipation of an eclipse that did not [end up] occur[ing] in the 8th Regnal Year of Cyrus also imply that the Saros was in use [and discovered] by at least the 6th century BCE (Julian).[781]

XX. Once the (8-7-8-7-8) scheme had been identified, the next problem facing the Babylonian astronomers was to decide when to begin the scheme (in other words, which eclipse possibility was to be defined as no. 1). Evidence for their solution to this problem comes from the records themselves. The texts LBAT *1414, LBAT 1415 + 1416 + 1417, and LBAT *1419 all appear to come from a large compilation of eclipse records that probably originally stretched from 747 to 315 BCE (Julian).[782]

XX. Running through these texts is the expected (8-7-8-7-8) grouping of eclipses, and where an eclipse is predicted at a five-(5)-month interval, the record will explicitly state 5 ITU "5 months". Unfortunately, however, these texts are somewhat fragmentary and so do not fully define the placing of the five-(5)-month intervals.[783]

XX. Eclipse reports are among the earliest astronomical cuneiform texts available. The [Astronomical] Diaries include eclipses – [both] predicted and observed. In the course of time the reports became more and more detailed.[784] Late Babylonian timings of lunar eclipse contacts provide interesting evidence on the accuracy of time measurement by astronomers of the period.[785] The following items can be listed in eclipse reports:[786]

778 Steele 2000a: 432
779 Steele 2000a: 432
780 i.e. the time of Daniel.
781 Steele 2000a: 432
782 Steele 2000a: 432
783 Steele 2000a: 432
784 Sachs and Hunger 1988: 23
785 Stephenson and Fatoohi 1993: 267
786 Sachs and Hunger 1988: 23-24

1.) Date;

2.) Time between Moonrise and Sunset (ME);

3.) Time of beginning of eclipse in terms of a culminating star (*ziqpu*);

4.) Entrance direction of shadow;

5.) Time to maximal phase;

6.) Magnitude of maximal phase;

7.) Duration of maximal phase;

8.) Time from the end of the maximal phase to the end of the eclipse;

9.) Direction in which the shadow crossed the disk;

10.) Duration of the eclipse;

11.) Meteorological remarks (especially wind);

12.) Presence and absence of planets and Sirius;

13.) Position of the Moon relative to a Normal Star;

14.) Time of the beginning of the eclipse relative to Sunset and Sunrise;

15.) Time between Sunrise and Moonset (NA);

XX. Customarily the Babylonians reported the time of the eclipse, the duration of the phases of the eclipse, and an estimate of the magnitude of the eclipse. The time of the eclipse relates to the moment that the eclipse began measured relative to either Sunset or Sunrise, whichever is nearer. However, most cultures have defined Sunrise or Sunset to be the moment when the upper limb of the Sun crossed the horizon, and it seems more likely that this was the definition used in Babylon.[787]

XX. If it were a total eclipse, examples show that the durations of each of the three (3) phases [of the

787 Sachs and Hunger 1988: 24

eclipse] were usually reported:[788]

1.) from 1st contact to 2nd contact (immersion);

2.) totality itself;

3.) 3rd contact (emersion) to last contact.

XX. Often the complete duration from start to finish was also stated. This provides useful confirmation of the veracity of the individual sub-durations. Furthermore, if one of the phases was interrupted by Moonrise or Moonset, this is usually recorded.[789]

XX. In the case of a partial eclipse, three (3) separate times were usually noted: 1.) the interval from 1st contact to the moment when the eclipse appeared to reach its height; 2.) the interval around maximal phase when no change in the degree of obscuration of the moon could be detected; and 3.) the interval from the end of maximal phase to last contact. Again: the interval from beginning to end was frequently stated.[790]

XX. The possibility of a lunar (or solar) eclipse occurs at intervals of six (6) – or sometimes five (5) – months, and this principle and the patterns of 6- or 5-month intervals were well understood by the Babylonians already by the 8th century BCE (Julian).[791]

XX. Solar eclipses may occur at a ½-month's interval before or after a lunar eclipse, but the pattern of 6- or 5- month intervals differs from lunar eclipses (hence lunar and solar eclipse lists are easily distinguishable). In principle, a solar eclipse may occur both before and after a lunar eclipse, but this was not recognized by the Babylonians who only allowed one (1) solar eclipse possibility at each luni-solar conjunction.[792]

XX. Some of the eclipse lists give the data in a conventional list format in one (1) or more columns, whereas others make use of the fact that eclipses of the same character typically recur at intervals of 18 years (or more precisely: 223 lunar months), the so-called Saros period (or Saros cycle)[793] of 38 eclipses (or possibilities).[794] As the Babylonians could predict the time of a lunar eclipse with a fair degree of accuracy by the 7th century BCE (Julian) they would be able to organize their observations so that they would be unlikely to miss an eclipse.[795]

788 Stephenson and Fatoohi 1993: 259
789 Stephenson and Fatoohi 1993: 259
790 Stephenson and Fatoohi 1993: 259
791 Walker 1997: 19
792 Walker 1997: 19
793 Walker 1997: 19
794 Walker 1997: 20
795 Steele 1995: 95

XX. Prior to about 600 BCE (Julian),[796] timing of eclipses was in general fairly crude in Babylon [with] most intervals being estimated only to the nearest 10°. However, by early in the 6th century BCE (Julian),[797] time intervals were consistently expressed to the nearest one (1)° degree.[798]

XX. Eclipses of course can happen at any time of day or night, but only on certain dates which could be established in advance. In view of the rough time indication used with the passings of the Moon by Normal Stars, it seems doubtful – according to Sachs and Hunger – that they could be used in the construction of lunar ephemerides. Certainly eclipse observations could produce far more useful data [for construction of lunar ephemerides].[799]

XX. Because of the precision that can be obtained from the observation of occultations [through eclipses of the Moon and Sun], they are occassionally fixed in time more precisely by means of stars culminating at the same time.[800]

XX. Tables for lunar and solar eclipses [converted to Julian dates] exist. Sachs and Hunger mention P.V. Neugebauer, *Spezieller Kanon der Mondfinsternisse*...., *Astronomische Abhandlungen 9/2* (1934); and *Spezieller Kanon der Sonnenfinsternisse*...., *Astronomische Abhandlungen 8/4* (1931); and the work by H. Mucke and J. Meeus, *Canon of Solar Eclipses*, and *Canon of Lunar Eclipses* (Vienna 1983).[801]

XX. [Ancient Babylonian] lunar eclipse tables normally list eclipses at 18-year intervals [with] the various eclipses in each particular year being cited. This periodicity has enabled Sachs and Schaumberger – as well as Huber – to restore [purportedly] accurate sequences of dates, especially when certain historical events (for example the deaths of rulers) are noted in sequence.[802]

XX. Stephenson and Fatoohi used astronomical computation to check that the Julian dates of lunar eclipses listed by Sachs-Hunger and Huber are valid, and – according to them – they note very few errors indeed.[803]

0.32. Solar Eclipses

XX. Eclipses of the Sun or Moon occur when the centers of the Sun, Earth, and Moon are nearly in a

796 i.e. the time of Daniel.
797 i.e. the time of Daniel.
798 Stephenson and Fatoohi 1993: 259
799 Sachs and Hunger 1988: 23
800 Sachs and Hunger 1988: 23
801 Sachs and Hunger 1988: 19
802 Stephenson and Fatoohi 1993: 257
803 Stephenson and Fatoohi 1993: 257

straight line. This condition can be fulfilled only when Conjunction [i.e. No Moon] and Opposition [i.e. Full Moon] occurs in the vicinity of a node of the lunar orbit on the ecliptic.[804]

XX. A total solar eclipse takes place when the disk of the Moon appears to completely cover the disk of the Sun. Alternately, an annular solar eclipse takes place when the apparent disk of the Moon is smaller than the apparent disk of the Sun so that the light of the Sun is not completely blocked. A partial eclipse of the Sun takes place when the location on the Earth is not in the path of the umbral shadow cast by the Moon.[805]

XX. The path of the shadow on the surface of the Earth can be rather complicated due to the orbital motions of the Earth and Moon, and the [non-uniform] rotation of the Earth. Seven (7) minutes of total solar eclipse is a rather long period for totality. The actual length of time and width of the path varies for solar eclipses. Solar eclipses come in certain patterns that repeat after a long interval called a Saros. A Saros is approximately 223 lunations which is approximately 19 passages[806] of the Sun through a node, or approximately 6585 1/3 days.[807]

0.33. Lunar Eclipses

XX. A lunar eclipse takes place when the Moon passes through the shadow of the Earth. In this case, the times and circumstances are the same for all parts of the Earth from which the Moon is visible [i.e. a key advantage].[808]

XX. There are three (3) types of lunar eclipses:[809]

1.) A penumbral eclipse (also called an appulse) takes place when the Moon enters only the penumbra of the Earth;

2.) A partial eclipse occurs when the Moon enters the umbra without being entirely immersed in it;

3.) A total eclipse takes place when the Moon is entirely immersed within the umbra;

0.34. Saros Cycles

804 Seidelmann 1992: 35
805 Seidelmann 1992: 35
806 cf. the 19-year Metonic cycle
807 Seidelmann 1992: 35
808 Seidelmann 1992: 35
809 Seidelmann 1992: 35

XX. The Saros Cycle of 223 synodic months played an important role in Late Babylonian astronomy. It was used to predict the dates of future eclipse possibilities together with the times of those eclipses and underpinned the development of mathematical lunar theories.[810]

XX. The "Saros Cycle" is the period of 223 synodic months containing 38 eclipse possibilities. "Saros Series" refers to a collection of eclipse possibilities [with] each separated by one (1) Saros of 223 synodic months from the preceding eclipse possibility. A "Saros Scheme" is taken to mean the particular distribution of eclipse possibilities within a Saros Cycle at a given time.[811]

XX. The term "Saros" is modern. To the Babylonians this period was simply called 18 MU.MEŠ "18 years".[812] The excess length of the Saros [i.e. approximately ¼ to 1/3 day] over a whole number [i.e. 6585] of days varies due to solar and lunar anomaly between about six (6) and nine (9) hours[813] between this eclipse and that 223 months later.[814] Furthermore, the Saros cycle of two hundred twenty-three (223) synodic months lasts approximately 10 days longer than 18 solar years.[815]

XX. Throughout most, perhaps all, of the Late Babylonian period, the basic method of predicting eclipses of the Sun and Moon in Babylonia was based upon the so-called Saros cycle (SC) of 223 synodic months. This method was not restricted to simply identifying dates one (1) or more Saroi after a visible eclipse but could be used to identify the dates of all eclipse possibilities (EPs).[816]

XX. It almost certainly originated with lunar eclipses and was then applied to solar eclipses. Indeed, it is most likely that all Babylonian eclipse theories – for want of a better term – were developed from lunar eclipse observations and used for predicting lunar eclipses and then applied by analogy to solar eclipses.[817]

XX. The method for predicting lunar eclipses itself arises by combining the Saros with the realization that EPs generally occur every 6 months but occasionally after only 5 months. Therefore, if within a Saros of 223 months, there are (*a*) eclipses at a 6-month interval and (*b*) eclipses at a 5-month interval, simple mathematics shows from ($6a + 5b == 223$) that ($a == 33$) and ($b == 5$).[818]

XX. Within a Saros of 223 months, there are therefore 38 EPs, of which 33 are at a 6-month interval

810 Brack-Bernsen and Steele 2005: 181
811 Steele 2000a: 424
812 Steele 2000a: 424
813 Brack-Bernsen and Steele 2005: 181
814 Brack-Bernsen and Steele 2005: 189
815 Brack-Bernsen and Steele 2005: 188, 191
816 Brack-Bernsen and Steele 2005: 181
817 Brack-Bernsen and Steele 2005: 181
818 Brack-Bernsen and Steele 2005: 181

and the remaining 5 are at a 5-month interval.[819] Distributing the 5-month intervals as evenly as possible, we obtain five groups of eclipses in which the 1st eclipse of each group follows a 5-month interval and the remainder are at 6-month intervals.[820]

XX. The groups contain respectively eight (8), seven (7), eight (8), seven (7), and eight (8) eclipses. Since EPs repeat after one (1) Saros, this simple distribution results in a matrix-like scheme by which all EPs can be predicted. All that needs to be done is to decide which EP corresponds to the 1st eclipse in the 1st column of the matrix.[821]

XX. There are two (2) groups of theoretical texts which give information on the eclipse schemes of this period: the single tablet LBAT *1418; and a text which has become known as the "Saros Canon" – LBAT 1428 – together with two (2) related texts LBAT *1422 + *1423 + *1424 and LBAT *1425. Both of these two (2) groups of texts contain calculations for earlier eclipses. LBAT *1418 contains dates of eclipse possibilities from parts of the years between -646 and -573 BCE (Julian).[822]

XX. The other group of texts [the "Saros Canon", LBAT 1428, together with two (2) related texts LBAT *1422 + *1423 + *1424 and LBAT *1425], however, contain a variant scheme that gives rise to eclipse possibilities on different dates.[823]

XX. The earliest eclipse records we possess apparently begin in 747 BCE (Julian), and already they include predicted eclipses.[824] LBAT 1413, published as Text No. 1 in Hunger (2001).[825]

XX. Furthermore, we have three (3) tablets that originally formed part of a large compilation of lunar eclipse observations and predictions covering 24 Saroi:[826] LBAT *1414, 1415 + 1416 + 1417, and *1419, published as Texts Nos. 2-4 in Hunger (2001). See the discussion by Steele in the Appendix to that volume, Walker (1997), and Huber and de Meis (2004: 1-6).[827]

XX. The earliest preserved eclipse on these tablets was in 731 BCE (Julian), but it is very probable from the structure of the compilation that when complete, it covered the period from 747 BCE to 315 BCE (Julian). The compilation is set out in exactly the form of the Saros scheme matrix described above, and predictions are given of the times of all EPs that were not observed.[828]

819 Brack-Bernsen and Steele 2005: 181
820 Brack-Bernsen and Steele 2005: 181-182
821 Brack-Bernsen and Steele 2005: 182
822 Steele 2000a: 433
823 Steele 2000a: 433-434
824 Brack-Bernsen and Steele 2005: 182
825 Brack-Bernsen and Steele 2005: 206
826 Brack-Bernsen and Steele 2005: 182
827 Brack-Bernsen and Steele 2005: 206
828 Brack-Bernsen and Steele 2005: 182

XX. Of course, this compilation does not tell us anything about when the Saros scheme was brought into use, since it is quite possible that some of the "predicted" eclipses were calculated backwards when the text was compiled to fill it out. Nevertheless, we have the evidence from LBAT *1420 that the Saros scheme was being used by at least circa 575 BCE (Julian).[829] [830]

XX. It is not known exactly when this scheme for predicting eclipses was developed, but it was almost certainly before about 575 BCE (Julian).[831] The tablet LBAT *1420, published as Text No. 6 in Hunger (2001), contains a collection of eclipse observations for the period from at least 604 BCE to 576 BCE (Julian),[832] and there is nothing to suggest that the text was compiled significantly after the date of its contents. Interspersed between the observed eclipses, the months of EPs are also noted.[833]

XX. The distribution of EPs is exactly what we would expect from the Saros scheme described above and thus implies that scheme was in use by at least 575 BCE (Julian).[834] Furthermore, the early eclipse records (before ca. 570 BCE (Julian)) are themselves recorded with a precision of only 5 UŠ (and were timed with an accuracy several times worse than that).[835]

XX. Determining which months are EPs is not the end of the matter, however. Many of the extant records of eclipse predictions also include the time of day or night when the eclipse was expected to occur. As has been shown by Steele and Stephenson (1997), this time relates to the moment when the eclipse was expected to begin.[836]

XX. This in itself indicates that the times were probably calculated by means of the Saros. If mathematical astronomy had been used to determine the time of the eclipse, it would naturally result in the time of syzygy. To obtain the moment when the eclipse began, a correction to this time would have to be made based upon the expected duration of the eclipse, but with the possible exception of one column of numbers on a single text (the so-called Text S), no functions for the duration of an eclipse are known from the mathematical astronomical texts.[837]

XX. However, when an eclipse was observed, the time when it began was recorded. Adding on the Saros length to this time, the time of the predicted eclipse would also be for when it was expected to begin.[838] The situation is slightly more complicated since times were measured relative to Sunrise or

829 i.e. the time of Daniel
830 Brack-Bernsen and Steele 2005: 182
831 i.e. the time of Daniel.
832 i.e. the time of Daniel.
833 Brack-Bernsen and Steele 2005: 182
834 Brack-Bernsen and Steele 2005: 182
835 Brack-Bernsen and Steele 2005: 190
836 Brack-Bernsen and Steele 2005: 182
837 Brack-Bernsen and Steele 2005: 182
838 Brack-Bernsen and Steele 2005: 182

Sunset, and so corrections would have to be applied to take into account the change in daylength.[839]

0.35. The Rotation of the Earth

XX. The continued failure of successive lunar theories to represent the observed motion of the Moon gradually led to the realization that the rotation of the Earth is not uniform.[840] In ephemerides calculated from gravitational theories, the tabular times are the values of a [theoretical] uniform measure of time, i.e. quite independent of the [actual] variable rotation of the Earth.[841]

XX. Consequently, the observed hour angle of a celestial body (which increases with the Earth's rotation) differs from the position tabulated in the ephemeris [that is based on theoretical, uniform mean motions and positions]. An analysis of the discrepancies between observed and computed positions reveals the difference [i.e. Δ (Delta)] between a measure of the Earth's rotation and the measure of a uniform gravitational time [i.e. with this difference producing the value of ΔT (Delta T)].[842]

XX. The discrepancies are most evident in the mean motion of the Moon due to the rapidity of its motion and the accuracy with which the inequalities of motion can be observed. However, variations in Earth's rotation also produce discrepancies in the observed motions of the other bodies in the solar system in proportion to the magnitude of their respective mean motions.[843]

XX. The first variation to be recognized was a secular retardation of the rate of rotation due to tidal interaction between the Earth and Moon. Adams (1853) showed that the observed Secular [Lunar Orbital Tidal Friction] Acceleration (ṅ) [i.e. rate of change] of the Mean motion of the Moon could not totally be produced by gravitational perturbations. However, the validity of his results was hotly contested.[844]

XX. The issue was resolved when Ferrel (1864) and Delaunay (1865) showed from dynamical principles that the tides [of the Earth's oceans] exert a retarding action on the rotation of the Earth accompanied by a variation of the orbital velocity of the Moon in accordance with the conservation of momentum.[845]

XX. Newcomb (1878) considered the possibility of irregular variations of the Earth's rotation as an

839 Brack-Bernsen and Steele 2005: 206
840 Seidelmann 1992: 76
841 Seidelmann 1992: 76-77
842 Seidelmann 1992: 77
843 Seidelmann 1992: 77
844 Seidelmann 1992: 77
845 Seidelmann 1992: 77

explanation of lunar residuals. However, in the end he could not find collaboration from planetary data.[846] Some of the difficulties faced by Newcomb (and prior astronomers) resulted from inadequacies in the lunar theory.[847]

XX. This situation was rectified when Brown (1919) introduced his new tables. A number of studies offered increasing evidence of irregular variations in the Earth's rotation. Most important were papers by de Sitter (1927) and Spencer Jones (1939) which correlated irregularities in the Moon's motions with irregularities of the inner planets.[848]

XX. Meanwhile, the accuracy of crystal-controlled clocks was becoming comparable with that of the rotation of the Earth. By comparing the observed rates of the clocks of national time services, Stoyko (1937) found a periodic seasonal variation in the rate of [Earth's] rotation.[849]

XX. The Earth's rotation rate was therefore understood to have secular, irregular, and periodic variations. Since Mean Solar Time [MST] was determined directly from sidereal observations, it could no longer be considered uniform [time].[850]

XX. The critical issue of ΔT (Delta T) will be discussed in detail in later sections.

0.36. The Hebrew Calendar is Identical (and Equivalent) with the Babylonian Calendar

XX. As will be unfolded, the Hebrew calendar is essentially identical to – and equivalent with – the Babylonian calendar, albeit the modern Hebrew calendar is the perfected refinement of the observed mathematical preciseness of this calendar.

XX. In his work in 1593 CE (Gregorian), Joseph Justus Scaliger wrote (translated from Latin): "Of all methods of intercalation that exist today, the Jewish calculation is the oldest, the most skillful, and the most elegant."[851] As was pointed out in 1000 CE by the celebrated Persian Muslim writer, al-Biruni, the Hebrew calendar repeats its pattern only after 689,472 years.[852] Therefore this makes the Hebrew calendar extremely accurate and thus the optimum calendar for chronology within our era of recorded history going back to the time of the Sumerians ca. 3000 BCE (Julian).

XX. In the Hebrew lunisolar calendar, days begin at Sunset, months begin with the New Moon, and

846 Newcomb 1912 – cited by Seidelmann 1992: 77
847 Seidelmann 1992: 77
848 Seidelmann 1992: 77
849 Seidelmann 1992: 77
850 Seidelmann 1992: 77
851 Dershowitz and Reingold 2013: 21
852 Dershowitz and Reingold 2013: 28, 29

years are kept in tune with the seasons by the intercalation of a leap month every 2-3 years [in years 3, 6, 8, 11, 14, 17, 19 (Year 20 == Year 1)][853] known as the Metonic cycle.

XX. A point of particular interest is when the standard 19-year cycle of seven (7) intercalations – known as the "Metonic cycle" after the Greek astronomer Meton who introduced it to the Athenians in the late 5th century (c. 432/33) BCE (Julian) – was introduced in Babylonia.[854] According to Finegan, this Metonic cycle, bearing the name of the astronomer Meton, was introduced to Greece in 433 BCE (Julian), having been already known in eastern countries.[855]

XX. Writing in 2013, Ossendrijver argues that this intercalation cycle had been implemented by the 6th century [i.e. 599 – 500] BCE (Julian) by Babylonian [priest-]astronomers [i.e. the Neo-Babylonian period precisely during the time of Daniel].[856] However, writing in 2017, he refines his position to state that it was near 475 BCE (Julian) that this 19-year cycle was introduced. Henceforth, a 13th month was inserted in seven (7) out of 19 years according to a fixed pattern, resulting in a cycle of 235 months.[857]

XX. Neugebauer (1957 / 1969) notes that although it was incorporated into the Babylonian calendar, Meton's proposal was rejected by Athens in the 5th century BCE (Julian).[858] In fact, the Greeks did not use calculated calendars before the 1st century BCE (Julian), [instead] relying mostly on observation. This cycle was part of the entry of Babylonian astronomy and mathematics into the Greco-Roman world that took place as cuneiform culture was declining.[859] Common perceptions of a fundamental difference between scientific cultures in the ancient Near East and classical Greece turned out to be untenable.[860]

XX. This [Metonic] system – still in use [today] in the current Jewish calendar – requires the addition of a 13th [intercalational leap lunar] month in seven (7) out of 19 years, and also entails specific alternations between 29 and 30-day months.[861] The process of applying astronomical knowledge to calendar procedures happened to the Jews long before the Greeks – most probably through the medium of Aramaic (the linguistic and cultural lingua franca of the later ancient Near East).[862]

XX. The Metonic cycle was retained in the Seleucid empire of Babylonia and is still employed in both Jewish and East Indian religious calendars. Toomer (1984) has argued that Meton's purpose was not the

853 Dershowitz and Reingold 2013: 22
854 Walker 1997: 23; Dershowitz and Reingold 1997: 9
855 Finegan 1998: 406
856 Ossendrijver 2013a: 6071
857 Ossendrijver 2016a: 131
858 Kelley and Milone 2011: 100
859 Schiffman 2012: 234
860 Ossendrijver 2015b: 1869
861 Schiffman 2012: 234
862 Schiffman 2012: 234

reform of the Athenian calendar, but [rather] the establishment of an astronomical chronology.[863]

XX. This 19-year cycle determined the intercalation of seven (7) individual [lunar] months every nineteen (19) [lunar] years, i.e. during the course of nineteen (19) years, there will be seven (7) years when an intercalary month will be added: in years 1, 3, 6, 9, 11, 14, 17. All years with intercalary months except year one (1) had an extra 12th month (Akkadian: *Addāru arkû*; Hebrew: *Adar 2*). The 1st year in the cycle had an extra 6th month (Akkadian: *Ulūlu arkû*; Hebrew: *Elul 2*). The 19-year cycle of intercalation, begun – according to Rochberg – under the Achaemenids, remained standard for the succeeding Seleucid and Arsacid periods, [and] until the end of the cuneiform tradition.[864]

XX. The so-called Metonic cycle is based on the observation that 19 solar years contain almost exactly 235 [lunar] months. This cycle makes a relatively simple and accurate fixed solar / lunar [i.e. lunar-solar] calendar feasible. The Metonic cycle is used in the Hebrew calendar as well as for the ecclesiastical calculation of Easter, and can be defined as follows:[865]

(19-year Metonic cycle) == (235 months) == ((twelve (12) 12-month years) + (seven (7) 13-month leap-years))

XX. Therefore:

(12 x 12) + (7 x 13) == (235 months)

XX. The more precise the mean year [of any particular calendar system], the larger the underlying constants must be. For example, the Metonic cycle [of the Hebrew calendar system] is currently accurate to within 6.5 minutes per year.[866]

XX. Dershowitz and Reingold note that the mean year (in days) [in the Hebrew calendar system] is 365.24682 [days per year], and a mean month (in days) [of the same Hebrew calendar system] is 29.530594 [days per lunar month].[867] However, two (2) pages prior, Dershowitz and Reingold offer two (2) slightly different numbers for these two (2) values. They note that in the case of lunisolar calendars [as is the Hebrew calendar system], the solar year is about 365.24244 days, while the lunar month is about 29.53059 days.[868]

XX. It is perhaps ironic that this is often called the Metonic cycle, after the Athenian astronomer who introduced it to the West in 432 BCE (Julian). Historians of Babylonian astronomy in recent decades

863 Kelley and Milone 2011: 100
864 Rochberg 1995: 1938
865 Dershowitz and Reingold 1997: 9; Dershowitz and Reingold 2008: 7
866 Dershowitz and Reingold 2008: 7
867 Dershowitz and Reingold 2008: 10
868 Dershowitz and Reingold 2008: 8

have come to the conclusion that the cycle was known to the Babylonians by about 500 BCE (Julian), and indeed [scholars] have regarded its recognition as one of the significant factors in the early development of [Babylonian] mathematical astronomy in the 5th century BCE (Julian).[869]

XX. Prior to this point, although the New Year never shifted far from the Spring Equinox, it is not clear that any mathematical principle of intercalation was applied.[870] Originally this [intercalation of a leap month] was done at irregular intervals[871] [according to edict of the king in consultation with his royal priest-astronomers, as Daniel would have served as astronomer – among other responsibilities – in the Royal Court of Babylon].

XX. Steele notes that though it is possible that Meton learned of the 19-year cycle (frequently called the "Metonic cycle" in modern scholarship) from Babylon, its use in China illustrates that the cycle is sufficiently simple to have been identified independently,[872] [However, perhaps the knowledge of this practice spread from the ancient Babylonians to the ancient Chinese].

XX. In ancient and classical times the month began with the observation by at least two (2) witnesses of the crescent Moon [in the western evening sky at Sunset on the critical 30th day of the month].[873]

XX. Leap months [at the end of a year in a leap year] were added by the Jerusalem authorities as the need arose. The fixed [Hebrew] calendar [as we have it today] – attributed to the 4th-century [CE] patriarch, Hillel II – is based on a mean month of $29^{d}\ 12^{h}\ 44^{m}\ 3\frac{1}{3}^{s}$ and on the 19-year Metonic cycle comprising seven (7) leap years (each containing thirteen (13) months), and twelve (12) [common] years (each containing twelve (12) months):[874]

(7 x 13) + (12 x 12) == (235 months in 19-year Metonic cycle)

XX. The average [lunisolar-]year length [of the Hebrew 19-year Metonic cycle of lunar years] is 365.24682 days,[875] which is slightly longer than the mean Gregorian solar year of 365.2425[876] days as well as the mean tropical [solar] year [of 365.242177 days]. On account of the accumulated discrepancy, Passover often occurs nowadays more than a month after the vernal equinox[877] [which itself also shifts / slips / moves / drifts / wanders because of precession (of the equinoxes), and thus was at a different position ca. 2550 – 2625 years ago during the Neo-Babylonian period, i.e. (the time of Daniel)].

869 Walker 1997: 23
870 Walker 1997: 23
871 Ossendrijver 2016a: 131
872 Steele 2011a: 477
873 Dershowitz and Reingold 2013: 22
874 Dershowitz and Reingold 2013: 22
875 Dershowitz and Reingold 2018: 11
876 Dershowitz and Reingold 2018: 11
877 Dershowitz and Reingold 2013: 22

XX. Because the average Hebrew year length of 365.24682 days is slightly too long, the Hebrew year will drift slowly through the Gregorian year (which closely approximates the mean tropical year). This drift means that Passover will get later and later in the Gregorian year.[878]

XX. The fixed Hebrew lunisolar calendar incorporates several rules for delaying the onset of the year [i.e. Rosh HaShanah ("Head of the Year") New Year] – as a consequence of which common years have 353, 354, or 355 days, and leap years have 383, 384, or 385 days.[879] Passover never begins on Monday, Wednesday, or Friday [and Rosh HaShanah (Civil New Year) never begins on Sunday, Wednesday, or Friday]. The details of the fixed [Hebrew] calendar were finalized by the 10th century [CE].[880]

XX. Since the Middle Ages, the [Hebrew] year number has almost always been Anno Mundi [(AM), i.e. Latin for "Year of [the Creation of] the World"], starting with year 1 AM which began on Monday 07 October 3761 BCE (Julian) or Monday 07 September -3760 (Gregorian Astronomical).[881] The historical use of several other "eras of creation" has created much confusion, notably with regard to the date of destruction of the Second Temple in 70 CE.[882]

XX. To highlight this asymmetry [between Julian BCE and Non-Calibrated Gregorian Astronomical BCE years], in the prose of their book, Dershowitz and Reingold append "BCE" only to Julian calendar years, i.e. reserving the minus sign [" – "] for Gregorian calendar years. However, in their internal computer representation of Julian BCE years (i.e. their mathematical algorithms for computer programming), Dershowitz and Reingold use negative numbers [also for Julian BCE years] with the convention that year Q BCE (Julian) [where there is no year zero (0) which causes the year before year 1 CE (Julian) to be 1 BCE (Julian)] is represented as -Q[883] [just as year Q BCE (Gregorian) – where year zero (0) BCE (Gregorian) which is the year before year one (1) CE (Gregorian) – is represented as -Q. However, there is a difference...

XX. Therefore, for sake of clarity, when discussing ancient dates BCE, we will mostly use the convention of not using the minus sign for both Gregorian and Julian BCE years, as well as denoting all years (including year zero (0)) before 1 CE as being year zero (0) "BCE". Furthermore, it will also be introduced here the concept of "calibrated Gregorian years", i.e. years that we term as "Gregorian Calibrated" which is simply to use the conventional Julian BCE year in most cases (except close to 01 January of every year when the Julian and Gregorian are in different years) along with the calculated equivalent Gregorian month and day. The only exception will be the few days (possibly) between,

878 Dershowitz and Reingold 2018: 133
879 cf. Appendix Table X.0
880 Dershowitz and Reingold 2013: 22
881 Monday 07 October 3761 BCE (Julian); Monday 07 September 3761 BCE (Gregorian Calibrated); 01 Tashritu / Tishri 1 AM; Astronomical Julian Day JD_X: 347997.5; RD_Y: 1; RD_X: -1373427; Sabbatical Cycle: 1; Metonic Cycle: 1;
882 Dershowitz and Reingold 2013: 25
883 Dershowitz and Reingold 2008: 14

during, before, and after the change of Julian / Gregorian year at Midnight (00:00) of 01 January of every year (i.e. in December / January) if these Julian and Gregorian dates are in different years.[884]

XX. For dates near the present, the Julian Day (JD) number [can be] inconvenient because at least 7-digit accuracy is needed. [Thus for more modern dates] astronomers occasionally use Modified Julian Day numbers (MJD) which counts days from Midnight (00:00) Wednesday 17 November 1858 CE (Gregorian). We do not use Modified Julian Days (MJD) directly because we want positive numbers for dates within recent [and ancient] history [prior to 17 November 1858 CE].(Gregorian).[885]

0.37. Calendrical Calculations of Chronological Data using a Custom Computer Program with the Algorithms of Dershowitz and Reingold

XX. It is possible to write a computer program to calculate Babylonian / Hebrew dates from Julian ones and vice versa.[886] Modern equivalents of [ancient] Babylonian / Hebrew dates are expressed in the Julian [and/or Gregorian] calendar. Since the Julian day begins at Midnight (00:00), and since a Babylonian / Hebrew day begins at Sunset [18:00+/-], a Babylonian / Hebrew date covers parts of two (2) successive Julian dates.[887]

XX. The algorithms and conventions utilized here for calendrical calculations of chronological data are those provided by Dershowitz and Reingold.[888]

XX. Note that in publications on ancient astronomy, the BCE year count [that is in Julian years] – which omits the year zero (0) – is often replaced by negative Julian year numbers such that 1 BCE (Julian) is counted as year zero (0) and – analogously:[889]

$-n$ BCE as: $-(n - 1)$

e.g. -586 BCE (Julian) as: -(586 – 1) == -585 (Gregorian Astronomical)

OR:

$-n$ BCE == $-(n - 1)$

e.g. -586 BCE (Julian) == -(586 – 1) == -585 (Gregorian Astronomical)

884 cf. Footnote #9
885 Dershowitz and Reingold 2008: 17
886 De Meis 2004: 119
887 Ossendrijver 2016a: 131
888 Dershowitz and Reingold 1997; Dershowitz and Reingold 2008; Dershowitz and Reingold 2018;
889 Ossendrijver 2016a: 131-132

OR:

$-n$ BCE == $-(n - 1)$

e.g. 586 BCE (Julian) == 586 BCE (Gregorian Calibrated)

XX. As noted by De Meis, modern astronomers are much more expert in big bang theories or other astrophysical questions than in calculating heliacal phenomena or lunar occultations. Hence it is necessary to find one who is conversant with the calculations of the ancient astronomical phenomenology.[890] [Ancient Babylonian] astronomical texts usually deal with datable and relatively well-defined phenomena that can be accurately reconstructed for ancient dates with the help of modern computations.[891]

XX. In the study of the astronomical part of cuneiform texts, amongst the main purposes is the dating of a text. As it is well known, even if a date was written in a tablet, often it is broken or otherwise illegible.[892] If we are lucky enough to find a relatively clear date, this might refer to a regnal or accession year of a king, [or perhaps] to a year of the Seleucid Era.[893] In this case, we have tools such as Parker-Dubberstein (1956), with extensions by Parpola (1983), or Peter Huber (1982, 1999/2000).[894]

XX. In regards to commercial [astronomy] computer programs, according to De Meis, no program can properly calculate all the phenomena described in Babylonian astronomy, as several programs are needed and linked, and commercial programs are far to meet the requirements of Assyriologists.[895] A word of warning includes even very recent ones.[896]

XX. De Meis cites Jean Meeus (1991) who advises that ideally one should write his [or her] own programs especially [in order to know] to know the limitations of the programs, the proper use of certain details and parameters, as well as the necessity of including special data or conditions. In other words [one who writes his/her own programs will] know what to expect, [and will] not have a blind trust on the program results which may be wrong instead.[897]

XX. Hence we can conclude how misleading could be the simple browsing into eclipse catalogs or Canons, or to perform calculations without the necessary accuracy and care.[898] To omit in the calculations the value of ΔT is another cause of errors [as will be examined later in great detail].[899]

890 De Meis 2004: 119
891 Ossendrijver 2016a: 125
892 De Meis 2004: 118
893 De Meis 2004: 118-119
894 De Meis 2004: 119
895 De Meis 2004: 119
896 De Meis 2004: 120
897 De Meis 2004: 120
898 De Meis 2004: 121
899 De Meis 2004: 121

XX. Another warning concerns the difficult motion of the Moon which is quite fast.[900] As previously mentioned, De Meis notes that [Isaac] Newton said that the Moon caused his head to ache, which is something for such a genius![901]

XX. Because of the Moon's fast motion, its calculation needs thousands of correcting parameters [i.e. variables], and they might not be known exactly or not all incorporated in a commercial [computer] program, so that the Moon [position] results somewhere else when the Sun or other bodies are supposed to be eclipsed or worse: occulted.[902]

XX. The positions of stars [i.e. variables: further complicates matters]: if calculated without precession and proper motion [i.e. variables], they may give surprises, and lunar conjunctions – or worse: occultations – fail the result, and [thus] we are tempted to exercise the usual attack to careless [ancient] scribes who really deserve no such insult.[903]

XX. A further advantage [of developing one's own custom program] is in the calculations of events for which no definite dating is immediately possible, and in this case one will be able to put forward several proposals which can be confirmed by further studies.[904]

XX. As we shall see, Babylonians could predict eclipses but not to the extent of their complete appearance evolution as seen from a given site.[905] We find that such eclipses rated [Sumerian:] "NU PAP", that is eclipses known to occur but not seen, were visible elsewhere [on the Earth other] than Babylon, but the times of the event are correct.[906]

XX. Due to lack of geographical knowledge and relative sizes of Sun and Moon, and in the absence of physical theories, the prediction of visibility from a given place was not possible. Still what is important is to document that predictions were made with a remarkable accuracy, [and only] obtained by other astronomers much later.[907]

0.38. Fixed Day Numbers (RD_X, RD_Y) and Epochs

XX. For a computer implementation, the easiest way to reckon time is simply to count days: Fix an arbitrary starting point as day one (1) and specify a date by giving a day number relative to that starting

900 De Meis 2004: 121
901 De Meis 2004: 121
902 De Meis 2004: 121
903 De Meis 2004: 121
904 De Meis 2004: 122
905 De Meis 2004: 122
906 De Meis 2004: 123
907 De Meis 2004: 123

point. A single 32-bit integer allows the representation of more than 11.7 million years. Evidently such a reckoning of time is extremely awkward for human beings and is not in common use, except among astronomers who use *Julian Day* (JD [or JD_X]) *numbers* to specify dates, as well as calendarists who use them to facilitate conversion among calendars.[908]

XX. Following Dershowitz and Reingold's outstanding scholarship of mathematical algorithms and conventions for these calendrical calculations, we will affirm their choice of the onset of Midnight [00:00] on Monday 01 January 0001 CE (Julian) as our fixed day one (1), which we abbreviate [as variable] (RD) or (RDX) – i.e. *Rata Die* meaning "fixed date" – [of variable X, i.e. (RD_X), e.g. (RD_1) for day 1, (RD_2) for day 2, (RD_3) for day 3, etc.].[909]

XX. Extrapolating Dershowitz and Reingold's (RDX) methodology and value for the 1st day of the Hebrew Epoch (i.e. $RD_{-1373427}$), we will create and coin another fixed-day variable, (RDY) – i.e. *Rata Die Yehudi* – i.e. "fixed Jewish date", with (RDY_1) being defined as the 1st day of the Hebrew Epoch, and (RDY_1) being equivalent to ($RDX_{-1373427}$):[910]

(RDY_1) == ($RDX_{-1373427}$) == Monday 07 October 3761 BCE (Julian)

(RDY_1) == ($RDX_{-1373427}$) == Monday 07 September -3760 (Gregorian Astronomical).

XX. However, as there is no Year Zero (0) in the Julian calendar, often it will be convenient to calibrate the Gregorian day and month with (and thus use) the Julian year to express Gregorian dates in ancient history:[911]

(RDY_1) == ($RDX_{-1373427}$) == Monday 07 September -3761 BCE (Gregorian Calibrated)

XX. Unless specified with the term "Gregorian Astronomical", all Gregorian BCE dates provided here are "Gregorian Calibrated", and shortened to "Gregorian" for convenience.

XX. Each day also has a Julian Day number ((JD [JDX, or JD_X])) equivalent:

(RDY_1) == ($RDX_{-1373427}$) == ($JDX_{347997.5}$)

XX. Thus we can define, equate, and solve for an unknown (RDY) number, (RDY_Q), when given a known (RDX) number:

908 Dershowitz and Reingold 1997: 9; Dershowitz and Reingold 2008: 9
909 Dershowitz and Reingold 1997: 9; Dershowitz and Reingold 2008: 10
910 Monday 07 October 3761 BCE (Julian); Monday 07 September 3761 BCE (Gregorian Calibrated); 01 Tashritu / Tishri 1 AM; Astronomical Julian Day JD_X: 347997.5; RD_Y: 1; RD_X: -1373427; Sabbatical Cycle: 1; Metonic Cycle: 1;
911 Monday 07 October 3761 BCE (Julian); Monday 07 September 3761 BCE (Gregorian Calibrated); 01 Tashritu / Tishri 1 AM; Astronomical Julian Day JD_X: 347997.5; RD_Y: 1; RD_X: -1373427; Sabbatical Cycle: 1; Metonic Cycle: 1;

$(RDY_Q) == (RD_X + 1373428)$

$(RDY_Q) == (RDX_{-1373427} + 1373428)$

$(RDY_Q) == (1.0)$

$(RDY_Q) == (RDY_1)$

XX. We can also define, equate, and solve for an unknown (RDX) number, (RDX_Q), when given a known (RDY) number:

$(RDX_Q) == ((RD_Y) – 1373428)$

$(RDX_Q) == ((RDY_1) – 1373428)$

$(RDX_Q) == (-1373427.0)$

$(RDX_Q) == (RDX_{-1373427})$

XX. Since the Gregorian calendar was devised only in the 16th century CE (Julian / Gregorian), there was no year 1 CE on the Gregorian calendar.[912] So by (RD_1) Monday 01 January 1 CE (Gregorian) we mean the day (RD_1) we get if we extrapolate backwards from the present, and this day turns out to be equivalent to (RD_1) Monday 03 January 1 CE (Julian).[913] However, this too is anachronistic.[914]

XX. Each calendar has an *epoch*, i.e. the 1st day of the 1st year of that calendar.[915] [Therefore (RD_1) or (RDX_1) Monday 01 January 1 CE (Gregorian) will be the epoch of the Gregorian calendar that we are using.][916] For the epoch, we know that (RD_1) is Monday 03 January 1 CE (Julian),[917] and thus the first day of the Julian calendar – Monday 01 January 1 CE (Julian) – must be 30 December 0 BCE (Gregorian Astronomical), that is: (RD_{-1}).[918] [919]

912 Dershowitz and Reingold 1997: 9; Dershowitz and Reingold 2008: 10

913 Monday 03 January 1 CE (Julian); Monday 01 January 1 CE (Gregorian); 18 Tebetu / Tevet 3761 AM; Astronomical Julian Day JD_X: 1721425.5; RD_Y: 1373429 ; RD_X: 1; Sabbatical Cycle: 2; Metonic Cycle: 18;

914 Dershowitz and Reingold 1997: 10; Dershowitz and Reingold 2008: 10

915 Dershowitz and Reingold 1997: 11; Dershowitz and Reingold 2008: 14

916 Dershowitz and Reingold 1997: 13; Dershowitz and Reingold 2008: 14

917 Monday 03 January 1 CE (Julian); Monday 01 January 1 CE (Gregorian); 18 Tebetu / Tevet 3761 AM; Astronomical Julian Day JD_X: 1721425.5; RD_Y: 1373429 ; RD_X: 1; Sabbatical Cycle: 2; Metonic Cycle: 18;

918 Saturday 01 January 1 CE (Julian); Saturday 30 December 0 BCE (Gregorian Astronomical); 16 Tebetu / Tevet 3761 AM; Astronomical Julian Day JD_X: 1721423.5; RD_Y: 1373427; RD_X: -1; Sabbatical Cycle: 2; Metonic Cycle: 18;

919 Dershowitz and Reingold 2008: 64

XX. Every calendar's epoch, which is its starting date, is virtually never the [same] date that the calendar was adopted, but rather a hypothetical starting point for the 1st day [of that calendar system].[920] As noted by Seidelmann, a count of years from an initial epoch is the most successful way of maintaining a consistent chronology.[921]

0.39. Epoch Numbers of Common Calendars

XX. With the exception of the Julian Day (JD) number (JD_X), Dershowitz and Reingold express the epochs of all [their calculated] calendars as (RD_X) day [numbers of whole] integer [type], i.e. the (RD_X) day [expressed as a whole] integer number (RD_X) at Noon (12:00) [relative to] the 1st day of the calendar, i.e. e.g.:[922] [923]

(RD_1) == Monday 03 January 1 CE (Julian)

(RD_1) == Monday 01 January 1 CE (Gregorian)

XX. Following Dershowitz and Reingold's convention, an integer [whole number] (RD_X) date is assigned to an epoch, even if the calendar in question [e.g. the Hebrew calendar system] begins its days at a time other than Midnight (00:00) – as does the Hebrew calendar system that begins its days at Sunset [at approximately (18:00+/-)].[924]

XX. Thus the epoch for the Gregorian calendar [i.e. variable: EpochGregorian] is (RD_1), and the epoch for the Hebrew calendar [i.e. variable: EpochHebrew] is ($RD_{-1373427}$).[925]

XX. That is to say the Gregorian epoch (i.e. variable: EpochGregorian) is defined as follows:[926] [927]

(RD_1) == Monday 03 January 1 CE (Julian)

(RD_1) == Monday 01 January 1 CE (Gregorian)

(RD_1) == Monday 18 Tebetu / Tevet (i.e. 10th month) 3761 AM (Hebrew)

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920 Dershowitz and Reingold 1997: 13; Dershowitz and Reingold 2008: 14
921 Seidelmann 1992: 579
922 Dershowitz and Reingold 2008: 14
923 Monday 03 January 1 CE (Julian); Monday 01 January 1 CE (Gregorian); 18 Tebetu / Tevet 3761 AM; Astronomical Julian Day JD_X: 1721425.5; RD_Y: 1373429 ; RD_X: 1; Sabbatical Cycle: 2; Metonic Cycle: 18;
924 Dershowitz and Reingold 2008: 12
925 Dershowitz and Reingold 2008: 14
926 Monday 03 January 1 CE (Julian); Monday 01 January 1 CE (Gregorian); 18 Tebetu / Tevet 3761 AM; Astronomical Julian Day JD_X: 1721425.5; RD_Y: 1373429 ; RD_X: 1; Sabbatical Cycle: 2; Metonic Cycle: 18;
927 Dershowitz and Reingold 1997: 14; Dershowitz and Reingold 2008: 15

XX. Within the context of – and relative to – this Gregorian calendar base, the epoch for the Julian calendar (i.e. variable: EpochJulian) is defined as follows:[928] [929]

(RD_{-1}) == Saturday 01 January 1 CE (Julian)

(RD_{-1}) == Saturday 30 December 0 BCE (Gregorian Astronomical)

(RD_{-1}) == Saturday 30 December 1 BCE (Gregorian Calibrated)

(RD_{-1}) == Saturday 16 Tebetu / Tevet (i.e. 10th month) 3761 AM (Hebrew)

XX. Within the context of – and relative to – this Gregorian calendar base, the epoch for the Hebrew calendar (i.e. variable: EpochHebrew) is defined as follows:[930] [931]

($RD_{-1373427}$) == Monday 07 October 3761 BCE (Julian)

($RD_{-1373427}$) == Monday 07 September -3760 (Gregorian Astronomical)

($RD_{-1373427}$) == Monday 07 September 3761 BCE (Gregorian Calibrated)

($RD_{-1373427}$) == Monday 01 Tashritu / Tishri (i.e. 7th month) 1 AM (Hebrew)

XX. Within the context of – and relative to – this Gregorian calendar base, the epoch for the Islamic calendar (i.e. variable: EpochIslamic) is defined as follows:[932] [933]

(RD_{227015}) == Friday 16 July 622 CE (Julian)

(RD_{227015}) == Friday 19 July 622 CE (Gregorian)

(RD_{227015}) == Friday 03 Abu / Av (i.e. 5th month) 4382 AM (Hebrew)

(RD_{227015}) == Friday 01 Muharram (i.e. 1st month) 0001 AI (Islamic)

928 Saturday 01 January 1 CE (Julian); Saturday 30 December 0 BCE (Gregorian Astronomical); 16 Tebetu / Tevet 3761 AM; Astronomical Julian Day JD_X: 1721423.5; RD_Y: 1373427; RD_X: -1; Sabbatical Cycle: 2; Metonic Cycle: 18;

929 Dershowitz and Reingold 1997: 14; Dershowitz and Reingold 2008: 15

930 Monday 07 October 3761 BCE (Julian); Monday 07 September 3761 BCE (Gregorian Calibrated); 01 Tashritu / Tishri 1 AM; Astronomical Julian Day JD_X: 347997.5; RD_Y: 1; RD_X: -1373427; Sabbatical Cycle: 1; Metonic Cycle: 1;

931 Dershowitz and Reingold 1997: 14; Dershowitz and Reingold 2008: 15

932 Friday 16 July 622 CE (Julian); Friday 19 July 622 CE (Gregorian); 03 Abu / Av 4382 AM; Astronomical Julian Day JD_X: 1948439.5; RD_Y: 1600443; RD_X: 227015; Sabbatical Cycle: 7; Metonic Cycle: 12;

933 Dershowitz and Reingold 1997: 14; Dershowitz and Reingold 2008: 15

0.40. How to Think of the Passage of Time

XX. We should thus think of the passage of time as a sequence of days numbered as -3, -2, -1, 0, 1, 2, 3...[or rather: (RD_{-3}), (RD_{-2}), (RD_{-1}), (RD_{0}), (RD_{1}), (RD_{2}), (RD_{3})...]. Each of the various human-oriented calendars label this sequence of days differently.[934] All that is required for calendrical conversion is to be able to convert each calendar to and from this fixed calendar.[935]

XX. Since some calendars begin their day at Midnight (00:00) and others at [variable] Sunrise (06:00+/-) or [variable] Sunset (18:00+/-), for our calculations we will follow Dershowitz and Reingold's convention to fix the time of day at which conversions are performed at Noon (12:00).[936]

XX. For example, Midnight (00:00) (RD_{710347}) would be formatted as follows:[937] [938]

1.) Monday Noon (12:00) 12 November 1945 CE (Gregorian).

2.) Monday Noon (12:00) 30 October 1945 CE (Julian).

3.) Monday Noon (12:00) 06 Dhu al-Ḥijja [i.e. 12th month] 1364 AI on the Islamic calendar (until Sunset).[939]

4.) Monday Noon (12:00) 07 Kislimu / Kislev [i.e. 9th month] 5706 AM on the Hebrew calendar (until Sunset).[940]

5.) Monday Noon (12:00) Julian Day (JDX) number JD 2,431,772.0 (i.e. $JD_{2431772.0}$).[941] [942]

6.) Monday Noon (12:00) Rata Die (RDX) number RD 710,347.5 (i.e. $RD_{710347.5}$)[943]

7.) Monday Noon (12:00) Rata Die (RDY) number RDY 2,083,775.5 (i.e. $RDY_{2083775.5}$)[944]

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934 Dershowitz and Reingold 1997: 10; Dershowitz and Reingold 2008: 11
935 Dershowitz and Reingold 1997: 10; Dershowitz and Reingold 2008: 12
936 Dershowitz and Reingold 1997: 10; Dershowitz and Reingold 2008: 12
937 Monday 30 October 1945 CE (Julian); Monday 12 November 1945 CE (Gregorian); 07 Kisilimu / Kislev 5706 AM; Astronomical Julian Day JD_X: 2431771.5; RD_Y: 2083775; RD_X: 710347; Sabbatical Cycle: 1; Metonic Cycle: 6;
938 Dershowitz and Reingold 1997: 10; Dershowitz and Reingold 2008: 11
939 Dershowitz and Reingold 1997: 10; Dershowitz and Reingold 2008: 11
940 Dershowitz and Reingold 1997: 10; Dershowitz and Reingold 2008: 11
941 Dershowitz and Reingold 2008: 11
942 The fractional decimal number here will be explained below.
943 The fractional decimal number here will be explained below.
944 The fractional decimal number here will be explained below.

0.41. Ancient Dates before the Common Era (CE)

XX. We cannot avoid dealing with dates before the common era CE.[945] [However, there is no year zero (0) on the Julian calendar.] On the Julian calendar it is customary to refer to the year preceding 1 CE (Julian) as 1 BCE (Julian).[946]

XX. Therefore it is computationally convenient – and mathematically sensible – to [also] label years with the sequence of integers such as -3, -2, -1, 0, 1, 2, 3, etc. so that year zero (0) BCE (Gregorian Astronomical) precedes year 1 CE (Julian / Gregorian). We do this when extrapolating backward on the Gregorian calendar so that the same [Gregorian] leap-year rule will apply based on divisibility by 4, 100, and 400.[947]

XX. Although the Gregorian calendar did not exist prior to the 16th century [i.e. October 1582 CE], we can extrapolate backwards using its rules to obtain what is sometimes called the "proleptic Gregorian calendar". However, the name is really a misnomer because "proleptic" refers to the future – not the past. Unlike the Julian calendar, we implement this proleptic [Gregorian] calendar with a year zero (0) [BCE] as is common among astronomers[948] who do use a year zero (0) [BCE] preceding year 1 CE on the Gregorian [i.e. not Julian] calendar.[949]

0.42. Julian Days (JD) or Julian Astronomical Days (JAD) and their RD_X and RD_Y Equivalents

XX. To facilitate chronological reckoning, astronomical days beginning at Greenwich Noon (12:00) are numbered consecutively from an epoch intended to be sufficiently far in the past to precede the historical period. The number assigned to a day in this continuous count is the Julian day number [(JD) or (JD_X)] which is defined to be zero (0) [i.e. ($JD_{0.0}$) for the day starting at Greenwich Mean Noon (12:00) on Monday 01 January 4713 BCE (Julian),[950] [951] [952] Julian proleptic calendar.[953]

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945 Dershowitz and Reingold 2008: 12
946 Dershowitz and Reingold 1997: 11; Dershowitz and Reingold 2008: 14
947 Dershowitz and Reingold 2008: 12
948 Dershowitz and Reingold 2008: 48
949 Dershowitz and Reingold 2008: 46
950 Monday 01 January 4713 BCE (Julian); Monday 24 November 4713 BCE (Gregorian Astronomical); 20 Tebetu / Tevet -952 AM; Astronomical Julian Day JD_X: -0.5; RD_Y: -347997; RD_X: -1721425; Sabbatical Cycle: 7; Metonic Cycle: 17;
951 Tuesday 02 January 4713 BCE (Julian); Tuesday 25 November 4713 BCE (Gregorian Astronomical); 21 Tebetu / Tevet -952 AM; Astronomical Julian Day JD_X: 0.5; RD_Y: -347996; RD_X: -1721424; Sabbatical Cycle: 7; Metonic Cycle: 17;
952 Wednesday 03 January 4713 BCE (Julian); Wednesday 26 November 4713 BCE (Gregorian Astronomical); 22 Tebetu / Tevet -952 AM; Astronomical Julian Day JD_X: 1.5; RD_Y: -347995; RD_X: -1721423; Sabbatical Cycle: 7; Metonic Cycle: 17;
953 Seidelmann 1992: 55

XX. Indeed, Dershowitz and Reingold define the Epoch ($JD_{0.0}$) of the Julian day number system to be Noon (12:00) Monday 01 January 4713 BCE (Julian), i.e. ($RD_{-1721424.5}$) which is equivalent to ($RDY_{-347996.5}$)[954] [955]

XX. The system of Julian day [(JD) or (JD_X)] numbers is a continuous count of days elapsed since the beginning of the Julian period as defined by the 16th-century CE (Julian / Gregorian) chronologist J.J. Scaliger. Although Scaliger's original idea was to introduce a count of years, 19th-century CE (Gregorian) astronomers adapted this system to create a count of days.[956] Astronomers avoid the confusing situation of specifying moments in time by giving them in "Julian Days" of JD [or (JD_X)] (sometimes [called] "Julian Astronomical Days" or JAD).[957]

XX. The "Julian" period was published in 1583 CE (Gregorian) by Joseph Justus Scaliger, and was originally a counting of years in a cycle of 7980 years starting from 4713 BCE (Julian). The roots of the 7980-year cycle are much earlier than Scaliger, dating back to the 12th century CE (Julian). In the mid-19th century CE (Gregorian), Herschel adapted the system into a strict counting of days backward and forward from:[958]

($JD_{0.0}$) == ($RDX_{-1721424.5}$) == ($RDY_{-347996.5}$) == Noon (12:00) Monday 01 January 4713 BCE (Julian)

($JD_{0.0}$) == ($RDX_{-1721424.5}$) == ($RDY_{-347996.5}$) == Noon (12:00) Monday 24 November -4713 (Gregorian Astronomical)

($JD_{0.0}$) == ($RDX_{-1721424.5}$) == ($RDY_{-347996.5}$) == Noon (12:00) Monday 24 November 4714 BCE (Gregorian Calibrated)[959] [960]

XX. Julian day (JD) 0 [or (JD_0)] commenced at Greenwich Noon (12:00) Monday 01 January -4713 BCE (Julian),[961] Julian proleptic calendar. The Julian day [JD or JD_X] number (expressed as an integer) denotes the number of complete days elapsed (at Greenwich Noon (12:00) on the day designated) since

954 Monday 01 January 4713 BCE (Julian); Monday 24 November 4713 BCE (Gregorian Astronomical); 20 Tebetu / Tevet -952 AM; Astronomical Julian Day JD_X: -0.5; RD_Y: -347997; RD_X: -1721425; Sabbatical Cycle: 7; Metonic Cycle: 17;

955 Dershowitz and Reingold 2018: 17

956 Seidelmann 1992: 600

957 Dershowitz and Reingold 1997: 12; Dershowitz and Reingold 2008: 16

958 Dershowitz and Reingold 1997: 12; Dershowitz and Reingold 2008: 16; Dershowitz and Reingold 2018: 16-18

959 This date is an instance in which both the Gregorian Astronomical (in November) and the Julian date (in January) have their respective years with the same integer value, i.e. -4713, and therefore it is necessary to calibrate the Gregorian Calibrated year with the previous Julian year, i.e 4714 BCE.

960 cf. Footnote #9

961 On page 600, Seidelmann erroneously records a date of 01 January -4712 (evidently intending a mistaken Gregorian Astronomical Year as the correct Gregorian Astronomical Year for this date is -4713) even though he writes "Julian proleptic calendar" here as well as on page 55. On page 55, Seidelmann records the correct date of 01 January -4713 BCE (Julian).

the initial epoch [i.e. the start of the calendar count].[962]

XX. The Julian date (JD) [or (JD_X)] corresponding to any instant [of the day other than Noon (12:00)] is – by a simple extension of this concept – the Julian Day Number (JD) [or (JD_X)] followed by a fraction of the day elapsed since the preceding Noon (12:00).[963] The Julian date (JD) [or (JD_X)] specifies a particular instant of a day by ending the Julian day (JD) [or (JD_X)] number with a decimal fraction.[964]

XX. For example, the Julian day (JD) [or (JD_X)] number of Monday 25 June 1990 CE (Gregorian)[965] [starting at Noon (12:00)] is 2448068 [i.e. the whole number without a decimal fraction], whereas the Julian date (JD) [or (JD_X)] at [exactly] Noon (12:00) is 2448068.0 [i.e. with the exact decimal fraction].[966]

XX. [Twelve (12) hours later on Tuesday 26 June 1990 CE (Gregorian)[967] starting at Midnight (00:00) is (JD) [or (JD_X)] 2448068.5.] The Midnight (00:00) that begins the civil day [in the Julian or Gregorian calendars – i.e. not to be confused with "Julian day numbers" (JD) [or (JD_X)] that begins at Noon (12:00) –] is specified [i.e. calculated] by subtracting 0.5 from the Julian date (JD) [or (JD_X)] at Noon (12:00).[968]

XX. That is to say: we can define, equate, and solve for the unknown Julian Day (JD) number, (JD_Q), for Midnight (00:00) when given a known Julian Day (JD) number, (JD_X), of Noon (12:00):

Midnight (00:00) (JD_Q) that precedes a known (JD_X) day == ((JD_X) - 0.5)

OR

Midnight (00:00) (JD_Q) == Noon (12:00) ((JD_X) – 0.5)

Midnight (00:00) (JD_Q) == ((JD_1) – 0.5)

Midnight (00:00) (JD_Q) == ($JD_{0.5}$)

962 Seidelmann 1992: 55, 600
963 Seidelmann 1992: 55-56
964 Seidelmann 1992: 600
965 Monday 12 June 1990 CE (Julian); Monday 25 June 1990 CE (Gregorian); 02 Du'uzu / Tammuz 5750 AM; Astronomical Julian Day JD_X: 2448067.5; RD_Y: 2100071; RD_X: 726643; Sabbatical Cycle: 3; Metonic Cycle: 12;
966 Seidelmann 1992: 600
967 Tuesday 13 June 1990 CE (Julian); Tuesday 26 June 1990 CE (Gregorian); 03 Du'uzu / Tammuz 5750 AM; Astronomical Julian Day JD_X: 2448068.5; RD_Y: 2100072; RD_X: 726644; Sabbatical Cycle: 3; Metonic Cycle: 12;
968 Seidelmann 1992: 600

XX. Therefore:[969]

$(JD_{0.5})$ == Midnight (00:00) Tuesday 02 January 4713 BCE (Julian)

$(JD_{0.5})$ == Midnight (00:00) Tuesday 25 November -4713 (Gregorian Astronomical)

$(JD_{0.5})$ == Midnight (00:00) Tuesday 25 November 4714 BCE (Gregorian Calibrated)

$(JD_{0.5})$ == Midnight (00:00) Tuesday 21 Tebetu / Tevet (i.e. 10th month) -952 AM (Hebrew)

XX. Switching dates at Noon (12:00) makes sense for astronomers who work through the night. A fractional part of a Julian Date[/Day with (JD) number, (JD_X)] gives the fraction of a day beyond Noon (12:00). In this system, for example, Sunset (18:00+/-) on the first day [i.e. Monday in which commenced the beginning of the second day Tuesday] of the Hebrew calendar occurred at about $(JD_{347997.25})$ (local [Jerusalem] time) which is 1/4 of a [24-hour] day [i.e. six (6) hours] after Noon (12:00) [i.e. $(JD_{347997.0})$].[970] [971] [972]

XX. Therefore the following formula can be derived where "F" represents the decimal fraction of the day past Noon (00:00):[973] [974]

Fractional part of a Julian Date/Day, $((JD_Q) + 0.0) == ((JD_X) + 0.0 + 0.F)$

Fractional part of a Julian Date/Day, $(JD_Q) == ((JD_X) + 0.F)$[975]

XX. For example, to calculate the Midnight (00:00) – i.e. which is ½ a day past Noon (12:00) – that immediately follows (i.e. is past) Noon (12:00) of $((JD_X) + 0.0)$, and expressed as a (JD_X) number), this moment of the (JD_X) day past Noon (12:00) would be specified by the formula:

XX. Unknown Midnight (00:00), (JD_Q), of a known Julian Day (JD) number, (JD_X), that follows that known Noon (12:00) of day (JD_X) has the following formula:[976]

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969 Tuesday 02 January 4713 BCE (Julian); Tuesday 25 November 4713 BCE (Gregorian Astronomical); 21 Tebetu / Tevet -952 AM; Astronomical Julian Day JD_X: 0.5; RD_Y: -347996; RD_X: -1721424; Sabbatical Cycle: 7; Metonic Cycle: 17;

970 Monday 07 October 3761 BCE (Julian); Monday 07 September 3761 BCE (Gregorian Calibrated); 01 Tashritu / Tishri 1 AM; Astronomical Julian Day JD_X: 347997.5; RD_Y: 1; RD_X: -1373427; Sabbatical Cycle: 1; Metonic Cycle: 1;

971 Tuesday 08 October 3761 BCE (Julian); Tuesday 08 September 3761 BCE (Gregorian Calibrated); 02 Tashritu / Tishri 1 AM; Astronomical Julian Day JD_X: 347998.5; RD_Y: 2; RD_X: -1373426; Sabbatical Cycle: 1; Metonic Cycle: 1;

972 Dershowitz and Reingold 1997: 12; Dershowitz and Reingold 2008: 16

973 Dershowitz and Reingold 1997: 12; Dershowitz and Reingold 2008: 16

974 Dershowitz and Reingold 2008: 10

975 Dershowitz and Reingold 2008: 10

976 Dershowitz and Reingold 2008: 10

$((JD_Q) + 0.0) == ((JD_X) + 0.0 + 0.5)$

$(JD_Q) == ((JD_X) + 0.5)$

$(JD_Q) == ((JD_1) + 0.5)$

$(JD_Q) == (JD_{1.5})$

$(JD_{1.5})$ == Midnight (00:00) Wednesday 03 January 4713 BCE (Julian)[977]

$(JD_{1.5})$ == Midnight (00:00) Wednesday 26 November -4713 (Gregorian Astronomical)

$(JD_{1.5})$ == Midnight (00:00) Wednesday 26 November 4714 BCE (Gregorian Calibrated)

$(JD_{1.5})$ == Midnight (00:00) Wednesday 22 Tebetu / Tevet (i.e. 10th month) -952 AM (Hebrew)

XX. The epoch for the Julian period and its Julian Day (JD) number system is as follows:[978] [979]

$(JD_{0.0})$ == $(RDX_{-1721424.5})$ == $(RDY_{-347996.5})$ == Noon (12:00) Monday 01 January 4713 BCE (Julian)

$(JD_{0.0})$ == $(RDX_{-1721424.5})$ == $(RDY_{-347996.5})$ == Noon (12:00) Monday 24 November -4713 (Gregorian Astronomical)

$(JD_{0.0})$ == $(RDX_{-1721424.5})$ == $(RDY_{-347996.5})$ == Noon (12:00) Monday 24 November 4713 BCE (Gregorian Calibrated)

$(JD_{0.0})$ == $(RDX_{-1721424.5})$ == $(RDY_{-347996.5})$ == Noon (12:00) Monday 20 Tebetu / Tevet (i.e. 10th month) -952 AM (Hebrew), i.e. 953 Years before the Creation of Adam in year 1 AM (Hebrew)

XX. As will be explained below, the following two (2) tables summarize a selection of chronological equivalencies:

Table 0.2. A First Selection of Chronological Equivalencies

JD_X	RD_X	RD_Y	Time of	Date (Gregorian)	Date (Julian)

977 Wednesday 03 January 4713 BCE (Julian); Wednesday 26 November 4713 BCE (Gregorian Astronomical); 22 Tebetu / Tevet -952 AM; Astronomical Julian Day JD_X: 1.5; RD_Y: -347995; RD_X: -1721423; Sabbatical Cycle: 7; Metonic Cycle: 17;

978 Dershowitz and Reingold 2018: 17

979 Dershowitz and Reingold 1997: 12; Dershowitz and Reingold 2008: 16

			Day		
$JD_{-1.0}$	$RDX_{-1721425.5}$	$RDY_{-347997.5}$	Noon (12:00)	Sunday 23 November -4713 (Gregorian Astronomical)	Sunday 31 December -4714 BCE (Julian)
$JD_{-0.5}$	$RDX_{-1721425.0}$	$RDY_{-347997.0}$	Midnight (00:00)	Monday 24 November -4713 (Gregorian Astronomical)	Monday 01 January -4713 BCE (Julian)
$JD_{0.0}$	$RDX_{-1721424.5}$	$RDY_{-347996.5}$	Noon (12:00)	Monday 24 November -4713 (Gregorian Astronomical)	Monday 01 January -4713 BCE (Julian)
$JD_{0.5}$	$RDX_{-1721424.0}$	$RDY_{-347996.0}$	Midnight (00:00)	Tuesday 25 November -4713 (Gregorian Astronomical)	Tuesday 02 January -4713 BCE (Julian)
$JD_{1.0}$	$RDX_{-1721423.5}$	$RDY_{-347995.5}$	Noon (12:00)	Tuesday 25 November -4713 (Gregorian Astronomical)	Tuesday 02 January -4713 BCE (Julian)
$JD_{1.5}$	$RDX_{-1721423.0}$	$RDY_{-347995.0}$	Midnight (00:00)	Wednesday 26 November -4713 (Gregorian Astronomical)	Wednesday 03 January -4713 BCE (Julian)
$JD_{2.0}$	$RDX_{-1721422.5}$	$RDY_{-347994.5}$	Noon (12:00)	Wednesday 26 November -4713 (Gregorian Astronomical)	Wednesday 03 January -4713 BCE (Julian)
$JD_{2.5}$	$RDX_{-1721422.0}$	$RDY_{-347994.0}$	Midnight (00:00)	Thursday 27 November -4713 (Gregorian Astronomical)	Thursday 04 January -4713 BCE (Julian)
$JD_{3.0}$	$RDX_{-1721421.5}$	$RDY_{-347993.5}$	Noon (12:00)	Thursday 27 November -4713 (Gregorian Astronomical)	Thursday 04 January -4713 BCE (Julian)
$JD_{3.5}$	$RDX_{-1721421.0}$	$RDY_{-347993.0}$	Midnight (00:00)	Friday 28 November -4713 (Gregorian Astronomical)	Friday 05 January -4713 BCE (Julian)
$JD_{4.0}$	$RDX_{-1721420.5}$	$RDY_{-347992.5}$	Noon (12:00)	Friday 28 November -4713 (Gregorian Astronomical)	Friday 05 January -4713 BCE (Julian)
$JD_{4.5}$	$RDX_{-1721420.0}$	$RDY_{-347992.0}$	Midnight	Saturday 29 November -	Saturday 06 January -4713

			(00:00)	4713 (Gregorian Astronomical)	BCE (Julian)
$JD_{5.0}$	$RDX_{-1721419.5}$	$RDY_{-347991.5}$	Noon (12:00)	Saturday 29 November -4713 (Gregorian Astronomical)	Saturday 06 January -4713 BCE (Julian)
$JD_{5.5}$	$RDX_{-1721419.0}$	$RDY_{-347991.0}$	Midnight (00:00)	Sunday 30 November -4713 (Gregorian Astronomical)	Sunday 07 January -4713 BCE (Julian)
$JD_{6.0}$	$RDX_{-1721418.5}$	$RDY_{-347990.5}$	Noon (12:00)	Sunday 30 November -4713 (Gregorian Astronomical)	Sunday 07 January -4713 BCE (Julian)

Table 0.3. A Second Selection of Chronological Equivalencies

JD_X	**RD_X**	**RD_Y**	**Time of Day**	**Date (Gregorian)**	**Date (Julian)**
$JD_{1721423.5}$	$RDX_{-1.0}$	$RDY_{1373427.0}$	Midnight (00:00)	Saturday 30 December 0 (Gregorian Astronomical)	Saturday 01 January 1 CE (Julian)
$JD_{1721424.0}$	$RDX_{-0.5}$	$RDY_{1373427.5}$	Noon (12:00)	Saturday 30 December 0 (Gregorian Astronomical)	Saturday 01 January 1 CE (Julian)
$JD_{1721424.5}$	$RDX_{0.0}$	$RDY_{1373428.0}$	Midnight (00:00)	Sunday 31 December 0 (Gregorian Astronomical)	Sunday 02 January 1 CE (Julian)
$JD_{1721425.0}$	$RDX_{0.5}$	$RDY_{1373428.5}$	Noon (12:00)	Sunday 31 December 0 (Gregorian Astronomical)	Sunday 02 January 1 CE (Julian)
$JD_{1721425.5}$	$RDX_{1.0}$	$RDY_{1373429.0}$	Midnight (00:00)	Monday 01 January 1 CE (Gregorian)	Monday 03 January 1 CE (Julian)
$JD_{1721426.0}$	$RDX_{1.5}$	$RDY_{1373429.5}$	Noon (12:00)	Monday 01 January 1 CE (Gregorian)	Monday 03 January 1 CE (Julian)
$JD_{1721426.5}$	$RDX_{2.0}$	$RDY_{1373430.0}$	Midnight (00:00)	Tuesday 02 January 1 CE (Gregorian)	Tuesday 04 January 1 CE (Julian)
$JD_{1721427.0}$	$RDX_{2.5}$	$RDY_{1373430.5}$	Noon (12:00)	Tuesday 02 January 1 CE (Gregorian)	Tuesday 04 January 1 CE (Julian)
$JD_{1721427.5}$	$RDX_{3.0}$	$RDY_{1373431.0}$	Midnight (00:00)	Wednesday 03 January 0001 CE (Gregorian)	Wednesday 05 January 1 CE (Julian)

$JD_{1721428.0}$	$RDX_{3.5}$	$RDY_{1373431.5}$	Noon (12:00)	Wednesday 03 January 0001 CE (Gregorian)	Wednesday 05 January 1 CE (Julian)
$JD_{1721428.5}$	$RDX_{4.0}$	$RDY_{1373432.0}$	Midnight (00:00)	Thursday 04 January 0001 CE (Gregorian)	Thursday 06 January 1 CE (Julian)
$JD_{1721429.0}$	$RDX_{4.5}$	$RDY_{1373432.5}$	Noon (12:00)	Thursday 04 January 0001 CE (Gregorian)	Thursday 06 January 1 CE (Julian)
$JD_{1721429.5}$	$RDX_{5.0}$	$RDY_{1373433.0}$	Midnight (00:00)	Friday 05 January 0001 CE (Gregorian)	Friday 07 January 1 CE (Julian)
$JD_{1721430.0}$	$RDX_{5.5}$	$RDY_{1373433.5}$	Noon (12:00)	Friday 05 January 0001 CE (Gregorian)	Friday 07 January 1 CE (Julian)
$JD_{1721430.5}$	$RDX_{6.0}$	$RDY_{1373434.0}$	Midnight (00:00)	Saturday 06 January 0001 CE (Gregorian)	Saturday 08 January 1 CE (Julian)

XX. The meaning of "day" [depends upon which of the] various calendars [is being discussed].[980] The beginning and end points of a day differ depending upon which calendar we are using.

XX. Gregorian days are the same as Julian days and begin at Midnight (00:00) of fixed date ((RD_X) + 0.0), and continue until Midnight (00:00) of fixed date ((RD_X) + 1.0).[981]

XX. The Hebrew [or Islamic] date corresponding to fixed date ((RD_X) + 0.0) is the Hebrew [or Islamic] day that begins at Sunset (18:00+/-) of the evening of fixed date ((RD_X) – 1.0) and ends at Sunset (18:00+/-) of the evening of fixed date ((RD_X) + 0.0).[982]

XX. [Comparing (JD_X) to fixed date (RD_X)], the (JD_X) (corresponding to [– and included within –] fixed date ((RD_X) + 0.0)) begins at Noon (12:00) of fixed date ((RD_X) + 0.0) and ends at Noon (12:00) of fixed date ((RD_X) + 1.0).[983]

XX. The day-count [of either (JD_X) days or (RD_X) days] can be augmented by a fractional part to give a specific moment during the day. For example, Noon (12:00) on day ((RD_X) + 0.0) – a [whole number] integer – would be specified by [the following formula]:[984]

Noon (12:00) on day ((RD_X) + 0.0) == ((RD_X) + 0.0 + 0.5)

980 Dershowitz and Reingold 2008: 13
981 Dershowitz and Reingold 2008: 13
982 Dershowitz and Reingold 2008: 13
983 Dershowitz and Reingold 2008: 13
984 Dershowitz and Reingold 2008: 9-10

Noon (12:00) on day (RD_X) == ((RD_X) + 0.5)

XX. In other words: the Noon (12:00) of Julian Day (JD) day, ((JD_X) + 0.0), is contained within that fixed day (RD_X) (Gregorian/Julian) day that began at Midnight (00:00) twelve (12) hours earlier:

Midnight (00:00) on day (RD_X) == ((JD_X) + 0.0 – 0.5)

XX. [Just as can be done with the (JD_X) number] the (RD_X) day count can [also] be augmented by a fractional part to give a specific moment during the day.[985] Following Dershowitz and Reingold's convention, we call a [variable] (RD_X) [whole integer number] that has a fractional [decimal] part that gives the time of day: a "moment" [of the day past Midnight (00:00)].[986] [Thus Noon (12:00) on a known day ((RD_X) + 0.0) – i.e. a whole integer number – would be specified by the formula:]

Noon (12:00) of a known day ((RD_X) + 0.0) == ((RD_X) + 0.0 + 0.5)

Noon (12:00) of a known day (RD_X) == ((RD_X) + 0.5)

XX. For example, as will be discussed in a later section in more detail, according to the Babylonian record, the 1st capture of Jerusalem by Nebuchadnezzar fell on the 2nd day of the 12th Babylonian month (Addaru / Adar) in the 7th Regnal Year of Nebuchadnezzar as king of Babylon. This works out to be Saturday 16 March 597 BCE (Julian).[987] [988]

XX. However, in the proleptic Hebrew calendar, this works out to be Thursday 15 February 597 BCE (Julian).[989] [990] Because of the Babylonian intercalary second 6th month Ululu 2 / Elul 2 beginning 21 September 598 BCE (Julian) attested in Nebuchadnezzar's 7th Regnal Year,[991] the corresponding Babylonian months in Parker and Dubberstein's tables fall one (1) month later until the proleptic Hebrew calendar resynchronizes with the Babylonian calendar when it intercalates its 13th month Addaru 2 / Adar 2 of 3165 AM that began Wednesday 05 March 596 BCE (Julian). The implications of the differences of this date of the 2nd day of 12th month (Addaru / Adar) 597 BCE (Julian) will be discussed in more detail in Section 14.

XX. As will be explained in detail later, in the proleptic Hebrew calendar the 2nd day of the 12th month (Addaru / Adar) works out to be the following times and dates for (JD_X) number ($JD_{1503413.5}$) – i.e. (JD_X)

985 Dershowitz and Reingold 2008: 9-10
986 Dershowitz and Reingold 2008: 10
987 Saturday 16 March 597 BCE (Julian); Saturday 10 March 597 BCE (Gregorian); 03 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503443.5; RD_Y: 1155447; RD_X: -217981; Sabbatical Cycle: 7; Metonic Cycle: 10;
988 cf. Appendix Table X.13, X.32, X.33, X.34
989 Thursday 15 February 597 BCE (Julian); Thursday 09 February 597 BCE (Gregorian); 02 Addaru / Adar 3164 AM; Astronomical Julian Day JD_X: 1503413.5; RD_Y: 1155417; RD_X: -218011; Sabbatical Cycle: 7; Metonic Cycle: 10;
990 cf. Appendix Table X.13, X.32
991 Parker and Dubberstein 1971: 5, 27

number at Midnight (00:00) which began twelve (12) hours after the previous Noon (12:00) at ($JD_{1503413.0}$) – and for (RD_X) number ($RD_{-218011.0}$) which begins at Midnight (00:00), ((RD_X) + 0.0), and marks the beginning of the Julian / Gregorian civil 24-hour day:[992] [993]

($JD_{1503413.5}$) == ($RDX_{-218011.0}$) == ($RDY_{1155417.0}$) == Midnight (00:00) Thursday 15 February 597 BCE (Julian)

($JD_{1503413.5}$) == ($RDX_{-218011.0}$) == ($RDY_{1155417.0}$) == Midnight (00:00) Thursday 09 February -596 (Gregorian Astronomical)

($JD_{1503413.5}$) == ($RDX_{-218011.0}$) == ($RDY_{1155417.0}$) == Midnight (00:00) Thursday 09 February 597 BCE (Gregorian Calibrated)

($JD_{1503413.5}$) == ($RDX_{-218011.0}$) == ($RDY_{1155417.0}$) == Midnight (00:00) Thursday 02 Addaru / Adar (12th month) 3164 AM (Babylonian / Hebrew)

XX. In order to find the exact fractional / decimal (RD_X) number for Noon (12:00) for later that (RD_X) day – i.e. that commences twelve (12) hours later after Midnight (00:00) of ((RD_X) + 0.0) day ($RDX_{-218011.0}$) – we use the formula:

Noon (12:00) for ((RD_X) + 0.0) == ((RD_X) + 0.0 + 0.5)

Noon (12:00) for (RD_X) == ((RD_X) + 0.5)

Noon (12:00) for ($RDX_{-218011.0}$) == (($RDX_{-218011.0}$) + 0.5)

Noon (12:00) for ($RDX_{-218011.0}$) == ($RDX_{-218010.5}$)

XX. Therefore (JD) day with (JD_X) number ($JD_{1503414.0}$) that begins at Noon (12:00)) and fixed day (RDX) day with (RD_X) number ($RDX_{-218010.5}$) – which is exactly twelve (12) hours after Midnight (00:00) of (RD_X) day ($RDX_{-218011.0}$) – provide the following times and dates:[994] [995]

($JD_{1503414.0}$) == ($RDX_{-218010.5}$) == ($RDY_{1155417.5}$) == Noon (12:00) Thursday 15 February 597 BCE (Julian)

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992 Thursday 15 February 597 BCE (Julian); Thursday 09 February 597 BCE (Gregorian); 02 Addaru / Adar 3164 AM; Astronomical Julian Day JD_X: 1503413.5; RD_Y: 1155417; RD_X: -218011; Sabbatical Cycle: 7; Metonic Cycle: 10;

993 cf. Appendix Table X.13, X.32

994 Thursday 15 February 597 BCE (Julian); Thursday 09 February 597 BCE (Gregorian); 02 Addaru / Adar 3164 AM; Astronomical Julian Day JD_X: 1503413.5; RD_Y: 1155417; RD_X: -218011; Sabbatical Cycle: 7; Metonic Cycle: 10;

995 cf. Appendix Table X.13, X.32

($JD_{1503414.0}$) == ($RDX_{-218010.5}$) == ($RDY_{1155417.5}$) == Noon (12:00) Thursday 09 February -596 (Gregorian Astronomical)

($JD_{1503414.0}$) == ($RDX_{-218010.5}$) == ($RDY_{1155417.5}$) == Noon (12:00) Thursday 09 February 597 BCE (Gregorian Calibrated)

($JD_{1503414.0}$) == ($RDX_{-218010.5}$) == ($RDY_{1155417.5}$) == Noon (12:00) Thursday 02 Addaru / Adar (12th month) 3164 AM (Babylonian / Hebrew)

XX. Although it is technically correct – according to Dershowitz and Reingold's convention – to use the whole integer (RD_X) numbers that begin at Midnight (00:00) to specify the date that includes also Noon (12:00) later that day, if we wanted to be more precise about the moment of the day past Midnight (00:00) to specify exactly Noon (12:00) vs. Midnight (00:00), we would specify as follows:[996] [997]

($JD_{1503413.5}$) == ($RDX_{-218011.0}$) == ($RDY_{1155417.0}$) == Midnight (00:00) Thursday 15 February 597 BCE (Julian)

($JD_{1503413.5}$) == ($RDX_{-218011.0}$) == ($RDY_{1155417.0}$) == Midnight (00:00) Thursday 09 February -596 (Gregorian Astronomical)

($JD_{1503413.5}$) == ($RDX_{-218011.0}$) == ($RDY_{1155417.0}$) == Midnight (00:00) Thursday 09 February 597 BCE (Gregorian Calibrated)

($JD_{1503413.5}$) == ($RDX_{-218011.0}$) == ($RDY_{1155417.0}$) == Midnight (00:00) Thursday 02 Addaru / Adar 3164 AM (Hebrew)

($JD_{1503414.0}$) == ($RDX_{-218010.5}$) == ($RDY_{1155417.5}$) == Noon (12:00) Thursday 15 February 597 BCE (Julian)

($JD_{1503414.0}$) == ($RDX_{-218010.5}$) == ($RDY_{1155417.5}$) == Noon (12:00) Thursday 09 February -596 (Gregorian Astronomical)

($JD_{1503414.0}$) == ($RDX_{-218010.5}$) == ($RDY_{1155417.5}$) == Noon (12:00) Thursday 09 February 597 BCE (Gregorian Calibrated)

($JD_{1503414.0}$) == ($RDX_{-218010.5}$) == ($RDY_{1155417.5}$) == Noon (12:00) Thursday 02 Addaru / Adar 3164

996 Thursday 15 February 597 BCE (Julian); Thursday 09 February 597 BCE (Gregorian); 02 Addaru / Adar 3164 AM; Astronomical Julian Day JD_X: 1503413.5; RD_Y: 1155417; RD_X: -218011; Sabbatical Cycle: 7; Metonic Cycle: 10;

997 cf. Appendix Table X.13, X.32

AM (Babylonian / Hebrew)

XX. Julian dates [i.e. (JD) dates in the Julian calendar] can be expressed in Universal Time (UT) or dynamical time (TDT), though timescale should be specified if precision is of concern. A Julian date expressed in UT will differ by ΔT (i.e. Delta T) from the same date expressed in TDT. According to Seidelmann, it is not necessary to know the exact difference between dynamical (TDT) and Universal Time (UT) when specifying the Julian day (JD) [or (JDX)] number in dynamical time (TDT).[998]

XX. The [Julian Day (JD)] day number is extended by the addition of the time of day, expressed as a decimal fraction of day to give the Julian date (JD) [number]. It is important to note, however, that the integral values of the Julian date (JD) [number] refer to the instants of Greenwich mean Noon (12:00) (since the system was introduced when the astronomical day began at Noon (12:00) rather than at Midnight (00:00)); correspondingly, the Julian date (JD) [number] for 0h [Zero Hour, i.e. Midnight (00:00)] UT always ends in [the fractional decimal] ".5".[999]

XX. This – and the fact that current Julian dates require seven (7) digits [not including the ".5" for Midnight (00:00) or ".0" for Noon (Noon)] to express the integral parts – make the Julian (JD) [number] date system inconvenient for some purposes. The value of (JD) [or (JD_X)] minus 2400000.5 is sometimes used for current dates, and it is known as the Modified Julian Date (MJD). The Julian (JD) [number] date system may be used with TAI, TDT, or UTC, so when the difference is significant, the particular timescale should be indicated.[1000]

0.43. Noon (12:00) in Julian Day Numbers (JD_X) and the RD_X and RD_Y Days that Contain them

XX. Since Noon (12:00) [for the fixed day] of (RD_0) is [equivalent with] ($JD_{1721425}$), it follows that:[1001]

XX. Given a known (JD_X) number, in order to find an unknown (RD_X) number, (RD_Q), for Midnight (00:00) of the day that includes this given known (JD_X) number (i.e. that is later that same day at Noon (12:00) of earlier (RD_Q)), we will calculate as follows:

XX. An unknown fixed day (RD_X), (RD_Q), that begins Midnight (00:00) and includes a known (JD_X) can be expressed by the following formula:[1002]

998 Seidelmann 1992: 56
999 Seidelmann 1992: 8
1000 Seidelmann 1992: 8
1001 Dershowitz and Reingold 1997: 12; Dershowitz and Reingold 2008: 16
1002 Dershowitz and Reingold 1997: 12; Dershowitz and Reingold 2008: 16; Dershowitz and Reingold 2018: 18

Unknown (RD_Q) == ((JD_X) – 1721425)[1003]

XX. For example, as will be discussed in a later section in more detail, Nebuzaradan's arrival to Jerusalem after which he burned the Jewish temple and smashed the wall of Jerusalem fell on the 7th day of the 5th proleptic Hebrew month (Abu / Av) of the 19th Regnal Year of Nebuchadnezzar as king of Babylon.[1004] Jewish tradition marks the date of the burning of the temple to be the 9th day of the 5th month, i.e. the 9th of Abu / Av.

XX. The 7th day of the 5th month Abu / Av noted in the Book of 2 Kings works out to be 15 July 586 BCE (Julian) [1005] [1006] [1007] with (RD_X) number ($RDX_{-213843.0}$) which begins at Midnight (00:00), and is exactly equivalent to (JD_X) number ($JD_{1507581.5}$) also at Midnight (00:00).

XX. This day ($RDX_{-213843.0}$) contains and includes the later Noon (12:00) (JD_X) number ($JD_{1507582.0}$) which itself (i.e. this (JD_X) number) is exactly equivalent with (RD_X) number ($RDX_{-213842.5}$):[1008] [1009] [1010]

($JD_{1507581.5}$) == ($RDX_{-213843.0}$) == ($RDY_{1159585.0}$) == Midnight (00:00) Sunday 15 July 586 BCE (Julian)

($JD_{1507581.5}$) == ($RDX_{-213843.0}$) == ($RDY_{1159585.0}$) == Midnight (00:00) Sunday 09 July -585 (Gregorian Astronomical)

($JD_{1507581.5}$) == ($RDX_{-213843.0}$) == ($RDY_{1159585.0}$) == Midnight (00:00) Sunday 09 July 586 BCE (Gregorian Calibrated)

($JD_{1507581.5}$) == ($RDX_{-213843.0}$) == ($RDY_{1159585.0}$) == Midnight (00:00) Sunday 07 Abu / Av 3175 AM (Babylonian / Hebrew)

($JD_{1507582.0}$) == ($RDX_{-213842.5}$) == ($RDY_{1159585.5}$) == Noon (12:00) Sunday 15 July 586 BCE (Julian)

($JD_{1507582.0}$) == ($RDX_{-213842.5}$) == ($RDY_{1159585.5}$) == Noon (12:00) Sunday 09 July -585 (Gregorian Astronomical)

1003 Dershowitz and Reingold solve for an unknown Noon (12:00) on RD_Q from a given JD_X because Noon (12:00) is included in the RD_X day that begins at Midnight (00:00).
1004 cf. 2 Kings 25:8-11
1005 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;
1006 cf. Appendix Table X.59
1007 cf. 2 Kings 25:8-11
1008 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;
1009 cf. Appendix Table X.59
1010 cf. 2 Kings 25:8-11

$(JD_{1507582.0})$ == $(RDX_{-213842.5})$ == $(RDY_{1159585.5})$ == Noon (12:00) Sunday 09 July 586 BCE (Gregorian Calibrated)

$(JD_{1507582.0})$ == $(RDX_{-213842.5})$ == $(RDY_{1159585.5})$ == Noon (12:00) Sunday 07 Abu / Av 3175 AM (Babylonian / Hebrew)

XX. Given this known (JD_X) number $(JD_{1507582.0})$ for Noon (12:00), to find the unknown fixed day (RD_Q) at Midnight (00:00) that contains (i.e. includes) this given known (JD_X) number Noon (12:00) – i.e. exactly 12 hours later than the unknown (RD_Q) – we use the formula:[1011] [1012] [1013]

$(RD_Q) == ((JD_X) - 1721425.0)$

$(RD_Q) == ((JD_{1507582.0}) - 1721425.0)$

$(RD_Q) == (RDX_{-213843.0})$

$(RD_Q) == (RDX_{-213843.0}) == (RDY_{1159585.0})$ == Midnight (00:00) Sunday 15 July 586 BCE (Julian)

$(RD_Q) == (RDX_{-213843.0}) == (RDY_{1159585.0})$ == Midnight (00:00) Sunday 09 July -585 (Gregorian Astronomical)

$(RD_Q) == (RDX_{-213843.0}) == (RDY_{1159585.0})$ == Midnight (00:00) Sunday 09 July 586 BCE (Gregorian Calibrated)

$(RD_Q) == (RDX_{-213843.0}) == (RDY_{1159585.0})$ == Midnight (00:00) Sunday 07 Abu / Av 3175 AM (Babylonian / Hebrew)

XX. This (RD_X) date $(RDX_{-213843.0})$ contains the later $(JD_{1507582.0})$ Noon (12:00) Sunday 15 July 586 BCE (Julian). This Noon (12:00) can also be expressed according to its precise "moment" at Noon (12:00) as an (RD_X) number $(RDX_{-213842.5})$ contained within the fixed day $(RDX_{-213843.0})$ that began twelve (12) hours earlier at precisely $(RDX_{-213843.0})$ Midnight (00:00) Sunday 15 July 586 BCE (Julian).[1014] [1015] [1016]

1011 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;

1012 cf. Appendix Table X.59

1013 cf. 2 Kings 25:8-11

1014 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;

1015 cf. Appendix Table X.59

1016 cf. 2 Kings 25:8-11

0.44. Midnight (00:00) in Julian Day Numbers (JD_X) and their RD_X and RD_Y Equivalents

XX. Because we want [i.e. need] our days to begin at civil Midnight (00:00), we do not use Julian Days (JD) directly as suggested [by O.L. Harvey in 1983 CE (Gregorian)]. We begin each day at Midnight (00:00), and use fractional [(JD_X) and/or (RD_X)] days when we need to calculate with time.[1017]

XX. [Given a known fixed day (RD_X) which begins at] Midnight (00:00), the equivalent of Midnight (00:00) (RD_X) expressed as (JD_X) [i.e. an unknown (JD_X), i.e. (JD_Q)] can be calculated as follows:[1018] [1019] [1020] [1021]

$$(JD_Q) == ((RD_X) + 1721424.5)$$

$$(JD_Q) == ((RDX_{-213843.0}) + 1721424.5)$$

$$(JD_Q) == (1507581.5)$$

$(JD_Q) == (JD_{1507581.5}) == (RDX_{-213843.0}) == (RDY_{1159585.0})$ == Midnight (00:00) Sunday 15 July -586 BCE (Julian).

$(JD_Q) == (JD_{1507581.5}) == (RDX_{-213843.0}) == (RDY_{1159585.0})$ == Midnight (00:00) Sunday 09 July -585 (Gregorian Astronomical)

$(JD_Q) == (JD_{1507581.5}) == (RDX_{-213843.0}) == (RDY_{1159585.0})$ == Midnight (00:00) Sunday 09 July 586 BCE (Gregorian Calibrated)

$(JD_Q) == (JD_{1507581.5}) == (RDX_{-213843.0}) == (RDY_{1159585.0})$ == Midnight (00:00) Sunday 07 Abu / Av 3175 AM (Babylonian / Hebrew)

0.45. Converting from Julian Day Numbers (JD_X) to their RD_X and RD_Y Equivalents

XX. It is possible to solve for an unknown (RD_X) or (RD_Y) number by using a permutation of the same basic algebraic formula above:

XX. Given a known (JD_X) number (any time of day), its equivalent expressed as an (RD_X) number (i.e.

1017 Dershowitz and Reingold 2008: 17
1018 Dershowitz and Reingold 1997: 12; Dershowitz and Reingold 2008: 17
1019 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;
1020 cf. Appendix Table X.59
1021 cf. 2 Kings 25:8-11

an unknown (RD_X) number, (RDX_Q)) or as an (RD_Y) number (i.e. an unknown (RD_Y) number, (RDY_Q)) can be calculated as follows:[1022] [1023] [1024]

XX. Solving for (RDX):

(RDX) == ((JDX) – 1721424.5)

(RDX_Q) == ((JD_X) – 1721424.5)

(RDX_Q) == (($JD_{1507581.5}$) – 1721424.5)

(RDX_Q) == (-213843.0)

(RDX_Q) == ($RDX_{-213843.0}$) == ($JD_{1507581.5}$) == Midnight (00:00) Sunday 15 July 586 BCE (Julian)

(RDX_Q) == ($RDX_{-213843.0}$) == ($JD_{1507581.5}$) == Midnight (00:00) Sunday 09 July -585 (Gregorian Astronomical)

(RDX_Q) == ($RDX_{-213843.0}$) == ($JD_{1507581.5}$) == Midnight (00:00) Sunday 09 July 586 BCE (Gregorian Calibrated)

(RDX_Q) == ($RDX_{-213843.0}$) == ($JD_{1507581.5}$) == Midnight (00:00) Sunday 07 Abu / Av 3175 AM (Babylonian / Hebrew)

XX. Another example solving for (RDX):[1025] [1026] [1027]

(RDX) == ((JDX) – 1721424.5)

(RDX_Q) == ((JD_X) – 1721424.5)

(RDX_Q) == (($JD_{1507582.0}$) – 1721424.5)

(RDX_Q) == (-213842.5)

1022 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;
1023 cf. Appendix Table X.59
1024 cf. 2 Kings 25:8-11
1025 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;
1026 cf. Appendix Table X.59
1027 cf. 2 Kings 25:8-11

$(RDX_Q) == (RDX_{-213842.5}) == (JD_{1507582.0})$ == Noon (12:00) Sunday 15 July 586 BCE (Julian)

$(RDX_Q) == (RDX_{-213842.5}) == (JD_{1507582.0})$ == Noon (12:00) Sunday 09 July -585 (Gregorian Astronomical)

$(RDX_Q) == (RDX_{-213842.5}) == (JD_{1507582.0})$ == Noon (12:00) Sunday 09 July 586 BCE (Gregorian Calibrated)

$(RDX_Q) == (RDX_{-213842.5}) == (JD_{1507582.0})$ == Noon (12:00) Sunday 07 Abu / Av 3175 AM (Babylonian / Hebrew)

XX Solving for (RDY):[1028] [1029] [1030]

(RDY) == ((JDX) – 347996.5)

$(RDY_Q) == ((JD_X) – 347996.5)$

$(RDY_Q) == ((JD_{1507581.5}) – 347996.5)$

$(RDY_Q) == (1159585.0)$

$(RDY_Q) == (RDY_{1159585.0}) == (JD_{1507581.5})$ == Midnight (00:00) Sunday 15 July 586 BCE (Julian)

$(RDY_Q) == (RDY_{1159585.0}) == (JD_{1507581.5})$ == Midnight (00:00) Sunday 09 July -585 (Gregorian Astronomical)

$(RDY_Q) == (RDY_{1159585.0}) == (JD_{1507581.5})$ == Midnight (00:00) Sunday 09 July 586 BCE (Gregorian Calibrated)

$(RDY_Q) == (RDY_{1159585.0}) == (JD_{1507581.5})$ == Midnight (00:00) Sunday 07 Abu / Av 3175 AM (Babylonian / Hebrew)

XX. Another example solving for (RDY):[1031] [1032] [1033]

1028 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;

1029 cf. Appendix Table X.59

1030 cf. 2 Kings 25:8-11

1031 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;

1032 cf. Appendix Table X.59

1033 cf. 2 Kings 25:8-11

(RDY) == ((JDX) – 347996.5)

$(RDY_Q) == ((JD_X) – 347996.5)$

$(RDY_Q) == ((JD_{1507582.0}) – 347996.5)$

$(RDY_Q) == (1159585.5)$

$(RDY_Q) == (RDY_{1159585.5}) == (JD_{1507582.0})$ == Noon (12:00) Sunday 15 July 586 BCE (Julian)

$(RDY_Q) == (RDY_{1159585.5}) == (JD_{1507582.0})$ == Noon (12:00) Sunday 09 July -585 (Gregorian Astronomical)

$(RDY_Q) == (RDY_{1159585.5}) == (JD_{1507582.0})$ == Noon (12:00) Sunday 09 July 586 BCE (Gregorian Calibrated)

$(RDY_Q) == (RDY_{1159585.5}) == (JD_{1507582.0})$ == Noon (12:00) Sunday 07 Abu / Av 3175 AM (Babylonian / Hebrew)

XX. Therefore the following times are expressed in equivalent (JDX), (RDX), and (RDY) numbers:[1034] [1035] [1036]

$(JD_{1507581.5}) == (RDX_{-213843.0}) == (RDY_{1159585.0})$ == Midnight (00:00) Sunday 15 July 586 BCE (Julian)

$(JD_{1507581.5}) == (RDX_{-213843.0}) == (RDY_{1159585.0})$ == Midnight (00:00) Sunday 09 July -585 (Gregorian Astronomical)

$(JD_{1507581.5}) == (RDX_{-213843.0}) == (RDY_{1159585.0})$ == Midnight (00:00) Sunday 09 July 586 BCE (Gregorian Calibrated)

$(JD_{1507581.5}) == (RDX_{-213843.0}) == (RDY_{1159585.0})$ == Midnight (00:00) Sunday 07 Abu / Av 3175 AM (Babylonian / Hebrew)

$(JD_{1507582.0}) == (RDX_{-213842.5}) == (RDY_{1159585.5})$ == Noon (12:00) Sunday 15 July 586 BCE (Julian)

1034 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;

1035 cf. Appendix Table X.58

1036 cf. 2 Kings 25:8-11

($JD_{1507582.0}$) == ($RDX_{-213842.5}$) == ($RDY_{1159585.5}$) == Noon (12:00) Sunday 09 July -585 (Gregorian Astronomical)

($JD_{1507582.0}$) == ($RDX_{-213842.5}$) == ($RDY_{1159585.5}$) == Noon (12:00) Sunday 09 July 586 BCE (Gregorian Calibrated)

($JD_{1507582.0}$) == ($RDX_{-213842.5}$) == ($RDY_{1159585.5}$) == Noon (12:00) Sunday 07 Abu / Av 3175 AM (Babylonian / Hebrew)

0.46. Converting to and from RD_X and RD_Y Numbers

XX. Similarly, it is possible to derive the formulas to convert to and from (RD_X) and (RD_Y) Numbers:

XX. Solving for (RDX):[1037] [1038] [1039]

(RDX) == ((RDY) – 1373428)

(RDX_Q) == (($RDY_{1159585.0}$) – 1373428)

(RDX_Q) == (-213843.0)

(RDX_Q) == ($RDX_{-213843.0}$)

(RDX_Q) == ($RDX_{-213843.0}$) == ($RDY_{1159585.0}$) == ($JD_{1507581.5}$) == Midnight (00:00) Sunday 15 July 586 BCE (Julian)

(RDX_Q) == ($RDX_{-213843.0}$) == ($RDY_{1159585.0}$) == ($JD_{1507581.5}$) == Midnight (00:00) Sunday 09 July -585 (Gregorian Astronomical)

(RDX_Q) == ($RDX_{-213843.0}$) == ($RDY_{1159585.0}$) == ($JD_{1507581.5}$) == Midnight (00:00) Sunday 09 July 586 BCE (Gregorian Calibrated)

(RDX_Q) == ($RDX_{-213843.0}$) == ($RDY_{1159585.0}$) == ($JD_{1507581.5}$) == Midnight (00:00) Sunday 07 Abu / Av 3175 AM (Babylonian / Hebrew)

1037 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;
1038 cf. Appendix Table X.59
1039 cf. 2 Kings 25:8-11

XX. Solving for (RDY):[1040] [1041] [1042]

(RDY) == ((RDX) + 1373428)

(RDY_Q) == (($RDX_{-213842.5}$) + 1373428)

(RDY_Q) == (1159585.5)

(RDY_Q) == ($RDY_{1159585.5}$)

(RDY_Q) == ($RDX_{-213842.5}$) == ($RDY_{1159585.5}$) == ($JD_{1507582.0}$) == Noon (12:00) Sunday 15 July 586 BCE (Julian)

(RDY_Q) == ($RDX_{-213842.5}$) == ($RDY_{1159585.5}$) == ($JD_{1507582.0}$) == Noon (12:00) Sunday 09 July -585 (Gregorian Astronomical)

(RDY_Q) == ($RDX_{-213842.5}$) == ($RDY_{1159585.5}$) == ($JD_{1507582.0}$) == Noon (12:00) Sunday 09 July 586 BCE (Gregorian Calibrated)

(RDY_Q) == ($RDX_{-213842.5}$) == ($RDY_{1159585.5}$) == ($JD_{1507582.0}$) == Noon (12:00) Sunday 07 Abu / Av 3175 AM (Babylonian / Hebrew)

0.47. The First Day of the Hebrew Calendar as JD_X, RD_X, and RD_Y Numbers

XX. As noted above, using this system [of Julian Day (JD_X) Numbers] Sunset (18:00+/-) on the first [official] day of the Hebrew calendar [i.e. its epoch] begins [officially] at about [(JD_X) number] ($JD_{347997.25}$) (local [Jerusalem] time), which is ¼ of a [24-hour] day [i.e. 6 hours] after Noon (12:00) ($JD_{347997.0}$).[1043] [1044] [1045]

XX. This would make Sunset (18:00+/-) at ($JD_{347997.25}$) exactly equivalent to Sunset (18:00+/-) at fixed day ($RDX_{-1373427.25}$) as well as exactly equivalent to Sunset (18:00+/-) at fixed day ($RDY_{0.75}$) since both fixed day ($RDX_{-1373427.0}$) and fixed day ($RDY_{1.0}$) begin at Midnight (00:00).

1040 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;

1041 cf. Appendix Table X.58

1042 cf. 2 Kings 25:8-11

1043 Sunday 06 October 3761 BCE (Julian); Sunday 06 September 3761 BCE (Gregorian Calibrated); 29 Ululu / Elul 0 AM; Astronomical Julian Day JD_X: 347996.5; RD_Y: 0; RD_X: -1373428; Sabbatical Cycle: 7; Metonic Cycle: 19;

1044 Monday 07 October 3761 BCE (Julian); Monday 07 September 3761 BCE (Gregorian Calibrated); 01 Tashritu / Tishri 1 AM; Astronomical Julian Day JD_X: 347997.5; RD_Y: 1; RD_X: -1373427; Sabbatical Cycle: 1; Metonic Cycle: 1;

1045 Dershowitz and Reingold 1997: 12; Dershowitz and Reingold 2008: 16

XX. That is to say (in other words) in mathematical notation the following are equivalent:[1046] [1047] [1048]

$(JD_{347997.25})$ == $(RDX_{-1373427.25})$ == $(RDY_{0.75})$ == Sunset (18:00+/-) Sunday 06 October 3761 BCE (Julian)

$(JD_{347997.25})$ == $(RDX_{-1373427.25})$ == $(RDY_{0.75})$ == Sunset (18:00+/-) Sunday 06 September -3760 (Gregorian Astronomical)

$(JD_{347997.25})$ == $(RDX_{-1373427.25})$ == $(RDY_{0.75})$ == Sunset (18:00+/-) Sunday 06 September 3761 BCE (Gregorian Calibrated)

$(JD_{347997.25})$ == $(RDX_{-1373427.25})$ == $(RDY_{0.75})$ == Sunset (18:00+/-) Sunday 01 Tashritu / Tishri 1 AM (Babylonian / Hebrew)

XX. The 1st day of the epoch of Hebrew calendar [technically exactly] begins at fixed day $(RDX_{-1373427.25})$ Sunset (18:00+/-) Sunday 06 October 3761 BCE (Julian) [for Julian / Gregorian civil day that began eighteen (18) hours earlier at fixed Julian / Gregorian civil day $(RDX_{-1373428.0})$ Midnight (00:00) Sunday 06 October 3761 BCE (Julian)].[1049] [1050] [1051]

0.48. The First Day of the Gregorian Calendar as JD_X, RD_X, and RD_Y Numbers

XX. The first day of the Gregorian Calendar can be expressed as the following (JD_X), (RD_X), and (RD_Y) number and date equivalencies:[1052]

$(JD_{1721425.5})$ == $(RDX_{1.0})$ == $(RDY_{1373429.0})$ == Midnight (00:00) Monday 03 January 1 CE (Julian)

$(JD_{1721425.5})$ == $(RDX_{1.0})$ == $(RDY_{1373429.0})$ == Midnight (00:00) Monday 01 January 1 CE (Gregorian)

1046 Sunday 06 October 3761 BCE (Julian); Sunday 06 September 3761 BCE (Gregorian Calibrated); 29 Ululu / Elul 0 AM; Astronomical Julian Day JD_X: 347996.5; RD_Y: 0; RD_X: -1373428; Sabbatical Cycle: 7; Metonic Cycle: 19;

1047 Monday 07 October 3761 BCE (Julian); Monday 07 September 3761 BCE (Gregorian Calibrated); 01 Tashritu / Tishri 1 AM; Astronomical Julian Day JD_X: 347997.5; RD_Y: 1; RD_X: -1373427; Sabbatical Cycle: 1; Metonic Cycle: 1;

1048 The Babylonian / Hebrew day begins at Sunset.

1049 Sunday 06 October 3761 BCE (Julian); Sunday 06 September 3761 BCE (Gregorian Calibrated); 29 Ululu / Elul 0 AM; Astronomical Julian Day JD_X: 347996.5; RD_Y: 0; RD_X: -1373428; Sabbatical Cycle: 7; Metonic Cycle: 19;

1050 Monday 07 October 3761 BCE (Julian); Monday 07 September 3761 BCE (Gregorian Calibrated); 01 Tashritu / Tishri 1 AM; Astronomical Julian Day JD_X: 347997.5; RD_Y: 1; RD_X: -1373427; Sabbatical Cycle: 1; Metonic Cycle: 1;

1051 Dershowitz and Reingold 1997: 11; Dershowitz and Reingold 2008: 12, 14

1052 Monday 03 January 1 CE (Julian); Monday 01 January 1 CE (Gregorian); 18 Tebetu / Tevet 3761 AM; Astronomical Julian Day JD_X: 1721425.5; RD_Y: 1373429; RD_X: 1; Sabbatical Cycle: 2; Metonic Cycle: 18;

($JD_{1721425.5}$) == ($RDX_{1.0}$) == ($RDY_{1373429.0}$) == Midnight (00:00) Monday 18 Tebetu / Tevet (10th month) 3761 AM (Babylonian / Hebrew)

XX. The date ($RD_{1.0}$) Monday 01 January 1 CE (Gregorian) – though arbitrarily chosen as starting point, has a desirable characteristic: it is early enough that almost all [modern] dates of interest are represented by positive integers of moderate size.[1053]

XX. All that is required for calendrical conversion is to be able to convert each calendar to and from this fixed [Gregorian] calendar. Because some calendars begin their day at Midnight (00:00) and others at Sunrise (06:00+/-) or Sunset (18:00+/-), Dershowitz and Reingold fix the time of day at which [calendrical] conversions are performed [to be at] Noon (12:00) [i.e. ((RD_X) + 0.5) of fixed day ((RD_X) + 0.0) which begins at Midnight (00:00) of ((RD_X) + 0.0)].[1054]

XX. Because days begin at different hours on different calendars, we follow Dershowitz and Reingold's convention that a calendar's epoch is the onset of the (civil) day [i.e. Midnight (00:00) starting at ((RD_X) + 0.0)] containing the 1st Noon (12:00) [i.e. ((RD_X) + 0.5) later that same civil day of fixed day ((RD_X) + 0.0)].[1055]

XX. For example, we take Midnight (00:00) at the onset of ($RDX_{-1373427.0}$) Monday 07 October 3761 (Julian)[1056] [1057] as the epoch of the Hebrew calendar [because it contains Noon (12:00) ($RDX_{-1373426.5}$) later that same fixed day that began at ($RDX_{-1373427.0}$), although the 1st Hebrew day [technically exactly] began at Sunset (18:00+/-) the previous evening [during Julian / Gregorian day] ($RDX_{-1373428.0}$) Sunday 06 October 3761 BCE (Julian) at precisely the "moment-in-time" of Sunset (18:00+/-) ($JD_{347997.25}$) == ($RDX_{-1373427.25}$) == ($RDY_{0.75}$)][1058]

0.49. Parker and Dubberstein's Calculated Dates and Their Admitted Uncertainties

XX. In Parker and Dubberstein's tables, none of the equivalent Julian dates on which each Babylonian month began were observed ancient dates, but rather were calculated by the New Moon tables of Karl Schoch (1928) within *The Venus Tablets of Ammizaduga* by S. Langdon and J.K. Fotheringham. As all dates are calculated, their accuracy depends upon the accuracy of Schoch's tables.[1059]

1053 Dershowitz and Reingold 1997: 11; Dershowitz and Reingold 2008: 10
1054 Dershowitz and Reingold 1997: 10; Dershowitz and Reingold 2008: 12
1055 Dershowitz and Reingold 1997: 13; Dershowitz and Reingold 2008: 14
1056 Sunday 06 October 3761 BCE (Julian); Sunday 06 September 3761 BCE (Gregorian Calibrated); 29 Ululu / Elul 0 AM; Astronomical Julian Day JD_X: 347996.5; RD_Y: 0; RD_X: -1373428; Sabbatical Cycle: 7; Metonic Cycle: 19;
1057 Monday 07 October 3761 BCE (Julian); Monday 07 September 3761 BCE (Gregorian Calibrated); 01 Tashritu / Tishri 1 AM; Astronomical Julian Day JD_X: 347997.5; RD_Y: 1; RD_X: -1373427; Sabbatical Cycle: 1; Metonic Cycle: 1;
1058 Dershowitz and Reingold 1997: 13; Dershowitz and Reingold 2008: 14
1059 Parker and Dubberstein 1971: 25

XX. O. Neugebauer cast some doubt upon these tables as a result of his own investigations into the visibility of Mercury and his discovery that Schoch's [set of values for] arcus visionis was in part derived from calculated rather than observed Babylonian data. How this error might affect Schoch's computations for the number of hours which must elapse after conjunction for the crescent to be visible has not yet been determined. Therefore, Parker and Dubberstein admit that until this can be done, they cannot be fully confident of their results in critical cases.[1060]

XX. Accordingly, they acknowledge that it is possible that a certain number of dates in their tables may be wrong by one day [or perhaps more]. It should be noted that there is a slight additional element of uncertainty for biblical dates given in the Babylonian calendar since the New Moon was visible at Jerusalem thirty-seven (37) minutes before it was visible in Babylon, and therefore upon occasion the new month could begin a day earlier in Jerusalem.[1061]

XX. According to Parker and Dubberstein, their tables are accurate as stated for sites in Babylonia between 31.5 degrees and 33.5 degrees in latitude, and tend to grow slightly less exact as these limits are exceeded. Ancient errors of observation are of course also possible. There are also degrees of uncertainty of the unattested intercalary months.[1062]

XX. The Babylonians did not always insert the same intercalary month. In some years they added a second Ululu (Ululu II, after month VI, i.e. a second 6th month). In others they added a second Addaru (Addaru II, after month XII, i.e. a second 12th month).[1063]

XX. Tablets attest the Babylonian leap months only in some years. In other years Parker and Dubberstein reconstruct them, and before 626 BCE (Julian) they consider reconstruction must await further knowledge.[1064]

XX. As Walker notes, the fundamental tables were printed in 1956 by Parker and Dubberstein and have never seriously been questioned. The principal problem involved is to determine the years in which a 13th lunar month – the "intercalary" month – was added to bring the lunar year back into line with the solar year, and which intercalary month was used: Ululu 2 / Elul 2 (month 6b), or Addaru 2 / Adar 2 (month 12b).[1065]

0.50. The Proleptic Hebrew calendar as a Reference Scale for Ancient Babylonian Dates

1060 Parker and Dubberstein 1971: 25
1061 Parker and Dubberstein 1971: 25
1062 Parker and Dubberstein 1971: 25
1063 Parker and Dubberstein 1971: 4-9; 26-47
1064 Parker and Dubberstein 1971: 2; 4-9; 26-47
1065 Walker 1997: 22

XX. The Hebrew calendar is fixed by arithmetic rules. Its mean month is 29 days, 12 hours and 793 parts, with 1080 parts to the hour.[1066] This differs from the mean synodic month, 29.530588 days at the year 2000, by about half a second.[1067] Its 19-year cycle has seven (8) leap years, in years 3, 6, 8, 11, 14, 17 and 19,[1068] so that 235 lunar months make 19 solar years. Its mean solar year is therefore 365.24682[1069] days which is about 6.7 minutes longer than the conventionally accepted tropical year, 365.24219878 days,[1070] a difference of one day in about 216 years.

XX. The proof is as follows:

(365.24682 days) − (365.24219878 days) == (0.00462 days per year)

(0.00462 days) × (1440 minutes per day) == (6.65 minutes) i.e. about 6.7 minutes per year.

(1 day) ÷ (0.00462 days per year) == (216.4 years)

XX. Applied proleptically, the Hebrew calendar provides a fixed, absolute, precise, mathematically rule-based reference scale against which recorded Babylonian dates and their Julian conversions can be compared and calibrated. Because its rules do not change, any departure of a Babylonian date from it can be located in a specific year and month.

XX. The departures are of two (2) kinds. A difference of one (1) or two (2) days arises from the different rules for beginning a month and/or the variable of location (i.e. Babylon vs. Jerusalem). The Babylonian month depended on when the new crescent became visible after the conjunction, which Parker and Dubberstein compute from Schoch's tables.[1071] The Hebrew month is set by the molad (i.e. the calculated mean conjunction, at fixed intervals of 29d 12h 793 parts) and its postponement rules.[1072] A difference of about one (1) month between the proleptic Hebrew and ancient Babylonian calendar arises from an intercalary month inserted in a different year in each calendar.

XX. Where tablets attest the Babylonian intercalary month, the comparison with the mathematically precise proleptic Hebrew calendar can be checked against them. Where no tablet survives, the comparison indicates the year in which the Babylonian intercalation most probably fell.

XX. Therefore, this proposed methodology of applying this proleptic Hebrew calendar to ancient Babylonian dates in this way may help to determine when the 19-year cycle was introduced, which

1066 Maimonides / Rambam, *Mishneh Torah Sefer Z'manim Kiddush HaChodesh*, Chapter 6, Halacha 2-3
1067 Espenak (NASA): *Six Millennium Catalog of Phases of the Moon*, https://eclipse.gsfc.nasa.gov/phase/phasecat.html
1068 Maimonides / Rambam, *Mishneh Torah Sefer Z'manim Kiddush HaChodesh*, Chapter 6, Halacha 10-11
1069 Dershowitz and Reingold 2008: 10
1070 Steele 2011a: 471; Kelley and Milone 2011: 100
1071 Parker and Dubberstein 1971: 25
1072 Maimonides / Rambam, *Mishneh Torah Sefer Z'manim Kiddush HaChodesh*, Chapter 7, Halacha 1-6

intercalary months were used by the Babylonians and in which years, and when the cycle became standardized, a date on which scholars still disagree.

XX. Throughout this work, the Julian dates, Gregorian dates, and Julian Day numbers given for Babylonian dates are their proleptic Hebrew equivalents, computed by the rules set out above. They are exact and precise as conversions within the mathematical Hebrew calendar, but they do not necessarily coincide with the Julian day on which the event took place in the (not yet mathematically refined perfectly) historical Babylonian calendar.

XX. The proleptic Hebrew calendar thus serves as a fixed, mathematically perfect reference standard, a control against which the historical Babylonian calendar can be measured (and refined) month by month and year by year. Because the proleptic Hebrew calendar's intercalations follow an invariable mathematical rule, every divergence of about one (1) month marks the point at which the Babylonians inserted an intercalary month in a different year, or chose a different intercalary month (Ululu 2 / Elul 2 rather than Addaru 2 / Adar 2).

XX. Where tablets attest the Babylonian intercalation, the divergence confirms it; where no tablet survives, the divergence locates it. Divergences of one (1) or two (2) days arise from the different rules by which each calendar began the month, as well as the variable of locations (Babylon vs. Jerusalem) which may have affected uncertainty in the computed visibility of the crescent, owing to Schoch's [values for] arcus visionis, by which Parker and Dubberstein admit that a certain number of their dates may be wrong by one day.[1073] In this way, the mathematically precise and perfect proleptic Hebrew calendar makes it possible to synchronize ancient Babylonian dated events, as well as to determine the years and months in which the Babylonians intercalated.

XX. In the Hebrew tradition, only the 12th month Addaru / Adar was intercalated (Tosefta Sanhedrin 2:4), since a second Ululu / Elul would have displaced the festivals of the 7th month: Rosh Hashanah and Yom Kippur. Therefore, it is reasonable to infer that Israel and Judah did not follow the Babylonian practice of intercalating a second Ululu / Elul, and thus the biblical dates of this period that are given here and in all sections follow their proleptic Hebrew equivalents.

XX. In regard to the work comparing Parker and Dubberstein's tables of intercalary months of the Babylonian calendar with the mathematically precise proleptic Hebrew calendar, Claude (Anthropic) model Opus 5.5 at medium effort was utilized. It was used to peruse, analyze, and verify the table data, compute the corresponding Hebrew dates, compile the comparison table, and assist in drafting the related text. All other work presented here is the author's own human effort, carried out without any AI assistance. Readers would therefore be prudent to independently check the accuracy of the work produced with Claude.

1073 Parker and Dubberstein 1971: 25

1.0. Ancient Biblical vs. Babylonian Chronologies: The Issues, the Reasoning, and the Academic Claims

XX. The date of the destruction of Jerusalem by the armies of Nebuchadnezzar is one of the most important dates in world history.[1074]

XX. There are known problems with the traditional, religious Jewish timeline of history as preserved in Jewish religious tradition that was put to writing in the 2nd century CE Hebrew book, *Seder Olam Rabbah* (*The Great Order of the World*), as it dates the destruction of the first Jewish temple to be Hebrew Year 3338 AM (i.e. 422 BCE (Julian)) – which conflicts with the academically-established date of 586 BCE (Julian), i.e. Hebrew Year 3175 AM.

XX. This discrepancy of 164 years is well known among scholars and researchers of Jewish history. The solution to this problem of how to find and reconcile these so-called "missing ~165 years" will be made possible here as a direct result of this research. It will be realized with the observation that this ancient Jewish religious timeline is sometimes precisely accurate, and sometimes extremely inaccurate.

XX. In other words: it fluctuates in its accuracy (depending on the event / period recorded) when compared to current academic consensus for specific events in specific years. The mathematically precise dates provided here are the keys to unlock and correct all dates throughout Jewish history. Further study and research in this field of Jewish Studies scholarship is required.

XX. According to this timeline, the 1st Regnal Year of Yehoyakim (Jehoiakim) – one of the last kings of Judah – was Hebrew Year 3317 – 3318 AM (i.e. 443 – 442 BCE (Julian)). However, this would be impossible as it would require that the 1st Regnal Year of Nebuchadnezzar (i.e. the 4th Regnal Year of Yehoyakim (Jehoiakim))[1075] to be Hebrew Year 3320 – 3321 AM (i.e. 440 – 439 BCE (Julian)) – when in fact, according to established academic consensus, the 1st Regnal Year of Nebuchadnezzar began Spring 604 – 603 BCE (i.e. Hebrew Year 3157 – 3158 AM).

XX. Nebuchadnezzar's father, Nabopolassar, began his 21st (FINAL) Regnal Year in Spring 605 – 604 BCE (Julian),[1076] [1077] which transitioned into the Year 0 "Accession Year" of Nebuchadnezzar before his accession to his 1st Regnal Year the next year that began Spring 604 – 603 BCE (Julian).[1078] [1079]

XX. Thus we must distinguish between Year 0 "Accession Year" of Nebuchadnezzar and his 1st Regnal

1074 Young 2005: 230
1075 cf. Jeremiah 25:1
1076 3156 – 3157 AM
1077 cf. Appendix Table X.4
1078 3157 – 3158 AM
1079 cf. Appendix Table X.4

Year, as Nebuchadnezzar's Accession Year was also the same final 21st Regnal Year of the previous king (i.e. his father Nabopolassar) – a sort of transition period between the previous king's last year and the 1st Regnal Year of the new king after the official coronation at the Akitu festival in the next Spring month of Nisannu / Nisan that officially marks the new king's 1st Regnal Year, as years for kings are counted from this Nisannu / Nisan Akitu festival.

XX. It is uncertain exactly where Nebuchadrezzar himself was at the time of the death of his father Nabopolassar on the 8th of Abu / Av 605 BCE (Julian). Basing his dates on Parker and Dubberstein's tables, Wiseman gives their date of 15 / 16 August 605 BCE (Julian).[1080] The proleptic Hebrew equivalent is Friday 15 July 605 BCE (Julian).[1081] [1082]

XX. The one-(1)-month difference arises because a Babylonian intercalary Addaru 2 / Adar 2 is attested at the end of Nabopolassar's 20th Regnal Year (i.e. 606 – 605 BCE (Julian)),[1083] whereas the proleptic Hebrew calendar does not insert an intercalary month until the 13th month of Addaru 2 / Adar 2 of 3157 AM beginning 03 March 604 BCE (Julian).

2.0. Death of Nabopolassar == Summer 605 BCE
== 21st (FINAL) Regnal Year of Nabopolassar
== Year 0 Accession Year of Nebuchadnezzar II
== 3rd Regnal Year of Yehoyakim (Jehoiakim)
== YEAR 4 OF SERVITUDE

XX. Babylonian Chronicles in the British Museum cover much of what was the closing period of the kingdom of Judah. By the correlation of such extra-biblical evidence with biblical data, it is possible to state dates in this period with relatively great precision and assurance.[1084] Babylonian tablets in the British Museum now provide records for the following years in this period here under review;[1085] The Babylonian tablets provide – among others – the following fixed dates for the period we are about to discuss:[1086]

– Egyptian campaign to the Euphrates [in the 17th Regnal Year of Nabopolassar], against Haran – 609 BCE (Julian)

– Babylonian victory at Carchemish [over the Egyptians in the 21st (FINAL) Regnal Year of

1080 Wiseman 1956: 26
1081 Friday 15 July 605 BCE (Julian); Friday 08 July 605 BCE (Gregorian); 08 Abu / Av 3156 AM; Astronomical Julian Day JD_X: 1500642.5; RD_Y: 1152646; RD_X: -220782; Sabbatical Cycle: 6; Metonic Cycle: 2;
1082 cf. Appendix Table X.11
1083 Parker and Dubberstein 1971: 4
1084 Finegan 1998: 251
1085 Thiele 1983: 179
1086 Thiele 1983: 180

Nabopolassar that is equivalent to the Year 0 Accession Year of Nebuchadnezzar II], Hatti-land taken – 605 BCE (Julian)

– Death of Nabopolassar – 8th day of 5th month (Abu / Av) 605 BCE (Julian)

XX. The new year (for kings) that began Spring (Nisannu / Nisan) March 605 BCE (Julian), i.e. two (2) Hebrew civil years, *Anno Mundi* (pl. *Anni Mundi*), that overlap the last six (6) months of 3156 AM, and the first six (6) months of 3157 AM:[1087]

== Nabopolassar's death in his 21st (FINAL) Regnal Year
== 605 – 604 BCE (Julian)
== 3156 – 3157 AM
== YEAR FOUR (4) OF SERVITUDE.

XX. Nabopolassar's son, Nebuchadnezzar II, ascended to the throne in place of his father, and thus this year became:[1088]

== Year 0 Accession Year of Nebuchadnezzar
== 605 – 604 BCE (Julian)
== 3156 – 3157 AM
== YEAR FOUR (4) OF SERVITUDE.

XX. This same year is equivalent to:[1089]

== 3rd Regnal Year of the Judean king, Yehoyakim (Jehoiakim)
== 605 – 604 BCE (Julian)
== 3156 – 3157 AM
== YEAR FOUR (4) OF SERVITUDE.

XX. Using Parker and Dubberstein's tables (1956), Thiele gives the following date for the death of Nabopolassar in his 21st (FINAL) Regnal Year: 15 August 605 BCE (Julian).[1090] The propleptic Hebrew equivalent date is Friday 15 July 605 BCE (Julian).[1091] [1092]

XX. The one-(1)-month difference arises because a Babylonian intercalary Addaru 2 / Adar 2 is attested

1087 cf. Appendix Table X.11, X.14
1088 cf. Appendix Table X.12, X.14
1089 cf. Appendix Table X.14
1090 Finegan 1998: 253
1091 Friday 15 July 605 BCE (Julian); Friday 08 July 605 BCE (Gregorian); 08 Abu / Av 3156 AM; Astronomical Julian Day JD_X: 1500642.5; RD_Y: 1152646; RD_X: -220782; Sabbatical Cycle: 6; Metonic Cycle: 2;
1092 cf. Appendix Table X.11

at the end of Nabopolassar's 20th Regnal Year.[1093] whereas the proleptic Hebrew calendar does not insert an intercalary month until the 13th month of Addaru 2 / Adar 2 of AM 3157 in March 604 BCE (Julian).

– Accession of Nebuchadnezzar – 1st day of 6th month (Ululu / Elul) 605 BCE (Julian). Using Parker and Dubberstein's tables (1956), Thiele and Wiseman[1094] give a Julian date of 07 September 605 BCE.[1095] The proleptic Hebrew equivalent date is Sunday 07 August 605 BCE (Julian).[1096] [1097] [1098] The one-(1)-month difference between these dates has the same cause as for the difference in the dates of the death of Nabopolassar (see above).

XX. It must be taken into consideration the time necessary for the transmission of this news from Babylon to Syria and the Levant by fast courier, as well as for Nebuchadrezzar's necessity to settle local affairs before his return journey with a small mounted party by the shortest desert route to Babylon. Since the crown-prince reached the capital twenty-three (23) days after his father's death, the [Babylonian] Chronicle supports the tradition of a swift return to Babylon that was preserved by Berossus.[1099]

– Jerusalem taken by Nebuchadnezzar – 2nd day of 12th month (Addaru / Adar) 597 BCE (Julian). Using Parker and Dubberstein's tables (1956), Thiele gives a Julian date of 16 March 597 BCE (Julian).[1100] The proleptic Hebrew date is Thursday 15 February 597 BCE (Julian).[1101] [1102] The one-(1)-month difference arises because a Babylonian intercalary Ululu 2 / Elul 2 is attested in Nebuchadnezzar's 7th Regnal Year,[1103] whereas the proleptic Hebrew calendar does not insert an intercalary month until the 13th month of Addaru 2 / Adar 2 of 3165 AM, i.e. March / April 596 BCE (Julian)). The implications of the date of the 2nd day of 12th month (Addaru / Adar) 597 BCE (Julian) will be discussed in more detail in Section 14.

– Jehoiakim was killed [Hebrew: Yehoyakim]; Jehoiachin [Hebrew: Yehoyachin] was captured; Zedekiah [Hebrew: Tzidkiyahu] was made vassal-puppet-slave-king over Jerusalem – 597 BCE (Julian).[1104]

1093 Parker and Dubberstein 1971: 4
1094 Wiseman 1956: 27
1095 Thiele 1983: 180
1096 Sunday 07 August 605 BCE (Julian); Sunday 31 July 605 BCE (Gregorian); 01 Ululu / Elul 3156 AM; Astronomical Julian Day JD_X: 1500665.5; RD_Y: 1152669; RD_X: -220759; Sabbatical Cycle: 6; Metonic Cycle: 2;
1097 cf. Appendix Table X.12
1098 Wiseman 1956: 25
1099 Wiseman 1956: 25
1100 Thiele 1983: 180
1101 Thursday 15 February 597 BCE (Julian); Thursday 09 February 597 BCE (Gregorian); 02 Addaru / Adar 3164 AM; Astronomical Julian Day JD_X: 1503413.5; RD_Y: 1155417; RD_X: -218011; Sabbatical Cycle: 7; Metonic Cycle: 10;
1102 cf. Appendix Table X.13, X.32
1103 Parker and Dubberstein 1971: 5
1104 Thiele 1983: 180

3.0. 4th Regnal Year of Yehoyakim (Jehoiakim) == Spring 604 – 603 BCE == 1st Regnal Year of Nebuchadnezzar II == YEAR 5 OF SERVITUDE

XX. The Prophet Jeremiah equates the 4th Regnal Year of Yehoyakim (Jehoiakim / Elyakim) with the 1st Regnal Year of Nebuchadnezzar:[1105] (i.e. Year 1 == Spring 604 – 603 BCE (Julian))[1106] which should not be confused with Nebuchadnezzar's Year 0 Accession Year (i.e. Year 0 == Spring 605 – 604 BCE (Julian))[1107] – this is our 1st Chronological Anchor Point (i.e. synchronism).

XX. Indeed this verse in Jeremiah 25:1 is the key, i.e. our "Rosetta Stone", from which we can derive all other dates of our chronology allowing us to correct, as well as recalibrate, the currently accepted (yet often incorrect and inaccurate) dates of chronologies in Assyriology and Egyptology. This research makes it possible to correct these chronologies precisely as we synchronize them by recalibrating them to the correct Biblical dates (to the very exact day) as will be mathematically proven.

XX. In the 4th Regnal Year of Yehoyakim (Jehoiakim / Elyakim) Jeremiah receives a word of prophecy in which he declares the date of that revelation as the 4th Regnal Year of Yehoyakim (Jehoiakim / Elyakim) to be equivalent to the 1st Regnal Year of Nebuchadnezzar, and that from the 13th Regnal Year of Yoshiyahu (Josiah)[1108] until that very day[1109] as equating to 23 total years.[1110] [1111] However 23 years is only the number of years from the 13th Regnal Year of Yoshiyahu (Josiah) until the 4th Regnal Year of Yehoyakim (Jehoiakim / Elyakim) == 1st Regnal Year of Nebuchadnezzar, and not until the 11th (FINAL) Regnal Year of Tzidkiyahu (Zedekiah) the last king of Judah who ruled later.

XX. Indeed, counting from the 13th Regnal Year of Yoshiyahu (Josiah) up to (and including) the 4th Regnal Year of Yehoyakim (Jehoiakim / Elyakim) is exactly 23 years – thus providing further mathematical evidence that the chronological synchronization tables that will be unfolded are mathematically correct at every Chronological Anchor Point (synchronism) – albeit they upend the status quo of Academia regarding synchronisms with the relative chronology of Assyriology, as well as that of Egyptology.

XX. It is evident from the 1st chapter of the Book of Jeremiah that – according to Jeremiah – he began to receive the word of prophecy in the 13th Regnal Year of Yoshiyahu (Josiah), and that this word of prophecy continued to be revealed to him until the end of the 11th (FINAL) Regnal Year of Tzidkiyahu

1105 cf. Jeremiah 25:1
1106 3157 – 3158 AM
1107 3156 – 3157 AM
1108 13th Regnal Year of Yoshiyahu (Josiah) == Year 0 Accession Year of Nabopolassar == 626 – 625 BCE (Julian)
1109 i.e. until that very day in the 4th Regnal Year of Yehoyakim (Jehoiakim / Elyakim) == 1st Regnal Year of Nebuchadnezzar
1110 cf. Appendix Table X.4
1111 cf. Jeremiah 25:1-3

(Zedekiah) son of Yoshiyahu (Josiah), king of Judah, until the carrying away of Jerusalem captive into the EXILE OF JERUSALEM[1112] in the 5th month Abu / Av (i.e of that 11th (FINAL) Regnal Year of Tzidkiyahu (Zedekiah)).[1113]

XX. We are told that Jeremiah continued to receive the word of YHVH until the 5th month (Abu / Av) of the 11th (FINAL) Regnal Year of Tzidkiyahu (Zedekiah), the last king of Judah.[1114] In order to calculate the total years of Jeremiah's career as prophet through the reign of the last four (4) kings of Judah, we need to calculate and total the years of their respective reigns.

XX. Both the Book of Kings and the Book of Chronicles tell us that Yoshiyahu (Josiah) took the throne at 8 years old, and reigned for 31 years in Jerusalem:

22:1 Eight (8) years old was Yoshiyahu at the [beginning of] his reign, and 31 years [he] reigned in Jerusalem, and the name of his mother [was] Yedidah daughter of 'Adayah from Batzkat.[1115]

34:1 Eight (8) years old was Yoshiyahu at the [beginning of] his reign, and 31 years [he] reigned in Jerusalem.[1116]

XX. Thus we can calculate:

31 Regnal Years (i.e. total length of the reign of Yoshiyahu (Josiah) excluding Year 0 Accession Year)[1117]

– 13th Regnal Year of Yoshiyahu (Josiah), i.e. the 1st year of Jeremiah's prophesying career[1118]

== 18 years + 1 (to include the 31st (or the 13th) Year in the calculation)

== 19 years (i.e. the number of years during the reign of Yoshiyahu (Josiah) that Jeremiah prophesied from the 13th Regnal Year of Yoshiyahu (Josiah) up to (and including) his 31st (FINAL) Regnal Year)

XX. As can be observed and confirmed in our unfolding of a Chronological Synchronization table,[1119] this period of time from the 13th Regnal Year of Yoshiyahu (Josiah) to the 5th month (Abu / Av) of the

1112 i.e. This is not to be confused with the EXILE OF YEHOYACHIN (JEHOIACHIN) which began in 597 BCE (Julian); The EXILE OF JERUSALEM == 586 BCE (Julian) == YEAR ONE (1) OF DESOLATION
1113 cf. Jeremiah 1:1-3
1114 cf. Jeremiah 1:1-3
1115 cf. 2 Kings 22:1
1116 cf. 2 Chronicles 34:1
1117 cf. 2 Kings 22:1; 2 Chronicles 34:1
1118 cf. Jeremiah 1:1-4; 25:3
1119 cf. Appendix Table X.4

11th (FINAL) Regnal Year of Tzidkiyahu (Zedekiah) works out perfectly mathematically, and thus we can calculate that Jeremiah received the word of YHVH for 41 years.

XX. For the same year, i.e. the 4th Regnal Year of Yehoyakim (Jehoiakim / Elyakim) == 1st Regnal Year of Nebuchadnezzar, (i.e. Spring 604 – 603 BCE (Julian)),[1120] Jeremiah – writing sometime in the 1st Regnal Year of Nebuchadnezzar (i.e. after the Battle of Carchemish which had already taken place one (1) year earlier in Spring or Summer 605 BCE (Julian)[1121] prior to the beginning of (and within) the 21st (FINAL) Regnal Year of Nabopolassar == Year 0 Accession Year of Nebuchadnezzar), and while prophesying to Egypt by dictating to his scribe-assistant, Baruch Ben Neryah – writes his prophecy addressed to Egypt:

45:1 The Word that Jeremiah the Prophet spoke to Baruch Ben Neryah in his writing these words on a book from the mouth of Jeremiah in the 4th year of Yehoyakim[1122] son of Yoshiyahu king of Yehudah saying:[1123]

46:1 That the Word of YHVH was unto Jeremiah the Prophet regarding the nations: 46:2 To Egypt regarding the Army of Pharaoh Necho king of Egypt – who was on the River Euphrates at Carchemish whom Nebuchadnezzar struck[1124] – the Word of YHVH was unto him, to Jeremiah, in the 4th year of Yehoyakim[1125] son of Yoshiyahu king of Yehudah.[1126] [1127]

XX. Thus we can reason that Jeremiah wrote the above mention of the Battle of Carchemish after the Battle of Carchemish had already taken place, i.e. as a historical mention, and not a prophetic mention of the future. It would be incorrect to translate and interpret this verse to mean that the Battle of Carchemish occurred in the 4th Regnal Year of Yehoyakim (Jehoiakim / Elyakim) == 1st Regnal Year of Nebuchadnezzar, i.e. Spring 604 – 603 BCE (Julian)[1128] since we know to be fact that the Battle of Carchemish happened in the 21st (FINAL) Regnal Year of Nabopolassar == Year 0 Accession Year of Nebuchadnezzar, i.e. Spring 605 – 604 BCE (Julian).[1129]

XX. Based upon these initial verses from the Hebrew Scriptures which provide the 1st Chronological

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1120 3157 – 3158 AM
1121 3156 AM
1122 i.e. in the 1st Regnal Year of Nebuchadnezzar, i.e. Spring 604 – 603 BCE (Julian) == 3157 – 3158 AM.
1123 cf. Jeremiah 45:1
1124 i.e. "struck" in the past tense one (1) year earlier, i.e. Spring 605 – 604 BCE (Julian) == 3156 – 3157 AM, after-the-fact.
1125 i.e. in the 1st Regnal Year of Nebuchadnezzar, i.e. Spring 604 – 603 BCE (Julian) == 3157 – 3158 AM.
1126 i.e. In the 4th Regnal Year of Yehoyakim (Jehoiakim / Elyakim), the Word of YHVH was unto Jeremiah; i.e. NOT that Nebuchadnezzar struck Carchemish in the 4th Regnal Year of Yehoyakim (Jehoiakim / Elyakim) – that happened in the prior 3rd Regnal Year of Yehoyakim (Jehoiakim / Elyakim) during Nebuchadnezzar's Year 0 Accession Year, 605 – 604 BCE (Julian) == 3156 – 3157 AM.
1127 cf. Jeremiah 46:1-2
1128 3157 – 3158 AM
1129 3156 – 3157 AM

Anchor Point (synchronism),[1130] we have our Rosetta Stone from which we can mathematically and logically derive all other exactly precise, correct dates in our chronology(ies). We can thus begin to construct a Chronological Synchronization table which compares, equates, and synchronizes the reigns of the kings of Babylon with the kings of Judah, beginning with specifically Nebuchadnezzar and Yehoyakim (Jehoiakim / Elyakim):[1131]

Table 3.1. Chronological Synchronization of Babylonian and Judean Kings (Divided Kingdom)

Year	Babylonia	Judea / Judah / Yehudah
608 – 607 BCE 3153 – 3154 AM	18th Regnal Year of Nabopolassar	31st (FINAL) Year of Yoshiyahu (Josiah) == Year 0 Accession Year of Yeho-ahaz (Jeho-ahaz / Shalum) (i.e. 3 months) == Year 0 Accession Year of Yehoyakim (Jehoiakim / Elyakim) == YEAR 1 OF SERVITUDE
607 – 606 BCE 3154 – 3155 AM	19th Regnal Year of Nabopolassar	1st Regnal Year of Yehoyakim (Jehoiakim) == YEAR 2 OF SERVITUDE
606 – 605 BCE 3155 – 3156 AM	20th Regnal Year of Nabopolassar	2nd Regnal Year of Yehoyakim (Jehoiakim) == YEAR 3 OF SERVITUDE
605 – 604 BCE 3156 – 3157 AM	21st Regnal Year of Nabopolassar == Year 0 Accession Year of Nebuchadnezzar II	3rd Regnal Year of Yehoyakim (Jehoiakim) == YEAR 4 OF SERVITUDE
604 – 603 BCE 3157 – 3158 AM	1st Regnal Year of Nebuchadnezzar II	4th Regnal Year of Yehoyakim (Jehoiakim) == YEAR 5 OF SERVITUDE

XX. Finegan cites Thiele who notes that two (2) lunar eclipses confirm 605 BCE (Julian) as the year of accession for Nebuchadnezzar. The 1st eclipse[1132] occurred in the 5th Regnal Year of Nabopolassar in 621 BCE (Julian) which puts Nabopolassar's death in his 21st (FINAL) Regnal Year in 605 BCE (Julian) and thus was the beginning of Nebuchadnezzar's reign [i.e. Year 0 Accession Year] in that same year 605 BCE (Julian).[1133]

XX. According to Huber and De Meis, this lunar eclipse in the 5th Regnal Year of Nabopolassar

1130 cf. Jeremiah 25:1
1131 cf. Appendix Table X.4
1132 i.e. 1st Astronomical Anchor Point, 2nd Chronological Anchor Point
1133 Finegan 1998: 253

occurred in the 2nd month, i.e. Ayyaru / Iyar. However, this Babylonian date is not independently attested in a Babylonian source – it is derived from Ptolemy's *Almagest* (Book V, Chapter 14), where the eclipse is dated via the Egyptian civil calendar.[1134] This date had been converted from the original Babylonian calendar reckoning in ancient times, traditionally attributed to Hipparchus.

XX. This Egyptian calendar date has since been converted to the Julian calendar using the fixed, unvarying structure of the Egyptian civil calendar[1135] which permits exact arithmetic conversion to a Julian day count once a single anchor date[1136] is established, i.e. the 1st day of the Era of Nabonassar (*Anno Nabonassari*, AN).[1137] [1138] Because Ptolemy also records the eclipse's regnal year[1139] and observational details, modern astronomers calculate that the resulting date is 21 / 22 April 621 BCE (Julian).

XX. According to Huber and De Meis, this lunar eclipse occurred on Saturday 22 April 621 BCE (Julian)[1140] which accords perfectly with the Babylonian / Hebrew date of the 15th day of the 2nd month, Ayyaru / Iyar. This is a plausible date because lunar eclipses can only occur on days when there is a Full Moon which only can occur on the Babylonian / Hebrew calendar on the 14th or 15th days of the month depending upon whether it is a 29- or 30-day month.[1141] [1142] [1143]

XX. This eclipse is also one (1) of ten (10) Babylonian lunar eclipses recorded in Ptolemy's *Almagest*:

(4) BC 621 April 21/22 (mag. = 0.15)

In the fifth year of Nabopolassar, which is the 127th year from Nabonassar, Athyr [month III] 27/28 in the Egyptian calendar [-620 April 21/22], at the end of the 11th hour in Babylon, the Moon

1134 Day 27/28 in the 3rd Egyptian month of Athyr

1135 i.e. exactly 365 days of twelve (12) months consisting of 30 days each plus five (5) epagomenal days with no intercalation. Thus this Egyptian calendar drifts backward (i.e. earlier) relative to the Julian calendar as time moves forward. This means that the Egyptian calendar loses approximately ¼ day every Julian year, or one (1) day every four (4) years relative to the Julian year. For example, if the Egyptian New Year (01 Thoth) falls on a given Julian date in one (1) year, four (4) years later it will fall about one (1) day earlier on the Julian calendar. Day 1 of the 1st month Thoth cycles through every season and every Julian calendar date once every 1460 Julian years as one (1) Sothic cycle.

1136 i.e. 1st day of the 1st Egyptian month Thoth == Year 1 of the Nabonassar Era, i.e. 26 February 747 BCE (Julian)

1137 Wednesday 26 February 747 BCE (Julian); Wednesday 18 February 747 BCE (Gregorian); 06 Addaru / Adar 3014 AM; Astronomical Julian Day JD_X: 1448637.5; RD_Y: 1100641; RD_X: -272787; Sabbatical Cycle: 4; Metonic Cycle: 12; 01 (Egyptian Month 01 Thoth) 1 AN; 0 days since beginning of Epoch;

1138 cf. Appendix Table X.6

1139 i.e. the 127th year from Nabonassar

1140 Huber and De Meis 2004: 185

1141 Friday 21 April 621 BCE (Julian); Friday 14 April 621 BCE (Gregorian); 14 Ayyaru / Iyar 3140 AM; Astronomical Julian Day JD_X: 1494713.5; RD_Y: 1146717; RD_X: -226711; Sabbatical Cycle: 4; Metonic Cycle: 5; 27 (Egyptian Month 03 Athyr) 127 AN; 46076 days since beginning of Epoch

1142 Saturday 22 April 621 BCE (Julian); Saturday 15 April 621 BCE (Gregorian); 15 Ayyaru / Iyar 3140 AM; Astronomical Julian Day JD_X: 1494714.5; RD_Y: 1146718; RD_X: -226710; Sabbatical Cycle: 4; Metonic Cycle: 5; 28 (Egyptian Month 03 Athyr) 127 AN; 46077 days since beginning of Epoch

1143 cf. Appendix Table X.7

began to be eclipsed; the maximum obscuration was 1/4 of the diameter from the south. Now since the beginning of the eclipse occurred 5 seasonal hours after midnight...[1144]

XX. The 2nd eclipse[1145] was in 568 BCE (Julian) in the 37th Regnal Year of Nebuchadnezzar which also counts correctly from his accession [Year 0 beginning] in [Spring] 605 [– 604 BCE] (Julian). Thiele concludes that no date in ancient history is more firmly established than this [i.e. the Year 0 Accession Year of Nebuchadnezzar in 605 BCE (Julian)].[1146]

XX. This 2nd eclipse is detailed in astronomical tablet VAT 4956 which is indeed the oldest known text of astronomical observations written in the detailed style of the late Babylonian Empire. It dates back to the 37th Regnal Year of Nebuchadnezzar II (568 BCE (Julian)), and is consequently the 1st major purely astronomical document from the time before the Neo-Babylonian Empire's fall.[1147]

XX. VAT 4956 is a fragment of [Astronomical] Diary for year 37 of Nebuchadnezzar II (i.e. Spring 568 – 567 BCE (Julian)). It is the 2nd oldest extant Astronomical Diary, dating 84 years after the oldest one (652 BCE (Julian)) and predating the bulk of the later diaries by about 175 years.[1148]

XX. This eclipse is recorded to have occurred on the 15th day of the 3rd month, i.e. Simannu / Sivan,[1149] which Huber and De Meis date to be equivalent to Friday 04 July 568 BCE (Julian).[1150] [1151] [1152] However, this Julian date given by Huber and De Meis in the proleptic Hebrew calendar is equivalent to the 13th day of the 4th month, i.e. Du'uzu / Tammuz.

XX. Therefore, the equivalent Julian date for the given proleptic Hebrew date of the 15th day of the 3rd month of Simannu / Sivan for the 37th Regnal Year of Nebuchadnezzar recorded in VAT 4956 is Friday 06 June 568 BCE (Julian).[1153] [1154] The one-(1)-month difference arises because a Babylonian intercalary Addaru 2 / Adar 2 that began Tuesday 25 March 568 BCE (Julian) is attested at the end of Nebuchadnezzar's 36th Regnal Year,[1155] whereas the proleptic Hebrew calendar inserted its intercalary 13th month of Addaru 2 / Adar 2 that began Wednesday 06 March 569 BCE (Julian), a year earlier in 3192 AM.

1144 *Almagest*, V, 14; trans. Toomer (1984: 253) – cited by Stephenson 1997: 100-101
1145 i.e. 2nd Astronomical Anchor Point, 3rd Chronological Anchor Point
1146 Finegan 1998: 253
1147 O'Maly 2011: 30
1148 Ossendrijver 2016a: 141
1149 Sachs and Hunger 1988: 48-49
1150 Friday 04 July 568 BCE (Julian); Friday 28 June 568 BCE (Gregorian); 13 Du'uzu / Tammuz 3193 AM; Astronomical Julian Day JD_X: 1514145.5; RD_Y: 1166149; RD_X: -207279; Sabbatical Cycle: 1; Metonic Cycle: 1;
1151 cf. Appendix Table X.9
1152 Huber and De Meis 2004: 69, 186
1153 Friday 06 June 568 BCE (Julian); Friday 31 May 568 BCE (Gregorian); 15 Simannu / Sivan 3193 AM; Astronomical Julian Day JD_X: 1514117.5; RD_Y: 1166121; RD_X: -207307; Sabbatical Cycle: 1; Metonic Cycle: 1;
1154 cf. Appendix Table X.8
1155 Parker and Dubberstein 1971: 5, 28

XX. Therefore, Huber and De Meis's date of 04 July 568 BCE (Julian) is correct for the Babylonian calendar, which ran one (1) month behind the proleptic Hebrew calendar in that year.

XX. The date of the 37th Regnal Year of Nebuchadnezzar that is recorded in VAT 4956 falls precisely during the time when Daniel purportedly would have lived and served in the Babylonian Royal Court. It is notable that it is precisely during this Neo-Babylonian and Achaemenid Perisan Era that the advent of a calculated, mathematically predictive system of a fixed 19-year intercalary, "Metonic" cycle for the Babylonian / Hebrew calendar became standardized.

XX. Until the Seleucid Era (311 BCE (Julian)), Babylonian dates were counted from the accession of each monarch, but continuous counting of years was adopted thereafter.[1156] According to Ossendrijver, year one (1) of the Seleucid Era began Spring 311 BCE (Julian),[1157] and year 12 began Spring 299 BCE (Julian).[1158]

XX. For the period covered by the [Babylonian Astronomical Diary] texts (from about 700 BCE (Julian) onwards), the [Babylonian] dates of accession for each ruler are accurately known.[1159]

4.0. The Last Four (4) Kings of Yehudah (Judah)

XX. According to Wiseman, the year in which the Battle of Carchemish took place is [astronomically] certain (605 BCE) though no date is expressly given in the text.[1160] We will accept this as our 4th Chronological Anchor Point (i.e. synchronism) from which to attempt to reconcile ancient biblical events to an accurate chronology.

XX. Tzidkiyahu (Zedekiah / Matanyah), the last king of Judah, reigned for 11 years (Spring 597 through Summer 586 BCE (Julian)),[1161] and replaced his half-nephew Yehoyachin (Jehoiachin) – the son of Yehoyakim (Jehoiakim / Elyakim). Yehoyachin (Jehoiachin) had replaced and/or evidently (as will be proven mathematically) coreigned as co-regent with his father, Yehoyakim (Jehoiakim / Elyakim), for exactly three (3) months and 10 days acceding to the co-regency in December 598 BCE

1156 Stephenson and Fatoohi 1993: 257
1157 Ossendrijver 2015b: 1867
1158 Ossendrijver 2013b: 11
1159 Stephenson and Fatoohi 1993: 257
1160 Wiseman 1956: 25
1161 cf. 2 Kings 24:18; 2 Chronicles 36:11

(Julian) (i.e. Late-Autumn 598 through Late-Winter 597 BCE (Julian)).[1162] [1163] [1164] [1165] Yehoyakim (Jehoiakim / Elyakim) had reigned prior to him for 11 years (i.e. ~Early / Mid-to-Late Autumn 608 through Mid-Winter 597 BCE (Julian).[1166] [1167] [1168]

XX. Yehoyakim (Jehoiakim / Elyakim) – who was the older half-brother of the later last king Tzidkiyahu (Zedekiah / Matanyah) – himself had replaced his younger half-brother Yeho-ahaz (Jeho-ahaz / Shalum): i.e. Tzidkiyahu (Zedekiah / Matanyah) and Yehoyakim (Jehoiakim / Elyakim) were half-brothers; Tzidkiyahu (Zedekiah / Matanyah) and Yeho-ahaz (Jeho-ahaz / Shalum) were full brothers:

15 And the sons of Yoshiyahu: the oldest [was] Yohanan, the second [was] Yehoyakim, the third [was] Tzidkiyahu, the fourth [was] Shalum.[1169]

Table 4.1. Last Four (4) Kings of Judah, Ages, Lengths of Reign, Names of Mothers, Bible Source(s)

King	Age	Length of Reign	Name of Mother	Bible Source(s)
Yeho-ahaz (Jeho-ahaz / Shalum)	23	3 months	Hamutal d. of Yirmeyahu (Jeremiah) from Livnah	2 Kings 23:31; 2 Chronicles 36:1-2
Yehoyakim (Jehoiakim / Elyakim)	25	11 years	Z'vudah d. of P'dayah from Rumah	2 Kings 23:36; 2 Chronicles 36:5
Yehoyachin (Jehoiachin)	18; 8[1170]	3 months; 3 months, 10 days	Nehushta d. of Elnatan from Jerusalem	2 Kings 24:8; 2 Chronicles 36:9
Tzidkiyahu (Zedekiah / Matanyah)	21	11 years	Hamutal d. of Yirmeyahu (Jeremiah) from Livnah	2 Kings 24:18; 2 Chronicles 36:11 Jeremiah 52:1

XX. The short reign of Yeho-ahaz (Jeho-ahaz / Shalum) who was taken captive by Pharaoh Necho II lasted only three (3) months: Yeho-ahaz (Jeho-ahaz / Shalum) took the throne at 23 years old, and three

1162 Sunday 17 December 598 BCE (Julian); Sunday 11 December 598 BCE (Gregorian); 01 Tebetu / Tevet 3164 AM; Astronomical Julian Day JD_X: 1503353.5; RD_Y: 1155357; RD_X: -218071; Sabbatical Cycle: 7; Metonic Cycle: 10;
1163 Saturday 23 March 597 BCE (Julian); Saturday 17 March 597 BCE (Gregorian); 10 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503450.5; RD_Y: 1155454; RD_X: -217974; Sabbatical Cycle: 7; Metonic Cycle: 10;
1164 cf. Appendix Table X.34
1165 cf. 2 Kings 24:8; 2 Chronicles 36:9
1166 Thursday 15 February 597 BCE (Julian); Thursday 09 February 597 BCE (Gregorian); 02 Addaru / Adar 3164 AM; Astronomical Julian Day JD_X: 1503413.5; RD_Y: 1155417; RD_X: -218011; Sabbatical Cycle: 7; Metonic Cycle: 10;
1167 cf. Appendix Table X.32
1168 cf. 2 Kings 23:36; 2 Chronicles 36:5
1169 cf. 1 Chronicles 3:15
1170 The age of 8 is clearly a simple scribal error that is evident to any reader of Hebrew: 18 is the correct age intended.

(3) months he reigned in Jerusalem.[1171]

XX. Since Yeho-ahaz (Jeho-ahaz / Shalum) was only 23 years old when he was crowned king (ruling for only three (3) months), and since Yehoyakim (Jehoiakim / Elyakim, the 2nd oldest son of Yoshiyahu (Josiah)) at the age of 25 was made vassal-puppet-slave-king by Pharaoh Necho II immediately upon (or soon after) his deposing of Yeho-ahaz (Jeho-ahaz / Shalum), then logically (mathematically) Yeho-ahaz (Jeho-ahaz / Shalum) could not have been Yohanan (Johanan, the 1st-born oldest son of Yoshiyahu (Josiah)).

XX. Furthermore, since Tzidkiyahu (Zedekiah / Matanyah) was the 3rd son of Yoshiyahu (Josiah) and (later) last king of Judah, the only possibility left for the identity of Yeho-ahaz (Jeho-ahaz) is Shalum, i.e. the 4th (and youngest) son of Yoshiyahu (Josiah).

XX. This logic is confirmed by Jeremiah's prophecy to Shalum son of Yoshiyahu (Josiah) king of Judah:

11 Because thus says YHVH to Shalum, son of Yoshiyahu king of Yehudah, the [one who] rules in place of[1172] Yoshiyahu his father: "[He] who [has] left from this place will not return [to] there [any] more. 12 Because in the place where [they] exiled him, there [he] will die, and this land [he] will not see [any] more."[1173]

XX. Egyptian Pharaoh Necho II had deposed Yeho-ahaz (Jeho-ahaz / Shalum) after only three (3) months reign (i.e. Year 0 Accession Year) and took him captive to Egypt. There in Egypt, Yeho-ahaz (Jeho-ahaz / Shalum) died.[1174] Pharaoh Necho II had earlier killed the mutual father (i.e. Yoshiyahu (Josiah)) of all three: Yeho-ahaz (Jeho-ahaz / Shalum), Yehoyakim (Jehoiakim / Elyakim), and Tzidkiyahu (Zedekiah / Matanyah) – i.e. Necho II had previously killed their mutual father, Yoshiyahu (Josiah),[1175] in Spring / Summer 608 BCE – not 609 BCE (Julian) as held by current academic consensus.[1176] After having deposed and taken captive Yeho-ahaz (Jeho-ahaz / Shalum), Pharaoh Necho II installed Yehoyakim (Jehoiakim / Elyakim) as his vassal-puppet-slave-king over the land of Judah.[1177]

5.0. 31st (FINAL) Year of Yoshiyahu (Josiah) == Spring 608 – 607 BCE
== 18th Regnal Year of Nabopolassar
== 395th (FINAL) YEAR OF INDEPENDENT JUDEAN KINGDOM

1171 cf. 2 Kings 23:31; 2 Chronicles 36:2-3
1172 Literally: "under"
1173 cf. Jeremiah 22:11-12
1174 cf. Jeremiah 22:10-12; 2 Kings 23:31; 2 Chronicles 36:1-2
1175 cf. 2 Kings 23:29; 2 Chronicles 35:20-24
1176 Finegan 1998: 252
1177 cf. 2 Kings 23:29-36; 2 Chronicles 36:1-4

== Year 0 Accession Year of Yeho-ahaz (Jeho-ahaz)
== Year 0 Accession Year of Yehoyakim (Jehoiakim)
== YEAR 1 OF SERVITUDE

XX. From the mathematics, it becomes clear that the 31st (FINAL) Regnal Year of Yoshiyahu (Josiah) coincided with the 18th Regnal Year of Nabopolassar that began in Spring 608 – 607 BCE (Julian)[1178] – not with his 17th Regnal Year that began Spring 609 – 608 BCE (Julian);[1179] this has implications that allows one to unlock a more accurate understanding of both Assyrian / Babylonian (i.e. Mesopotamian) as well as Egyptian chronologies and histories.

XX. British Museum Tablet BM 21901 records that in his 16th Regnal Year, i.e. 610 – 609 BCE (Julian)[1180] Neo-Babylonian king Nabopolassar drove Assyrian king Aššur-uballit out of Haran [Harran].

XX. In the next year [i.e. the 17th Regnal Year of Nabopolassar king of Babylon] 609 – 608 BCE (Julian) between Du'uzu / Tammuz (i.e. June / July) and Ululu / Elul (i.e. August / September) 609 BCE (Julian), Aššur-uballit and "a large army of Egypt" tried to reconquer Haran. The "army of Egypt" is surely the expeditionary force of Pharaoh Necho II (r. 609 – 595 BCE (Julian))[1181] whose purpose was evidently to assist the Assyrians and resist the Babylonians. This event [in the 17th Regnal Year of Nabopolassar king of Babylon] in 609 BCE (Julian)[1182] would be the final end of the Assyrian empire.[1183]

XX. Therefore, scholars have thus incorrectly reasoned that the 31st (FINAL) Regnal Year of Yoshiyahu (Josiah) king of Judah equates with these events in the Spring / Summer 609 BCE (Julian) in the 17th Regnal Year of Nabopolassar king of Babylon. However, this is not mathematically possible.

XX. Although Finegan reasonably argues that the intent of Yoshiyahu (Josiah) [i.e. in his 31st (FINAL) Regnal Year as recounted to us in the Hebrew Scriptures when he went to confront Pharaoh Necho II at Megiddo] was to keep [additional Egyptian] help from reaching the Assyrians (the old enemy of Judah), he incorrectly assumes (following scholars before him) that this happened in the 17th Regnal Year of Nabopolassar in Spring / Summer 609 BCE (Julian).[1184] According to Finegan, the death of Yoshiyahu (Josiah) in his own 31st (FINAL) Regnal Year is to be dated to the 17th Regnal Year of Nabopolassar in the month of Du'uzu / Tammuz (25 June – 23 July) 609 BCE (Julian).[1185] [1186]

1178 3153 – 3154 AM
1179 3152 – 3153 AM
1180 Finegan 1998: 252
1181 Dates given by Finegan 1998
1182 Dates given by Finegan 1998
1183 Finegan 1998: 252
1184 Finegan 1998: 252
1185 Dates given by Finegan 1998
1186 Finegan 1998: 252

XX. There is no reason to assume that Pharaoh Necho II didn't leave the majority of his troops in Harran in Winter 609 BCE to Spring 608 BCE (Julian) while he traveled back to Egypt (assuming he had even led his forces to battle in Spring / Summer 609 BCE (Julian)). Especially in ancient times, Winter is the natural season to rest from active warfare because of its harsh conditions. As recorded in the Hebrew Bible, in the 31st (FINAL) Regnal Year of Yoshiyahu (Josiah) – in Spring / Summer of 608 BCE (Julian) – Egyptian Pharaoh Necho II was presumably coming back from Egypt to rejoin the front line while his troops had spent the entire Winter of 609 – 608 BCE (Julian) on the east bank of the Euphrates after having re-conquered Harran earlier that year Spring/Summer 609 BCE (Julian) in the 17th Regnal Year of Nabopolassar in Spring 609 – 608 BCE (Julian).

XX. Instead it is more logical to reason that Yoshiyahu (Josiah) king of Judah went out to confront a smaller Egyptian contingent force (and not the massive Egyptian army going to the front line) including Pharaoh Necho II at the Battle of Megiddo only in the 18th Regnal Year of Nabopolassar in Spring / Summer 608 – 607 BCE (Julian). It would only be logical that Yoshiyahu (Josiah) felt reasonably confident pursuing and attacking a much smaller (perhaps reinforcement) Egyptian force which included the Egyptian Head of State, Pharaoh Necho II, who was rejoining his troops after an unspecified time in Egypt during Winter 609 to Spring/Summer 608 BCE (Julian), i.e. within the 17th Regnal Year of Nabolassar from Spring 609 to 608 BCE (Julian) – indeed the mathematics will have it no other way.

XX. It is entirely natural that Yoshiyahu (Josiah) should have attempted to intercept [Egyptian Pharaoh Necho II] at the strategic pass of Megiddo.[1187] Mitchell argues that it is probable that Yeho-ahaz (Jeho-ahaz / Shalum) had become king in the Summer of 609 BCE (Julian).[1188] [1189]

XX. However, as presented, the only mathematically possible correct year is 608 BCE (Julian), and thus Yehoyakim (Jehoiakim / Elyakim) succeeded Yeho-ahaz (Jeho-ahaz / Shalum) in (or by) the Autumn,[1190] beginning a reign of eleven (11) years [in addition to Year 0 Accession Year] which ended with the fall (i.e. not the destruction) of Jerusalem to Nebuchadrezzar[1191] in 597 BCE (Julian).

6.0. Year 0 Accession Year of Yehoyakim (Jehoiakim) == Spring 608 – 607 BCE
== 18th Regnal Year of Nabopolassar
== YEAR 1 OF SERVITUDE

1187 Mitchell 1991: 391
1188 Mitchell 1991: 392
1189 Date given by Mitchell 1991
1190 i.e. Autumn that begins 21 September (Gregorian) of the same year 608 BCE (Julian), after three (3) months in which Yeho-ahaz (Jeho-ahaz / Shalum) reigned immediately after Yoshiyahu (Josiah) prior to Yehoyakim (Jehoiakim / Elyakim), i.e. Summer – Autumn 608 BCE (Julian)
1191 Mitchell 1991: 392

XX. Pharaoh Necho II removed Yeho-ahaz (Jeho-ahaz / Shalum) three (3) months after his being crowned the last independent king of Judah in Jerusalem. Logically it is reasonable that Pharaoh Necho II deposed Yeho-ahaz (Jeho-ahaz / Shalum) upon his return to Egypt from the front line of Assyria (i.e. after having had killed Yoshiyahu (Josiah) king of Yehudah three (3) months earlier on Pharaoh Necho II's way to Assyria).

XX. Pharaoh punished the land and its people by imposing an obligation to pay tribute of 100 talents of silver, and one (1) talent of gold. In place of Yeho-ahaz (Jeho-ahaz / Shalum), Pharaoh Necho II installed his own choice as vassal-puppet-slave-king of Judah: Elyakim – son of Yoshiyahu (Josiah) and older half-brother of Yeho-ahaz (Jeho-ahaz / Shalum) – whom he renamed to Yehoyakim (Jehoiakim):

33 And Pharaoh Necho imprisoned him[1192] at Riblah, in the land of Hamat, from reigning in Jerusalem, and he gave a punishment upon the land [of] 100 talents of silver, and one talent of gold. 34 And Pharaoh Necho made king Elyakim son of Yoshiyahu in place of[1193] Yoshiyahu his father, and [he] changed his name [to] Yehoyakim, and [he] took Yeho-ahaz, and [he] came to Egypt, and [he] died there.[1194]

XX. The Book of Chronicles confirms the account of the Book of Kings, albeit with fewer details:

4 And Pharaoh king of Egypt made king Elyakim, his [half-]brother,[1195] over Yehudah and Jerusalem, and [he] changed his name [to] Yehoyakim, and Yeho-ahaz – his [half-]brother – Necho took, and [he] brought him to Egypt.[1196]

XX. This fulfilled the prophecy of Jeremiah:

11 Because thus says YHVH to Shalum, son of Yoshiyahu king of Yehudah, the [one who] rules in place of[1197] Yoshiyahu his father: "[He] who [has] left from this place will not return [to] there [any] more. 12 Because in the place where [they] exiled him, there [he] will die, and this land [he] will not see [any] more."[1198]

XX. Therefore it follows that Yehoyakim (Jehoiakim / Elyakim) who succeeded Yeho-ahaz (Jeho-ahaz / Shalum) after three (3) months, did so in (or by) Autumn of 608 (not 609) BCE (Julian). However, even

1192 i.e. Yeho-ahaz (Jeho-ahaz / Shalum)
1193 Literally: "under"
1194 cf. 2 Kings 23:33-34
1195 i.e. The half-brother of Yeho-ahaz (Jeho-ahaz / Shalum)
1196 cf. 2 Chronicles 36:4
1197 Literally: "under"
1198 cf. Jeremiah 22:11-12

if there was an interregnum of a few weeks or months before Yehoyakim (Jehoiakim / Elyakim) acceded to the throne of Judah – and if Yehoyakim (Jehoiakim / Elyakim) wasn't crowned vassal-puppet-slave-king by Pharaoh Necho II until Winter 608 – 607 BCE (Julian) – that year (3153 – 3154 AM) was a leap year that contained 13 months and so there is nothing that conflicts chronologically.[1199] Thus the 1st Regnal Year of Yehoyakim (Jehoiakim / Elyakim) would still (nonetheless) have begun the next 1st day of Nisannu / Nisan in the following Spring 607 BCE (Julian).[1200]

XX. Thus Yeho-ahaz (Jeho-ahaz / Shalum) was the last independent Judean king of Judah in 608 BCE (Julian), and therefore the 70 YEARS OF SERVITUDE are to be reckoned and counted from the deposition of Yeho-ahaz (Jeho-ahaz / Shalum) and/or the Year 0 Accession Year of Yehoyakim (Jehoiakim / Elyakim) king of Judah – which makes both logical and (as will be proven) also mathematical sense as the latter was a vassal-puppet-slave-king installed by the Egyptian Pharaoh Necho II.

7.0. Year 0 Accession Year of Nebuchadnezzar II == Spring 605 – 604 BCE
== 3rd Regnal Year of Yehoyakim (Jehoiakim)
== YEAR 4 OF SERVITUDE

XX. Upon Nebuchadnezzar's accession to the throne on 01 Ululu / Elul, the proleptic Hebrew equivalent of which is Sunday 07 August 605 BCE (Julian).[1201] [1202] his position was sufficiently secure for him to return to the land of Hatti in the Autumn of 605 BCE (Julian) and to join his army for a lengthy campaign which lasted into the 11th month of Shabatu / Shvat. This expedition (the first of many similar incursions) was an unopposed display of the military might of Babylon, no doubt intended to facilitate the collection of the heavy tribute which Nebuchadrezzar brought back to Babylon.[1203]

XX. Although Wiseman (following Parker and Dubberstein) equates Nebuchadnezzar's accession with 07 September 605 BCE (Julian) and this 11th month of Shabatu / Shvat with February 604 BCE (Julian), their proleptic Hebrew equivalents fall one (1) month earlier, in August 605 and January 604 BCE (Julian).[1204]

XX. Both dates fall within a single period of one-(1)-month difference between the two (2) calendars. It

1199 cf. Appendix Table X.10
1200 3154 AM
1201 Sunday 07 August 605 BCE (Julian); Sunday 31 July 605 BCE (Gregorian); 01 Ululu / Elul 3156 AM; Astronomical Julian Day JD_X: 1500665.5; RD_Y: 1152669; RD_X: -220759; Sabbatical Cycle: 6; Metonic Cycle: 2;
1202 cf. Appendix Table X.12, X.14
1203 Wiseman 1956: 27, 47, 69
1204 cf. Appendix Table X.15

began with the Babylonian intercalary Addaru 2 / Adar 2 attested at the end of Nabopolassar's 20th Regnal Year (606 – 605 BCE (Julian)), beginning 14 March 605 BCE (Julian), for which the proleptic Hebrew calendar has no counterpart, and it ended only when the proleptic Hebrew calendar inserted its own intercalary month, Addaru 2 / Adar 2 of 3157 AM beginning 03 March 604 BCE (Julian). In the 1st Regnal Year of Nebuchadnezzar II, beginning Nisan 604 BCE (Julian), the two (2) calendars again agree to within a day.

XX. Wiseman argues that the only possible evidence for a siege of Jerusalem in this year [i.e. 605 – 604 BCE (Julian) being the 3rd Regnal Year of Yehoyakim (Jehoiakim)] is Daniel 1:1. However, he asserts that the exegesis of this verse is difficult and uncertain.[1205] The Babylonian Chronicle seems to preclude any action by Nebuchadrezzar in Judah in 606 BCE (Julian), and moreover makes no [explicit] reference to any siege of Jerusalem in that or any year before 597 BCE (Julian).[1206] However, as will be unfolded below, this is an incorrect understanding of the Babylonian Cuneiform record as well as of the Biblical account.

XX. Wiseman reasons that the Battle of Carchemish most likely took place around May / June of 605 BCE (Julian), and the effects of the Babylonian victory were immediate and far-reaching. At that time, Nebuchadrezzar conquered the whole area of Hatti, the geographical term for the whole of Syria and Palestine. Nebuchadrezzar's victorious troops had marched through Syria and reached Egypt early in the 5th month of Abu / Av of the same year (July / August 605 BCE (Julian)). The battle took place between the commencement of Nabopolassar's 21st (FINAL) Regnal Year in the 1st month of Nisannu / Nisan (March / April 605 BCE (Julian))[1207] and his death on 08 Abu / Av,[1208] i.e. 15 August 605 BCE (Julian) according to the Babylonian calendar, and Friday 15 July 605 BCE (Julian) according to the proleptic Hebrew calendar.[1209] [1210]

XX. Because of the Babylonian intercalary Addaru 2 / Adar 2 attested at the end of Nabopolassar's 20th Regnal Year (606 – 605 BCE (Julian)), the corresponding Babylonian months in Parker and Dubberstein's tables fall one (1) month later until the proleptic Hebrew calendar resynchronizes with the Babylonian calendar when it intercalates its 13th month Addaru 2 / Adar 2 of 3157 AM that began 03 March 604 BCE (Julian).

XX. According to Wiseman's reading of the Old Testament (OT), both the OT and Josephus describe that Nebuchadrezzar took all Syria from the Euphrates to the Egyptian border without entering the hilly terrain of Judah itself.[1211]

1205 Wiseman 1956: 26
1206 Wiseman 1965: 16
1207 cf. Appendix Table X.14
1208 Wiseman 1956: 25
1209 Friday 15 July 605 BCE (Julian); Friday 08 July 605 BCE (Gregorian); 08 Abu / Av 3156 AM; Astronomical Julian Day JD_X: 1500642.5; RD_Y: 1152646; RD_X: -220782; Sabbatical Cycle: 6; Metonic Cycle: 2;
1210 cf. Appendix Table X.11
1211 Wiseman 1956: 26

XX. [However, the Book of Daniel specifically states that in the 3rd Regnal Year of Yehoyakim (Jehoiakim), Nebuchadnezzar came to Jerusalem and placed siege against the city, and the LORD gave the king of Judah into his hand.[1212] Therefore, it is incorrect to assume that Nebuchadnezzar did not enter Judah itself as] the effect on Judah was that Yehoyakim (Jehoiakim), a vassal of Egyptian Pharaoh Necho II, submitted voluntarily to Nebuchadrezzar, and some Jews [including the prophetic seer Daniel, Hananyah, Misha'el, and Azaryah], were taken as captives or hostages to Babylon.[1213]

XX. According to Wiseman, none of the sources implies that Nebuchadrezzar himself moved far south at this time.[1214] However, the Hebrew Bible records that the king of Egypt did not continue to leave Egypt because Nebuchadnezzar took from the "brook" of Egypt until the Euphrates River: all that was to the king of Egypt.[1215]

XX. We can reason that sometime between the date of Nebuchadnezzar's accession to the throne 01 Ululu / Elul 605 BCE (Julian)[1216] and the month of Shabatu / Shevat 604 BCE (Julian)[1217] – when he returned to Babylon with the heavy tribute including captives from the land of Hatti which geographically includes the land of Judah.

XX. Since Daniel is recorded to have interpreted Nebuchadnezzar's dream in his 2nd Regnal Year,[1218] the interpretation that the date mentioned in Daniel 1:1 of the 3rd Regnal Year of Yehoyakim (Jehoiakim) as being Spring 605 – 604 BCE (Julian) turns out to be the most logical and consistent with the literal reading of the Book of Daniel.[1219] It also is mathematically possible and reasonably feasible: assuming Daniel was between the ages of 10-13 years old in 605 BCE (Julian), he would have been between the ages of 95-98 in the 1st Regnal Year of Darius[1220] in 521 – 520 BCE (Julian).

8.0. 1st Regnal Year of Nebuchadnezzar II == Spring 604 – 603 BCE
== 4th Regnal Year of Yehoyakim (Jehoiakim)
== YEAR 5 OF SERVITUDE

XX. British Museum Tablet BM 21946 records that in the Spring of his 1st full year of reign (i.e. beginning Nisannu / Nisan 604 – 603 BCE (Julian))[1221] in the 3rd month of Simannu / Sivan 604 BCE

1212 cf. Daniel 1:1-2
1213 Wiseman 1956: 26
1214 Wiseman 1956: 26
1215 cf. 2 Kings 24:7
1216 cf. Appendix Table X.12, X.14
1217 cf. Appendix Table X.14, X.15
1218 cf. Daniel 2:1
1219 cf. Daniel 1:1
1220 cf. Daniel 9:1
1221 cf. Appendix Table X.16

(Julian),[1222] Nebuchadrezzar called out his army for an expedition to the land of Hatti where for six (6) months they marched about unopposed until the 9th month of Kisilimu / Kislev 604 BCE (Julian).[1223]

XX. During this time the heads of the various small states of Syria and the Levant – which at this time are included in the term (*māt*) *Ḫatti*, i.e. the "(land of) Ḫatti" – were forced to submit to Babylonian rule and to bring in their tribute. The [Babylonian] chronicler claims that "all the kings of Ḫatti" came before Nebuchadrezzar. It is likely that among the kings who submitted at this time were the rulers of Damascus, Tyre, and Sidon, and Yehoyakim (Jehoiakim / Elyakim) of Judah who was to be a faithful vassal[1224] for three (3) years.[1225] [1226]

XX. It must be noted that Parker and Dubberstein's dates cited for the 3rd month of Simannu / Sivan (i.e. 30 May – 28 June 604 BCE (Julian)) agree with the proleptic Hebrew dates to the very day.[1227] This is because the proleptic Hebrew calendar resynchronized with the Babylonian calendar when it intercalated its 13th month Addaru 2 / Adar 2 of 3157 AM that began 03 March 604 BCE (Julian).

XX. In Nebuchadnezzar's 1st Regnal Year neither calendar inserts an intercalary month, so the two (2) calendars remain synchronized throughout the year: the 1st day of each month agrees exactly in seven (7) months and differs by one (1) to three (3) days in the other five (5). Thus a phenomenon continues to be observed that Parker and Dubberstein's dates sometimes agree with the proleptic Hebrew dates to the very day, sometimes differ by a day (or a few days), and sometimes differ by exactly a 29 / 30-day lunar month, the last being due to intercalary months inserted in different years.

XX. In that same 1st Regnal Year of Nebuchadnezzar (Nisannu / Nisan 604 – 603 BCE (Julian)),[1228] in the 9th month of Kisilimu / Kislev of 604 BCE (Julian),[1229] Nebuchadrezzar marched against the Philistine city of Ashkelon, captured it, sacked it, and reduced it to rubble. According to British Museum Tablet BM 21946 Nebuchadnezzar "captured its king and plundered it and carried off [spoil from it . . .], he turned the city into a mound and heaps of ruins and then in the month of Shabatu / Shvat 603 BCE (Julian)[1230] he marched back to Babylon".[1231]

9.0. 2nd Regnal Year of Nebuchadnezzar II == Spring 603 – 602 BCE == 5th Regnal Year of Yehoyakim (Jehoiakim)

1222 cf. Appendix Table X.17
1223 cf. Appendix Table X.18
1224 Literally: English translation of the Hebrew: "slave"
1225 cf. 2 Kings 24:1
1226 Wiseman 1956: 28, 47, 69; Finegan 1998: 255
1227 cf. Appendix Table X.17
1228 cf. Appendix Table X.16
1229 cf. Appendix Table X.18
1230 cf. Appendix Table X.19
1231 Wiseman 1956: 28, 47, 69

== YEAR 6 OF SERVITUDE

XX. Unfortunately, the Babylonian account for the events recorded in the 2nd Regnal Year of Nebuchadnezzar (Nisannu / Nisan 603 – 602 BCE (Julian))[1232] [1233] is broken, and therefore crucial details for our understanding are missing. According to British Museum Tablet BM 21946, in the 2nd month of the year, in the Spring [2nd] month of Ayyaru / Iyar [i.e. April / May] 603 BCE (Julian),[1234] "the king of Akkad (i.e. the Babylonian king) gathered together a powerful army and [marched to the land of Hatti]. [. . . . . .] he threw down, great siege towers he [. . . . . .] from the month of Aiaru / Iyyar until the mon[th of . . .] he marched about unopposed in the land of Hatti."

XX. The remainder of the text for this 2nd Regnal Year of Nebuchadnezzar is – unfortunately – broken. Notably, siege towers and other heavy equipment are mentioned in the text, a fact which implies that the Babylonian troops were in the field for some months and were presumably engaged in the siege of a city (or cities) – though the name of the city (or cities) is lost.[1235]

XX. In this 2nd Regnal Year of Nebuchadnezzar, the 2nd month of Ayyaru / Iyar precedes any intercalation, and the proleptic Hebrew and Babylonian calendars remain synchronized. However, later in this same year, the Babylonians intercalated a 2nd Ululu / Elul (Ululu 2 / Elul 2) that began 15 September 603 BCE (Julian),[1236] which the proleptic Hebrew calendar does not.

XX. Because of the Babylonian intercalary 6th month of Ululu 2 / Elul 2 attested in Nebuchadnezzar's 2nd Regnal Year, the corresponding Babylonian months in Parker and Dubberstein's tables fall one-(1)-month later than the proleptic Hebrew calendar until it resynchronizes with the Babylonian calendar when it intercalates its 13th month Addaru 2 / Adar 2 of 3160 AM beginning 28 February 601 BCE (Julian)

XX. Daniel tells us that in the 2nd Regnal Year of Nebuchadnezzar, the king of Babylon has a disturbing dream that Daniel correctly interprets.[1237]

XX. Jeremiah tell us that in the 5th Regnal Year of Yehoyakim (Jehoiakim / Elyakim), i.e. the 2nd Regnal Year of Nebuchadnezzar,[1238] Nisannu / Nisan 603 – 602 BCE, in the 9th month (Kisilimu / Kislev == November / December) 603 BCE (Julian),[1239] a fast is called for Jerusalem,[1240] king Yehoyakim (Jehoiakim / Elyakim) is in his winter house, and the fireplace before him is blazing as the words of

1232 cf. Appendix Table X.20
1233 cf. Daniel 2:1
1234 cf. Appendix Table X.21
1235 Wiseman 1956: 28, 29, 70, 71
1236 Parker and Dubberstein 1971: 4, 27
1237 cf. Daniel 2
1238 cf. Appendix Table X.20
1239 cf. Appendix Table X.22
1240 cf. Jeremiah 36:9

Jeremiah are being read to the king by his scribes:[1241]

9 And it was in the 5th year of Yehoyakim son of Yoshiyahu king of Yehudah in the 9th month, [they] called a fast before YHVH – all the people in Jerusalem and all the people who came from the cities of Yehudah – in Jerusalem.[1242]

22 And the King [was] sitting [in] the Winter house in the 9th month, and the fireplace before him [was] blazing. 23 And it was at Yehudi's reading three paragraphs,[1243] *and four [that he]*[1244] *tore it with the scribe's razor and [he] cast [it] to the fire that [was] to [within] the fireplace, until the end of the entire Megillah on the fire that [was] on the fireplace.*[1245]

XX. Especially given the Babylonian account for the 2nd Regnal Year of Nebuchadnezzar which describes siege towers and heavy equipment, it is reasonable to argue that the reason why the people of the cities of Judah had fled to the refuge of Jerusalem (and together called a fast with all the people of Jerusalem was because of the invading Babylonian army) – which had destroyed the Philistine city of Ashkelon in the previous 1st Regnal Year of Nebuchadnezzar in the 9th month Kisilimu / Kislev (i.e. November / December) 604 BCE (Julian)[1246] – it would seem reasonable that upon Nebuchadnezzar's invasion of Judah again in his 2nd Regnal Year, that all the people of Judah who could do so fled to the refuge of the walled capital city of Jerusalem.

XX. The account in the Book of Chronicles elaborates and makes clear that Yehoyakim (Jehoiakim / Elyakim) was taken to Babylon by Nebuchadnezzar to be a slave:

6 Upon him ascended Nebuchadnezzar king of Babylon, and [he] bound him in fetters [in order] to march him to Babylon. 7 And from the vessels of the House of YHVH Nebuchadnezzar brought to Babylon, and [he] gave them [to be] in his hall [of his palace] in Babylon.[1247]

XX. The account in the Book of Kings confirms this:

1 In his [Yehoyakim's] days, Nebuchadnezzar king of Babylon ascended, and Yehoyakim was a slave to him three years, and [he Yehoyakim] returned [from Babylon], and [he] rebelled against him.[1248]

XX. Thus we can reason that in the 5th Regnal Year of Yehoyakim (Jehoiakim), the king of Babylon,

1241 cf. Jeremiah 36:22-23
1242 cf. Jeremiah 36:9
1243 Literally: "doors"
1244 i.e. Yehoyakim (Jehoiakim / Elyakim)
1245 cf. Jeremiah 36:22-23
1246 cf. Appendix Table X.18
1247 cf. 2 Chronicles 36:6-7
1248 cf. 2 Kings 24:1

Nebuchadnezzar, came to Jerusalem, placed a siege upon the city and/or other Judean cities (and thus surrounded the city with his massive army), and king Yehoyakim (Jehoiakim / Elyakim) surrendered to Nebuchadnezzar. He was marched to Babylon bound in fetters,[1249] where he was slave to Nebuchadnezzar for three (3) years, and then returned to Jerusalem by his (i.e. Yehoyakim's (Jehoiakim's / Elyakim's)) 8th Regnal Year (i.e. the 5th Regnal Year of Nebuchadnezzar), after which he rebelled against Nebuchadnezzar.

XX. From available information, the exact length of the time Yehoyakim (Jehoiakim / Elyakim) spent in Babylon is unknown. Assuming: 3-4 months march[1250] back to Babylon as a captured slave, 2.33 – 2.5 years of slavery in Babylon, then another 3-4 months march[1251] back to Jerusalem as a possessed slave before his release.

10.0. 3rd Regnal Year of Nebuchadnezzar II == Spring 602 – 601 BCE
== 6th Regnal Year of Yehoyakim (Jehoiakim)
== YEAR 7 OF SERVITUDE

XX. Returning to the events of the Babylonian Chronicle, British Museum Tablet BM 21946 records, "In the third year[1252] [1253] [. . . . . .] the king of Akkad (i.e. king of Babylon) mustered his troops and [marched] to the land of Hatti and brought (back) much [spoil] from the land of Hatti to the land of Akkad". By this time these annual expeditions and campaigns in Syria seem to have had as their primary aim the maintenance of political pressure with the economic gains resulting from the collection of dues.[1254]

XX. As this record gives no month, the campaign of the 3rd Regnal Year of Nebuchadnezzar (Nisannu / Nisan 602 – 601 BCE (Julian)) cannot be dated more precisely. Calendrically, however, this year is significant: the proleptic Hebrew calendar resynchronizes with the Babylonian calendar when it intercalates its 13th month Addaru 2 / Adar 2 of 3160 AM that began 28 February 601 BCE (Julian) at the end of this Babylonian / Hebrew year for kings.

11.0. 4th Regnal Year of Nebuchadnezzar II == Spring 601 – 600 BCE
== 7th Regnal Year of Yehoyakim (Jehoiakim)
== YEAR 8 OF SERVITUDE

1249 i.e. Thus beginning Yehoyakim (Jehoiakim / Elyakim) king of Judah's 1st year of slavery in Babylon
1250 cf. Ezra 7:7-9
1251 cf. Ezra 7:7-9
1252 cf. Appendix Table X.23
1253 i.e. Yehoyakim (Jehoiakim) king of Judah's 2nd year of slavery in Babylon
1254 Wiseman 1956: 29, 47, 70, 71

XX. Following the precedent of the previous five (5) years, the Babylonian army was called out again in the 4th Regnal Year of Nebuchadnezzar (Nisannu / Nisan 601 – 600 BCE (Julian))[1255] [1256] and sent to the land of Hatti where it once more moved about unopposed. Its presence helped the district-governors to collect and dispatch to Babylon the annual tribute which it was their duty to exact. Later in the year Nebuchadrezzar took over command of the army in person and marched to Egypt. On hearing of his approach, the Egyptian king Necho II called out his army and both forces met in open battle.[1257]

XX. With characteristic frankness, the chronicler states that both sides suffered heavy losses, resulting in an inconclusive open battle. Nebuchadrezzar and his army withdrew immediately afterwards and returned to Babylon. Other than knowing that the battle took place after the month of Kisilimu / Kislev (November / December) 601 BCE (Julian),[1258] no date or place for this clash between the major powers is given in this text, and there are no references in Egyptian sources with which the Babylonian record may be compared.[1259]

XX. In the 4th Regnal Year of Nebuchadnezzar (Nisannu / Nisan 601 – 600 BCE (Julian)), neither calendar intercalates a month. Having resynchronized at the end of the previous year, when the proleptic Hebrew calendar intercalated its 13th month Addaru 2 / Adar 2 of 3160 AM beginning 28 February 601 BCE (Julian), the proleptic Hebrew and Babylonian calendars remain synchronized throughout this year, and the 9th month of Kisilimu / Kislev falls in November / December 601 BCE (Julian) in both.

12.0. 5th Regnal Year of Nebuchadnezzar II == Spring 600 – 599 BCE == 8th Regnal Year of Yehoyakim (Jehoiakim) == YEAR 9 OF SERVITUDE

XX. It can be judged from the entry in the chronicle (BM 21946) for the following year (i.e. 5th Regnal Year, Nisannu / Nisan 600 – 599 BCE (Julian))[1260] that the outcome of the battle [in the previous 4th Regnal Year] with the Egyptians was a severe set-back for the Babylonians. The fact that Nebuchadrezzar stayed in Babylon and was forced to re-equip his army with chariots and horses may indicate that it was deficiency in these which had been a primary factor of the Babylonian defeat.[1261]

XX. It is thus here (after having had served the king of Babylon as a slave in Babylon in the 2nd, 3rd, 4th Regnal Years of Nebuchadnezzar – i.e. the 5th, 6th, and 7th Regnal Years of Yehoyakim (Jehoiakim /

1255 cf. Appendix Table X.24
1256 i.e. Yehoyakim (Jehoiakim) king of Judah's 3rd year of slavery in Babylon
1257 Wiseman 1956: 29, 30, 47, 70, 71
1258 cf. Appendix Table X.25
1259 Wiseman 1956: 29, 30, 47, 70, 71
1260 cf. Appendix Table X.26
1261 Wiseman 1956: 29, 30

Elyakim) – and then having returned to Jerusalem) that Yehoyakim (Jehoiakim / Elyakim) rebelled against Nebuchadnezzar beginning in the 8th Regnal Year of Yehoyakim (Jehoiakim / Elyakim) while the Babylonian king remained in Babylon to rebuild and re-amass his army.[1262]

XX. Recalling the account in the Book of Kings:

1 In his [Yehoyakim's] days, Nebuchadnezzar king of Babylon ascended, and Yehoyakim was a slave to him three (3) years, and [he Yehoyakim] returned [from Babylon], and [he] rebelled against him.[1263]

XX. The Book of Kings continues:

2 And YHVH sent against him[1264] *the battalions of [soldiers of] Chaldeans, and the battalions of [soldiers from] Aram, and the battalions of [soldiers from] Mo'av, and the battalions [soldiers of] of the sons of Ammon, and [he] sent them against Yehudah to annihilate him*[1265] *as the word of YHVH that [He] spoke through the hand[s] of His servants the Prophets.*[1266] [1267]

XX. The 5th Regnal Year of Nebuchadnezzar (Nisannu / Nisan 600 – 599 BCE (Julian)) begins with the proleptic Hebrew and Babylonian calendars still synchronized. Later in the year, however, the Babylonians intercalated a 2nd Ululu / Elul (Ululu 2 / Elul 2) that began 11 September 600 BCE (Julian),[1268] which the proleptic Hebrew calendar does not.

XX. Because of the Babylonian intercalary Ululu 2 / Elul 2 attested in Nebuchadnezzar's 5th Regnal Year, the corresponding Babylonian months in Parker and Dubberstein's tables fall one (1) month later until the proleptic Hebrew calendar resynchronizes with the Babylonian calendar when it intercalates its 13th month Addaru 2 / Adar 2 of 3162 AM that began 07 March 599 BCE (Julian) at the end of this Babylonian / Hebrew year.

13.0. 6th Regnal Year of Nebuchadnezzar II == Spring 599 – 598 BCE == 9th Regnal Year of Yehoyakim (Jehoiakim) == YEAR 10 OF SERVITUDE

1262 cf. Wiseman 1956: 29, 30, 31, 47, 70, 71
1263 cf. 2 Kings 24:1
1264 i.e. Yehoyakim (Jehoiakim)
1265 i.e. Yehoyakim (Jehoiakim)
1266 cf. 2 Kings 24:2
1267 cf. Wiseman 1956: 29, 30, 31, 47, 70, 71
1268 Parker and Dubberstein 1971: 5, 27 (the 1956 edition misprints 12 September, which would give a 28-day month)

XX. In his 6th Regnal Year (Nisannu / Nisan 599 – 598 BCE (Julian)),[1269] in the month of Kisilimu / Kislev (i.e. November / December 599 BCE (Julian))[1270] Nebuchadrezzar called up his army and marched to the land of Hatti. Then from one of the cities under his control (Hamath, Riblah, or possibly Kadesh), he sent raiding parties into the adjacent desert, thus collecting much spoil from the Arab tribes.[1271]

XX. Nebuchadrezzar, therefore, had a two-(2)-fold aim in undertaking the campaign of his 6th Regnal Year:[1272]

1.) to regain control of the western Syrian desert as part of the Assyrian empire that he now sought to inherit;

2.) to guard those areas such as Hamath and Riblah that he already held, and from which he could thrust south-westwards toward Egypt.

XX. It may be significant that the chronicler mentions the personal return of the king to Babylon after this campaign in the month of Addaru / Adar (i.e. February / March 598 BCE),[1273] but omits any reference to the homeward march of the troops.[1274]

XX. In the 6th Regnal Year of Nebuchadnezzar (Nisannu / Nisan 599 – 598 BCE (Julian)), neither calendar intercalates a month. Having resynchronized at the end of the previous year, when the proleptic Hebrew calendar intercalated its 13th month Addaru 2 / Adar 2 of 3162 AM beginning 07 March 599 BCE (Julian), the proleptic Hebrew and Babylonian calendars remain synchronized throughout this year, and thus the 9th month of Kisilimu / Kislev (November / December 599 BCE (Julian)) and the 12th month of Addaru / Adar (February / March 598 BCE (Julian)) fall in the same Julian months in both calendars.

14.0. 7th Regnal Year of Nebuchadnezzar II == Spring 598 – 597 BCE == 10th Regnal Year of Yehoyakim (Jehoiakim) == YEAR 11 OF SERVITUDE

XX. The 7th Regnal Year of Nebuchadnezzar began 01 Nisannu / Nisan 598 BCE (Julian)[1275] which according to Finegan and Wiseman (who follow Parker and Dubberstein) works out to be 27 March

1269 cf. Appendix Table X.27
1270 cf. Appendix Table X.28
1271 Wiseman 1956: 31, 32, 48, 70, 71
1272 Wiseman 1956: 31, 32, 48, 70, 71
1273 cf. Appendix Table X.29
1274 Wiseman 1956: 31, 32, 48, 70, 71
1275 cf. Appendix Table X.30

598 BCE (Julian). However, the proleptic Hebrew equivalent is Saturday 25 March 598 BCE (Julian),[1276] [1277] i.e. a difference of only two (2) days.

XX. According to Parker and Dubberstein, the 9th month of Kisilimu / Kislev began on 18 December 598 BCE (Julian). However, the proleptic Hebrew equivalent is Friday 17 November 598 BCE (Julian).[1278] [1279] According to them, the 2nd day of the 12th month Addaru / Adar was 16 March 597 BCE (Julian). However, the proleptic Hebrew equivalent date is Thursday 15 February 597 BCE (Julian).[1280] [1281]

XX. Because of the Babylonian intercalary second 6th month Ululu 2 / Elul 2 beginning 21 September 598 BCE (Julian) attested in Nebuchadnezzar's 7th Regnal Year,[1282] the corresponding Babylonian months in Parker and Dubberstein's tables fall one (1) month later until the proleptic Hebrew calendar resynchronizes with the Babylonian calendar when it intercalates its 13th month Addaru 2 / Adar 2 of 3165 AM that began Wednesday 05 March 596 BCE (Julian).

XX. British Museum Tablet BM 21946 provides the following record for the 7th Regnal Year of Nebuchadnezzar:[1283]

In the 7th year, the month of Kisilimu, the king of Akkad mustered his troops, marched into the land of Hatti, and encamped against[1284] the city of Judah, and on the 2nd day of the month of Addaru he seized the city and captured the king. He appointed there a king of his own choice,[1285] received its heavy tribute, and sent them to Babylon.[1286]

XX. The "king of Akkad" is Nebuchadnezzar, the "city of Judah" must be Jerusalem, the captured king – according to Finegan and most scholars – is Yehoyachin (Jehoiachin), and the newly chosen king must be Tzidkiyahu (Zedekiah / Matanyah), so this is unmistakably the Babylonian record of the fall of Jerusalem to Nebuchadnezzar in 597 BCE (Julian), corresponding to the biblical accounts.[1287] [1288]

1276 Saturday 25 March 598 BCE (Julian); Saturday 19 March 598 BCE (Gregorian); 01 Nisannu / Nisan 3163 AM; Astronomical Julian Day JD_X: 1503086.5; RD_Y: 1155090; RD_X: -218338; Sabbatical Cycle: 6; Metonic Cycle: 9;
1277 cf. Appendix Table X.30
1278 Friday 17 November 598 BCE (Julian); Friday 11 November 598 BCE (Gregorian); 01 Kisilimu / Kislev 3164 AM; Astronomical Julian Day JD_X: 1503323.5; RD_Y: 1155327; RD_X: -218101; Sabbatical Cycle: 7; Metonic Cycle: 10;
1279 cf. Appendix Table X.31
1280 Thursday 15 February 597 BCE (Julian); Thursday 09 February 597 BCE (Gregorian); 02 Addaru / Adar 3164 AM; Astronomical Julian Day JD_X: 1503413.5; RD_Y: 1155417; RD_X: -218011; Sabbatical Cycle: 7; Metonic Cycle: 10;
1281 cf. Appendix Table X.32
1282 Parker and Dubberstein 1971: 5, 27
1283 Finegan 1998: 256; Wiseman 1956: 32, 72, 73
1284 i.e. besieged
1285 Literally: "his own heart"
1286 Finegan 1998: 256; Wiseman 1956: 32, 72, 73
1287 Finegan 1998: 256
1288 cf. 2 Kings 24; 2 Chronicles 36

XX. The Hebrew Scriptures do not give us details how Yehoyakim (Jehoiachim/Elyakim) died, but as a result of the burning of Jeremiah's scroll by Yehoyakim (Jehoiakim / Elyakim),[1289] Jeremiah is instructed to prepare another scroll to replace the one that was burned by Yehoyakim (Jehoiakim / Elyakim), and prophesies that Yehoyakim (Jehoiakim / Elyakim) would not have someone to sit on the throne of David, and that his body would be cast to the heat of day, and the ice of night.[1290] Ice – of course – is plausible and possible for Jerusalem mostly in winter months, i.e. before the Vernal Equinox, circa 20 / 21 March 597 BCE (Gregorian).

15.0. 8th Regnal Year of Nebuchadnezzar II == Spring 597 – 596 BCE
== 11th (FINAL) Regnal Year of Yehoyakim (Jehoiakim)
== Year 0 Accession Year of Tzidkiyahu (Zedekiah)
== YEAR 12 OF SERVITUDE
== YEAR 1 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. In his 8th Regnal Year (Nisannu / Nisan 597 – 596 BCE (Julian),[1291] having returned to Babylon from Judah arriving with the exiles from Judah [which likely set out 10 Nisannu / Nisan 597 BCE (Julian),[1292] see below] and heavy tribute, Nebuchadrezzar did not venture so far afield as in the previous year's campaign against Judah. The broken and brief entry in the Babylonian Chronicle makes no mention of any use of the army, but simply states that in the 10th month of Tebetu / Tebet 597 – 596 BCE (Julian) the king went to the land of Hatti, traveled as far as Carchemish, and then returned to Babylonia the next month.[1293]

XX. Finegan and Wiseman (following Parker and Dubberstein) give the following Julian equivalents for 10 Nisannu / Nisan as well as for the later 10th month of Tebetu/Tebet 597 – 596 BCE (Julian).

- 10 Nisannu / Nisan 597 BCE == 22 April 597 BCE (Julian)[1294] [1295] [1296]

- Tebetu / Tebet 597 – 596 BCE == c. January 596 BCE (Julian)[1297] [1298] [1299]

1289 cf. Jeremiah 36:22-23
1290 cf. Jeremiah 36:30-31
1291 cf. Appendix Table X.33
1292 cf. Appendix Table X.34
1293 Wiseman 1956: 35, 48, 72, 73
1294 Saturday 23 March 597 BCE (Julian); Saturday 17 March 597 BCE (Gregorian); 10 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503450.5; RD_Y: 1155454; RD_X: -217974; Sabbatical Cycle: 7; Metonic Cycle: 10;
1295 cf. Appendix Table X.34
1296 Finegan 1998: 257
1297 December 597 – January 596 BCE (Julian); November – December 597 BCE (Gregorian); 10th Month of Tebetu / Tevet 3165 AM; Sabbatical Cycle: 1; Metonic Cycle: 11;
1298 cf. Appendix Table X.35
1299 Wiseman 1956: 35

XX. As is evident, the Julian dates provided by Parker and Dubberstein through Finegan and Wiseman fall exactly one (1) lunar month later than their proleptic Hebrew equivalents: 10 Nisannu / Nisan 597 BCE is Saturday 23 March 597 BCE (Julian), thirty (30) days before 22 April BCE (Julian), and the 10th month of Tebetu / Tevet runs from Friday 06 December 597 BCE (Julian) to Friday 03 January 596 BCE (Julian)

XX. Because of the Babylonian intercalary second 6th month Ululu 2 / Elul 2 beginning 21 September 598 BCE (Julian) attested in Nebuchadnezzar's 7th Regnal Year, the corresponding Babylonian months in Parker and Dubberstein's tables fall one (1) month later until the proleptic Hebrew calendar resynchronizes with the Babylonian calendar when it intercalates its 13th month Addaru 2 / Adar 2 of 3165 AM beginning 05 March 596 BCE (Julian) at the end of this Babylonian / Hebrew year.

XX. The purpose of this journey [in the 8th Regnal Year of Nebuchadnezzar] is now lost to us, but it was probably to receive homage and to collect tribute from those Syrian vassals who formerly would not have made these payments without the presence of the Babylonian army to enforce them. If this is so, Wiseman reasons, then it would show that the Babylonian hold over these territories was now firm and the administration settled.[1300]

XX. Finegan's argument placing both the end of the reign of Yehoyachin (Jehoiachin) as well as the beginning of the reign of Tzidkiyahu (Zedekiah) on the same day is the most logical and plausible explanation. Thus the day in which YEAR ONE (1) OF THE EXILE OF YEHOYACHIN (JEHOIACHIN) began was likely the 10th day of Nisannu / Nisan 597 BCE (Julian), and it was the last day of Yehoyachin's (Jehoiachin's) reign of three (3) months and ten (10) days while simultaneously being the 1st day of the 1st month of the Year 0 Accession Year of Tzidkiyahu (Zedekiah).[1301] [1302]

XX. Thus counting backwards three (3) months and ten (10) days means that Yehoyachin (Jehoiachin) acceded to the co-regency with his father Yehoyakim (Jehoiakim) on Sunday 17 December 598 BCE (Julian) in the proleptic Hebrew calendar.[1303]

XX. Therefore according to the mathematics as recorded in the Hebrew Scriptures (and according to the proleptic Hebrew calendar), Yehoyakim (Jehoiakim) survived until at least the 1st day of Nisannu / Nisan in his 11th Regnal Year (i.e. Spring 598 – 597 BCE (Julian)) and then was executed. His body was then cast to the heat of the day and the ice of night.[1304] Although rare in Jerusalem, especially at such a time in mid-to-late March (Julian) or early-to-mid March 597 BCE (Gregorian),[1305] it is possible

1300 Wiseman 1956: 35-36

1301 In other words: the 1st day of the 1st month of the Year 0 Accession Year of Tzidkiyahu (Zedekiah) == the 1st day of the 1st month of the YEAR ONE (1) OF THE EXILE OF YEHOYACHIN (JEHOIACHIN).

1302 Finegan 1998: 257

1303 Sunday 17 December 598 BCE (Julian); Sunday 11 December 598 BCE (Gregorian); 01 Tebetu / Tevet 3164 AM; Astronomical Julian Day JD_X: 1503353.5; RD_Y: 1155357; RD_X: -218071; Sabbatical Cycle: 7; Metonic Cycle: 10;

1304 cf. Jeremiah 36:30-31

1305 cf. Appendix Table X.34

(and indeed feasible) for the weather to have been unseasonably cold at this time including snow with freezing temperatures at night enough to produce ice.

XX. Indeed, comparing the equivalent proleptic Julian (and Gregorian) date for 01 Nisannu / Nisan for 597 BCE (Julian) confirms that this date would have fallen on Thursday 14 March 597 BCE (Julian),[1306] [1307] i.e. immediately before the Vernal Equinox (i.e. 20 / 21 March (Gregorian)) of approximately Tuesday 26 March[1308] [1309] or Wednesday 27 March 597 BCE (Julian).[1310] [1311] Therefore a late Winter freeze with the heat of the Sun during the day and the accumulated ice during the night, as described in the Biblical account, is entirely plausible.

XX. It is noteworthy that the Passover for this year occurs immediately after the Vernal Equinox. Furthermore, this would be a Passover that they would not be celebrating since they would have begun their march into exile and captivity on Sabbath 10 Nisannu / Nisan 3164 AM, i.e. Saturday 23 March 597 BCE (Julian).[1312] [1313]

XX. We must recall the system of Year 0 Accession Years for kings in most of the ancient Near East. If we reason that (for our purposes of calculation) that the 1st official day of Tzidkiyahu (Zedekiah / Matanyah) as king of Judah began on the same day that the exiles began their march to Babylon (i.e. 10 Nisannu / Nisan 597 BCE (Julian)),[1314] then since he had not acceded to the throne before 01 Nisannu / Nisan 597 BCE (Julian), his time of accession to the throne from 10 Nisannu / Nisan 597 BCE (Julian) through 01 Nisannu / Nisan 596 BCE (Julian) would have been considered his Year 0 Accession Year to the throne of Judah, and thus not yet his 1st official Regnal Year.

XX. However, it should be noted that the calendar used in Judah in 597 BCE (Julian) is not documented, and the sequence of events after the fall of Jerusalem depends on it. Nevertheless, from the perspective of the biblical account, we can reason that it was according to the Hebrew calendar.

XX. The Babylonian Chronicle dates the capture of the city to 02 Addaru / Adar in the 7th Regnal Year of Nebuchadnezzar, which in Parker and Dubberstein's tables falls on 16 March 597 BCE (Julian), owing to the Babylonian intercalary Ululu 2 / Elul 2 earlier in that year.

1306 Thursday 14 March 597 BCE (Julian); Thursday 08 March 597 BCE (Gregorian); 01 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503441.5; RD_Y: 1155445; RD_X: -217983; Sabbatical Cycle: 7; Metonic Cycle: 10;
1307 cf. Appendix Table X.34
1308 Tuesday 26 March 597 BCE (Julian); Tuesday 20 March 597 BCE (Gregorian); 13 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503453.5; RD_Y: 1155457; RD_X: -217971; Sabbatical Cycle: 7; Metonic Cycle: 10;
1309 cf. Appendix Table X.34
1310 Wednesday 27 March 597 BCE (Julian); Wednesday 21 March 597 BCE (Gregorian); 14 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503454.5; RD_Y: 1155458; RD_X: -217970; Sabbatical Cycle: 7; Metonic Cycle: 10;
1311 cf. Appendix Table X.34
1312 Saturday 23 March 597 BCE (Julian); Saturday 17 March 597 BCE (Gregorian); 10 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503450.5; RD_Y: 1155454; RD_X: -217974; Sabbatical Cycle: 7; Metonic Cycle: 10;
1313 cf. Appendix Table X.34
1314 cf. Appendix Table X.34

XX. If the months in Judah ran with the proleptic Hebrew calendar, this day was already 03 Nisannu / Nisan, so that Yehoyakim (Jehoiakim) had begun his 11th Regnal Year before the city fell, and the capture took place in the 8th Regnal Year of Nebuchadnezzar by Nisannu / Nisan reckoning in agreement with 2 Kings 24:12. The march into exile and the accession of Tzidkiyahu (Zedekiah) followed only one (1) week later on 10 Nisannu / Nisan (Saturday 23 March 597 BCE (Julian)).

XX. If, however, the months in Judah ran with the Babylonian calendar, the city fell on 02 Addaru / Adar before the beginning of Yehoyakim's (Jehoiakim's) 11th Regnal Year, so that he must have been kept alive after the capture until at least 01 Nisannu / Nisan (13 April 597 BCE (Julian)). If so, then the march into exile and the accession of Tzidkiyahu (Zedekiah) on 10 Nisannu / Nisan fell on 22 April 597 BCE (Julian), some five (5) weeks after the capture, allowing time to organize the deportation. The proleptic Hebrew equivalent of this date is 10 Ayyaru / Iyar.

XX. Furthermore, if intercalation in Judah was still determined by observation rather than by fixed rule, either sequence is possible. In each case the proleptic Hebrew calendar makes these alternatives visible, dated exactly, so that each can be weighed on its merits, and thoughtfully considered.

XX. Thus if we summarize what can be known so far:

- 10 Nisannu / Nisan 597 BCE (Julian) == 1st day of the YEAR ONE (1) OF THE EXILE OF YEHOYACHIN (JEHOIACHIN) 3164 AM

- 8th Regnal Year of Nebuchadnezzar == Nisannu / Nisan 597 – 596 BCE (Julian) 3164 – 3165 AM

XX. Applying logic, we can reason that:

- YEAR ONE (1) OF THE EXILE OF YEHOYACHIN (JEHOIACHIN) 3164 – 3165 AM == 8th Regnal Year of Nebuchadnezzar 3164 – 3165 AM

- YEAR ONE (1) OF THE EXILE OF YEHOYACHIN (JEHOIACHIN) 3164 – 3165 AM == Year 0 Accession Year of Tzidkiyahu (Zedekiah) 3164 – 3165 AM

XX. Therefore according to this refined reasoning:

- Year 0 Accession Year of Tzidkiyahu (Zedekiah) 3164 – 3165 AM == 8th Regnal Year of Nebuchadnezzar 3164 – 3165 AM

- 1st Regnal Year of Tzidkiyahu (Zedekiah) 3165 – 3166 AM == 9th Regnal Year of

Nebuchadnezzar 3165 – 3166 AM

XX. As noted above, the discrepancies between Parker and Dubberstein's dates and the proleptic Hebrew dates provided here range from agreement to the exact day, to a difference of a day or a few, to exactly the length of a 29 / 30-day lunar month. It is precisely these measurable discrepancies which provide the key to deciphering ancient observational Babylonian astronomy (i.e. as they recorded it) – particularly in relation to which months they intercalated, whether they chose Ululu (Hebrew: Elul) or Addaru (Hebrew: Adar) for a particular year – until the later advent of precise, calculated, predictable, mathematical intercalation with the 19-year Metonic cycle

XX. It is evident that the transition from irregular intercalation toward that fixed cycle appears to have begun in the period in which Daniel is recorded to have lived in Babylon and Persia, 605 – 521 BCE (Julian). The intercalations "remained irregular until the sixth century BCE," and in the decades before the 10th Regnal Year of Xerxes I (476 BCE (Julian)) they form "partly repeating patterns," although the 19-year cycle itself is not attested before that year.[1315] This period of transition coincides with the time of:

XX. Neo-Babylonian kings:

Nebuchadnezzar II (604 – 562 BCE)

Amel-Marduk (561 – 559 BCE) == Hebrew: Evil-Merodach

Neriglissar (559 – 556 BCE) == Hebrew: Nergal Sar Etzer (i.e. the Rab Mag of Jeremiah 39)

Labashi-Marduk (556 BCE)

Nabonidus (555 – 539 BCE) with son Belshazzar as Co-Regent from the 3rd to 13th Regnal Year – cf. Daniel

XX. Persian kings (over Babylon):

Cyrus II (538 – 530 BCE) == Hebrew: Koresh – cf. Ezra, Chronicles, Daniel

Cambyses II (529 – 522 BCE)

Bardia/Smerdis (522 BCE)

Darius (521 – 486 BCE) == Hebrew: Daryavesh – cf. Ezra, Haggai, Zechariah, Daniel

1315 Ossendrijver 2018a: 139, 148-149

Xerxes I (485 – 465 BCE) == Hebrew: Ahashverosh (Greek / English: Ahasuerus) – cf. Esther, Ezra

Artaxerxes I (464 – 424 BCE) == Hebrew: Artachshasta – cf. Ezra, Nehemiah

XX. According to Finegan, Wiseman, and other scholars who follow Parker and Dubberstein, the 2nd day of Addaru / Adar (12th month) was Saturday 16 March 597 BCE (Julian).[1316] However, as is evident from the data, the proleptic Hebrew date for 02 Addaru / Adar is Thursday 15 February 597 BCE (Julian).[1317] [1318] Finegan notes that this date of 02 Addaru / Adar in the 7th Regnal Year of Nebuchadnezzar is the most exact information to come from cuneiform records for an event [that is also] recorded in the Bible, and gives us a precise day for the fall of Jerusalem. According to Finegan and most other scholars, this is also the same exact day for the capture of Yehoyachin (Jehoiachin).[1319]

XX. From the mathematical evidence, reasoning, and logic (as recorded in the Hebrew Scriptures), Yehoyakim (Jehoiakim) had to have survived until at least 01 Nisannu / Nisan of 597 BCE (Julian) because it is recorded that he ruled for 11 years, and this would be the date of the beginning of his 11th Regnal Year.[1320] [1321] By the Hebrew calendar, 02 Addaru / Adar (Babylonian reckoning) Saturday 16 March 597,[1322] [1323] on which Jerusalem fell was already Saturday 03 Nisannu / Nisan (proleptic Hebrew reckoning) 597 BCE (Julian), two (2) days after the 11th Regnal Year of Yehoyakim (Jehoiakim) had begun.

XX. Therefore, it is evident that Nebuchadnezzar captured Yehoyakim (Jehoiakim) on or immediately after the fall of Jerusalem on 02 Addaru / Adar and subsequently killed him. The Babylonian Chronicle states only that Nebuchadnezzar seized the city on 02 Addaru / Adar and captured "the king" without naming him. Since by the proleptic Hebrew reckoning Yehoyakim (Jehoiakim) was still reigning in his 11th Regnal Year when the city fell, the king captured on that day was most definitively Yehoyakim (Jehoiakim), while his son Yehoyachin (Jehoiachin) was called out to Nebuchadnezzar only afterwards, as the scriptures record.

XX. At some point after the fall of Jerusalem, Nebuchadnezzar learned of Yehoyakim's (Jehoiakim's)

1316 Wiseman specifies [the day beginning from evening to evening] 15/16 March 597 BCE – Wiseman 1956: 33; Finegan 1998: 256.

1317 Thursday 15 February 597 BCE (Julian); Thursday 09 February 597 BCE (Gregorian); 02 Addaru / Adar 3164 AM; Astronomical Julian Day JD_X: 1503413.5; RD_Y: 1155417; RD_X: -218011; Sabbatical Cycle: 7; Metonic Cycle: 10;

1318 cf. Appendix Table X.32

1319 Finegan 1998: 256

1320 Thursday 14 March 597 BCE (Julian); Thursday 08 March 597 BCE (Gregorian); 01 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503441.5; RD_Y: 1155445; RD_X: -217983; Sabbatical Cycle: 7; Metonic Cycle: 10;

1321 cf. Appendix Table X.34

1322 Thursday 15 February 597 BCE (Julian); Thursday 09 February 597 BCE (Gregorian); 02 Addaru / Adar 3164 AM; Astronomical Julian Day JD_X: 1503413.5; RD_Y: 1155417; RD_X: -218011; Sabbatical Cycle: 7; Metonic Cycle: 10;

1323 cf. Appendix Table X.32

co-regent son, Yehoyachin (Jehoiachin), and thus subsequently called this co-regent out to him as recorded in the scriptures.[1324] Then Nebuchadnezzar would have installed his choice for vassal king, Tzidkiyahu (Zedekiah). Mathematically, all of this could have happened only between 01 to 10 Nisan 597 BCE (Julian) according to the proleptic Hebrew calendar, i.e. between 16 and 23 March 597 BCE (Julian).

XX. Realizing that the month of Addaru / Adar is the 12th and final month (except for leap years) of both the ancient Babylonian and Hebrew calendars, and that the month following Addaru / Adar (i.e. Nisannu / Nisan) is the 1st month of a New Year (for kings), we will recall the passage of the capture of Yehoyachin (Jehoiachin) in 2 Chronicles 36:10:

10 And at the return[1325] of the year, the king Nebuchadnezzar sent [for], and brought him [Yehoyachin] to Babylon with the vessels of the desire of the House of YHVH, and [he] [Nebuchadnezzar] made Tzidkiyahu his brother[1326] king over Yehudah and Jerusalem.[1327]

XX. If it is correct that the total number of persons selected for deportation was 10000,[1328] and if much booty was taken and prepared for transport – even to the point of the cutting in pieces of the vessels of gold from the temple as stated in the Hebrew Scriptures[1329] – then according to Wiseman it may readily be supposed that it was a number of weeks before the captives and goods were assembled, and the caravan could depart.[1330] Indeed, Wiseman notes that the Babylonians do not seem to have collected this large group quickly, and the Jewish captives only move off to exile some weeks after the city itself had fallen.[1331]

XX. Because of the discrepancy between the Babylonian and the proleptic Hebrew calendars in this year, arising from the Babylonian intercalary Ululu 2 / Elul 2 of Nebuchadnezzar's 7th Regnal Year, the Babylonian 02 Addaru / Adar (16 March 597 BCE (Julian)) fell on 03 Nisannu / Nisan in the proleptic Hebrew calendar, so that the capture of Jerusalem and the march into exile likely on 10 Nisannu / Nisan were separated by only one (1) week.

XX. Such a one-(1)-week interval is entirely reasonable. Neither the biblical account nor the Babylonian Chronicle describes a prolonged sack of the city in 597 BCE (Julian). The biblical account records that Nebuchadnezzar carried off all the treasures of the house of YHVH and of the king's house as well as cut in pieces the vessels of gold, but it describes no destruction of the city. The Chronicle records only that Nebuchadnezzar seized the city, captured its king, appointed a king of his own choice,

1324 cf. 2 Chronicles 36:10
1325 i.e. "the return of the New Year", i.e. "the beginning of the year" for kings that starts with Nisannu / Nisan
1326 Figuratively his brother kinsman, but literally his uncle, i.e. the [half-]brother of his father Yehoyakim (Jehoiakim)
1327 cf. 2 Chronicles 36:10
1328 cf. 2 Kings 24:14
1329 cf. 2 Kings 24:13
1330 Finegan 1998: 257
1331 Wiseman 1956: 34

and received its heavy tribute, while the biblical account records that Yehoyachin (Jehoiachin) went out to the king of Babylon together with his mother, his servants, his princes, and his officers.[1332]

XX. Wiseman's interval of some weeks corresponds to the time between the Babylonian 02 Addaru / Adar (16 March 597 BCE (Julian)) and the Babylonian New Year, 01 Nisannu / Nisan (13 April 597 BCE (Julian)). According to 2 Chronicles 36:10, however, Yehoyachin (Jehoiachin) was brought to Babylon "at the turn of the year" together with the precious vessels, and by the proleptic Hebrew calendar the New Year, 01 Nisannu / Nisan (14 March 597 BCE (Julian)), had already begun two (2) days before the city fell.

XX. This describes a surrender rather than a destruction of the city, which did not take place until the later siege under Tzidkiyahu (Zedekiah) in 587 – 586 BCE (Julian). With the treasuries of a surrendered city open to the victor, the removal of the king, the royal household, the officials, and the treasure – together with the deportation of the exiles – need not have required several weeks, and could well have been accomplished within the week between 03 and 10 Nisannu / Nisan (16 – 23 March 597 BCE (Julian)) in the proleptic Hebrew calendar.

XX. This corresponds to 02 – 09 Addaru / Adar in the 7th Regnal Year of Nebuchadnezzar in the Babylonian calendar. This also explains why the Hebrew Scriptures record that Yehoyachin (Jehoiachin) was taken in the 8th Regnal Year of Nebuchadnezzar, since by the proleptic Hebrew Nisannu / Nisan reckoning that week already belonged to his 8th Regnal Year.[1333]

XX. As mentioned, the proleptic Hebrew date for 10 Nisannu / Nisan for the 8th Regnal Year of Nebuchadnezzar is Saturday 23 March 597 BCE (Julian)[1334] [1335] with this Babylonian / Hebrew day having begun the previous Sabbath evening on the day of Friday 22 March 597 BCE (Julian).[1336] [1337] Because potential problems with marching in the heat during the day, it is entirely plausible that the march into captivity was conducted at night into order to avoid losing captives to heat exhaustion and exposure to the Sun.

XX. It is thus possible that the march into exile began that Sabbath Friday evening itself at the beginning of the proleptic Hebrew day 10 Nisannu / Nisan. However, even if the march into captivity didn't begin until the next early morning hours of Saturday 23 March 597 BCE (Julian), it nonetheless was on the Sabbath. Therefore the Hebrew exiles were forced to break the Sabbath by marching from their place in violation of one (1) of the three (3) commandments for Sabbath:

1332 cf. 2 Kings 24:12
1333 cf. 2 Kings 24:12
1334 Saturday 23 March 597 BCE (Julian); Saturday 17 March 597 BCE (Gregorian); 10 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503450.5; RD_Y: 1155454; RD_X: -217974; Sabbatical Cycle: 7; Metonic Cycle: 10;
1335 cf. Appendix Table X.34
1336 Friday 22 March 597 BCE (Julian); Friday 16 March 597 BCE (Gregorian); 09 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503449.5; RD_Y: 1155453; RD_X: -217975; Sabbatical Cycle: 7; Metonic Cycle: 10;
1337 cf. Appendix Table X.34

1.) Don't do labor;

2.) Don't kindle fire;

3.) Don't travel from your place because on Sabbath everyone must stay in their place.[1338]

XX. This day of the proleptic Hebrew 10 Nisannu / Nisan was little more than a week after the fall of Jerusalem on the Babylonian 02 Addaru / Adar 597 BCE (i.e. in the 12th month of the 7th Regnal Year of Nebuchadnezzar).[1339] 10 Nisannu / Nisan 597 BCE in the 8th Regnal Year of Nebuchadnezzar (proleptic Hebrew reckoning) is therefore the most probable date of the actual deportation of Yehoyachin (Jehoiachin) and the beginning of the Exile.[1340]

XX. It would also be poignantly (and painfully) symbolic since the date of Israel's crossing the Jordan River and entering the land of Israel was also on 10 Nisannu / Nisan.[1341] The proleptic Hebrew date of Entering the Promised Land on Monday 22 March 1457 BCE (Julian)[1342] [1343] [1344] [1345] marks the beginning of exactly 860 Hebrew years (to the very day) until the EXILE OF YEHOYACHIN (JEHOIACHIN) on 10 Nisannu / Nisan, i.e. Saturday 23 March 597 BCE (Julian).[1346] [1347]

XX. In other words, Israel sojourned exactly 860 Babylonian / Hebrew years to the very day (i.e. exactly 314116 days)[1348] in the land of Israel from 1457 BCE (2304 AM) until being exiled to Babylon in 597 BCE (3164 AM). Furthermore, this date of 10 Nisannu / Nisan is also the date commanded for the ancient Israelites to take the sacrificial Passover lamb to guard and hold until the eve of Passover on

1338 cf. Deuteronomy 5:15
1339 cf. Appendix Table X.32
1340 Finegan 1998: 257
1341 cf. Joshua 4:19
1342 Monday 22 March 1457 BCE (Julian); Monday 09 March 1457 BCE (Gregorian); 10 Nisannu / Nisan 2304 AM; Astronomical Julian Day JD_X: 1189334.5; RD_Y: 841338; RD_X: -532090; Sabbatical Cycle: 1; Metonic Cycle: 5;
1343 cf. Appendix Table X.148
1344 cf. Joshua 4:19
1345 The date of Entering the Promised Land in the Hebrew year 2304 AM (1457 BCE) marks the beginning of exactly 860 Hebrew years (to the very day) until the Exile of Yehoyachin (Jehoiachin) on 10 Nisan, i.e. Saturday 23 March 597 BCE (Julian) / Saturday 17 March 597 BCE (Gregorian) (3164 AM). In other words, Israel sojourned exactly 860 years to the very day (i.e. exactly 314116 days) in the land of Israel from 1457 BCE (2304 AM) until being exiled to Babylon in 597 BCE (3164 AM).
1346 Saturday 23 March 597 BCE (Julian); Saturday 17 March 597 BCE (Gregorian); 10 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503450.5; RD_Y: 1155454; RD_X: -217974; Sabbatical Cycle: 7; Metonic Cycle: 10;
1347 cf. Appendix Table X.34
1348 A friend of the author (who wishes to remain anonymous) is a researcher of Gematria in the Torah. He has the following discoveries and insights to share about this number: 1.) It is a very close approximation to the Greek number (π) pi (3.1415); 2.) 314116: 314 + 116 == 430; 430 x 2 == 860; This number 860 is the Gematria numerical value of the following Hebrew terms: 386 (ישוע) Yeshua + 358 (משיח) Mashiach + 116 (ינון) Yinon which translates to: "The Messiah Yeshua will sprout / flourish / continue / increase / propagate / endure"; cf. Psalms 72:17

14 Nisannu / Nisan.[1349] Moreover, 10 Nisannu / Nisan is notably also the date of Jesus' Last Supper with his subsequent arrest and detention until he was crucified on 14 Nisannu / Nisan, as will be unfolded below in section 75.

XX. Recalling that the Babylonian account (BM 21946) reports that Jerusalem was captured in the 7th Regnal Year of Nebuchadnezzar in the last month of that year (i.e. 02 Addaru / Adar 597 BCE (Julian)), Jeremiah's count of this Exile (i.e. the fall of Jerusalem and Exile one (1) week later) as having had occurred in the 7th Regnal Year of Nebuchadnezzar[1350] no longer is as seemingly contradictory with the account in the book of 2 Kings that declares that Yehoyachin (Jehoiachin) was captured in the 8th Regnal Year of Nebuchadnezzar (by proleptic Hebrew reckoning). Indeed both accounts are precisely accurate, and both biblical accounts concord exactly with the account in the Babylonian Chronicle.[1351]

XX. The biblical accounts do not give the day or month of the capture of Jerusalem in 597 BCE (Julian). This precise date, 02 Addaru / Adar in the 7th Regnal Year of Nebuchadnezzar, comes only from the Babylonian Chronicle (BM 21946). The biblical accounts give only the year: Jeremiah counts the Judeans exiled by Nebuchadrezzar in his 7th Regnal Year,[1352] whereas 2 Kings records that the king of Babylon took Yehoyachin (Jehoiachin) captive in the 8th Regnal Year of Nebuchadnezzar's reign,[1353] and 2 Chronicles places his removal to Babylon and the accession of Tzidkiyahu (Zedekiah) at the turn of the year.[1354]

XX. These accounts are reconciled once the Babylonian and proleptic Hebrew calendars are compared: the city fell at the end of the 7th Regnal Year of Nebuchadnezzar by Babylonian reckoning (02 Addaru / Adar, 16 March 597 BCE (Julian)), which was already 03 Nisannu / Nisan in the proleptic Hebrew calendar, and the march into exile and the accession of Tzidkiyahu (Zedekiah) followed one (1) week later at the turn of the year on 10 Nisannu / Nisan (23 March 597 BCE (Julian)) in the 8th Regnal Year of Nebuchadnezzar by proleptic Hebrew Nisannu / Nisan reckoning.

XX. Thus Jeremiah dates the first Exile by the fall of the city to the king of Babylon in his 7th Regnal Year (according to Babylonian reckoning),[1355] in agreement with the Babylonian Chronicle. This marks the moment when freedom ends and Exile begins. This contrasts with Ezekiel's chronology (according to the proleptic Hebrew reckoning) who evidently dates the Exile from the day he and the other captives were actually forced to leave their native land of Judah and begin the march to Babylon.[1356]

1349 cf. Exodus 12:3-6
1350 cf. Jeremiah 52:28
1351 cf. Jeremiah 52:28 (i.e. that Nebuchadnezzar exiled in years 7, 18, and 23) vs. 2 Kings 24:12 (i.e. Nebuchadnezzar took the king Yehoyachin (Jehoiachin) in year 8 of his reign).
1352 cf. Jeremiah 52:28
1353 cf. 2 Kings 24:12
1354 cf. 2 Chronicles 36:10
1355 cf. Jeremiah 52:28
1356 cf. Ezekiel 1:1-2; 8:1; 20:1; 24:1; 26:1; 29:1, 17; 30:20; 31:1; 32:1, 17; 33:21; 40:1

XX. By the proleptic Hebrew reckoning, Ezekiel's exile fell on 10 Nisannu / Nisan, at the beginning of the 8th Regnal Year of Nebuchadnezzar.[1357] The fall of the city (02 Addaru / Adar, 16 March 597 BCE (Julian)) and the march into Exile (10 Nisannu / Nisan, 23 March 597 BCE (Julian)) were thus only one (1) week apart, but are reckoned in different Regnal Years: the 7th by the Babylonian calendar, the 8th by the proleptic Hebrew calendar.

XX. The apparent discrepancy between the 7th Regnal Year of Nebuchadnezzar in Jeremiah 52:28 and the 8th Regnal Year in 2 Kings 24:12 is explained by the difference between the Babylonian and the proleptic Hebrew calendars in that year. Because of the Babylonian intercalary Ululu 2 / Elul 2 of the 7th Regnal Year, the capture of Jerusalem and its king on 02 Addaru / Adar (16 March 597 BCE (Julian)) still fell in the 7th Regnal Year of Nebuchadnezzar by Babylonian reckoning, in agreement with the Babylonian Chronicle.

XX. By comparison, the proleptic Hebrew reckoning of this same day of the fall of Jerusalem was already 03 Nisannu / Nisan, the beginning of his 8th Regnal Year. The deportation list of Jeremiah 52:28-30 which notes deportations in the 7th, 18th, and 23rd years thus evidently follows the Babylonian reckoning of Nebuchadnezzar's years,[1358] while the narrative of 2 Kings reckons his years from the Nisannu / Nisan of Judah. The same list of Jeremiah dates a deportation from Jerusalem to the 18th Regnal Year of Nebuchadnezzar before the arrival of Nebuzaradan at Jerusalem in his 19th Regnal Year.[1359]

XX. Since Jerusalem was destroyed in the 19th Regnal Year of Nebuchadnezzar by both the Babylonian and the proleptic Hebrew reckoning, the deportation of the 18th Regnal Year must precede its destruction. The mathematics thus indicates that this deportation refers to those who had surrendered or been captured during the siege of the city, before its destruction.

XX. At the end of the 8th Regnal Year of Nebuchadnezzar, the proleptic Hebrew calendar resynchronizes with the Babylonian calendar when it intercalates its 13th month Addaru 2 / Adar 2 of 3165 AM that began 05 March 596 BCE (Julian), which begins on the same day as the Babylonian month of Addaru / Adar.[1360] Thus the one-(1)-month difference caused by the Babylonian intercalary Ululu 2 / Elul 2 that began 21 September 598 BCE (Julian) in the 7th Regnal Year of Nebuchadnezzar comes to an end, and both calendars begin the 9th Regnal Year of Nebuchadnezzar on the same day, 01 Nisannu / Nisan (03 April 596 BCE (Julian)).

16.0. 9th Regnal Year of Nebuchadnezzar II == Spring 596 – 595 BCE

1357 cf. 2 Kings 24:12
1358 cf. Jeremiah 52:28-30
1359 cf. Jeremiah 52:12; 2 Kings 25:8
1360 Parker and Dubberstein 1971: 27

== 1st Regnal Year of Tzidkiyahu (Zedekiah)
== YEAR 13 OF SERVITUDE
== YEAR 2 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. In his 9th Regnal Year (Nisannu / Nisan 596 – 595 BCE),[1361] Nebuchadrezzar was threatened by a new enemy and is found with his army marching along the Tigris River and then pitching a camp at a place that the name of which is now lost in the broken cuneiform text. The name of his opponent is also regrettably lost, although the signs which remain may point to the (unnamed) king of Elam.[1362]

XX. Supporting this theory, the power of the Medes, who at this time who were allies of Babylon to its Northeast, was so great that the Babylonians did not have their attention diverted to their northern or eastern borders, and were usually free therefore to concentrate on the west. The Chronicle implies that the enemy, whether Elamite or not, took fright and turned back to his own country when [sic: within] a day's march from the Babylonian forces.[1363]

XX. In the 9th Regnal Year of Nebuchadnezzar (Nisannu / Nisan 596 – 595 BCE (Julian)), the Babylonians intercalated a 2nd Ululu / Elul (Ululu 2 / Elul 2) beginning 28 September 596 BCE (Julian)[1364] which the proleptic Hebrew calendar does not. From this point onward, the corresponding Babylonian months in Parker and Dubberstein's tables again fall one (1) month later than those of the proleptic Hebrew calendar.

17.0. 10th Regnal Year of Nebuchadnezzar II == Spring 595 – 594 BCE
== 2nd Regnal Year of Tzidkiyahu (Zedekiah)
== YEAR 14 OF SERVITUDE
== YEAR 3 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. In his 10th Regnal Year (Nisannu / Nisan 595 – 594 BCE (Julian)),[1365] Nebuchadrezzar remained at home in Babylonia for the greater part of it. During this year, there was a revolt in the country which was only suppressed by the slaughter of many of his own army and by the capture of the rebel leader.[1366] The strain put on the standing army by the frequent and lengthy campaigns in the west may have been a factor that contributed to the disturbances.[1367]

XX. According to the Babylonian reckoning this revolt took place during the 9th month of Kisilimu /

1361 cf. Appendix Table X.36
1362 Wiseman 1956: 36
1363 Wiseman 1956: 36
1364 Parker and Dubberstein 1971: 5, 27
1365 cf. Appendix Table X.37
1366 Wiseman 1956: 36, 48
1367 Wiseman 1956: 37

Kislev (15 December 595 – 12 January 594 BCE (Julian)) and the 10th month of Tebetu / Tebet (13 January – 11 February 594 BCE (Julian)) – i.e. one (1) month later than the proleptic Hebrew months of the same name, Kisilimu / Kislev (13 November – 12 December 595 BCE (Julian))[1368] and Tebetu / Tevet (13 December 595 – 10 January 594 BCE (Julian)).[1369]

XX. This one-(1)-month difference is due to the Babylonian intercalary Ululu 2 / Elul 2 that began 28 September 596 BCE (Julian) in the 9th Regnal Year, because of which the Babylonian months in Parker and Dubberstein's tables fall one (1) month later than the proleptic Hebrew months of the same name.

XX. Order must soon have been restored, because the king was able to go in person to the land of Hatti before the end of the year to receive the tribute brought by the vassal kings and other officials, and to bring it back to the capital. This journey was apparently made – just as two (2) years earlier – without the presence of the army, for the latter is not specifically mentioned. It seems, therefore, that the west remained undisturbed.[1370]

XX. In the 10th Regnal Year of Nebuchadnezzar neither calendar intercalates a month, so the one-(1)-month difference caused by the Babylonian intercalary Ululu 2 / Elul 2 of the 9th Regnal Year continues throughout the year. Consequently, the 11th Regnal Year of Nebuchadnezzar begins on 01 Nisannu / Nisan 11 March 594 BCE (Julian) in the proleptic Hebrew calendar but on 12 April 594 BCE (Julian) in the Babylonian calendar, still one (1) month apart.

18.0. 11th Regnal Year of Nebuchadnezzar II == Spring 594 – 593 BCE
== 3rd Regnal Year of Tzidkiyahu (Zedekiah)
== YEAR 15 OF SERVITUDE
== YEAR 4 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. The catch-line which ends the text of British Museum Tablet BM 21946 gives us only a brief glimpse of Nebuchadrezzar's activity at the beginning of his 11th Regnal Year[1371] which in the proleptic Hebrew calendar began on 01 Nisannu / Nisan 3167 AM (11 March 594 BCE (Julian)),[1372] one (1) month earlier than the Babylonian 01 Nisannu / Nisan (12 April 594 BCE (Julian)). This tablet records that he called out his army in the 9th month of Kisilimu / Kislev for a further expedition to the land of Hatti.[1373]

XX. According to the Babylonian reckoning, this month ran from 04 December 594 to 01 January 593

1368 cf. Appendix Table X.38
1369 cf. Appendix Table X.39
1370 Wiseman 1956: 36, 48, 72, 73
1371 cf. Appendix Table X.40
1372 cf. Appendix Table X.41
1373 Wiseman 1956: 37

BCE (Julian) – i.e. one (1) month later than the proleptic Hebrew month of the same name, Kisilimu / Kislev (03 November – 02 December 594 BCE (Julian)),[1374] owing to the Babylonian intercalary Ululu 2 / Elul 2 that began 28 September 596 BCE (Julian) in the 9th Regnal Year.

XX. At the end of this year both calendars intercalate: the proleptic Hebrew calendar its 13th month Addaru 2 / Adar 2 of 3168 AM beginning 01 March 593 BCE (Julian), and the Babylonian calendar an Addaru 2 / Adar 2 beginning 31 March 593 BCE (Julian), one (1) month later.[1375] Because the proleptic Hebrew calendar intercalates first, the two (2) calendars are synchronized for that single month only: its Addaru 2 / Adar 2 coinciding with the Babylonian Addaru / Adar. Then when the Babylonians subsequently intercalate their own Addaru 2 / Adar 2, the one-(1)-month difference resumes and continues into the 12th Regnal Year of Nebuchadnezzar.

XX. Unfortunately, events for the remainder of this year, as well as the remaining 33 years of Nebuchadrezzar are lost, as the long gap in the Babylonian Chronicle between Nebuchadnezzar's 11th Regnal Year (Nisannu / Nisan 594 – 593 BCE (Julian)), and the record for the reign of Nabonidus (555 – 539 BCE (Julian)) is filled for a single year only by the British Museum Tablet BM 25124.[1376] This text, giving details of the 3rd Regnal Year of Neriglissar (*Nergal-šar-usur* 559 – 556 BCE (Julian)), is the only strictly historical text extant for his reign.[1377]

19.0. 12th Regnal Year of Nebuchadnezzar II == Spring 593 – 592 BCE
== 4th Regnal Year of Tzidkiyahu (Zedekiah)
== YEAR 16 OF SERVITUDE
== YEAR 5 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. In the 4th Regnal Year of Tzidkiyahu (Zedekiah),[1378] i.e. YEAR FIVE (5) OF THE EXILE OF YEHOYACHIN (JEHOIACHIN), Jeremiah's scribe, S'raya son of Neryah son of Mahseyah, goes with Tzidkiyahu (Zedekiah) king of Judah to Babylon, and delivers a message to the Hebrew exiles in Babylon.[1379] It is entirely plausible that this was the occasion to which Nebuchadnezzar called all vassal kings of his empire to Babylon in order to worship the golden image described in the Book of Daniel.[1380]

XX. The Prophet Ezekiel reckons his chronology using the EXILE OF YEHOYACHIN (JEHOIACHIN) as the epoch.[1381] However, when reading through the Book of Ezekiel, it becomes

1374 cf. Appendix Table X.42
1375 Parker and Dubberstein 1971: 5, 28
1376 Wiseman 1956: 37
1377 Wiseman 1956: 37-38
1378 cf. Appendix Table X.43
1379 cf. Jeremiah 51:59
1380 cf. Daniel 3:1-30
1381 cf. Ezekiel 1:2; 33:21; 40:1

evident that the book is not entirely chronological in its redacted order:

Table 19.1. Chronological Time Indicators in the Book of Ezekiel

Time Indicator	Book Chapter:Verse
In 30th Year, in 4th [Month], [Day] five (5) of Month	Ezekiel 1:1
On [Day] five (5) of Month that is the 5th Year of Exile of the King Yehoyachin (Jehoiachin)	Ezekiel 1:2
In 6th Year, in 6th [Month], on [Day] five (5) of Month	Ezekiel 8:1
In 7th Year, in 5th [Month], on [Day] ten (10) of Month	Ezekiel 20:1
In 9th Year, in 10th Month, on [Day] ten (10) of Month	Ezekiel 24:1
In 11th Year, [in 1st Month,] on [Day] one (1) of Month	Ezekiel 26:1
In 10th Year, in 10th [Month], on [Day] 12 of Month	Ezekiel 29:1
In 27th Year, in 1st [Month], on [Day] one (1) of Month	Ezekiel 29:17
In 11th Year, in 1st [Month], on [Day] seven (7) of Month	Ezekiel 30:20
In 11th Year, in 3rd [Month], on [Day] one (1) of Month	Ezekiel 31:1
In 12th Year, in 12th Month, on [Day] one (1) of Month	Ezekiel 32:1
In 12th Year, [in 12th Month,] on [Day] 15 of Month	Ezekiel 32:17
In 12th Year, in 10th [Month], on [Day] five (5) of Month of our Exile	Ezekiel 33:21
In 25th Year of our Exile, at the Head of the Year (i.e. Rosh HaShanah), on the [Day] ten (10) of the Month (i.e. Yom Kippur) – in 14th Year after the City was Struck – on this very day...	Ezekiel 40:1

XX. The Book of Ezekiel opens with Ezekiel's vision of GOD in the 5th year of the EXILE OF YEHOYACHIN (JEHOIACHIN), in the 4th month, on [day] five (5) of the month. The prophet Ezekiel records that in the 30th year (of some unspecified event) in the 4th [month] on [day] five (5) of the month, Ezekiel is in Exile upon the River Kebar and the skies open [up] and he sees appearances of ELOHIM.[1382] [1383]

XX. Ezekiel specifies that this [day] five (5) of the [4th] month is the 5th year of the EXILE OF YEHOYACHIN (JEHOIACHIN).[1384] This works out to be Saturday 01 July 593 BCE (Julian).[1385] [1386] In other words, Ezekiel's vision of GOD was experienced on the Jewish Sabbath (Shabbat).

XX. Evidently, Ezekiel notes this date as being the 30th year of some previous event. If 593 BCE (3168 AM) in the 4th month in YEAR FIVE (5) OF EXILE OF YEHOYACHIN (JEHOIACHIN) is the 30th year of that event, then the 1st year works out to be 622 – 621 BCE (3139 – 3140 AM), i.e. the 17th Regnal Year of Yoshiyahu (Josiah) ben Amon which is equivalent to the 4th Regnal Year of

1382 cf. Ezekiel 1:1-2
1383 cf. Appendix Table X.44
1384 cf. Ezekiel 1:1-2
1385 Saturday 01 July 593 BCE (Julian); Saturday 25 June 593 BCE (Gregorian); 05 Du'uzu / Tammuz 3168 AM; Astronomical Julian Day JD_X: 1505011.5; RD_Y: 1157015; RD_X: -216413; Sabbatical Cycle: 4; Metonic Cycle: 14;
1386 cf. Appendix Table X.44

Nabopolassar.[1387]

20.0. 13th Regnal Year of Nebuchadnezzar II == Spring 592 – 591 BCE
== 5th Regnal Year of Tzidkiyahu (Zedekiah)
== YEAR 17 OF SERVITUDE
== YEAR 6 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. In the 6th year of the EXILE OF YEHOYACHIN (JEHOIACHIN)[1388] in the 6th [month] on the [day] five (5) of the month… the hand of ADONAI YHVH fell upon Ezekiel.[1389] This works out to be Sunday 19 August 592 BCE (Julian).[1390] [1391]

21.0. 14th Regnal Year of Nebuchadnezzar II == Spring 591 – 590 BCE
== 6th Regnal Year of Tzidkiyahu (Zedekiah)
== YEAR 18 OF SERVITUDE
== YEAR 7 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. In the 7th year of the EXILE OF YEHOYACHIN (JEHOIACHIN)[1392] in the 7th [month] on the [day] ten (10) of the month they came from the elders of Israel to require of YHVH and sat before Ezekiel.[1393] This works out to be Sunday 14 July 591 BCE (Julian).[1394] [1395]

22.0. 15th Regnal Year of Nebuchadnezzar II == Spring 590 – 589 BCE
== 7th Regnal Year of Tzidkiyahu (Zedekiah)
== YEAR 19 OF SERVITUDE
== YEAR 8 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. No events are recorded in the Hebrew Bible or in the Babylonian Chronicles for the 7th Regnal Year of Tzidkiyahu (Zedekiah), i.e. YEAR EIGHT (8) OF EXILE OF YEHOYACHIN (JEHOIACHIN).[1396]

1387 cf. Appendix Table X.4
1388 cf. Appendix Table X.45
1389 cf. Ezekiel 8:1
1390 Sunday 19 August 592 BCE (Julian); Sunday 13 August 592 BCE (Gregorian); 05 Ululu / Elul 3169 AM; Astronomical Julian Day JD_X: 1505425.5; RD_Y: 1157429; RD_X: -215999; Sabbatical Cycle: 5; Metonic Cycle: 15;
1391 cf. Appendix Table X.46
1392 cf. Appendix Table X.47
1393 cf. Ezekiel 20:1
1394 Sunday 14 July 591 BCE (Julian); Sunday 08 July 591 BCE (Gregorian); 10 Abu / Av 3170 AM; Astronomical Julian Day JD_X: 1505754.5; RD_Y: 1157758; RD_X: -215670; Sabbatical Cycle: 6; Metonic Cycle: 16;
1395 cf. Appendix Table X.48
1396 cf. Appendix Table X.49

23.0. 16th Regnal Year of Nebuchadnezzar II == Spring 589 – 588 BCE
== 8th Regnal Year of Tzidkiyahu (Zedekiah)
== YEAR 20 OF SERVITUDE
== YEAR 9 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. In the 9th year of the EXILE OF YEHOYACHIN (JEHOIACHIN)[1397] in the 10th month on [day] ten (10) of the month the word of YHVH was to Ezekiel:

"*Son of Adam, write for yourself the name of this day, this very day the king of Babylon will reach (SAMACH EL (סמך אל)) Jerusalem — on this very day.*"[1398]

XX. Since Ezekiel's chronology counts from the EXILE OF YEHOYACHIN (JEHOIACHIN), this can be understood as a prophecy to Ezekiel that he receives in the 8th Regnal Year of Tzidkiyahu (Zedekiah) on Friday 17 December 589 BCE (Julian)[1399] [1400] which is exactly one (1) Babylonian / Hebrew year (precisely to the very day) before the day that the Babylonian Siege of Jerusalem begins on Thursday 05 January 587 BCE (Julian).[1401] [1402] [1403]

24.0. 17th Regnal Year of Nebuchadnezzar II == Spring 588 – 587 BCE
== 9th Regnal Year of Tzidkiyahu (Zedekiah)
== YEAR 21 OF SERVITUDE
== YEAR 10 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. The Book of Kings and the Book of Jeremiah both tell us that Nebuchadnezzar and his army came (i.e. returned) in the 9th Regnal Year of Tzidkiyahu (Zedekiah),[1404] in the 10th month, on [day] 10 of the month, i.e. Thursday 05 January 587 BCE (Julian),[1405] [1406] Nebuchadnezzar king of Babylon and all his army came against Jerusalem and encamped against her, and they built a siege rampart all around

1397 cf. Appendix Table X.50
1398 cf. Ezekiel 24:1-2
1399 Friday 17 December 589 BCE (Julian); Friday 11 December 589 BCE (Gregorian); 10 Tebetu / Tevet 3173 AM; Astronomical Julian Day JD_X: 1506641.5; RD_Y: 1158645; RD_X: -214783; Sabbatical Cycle: 2; Metonic Cycle: 19;
1400 cf. Appendix Table X.51
1401 Thursday 05 January 587 BCE (Julian); Thursday 30 December 588 BCE (Gregorian); 10 Tebetu / Tevet 3174 AM; Astronomical Julian Day JD_X: 1507025.5; RD_Y: 1159029; RD_X: -214399; Sabbatical Cycle: 3; Metonic Cycle: 1;
1402 cf. Appendix Table X.53
1403 cf. Footnote #9
1404 cf. Appendix Table X.52
1405 Thursday 05 January 587 BCE (Julian); Thursday 30 December 588 BCE (Gregorian); 10 Tebetu / Tevet 3174 AM; Astronomical Julian Day JD_X: 1507025.5; RD_Y: 1159029; RD_X: -214399; Sabbatical Cycle: 3; Metonic Cycle: 1;
1406 cf. Appendix Table X.53

her.[1407] The city was besieged until the 11th Regnal Year of King Tzidkiyahu (Zedekiah).[1408] [1409] [1410]

XX. Jeremiah describes these same events of the beginning of the siege as well as its conclusion and mentions various Babylonian officials. Among those mentioned are the Chief of the Slaughterers (Nebuzaradan), the Chief Eunich (who seems to have been replaced between verse 3 and verse 13), the Rab Mag (Nergal Sar Etzer).[1411] [1412] Notably, in verse 3 it mentions "and all the remainder of the ministers of the king of Babylon", and in verse 13 "and all the chiefs of the king of Babylon".

XX. Therefore, since Daniel had been promoted to be number two (2) in the kingdom after Nebuchadnezzar,[1413] unless he was back in Babylon managing the empire during Nebuchadnezzar's absence, these verses in Jeremiah would seem to suggest that Daniel was also included among all these Babylonian officials who presided over (and witnessed) the siege and destruction of Jerusalem.

XX. In this 10th year of the EXILE OF YEHOYACHIN (JEHOIACHIN) in the 10th [month] on the 12th [day] of the month, a word of YHVH was to Ezekiel against the king of Egypt and upon all of Egypt (40 years exile of Egypt).[1414] This works out to be Saturday 07 January 587 BCE (Julian).[1415] [1416] [1417]

XX. In this 17th Regnal Year of Nebuchadnezzar (Nisannu / Nisan 588 – 587 BCE (Julian)), the proleptic Hebrew and Babylonian calendars are synchronized with the proleptic Hebrew calendar having intercalated its 13th month Addaru 2 / Adar 2 of 3173 AM that began 07 March 588 BCE (Julian). The beginning of the siege on 10 Tebetu / Tevet therefore falls on the same day in both calendars, Thursday 05 January 587 BCE (Julian).[1418] The two (2) calendars diverge again at the end of this year when the Babylonians intercalate its 13th month Addaru 2 / Adar 2 that began 25 March 587 BCE (Julian),[1419] while the proleptic Hebrew calendar does not.

25.0. 18th Regnal Year of Nebuchadnezzar II == Spring 587 – 586 BCE == 10th Regnal Year of Tzidkiyahu (Zedekiah)

1407 2 Kings 25:1; Jeremiah 52:4
1408 cf. Appendix Table X.57, X.58, X.59
1409 cf. 2 Kings 25:2; Jeremiah 1:3; 52:5
1410 cf. Jeremiah 38:28; 39:1-14
1411 Babylonian (Akkadian): *Nergal-šar-usur*, i.e. Neriglissar who was Nebuchadnezzar's son-in-law, and the later Babylonian king (r. 559 – 556 BCE) who succeeded Nebuchadnezzar's son, Amel-Marduk (r. 561 – 560 BCE).
1412 cf. Jeremiah 38:28; 39:1-14
1413 cf. Daniel 2
1414 cf. Ezekiel 29:1
1415 Saturday 07 January 587 BCE (Julian); Saturday 01 January 587 BCE (Gregorian); 12 Tebetu / Tevet 3174 AM; Astronomical Julian Day JD_X: 1507027.5; RD_Y: 1159031; RD_X: -214397; Sabbatical Cycle: 3; Metonic Cycle: 1;
1416 cf. Appendix Table X.53
1417 cf. Footnote #9
1418 Parker and Dubberstein 1971: 28
1419 Parker and Dubberstein 1971: 5, 28

== YEAR 22 OF SERVITUDE
== YEAR 11 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. Jeremiah declares the 10th Regnal Year of Tzidkiyahu (Zedekiah) king of Judah to be equivalent to the 18th Regnal Year of Nebuchadnezzar.[1420] [1421] The Book of Kings confirms Jeremiah's chronology, evidently equating the 11th Regnal Year of Tzidkiyahu (Zedekiah) with the 19th Regnal Year of the Babylonian king Nebuchadnezzar.[1422] During this time, i.e. the 10th Regnal Year of Tzidkiyahu (Zedekiah), the army of the king of Babylon was sieging Jerusalem.[1423]

XX. In the 11th year of the EXILE OF YEHOYACHIN (JEHOIACHIN) on [day] one (1) of the month the word of YHVH was to Ezekiel: a prophecy against Tyre.[1424] Since there is no indication of the exact month, it is reasonable to assume that this is [day] one (1) of the 1st month, i.e. Nisannu / Nisan. This works out to be Saturday 25 March 587 BCE (Julian).[1425] [1426]

XX. In the 11th year of the EXILE OF YEHOYACHIN (JEHOIACHIN) in the 1st [month] on [day] seven (7) of the month the word of YHVH was to Ezekiel: a prophecy against Egypt.[1427] This works out to be Friday 31 March 587 BCE (Julian).[1428] [1429]

XX. In the 11th year of the EXILE OF YEHOYACHIN (JEHOIACHIN) in the 3rd [month] on [day] one (1) of the month the word of YHVH was to Ezekiel: a prophecy against the king of Egypt and his mass [of people].[1430] This works out to be Tuesday 23 May 587 BCE (Julian).[1431] [1432]

XX. In the 18th Regnal Year of Nebuchadnezzar, the proleptic Hebrew calendar is again one (1) month ahead of the Babylonian calendar, because of the Babylonian intercalary Addaru 2 / Adar 2 that began 25 March 587 BCE (Julian) at the end of his 17th Regnal Year[1433] which the proleptic Hebrew calendar does not intercalate. The proleptic Hebrew 18th Regnal Year thus began on 01 Nisannu / Nisan 3174 AM (25 March 587 BCE (Julian)), whereas the Babylonian 18th Regnal Year began one (1) month later,

1420 cf. Appendix Table X.54
1421 cf. Jeremiah 32:1
1422 cf. 2 Kings 25:8-11
1423 cf. Jeremiah 32:1-2; 38:28; 39:1-14
1424 cf. Ezekiel 26:1
1425 Saturday 25 March 587 BCE (Julian); Saturday 19 March 587 BCE (Gregorian); 01 Nisannu / Nisan 3174 AM; Astronomical Julian Day JD_X: 1507104.5; RD_Y: 1159108; RD_X: -214320; Sabbatical Cycle: 3; Metonic Cycle: 1;
1426 cf. Appendix Table X.54, X.55
1427 cf. Ezekiel 30:20
1428 Friday 31 March 587 BCE (Julian); Friday 25 March 587 BCE (Gregorian); 07 Nisannu / Nisan 3174 AM; Astronomical Julian Day JD_X: 1507110.5; RD_Y: 1159114; RD_X: -214314; Sabbatical Cycle: 3; Metonic Cycle: 1;
1429 cf. Appendix Table X.55
1430 cf. Ezekiel 31:1
1431 Tuesday 23 May 587 BCE (Julian); Tuesday 17 May 587 BCE (Gregorian); 01 Simannu / Sivan 3174 AM; Astronomical Julian Day JD_X: 1507163.5; RD_Y: 1159167; RD_X: -214261; Sabbatical Cycle: 3; Metonic Cycle: 1;
1432 cf. Appendix Table X.55
1433 Parker and Dubberstein 1971: 5, 28

on 23 April 587 BCE (Julian).

XX. Thus Ezekiel's prophecies on 01 and 07 Nisannu / Nisan (according to the proleptic Hebrew reckoning) therefore fell in the 13th intercalary Babylonian month of Addaru 2 / Adar 2 at the end of the 17th Regnal Year of Nebuchadnezzar, and his prophecy on 01 Simannu / Sivan fell on the Babylonian 01 Ayyaru / Iyar of his 18th Regnal Year.[1434]

XX. It is in this 18th Regnal Year of Nebuchadnezzar that the deportation list of Jeremiah records the deportation of 832 persons from Jerusalem.[1435] By Babylonian reckoning this year ran from 23 April 587 to 12 April 586 BCE (Julian), entirely within the recorded siege of Jerusalem, which began on 10 Tebetu / Tevet (05 January 587 BCE (Julian)), and ended when the city was breached on 09 Du'uzu / Tammuz (18 June 586 BCE (Julian)) according to the proleptic Hebrew calendar. These deportees were therefore evidently those who had surrendered or been captured during the siege, among them the Judeans who had defected to the Chaldeans.[1436]

XX. At the end of the 18th Regnal Year of Nebuchadnezzar neither calendar intercalates a month, so the one-(1)-month difference between them continues into his 19th Regnal Year. The proleptic Hebrew calendar thus began the 19th Regnal Year on 01 Nisannu / Nisan 3175 AM (13 March 586 BCE (Julian)), whereas the Babylonian calendar began it one (1) month later on 13 April 586 BCE (Julian).[1437] The two (2) calendars resynchronize only at the end of the 19th Regnal Year when the proleptic Hebrew calendar intercalates its 13th month Addaru 2 / Adar 2 of 3176 AM that began 03 March 585 BCE (Julian).

26.0. 19th Regnal Year of Nebuchadnezzar II == Spring 586 – 585 BCE
== 11th Regnal Year of Tzidkiyahu (Zedekiah)
== YEAR 23 OF SERVITUDE
== YEAR 12 OF EXILE OF YEHOYACHIN (JEHOIACHIN)
== YEAR 1 OF DESOLATION

XX. Jerusalem was besieged until the 11th Regnal Year of King Tzidkiyahu (Zedekiah),[1438] [1439] and by the 9th day of the 4th month, i.e. Monday 18 June 586 BCE (Julian),[1440] [1441] the famine had become so

1434 Parker and Dubberstein 1971: 28
1435 cf. Jeremiah 52:29
1436 cf. Jeremiah 38:19
1437 Parker and Dubberstein 1971: 28
1438 cf. Appendix Table X.57
1439 cf. 2 Kings 25:2; Jeremiah 1:3; 38:28; 39:1-2; 52:5
1440 Monday 18 June 586 BCE (Julian); Monday 12 June 586 BCE (Gregorian); 09 Du'uzu / Tammuz 3175 AM; Astronomical Julian Day JD_X: 1507554.5; RD_Y: 1159558; RD_X: -213870; Sabbatical Cycle: 4; Metonic Cycle: 2;
1441 cf. Appendix Table X.58

severe in the city that there was no food for the people of the land,[1442] and the city was split [open].[1443] All the Jewish men-of-war fled the city with the king Tzidkiyahu (Zedekiah), were chased by the Chaldeans, the king then captured, and his army dispersed from upon him.[1444]

XX. In the 19th Regnal Year of Nebuchadnezzar the proleptic Hebrew calendar is still one (1) month ahead of the Babylonian calendar, so that the 9th day of the 4th month, Du'uzu / Tammuz (18 June 586 BCE (Julian)), corresponds to the 8th day of the 3rd month, Simannu / Sivan, in the Babylonian calendar.[1445] Since the biblical account dates this event by the 11th Regnal Year of the Judean king Tzidkiyahu (Zedekiah),[1446] it evidently follows the proleptic Hebrew reckoning. In either case, both calendars had already begun the 19th Regnal Year of Nebuchadnezzar three (3) and two (2) months prior respectively: the proleptic Hebrew on 13 March 586 BCE (Julian) and the Babylonian on 13 April 586 BCE (Julian), so that the breach of the city falls definitively in the 19th Regnal Year of Nebuchadnezzar by either reckoning.

XX. Tzidkiyahu (Zedekiah) was then brought to the king of Babylon who was at Riblah,[1447] in modern-day Syria, where the sons of Tzidkiyahu (Zedekiah) were "slaughtered before his eyes", and then Nebuchadnezzar blinded him, bound him in fetters, and brought him to Babylon.[1448]

XX. The Book of Jeremiah recounts this episode, supplying the missing detail of the exact month (i.e. the 4th month) that is missing from the account in the Book of Kings:[1449]

4 And it was in the 9th year of his reign, in the 10th month, on [day] 10 of the month[1450] [1451] [1452] *Nebuchadnezzar King of Babylon came – he and all his army – upon Jerusalem and encamped against her, and built a siege wall against her [all] around. 5 And the city came under siege until the 11th year of the King Tzidkiyahu. 6 In the 4th month, on [day] 9 of the month,*[1453] [1454] *the famine [had] become severe in the city and there was no bread for the people of the land. 7 And the city was split [open] and all the men-of-war fled and left from the city [at] night [by] way of the gate between the double-wall that was on the garden of the King, and the Chaldeans [were] upon the city [all] around, and they went*

1442 cf. 2 Kings 25:3; Jeremiah 52:6
1443 cf. 2 Kings 25:4; Jeremiah 52:7
1444 cf. 2 Kings 25:4-5; Jeremiah 39:4-5; 52:7-8
1445 Parker and Dubberstein 1971: 28
1446 cf. Jeremiah 39:2; 52:5–6; 2 Kings 25:2–3
1447 cf. 2 Kings 25:6; Jeremiah 39:5; 52:9
1448 cf. 2 Kings 25:6-8; Jeremiah 39:6-7; 52:10-11
1449 Jeremiah 52:4-11
1450 Thursday 05 January 587 BCE (Julian); Thursday 30 December 588 BCE (Gregorian); 10 Tebetu / Tevet 3174 AM; Astronomical Julian Day JD_X: 1507025.5; RD_Y: 1159029; RD_X: -214399; Sabbatical Cycle: 3; Metonic Cycle: 1;
1451 cf. Appendix Table X.53
1452 cf. Footnote #9
1453 Monday 18 June 586 BCE (Julian); Monday 12 June 586 BCE (Gregorian); 09 Du'uzu / Tammuz 3175 AM; Astronomical Julian Day JD_X: 1507554.5; RD_Y: 1159558; RD_X: -213870; Sabbatical Cycle: 4; Metonic Cycle: 2;
1454 cf. Appendix Table X.58

by way of the plain. 8 And the army of the Chaldeans chased after the King and they caught Tzidkiyahu in the plains of Jericho, and all his army was scattered from upon him. 9 And they grabbed the King and they brought him up to the King of Babylon to Rivlah in the Land of Hamat and he spoke with him sentences.[1455] 10 And the King of Babylon slaughtered the sons of Tzidkiyahu before his eyes, and also all the Ministers of Yehudah [he] slaughtered at Rivlah. 11 And the eyes of Tzidkiyahu [he] blinded and [he] imprisoned him in bronze-snakelike-fetters[1456] and the King of Babylon brought him to Babylon, and [he] gave him [over to] the prison house until the day of his death.

XX. Jeremiah tells us that during the final days before Jerusalem's fall to Nebuchadnezzar's army, he sat in prison until the day that Jerusalem was captured by the Babylonians:

38:28 And Yirmeyahu sat in the courtyard of the prison until the day that Jerusalem was captured. And it was when Jerusalem was captured: 39:1 In the 9th year of Tzidkiyahu King of Yehudah in the 10th month[1457] [1458] [1459] *Nebuchadrezzar King of Babylon came and all his army to Jerusalem and they placed siege upon her. 2 In the 11th year of Tzidkiyahu in the 4th month on [day] 9 of the month,*[1460] [1461] *the city was split [open].*[1462]

XX. As noted above, Jeremiah equates the 10th Regnal Year of Tzidkiyahu (Zedekiah) with the 18th Regnal Year of Nebuchadnezzar. Then (during this time, i.e. the 10th Regnal Year) the army of the king of Babylon had been besieging Jerusalem[1463] since the 10th day of the 10th month of the 9th Regnal Year of Tzidkiyahu (Zedekiah).[1464] [1465] [1466] [1467] The city was besieged until the 11th Regnal Year of King Tzidkiyahu (Zedekiah).[1468] [1469]

XX. The Book of Kings confirms Jeremiah's chronology, evidently equating this 7th day of the 5th month of the 11th Regnal Year of Tzidkiyahu with the 19th Regnal Year of the Babylonian king

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1455 Literally: "sentences", but also "legal judgments".
1456 Hebrew: *Nehushtaim*
1457 Thursday 05 January 587 BCE (Julian); Thursday 30 December 588 BCE (Gregorian); 10 Tebetu / Tevet 3174 AM; Astronomical Julian Day JD_X: 1507025.5; RD_Y: 1159029; RD_X: -214399; Sabbatical Cycle: 3; Metonic Cycle: 1;
1458 cf. Appendix Table X.53
1459 cf. Footnote #9
1460 Monday 18 June 586 BCE (Julian); Monday 12 June 586 BCE (Gregorian); 09 Du'uzu / Tammuz 3175 AM; Astronomical Julian Day JD_X: 1507554.5; RD_Y: 1159558; RD_X: -213870; Sabbatical Cycle: 4; Metonic Cycle: 2;
1461 cf. Appendix Table X.58
1462 Jeremiah 38:28-39:2
1463 Jeremiah 32:1-2
1464 Thursday 05 January 587 BCE (Julian); Thursday 30 December 588 BCE (Gregorian); 10 Tebetu / Tevet 3174 AM; Astronomical Julian Day JD_X: 1507025.5; RD_Y: 1159029; RD_X: -214399; Sabbatical Cycle: 3; Metonic Cycle: 1;
1465 cf. Appendix Table X.53
1466 cf. Footnote #9
1467 2 Kings 25:1-2; Jeremiah 1:3; 52:4-5
1468 Monday 18 June 586 BCE (Julian); Monday 12 June 586 BCE (Gregorian); 09 Du'uzu / Tammuz 3175 AM; Astronomical Julian Day JD_X: 1507554.5; RD_Y: 1159558; RD_X: -213870; Sabbatical Cycle: 4; Metonic Cycle: 2;
1469 cf. Appendix Table X.58

Nebuchadnezzar:[1470] [1471]

8 And in the 5th month on [day] 7 of the month – that is the year 19 of the King Nebuchadnezzar, King of Babylon – Nebuzaradan came (the chief of the slaughterers, slave of the King of Babylon) to Jerusalem. 9 And He burned the house of YHVH and the house of the King, and all the houses of Jerusalem, and every large house he burned with fire. 10 And the walls of Jerusalem around they shattered – the entire army of the Chaldeans who were great slaughterers. 11 And the rest of the people who remained in the city, and those who were falling who fell upon the King of Babylon, and the rest of the mass [of people] Nebuzaradan, Chief of the Slaughterers, exiled.[1472]

XX. The Book of Jeremiah recounts the same episode, but it will be noticed that one detail (i.e the day of the month) is not the same as mentioned in 2 Kings 25:8 (i.e. 7th day of the month in 2 Kings vs. the 10th day of the month in Jeremiah).[1473] [1474]

12 And in the 5th month on [day] 10 of the month – that is year 19, year [19] of king Nebuchadnezzar king of Babylon – Nebuzaradan, the chief of the slaughterers, came [and] stood before the king of Babylon in Jerusalem. 13 And he burned the house of YHVH and the house of the king, and all the houses of Jerusalem, and every large house he burned in fire. 14 And all the walls of Jerusalem surrounding [the city] they shattered – all the army of the Chaldeans who were with the chief of the slaughterers.[1475]

XX. In the same Hebrew paragraph[1476] that mentions Tzidkiyahu's (Zedekiah's) replacement of Yehoyachin (Jehoiachin) as king of Judah, Jeremiah tells us that during this time [i.e. after the accession of Tzidkiyahu (Zedekiah)] the Chaldeans were besieging Jerusalem, and disengaged from the siege upon hearing that the army of Egypt had left Egypt.[1477]

XX. Although the exact date of the temporary lifting of the siege is not recorded in the Hebrew Scriptures, it is reasonable to place this event in the 11th Year of the EXILE OF YEHOYACHIN (JEHOIACHIN) == 10th Regnal Year of Tzidkiyahu (Zedekiah) == 18th Regnal Year of Nebuchadnezzar == (Spring 587 – 586 BCE (Julian)).[1478]

1470 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;

1471 cf. Appendix Table X.59

1472 2 Kings 25:8-11

1473 Wednesday 18 July 586 BCE (Julian); Wednesday 12 July 586 BCE (Gregorian); 10 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507584.5; RD_Y: 1159588; RD_X: -213840; Sabbatical Cycle: 4; Metonic Cycle: 2;

1474 cf. Appendix Table X.59

1475 Jeremiah 52:12-14

1476 Literally: "doors"; cf. Jeremiah 36:23

1477 cf. Jeremiah 37:1-5

1478 cf. Jeremiah 32:1-2

5 And the army of Pharaoh left from Egypt, and the Chaldeans [who were] sieging Jerusalem heard, and upon their hearing, they ascended from upon Jerusalem.[1479]

XX. Jeremiah then prophesies to the king of Judah (i.e. Tzidkiyahu (Zedekiah)) and declares that the army of Pharaoh will return back to Egypt, and that the Chaldeans will return to war against the city of Jerusalem, capture her, and burn her in fire.[1480]

XX. After the destruction of Jerusalem in the 4th and 5th proleptic Hebrew months in the 11th Regnal Year of Tzidkiyahu (Zedekiah) – which is equivalent to the 12th year of the EXILE OF YEHOYACHIN (JEHOIACHIN) – in the 10th month on [day] five (5) of the month a refugee from Jerusalem came to Ezekiel saying, "The city has been struck!" Ezekiel mentions that the previous evening before the coming of the refugee that GOD opened Ezekiel's mouth, and he was no longer mute.[1481] The day of the refugee's arrival works out to be Sunday 09 December 586 BCE (Julian),[1482] [1483] and thus the opening of Ezekiel's mouth took place on Saturday evening the night before the refugee's arrival.

XX. In the 12th year of the EXILE OF YEHOYACHIN (JEHOIACHIN) in the 12th month on [day] one (1) of the month the word of YHVH was to Ezekiel: a prophecy against Pharaoh king of Egypt.[1484] This works out to be Saturday 02 February 585 BCE (Julian).[1485] [1486]

XX. In the 12th year of the EXILE OF YEHOYACHIN (JEHOIACHIN) [in the 12th Month] on [day] 15 of the month the word of YHVH was to Ezekiel: a prophecy against the mass [of people] of Egypt.[1487] This works out to be Saturday 16 February 585 BCE (Julian).[1488] [1489]

XX. Throughout the remainder of the 19th Regnal Year of Nebuchadnezzar the proleptic Hebrew calendar remains one (1) month ahead of the Babylonian calendar, so that Ezekiel's dates in the 10th and 12th months fall in the Babylonian 9th, 10th, and 11th months.[1490] Since 3176 AM is a leap year in the proleptic Hebrew calendar, Ezekiel's 12th month is Addaru 1 / Adar 1,[1491] followed by the 13th intercalary month Addaru 2 / Adar 2 of 3176 AM that began 03 March 585 BCE (Julian), i.e. the same

1479 cf. Jeremiah 37:5
1480 cf. Jeremiah 37:6-11
1481 cf. Ezekiel 33:21-22
1482 Sunday 09 December 586 BCE (Julian); Sunday 03 December 586 BCE (Gregorian); 05 Tebetu / Tevet 3176 AM; Astronomical Julian Day JD_X: 1507728.5; RD_Y: 1159732; RD_X: -213696; Sabbatical Cycle: 5; Metonic Cycle: 3;
1483 cf. Appendix Table X.60
1484 cf. Ezekiel 32:1
1485 Saturday 02 February 585 BCE (Julian); Saturday 27 January 585 BCE (Gregorian); 01 Addaru / Adar I 3176 AM; Astronomical Julian Day JD_X: 1507783.5; RD_Y: 1159787; RD_X: -213641; Sabbatical Cycle: 5; Metonic Cycle: 3;
1486 cf. Appendix Table X.61
1487 cf. Ezekiel 32:17
1488 Saturday 16 February 585 BCE (Julian); Saturday 10 February 585 BCE (Gregorian); 15 Addaru / Adar I 3176 AM; Astronomical Julian Day JD_X: 1507797.5; RD_Y: 1159801; RD_X: -213627; Sabbatical Cycle: 5; Metonic Cycle: 3;
1489 cf. Appendix Table X.61
1490 Parker and Dubberstein 1971: 28
1491 cf. Ezekiel 32:1; 33:21

day as the Babylonian 12th month Addaru / Adar (there being no Babylonian intercalation that year). Thus the one-(1)-month difference caused by the Babylonian intercalary Addaru 2 / Adar 2 that began 25 March 587 BCE (Julian) in the 17th Regnal Year comes to an end, and both calendars begin the 20th Regnal Year of Nebuchadnezzar synchronized.

27.0. 32nd Regnal Year of Nebuchadnezzar II == Spring 573 – 572 BCE
== YEAR 36 OF SERVITUDE
== YEAR 25 OF EXILE OF YEHOYACHIN (JEHOIACHIN)
== YEAR 14 OF DESOLATION

XX. In the 25th year of the EXILE OF YEHOYACHIN (JEHOIACHIN)[1492] at "the Head of the Year" (i.e. *Rosh HaShanah*), on [day] ten (10) of the Month (i.e. *Yom Kippur*) – i.e. at the beginning of the Hebrew civil year (*Anno Mundi* AM) in the 7th month – in Year 14 after the City was Struck, "on this very day" the hand of YHVH upon Ezekiel, and in "visions of Elohim" He brings Ezekiel to the land of Israel.[1493] This works out to be Monday 21 September 573 BCE (Julian).[1494] [1495]

XX. According to Ezekiel, this date was in Year 14 after the city was stuck (i.e. 14 years after the city was destroyed in 586 BCE (Julian)) in Year 25 of the EXILE OF YEHOYACHIN (JEHOIACHIN) that began Saturday 23 March (Julian) 597 BCE and took place in the 8th Regnal Year of Nebuchadnezzar 10 Nisannu / Nisan,[1496] and according to Finegan, Wiseman, and Parker & Dubberstein works out to be 22 April 597 BCE.[1497]

XX. However, as noted previously, the most likely date for the beginning of YEAR ONE (1) OF THE EXILE OF YEHOYACHIN (JEHOIACHIN) was the proleptic Hebrew, i.e. Saturday 23 March 597 BCE (Julian).[1498] [1499]

XX. At the beginning of the 31st Regnal Year of Nebuchadnezzar, the proleptic Hebrew and Babylonian calendars were synchronized, beginning the year on 30 March and 01 April 574 BCE (Julian) respectively, but later that year the Babylonians intercalated a 2nd Ululu / Elul (Ululu 2 / Elul 2) that began 25 September 574 BCE (Julian) which the proleptic Hebrew calendar does not.[1500] Consequently,

1492 cf. Appendix Table X.62
1493 cf. Ezekiel 40:1-2
1494 Monday 21 September 573 BCE (Julian); Monday 15 September 573 BCE (Gregorian); 10 Tashritu / Tishri 3189 AM; Astronomical Julian Day JD_X: 1512398.5; RD_Y: 1164402; RD_X: -209026; Sabbatical Cycle: 4; Metonic Cycle: 16;
1495 cf. Appendix Table X.63
1496 cf. 2 Kings 24:12
1497 Finegan 1998: 25
1498 Saturday 23 March 597 BCE (Julian); Saturday 17 March 597 BCE (Gregorian); 10 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503450.5; RD_Y: 1155454; RD_X: -217974; Sabbatical Cycle: 7; Metonic Cycle: 10;
1499 cf. Appendix Table X.34
1500 Parker and Dubberstein 1971: 5, 28

throughout the 32nd Regnal Year of Nebuchadnezzar the proleptic Hebrew calendar is one (1) month ahead of the Babylonian calendar, beginning the year on 01 Nisannu / Nisan 3188 AM (19 March 573 BCE (Julian)), whereas the Babylonian calendar began it one (1) month later, on 19 April 573 BCE (Julian). Since neither calendar intercalates during this year,[1501] the one-(1)-month difference continues into the 33rd Regnal Year of Nebuchadnezzar.[1502]

28.0. 34th Regnal Year of Nebuchadnezzar II == Spring 571 – 570 BCE
== YEAR 38 OF SERVITUDE
== YEAR 27 OF EXILE OF YEHOYACHIN (JEHOIACHIN)
== YEAR 16 OF DESOLATION

XX. In the 27th year of the EXILE OF YEHOYACHIN (JEHOIACHIN),[1503] in the 1st month, on [day] one (1) of the month, the word of YHVH was to Ezekiel.[1504] This works out to be Sunday 27 March 571 BCE (Julian).[1505] [1506]

XX. At the beginning of the 33rd Regnal Year of Nebuchadnezzar, the proleptic Hebrew calendar was one (1) month ahead of the Babylonian calendar because of the Babylonian intercalary Ululu 2 / Elul 2 that began 25 September 574 BCE (Julian) in his 31st Regnal Year.[1507] At the end of the 33rd Regnal Year both calendars intercalated, the proleptic Hebrew calendar in its 13th month Addaru 2 / Adar 2 of 3190 AM that began 26 February 571 BCE (Julian), and the Babylonian calendar in its 13th month Addaru 2 / Adar 2 that began 28 March 571 BCE (Julian), one (1) month later.[1508]

XX. Because the proleptic Hebrew calendar intercalated first, the two (2) calendars were synchronized for that single month only.[1509] Consequently, throughout the 34th Regnal Year of Nebuchadnezzar the proleptic Hebrew calendar remains one (1) month ahead of the Babylonian calendar having begun the year on 01 Nisannu / Nisan 3190 AM (27 March 571 BCE (Julian)), whereas the Babylonian calendar began it on 26 April 571 BCE (Julian). Since neither calendar intercalates during this year, the one-(1)-month difference continues into the 35th Regnal Year of Nebuchadnezzar.

1501 Parker and Dubberstein reject a claimed Babylonian Ululu 2 / Elul 2 in the 32nd Regnal Year
1502 Parker and Dubberstein 1971: 5, 28
1503 cf. Appendix Table X.64
1504 cf. Ezekiel 29:17
1505 Sunday 27 March 571 BCE (Julian); Sunday 21 March 571 BCE (Gregorian); 01 Nisannu / Nisan 3190 AM; Astronomical Julian Day JD_X: 1512950.5; RD_Y: 1164954; RD_X: -208474; Sabbatical Cycle: 5; Metonic Cycle: 17;
1506 cf. Appendix Table X.65
1507 Parker and Dubberstein 1971: 5, 28
1508 Parker and Dubberstein 1971: 5, 28
1509 Parker and Dubberstein 1971: 5, 28

29.0. 1st Regnal Year of Amel-Marduk == Spring 561 – 560 BCE
== YEAR 48 OF SERVITUDE
== YEAR 37 OF EXILE OF YEHOYACHIN (JEHOIACHIN)
== YEAR 26 OF DESOLATION

XX. The Book of Jeremiah records the date of the freeing of Yehoyachin (Jehoiachin) from Babylonian prison in the 37th year of the EXILE OF YEHOYACHIN (JEHOIACHIN):[1510]

In the 37th year of Exile of Yehoyachin king of Yehudah, in 12th month, on [day] 25 of the month, Evil-Merodach king of Babylon in the [1st Regnal] Year of his reign lifted the head of Yehoyachin king of Yehudah and took him out of the House of the Jailed.[1511]

XX. This text is slightly different than the one found in the Book of 2 Kings that states:

In the 37th year of Exile of Yehoyachin king of Yehudah, in the 12th month, on [day] 27 of the month, Evil-Merodach king of Babylon in the [1st Regnal] Year of his reign lifted the head of Yehoyachin king of Yehudah from the Jailhouse.[1512]

XX. These dates work out to be:

- Day 25 of the 12th Month (Addaru / Adar), i.e. Monday 21 March 560 BCE (Julian)[1513] [1514]

- Day 27 of the 12th Month (Addaru / Adar), i.e. Wednesday 23 March 560 BCE (Julian)[1515] [1516]

XX. Throughout the 43rd and final Regnal Year of Nebuchadnezzar, the proleptic Hebrew calendar was one (1) month ahead of the Babylonian calendar because of the Babylonian intercalary Addaru 2 / Adar 2 that began 19 March 562 BCE (Julian) at the end of his 42nd Regnal Year.[1517] The proleptic Hebrew calendar then resynchronized with the Babylonian calendar when it intercalated its 13th month Addaru 2 / Adar 2 of 3200 AM that began 08 March 561 BCE (Julian), so that both calendars began the 1st Regnal Year of Amel-Marduk on the same day, 01 Nisannu / Nisan 3200 AM (06 April 561 BCE (Julian)).

1510 cf. Appendix Table X.66
1511 cf. Jeremiah 52:31
1512 cf. 2 Kings 25:27
1513 Monday 21 March 560 BCE (Julian); Monday 15 March 560 BCE (Gregorian); 25 Addaru / Adar 3201 AM; Astronomical Julian Day JD_X: 1516962.5; RD_Y: 1168966; RD_X: -204462; Sabbatical Cycle: 2; Metonic Cycle: 9;
1514 cf. Appendix Table X.67
1515 Wednesday 23 March 560 BCE (Julian); Wednesday 17 March 560 BCE (Gregorian); 27 Addaru / Adar 3201 AM; Astronomical Julian Day JD_X: 1516964.5; RD_Y: 1168968; RD_X: -204460; Sabbatical Cycle: 2; Metonic Cycle: 9;
1516 cf. Appendix Table X.67
1517 Parker and Dubberstein 1971: 5, 28

XX. Consequently, the release of Yehoyachin (Jehoiachin) in the 37th year of his exile in the 12th month falls on exactly the same Julian day in both calendars – whether on the 25th day, Monday 21 March 560 BCE (Julian),[1518] or on the 27th day, Wednesday 23 March 560 BCE (Julian).[1519]

30.0. 3rd Regnal Year of Nabonidus == Spring 553 – 552 BCE
== 1st Year of Coregency of Nabonidus' son, Belshazzar
== YEAR 56 OF SERVITUDE
== YEAR 45 OF EXILE OF YEHOYACHIN (JEHOIACHIN)
== YEAR 34 OF DESOLATION

XX. According to the Verse Account of Nabonidus, when the 3rd Regnal Year of Nabonidus was about to begin, he entrusted the army[?] to his oldest son, his first born [Belshazzar], and the troops in the country he ordered under his command. He let everything go, and entrusted the kingship to him.[1520] We can reasonably assume that since Belshazzar was only coregent, that there was no Year 0 Accession Year, i.e. that his 1st Regnal Year began immediately upon his appointment, or at the beginning of the 3rd Regnal Year of Nabonidus.[1521] Daniel's dream vision described in Daniel chapter 7 should be dated to this year 1st Regnal Year of Belshazzar, 553 – 552 BCE (Julian).[1522]

XX. By the middle of the 4th Regnal Year (i.e. Summer 552 BCE (Julian)) of Nabonidus, the double kingship instituted by Nabonidus was already in effect as Nabonidus had left to reside in Teima, where – according to Beaulieu – he must have arrived no later than the 4th month of his 4th Regnal Year.[1523]

XX. Several texts prove that Belshazzar was performing official duties in the king's stead from the 4th Regnal Year to the beginning of the 13th Regnal Year of Nabonidus. Therefore, one can easily determine with the help of archival texts when Nabonidus was replaced by Belshazzar as the acting ruler in Babylonia.[1524]

XX. Only the Verse Account of Nabonidus states when the coregency of Belshazzar began, namely when the 3rd Regnal Year of Nabonidus was about to begin, at which time Nabonidus entrusted the kingship to his firstborn son and set out for Teima.[1525] The earliest surviving archival texts attesting

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1518 cf. Jeremiah 52:31
1519 cf. 2 Kings 25:27
1520 Oppenheim; "Verse Account of Nabonidus" (Column ii.5 – Column ii.6); Livius.org; https://www.livius.org/sources/content/anet/verse-account-of-nabonidus/
1521 cf. Appendix Table X.68
1522 cf. Daniel 7:1
1523 Beaulieu 1989: 159-160
1524 Beaulieu 1989: 159-160
1525 Oppenheim; "Verse Account of Nabonidus" (Column ii.5 – Column ii.6); Livius.org; https://www.livius.org/sources/content/anet/verse-account-of-nabonidus/

Belshazzar performing official duties date from the 4th Regnal Year of Nabonidus.[1526] This does not contradict the Verse Account since Belshazzar may well have performed such duties in the 3rd Regnal Year, and texts from that year simply may not have survived.

XX. However, the matter is not so easy for the last years of the reign since Belshazzar abruptly disappears as regent in the beginning of the 13th Regnal Year, but Nabonidus is not attested in dated archival texts after this period either.[1527]

XX. In other words, neither the king nor his son is attested performing official duties after the beginning of the 13th Regnal Year. This might be due to a gap in the documentation. Yet the number of available texts considerably increases in that period, making it difficult to attribute the absence of Nabonidus and Belshazzar to mere coincidence.[1528]

XX. From the study of official letters, which can be dated approximately upon prosopographical evidence, they show that Nabonidus had resumed rule by the middle of his 13th Regnal Year.[1529]

XX. Belshazzar's last attestation in his capacity as co-regent is dated to the 3rd month of the 13th Regnal Year. According to Beaulieu, since up to that point Belshazzar was regularly performing official duties, one may conclude that Nabonidus returned to Babylon shortly after [the 7th month of] Tashritu / Tishri of the 13th Regnal Year.[1530]

XX. At the beginning of the 3rd Regnal Year of Nabonidus (i.e. the 1st Regnal Year of co-regent Belshazzar), the proleptic Hebrew calendar was one (1) month ahead of the Babylonian calendar because of the Babylonian intercalary Addaru 2 / Adar 2 that began 20 March 554 BCE (Julian) at the end of his 1st Regnal Year, with no intercalation in his 2nd Regnal Year.[1531] The proleptic Hebrew calendar thus began the 3rd Regnal Year on 01 Nisannu / Nisan 3208 AM (08 March 553 BCE (Julian)), whereas the Babylonian calendar began it one (1) month later, on 07 April 553 BCE (Julian).[1532]

XX. At the end of this year both calendars intercalated, the proleptic Hebrew calendar in its 13th month Addaru 2 / Adar 2 of 3209 AM that began 25 February 552 BCE (Julian), and the Babylonian calendar in its 13th month Addaru 2 / Adar 2 that began 28 March 552 BCE (Julian), one (1) month later. Because the proleptic Hebrew calendar intercalated first, the two (2) calendars were synchronized for that single month only, and the one-(1)-month difference continued into the 4th Regnal Year of Nabonidus.[1533]

1526 Beaulieu 1989: 159-160
1527 Beaulieu 1989: 159-160
1528 Beaulieu 1989: 159-160
1529 Beaulieu 1989: 159-160
1530 Beaulieu 1989: 159-160
1531 Parker and Dubberstein 1971: 5, 7, 29
1532 Parker and Dubberstein 1971: 29
1533 Parker and Dubberstein 1971: 7, 29

31.0. 5th Regnal Year of Nabonidus == Spring 551 – 550 BCE
== 3rd Year of Coregency of Nabonidus' son, Belshazzar
== YEAR 58 OF SERVITUDE
== YEAR 47 OF EXILE OF YEHOYACHIN (JEHOIACHIN)
== YEAR 36 OF DESOLATION

XX. Daniel's vision described in Daniel chapter 8 should be dated to this 3rd Regnal Year of Belshazzar, i.e. the 5th Regnal Year of Nabonidus, 551 – 550 BCE (Julian).[1534] [1535]

XX. At the beginning of the 5th Regnal Year of Nabonidus, the proleptic Hebrew calendar was one (1) month ahead of the Babylonian calendar because of the Babylonian intercalary Addaru 2 / Adar 2 that began 28 March 552 BCE (Julian) at the end of his 3rd Regnal Year with no intercalation in either calendar in his 4th Regnal Year.[1536] The proleptic Hebrew calendar thus began the 5th Regnal Year on 01 Nisannu / Nisan 3210 AM (16 March 551 BCE (Julian)), whereas the Babylonian calendar began it one (1) month later, on 16 April 551 BCE (Julian).[1537]

XX. At the end of this year the proleptic Hebrew calendar resynchronized with the Babylonian calendar when it intercalated its 13th month Addaru 2 / Adar 2 of 3211 AM that began 06 March 550 BCE (Julian), so that both calendars began the 6th Regnal Year of Nabonidus synchronized on 04 and 05 April 550 BCE (Julian) respectively.[1538]

32.0. 1st Regnal Year of Cyrus II == Spring 538 – 537 BCE
== YEAR 71 OF SERVITUDE
== 70 YEARS OF SERVITUDE COMPLETE
== YEAR 60 OF EXILE OF YEHOYACHIN (JEHOIACHIN)
== YEAR 49 OF DESOLATION

XX. According to Parker and Dubberstein,[1539] the first tablets of Persian King Cyrus II after having conquered Babylon are from Autumn 539 BCE (Julian). Thus Cyrus II ascended to the throne of Babylon in his Year 0 Accession Year from Spring 539 – 538 BCE (Julian), and then subsequently

1534 cf. Appendix Table X.69
1535 cf. Daniel 8:1
1536 Parker and Dubberstein 1971: 7, 29
1537 Parker and Dubberstein 1971: 29
1538 Parker and Dubberstein 1971: 29
1539 Parker and Dubberstein 1956: 13-14

began his 1st Regnal Year over Babylon from Spring 538 – 537 BCE (Julian).[1540] In this 1st Regnal Year of Cyrus II (over Babylon), the Persian emperor issues an edict that all Jews can return home.[1541]

XX. Counting exactly 70 Hebrew years from the date of ~Late-Summer/Autumn 608 BCE (Julian) – when Yehoyakim (Jehoiakim / Elyakim) had been enthroned as a vassal-puppet-slave-king by Egyptian Pharaoh Necho II – brings us to the 1st Regnal Year of Cyrus (Spring 538 – 537 BCE (Julian)). At this time, Cyrus issued a proclamation in this 1st Regnal Year to free the Jews to allow them to return to Jerusalem to build their temple, thus completing 70 YEARS OF SERVITUDE in YEAR 60 OF THE EXILE OF YEHOYACHIN (JEHOIACHIN).[1542]

XX. Although it is unclear exactly when Cyrus issued this edict – but assuming it was in his 1st month of his 1st Regnal Year (over Babylon) – and reasoning that the Jews who chose to return quickly organized and left for Jerusalem, it is evident that by the 7th month (i.e. Tashritu / Tishri) they were already in Jerusalem. As this month approached, the people who had arrived by this time gathered "as one man to Jerusalem". Then Yeshua son of Yotzadak and his brothers the priests, as well as Zerubavel son of She'altiel and his brothers, built the altar of the GOD of Israel, and established it. From [day] one (1) of the 7th month [of that 1st year] they began to sacrifice rising offerings to YHVH. This date for *Rosh HaShanah* works out to be Thursday 16 September 538 BCE (Julian).[1543] [1544] [1545]

XX. They subsequently celebrated the Feast of *Sukkot* (i.e. Tabernacles) from day 15 through day 21, as well as the 8th day of Simchat Torah (i.e. *Shmini Atzeret*) on day 22,[1546] and afterwards made the monthly sacrifice, but the Hall [of the Temple] of YHVH had not yet been established with its foundation.[1547]

XX. If Daniel had been exiled in 605 – 604 BCE (Julian) in the 3rd Regnal Year of Yehoyakim (Jehoiakim) – as described in the 1st chapter of the Book of Daniel[1548] – at the age of 10-13, then he would have been 77-80 years old in the 1st Regnal Year of Cyrus (over Babylon) in 538 – 537 BCE (Julian), at which time he may have returned to Jerusalem, as suggested in Daniel 1:21,[1549] albeit briefly since Daniel has a vision on the Tigris river in the 3rd Regnal Year of Cyrus (i.e. Spring 536 – 535 BCE (Julian)) as recorded in Daniel 10-12,[1550] [1551] as well as interacts with the later Persian king, Darius (r.

1540 cf. Appendix Table X.70
1541 cf. Ezra 1:1-11; 2:1; 5:13; 6:3; 2 Chronicles 36: 20-23; Daniel 1:21
1542 cf. Ezra 1:1-11; 2:1; 5:13; 6:3; 2 Chronicles 36: 20-23; Daniel 1:21
1543 Thursday 16 September 538 BCE (Julian); Thursday 10 September 538 BCE (Gregorian); 01 Tashritu / Tishri 3224 AM; Astronomical Julian Day JD_X: 1525176.5; RD_Y: 1177180; RD_X: -196248; Sabbatical Cycle: 4; Metonic Cycle: 13;
1544 cf. Appendix Table X.70, X.71
1545 cf. Ezra 3:1-7
1546 cf. Appendix Table X.71
1547 cf. Ezra 3:1-7
1548 cf. Daniel 1:1
1549 cf. Daniel 1:21
1550 cf. Appendix Table X.74, X.75
1551 cf. Daniel 10-12

521 – 486 BCE (Julian)),[1552] and experiences his prophetic, messianic vision in the 1st Regnal Year of Darius (i.e. Spring 521 – 520 BCE (Julian))[1553] as recorded in Daniel 9.[1554]

XX. At the beginning of the 1st Regnal Year of Cyrus II, the proleptic Hebrew and Babylonian calendars were synchronized, the proleptic Hebrew calendar having intercalated its 13th month Addaru 2 / Adar 2 of 3222 AM that began 04 March 539 BCE (Julian) at the end of the 16th Regnal Year of Nabonidus.[1555] Both calendars thus began the 1st Regnal Year of Cyrus II in step: the proleptic Hebrew on 01 Nisannu / Nisan 3223 AM (23 March 538 BCE (Julian)) and the Babylonian on 24 March 538 BCE (Julian). Since neither calendar intercalated during or at the end of this year, they remained synchronized into the 2nd Regnal Year of Cyrus II.[1556]

33.0. 2nd Regnal Year of Cyrus II == Spring 537 – 536 BCE == YEAR 61 OF EXILE OF YEHOYACHIN (JEHOIACHIN) == YEAR 50 OF DESOLATION

XX. In the 2nd year of their return, reasonably in the 2nd Regnal Year of Cyrus,[1557] in the 2nd month (i.e. Ayyaru / Iyar), the exiles who had returned charged the Levites who were 20 years of age and above with the labor of rebuilding the Temple. The builders established the foundations of the Hall [of the Temple] of YHVH, and there was a celebration and happiness.[1558] This works out to be April / May 537 BCE (Julian) in both the proleptic Hebrew and the Babylonian calendars.[1559]

XX. According to Parker and Dubberstein's tables, the 2nd Regnal Year of Cyrus II began on 01 Nisannu / Nisan, 12 March 537 BCE (Julian), and its 2nd month of Ayyaru / Iyar began on 10 April 537 BCE (Julian).[1560] In the proleptic Hebrew calendar, the 2nd month of Ayyaru / Iyar of 3224 AM likewise began on 10 April 537 BCE (Julian) so that the two (2) calendars were synchronized at the beginning of this year, and the 2nd month in which the foundations of the Temple were laid[1561] was the same month as well as the same day in both.

XX. Later in this year Parker and Dubberstein's tables place an attested Babylonian intercalary Ululu 2 / Elul 2 that began 06 September 537 BCE (Julian), so that the proleptic Hebrew calendar became

1552 cf. Daniel 6
1553 cf. Appendix Table X.76
1554 cf. Daniel 9:1-27
1555 Parker and Dubberstein 1971: 29
1556 Parker and Dubberstein 1971: 7, 29
1557 cf. Appendix Table X.72
1558 cf. Ezra 3:8-13
1559 cf. Appendix Table X.73
1560 Parker and Dubberstein 1971: 29
1561 cf. Ezra 3:8

one (1) month ahead of the Babylonian calendar.[1562]

XX. The proleptic Hebrew calendar intercalates its 13th month Addaru 2 / Adar 2 of 3225 AM that began 28 February 536 BCE (Julian) at the end of this year. The two (2) calendars thus began the 3rd Regnal Year of Cyrus II synchronized.[1563]

34.0. 3rd Regnal Year of Cyrus II == Spring 536 – 535 BCE == YEAR 62 OF EXILE OF YEHOYACHIN (JEHOIACHIN) == YEAR 51 OF DESOLATION

XX. In the 3rd Regnal Year of Cyrus,[1564] Daniel has a vision on the Tigris river after having abstained from choice bread, wine, and meat for "three (3) weeks of days". On day 24 of the 1st month of Nisannu / Nisan, Daniel was on the Tigris river,[1565] and sees a vision as recorded in three (3) chapters of Daniel 10-12. This works out to be Saturday 21 April 536 BCE (Julian).[1566] [1567] In other words, Daniel experienced this vision on the Sabbath (Shabbat) 21 days after beginning his fast from choice bread, meat, and wine having fasted through the Passover holiday.

XX. In this same vision, the "man dressed in fabrics" continues speaking with Daniel:

20 And he said, "Do you know why I have come to you? And now I will return to battle with the Minister of Persia, and I am leaving, and behold the Minister of Greece is coming. 21 However, I will tell you the recorded [matter] in [the] Writing of Truth, and there is not [any] one who is becoming strong with me upon these [Ministers of Persia and Greece] except for Michael your[1568] Minister."[1569]

XX. The man dressed in fabrics continues speaking in chapter 11:

1 "And I, in year one of Daryavesh (Darius) the Mede,[1570] my position[1571] [will be] for maintaining and for a stronghold to him. 2 And now truth I will tell to you: behold another three

1562 Parker and Dubberstein 1971: 25, 29
1563 Parker and Dubberstein 1971: 29
1564 cf. Appendix Table X.74
1565 cf. Daniel 10:1-4
1566 Saturday 21 April 536 BCE (Julian); Saturday 15 April 536 BCE (Gregorian); 24 Nisannu / Nisan 3225 AM; Astronomical Julian Day JD_X: 1525759.5; RD_Y: 1177763; RD_X: -195665; Sabbatical Cycle: 5; Metonic Cycle: 14;
1567 cf. Appendix Table X.75
1568 The word "your" here is the plural tense.
1569 cf. Daniel 10:20-21
1570 i.e. Darius (r. 521 – 486 BCE (Julian))
1571 Most certainly it is clear from the Hebrew that this is not in the past tense.

kings[1572] *stand for Persia, and the fourth*[1573] *will make wealthy, a great wealth from all [before him], and at his strength in his wealth, he will arouse all [his kingdom to war] with the kingdom of Greece. 3 And a mighty-warrior king*[1574] *stood and ruled a great dominion, and did as [according to] his will. 4 And at his standing his kingdom will break and will be divided to the four winds*[1575] *of the skies, and not for his end, and not as his dominion that he ruled because his kingdom will be uprooted, and to others besides these."*[1576]

XX. According to Parker and Dubberstein's tables, which include the Babylonian Ululu 2 / Elul 2 of the previous year, the 3rd Regnal Year of Cyrus II began on 01 Nisannu / Nisan, 31 March 536 BCE (Julian), whereas in the proleptic Hebrew calendar it began on 01 Nisannu / Nisan 3225 AM (29 March 536 BCE (Julian)). Therefore the two (2) calendars were synchronized, and Daniel's vision on the 24th day of the 1st month falls in the same month in both.[1577] [1578]

XX. At the end of this year Parker and Dubberstein's tables place an attested Babylonian intercalary Addaru 2 / Adar 2 that began 20 March 535 BCE (Julian),[1579] while the proleptic Hebrew calendar does not intercalate in this year. The proleptic Hebrew calendar is thus one (1) month ahead of the Babylonian calendar at the beginning of the 4th Regnal Year of Cyrus II.

35.0. 1st Regnal Year of Darius I == Spring 521 – 520 BCE
== YEAR 77 OF EXILE OF YEHOYACHIN (JEHOIACHIN)
== YEAR 66 OF DESOLATION

XX. According to Parker and Dubberstein, the Persian king Darius the Great killed Bardiya in 522 BCE (Julian),[1580] and thus began his Year 0 Accession Year to the throne of Babylon through Spring 521 BCE (Julian). In the same Year 0 Accession Year within the same month after just a few days, Nidintu-Bel (Nebuchadnezzar III) then usurps the Babylonian cities Sippar, Babylon, and Borsippa before being defeated and killed by Darius in late 522 BCE (Julian).[1581]

1572 i.e. another three (3) kings counting from the time of speaking in the 3rd Regnal Year of Cyrus: 1.) Cyrus II (r. 538 – 530 BCE (Julian)); 2.) Cambyses II (r. 529 – 522 BCE (Julian)); 3.) Darius (r. 521 – 486 BCE (Julian));
1573 i.e. Xerxes I (r. 485 – 465 BCE (Julian))
1574 i.e. Alexander the Great (r. 336 – 323 BCE (Julian))
1575 Hebrew: "four spirits"
1576 cf. Daniel 11:1-4
1577 cf. Daniel 10:4
1578 Parker and Dubberstein 1971: 29
1579 Parker and Dubberstein 1971: 25, 29
1580 Parker and Dubberstein 1956: 15
1581 Parker and Dubberstein 1956: 15

XX. Later, in the year that followed, i.e. the 1st Regnal Year of Darius, he secures the throne by also defeating Nebuchadnezzar IV, and then he is reaccepted at Borsippa, Sippar, and Babylon in late 521 BCE (Julian).[1582]

XX. It is within this context of the 1st Regnal Year of Darius,[1583] i.e. 521 – 520 BCE (Julian), i.e. YEAR 77 OF THE EXILE OF YEHOYACHIN (JEHOIACHIN), i.e. YEAR 66 OF THE DESOLATION after the destruction of the Temple, that Daniel contemplates and understands the Word of the LORD to Jeremiah about the 70 years of desolation. Thus Daniel understands that the desolation of Jerusalem would soon end and be complete within five (5) years.[1584] The content of Daniel's prophetic, messianic vision will be discussed in Sections 64.0 – 73.0.

XX. If Daniel had been exiled in 605 – 604 BCE (Julian) in the 3rd Regnal Year of Yehoyakim (Jehoiakim) – as described in the 1st chapter of the Book of Daniel[1585] – at the age of 10-13, then he would have been 94-97 years old in the 1st Regnal Year of Darius in 521 – 520 BCE (Julian).

XX. At the beginning of the 1st Regnal Year of Darius I, the proleptic Hebrew calendar was one (1) month ahead of the Babylonian calendar because of the Babylonian intercalary Addaru 2 / Adar 2 that began 15 March 521 BCE (Julian) at the end of his accession year.[1586] The proleptic Hebrew calendar thus began the 1st Regnal Year of Darius I on 01 Nisannu / Nisan 3240 AM (13 March 521 BCE (Julian)), whereas the Babylonian calendar began it one (1) month later, on 14 April 521 BCE (Julian).[1587] At the end of this year the proleptic Hebrew calendar resynchronized with the Babylonian calendar when it intercalated its 13th month Addaru 2 / Adar 2 of 3241 AM that began 03 March 520 BCE (Julian), so that both calendars began the 2nd Regnal Year of Darius I synchronized.[1588]

36.0. 2nd Regnal Year of Darius I == Spring 520 – 519 BCE
== YEAR 78 OF EXILE OF YEHOYACHIN (JEHOIACHIN)
== YEAR 67 OF DESOLATION

XX. The Book of Ezra tells us that the Jews who had returned from Exile to Jerusalem had begun rebuilding the Temple with the permission of Cyrus II (r. 538 – 530 BCE (Julian)) given in his 1st Regnal Year,[1589] [1590] had had their reconstruction work frustrated by the people of the land all the days

1582 Parker and Dubberstein 1956: 16
1583 cf. Appendix Table X.76
1584 cf. Daniel 9
1585 cf. Daniel 1:1
1586 Parker and Dubberstein 1971: 7, 30
1587 Parker and Dubberstein 1971: 30
1588 Parker and Dubberstein 1971: 30
1589 cf. Ezra 1:1-11; 3:7; 4:3-5; 5:13-17; 6:3
1590 cf. Appendix Table X.70

of Cyrus until the reign of Darius,[1591] and had been canceled by Artachshasta (i.e. Artaxerxes I) until the 2nd Regnal Year of Darius I king of Persia (r. 521 – 486 BCE (Julian)).[1592] [1593]

XX. Since there were only two (2) Persian kings between Cyrus II and Darius I, i.e. Cambyses II (r. 529 – 522 BCE (Julian)) and Smerdis / Bardiya (r. 522 BCE (Julian)) – who never acceded to the throne in the time from the record of his first tablets until being killed by Darius after four (4) months and four (4) days – the cancellation of the work to rebuild the Temple evidently could only have been decreed by Cambyses II. Thus several mentions[1594] of Artachshasta (i.e. Artaxerxes I, r. 464 – 424 BCE (Julian)) within the Book of Ezra are either:

1.) unintentional scribal errors writing the name of the then-current Persian king Artachshasta (i.e. Artaxerxes I, r. 464 – 424 BCE (Julian)) during the time of Ezra and Nehemiah; or

2.) unintentional later scribal errors when copying the Book of Ezra; or

3.) intentional scribal errors as an attempt to erase the name of the Persian king (who canceled the rebuilding of the Temple) from Jewish history.

XX. In the 2nd Regnal Year of Darius the king,[1595] in the 6th month, on day one (1) of the month, a word of YHVH was in the hand of Haggai the Prophet to Zerubavel son of She'altiel, the governor of Judah, and to Yehoshua son of Yehotzadak, the High Priest, commanding them to rebuild the Temple. This works out to be Sunday 27 August 520 BCE (Julian).[1596] [1597] [1598]

14 And YHVH aroused the spirit of Zerubavel son of She'altiel, the governor of Judah, and the spirit of Yehoshua son of Yehotzadak the High Priest, and the spirit of all the remainder of the people, and [they] came and [they] did labor in the house of YHVH of Armies, their Elohim. 15 on day twenty four (24) of the 6th month, in year two (2) of Daryavesh the king.[1599]

XX. The day that the laborers began to complete the rebuilding of the Temple works out to be Tuesday 19 September 520 BCE (Julian).[1600] [1601]

1591 cf. Ezra 4:4-5
1592 cf. Ezra 4:23-24
1593 cf. Appendix Table X.77
1594 cf. Ezra 4:7, 8, 11, 23
1595 cf. Appendix Table X.77
1596 Sunday 27 August 520 BCE (Julian); Sunday 21 August 520 BCE (Gregorian); 01 Ululu / Elul 3241 AM; Astronomical Julian Day JD_X: 1531731.5; RD_Y: 1183735; RD_X: -189693; Sabbatical Cycle: 7; Metonic Cycle: 11;
1597 cf. Appendix Table X.78
1598 cf. Haggai 1:1-15
1599 cf. Haggai 1:14-15
1600 Tuesday 19 September 520 BCE (Julian); Tuesday 13 September 520 BCE (Gregorian); 24 Ululu / Elul 3241 AM; Astronomical Julian Day JD_X: 1531754.5; RD_Y: 1183758; RD_X: -189670; Sabbatical Cycle: 7; Metonic Cycle: 11;
1601 cf. Appendix Table X.78

XX. In the 7th month on [day] 21 of the month, a word of YHVH was in the hand of Haggai to Zerubavel son of She'altiel, the governor of Judah, and to Yehoshua son of Yehotzadak, the High Priest, and to the remainder of the people encouraging them to rebuild the Temple. This works out to be Sunday 15 October 520 BCE (Julian).[1602] [1603]

XX. In the 8th month in the 2nd Regnal Year of Darius, a word of YHVH was to Zecharyah son of Berechyah son of 'Ido the Prophet. No exact day is given. This works out to be October / November 520 BCE (Julian).[1604] [1605]

XX. On [day] 24 of the 9th [month] in the 2nd Regnal Year of Darius, a word of YHVH was in the hand of Haggai the Prophet.[1606] Haggai emphasizes several times that those hearing this word must pay attention. This works out to be Sunday 17 December 520 BCE (Julian).[1607] [1608] [1609] [1610]

15 And now, place your hearts: from this day and forward, from before placing a stone to [another] stone in the Hall [of the Temple] of YHVH. … 18 Place please your hearts from this day and forward from day twenty (24) of the 9th [month] from today that the Hall [of the Temple] of YHVH will be founded – place your hearts! … 19 …from today [I] will bless.[1611]

XX. On the same day 24 of the 9th month of the 2nd Regnal Year of Darius, a word of YHVH was to Haggai a second time.[1612]

XX. On day 24 of the 11th month, i.e. the month of Shabatu / Shvat, in the 2nd Regnal Year of Darius, a word of YHVH was to Zecharyah son of Berechyah son of 'Ido the Prophet. This works out to be Wednesday 14 February 519 BCE (Julian).[1613] [1614] [1615]

1602 Sunday 15 October 520 BCE (Julian); Sunday 09 October 520 BCE (Gregorian); 21 Tashritu / Tishri 3242 AM; Astronomical Julian Day JD_X: 1531780.5; RD_Y: 1183784; RD_X: -189644; Sabbatical Cycle: 1; Metonic Cycle: 12;
1603 cf. Appendix Table X.79
1604 cf. Appendix Table X.80
1605 cf. Zechariah 1:1
1606 cf. Haggai 2:10
1607 Sunday 17 December 520 BCE (Julian); Sunday 11 December 520 BCE (Gregorian); 24 Kisilimu / Kislev 3242 AM; Astronomical Julian Day JD_X: 1531843.5; RD_Y: 1183847; RD_X: -189581; Sabbatical Cycle: 1; Metonic Cycle: 12;
1608 cf. Appendix Table X.81
1609 It is notable that the holiday of Hanukkah that was instituted in 164 BCE (Julian) began 25 Kisilimu / Kislev 3598 AM;
1610 Sunday 13 December 164 BCE (Julian); Sunday 10 December 164 BCE (Gregorian); 25 Kisilimu / Kislev 3598 AM; Astronomical Julian Day JD_X: 1661868.5; RD_Y: 1313872; RD_X: -59556; Sabbatical Cycle: 7; Metonic Cycle: 7;
1611 cf. Haggai 2:15, 18, 19
1612 cf. Haggai 2:20
1613 Wednesday 14 February 519 BCE (Julian); Wednesday 08 February 519 BCE (Gregorian); 24 Shabatu / Shvat 3242 AM; Astronomical Julian Day JD_X: 1531902.5; RD_Y: 1183906; RD_X: -189522; Sabbatical Cycle: 1; Metonic Cycle: 12;
1614 cf. Appendix Table X.82
1615 cf. Zechariah 1:7

XX. Having resynchronized at the end of the 1st Regnal Year of Darius I, when the proleptic Hebrew calendar intercalated its 13th month Addaru 2 / Adar 2 of 3241 AM that began 03 March 520 BCE (Julian), the proleptic Hebrew and Babylonian calendars remained synchronized throughout his 2nd Regnal Year, so that the months of the prophecies of Haggai and Zecharyah fall in the same months in both calendars.[1616] Since neither calendar intercalated during or at the end of this year, they remained synchronized into the 3rd Regnal Year of Darius I.

37.0. 4th Regnal Year of Darius I == Spring 518 – 517 BCE
== YEAR 80 OF EXILE OF YEHOYACHIN (JEHOIACHIN)
== YEAR 69 OF DESOLATION

XX. In the 4th Regnal Year of Darius the king,[1617] there was a word of YHVH to Zecharyah on [day] four (4) of the 9th month in Kisilimu / Kislev.[1618] This prophetic word references the 70 years that the people fasted and mourned in the 5th and 7th months.[1619] In the proleptic Hebrew calendar, this works out to be Wednesday 06 November 518 BCE (Julian).[1620] [1621]

XX. At the beginning of the 4th Regnal Year of Darius I, the proleptic Hebrew calendar was one (1) month ahead of the Babylonian calendar because of the Babylonian intercalary Ululu 2 / Elul 2 that began 17 September 519 BCE (Julian) in his 3rd Regnal Year.[1622] The proleptic Hebrew calendar thus began the 4th Regnal Year on 01 Nisannu / Nisan 3243 AM (12 March 518 BCE (Julian)) whereas the Babylonian calendar began it one (1) month later on 11 April 518 BCE (Julian). Therefore Zecharyah's prophecy on 04 Kisilimu / Kislev (Wednesday 06 November 518 BCE (Julian)) fell in the Babylonian 8th month on 03 Arahsamna / Marheshvan.[1623]

XX. At the end of this year the proleptic Hebrew calendar resynchronized with the Babylonian calendar when it intercalated its 13th month Addaru 2 / Adar 2 of 3244 AM that began 29 February 517 BCE (Julian) so that both calendars began the 5th Regnal Year of Darius I synchronized.[1624]

38.0. 6th Regnal Year of Darius I == Spring 516 – 515 BCE
== YEAR 82 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

1616 Parker and Dubberstein 1971: 30
1617 cf. Appendix Table X.83
1618 cf. Zechariah 7:1-5
1619 cf. Zechariah 7:5
1620 Wednesday 06 November 518 BCE (Julian); Wednesday 31 October 518 BCE (Gregorian); 04 Kisilimu / Kislev 3244 AM; Astronomical Julian Day JD_X: 1532532.5; RD_Y: 1184536; RD_X: -188892; Sabbatical Cycle: 3; Metonic Cycle: 14;
1621 cf. Appendix Table X.84
1622 Parker and Dubberstein 1971: 7, 30
1623 Parker and Dubberstein 1971: 30
1624 Parker and Dubberstein 1971: 30

== YEAR 71 OF DESOLATION
== 70 YEARS OF DESOLATION COMPLETE

XX. In the 6th Regnal Year of Darius,[1625] the Jews who had returned to Jerusalem had continued working to rebuild the temple "*until day three (3) of the moon of Adar that is year six (6) of the reign of Daryavesh the king.*"[1626] In the proleptic Hebrew calendar, this works out to be Wednesday 09 February 515 BCE (Julian).[1627] [1628]

XX. If, as in the rest of the Hebrew Scriptures, the date in Ezra 6:15 is a Hebrew date, the completion of the Temple on 03 Addaru / Adar 3246 AM (Wednesday 09 February 515 BCE (Julian)) fell exactly 81 years and 11 months after the capture of Jerusalem (which took place on the Babylonian 02 Addaru / Adar), i.e. on the proleptic Hebrew 03 Nisannu / Nisan 3164 AM (Saturday 16 March 597 BCE (Julian)), and thus in the 82nd year after it.

XX. However, if the date in Ezra 6:15 – which belongs to the Aramaic section of the book – follows the Babylonian calendar, as did the dates of the Jews of Elephantine in the Persian period,[1629] then the completion of the Temple fell on the Babylonian 03 Addaru / Adar in the 6th Regnal Year of Darius I (12 March 515 BCE (Julian)). If so, then this works out to be exactly 82 years and one (1) day after the capture of Jerusalem on the Babylonian 02 Addaru / Adar in the 7th Regnal Year of Nebuchadnezzar (16 March 597 BCE (Julian)).

XX. In either case, the dates of the destruction of Jerusalem in Summer 07 to 10 Abu / Av 586 BCE (3175 AM) began 70 YEARS OF DESOLATION that were complete in this year 516 BCE (3245 AM).

XX. According to Parker and Dubberstein's tables, which include an attested Babylonian intercalary Addaru 2 / Adar 2 that began 20 March 516 BCE (Julian) at the end of the 5th Regnal Year of Darius I, the Babylonian calendar began his 6th Regnal Year on 01 Nisannu / Nisan, 18 April 516 BCE (Julian), and its 3rd day of Addaru / Adar fell on 12 March 515 BCE (Julian).[1630]

XX. In the proleptic Hebrew calendar, which did not intercalate in that year, the 6th Regnal Year began one (1) month earlier, on 01 Nisannu / Nisan 3245 AM (Thursday 18 March 516 BCE (Julian)), and 03 Addaru / Adar 3246 AM fell on Wednesday 09 February 515 BCE (Julian).

XX. The two (2) calendars were thus one (1) month apart in this year. In both calendars the 6th Regnal

1625 cf. Appendix Table X.85
1626 cf. Ezra 6:15
1627 Wednesday 09 February 515 BCE (Julian); Wednesday 03 February 515 BCE (Gregorian); 03 Addaru / Adar 3246 AM; Astronomical Julian Day JD_X: 1533358.5; RD_Y: 1185362; RD_X: -188066; Sabbatical Cycle: 5; Metonic Cycle: 16;
1628 cf. Appendix Table X.86
1629 Ossendrijver 2018a: 143–145
1630 Parker and Dubberstein 1971: 25, 30

Year of Darius I was a common year of twelve (12) months, so that the 3rd day of Addaru / Adar fell in the single, regular month of Addaru / Adar near the end of the year in both. Since neither calendar intercalated at its end, the one-(1)-month difference continued into the 7th Regnal Year of Darius I.

39.0. 7th Regnal Year of Darius I == Spring 515 – 514 BCE == YEAR 83 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. In the 7th Regnal Year of Darius,[1631] the Jews who had returned and completed the rebuilding of the Temple celebrated the Passover on [day] 14 of the 1st month (i.e. Nisannu / Nisan).[1632] In the proleptic Hebrew calendar, this works out to be Monday 21 March 515 BCE (Julian).[1633] [1634]

XX. However, in Parker and Dubberstein's tables, the Babylonian 7th Regnal Year began one (1) month later on 08 April 515 BCE (Julian) so that the Babylonian 14 Nisannu / Nisan fell on Thursday 21 April 515 BCE (Julian) because of the attested Babylonian intercalary Addaru 2 / Adar 2 that began 20 March 516 BCE (Julian) at the end of his 5th Regnal Year.[1635]

XX. At the end of this 7th Regnal Year, only the proleptic Hebrew calendar intercalated its 13th month Addaru 2 / Adar 2 of 3247 AM that began 27 February 514 BCE (Julian) – i.e. the same day as the Babylonian 12th month Addaru 1 / Adar 1 (there being no Babylonian intercalation that year). The one-(1)-month difference that had existed since the end of the 5th Regnal Year thus came to an end, and the two (2) calendars began the 8th Regnal Year of Darius I synchronized, and remained synchronized throughout.[1636]

40.0. 3rd Regnal Year of Xerxes I == Spring 483 – 482 BCE == YEAR 115 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. According to Parker and Dubberstein, there is evidence of the accession of the Persian king, Xerxes I (the son of Darius the Great), in 486 BCE (Julian).[1637] [1638] Dandamaev specifies the month of Xerxes I's accession to be November 486 BCE (Julian).[1639]

1631 cf. Appendix Table X.87
1632 cf. Ezra 6:19
1633 Monday 21 March 515 BCE (Julian); Monday 15 March 515 BCE (Gregorian); 14 Nisannu / Nisan 3246 AM; Astronomical Julian Day JD_X: 1533398.5; RD_Y: 1185402; RD_X: -188026; Sabbatical Cycle: 5; Metonic Cycle: 16;
1634 cf. Appendix Table X.88
1635 Parker and Dubberstein 1971: 25, 30
1636 Parker and Dubberstein 1971: 30
1637 3276 AM
1638 Parker and Dubberstein 1956: 17
1639 Dandamaev 1989: 181

XX. This Xerxes I is the same Persian king recorded in the Book of Esther, known in Hebrew as Ahashverosh, and transliterated to English as Ahasuerus. The 3rd Regnal Year of Xerxes I began in Spring (Nisannu / Nisan) 483 BCE.[1640] [1641] [1642]

XX. At the beginning of the 3rd Regnal Year of Xerxes I, the proleptic Hebrew calendar was one (1) month ahead of the Babylonian calendar because of the Babylonian intercalary Ululu 2 / Elul 2 that began 20 September 484 BCE (Julian) in his 2nd Regnal Year. According to Ossendrijver, this intercalation is attested at Persepolis and most probably observed in Babylonia as well.[1643]

XX. The proleptic Hebrew calendar thus began the 3rd Regnal Year on 01 Nisannu / Nisan 3278 AM (Saturday 14 March 483 BCE (Julian)), whereas the Babylonian calendar began it one (1) month later on 14 April 483 BCE (Julian).[1644]

XX. At the end of this year the proleptic Hebrew calendar resynchronized with the Babylonian calendar when it intercalated its 13th month Addaru 2 / Adar 2 of 3279 AM that began 05 March 482 BCE (Julian) so that both calendars began the 4th Regnal Year of Xerxes I on the same day, 03 April 482 BCE (Julian).[1645]

41.0. 7th Regnal Year of Xerxes I == Spring 479 – 478 BCE == YEAR 119 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. It is recorded in the Book of Esther that Esther was taken to the king Xerxes I in the 10th Month (i.e. Tebetu / Tevet) in his 7th Regnal Year. In the proleptic Hebrew calendar, this occurred between December 479 BCE and January 478 BCE (Julian).[1646] [1647] [1648]

XX. At the beginning of the 7th Regnal Year of Xerxes I, the proleptic Hebrew and Babylonian calendars were synchronized with the proleptic Hebrew calendar having intercalated its 13th month Addaru 2 / Adar 2 of 3282 AM that began 01 March 479 BCE (Julian), at the end of his 6th Regnal Year.[1649] Consequently the 10th month of Tebetu / Tevet, in which Esther was taken to the king, began

1640 3278 AM
1641 cf. Appendix Table X.89
1642 cf. Esther 1
1643 Parker and Dubberstein 1971: 8, 31; Ossendrijver 2018a: 141
1644 Parker and Dubberstein 1971: 31
1645 Parker and Dubberstein 1971: 31
1646 3283 AM
1647 cf. Appendix Table X.90, X.91
1648 cf. Esther 2:16
1649 Parker and Dubberstein 1971: 31

on the same day in both calendars, 22 December 479 BCE (Julian).[1650]

XX. At the end of this year Parker and Dubberstein's tables include a reconstructed Babylonian intercalary Addaru 2 / Adar 2, but it is not attested, and the proleptic Hebrew calendar does not intercalate in this year.[1651]

XX. At the end of this year, Parker and Dubberstein's tables include a reconstructed Babylonian intercalary Addaru 2 / Adar 2 that is not attested.[1652] Later evidence discards it in favor of a Babylonian Ululu 2 / Elul 2 in the 8th Regnal Year of Xerxes I that began 13 September 478 BCE (Julian).[1653] Since neither calendar intercalated at the end of the 7th Regnal Year, both began the 8th Regnal Year synchronized on 20 and 21 March 478 BCE (Julian) respectively.

42.0. 12th Regnal Year of Xerxes I == Spring 474 – 473 BCE == YEAR 124 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. The fateful event for the casting of lots (i.e. Hebrew *"Purim"*) for the date of the intended genocide extermination of all Jews within the Persian Empire was in the 12th Regnal Year of Xerxes I which began in Spring (Nisannu / Nisan) 474 BCE.[1654] [1655] [1656]

XX. This fateful day of the first recorded, decreed intention to genocide the Jews was announced 13 Nisannu / Nisan, i.e. Sunday 15 April 474 BCE (Julian).[1657] [1658] Notably, this day of the announcement of the forthcoming genocide of the Jews, Sunday 13 Nisannu / Nisan 3287 AM, was the day before Passover.[1659]

XX. In this 1st month of the 12th Regnal Year of Xerxes I,[1660] the lot was cast, and the planned day of genocide was set for [day] 13 of the 12th month, that is the month of Addaru / Adar, of the same 12th Regnal Year of Xerxes I,[1661] and the announcement was made: i.e. the date of the planned genocide was set to be 11 months later on day 13 of month 12 (i.e. Addaru / Adar) in Late Winter Thursday 06 March

1650 Parker and Dubberstein 1971: 31
1651 Parker and Dubberstein 1971: 8, 31
1652 Parker and Dubberstein 1971: 25, 31
1653 Walker 1997; Ossendrijver 2018a: 141-142
1654 3287 AM
1655 cf. Appendix Table X.92
1656 cf. Esther 3:6-7
1657 Sunday 15 April 474 BCE (Julian); Sunday 10 April 474 BCE (Gregorian); 13 Nisannu / Nisan 3287 AM; Astronomical Julian Day: JD_X 1548398.5; RD_Y: 1200402; RD_X: -173026; Sabbatical Cycle: 4; Metonic Cycle: 19;
1658 cf. Appendix Table X.93
1659 cf. Esther 3:7, 12
1660 cf. Esther 3:7, 12
1661 cf. Esther 3:7, 13

473 BCE (Julian).[1662] [1663] [1664]

XX. Queen Esther successfully obtains permission from Xerxes I to write a new edict that would allow the Jews to defend themselves on the day of the planned genocide.[1665] This new decree is read out to the public on the 23rd day of the 3rd month (i.e. Simannu / Sivan), i.e. Saturday 23 June 474 BCE (Julian).[1666] [1667] [1668]

XX. The fateful day of Thursday 13 Addaru / Adar 3288 AM (i.e. Thursday 06 March 473 BCE (Julian)) arrived,[1669] [1670] as did the days Friday 14 Addaru / Adar (i.e. Friday 07 March 473 BCE (Julian))[1671] [1672] and Saturday 15 Addaru / Adar (i.e. Saturday 08 March 473 BCE (Julian))[1673] [1674] which Jews commemorate today as the biblical holiday of Purim.[1675]

XX. At the beginning of the 12th Regnal Year of Xerxes I the proleptic Hebrew and Babylonian calendars were synchronized. The proleptic Hebrew calendar had intercalated its 13th month Addaru 2 / Adar 2 of 3287 AM that began 05 March 474 BCE (Julian) at the end of his 11th Regnal Year.[1676] Consequently, the months recorded in the Book of Esther for this year – i.e. the 1st month of Nisannu / Nisan, the 3rd month of Simannu / Sivan, and the 12th month of Addaru / Adar – fall in the same months in both calendars.[1677]

XX. Only after the days of Purim did the Babylonians intercalate its 13th month Addaru 2 / Adar 2 that began 24 March 473 BCE (Julian) at the end of this year. The proleptic Hebrew calendar did not intercalate, and therefore was one (1) month ahead of the Babylonian calendar at the beginning of the 13th Regnal Year of Xerxes I.[1678]

1662 Thursday 06 March 473 BCE (Julian); Thursday 01 March 473 BCE (Gregorian); 13 Addaru / Adar 3288 AM; Astronomical Julian Day: JD_X 1548724.5; RD_Y: 1200728; RD_X: -172700; Sabbatical Cycle: 5; Metonic Cycle: 1;
1663 cf. Appendix Table X.95
1664 cf. Esther 3:13
1665 cf. Esther 8:9, 12; 9:1
1666 Saturday 23 June (Julian) 474 BCE; Saturday 18 June 474 BCE (Gregorian); 23 Simannu / Sivan 3287 AM; Astronomical Julian Day: JD_X 1548467.5; RD_Y: 1200471; RD_X: -172957; Sabbatical Cycle: 4; Metonic Cycle: 19;
1667 cf. Appendix Table X.94
1668 cf. Esther 8:9-14
1669 Thursday 06 March 473 BCE (Julian); Thursday 01 March 473 BCE (Gregorian); 13 Addaru / Adar 3288 AM; Astronomical Julian Day: JD_X 1548724.5; RD_Y: 1200728; RD_X: -172700; Sabbatical Cycle: 5; Metonic Cycle: 1;
1670 cf. Appendix Table X.95
1671 Friday 07 March 473 BCE (Julian); Friday 02 March 473 BCE (Gregorian); 14 Addaru / Adar 3288 AM; Astronomical Julian Day JD_X: 1548725.5; RD_Y: 1200729; RD_X: -172699; Sabbatical Cycle: 5; Metonic Cycle: 1;
1672 cf. Appendix Table X.95
1673 Saturday 08 March 473 BCE (Julian); Saturday 03 March 473 BCE (Gregorian); 15 Addaru / Adar 3288 AM; Astronomical Julian Day JD_X: 1548726.5; RD_Y: 1200730; RD_X: -172698; Sabbatical Cycle: 5; Metonic Cycle: 1;
1674 cf. Appendix Table X.95
1675 cf. Esther 9:15-32
1676 Parker and Dubberstein 1971: 31
1677 Parker and Dubberstein 1971: 31
1678 Parker and Dubberstein 1971: 8, 31; Ossendrijver 2018a: 142

43.0. 7th Regnal Year of Artaxerxes I == Spring 458 – 457 BCE == YEAR 140 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. In the 7th Regnal Year of Artaxerxes I,[1679] on [day] one (1) of the 1st month (Nisannu / Nisan), Ezra and the group he was leading left Babylon. In the proleptic Hebrew calendar, this works out to be Thursday 09 March 458 BCE (Julian).[1680] [1681] [1682]

XX. Ezra records he and his group gathered on the river that flows to the Ahava river, and camped there three (3) days. On [day] 12 of this same 1st month (Nisannu / Nisan), Ezra records setting out from the Ahava river on the way to Jerusalem.[1683] This works out to be Monday 20 March 458 BCE (Julian).[1684] [1685]

XX. Therefore, Ezra and his group arrived to the Ahava river on Friday 17 March 458 BCE (Julian),[1686] and – evidently observing the Sabbath (Shabbat) rest – camped for the three (3) days of Friday evening, Saturday evening, and Sunday evening, and then set out for Jerusalem on Monday 20 March 458 BCE (Julian).[1687]

XX. Ezra records having traveled from the Ahava river for almost four (4) months until they arrived to Jerusalem on [day] one (1) of the 5th month (Abu / Av) of the 7th Regnal Year of Artaxerxes I.[1688] This works out to be Wednesday 05 July 458 BCE (Julian).[1689] [1690]

XX. Ezra records that upon their arrival to Jerusalem, they sat three (3) days, and on the 4th day they got to work.[1691] Thus they rested Thursday, Friday, and Saturday (i.e. Sabbath (Shabbat)), and then

1679 cf. Appendix Table X.96
1680 Thursday 09 March 458 BCE (Julian); Thursday 04 March 458 BCE (Gregorian); 01 Nisannu / Nisan 3303 AM; Astronomical Julian Day JD_X: 1554205.5; RD_Y: 1206209; RD_X: -167219; Sabbatical Cycle: 6; Metonic Cycle: 16;
1681 cf. Appendix Table X.97
1682 cf. Ezra 7:7-9
1683 cf. Ezra 8:31
1684 Monday 20 March 458 BCE (Julian); Monday 15 March 458 BCE (Gregorian); 12 Nisannu / Nisan 3303 AM; Astronomical Julian Day JD_X: 1554216.5; RD_Y: 1206220; RD_X: -167208; Sabbatical Cycle: 6; Metonic Cycle: 16;
1685 cf. Appendix Table X.97
1686 Friday 17 March 458 BCE (Julian); Friday 12 March 458 BCE (Gregorian); 09 Nisannu / Nisan 3303 AM; Astronomical Julian Day JD_X: 1554213.5; RD_Y: 1206217; RD_X: -167211; Sabbatical Cycle: 6; Metonic Cycle: 16;
1687 cf. Appendix Table X.97
1688 cf. Ezra 7:7-9
1689 Wednesday 05 July 458 BCE (Julian); Wednesday 30 June 458 BCE (Gregorian); 01 Abu / Av 3303 AM; Astronomical Julian Day JD_X: 1554323.5; RD_Y: 1206327; RD_X: -167101; Sabbatical Cycle: 6; Metonic Cycle: 16;
1690 cf. Appendix Table X.98
1691 cf. Ezra 8:32-33

began their labor on Sunday 09 July 458 BCE (Julian).[1692] [1693]

XX. Reasonably in the same year of their arrival, Ezra calls for all the people to gather in Jerusalem for three (3) days with the condition that anybody who did not come would have their property confiscated and that person would be excommunicated from the community of the returning exiles. This three (3) day gathering occurred in the 9th month (Kisilimu / Kislev) on [day] 20 of the month in which the people were gathered to Jerusalem, and they sat in the street of the House of the Elohim (i.e. the Temple) trembling over the decree and because of the rains.[1694] This works out to be Sunday 19 November 458 BCE (Julian).[1695] [1696]

XX. On day one (1) of the 10th month (Tebetu / Tevet) the community set out to question the men who had taken pagan wives.[1697] This works out to be Wednesday 29 November 458 BCE (Julian).[1698] [1699]

XX. They did this until day one (1) of the 1st month (Nisannu / Nisan) that begins the next year, i.e. the 8th Regnal Year of Artaxerxes I.[1700] This works out to be Tuesday 26 March 457 BCE (Julian).[1701] [1702]

XX. At the beginning of the 7th Regnal Year of Artaxerxes I, the proleptic Hebrew calendar was one (1) month ahead of the Babylonian calendar because of the Babylonian intercalary Addaru 2 / Adar 2 that began 20 March 459 BCE (Julian) at the end of his 5th Regnal Year.[1703] Consequently Ezra's departure from Babylon on 01 Nisannu / Nisan 3303 AM (09 March 458 BCE (Julian)) fell by Babylonian reckoning on 01 Addaru / Adar at the end of the 6th Regnal Year of Artaxerxes I. The Babylonian 7th Regnal Year did not begin until 08 April 458 BCE (Julian).[1704]

XX. By Babylonian reckoning, Ezra's departure on 01 Nisannu / Nisan would have fallen on a Sabbath, Saturday 08 April 458 BCE (Julian) – which is extremely strong evidence that he used the proleptic Hebrew calendar reckoning because of the religious commandment in the Torah to not travel on the Sabbath – whereas by the proleptic Hebrew calendar it fell on Thursday 09 March 458 BCE (Julian)

1692 Sunday 09 July 458 BCE (Julian); Sunday 04 July 458 BCE (Gregorian); 05 Abu / Av 3303 AM; Astronomical Julian Day JD_X: 1554327.5; RD_Y: 1206331; RD_X: -167097; Sabbatical Cycle: 6; Metonic Cycle: 16;

1693 cf. Appendix Table X.98

1694 cf. Ezra 10:7-9

1695 Sunday 19 November 458 BCE (Julian); Sunday 14 November 458 BCE (Gregorian); 20 Kisilimu / Kislev 3304 AM; Astronomical Julian Day JD_X: 1554460.5; RD_Y: 1206464; RD_X: -166964; Sabbatical Cycle: 7; Metonic Cycle: 17;

1696 cf. Appendix Table X.99

1697 cf. Ezra 10:16-17

1698 Wednesday 29 November 458 BCE (Julian); Wednesday 24 November 458 BCE (Gregorian); 01 Tebetu / Tevet 3304 AM; Astronomical Julian Day JD_X: 1554470.5; RD_Y: 1206474; RD_X: -166954; Sabbatical Cycle: 7; Metonic Cycle: 17;

1699 cf. Appendix Table X.100

1700 cf. Appendix Table X.101

1701 Tuesday 26 March 457 BCE (Julian); Tuesday 21 March 457 BCE (Gregorian); 01 Nisannu / Nisan 3304 AM; Astronomical Julian Day JD_X: 1554588.5; RD_Y: 1206592; RD_X: -166836; Sabbatical Cycle: 7; Metonic Cycle: 17;

1702 cf. Appendix Table X.102

1703 Parker and Dubberstein 1971: 8, 32

1704 Parker and Dubberstein 1971: 32

and the three (3) days at the Ahava river included the Sabbath. Therefore, Ezra evidently reckoned the 7th Regnal Year from the proleptic Hebrew Nisannu / Nisan.

XX. At the end of this year, the proleptic Hebrew calendar resynchronized with the Babylonian calendar when it intercalated its 13th month Addaru 2 / Adar 2 of 3304 AM that began 26 February 457 BCE (Julian) so that both calendars began the 8th Regnal Year of Artaxerxes I synchronized.[1705]

44.0. 20th Regnal Year of Artaxerxes I == Spring 445 – 444 BCE
== YEAR 153 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. In the 20th Regnal Year of Artaxerxes I,[1706] in the 9th month of Kisilimu / Kislev, Nehemiah was in Susa (Shushan) the capital of Persia.[1707] This works out to be November / December 445 BCE (Julian).[1708]

XX. At the beginning of the 20th Regnal Year of Artaxerxes I, the proleptic Hebrew calendar was one (1) month ahead of the Babylonian calendar because of the Babylonian intercalary 13th month Addaru 2 / Adar 2 that began 15 March 445 BCE (Julian) at the end of his 19th Regnal Year.[1709] The proleptic Hebrew calendar thus began the 20th Regnal Year on 01 Nisannu / Nisan 3316 AM (Thursday 13 March 445 BCE (Julian)), whereas the Babylonian calendar began it one (1) month later on 13 April 445 BCE (Julian).[1710]

XX. Therefore, the Babylonian month of Kisilimu / Kislev fell from 06 December 445 to 03 January 444 BCE (Julian), one (1) month after the proleptic Hebrew Kisilimu / Kislev from 05 November to 04 December 445 BCE (Julian).[1711]

XX. At the end of this year, the proleptic Hebrew calendar resynchronized with the Babylonian calendar when it intercalated its 13th month Addaru 2 / Adar 2 of 3317 AM that began 04 March 444 BCE (Julian) the same day as the Babylonian 12th month Addaru / Adar. Both calendars thus began the 21st Regnal Year of Artaxerxes I synchronized.[1712]

45.0. 21st Regnal Year of Artaxerxes I == Spring 444 – 443 BCE
== YEAR 154 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

1705 Parker and Dubberstein 1971: 32
1706 cf. Appendix Table X.103
1707 cf. Nehemiah 1:1
1708 cf. Appendix Table X.105
1709 Parker and Dubberstein 1971: 8, 32
1710 Parker and Dubberstein 1971: 32
1711 Parker and Dubberstein 1971: 32
1712 Parker and Dubberstein 1971: 32

XX. In the 21st Regnal Year of Artaxerxes I,[1713] in the 1st month of Nisannu / Nisan,[1714] Nehemiah stood before the king as his wine-bearer. Since regnal years of kings were counted from Nisannu / Nisan, it is clearly evident that the record of this 1st month of Nisannu / Nisan being after the 9th month of Kisilimu / Kislev in the 20th Regnal Year is impossible.[1715] Thus one can reasonably conclude that this is a simple scribal error lacking the word "one", i.e. "in year twenty [one] (21) of Artachshasta (Artaxerxes) the king".[1716] This works out to be April / May 444 BCE (Julian).[1717]

XX. At the beginning of the 21st Regnal Year of Artaxerxes I, the proleptic Hebrew and Babylonian calendars were synchronized. The proleptic Hebrew calendar had intercalated its 13th month Addaru 2 / Adar 2 of 3317 AM that began 04 March 444 BCE (Julian). The 1st month of Nisannu / Nisan in which Nehemiah stood before the king thus began on 02 and 03 April 444 BCE (Julian) respectively.[1718]

XX. At the end of this year the Babylonians intercalated an Addaru 2 / Adar 2 that began 23 March 443 BCE (Julian).[1719] However, the proleptic Hebrew calendar did not intercalate, and was therefore one (1) month ahead of the Babylonian calendar at the beginning of the 22nd Regnal Year of Artaxerxes I.

46.0. 32nd Regnal Year of Artaxerxes I == Spring 433 – 432 BCE == YEAR 165 OF EXILE OF YEHOYACHIN (JEHOIACHIN)

XX. In the Book of Nehemiah, it is recorded that Nehemiah was commanded to be the governor of Judah from the 20th Regnal Year to the 32nd Regnal Year of Artaxerxes I for a total of 12 years.[1720] This works out to be between 445 – 433 BCE (Julian).[1721]

XX. At the beginning of the 32nd Regnal Year of Artaxerxes I, the proleptic Hebrew and Babylonian calendars were synchronized. The proleptic Hebrew calendar had intercalated its 13th month Addaru 2 / Adar 2 of 3328 AM that began 02 March 433 BCE (Julian) at the end of his 31st Regnal Year. Both calendars thus began the 32nd Regnal Year on 31 March and 01 April 433 BCE (Julian) respectively.[1722]

XX. At the end of this year the Babylonians intercalated an Addaru 2 / Adar 2 that began 21 March 432 BCE (Julian).[1723] However, the proleptic Hebrew calendar did not intercalate, and was therefore one (1)

1713 cf. Appendix Table X.106
1714 cf. Appendix Table X.107
1715 cf. Appendix Table X.103, X.104, X.105
1716 cf. Nehemiah 2:1
1717 cf. Appendix Table X.106, X.107
1718 Parker and Dubberstein 1971: 32
1719 Parker and Dubberstein 1971: 8, 32
1720 cf. Nehemiah 5:14
1721 cf. Appendix Table X.103, X.104, X.105, X.106, X.107, X.108
1722 Parker and Dubberstein 1971: 32
1723 Parker and Dubberstein 1971: 8, 32

month ahead of the Babylonian calendar at the beginning of the 33rd Regnal Year of Artaxerxes I.

47.0. The Chronology of the Hebrew Kings

XX. Having established fixed, mathematically-synchronized dates for the reigns of the last five (5) Hebrew kings (r. 608 – 586 BCE (Julian)) – i.e. 1.) Yoshiyahu (Josiah); 2.) Yeho-ahaz (Jeho-ahaz / Shalum); 3.) Yehoyakim (Jehoiakim / Elyakim); 4.) Yehoyachin (Jehoiachin); 5.) Tzidkiyahu (Zedekiah / Matanyah) – as well as forward in time through the reign of Artaxerxes I (r. 464 – 424 BCE (Julian)), it will now be necessary to attempt test the accuracy of these calculated and reckoned dates proleptically into prior ancient history to determine if more ancient dates are feasible using the same calendar and methodology.

XX. Applying these calculations and reckonings back in time to the reigns of the Hebrew kings of Judah and Israel (with their purported synchonisms to the chronologies of Egyptology and Assyriology) will be just such the test necessary to confirm (or disprove) the accuracy of these chronological calculations to see if they accord (and seem feasibly reasonable) with recorded history.

XX. The best methodology to scientifically test these chronological calculations will be to attempt to reconstruct a table that synchronizes the relative chronologies of the Hebrew kings of Israel and Judah. According to computer science and mathematics, these chronologies would necessarily either be absolutely synchronized (i.e. 1 / TRUE), or they would not be absolutely synchronized (i.e. 0 / FALSE) with the condition that requires that they would need to be synchronized absolutely perfectly and exactly precise with absolutely no room for compromise or any error whatsoever.

XX. The chronology of the kings of Judah and Israel is one of the most complex issues in Biblical history. It has been discussed quite extensively in the scholarly literature, but no satisfactory solution has been offered to date, nor has the testimony of the Biblical and extra-Biblical sources received an exhaustive treatment.[1724]

XX. The determination of an absolute system of dates for the kings of Israel and Judah [purportedly only] became possible only in the 2nd half of the 19th century CE (Gregorian) following the deciphering of Assyrian royal inscriptions pertaining to the history of the two (2) Israelite kingdoms. However, the chronology of the kings of Israel and Judah was first discussed about two (2) and half millennia [ca. 2500 years] ago in the 5th century BCE (Julian) with the writing [of the final editorial redaction by Ezra the Scribe] of the Book of Kings.[1725]

XX. Since then countless efforts have been made to reconstruct the chronological framework of these

1724 Galil 1996: 1
1725 Galil 1996: 1

monarchs and to reconcile these many [seemingly apparent] contradictions between the confusing data regarding the duration of the reigns of these thirty-nine (39) kings. Some of these attempts have survived in the Septuagint and in the works of Josephus. A few proposals, mainly harmonious, appear in *Seder Olam Rabbah*, in the patristic literature, and in the medieval commentators.[1726]

XX. Beginning in the mid-19th century CE (Gregorian), researchers proposed a plethora of reconstructions of the chronological framework and of the principles underlying the chronology of Judean and Israelite history in this period. Researchers deny the accuracy of the majority of the Biblical chronological data arguing that these data are artificial and schematic, and cannot serve as a basis for the determination of the Israelite kings.[1727]

XX. In 1944 – 1945 CE (Gregorian) two studies were published in the United States which greatly influenced the research of the chronology of the period of the divided monarchy: the works of E.R. Thiele and W.F. Albright. Their proposals were accepted by many researchers, and to the present almost half a century later [1996 CE (Gregorian) – and now more than 80 years later in 2026 CE (Gregorian)] after their publication most scholars are of the opinion that the system of Thiele or that of Albright is the most plausible for reconstructing the chronological frame of the period under discussion.[1728]

XX. The two (2) proposed systems differ from each other. Thiele's view was accepted by many researchers including K.A. Kitchen and T.C. Mitchell, Hallo, H.S. Gehman, Rainey, de Vries, and others. Albright's reconstruction was adopted by Bright, Freedman, Coggins, cited in Ackroyd, and others.[1729]

XX. Like Thiele, Albright, Tadmor and others before him, Galil makes several presumptions about the principles of the chronology of the period of the divided monarchy of Israel[1730] – all of which after careful research are proven to contribute to mistaken and incorrect understandings when attempting to construct an absolute chronology.

XX. In order to take into account for the possibility of co-regencies and other overlaps in the synchronization of the chronologies of the Hebrew kings (of both the southern kingdom of Judah and the northern kingdom of Israel), it will be necessary to construct a table and divide it into four (4) main columns:

1.) Kings of Judah A

2.) Kings of Judah B

1726 Galil 1996: 1
1727 Galil 1996: 1
1728 Galil 1996: 3
1729 Galil 1996: 3
1730 Galil 1996: 9-10

3.) Kings of Israel A

4.) Kings of Israel B

XX. We will begin our reconstruction of the synchonization of absolute chronologies at the split of Shlomo's (Solomon's) United Kingdom after his death, i.e. at the beginning of the reign of his son Rehav'am (Rehoboam) ben Shlomo. We will make the reasonable working assumption (to be tested) that Rehav'am ben Shlomo's Year 0 Accession Year was the same year as the 40th (FINAL) Regnal Year of Shlomo ben David.[1731] [1732]

XX. The Hebrew Scriptures record that Rehav'am ben Shlomo was 41 years old upon his accession to the throne (i.e. in his Year 0 Accession Year), and he reigned for 17 years.[1733] [1734] Additionally we are told that there was war between Rehav'am ben Shlomo king of Judah and Yarav'am ben N'vat king of Israel all of their mutual days,[1735] i.e. all mutual 17 years of the reign of Rehav'am. Furthermore, a critical synchronism with Egyptian chronology is recorded in the 5th Regnal Year of Rehav'am ben Shlomo that the Egyptian Pharaoh Shishak (i.e. Shoshenq) ascended upon and invaded Jerusalem.[1736] [1737]

XX. As will become evident as the data unfolds, the Hebrew chroniclers (i.e. e.g. Moshe, Joshua, Samuel, Jeremiah, Daniel, and Ezra) who recorded the chronologies of the Hebrew kings in the Hebrew calendar did not (and indeed could not possibly) use a Year 0 Accession Year system for the Kings of Israel as they most certainly did with the Kings of Judah.

48.0. The Biblical Chronology of Edwin Thiele

XX. Edwin Thiele, a Seventh-Day Adventist teacher, authored *The Mysterious Numbers of the Hebrew Kings*, which – in the absence of a credible alternative – for the last 80+ years has been a standard reference for dating reigns in the 1st and 2nd Books of Kings in the Old Testament. However, Thiele based his theories only on the Hebrew Masoretic Text which only goes back to about 1000 CE (Julian). He dismissed the variant numbers recorded in the Septuagint – the Greek translation of the Hebrew Text – as it stood about 1200 years earlier, around 200 BCE (Julian).[1738]

1731 cf. Appendix Table X.1, X.2, X.194
1732 cf. 1 Kings 11:42-43; 2 Chronicles 9:30-31
1733 cf. Appendix Table X.2, X.3, X.194
1734 cf. 1 Kings 14:21; 2 Chronicles 12:13-14
1735 cf. 1 Kings 14:30; 2 Chronicles 12:15
1736 cf. Appendix Table X.2, X.194
1737 1 Kings 14:25-26; 2 Chronicles 12:2-12
1738 Tetley 2014: iii

XX. Thiele explained the "mysterious" numbers by invoking separate calendars for Judah and Israel, and many co-regencies, which find no mention in the records. He supported his theory by a continuous list of Assyrian Kings [i.e. the Assyrian Kings List (AKL) based on the Assyrian Eponym Canon (AEC)] which remains unproven.[1739]

XX. In order to give a date to the kings of Judah and Israel, Thiele had to find a starting date or a synchronism between a king of Assyria and a king of Israel or Judah. The first synchronism between Assyria and Israel in the divided kingdom period [chosen by Thiele] is provided by the battle of Qarqar fought between the Assyrians and a coalition of Levantine kings, including Ahab of Israel. This event is dated in Assyrian records to the 6th year of Shalmaneser III of Assyria.[1740]

XX. Assyrian records of course do not give Julian dates, but it was assigned the date of 853 BCE (Julian) on the assumption that the AEC [i.e. Assyrian Eponym Canon] contained all the eponyms from 910 to 612 BCE (Julian). However, by adding up the reign lengths given for the kings of Israel and Judah as recorded in the English translation of the Hebrew text, Thiele realized that the regnal year numbers given for the kings of Israel for the period of the divided kingdom (from the accession of Rehoboam of Judah and Jeroboam I of Israel until the fall of Samaria in the reigns of Hoshea of Israel and Hezekiah of Judah) were about 23 years higher for Israel than for the concurrent period of Assyrian history – and for the kings of Judah 46 years higher, based on the dates allocated to the AEC.[1741]

XX. Thiele had a choice: Either recognize that the AEC was deficient and try to reconstruct a chronology for the kings of Israel and Judah from the figures given in the Hebrew / English – and according to Tetley [who is in error], taking into account variants in the Greek texts – or compress the Hebrew / English data for the kings of Israel and Judah to bring them into line with the years assigned to the AEC.[1742]

XX. However, there is another more simple, elegant, and better option of just reevaluating the Masoretic Text (MT) to read the original Hebrew literally in order to attempt to refine our understanding simply by going with whatever the mathematics reveal.

XX. [Of the two (2) options reasoned by Tetley, Thiele] chose the latter option [i.e. he tried to calibrate the Biblical chronologies of the Hebrew Scriptures to the Assyrian Eponym Canon (AEC) instead of vice-versa], even though the kings of Judah and Israel had a dual system of cross-referenced reigns [i.e. a chronological synchronism control with a fail-safe], whereas the AEC was composed of one linear record of Assyrian kings whose chronology had never been corroborated in the period prior to the solar eclipse of 763 BCE (Julian). Thiele decided that the numbers [of the Hebrew Scriptures] were

1739 Tetley 2014: iii
1740 Tetley 2014: 6
1741 Tetley 2014: 6
1742 Tetley 2014: 6

"mysterious" and proceeded on the basis that the numbers could be harmonized if certain dating systems were applied.[1743]

XX. Having made this decision, Thiele overlapped the reigns of the kings of Israel and Judah by about 50 years overall to make the reigns fit the years indicated by the AEC. Thus he dated the 22nd (FINAL) Regnal Year of Ahab to 853 BCE [even though in actuality it was 903 – 902 BCE] (Julian) which was presumed to be the 6th Regnal Year of [Assyrian king] Shalmaneser III. By means of his dating systems, he arrived at the date of 931 BCE (Julian) for the commencement of the divided kingdom [even though in actuality it was 981 – 980 BCE (Julian)] and the 5th Regnal Year of Rehoboam in 925 BCE [even though in actuality it was 977 – 976 BCE] (Julian).[1744]

XX. A military confrontation in the Middle East occurred about 3000 years ago in the 5th Regnal Year of Judah's King Rehoboam [i.e. 977 – 976 BCE (Julian)] and the 20th Regnal Year of Egypt's King Shoshenq I (i.e. Shishak). Though the campaign was recorded in the annals of both nations, agreement on the actual [absolute] year – which could anchor the chronologies of each – has not been established, despite much effort over the last century.[1745]

XX. The excess years for the kings of Judah and Israel were explained away [by Thiele] by the use of various dating systems. Two such systems used were Antedating and Postdating (also called [Year 1] Non-Accession [Year] and [Year 0] Accession Year dating).[1746]

XX. In Antedating [i.e. Year 1 Non-Accession Year dating], a king's 1st Regnal Year [Year 1] is the year [i.e. counting from the 1st day] he comes to the throne, and his 1st full year is his 2nd year [Year 2]; in Postdating [i.e. Year 0 Accession Year dating], the king's [1st Regnal Year of] reign is dated from the beginning of the [subsequent] year [i.e. Year 1] after his Accession [Year 0].[1747]

XX. These dating systems give flexibility to the length of a king's reign. Judah's kings supposedly [according to Thiele] used Postdating [i.e. Year 0 Accession Year dating] for the 1st four (4) kings (Rehoboam to Jehoshaphat), then switched to Antedating [i.e. Year 1 Non-Accession Year dating] for Jehoram, Ahaziah, Queen Athaliah, and Joash, and then switched back to Postdating [i.e. Year 0 Accession Year dating] for the remaining kings. [According to Thiele,] Israel used Antedating [i.e. Year 1 Non-Accession Year dating] from Jeroboam I to Jehoahaz and then switched to Postdating [i.e. Year 0 Accession Year dating] from Jehoash to Hoshea.[1748]

XX. Thiele also started the calendar years six (6) months apart in the two (2) kingdoms: Israel starting

1743 Tetley 2014: 6
1744 Tetley 2014: 6
1745 Tetley 2014: iii
1746 Tetley 2014: 6
1747 Tetley 2014: 6
1748 Tetley 2014: 6

in the 1st month of Nisannu / Nisan and Judah in the 7th month of Tashritu / Tishri. He also states that Judah used its (Tishri-commencing) system for recording its years and for those of Israel, while Israel used its (Nisan-commencing) system for recording its years and those of Judah.[1749]

XX. However, Thiele's main resort to bring the Hebrew chronology into line with the Assyrian chronology ([i.e.] dated to the AEC) was by introducing co-regencies or overlapping reigns into both kingdoms – some of considerable length. Thiele also proposed that at times there were two (2) kings ruling contemporaneously in both kingdoms; that is, four (4) kings altogether.[1750]

XX. The Books of 1 [Kings] and 2 Kings are silent about these dating methods. The dating method that is stated is that a king began to reign in a certain year of the king of the other kingdom [i.e. a clear absolute synchronism], and that he reigned so many years. When that king died his son or successor began to reign.[1751]

XX. This is the only dating system given in the Books of 1 [Kings] and 2 Kings [as well as 1 Chronicles and 2 Chronicles]. The terms Postdating [i.e. Year 0 Accession Year dating] or Antedating [i.e. Year 1 Non-Accession Year dating] are never used, co-regencies are never stated [explicitly, although they are clear from a literal reading of the text], nor does it state that Israel and Judah started their calendar years six (6) months apart. Therefore, Thiele's dating systems are not exhibited in the Books of Kings [or the Books of Chronicles].[1752]

XX. As noted by Thiele, [when dealing with the study of biblical chronology] numerous difficulties must be faced. If harmony is to be achieved in bringing together the many data left by the various biblical writers, it is necessary to understand how each dealt with the reigns under discussion.[1753]

XX. Although Thiele is correct in some of his observations, he is mistaken in others, and these will be clarified and debunked throughout this study. For example, Thiele is half-correct (with a caveat) when he observes that a careful examination indicates that all biblical writers of this period used the [Year 0] Accession Year system for Hebrew, Babylonian, and Persian kings.[1754]

XX. This caveat is because of a strange phenomenon that is scientifically (mathematically) observed in which Year 0 Accession Years are used only for the legitimate kings of Judah, whereas Year 1 Non-Accession Years are used for all illegitimate rulers (including the illegitimate, breakaway, northern kingdom of Israel) including the illegitimate usurper Queen-Mother Atalyahu who usurped the throne of Judah from 892 – 891 BCE through 886 – 885 BCE (Julian). This phenomenon can only be

1749 Tetley 2014: 7
1750 Tetley 2014: 7
1751 Tetley 2014: 7
1752 Tetley 2014: 7
1753 Thiele 1983: 180
1754 Thiele 1983: 180

observed via the mathematical chronology of the kings of Judah and Israel with the numbers and dates that are recorded in various books of the Hebrew Scriptures (i.e. Old Testament) regarding these ancient kings of Judah and the breakaway northern kingdom of Israel.[1755]

49.0. Year 0 Accession Years vs. Year 1 Non-Accession Years

XX. As a direct insight from the mathematics – and subsequent applied logic and reason – one can observe that the writer(s) and editor(s) of the books of Kings, Chronicles, and Jeremiah use as a consistent and verifiable rule that is observed through the math: Year 0 Accession Years are used only for legitimate kings of Judah (as well as for those of Babylon and Persia) that are exclusively from the line of David. All others (e.g. Atalyah Queen-Mother of Ahazyahu ben Yehoram and all kings of Israel) were considered illegitimate rulers of Judah/Israel – thus disqualifying them from the evident legitimacy of kingship from the line of David expressed in Year 0 Accession Year counting.

XX. Mesopotamian chronological practice, as established from the 14th century BCE (Julian) onwards, counted the official regnal years of a king from the first [1st Regnal Year] New Year's Day after his [Year 0 Accession Year] enthronement.[1756] The period of time from the actual [Year 0 Accession Year] enthronement of a king to the next [1st Regnal Year] New Year's Day was known as the [Year 0] Accession Year (*rēš šarrūti, šurrû šarrūti*) – see Brinkman (1976: 403); Tadmor (1958: 27-28).[1757]

XX. This ends up being convenient (and perhaps precisely one of the reasons for its use) that the kings of Israel are recorded chronologically using a Year 1 Non-Accession Year counting system, i.e. because some of the kings of Israel ruled for less than one (1) year, one (1) month, and some only a few days, which would make using a Year 0 Accession Year counting system impossible for reigns less than one (1) year.

XX. Thus the use of a Year 0 Accession Year counting system was not practical (or possible) for recording the chronology of the kings of Israel. However, such was possible and practical for recording the chronology of the kings of Judah – all except for counting the reign of notably the usurper Queen-Mother Atalyah(u) (i.e. mother of Ahazyahu Ben Yehoram king of Judah) who usurped the throne of Judah from 892 – 891 BCE through 886 – 885 BCE (Julian).

XX. Notably, Queen-Mother Atalyah(u) is the only ruler of Judah whose counting years of reign did not use a Year 0 Accession-Year counting. That is to say that the mathematics reveal that the seven (7) years of Queen-Mother Atalyahu's reign over Judah were counted and recorded using a Year 1 Non-Accession Year counting system (exactly as with all kings of Israel whom Judah would have regarded

1755 cf. Appendix Table X.2, X.194
1756 Bloch 2012a: 55-56
1757 Bloch 2012a: 56

as illegitimate usurpers to the throne of David) in direct contrast to all other legitimate kings of Judah whose reigns were counted using a Year 0 Accession Year counting system.

XX. Such use of two (2) different systems of chronological dating that can be very precisely mathematically calibrated and scientifically verified testify to the precision of the writer(s) and editor(s) of the books of Kings, Chronicles, and Jeremiah. Both of these systems, for chronologically recording the dates of the reigns of kings in the ancient Near East, are discussed below.[1758]

XX. This phenomenon of use of a Year 1 Non-Accession Year system is observed for all recorded dates of the kings of Israel in the Divided Kingdom. Therefore, it is clear that the writer(s) and editor(s) of the books (i.e. from the perspective of Judah who survived and wrote the history) of Kings, Chronicles, and Jeremiah consider all kings of the breakaway northern kingdom of Israel to be illegitimate rulers of the one (1) legitimate Hebrew Judean (Jewish) people of Israel/Judah.

XX. Therefore, as will be be unfolded and proven mathematically, all dates (years, months, days) of all the legitimate kings of Judah (i.e. all excluding the usurper Queen-Mother Atalyah(u) who ruled Judah for seven (7) years from 892 – 891 BCE through 886 – 885 BCE (Julian)) were recorded with the Year 0 Accession Year counting system. All others were considered illegitimate usurpers to the throne of the line of David) with no Year 0 Accession Year (according to the mathematics).[1759]

XX. In the [Year 0] Accession Year system, the year in which the king comes to the throne is called his Accession Year [i.e. Year 0] and – for the purposes of counting his years of reign – his 1st official [Regnal] Year is that which begins with the [1st] New Year's Day [i.e. 1st day of Nisannu / Nisan in Spring] after his [Year 0] accession [to the throne].[1760] This is generally accepted on the basis of the Judeo-Babylonian synchronisms for that period.[1761]

XX. [This is in contradistinction to the Year 1] Non-Accession-Year system [whereby] the year in which the king comes [i.e. accedes] to the throne is counted as his 1st year (i.e. Year 1) [from his very 1st day as king] regardless of how many days or months remain in that year, and his 2nd year begins with the 1st New Year's Day [i.e. 1st day of Nisan][1762] after his accession [to the throne].[1763]

XX. To reiterate: Though it is not immediately clear from a casual reading of the Hebrew Scriptures, careful study and scientific mathematical analysis of the chronological dates of all biblical writers proves mathematically that the Year 0 Accession Year system was used for all legitimate kings of

1758 cf. Appendix Table X.2, X.194
1759 cf. Appendix Table X.2, X.194
1760 Finegan 1998: 247
1761 Tadmor 1956: 227
1762 Finegan writes "Tishri 1 or Nisan 1", but it will be proven mathematically that Nisan 1 is the only option or possibility.
1763 Finegan 1998: 247

Judah, and the mathematics allows us to also see clearly that a Year 1 Non-Accession Year system was used for all illegitimate rulers of Judah and Israel (i.e. Queen-Mother Atalyah(u) of Judah and every king of Israel).

XX. The only other central debate in modern scholarship (which will be mathematically proven and put to rest) is which (if any) of the biblical writers used a Tashritu / Tishri year instead of a Nisannu / Nisan year in their recording of dates of kings' regnal years in the Hebrew Scriptures. According to Galil, all the testimonies discovered by researchers attesting that the new year began in Israel or in Judah in the Spring or in the Fall are unconvincing, and cannot provide a decisive answer to this question.[1764]

XX. However, as will be clearly demonstrated and mathematically proven, the Nisannu / Nisan year is the only year-dating system used for the chronological dating of all kings including Judah, Israel, Babylon, and Persia.

50.0. Years of Kings Counted from 1st of Nisan

XX. The Babylonian Talmud (tractate *Rosh HaShanah* 1A) tells us definitively that the New Year's Day for dating of kings is the 1st day of Nisannu / Nisan, and indeed lists several New Years' Days – not just one (1):

XX. *"[The] New Years' Days[1765] are: On [day] one of Nisan is New Year's Day for [the counting of years of] kings and for pilgrimmage-festivals. On [day] one of Elul is New Year's Day for [the counting of] tithing of beast[s]. Rabbi Eliezer and Rabbi Shimon say, 'On [day] one of Tishri – on [day] one of Tishri – is New Year's Day for [the counting of] years and for [the counting of] sabbatical-years and for [the counting of] jubilee-years and for planting and for vegetables. On [day] one of Shvat is New Year's Day for [the return of Spring growth of] the tree as per words of the House of Shammai; the House of Hillel says [that the return of Spring growth for the tree is] on [day] fifteen in it [i.e. on day 15 in the month of Shvat].'"*

XX. Thus ancient Jewish scholars from the time of the Mishnah and Talmud codified many centuries of their accumulated understanding then in the 2nd – 4th centuries CE (Julian) that the counting of years of kings was from the 1st day of Nisannu / Nisan — not the 1st day of Tashritu / Tishri. Therefore modern scholars such as Thiele, who argue for a Tishri year system for – strangely – only some of the biblical writers, are simply ignorant of ancient Jewish scholarship that clearly, categorically, and unequivocally declares that the counting of the years of kings is from day one (1) of the month of Nisannu / Nisan using a Nisan-to-Nisan year dating system.

1764 Galil 1996: 10
1765 Literally:"heads of years".

XX. Thiele is correct that biblical writers such as Jeremiah, Ezekiel, Haggai, and Zechariah used Nisannu / Nisan years (i.e. years beginning in the Spring month of Nisannu / Nisan) for reckoning years for Hebrew kings, as well as for the rulers of Babylon and Persia.[1766] However, Thiele is incorrect (as are all other scholars who follow him) when he states that "there are wide diversities concerning the months with which regnal years began." Thiele is thus further mistaken when he asserts that "the recorders of [the book of] Kings continued to use Tishri years (i.e. years beginning in the Autumn month of Tashritu / Tishri) for Hebrew kings and apparently also for Babylonian rulers."[1767]

XX. Another example is that while Thiele admits that Persia itself had Nisannu / Nisan years, he incorrectly asserts that "Ezra-Nehemiah used Hebrew Tishri years for Persia." As evidence of his incorrect reckoning, Thiele uses as evidence Nehemiah 1:1 and 2:1, "for Nehemiah was in the palace with Artaxerxes in Kislev (9th month) of the 20th Regnal Year,[1768] and the following Nisan (1st month) was still in the 20th Regnal Year.[1769] That would have been the 21st Regnal Year if the regnal year had been reckoned from Nisan".[1770] However, as explained above, Nehemiah 2:1 was precisely in the 21st Regnal Year, and Thiele incorrectly reasons from the simple scribal error that here Tishri-to-Tishri years are used.

XX. This issue in Nehemiah 1:1 and 2:1 has perplexed even the most brilliant Jewish scholars throughout the course of history. Rashi admitted that he could offer no explanation for this issue with a solution. However the most simple solution of all has been overlooked by all scholars until now.

XX. Simple, logical, reasonable explanations are often the best, most accurate, most comprehensive, and most feasible of explanations. As will be explained in greater detail later in this study, this discrepancy and seeming contradiction – if we are to use Nisan-to-Nisan years for the seeming discrepancy in logic in Nehemiah 1:1 and 2:1 – led Thiele to the incorrect assumption that Tishri-to-Tishri years are used. As will be unfolded, this seeming discrepancy can be attributed to scribal error, writing year 20 instead of the correct 21.

XX. Additionally, Thiele also incorrectly applies this Tishri reckoning of dates for Hebrew kings to the Book of Daniel.[1771] Using faulty reasoning to support his position (that will be examined, unfolded, and refuted in detail later), Finegan follows Thiele to argue incorrectly that the Book of Daniel uses Tashritu / Tishri years[1772] when in fact the Book of Daniel uses Nisannu / Nisan years like every other book in the Hebrew Canon of Scriptures.

1766 Thiele 1983: 180
1767 Thiele 1983: 180
1768 cf. Nehemiah 1:1
1769 cf. Nehemiah 2:1
1770 Thiele 1983: 180
1771 Thiele 1983: 180
1772 Finegan 1998: 246, 254

XX. As observed by Gershon Galil, the author of the Book of Kings undoubtedly attributed great importance to the data concerning the reigns of all the kings of Israel and Judah, and without exception he included in his composition the data for all the kings of Israel and Judah including the kings who ruled only a few days or weeks. He also took pains to mention the synchronisms of all the kings of the Northern and Southern Kingdoms from the schism [i.e. the beginning of the Divided Kingdom] to the fall of Samaria, and even arranged his composition in the chronological-synchronistic order.[1773]

XX. According to the Babylonian Talmud (tractate *Bava Batra* 15), "Jeremiah wrote his [own] book [i.e. the Book of Jeremiah], and [the] Book of Kings, and [the Book of] Lamentations." Thus the question would logically arise why the Prophet Jeremiah – as the purported writer of the Book of Kings – would choose a different dating system (i.e. Tishri to Tishri years) than the one used in his own book (i.e. Nisan to Nisan years), the Book of Jeremiah? Furthermore, why would Jeremiah as the author of both the Book of Jeremiah and the Book of Kings (which both contain specific dates of Year, Month, Day) use a different dating system (i.e. Tishri years) than the one that appears elsewhere in other books of the Hebrew Scriptures (i.e. Nisan years), e.g. throughout the five (5) books of the Torah itself, the Books of Joshua, Ezekiel, Haggai, Zechariah, as well as Ezra, Nehemiah, Esther, Daniel, and Chronicles?

XX. Logically – and obviously – Jeremiah wouldn't use the different Tishri-year dating system for the Book of Kings as argued by Thiele and all other scholars who follow him. Therefore it will be put forth here the mathematically testable scientific theory (and mathematically proven) that all scholars who adhere to the Tishri-to-Tishri year system for any dates whatsoever recorded in Hebrew Scriptures for the recording of the dates of kings are simply mistaken and incorrect with faulty reasoning that becomes clear with good logic and reason applied to indisputable mathematics. Therefore all dates recorded in the Hebrew Scriptures that are relative to a king's regnal year absolutely, categorically use a Nisan-to-Nisan year dating system without exception as will be demonstrated and proven throughout the course of this work.

XX. Furthermore, why would Daniel and/or the editor(s) of the Book of Daniel use a different dating system than the one used in all other books of the Hebrew Scriptures where dates appear?

XX. Logically, he – and/or they – would not use a different dating system.

51.0. Biblical Chronology vs. Mesopotamian Astronomy

XX. [In contrast to the biblical chronology,] it has long been widely accepted that we have detailed control [i.e. understanding] of the lunar calendar of the Neo-Babylonian and [Persian] Achaemenid

1773 Galil 1996: 11

periods on the basis of the evidence of contemporary economic and astronomical records.[1774] In recent times, Parker and Dubberstein (1956) have made a theoretical study of the Babylonian calendar using crescent visibility computations.[1775] Their studies of the Babylonian calendar [produced] tables [that purportedly] allow Babylonian dates between 626 BCE and 75 CE (Julian) to be readily converted to the Julian calendar.[1776]

XX. [These] tables [that have been used among scholars since 1956 CE (Gregorian) and even prior to this with an earlier edition by Parker and Dubberstein] for the rapid and accurate conversion of dates from the Babylonian lunar calendar to the Julian calendar in the entire period from 626 BCE to 75 CE (Julian) – taking into account [the geometrical arcus visionis] visibility of the lunar crescent – have been constructed by Parker and Dubberstein.[1777] The Julian dates that are given in Wiseman's book (1956) follow the tables of R.A. Parker and W.H. Dubberstein in *Babylonian Chronology 626 B.C. – A.D. 45* (1942).[1778]

XX. As Walker notes, the fundamental tables were printed in 1956 by Parker and Dubberstein and have never seriously been questioned. The principal problem involved is to determine the years in which a 13th lunar month – the "intercalary" month – was added to bring the lunar year back into line with the solar year, and which intercalary month was used: Ululu 2 / Elul 2 (month 6b), or Addaru 2 / Adar 2 (month 12b).[1779]

52.0. Egyptian Chronology (Based on Thiele's Chronology) is Relative to Assyrian Chronology

XX. A key synchronism for the standard chronology of Egypt (and wider Near Eastern history) concerns the identification of the important Egyptian Pharaoh Shoshenq I with the Shishak attested in the Bible[1780] as invading Judah and Israel in the 5th Regnal Year of Rehabeam [Rehav'am/Rehoboam].[1781] According to current scholarship, Rehabeam Year 5 is in turn dated circa 926/925 BCE [even though in actuality it was 977 – 976 BCE] (Julian)[1782] by linking the attested names and reign lengths of the 10th – 9th century BCE (Julian) kings of Israel and Judah with recorded synchronisms in the 9th century BCE (Julian) between the Israelite kings Ahab and Jehu and the Assyrian king Shalmaneser III.[1783]

1774 Walker 1997: 22
1775 Steele 2000b: 96
1776 Steele, Stephenson, and Morrison 1997: 338; Steele 2000b: 96
1777 Stephenson and Fatoohi 1993: 257
1778 Wiseman 1956: 7, note 2
1779 Walker 1997: 22
1780 cf. 1 Kings 11:40; 14:25-26; 2 Chronicles 12:2-12
1781 Manning 2006: 350
1782 cf. Appendix Table X.2, X.194
1783 Manning 2006: 350-351

XX. Since the chronology of the Assyrian kings is [thought to be] effectively absolute back to the 10th century BCE (Julian), this enables precise calendar dates to be applied (of respectively circa 853 BCE (Julian) for the last year of Ahab's reign [even though in actuality it was 903 – 902 BCE (Julian)][1784] and circa 841 BCE (Julian) for the first year of Jehu's reign [even though in actuality it was 892 – 891 BCE (Julian)]).[1785] [1786]

XX. Naturally enough, Egyptologists assume that the scholars concerned with the chronology of Israel and Judah have established the correct dates for the Hebrew kings and that they can confidently use Rehoboam's 5th Regnal Year in 925 BCE (Julian) as the date for Shoshenq I's 20th Regnal Year [even though in actuality it was 977 – 976 BCE (Julian)].[1787] It seems that they have not investigated the textual evidence for themselves so they do not realize that Thiele's dates are based on many assumptions and not on the actual dating method indicated in the statements of accession given with the regnal years cited.[1788]

XX. Egyptologists have accepted Thiele's dates for Rehoboam's 5th Regnal Year, and they rely on synchronisms with Assyria dated to the AEC while at the same time limiting their use of lunar and astronomical data.[1789]

53.0. Conversion of Egyptian Dates to Julian Dates

XX. Astronomers reckon ancient astronomical events in terms of the Julian – not the Gregorian – calendar. The correlation between the Egyptian and the Julian calendars is implicit [yet not explicit enough] in Ptolemy's dating of astronomical observations according to the Egyptian calendar and to the 365.0-day Egyptian [Civil] year of the so-called Ptolemaic Royal Canon which began with [slightly before the beginning of] Year 1 [i.e. officially in the Year 0 Accession Year before 01 Nisannu / Nisan] of the Babylonian king Nabonassar.[1790] [1791] [1792]

XX. During this 1st year [Anno 1] of the Era Nabonassar [AN][1793] the correlation between the Egyptian [365.0-day civil calendar dates] and the corresponding Julian [365.25-day] calendar date was:

Royal Canon, Year 1: 01 [1st Egyptian Month] Thoth

1784 cf. Appendix Table X.2, X.194
1785 cf. Appendix Table X.2, X.194
1786 Manning 2006: 351
1787 cf. Appendix Table X.2, X.194
1788 Tetley 2014: 7
1789 Tetley 2014: 7
1790 1 Anno Nabonassar (AN), i.e. 26 February 747 BCE (Julian)
1791 cf. Appendix Table X.6
1792 Hornung, Krauss, and Warburton 2006: 49
1793 1 Anno Nabonassar (AN), i.e. 26 February 747 BCE (Julian)

== 26 February 747 BCE (Julian)[1794] [1795]

== Year (Anno) 1 of Nabonassar (AN)[1796]

XX. This date of 26 February 747 BCE (Julian) in the Julian 365.25-day calendar is equivalent to Day 1 of [Egyptian 1st Month] Thoth in the final days of the Year 0 Accession Year of Babylonian king Nabonassar (i.e. before the beginning of his 1st Regnal Year on 01 Nisannu / Nisan). In the proleptic Hebrew calendar, the 1st Regnal Year of Nabonassar would begin on 01 Nisannu / Nisan 3014 AM, Saturday 22 March 747 BCE (Julian).[1797]

XX. No record survives of whether the Babylonian calendar intercalated an Addaru 2 / Adar 2 that year. If it did intercalate, then the Babylonian 01 Nisannu / Nisan fell one (1) to two (2) days after the Hebrew, about 23 / 24 March 747 BCE (Julian). The proleptic Hebrew calendar would then match the Babylonian calendar to within one (1) to two (2) days, as it does in the 105 years in which the two calendars were in step from 626 to 424 BCE (Julian).

XX. If the Babylonian calendar did not intercalate, then the Babylonian 01 Nisannu / Nisan fell one (1) month earlier in late February 747 BCE (Julian). In that case 01 Thoth (26 February 747 BCE (Julian)) fell at the very beginning of the 1st Regnal Year of Nabonassar rather than in the final days of his Year 0 Accession Year

XX. This date 26 February 747 BCE (being equivalent to Year 1 A.N.) in the Egyptian 365.0-day calendar is a crucial calibration [and anchor] point to correct and fine-tune a more accurate calibration of chronologies of the Ancient Near East.

XX. The Year 0 Accession of Nabonassar is based upon the 1st lunar eclipse in his reign. Its report, preserved in a text published as LBAT 1413 (== BM 41985), is the earliest observation record of a [Babylonian] lunar eclipse that we possess. The eclipse was observed in Babylon in the 12th month of the Year 0 Accession Year of Nabonassar. Beaulieu and Britton, as well as Huber and De Meis date this to 06 February 747 BCE (Julian).[1798] [1799]

1794 Wednesday 26 February 747 BCE (Julian); Wednesday 18 February 747 BCE (Gregorian); 06 Addaru / Adar 3014 AM; Astronomical Julian Day JD_X: 1448637.5; RD_Y: 1100641; RD_X: -272787; Sabbatical Cycle: 4; Metonic Cycle: 12; 01 (Egyptian Month 01 Thoth) 1 AN; 0 days since beginning of Epoch;

1795 cf. Appendix Table X.6

1796 Hornung, Krauss, and Warburton 2006: 49

1797 Saturday 22 March 747 BCE (Julian); Saturday 14 March 747 BCE (Gregorian); 01 Nisannu / Nisan 3014 AM; Astronomical Julian Day JD_X: 1448661.5; RD_Y: 1100665; RD_X: -272763; Sabbatical Cycle: 4; Metonic Cycle: 12; 25 (Egyptian Month 01 Thoth) 1 AN; 24 days since beginning of Epoch;

1798 Beaulieu and Britton 1994: 79

1799 Huber and De Meis 2004: 68, 180

XX. In the proleptic Hebrew calendar, the eclipse of Thursday 06 February 747 BCE (Julian) falls on 16 Shevat 3014 AM,[1800] [1801] whereas the Babylonian record places it on day 14 of the 12th month Addaru / Adar. Both are correct for their own calendar. The proleptic Hebrew month is calculated from the molad, whereas the Babylonian month began with the observed first visible crescent about two (2) days later. In that year the Babylonian month count also ran one (1) month ahead of the Hebrew.[1802] [1803] The eclipse date itself is fixed astronomically at 06 February 747 BCE (Julian).

XX. On this basis it is [theoretically] possible to convert any Egyptian date in a given or chosen year into the Julian calendar equivalent – with due consideration that the Julian calendar [365.25 days per year] has a leap year and the Egyptian calendar [365.0 days per year] does not. Tables for conversion have been constructed by P. V. Neugebauer (1937), *Astronomische Nachrichten*, No. 6261 (Kiel, 1937).[1804]

XX. Thus the Julian calendar shifts relative to the Egyptian calendar at the rate of exactly one (1) day every four (4) years. However, relative to the sky the Julian calendar itself drifts slowly: about one (1) day every ~128 years against the seasons (tropical year), and about one (1) day every ~157 years against the fixed stars (sidereal year).

XX. Parker presented a circumstantial argument in favor of the beginning of the [Egyptian Civil] calendar day at Dawn: “It is obvious that, when the [lunar] month begins, the first day of the month also begins.” Most Egyptologists accepted the validity of Parker’s argument. However in the 1980s CE (Gregorian) Leitz and Luft revived the arguments presented by Sethe in the 1920s CE (Gregorian) that the calendar day began with Sunrise.[1805] They were countered by scholars like Grzybek, Spalinger and Wells who defended Parker’s standpoint.[1806]

XX. Actually – according to Hornung, Krauss, and Warburton – August Böckh had solved the question in 1863 CE (Gregorian) when he used Ptolemy’s *Almagest* to demonstrate that the Egyptian calendar day began at Dawn before Sunrise. In the *Almagest* Ptolemy twice recorded observations of Mercury, first with double dates:[1807]

1800 Thursday 06 February 747 BCE (Julian); Thursday 29 January 747 BCE (Gregorian); 16 Shabatu / Shvat 3014 AM; Astronomical Julian Day JD_X: 1448617.5; RD_Y: 1100621; RD_X: -272807; Sabbatical Cycle: 4; Metonic Cycle: 12; 16 (Egyptian Month 12 Mesori) 0 AN; -20 days since beginning of Epoch;

1801 cf. Appendix Table X.5

1802 Thursday 06 March 747 BCE (Julian); Thursday 26 February 747 BCE (Gregorian); 14 Addaru / Adar 3014 AM; Astronomical Julian Day JD_X: 1448645.5; RD_Y: 1100649; RD_X: -272779; Sabbatical Cycle: 4; Metonic Cycle: 12; 09 (Egyptian Month 01 Thoth) 1 AN; 8 days since beginning of Epoch;

1803 cf. Appendix Table X.6

1804 Hornung, Krauss, and Warburton 2006: 49

1805 Hornung, Krauss, and Warburton 2006: 49

1806 Hornung, Krauss, and Warburton 2006: 49-50

1807 Hornung, Krauss, and Warburton 2006: 50

Hadrian, year 18, [11th Month] Epiphi 18 to 19, at Dawn;[1808]

Antoninus, year 4, [7th Month] Phamenoth 18 to 19, at Dawn.[1809]

Subsequently, he gave single dates for the same observations:[1810]

Mercury as morning star: Hadrian, year 18, [11th Month] Epiphi 19[1811]

Mercury as morning star: Antoninus, year 4, [7th Month] Phamenoth 19[1812]

XX. Mercury is only observable at Dawn as morning star or at Dusk as evening star. Ptolemy could have substituted single dates for double dates only if he reckoned Dawn as the beginning of the Egyptian calendar days [11th Month] Epiphi 19 and [7th Month] Phamenoth 19. Accordingly – assert Egyptologists Hornung, Krauss, and Warburton – the Egyptian calendar day began during the observability of Mercury as morning star, i.e. at Dawn before Sunrise.[1813]

XX. Hieroglyphic sources also exist that imply the beginning of the calendar day at Dawn – notably an entry in the Calendar of Lucky and Unlucky Days describing the appearance of Seth in the bow of the solar bark which led Malinine to conclude that the calendar day began before Sunrise, i.e. the time when the god appeared at the bow of the solar bark. His argument is supported by the identification of the god Seth with planet Mercury here rising shortly before the Sun and thus visible at Dawn.[1814]

XX. Because an Egyptian calendar day (*D*) begins at Dawn before Sunrise, it overlaps two (2) Julian calendar days (*d*) and (*d + 1*) that begin at Midnight [00:00]. [P.V.] Neugebauer's tables (1937) yield Julian calendar day (*d*) as equivalent of Egyptian calendar day (*D*) [i.e. although they are not exactly equivalent].[1815]

XX. The assertion that an Egyptian calendar day (*D*) equals the Julian calendar day (*d*), means that the bright part of Egyptian calendar day (*D*) corresponds to the bright part of Julian calendar day (*d*) [because both the Egyptian and Julian day each start at different times, i.e. Dawn (06:00+/-) vs. Midnight (00:00) respectively]. [Thus] it is implied that the early hours of Julian calendar day (*d*) are not contained in Egyptian calendar day (*D*), and that the last hours of Egyptian calendar day (*D*) correspond to the first hours of the Julian calendar day (*d + 1*).[1816]

1808 Hornung, Krauss, and Warburton 2006: 50
1809 Hornung, Krauss, and Warburton 2006: 50
1810 Hornung, Krauss, and Warburton 2006: 50
1811 Hornung, Krauss, and Warburton 2006: 50
1812 Hornung, Krauss, and Warburton 2006: 50
1813 Hornung, Krauss, and Warburton 2006: 50
1814 Hornung, Krauss, and Warburton 2006: 50
1815 Hornung, Krauss, and Warburton 2006: 51
1816 Hornung, Krauss, and Warburton 2006: 51

XX. Therefore following the reasoning of Egyptologists that the Egyptian day began at Dawn before Sunrise we can attempt to be more precise with the calibration of the Egyptian calendar day (*D*) to the Julian calendar day (*d*) within the 1 AN by reasoning that Noon (12:00) is our Calibration Anchor point to synchronize Egyptian and Julian calendars:

> Royal Canon, Year 1: [Noon (12:00)] 01 [1st Egyptian Month] Thoth
>
> == [Noon (12:00)] 26 February 747 BCE (Julian)
>
> == Year (Anno) 1 of Nabonassar (AN)

XX. It must be recalled that the Julian calendar (i.e. 365.25 days per year) shifts relative to the Egyptian calendar (i.e. 365.0 days per year) at the rate of exactly one (1) day every four (4) years.

54.0. The Historical Chronology of Egypt

XX. Egyptian chronology is based on combining several different systems.[1817] The historical chronology of Egypt is [not an absolute chronology, but rather is a relative chronology that is] basically an interpretation of a complicated network of interlocked [mostly relative] data such as king lists, genealogical information, astronomical observations, and similar sources.[1818] During the historic/dynastic period, written sources in Egypt include king lists, royal annals, and biographic treatises.[1819] [1820] [Nonetheless] Egyptian chronology is still the [current] touchstone by which all of the other chronologies in the ancient world are measured and the issue of its reliability is thus central.[1821]

XX. [In Egyptian chronology] absolute (calendar) dates are mainly derived from so-called "dead reckoning" (calculating reign lengths backward from [i.e. relative to] an assumed fixed point in time) or from astronomical observations such as Sothic or lunar data. The textual sources, their interpretation, and the historical reconstructions based upon them, have been summarized several times in the recent literature.[1822] [1823]

XX. According to Krauss and Warburton, this "dead reckoning" is the most reliable system as it is based on counting regnal years documented from contemporaneous written sources. The principle is

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1817 Krauss and Warburton 2009: 125
1818 Höflmayer 2016: 2
1819 Redford 1986; Kantor 1992; Ward 1992 – cited by Hasel 2004
1820 Hasel 2004: 11
1821 Hornung, Krauss, and Warburton 2006: 8
1822 For recent assessments, see Kitchen 2013 – cited by Höflmayer 2016
1823 Höflmayer 2016: 2

simply establishing the sequence of kings and establishing a minimum number of years for the reign of each individual king. This produces a minimum basis for the estimation of the date of accession and death of each king, and thus from a given point one can simply count backwards providing a reasonably reliable [minimum] skeleton for a chronology.[1824]

XX. However, it is far from certain that the final year of a reign will be recorded in a dated inscription. Thus, these dates are only a minimum; other sources can be used to argue that reigns were actually longer than the final attested dates.[1825]

XX. There are also lists of kings preserved from various different periods, and some of these can be used to suggest possible sequences of kings and possible lengths of reigns. However, these lists were edited, and potentially contain copying errors. In general, the copies of such lists which survived until today are themselves copies of other copies, and the originals based on the actual documents are not preserved. Although not in themselves reliable sources, they can on occasion be used to see whether a particular solution might be compatible with the facts known from elsewhere.[1826]

XX. Thirdly, for certain periods one can also track the genealogy of private families through time, and relate these to the kings under whom they lived. Where members of such families can be related to one another and to regnal year dates, one can use these as a control for reigns where either the absolute chronology or the length of the reign is in dispute.[1827]

XX. The fourth system relies on synchronisms with rulers of Mesopotamia, Syria, and Anatolia. This method is based on philological sources and simply states that two rulers either ruled simultaneously (if they exchanged several letters, for example) or that they ruled at roughly the same time (if circumstances related to one king can be related to another).[1828]

XX. The fifth system depends upon archaeological typology and stratigraphy, linking objects which can be assigned relative positions in archaeological chronologies with the reigns of dated kings (e.g., Middle Bronze Age Western Asian axes known from archaeological excavations and from dated paintings). An abundance of monuments dating to a particular reign can also be used to argue that that reign may have been longer than the last date preserved if that date is rather low. Conversely, a paucity of remains can be used to support a shorter reign.[1829]

XX. In some cases, archaeological objects which can be linked to royal names can also be radiocarbon

1824 Krauss and Warburton 2009: 125
1825 Krauss and Warburton 2009: 125
1826 Krauss and Warburton 2009: 125
1827 Krauss and Warburton 2009: 125
1828 Krauss and Warburton 2009: 125
1829 Krauss and Warburton 2009: 125

dated, providing a range of possible radiocarbon years, and thus the sixth system.[1830]

XX. Together, these six systems provide some kind of foundation for a relative chronology.[1831]

XX. Absolute chronologies can only be established using astronomical data, which is the seventh system. Egyptian astronomical data preserved in the written documentation is of two (2) kinds: monthly lunar dates and Sothic dates.[1832]

XX. Astronomical events tend to repeat themselves at regular intervals, and thus the examination of ancient astronomical phenomena can provide a number of windows which can be assigned for certain kings. Dead-reckoning, synchronisms, archaeological evidence, and Carbon 14 dates can be exploited to suggest which windows are more reasonable than others. However, the repetitions of astronomical events are not necessarily exactly regular, allowing one to favor certain events and discard others.[1833]

XX. Kings assigned a number to each year of their reign. Over the millennia, there were a number of different methods of dealing with the actual year of the death of one king, and the accession of his successor. Thus in some epochs, the [365.0-day] civil year is simply divided between two (2) kings (as in the Middle Kingdom), so that neither the final year of one king's reign nor the 1st year of the next king's reign correspond to one entire Egyptian year.[1834]

XX. In others, the civil year and the regnal year are not identical (as in the New Kingdom), so that the 1st Regnal Year of a king is an entire year, while obviously the final "year" of the preceding king is not – since the new 1st Regnal Year is simply counted as 365 days from the accession, and is followed by a 2nd year based upon the date of accession, i.e. not the 1st day of the new year (as in the Middle Kingdom).[1835]

XX. Obviously, this means that Egyptian years have no relationship to sidereal years, and Egyptian months in the civil calendar have no relationship to lunar months. It also means that over the millennia, there is room for doubt about a certain number of years. According to Krauss and Warburton, the advantage of dead-reckoning is that it offers an absolutely minimal framework.[1836] However, one could argue that this is a disadvantage because it yields a low chronology and does not provide the accurate count of years.

XX. On the other hand, however, the very irregularities themselves give rise to disputes. Furthermore,

1830 Krauss and Warburton 2009: 125-126
1831 Krauss and Warburton 2009: 126
1832 Krauss and Warburton 2009: 126
1833 Krauss and Warburton 2009: 126
1834 Krauss and Warburton 2009: 126
1835 Krauss and Warburton 2009: 126
1836 Krauss and Warburton 2009: 126

there is always a risk of recording errors. However, in general, [by] using the basic framework to provide a minimum date, the choice of astronomical data does not leave much leeway, and thus – according to Krauss and Warburton – the chronology can be straightened out at certain points.[1837]

XX. The most important synchronisms for building up a chronology [especially] for the period under review [by Krauss and Warburton, i.e. the Pharaohs of the 18th Dynasty] are those between the Egyptian kings and their Near Eastern neighbors: the Hittites, Assyrians, and Babylonians. Unfortunately, the chronology of the Hittite kings is itself dependent upon the chronology of the Egyptian and Mesopotamian kings, so that effectively – for detailed chronological reconstructions – we can really only work with the Mesopotamian kings.[1838]

XX. Fortunately, the most important problem in Mesopotamian chronology lies before 1500 BCE (Julian) and thus before the New Kingdom, so that the king lists from Babylon and Assyria provide a sufficient basis to buttress the Egyptian chronology of the New Kingdom.[1839]

XX. It was during the reigns of [Egyptian Pharaohs] Akhenaten and his father Amenhotep III that the Amarna letters were received and stored. Those which have been recovered were found at Akhenaten's capital at Amarna.[1840] These chronologically important Amarna letters were written by the rulers of Babylonia, Mitanni, and Assyria,[1841] and were written to Amenhotep III, the father of Akhenaten, and to Akhenaten himself.[1842]

XX. According to the documentation in the archive, Burnaburiash II of Babylon became king and wrote his first letter to Egypt between Regnal Year 31 and 36 of Amenhotep III. For Burnaburiash II, Brinkman gives 1359 – 1333 BCE (Julian); Klinger, 1349 – 1323 ± 5 years BCE (Julian); and Gasche et al. 1354 – 1328 BCE (Julian). According to dead-reckoning, as presented above, Year 1 of Akhenaten would have been 1317 BCE (Julian). Fortunately, Burnaburiash II also wrote letters to Akhenaten, and thus it follows that the reigns need to be raised, possibly by as much as a couple of decades.[1843]

XX. Thus there can be little doubt that the reigns of the Egyptian kings up to the Amarna period must be adjusted from the dead-reckoning calculations presented above to match their Mesopotamian counterparts with whom they exchanged letters. This is hardly surprising, but merely an indication of the degree of variability. However, dead-reckoning and other calculations become important for the details again later, when specific dates are proposed. Thus, we must now turn to further internal control

1837 Krauss and Warburton 2009: 126
1838 Krauss and Warburton 2009: 130
1839 Krauss and Warburton 2009: 130
1840 Krauss and Warburton 2009: 130
1841 Krauss and Warburton 2009: 130-131
1842 Krauss and Warburton 2009: 131
1843 Krauss and Warburton 2009: 131

mechanisms.[1844]

XX. Thus our most useful documentation records dates in terms of regnal years using the [365.0-day Egyptian] civil calendar. Some dated documents also refer to the observation or prediction of the heliacal rise of Sothis, while other sources reveal that the Egyptians also observed the phases of the Moon, and used these to determine the dates of at least some of their festivals. On occasion, a lunar date can be related to an historical date, either due to the use of the civil calendar or because the date can be estimated for some other reason.[1845]

XX. In principal, a lunar month lasts 29 or 30 days. The Egyptian day one (1) of a lunar month is defined as the 1st day of the Moon's invisibility. On the last day of the lunar month (which is simultaneously the day before the 1st day of the following lunar month), the waning moon was seen for a last time, just before Sunrise. Thus if the moon was visible on the morning of Lunar Day 30, then the next lunar month would start the next day. If it was invisible, the new lunar month had already started.[1846]

XX. Thus in principle one should be able to examine a given lunar date tied to an historical date using the civil calendar, and establish whether or not a crescent was visible on that day. Furthermore, the lunar months can be arranged in 25-year cycles and thus even longer chronological units can be isolated. However, there are two (2) completely different complications which intervene into our chronological calculations.[1847]

XX. The first is the simple matter of erroneous observations: a crescent which was present could have been obscured by clouds or haze, or the observer might have made a recording error. Fortunately, aside from the Egyptian observations and astronomical calculations of lunar movements, Babylonian lunar observations are available, and remarkably, these confirm the Egyptian data. Significantly – according to Krauss – in all of the material he has analyzed, there is not one single case of an error resulting in the recording of the observation of a crescent which should not have been visible. According to him, the "error" that does appear in the documentation of the roughly 300 dates studied is that crescents which should have been visible were not recorded as "observed".[1848]

XX. The second complication is related to the fact that in principle 25 Egyptian [365.0-day Solar] years == 309 synodic Lunar months == 9125 days. Thus any given lunar date could be repeated after exactly 25 years, meaning for example that two different texts mentioning instances of the same lunar-based festival in different reigns can be related, and an interval proposed.[1849]

1844 Krauss and Warburton 2009: 131
1845 Krauss and Warburton 2009: 132
1846 Krauss and Warburton 2009: 132
1847 Krauss and Warburton 2009: 132
1848 Krauss and Warburton 2009: 132
1849 Krauss and Warburton 2009: 132

XX. The difference between cycles of 309 synodic Lunar months and 25 Egyptian [365.0-day Solar] years consists of 1 hour 7 minutes, so that most cycles last 9125 days. Because the difference is small, one might suppose that after 25 Egyptian years the same lunar phase falls on the same Egyptian calendar day.[1850]

XX. However, because of the irregularity of the movement of the Moon, in reality no Lunar month lasts 29.53059 days. Rather, there are Lunar months of a full 29 or a full 30 days. Over 309 synodic Lunar months, there are often 164 months of 30 days and 145 months of 29 days.[1851]

XX. However, it can [also] happen that in a sequence of 309 Lunar months there are 165 months with 30 days each and 144 months with 29 days. In such a case, a lunar day does not coincide with the same calendar day after exactly 25 Egyptian years, but rather it falls on the next calendar day. If, by contrast, there were 145 lunar months with 29 days and 164 lunar months with 30 days, then after 25 Egyptian years, the given lunar day fell on the previous day of the calendar (in comparison to the date 25 years earlier).[1852]

XX. Thus we can see that the complexity of the interaction of the Egyptian calendar systems is very problematic because of their imprecision and margins for error in attempts to reconstruct the ancient Egyptian chronologies.

XX. For a long time the Egyptian historical chronology was the sole chronological reference system not only for the Nile Valley itself but also for the Bronze and Iron Age eastern Mediterranean basin. It was indeed the backbone of history especially during the 2nd – and much of the 1st – millennium BCE.[1853] The chronology of Mesopotamia and much of the remainder of the ancient Near East has been linked with [so-called quote “] Egyptian absolute dating [” end-quote].[1854] [1855]

55.0. Egyptian Chronology: The Backbone of History?

XX. Vast amounts of literature have been devoted to ancient Egyptian history including the pursuit of its chronology. With all the resources available, it is remarkable that neither the relative nor the absolute chronology of ancient Egypt has yet been established.[1856] A relative chronology refers to the time-span between kings, whereas an absolute chronology refers to the [exact] dates applied to

1850 Krauss and Warburton 2009: 132
1851 Krauss and Warburton 2009: 132
1852 Krauss and Warburton 2009: 132-133
1853 Höflmayer 2016: 2
1854 Knapp 1992:716 – cited by Hasel 2004
1855 Hasel 2004: 11
1856 Tetley 2014: 1

kings.[1857]

XX. Egyptologists who adhere to the commonly assumed chronology (derived from written records) are fairly confident that their dates of ca. 1540 – 1530 BCE (Julian) for the beginning of the [New Kingdom] 18th Dynasty are accurate. However these dates are challenged by scientists who rely on recent radiocarbon and other science-based tests, who propose [raising/lengthening the chronology to] a date in the mid-to-late 17th century BCE (Julian), a disparity of 100 – 150 years.[1858]

XX. Scientists cannot explain how their dates can be so much higher than those derived from written records, and historical chronologists cannot see how one and a half (1.5) centuries can be added to the chronology based on the kings' regnal years.[1859]

XX. A starting date for the Egyptian chronology has to come from a king of Egypt who can be dated by the Julian calendar. The earliest certain (but late) date comes at the end of the 25th Dynasty when Taharqa acceded the throne in 690 BCE (Julian) and after a 26-year reign died in 664 BCE (Julian). He was succeeded by Psammetichus I who became the 1st king of the 26th Dynasty.[1860]

XX. From this date Egyptologists proceed backwards [in "dead-reckoning" of relative chronology] using "known" regnal years of the kings of Egypt. This system gives a minimal [i.e. a Low / Short] chronology. Since the final regnal years of most kings are not stated, additional years beyond their highest attested years [i.e. a Higher / Longer Chronology] need to be considered.[1861]

XX. In an effort to tie the minimal [Low / Short] chronology to an external date, Egyptologists look for a synchronism with a neighboring nation [i.e. a synchronism with another civilization's historical chronology].[1862]

XX. They utilize one (1) at the beginning of the 22nd Dynasty noting that the invasion of Shoshenq I (the biblical Shishak) of Israel and Judah is dated to the 5th Regnal Year of Rehoboam [Hebrew: Rehav'am] of Judah.[1863] [According to Egyptology] this [possibly and feasibly] equates in the Egyptian chronology with Shoshenq I's 20th year because his victory stelae describing the campaign dates it to his 21st year.[1864]

XX. The military campaign of Sheshonq I (biblical Shishak), the founder of the 22nd Dynasty in Egypt, to Canaan is reported as a list of conquered towns on a wall in the temple of Amun at Karnak in Upper

1857 Tetley 2014: 2
1858 Tetley 2014: 1
1859 Tetley 2014: 1
1860 Tetley 2014: 4
1861 Tetley 2014: 4
1862 Tetley 2014: 4-5
1863 cf. 1 Kings 14:25-26; 2 Chronicles 12:2-5
1864 Tetley 2014: 5

Egypt and mentioned in 1 Kings 14:25-28. The widely accepted dating of the campaign to [the purported 5th Regnal Year of Rehoboam in] 926 BCE (Julian)[1865] is based solely on the biblical reference. Taking these factors into consideration, the Sheshonq I campaign could have – as noted by Finkelstein – taken place almost any time in the mid- to late 10th century BCE (Julian). According to Finkelstein: One can hardly be more precise.[1866]

XX. [Since Egyptologists synchronize this 20th Regnal Year of Shoshenq I with the 5th Regnal Year of Rehoboam,] Egyptologists then look to the chronology of Judah and Israel to find when this invasion took place [i.e. Egyptian Chronology is ultimately based on Biblical Chronology]. They find that in 1944 CE (Gregorian) Edwin R. Thiele, a scholar of St. Andrews Seventh Day Adventist Seminary of Berrien Springs, MI, USA, placed Rehoboam's 5th Regnal Year in 925 BCE (Julian) with the commencement of the divided kingdom in 931 BCE (Julian),[1867] [1868] [even though it actuality Rehoboam's 5th Regnal Year was 977 – 976 BCE (Julian)].[1869]

XX. [However, the question arises:] Where did Thiele get this date from? He looked to records from Assyria, and specifically the AEC [*Assyrian Eponym Canon*], and derived his dates from it.[1870]

XX. This Assyrian canon [i.e. the *Assyrian Eponym Canon* (AEC)] is compiled from fragments of eponym lists found on tablets in the ruins of three (3) sites (Nineveh, Assur, and Sultantepe) that were copied in the 7th century BCE (Julian) from earlier records.[1871] Where one (1) list is broken away, one (1) or more other lists appear to overlap it, yielding a composite list of 261 or 262 eponyms.[1872]

XX. The pieces appear to overlap and are now made into one (1) long list that seems to be continuous apart from one (1) section where an eponym appears in one (1) list but not in others. This canon is a list of Assyrian kings and their officials with each year being named after the king or one of his subordinates, and called an eponym year (*limmu* in Assyrian [Akkadian]).[1873] If the list is complete, its extant portion extends back to 911 or 910 BCE (Julian).[1874]

XX. Despite uncertainties, the reliability of the AEC appears to be confirmed from the restoration of the Assyrian King List (itself composed from several kings lists). The reliability of Assyrian, Hebrew, Egyptian, and related chronologies depends upon the tenuous – and unverifiable – assumption that the lists copied in the 7th BCE (Julian) were then complete and without error. Despite Millard's confident

1865 The actual correct date for the 5th Regnal Year of Rehoboam is 977 – 976 BCE (Julian).
1866 Finkelstein 2013: 41
1867 Tetley 2014: 5
1868 The actual correct date for the 1st Regnal Year of Rehoboam is 981 – 980 BCE (Julian).
1869 cf. Appendix Table X.2, X.194
1870 Tetley 2014: 5
1871 Tetley 2014: 5
1872 Tetley 2005: 97
1873 Tetley 2014: 5
1874 Tetley 2005: 97

assertion, the kings lists cannot be used as a definite a confirmation since they may incorporate whatever faults their *Vorlage(n)* contained.[1875]

XX. Furthermore, the reliability of a composite list collated from fragments has been questioned in at least two (2) places: 1.) scholars acknowledge confusion among the eponyms for Adad-nirari III's reign; and 2.) aligning the reign of Joash-I with Adad-nirari III (to whom Joash-I paid tribute) is problematic.[1876]

XX. Significant for chronology is the note against the eponym of a certain Bur-Saggile "of Guzanna" about a "revolt in the citadel: in Siwan [i.e. Simannu / Sivan] the Sun had an eclipse." Scientists are able to date this eclipse [hitherto and heretofore] to 15/16 June in 763 BCE (Julian). Proceeding upward and downward from this date the surviving eponyms have been attributed [hitherto and heretofore] to the years from 910 to 649 BCE (Julian). Another list, the Babylonian king-list, begins in 747 BCE (Julian) and together with the [*Assyrian Eponym Canon*] AEC and other records the chronology of Assyria is securely linked to the Babylonian king-list from 747 BCE (Julian) forward [i.e. Year 1 of the Era of Nabonassar, or Anno Nabonassar (AN), whose epoch began 26 February 747 BCE (Julian)].[1877]

XX. The chronological framework for the history of the entire Near East in the second half of 2nd millennium BCE (Julian) is ultimately based on the Assyrian chronology of the relevant historical period (the Middle Assyrian period).[1878] This is due to the fact that the *Assyrian King List* (AKL) constitutes the only documentary source that presents a continuous chronology of the second half of the 2nd millennium BCE (Julian) and connects it to the chronology of a later period (i.e. the 9th – 8th centuries BCE (Julian)) for which the precise absolute regnal dates of the Assyrian kings can be established based on continuous lists of Assyrian yearly eponyms and on astronomical data.[1879]

XX. The key astronomical evidence enabling the precise absolute dating of the Assyrian eponyms – and hence also of the regnal dates of the Assyrian kings in the 9th – 7th centuries BCE (Julian) – is the solar eclipse in the month of Simānu [i.e. Simannu / Sivan] of the eponym year of Būr-saggilê which has been dated to Monday 15 June 763 BCE (Julian),[1880] [1881] i.e. day 30 of the 3rd month of Simannu / Sivan.

XX. Although no exact day is given for this solar eclipse, a solar eclipse can only occur at conjunction of the Sun and Moon (i.e. No Moon) which is only possible on the 29th or 30th day of a Babylonian /

1875 Tetley 2005: 97
1876 Tetley 2005: 97
1877 Tetley 2014: 5
1878 Bloch 2010: 41
1879 Bloch 2010: 41-42
1880 Monday 15 June 763 BCE (Julian); Monday 07 June 763 BCE (Gregorian); 30 Simannu / Sivan 2998 AM; Astronomical Julian Day JD_X: 1442902.5; RD_Y: 1094906; RD_X: -278522; Sabbatical Cycle: 2; Metonic Cycle: 15;
1881 Bloch 2010: 42

Hebrew month, i.e. the last day of the month and one (1) before the 1st of the new month, e.g. day 30 of the month of Simannu / Sivan before the next day one (1) of the 4th month of Du'uzu / Tammuz.

XX. Indeed, if this year can be confirmed as the eponym year of Būr-saggilê, then this date can be confirmed as a possibility for a solar eclipse in the month of Simannu / Sivan in year 763 BCE (Julian). However, Tuesday 16 June 763 BCE (Julian) can be ruled out because it is equivalent to day one (1) of the 4th month Du'uzu / Tammuz.[1882]

XX. However, the date of 15 June 763 BCE (Julian) for the solar eclipse of Bur-Sagale has been challenged several times in the past. Some have fixed this solar phenomena as that of Wednesday 24 June 791 BCE (Julian),[1883] [1884] and others[1885] have identified it with the eclipse of Friday 13 June 809 BCE (Julian).[1886] [1887] [1888] [1889]

XX. As is evident, both of these dates are equivalent with day 29 of the 3rd month Simannu / Sivan, and therefore are not possible. However, the following last day 30 for each of these suggested dates remain possible for a solar eclipse in this 3rd month of Simannu / Sivan. According to the NASA canon for solar eclipses, both of these eclipses would have been visible over Assyria as annular eclipses.[1890]

XX. [Hitherto and heretofore] most historical chronologists [notably excluding Tetley who uses the Septuagint text of the Bible] derive their dates for the ancient Near East including Assyria, Israel/Judah, and Egypt on the presumed accuracy of the *Assyrian Eponym Canon* (AEC). Indeed the entire Near Eastern chronology relies on the accuracy of the AEC for the years 910 – 612 BCE (Julian).[1891] Thus Egypt gets its dates from the Hebrew chronology of Israel and Judah, and the Hebrew chronological construction gets its dates from the AEC.[1892]

XX. Therefore it is clear to see that there are significant – if not huge – margins for error in the transmission (and synchronization) of these relative chronologies across civilizations: each with its own

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1882 Tuesday 16 June 763 BCE (Julian); Tuesday 08 June 763 BCE (Gregorian); 01 Du'uzu / Tammuz 2998 AM; Astronomical Julian Day JD_X: 1442903.5; RD_Y: 1094907; RD_X: -278521; Sabbatical Cycle: 2; Metonic Cycle: 15;

1883 Wednesday 24 June 791 BCE (Julian); Wednesday 16 June 791 BCE (Gregorian); 29 Simannu / Sivan 2970 AM; Astronomical Julian Day JD_X: 1432684.5; RD_Y: 1084688; RD_X: -288740; Sabbatical Cycle: 2; Metonic Cycle: 6;

1884 Thursday 25 June 791 BCE (Julian); Thursday 17 June 791 BCE (Gregorian); 30 Simannu / Sivan 2970 AM; Astronomical Julian Day JD_X: 1432685.5; RD_Y: 1084689; RD_X: -288739; Sabbatical Cycle: 2; Metonic Cycle: 6;

1885 Jules Oppert (1868); Martin Anstey (1913) and George Smith (1875) – cited by Jones 2002: 154

1886 Friday 13 June 809 BCE (Julian); Friday 05 June 809 BCE (Gregorian); 29 Simannu / Sivan 2952 AM; Astronomical Julian Day JD_X: 1426099.5; RD_Y: 1078103; RD_X: -295325; Sabbatical Cycle: 5; Metonic Cycle: 7;

1887 Saturday 14 June 809 BCE (Julian); Saturday 06 June 809 BCE (Gregorian); 30 Simannu / Sivan 2952 AM; Astronomical Julian Day JD_X: 1426100.5; RD_Y: 1078104; RD_X: -295324; Sabbatical Cycle: 5; Metonic Cycle: 7;

1888 Jules Oppert (1868); George Smith (1875); Martin Anstey (1913) – cited by Jones 2002: 154

1889 Jones 2002: 154

1890 NASA/TP–2006–214141: Espenak and Meeus (2006): *Five Millennium Canon of Solar Eclipses: –1999 to +3000 (2000 BCE to 3000 CE)*

1891 Tetley 2014: 5

1892 Tetley 2014: 6

various, variable "moving parts"/ variables with the increased possibility to be "Lost in Translation" and transmission.

XX. Indeed a simpler methodology with a better, simpler, more elegant solution would be preferable – and indeed it already exists.

56.0. The Chronology of the Exodus

XX. Having established fixed, mathematically-synchronized dates for the reigns of all the Hebrew kings in the Divided Monarchy (i.e. the kingdoms and kings of Judah and Israel), it is possible to proceed to count backwards to calculate (i.e. reverse engineer) the chronology of the Exodus by first establishing the 4th Regnal Year of the United Monarchy of Shlomo (Solomon) ben David, and then proceeding to count backwards with the reigns of Shlomo (Solomon) and David.[1893] [1894]

XX. By establishing and utilizing the key dates of the 4th and 11th Regnal Years of the reign of Shlomo (Solomon) as the dates of the foundation of (until completion of the construction of) the Temple of Shlomo (Solomon) ben David, it is possible to calculate the exact date – and subsequent chronology – of the Exodus.

XX. We will extend our reconstruction of the synchronization of absolute chronologies from the split of Shlomo's (Solomon's) United Monarchy after his death, i.e. at the beginning of the reign of his son Rehav'am ben Shlomo. We will make the reasonable working assumption (to be tested) that Rehav'am ben Shlomo's Year 0 Accession Year was the same year as the 40th (FINAL) Regnal Year of Shlomo (Solomon) ben David.[1895] [1896]

XX. The Hebrew Scriptures record that Rehav'am ben Shlomo was 41 years old upon his accession to the throne (i.e. in his Year 0 Accession Year), and he reigned for 17 years.[1897] [1898]

XX. Calculating back from the 40th (FINAL) Regnal Year of Shlomo (Solomon) that began Spring 982 – 981 BCE (Julian), his Year 0 Accession Year was 1022 – 1021 BCE (Julian). Therefore the 1st Regnal Year of Shlomo (Solomon) began 01 Nisannu / Nisan 1021 – 1020 BCE (Julian).

XX. This date of the 4th Regnal Year of Shlomo (Solomon) is critical to calculation of dates back through the Exodus and those prior. Therefore, counting forward four (4) years from his Year 0

1893 cf. Appendix Table X.1, X.2, X.3, X.4, X.194
1894 cf. 1 Kings 6:1; 37-38
1895 cf. Appendix Table X.1, X.2, X.3, X.194
1896 cf. 1 Kings 11:42-43; 2 Chronicles 9:30-31
1897 cf. Appendix Table X.2, X.3, X.194
1898 cf. 1 Kings 14:21; 2 Chronicles 12:13-14

Accession Year, we arrive to the 4th Regnal Year of Shlomo (Solomon) commencing 01 Nisannu / Nisan in Spring 1018 – 1017 BCE (Julian):[1899]

1 And [it] was in year 480 of the Exodus of the sons of Israel from the land of Egypt, in the 4th year, in the month of Ziv – that [is] the 2nd month[1900] – of the reign of Shlomo over Israel, and [he] built the House for YHVH.[1901]

37 In the 4th year, the House of YHVH was founded in the Moon of Ziv. 38 And in the year 11 in the Moon of Bul – that [is] the 8th month[1902] – the House was complete for all its things and for all its statutes, and [he] built it seven (7) years.[1903]

XX. Thus the construction of the 1st Temple took 7.5 years from the 2nd month of the 4th Regnal Year (1018 – 1017 BCE (Julian)) through the 8th month of the 11th Regnal Year (1011 – 1010 BCE (Julian)) of Shlomo (Solomon). It stood for 424 years from the Babylonian / Hebrew year 2751 AM through 3175 AM. This number 424 is the gematria numerical value for the Hebrew term for Messiah son of David (Mashiach ben David, משיח בן דוד).

XX. Counting backwards 480 years from the 4th Regnal Year of Shlomo (Solomon), we arrive to the 0th Year of the Exodus (i.e. No Such Thing) in Spring 1498 – 1497 BCE (Julian). Logically the Exodus is not a king acceding to the throne in a year of accession, and thus does not require a Year 0 Accession Year. Therefore YEAR ONE (1) OF THE EXODUS began immediately on the day of their Exodus, Saturday 15 Nisannu / Nisan 2264 AM in Spring 1497 BCE (Julian), and counting continued through YEAR 480 OF THE EXODUS that began Spring 1018 BCE (Julian) in the 4th Regnal Year of Shlomo (Solomon) ben David.

57.0. The Date of the Exodus

XX. The details of the first days of the Exodus are recorded in Numbers 33 along with each encampment of the Sons of Israel from day one (1) of the YEAR ONE (1) OF THE EXODUS[1904] until the YEAR 40 OF THE EXODUS[1905] with the death of Aharon.[1906]

XX. In this passage, there are recorded 33 encampments from the Exodus until the death of Aharon,

1899 cf. 1 Kings 6:1; 37
1900 Evidently, the 2nd month of Ziv is equivalent to the 2nd month of Ayyaru / Iyar.
1901 cf. 1 Kings 6:1
1902 Evidently, the 8th month of Bul is equivalent to the 8th month of Arahsamna / Marheshvan (Heshvan).
1903 cf. 1 Kings 6:37-38
1904 cf. Appendix Table X.109
1905 cf. Appendix Table X.135
1906 cf. Numbers 33:1-39

and then another eight (8) encampments until the arrival at Jericho,[1907] i.e. a total of 41 recorded encampments during the wandering in the desert by the Sons of Israel during those 40 years in the desert:

0.) *Ra'amses* (רעמסס)

1.) *Sukkot* (סכת)

2.) *Eitam* (אתם)

3.) The Mouth of the Freedoms, i.e. *Pi HaHirot* (פי החירת)

4.) *Marah* (מרה)

5.) *Eilimah* (אילמה)

6.) The Sea of Reeds, i.e. *Yam Suf* (ים סוף)

7.) The Desert of Sin, i.e. *Midbar Sin* (מדבר סין)

8.) *Dafkah* (דפקה)

9.) *Alush* (אלוש)

10.) *Refidim* (רפידם)

11.) The Desert of *Sinai*, i.e. *Midbar Sinai* (מדבר סיני)

12.) The Graves of Desire, i.e. *Kivrot HaTa'avah* (קברת התאוה)

13.) *Hatzerot* (חצרת)

14.) *Ritmah* (רתמה)

15.) *Rimon Paretz* (רמן פרץ)

16.) *Livnah* (לבנה)

17.) *Risah* (רסה)

1907 cf. Numbers 33:40-49

18.) *Kehelatah* (קהלתה)

19.) The Mountain of *Shafer*, i.e. *Har Shafer* (הר שפר)

20.) *Haradah* (חרדה)

21.) *Makhelot* (מקהלת)

22.) *Tahat* (תחת)

23.) *Tarah* (תרח)

24.) *Mitkah* (מתקה)

25.) *Hashmonah* (חשמנה)

26.) *Moserot* (מסרות)

27.) *Ya'akan* (יעקן)

28.) *Gidgad* (גדגד)

29.) *Yatvatah* (יטבתה)

30.) *'Avronah* (עברנה)

31.) *'Etzion Gaver* (עציון גבר)

32.) The Desert of *Tzin* that [is] *Kadesh*, i.e. *Midbar Tzin hu Kadesh* (מדבר צן, קדש)

33.) *Hor* the Mountain, i.e. *Hor HaHar* (הר ההר)

34.) *Tzalmonah* (צלמנה)

35.) *Funon* (פונן)

36.) *Ovot* (אבת)

37.) *'I'yei Ha'Avarim*, a.k.a. *'I'yim* (עיי העברים, עיים)

38.) *Divon Gad* (דיבן גד)

39.) *'Almon Divlataimah* (עלמן דבלתימה)

40.) The Mountains of the *'Avarim* before *Nevo*, i.e. *Harei Ha'Avarim lifnei Nevo* (הרי העברים לפני נבו)

41.) The Plains of Mo'av upon Jordan Jericho, i.e. *'Arvot Mo'av 'al Yarden Yereho* (ערבת מואב על ירדן ירחו)

XX. In the Book of Numbers 33:1-3 it is written that the Sons of Israel left Egypt on day 15 of the 1st month, i.e. Nisannu / Nisan, on the day after the Passover 2264 AM,[1908] by the hand of Moses and Aharon. This works about to be Saturday 18 April 1497 BCE (Julian),[1909] [1910] meaning that the Passover Night of the Seder (*Leil HaSeder*) occurred on the previous night of Friday evening, i.e. on the Eve of Sabbath (Erev Shabbat).

XX. Since the Exodus itself out of Egypt occurred on a Sabbath (Saturday), it is now possible to understand one of the several reasons mentioned for remembering the commandment to keep the Sabbath is to remember that we were slaves in Egypt.[1911]

XX. After having determined the precise date of the Exodus of Israel from Egypt, one can calculate the precise date for the beginning of the 430 years sojourn of Israel within Egypt that is mentioned in Exodus 12, i.e. 1927 BCE (Julian),[1912] although there is a tradition in Judaism that the actual sojourn lasted only 210 years.

XX. While the Egyptians were burying their dead, on Saturday 15 Nisannu / Nisan YEAR ONE (1) OF THE EXODUS, the Sons of Israel traveled from Ra'amses and camped in Sukkot.[1913] Then they traveled from Sukkot, and camped in Eitam on the edge of the desert.[1914] Then they traveled from Eitam and sat upon Pi-HaHirot that was upon the face of Ba'al Tzafon, and they camped before Migdol.[1915]

XX. All of this happened in the first six (6) days of the Exodus before the parting of the Sea which – according to Jewish tradition (although not specifically stated in the Hebrew Scriptures) – occurred on

1908 cf. Numbers 33:3
1909 Saturday 18 April 1497 BCE (Julian); Saturday 05 April 1497 BCE (Gregorian); 15 Nisannu / Nisan 2264 AM; Astronomical Julian Day JD_X: 1174751.5; RD_Y: 826755; RD_X: -546673; Sabbatical Cycle: 3; Metonic Cycle: 3;
1910 cf. Appendix Table X.110
1911 cf. Deuteronomy 5:12-15
1912 cf. Exodus 12:40-42
1913 cf. Numbers 33:3-5
1914 cf. Numbers 33:6
1915 cf. Numbers 33:7

the 7th day of the Exodus. Indeed, this is logical reasoning that this is the event that is commemorated in the observance of also the 7th day of the Passover as a Holy Day (Holiday) Convocation in addition to the 1st day.[1916]

XX. In the Book of Exodus, it becomes evident that the commandment for Moses and the Sons of Israel to sit and camp before Pi-HaHirot ("The Mouth of Freedoms") came after their arrival to camp in Eitam.[1917] Furthermore, it becomes clear that this was the encampment from which they crossed the Sea[1918] – i.e. from "the Mouth of [Many] Freedoms". Exodus describes Pi-HaHirot as being between Migdol ("Tower") and the Sea before Ba'al Tzafon ("Master of the North") [in] its presence, and He commands them to camp upon the Sea.[1919]

57.1. The Date of the Parting of the Sea of Reed[s]

XX. It took the entire night of the 7th Day of Passover to cross the Sea of Reed[s], i.e. the night of 20/21 Nisannu / Nisan. This works out to be the evening of Thursday 23 April 1497 BCE (Julian)[1920] and continuing to the morning of Friday 24 April 1497 BCE (Julian).[1921] [1922]

57.2. Three (3) Days to Marah

XX. Having arrived to the other side of the Sea of Reed[s] by the morning of Friday 24 April 1497 BCE (Julian), they traveled another three (3) days to (and through) the Desert of Shur/Eitam and did not find water, and then they arrived to camp in Marah ("Bitter") by 24 Nisannu / Nisan 2264 AM.[1923] This works out to be Monday 27 April 1497 BCE (Julian).[1924] [1925]

57.3. From Marah to Eilim, to the Sea of Reed[s], to the Desert of Sin

XX. Some time after the incident of the Bitter Waters made Fresh, the Sons of Israel traveled

1916 cf. Exodus 12:1-20; 13:3-10; 23:15; 34:25; Leviticus 23:4-8; Numbers 28:16-25; Deuteronomy 16:1-8
1917 cf. Exodus 13:17-22; 14:1-2
1918 cf. Exodus 14:1-2
1919 cf. Exodus 14:1-2
1920 Thursday 23 April 1497 BCE (Julian); Thursday 10 April 1497 BCE (Gregorian); 20 Nisannu / Nisan 2264 AM; Astronomical Julian Day JD_X: 1174756.5; RD_Y: 826760; RD_X: -546668; Sabbatical Cycle: 3; Metonic Cycle: 3;
1921 Friday 24 April 1497 BCE (Julian); Friday 11 April 1497 BCE (Gregorian); 21 Nisannu / Nisan 2264 AM; Astronomical Julian Day JD_X: 1174757.5; RD_Y: 826761; RD_X: -546667; Sabbatical Cycle: 3; Metonic Cycle: 3;
1922 cf. Appendix Table X.110
1923 cf. Exodus 15:22-23; Numbers 33:8
1924 Monday 27 April 1497 BCE (Julian); Monday 14 April 1497 BCE (Gregorian); 24 Nisannu / Nisan; Astronomical Julian Day JD_X: 1174760.5; RD_Y: 826764; RD_X: -546664; Sabbatical Cycle: 3; Metonic Cycle: 3;
1925 cf. Appendix Table X.110

from Marah and camped in Eilim;[1926] Then they traveled from Eilim and camped upon [the shore of] the Sea of Reed[s];[1927] Then they traveled from the Sea of Reed[s] and arrived to camp in the Desert of Sin – that is between Eilim and Sinai – on day 15 of the 2nd month (i.e. Ayyaru / Iyar) of their Exodus from the Land of Egypt.[1928] This works out to be Monday 18 May 1497 BCE (Julian).[1929] [1930]

57.4. From the Desert of Sin to Dakfa, to Alush, to Refidim

XX. The Sons of Israel traveled from the Desert of Sin and camped in Dafka;[1931] Then they traveled from Dafka and camped in Alush;[1932] Then they traveled from Alush and camped in Refidim where there was no water for the people to drink.[1933]

XX. The account in Exodus skips over providing descriptions for Dafka and Alush – perhaps because these were encampments that did not last very long, i.e. they were short-term transit points – between their sojourns at the Desert of Sin and Refidim which became known as Masah u-Merivah ("A Test and a Fight") because the people tested YHVH, fought with Moses, and demanded water.[1934]

57.5. The Battle with Amalek at Refidim

XX. During their stay at Refidim, Amalek came and fought with Israel, and Moses built an altar to YHVH.[1935]

57.6. From Refidim to the Mountain of Sinai

XX. The Sons of Israel traveled from Refidim directly to the Mountain of Sinai:

In the 3rd month of the Exodus of the Sons of Israel from the Land of Egypt – on this day – [they] arrived to the Desert of Sinai, and Israel camped there in front of the Mountain.[1936]

1926 cf. Exodus 15:27; Numbers 33:9
1927 cf. Numbers 33:10
1928 cf. Exodus 16:1; Numbers 33:11
1929 Monday 18 May 1497 BCE (Julian); Monday 05 May 1497 BCE (Gregorian); 15 Ayyaru / Iyar 2264 AM; Astronomical Julian Day JD_X: 1174781.5; RD_Y: 826785; RD_X: -546643; Sabbatical Cycle: 3; Metonic Cycle: 3;
1930 cf. Appendix Table X.111
1931 cf. Exodus 17:1; Numbers 33:12
1932 cf. Numbers 33:13
1933 cf. Exodus 17:1; Numbers 33:14
1934 cf. Exodus 17:1-7
1935 cf. Exodus 17:8-16
1936 cf. Exodus 19:1-2; Numbers 33:15

XX. Although it may seem that this could be understood as their having arrived exactly three (3) months after the day of the Exodus, i.e. day 15 of the 3rd month of Simannu / Sivan, the wording is not clear. However, the mathematics reveal that this verse should be understood to not imply three (3) full months, but rather as their having arrived in the 3rd month of the Exodus ("on this day"), i.e. [day] one (1) of the 3rd month of Simannu / Sivan. This works out to be Tuesday 02 June 1497 BCE (Julian).[1937] [1938]

XX. According to Jewish tradition, the receiving of the Torah on Mount Sinai is commemorated on the holiday of Shavuot (i.e. "Weeks"),[1939] and took place on day six (6) of the 3rd month of Simannu / Sivan. Indeed, this is exactly 50 days for the counting of the Omer, as commanded to count seven (7) Sabbaths:

15 And [you] will count for yourselves from the day of your bringing the 'Omer for waiving, seven (7) complete Sabbaths [they] will be. 16 Until the day after the 7th Sabbath [you] will count 50 days, and [you] will sacrifice a new offering.[1940]

XX. Counting seven (7) Sabbaths starting from Sunday 16 Nisannu / Nisan 2264 AM (i.e. the day after the Saturday 15 Nisannu / Nisan of the 1st Passover) brings us to Saturday 06 June 1497 BCE (Julian),[1941] [1942] i.e. 49 days. The next day, i.e. the 50th day, was Sunday 07 June 1497 BCE (Julian),[1943] [1944] which was commanded to be a Holy Convocation on which no labor of work can be done, i.e. the Feast of Weeks (*Shavuot*).[1945] This is the date on which GOD gave the Torah to Moses and the nation of Israel.

58.0. Determining the Pharaoh of the Exodus

XX. Having determined the exact year of the Exodus to be in 1497 BCE (Julian), we can now attempt to determine who was the Egyptian Pharaoh of the Exodus.

XX. Pending a final recalibration and correct synchronization of Egyptian chronologies with this mathematically accurate biblical chronology, we can theorize – assuming that the current Egyptian

1937 Tuesday 02 June 1497 BCE (Julian); Tuesday 20 May 1497 BCE (Gregorian); 01 Simannu / Sivan 2264 AM; Astronomical Julian Day JD_X: 1174796.5; RD_Y: 826800; RD_X: -546628; Sabbatical Cycle: 3; Metonic Cycle: 3;
1938 cf. Appendix Table X.112
1939 cf. Deuteronomy 16:9-10
1940 cf. Leviticus 23:21
1941 Saturday 06 June 1497 BCE (Julian); Saturday 24 May 1497 BCE (Gregorian); 05 Simannu / Sivan 2264 AM; Astronomical Julian Day JD_X: 1174800.5; RD_Y: 826804; RD_X: -546624; Sabbatical Cycle: 3; Metonic Cycle: 3;
1942 cf. Appendix Table X.118
1943 Sunday 07 June 1497 BCE (Julian); Sunday 25 May 1497 BCE (Gregorian); 06 Simannu / Sivan 2264 AM; Astronomical Julian Day JD_X: 1174801.5; RD_Y: 826805; RD_X: -546623; Sabbatical Cycle: 3; Metonic Cycle: 3;
1944 cf. Appendix Table X.118
1945 cf. Leviticus 23:15-16

chronology holds as reasonably stable during this period – that the most likely candidate for the Pharaoh of Egypt during this time was Thutmose I of the 18th Dynasty who ruled between 1507 – 1494 BCE (Julian) according to the conventional Middle Chronology, between 1504 – 1491 BCE (Julian) according to the Low Chronology, and between 1504 – 1490 BCE (Julian) according to Derstine's Modified Middle Chronology.[1946]

XX. However, the High Chronology of 1525 – circa 1512 BCE (Julian)[1947] would disqualify Thutmose I if this High Chronology for this Pharaoh holds to be true or more accurate. Further research is required to recalibrate and synchronize Egyptian chronology to this mathematically correct, precise biblical chronology before we can definitively determine the correct identity of the Pharaoh of the Exodus in 1497 BCE (Julian).

59.0. The 2nd Year of the Exodus

XX. Although we do not know the exact date in that 1st year of the Exodus that Moses received the command to sanctify and dedicate the Sanctuary, its altars, as well as Aharon as his sons,[1948] it is recorded to have taken place exactly as commanded in the 1st month in YEAR TWO (2) OF THE EXODUS[1949] on [day] 1 of the month, i.e. Nisannu / Nisan.[1950] This works out to be Tuesday 23 March 1496 BCE (Julian).[1951] [1952] In that same 1st month of that YEAR TWO (2) OF THE EXODUS, before the Passover of that 2nd year, YHVH spoke to Moses.[1953]

XX. As evident in the commandment,[1954] the same day of the completion of the Sanctuary was the same day that it was dedicated, with its altars as well as Aharon and his sons anointed.[1955] On that same day, the cloud by day and the fire by night began to cover the sanctuary.[1956]

XX. On that 1st day of Nisannu / Nisan of YEAR TWO (2) OF THE EXODUS, i.e. 1496 BCE (Julian) – which began a 12-day dedication[1957] of the Sanctuary with the anointing of it and its altars, as well as Aharon and his sons – Moses commands Aharon and his sons to not leave from the entrance to the

1946 cf. Derstine 2024: 33
1947 cf. Derstine 2024: 33
1948 cf. Exodus 40:1-16
1949 cf. Appendix Table X.122
1950 cf. Exodus 40:1, 17-35; Numbers 7:1
1951 Tuesday 23 March 1496 BCE (Julian); Tuesday 10 March 1496 BCE (Gregorian); 01 Nisannu / Nisan 2265 AM; Astronomical Julian Day JD_X: 1175090.5; RD_Y: 827094; RD_X: -546334; Sabbatical Cycle: 4; Metonic Cycle: 4;
1952 cf. Appendix Table X.123
1953 cf. Numbers 9:1-14
1954 cf. Exodus 40:1-16, 17-35
1955 cf. Leviticus 3:12-13; 7:35-36; Numbers 7:1, 10-11
1956 cf. Exodus 40:35-38; Numbers 9:15-16
1957 cf. Numbers 7:1, 10-11, 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, 72, 78, 84

Sanctuary for seven (7) days and nights.[1958]

XX. On the 8th day of the 1st month (i.e. Nisannu / Nisan) of YEAR TWO (2) OF THE EXODUS, i.e. 1496 BCE (Julian), Moses calls to Aharon, his sons, and the elders of Israel, and tells him to prepare themselves because today YHVH (and the Glory of YHVH) will appear to them.[1959] Later that day, both Moses and Aharon leave the tent of meeting (i.e. the sanctuary), and the Glory of YHVH appeared to the children of Israel, and a flame of fire went out from before the Glory of YHVH to eat their sacrifices.[1960] This works out to be Tuesday 30 March 1496 BCE (Julian).[1961] [1962]

XX. Immediately after, a second flame of fire went out from the Glory of YHVH to destroy the sons of Aharon, Nadav and Avihu, for having done something that was not commanded.[1963]

XX. On the [day] one (1) of the 2nd month (Ayyaru / Iyar) in YEAR TWO (2) OF THE EXODUS, the Book of Numbers (*BaMidbar*) opens with YHVH commanding Moses to order a census of the children of Israel.[1964] This works out to be Thursday 22 April 1496 BCE (Julian).[1965] [1966]

XX. On the [day] 20 of this 2nd month (i.e. Ayyaru / Iyar) in YEAR TWO (2) OF THE EXODUS, the cloud ascends from above the sanctuary, and leads the Sons of Israel from the mountain of Sinai in the desert of Sinai to the desert of Paran (*Kivrot HaTa'avah*), a three-(3)-day journey.[1967] This works out to be Tuesday 11 May 1496 BCE (Julian).[1968] [1969]

XX. At some point, the cloud ascends and leads the Sons of Israel from Kivrot HaTa'avah in the desert of Paran to Hatzerot, and then back to Ritma in the desert of Paran.[1970]

XX. Although the day that the spies left to scope out the land and its peoples is not explicitly specified, the Torah does specifically state that they returned at the end of 40 days.[1971] After hearing the spies'

1958 cf. Leviticus 8:33-36
1959 cf. Leviticus 9:1-6
1960 cf. Leviticus 9:23-24
1961 Tuesday 30 March 1496 BCE (Julian); Tuesday 17 March 1496 BCE (Gregorian); 08 Nisannu / Nisan 2265 AM; Astronomical Julian Day JD_X: 1175097.5; RD_Y: 827101; RD_X: -546327; Sabbatical Cycle: 4; Metonic Cycle: 4;
1962 cf. Appendix Table X.123
1963 cf. Leviticus 10:1-2
1964 cf. Numbers 1:1-16
1965 Thursday 22 April 1496 BCE (Julian); Thursday 09 April 1496 BCE (Gregorian); 01 Ayyaru / Iyar 2265 AM; Astronomical Julian Day JD_X: 1175120.5; RD_Y: 827124; RD_X: -546304; Sabbatical Cycle: 4; Metonic Cycle: 4;
1966 cf. Appendix Table X.124
1967 cf. Numbers 10:11-12, 33-36; 33:15-16
1968 Tuesday 11 May 1496 BCE (Julian); Tuesday 28 April 1496 BCE (Gregorian); 20 Ayyaru / Iyar 2265 AM; Astronomical Julian Day JD_X: 1175139.5; RD_Y: 827143; RD_X: -546285; Sabbatical Cycle: 4; Metonic Cycle: 4;
1969 cf. Appendix Table X.124
1970 cf. Numbers 11:35; 12:16; 33:17-18
1971 cf. Numbers 13:25

report, the Sons of Israel were dismayed, and cried that night.[1972] According to Jewish tradition, this occurred on the 9th day of the 5th month (Abu / Av) which began at Sundown, i.e. the 9th day of the month begins at evening.

XX. This 9th day of the month began the same evening after the spies had returned during the final daylight hours of the day eight (8) which ends at Sundown. Therefore, one could count backwards 40 days to calculate that (if this tradition is correct) the spies left for their 40-day journey on the day 28 of the 3rd month (Simannu / Sivan). This works out to be Thursday 17 June 1496 BCE (Julian).[1973] [1974]

XX. Counting 40 days forward from this date means that the spies returned on the day eight (8) of the 5th month (Abu / Av) in YEAR TWO (2) OF THE EXODUS. This works out to be Monday 26 July 1496 BCE (Julian).[1975] [1976]

XX. Therefore, day nine (9) of the 5th month (i.e. Abu / Av) works out to be Tuesday 27 July 1496 BCE (Julian).[1977] [1978]

60.0. The 40th Year of the Exodus

XX. In YEAR 40 OF THE EXODUS,[1979] in the 1st month (Nisannu / Nisan), the entire congregation travels to Kadesh in the Desert of Tzin, and there Miryam dies, and is buried.[1980] This works out to be March / April 1458 BCE (Julian).[1981]

XX. From Kadesh, the congregation travels to Hor HaHar where Aharon dies at 123 years of age in YEAR 40 OF THE EXODUS in the 5th month (Abu / Av) on [day] one (1) of the month.[1982] This works out to be Monday 21 July 1458 BCE (Julian).[1983] [1984]

1972 cf. Numbers 14:1
1973 Thursday 17 June 1496 BCE (Julian); Thursday 04 June 1496 BCE (Gregorian); 28 Simannu / Sivan 2265 AM; Astronomical Julian Day JD_X: 1175176.5; RD_Y: 827180; RD_X: -546248; Sabbatical Cycle: 4; Metonic Cycle: 4;
1974 cf. Appendix Table X.125
1975 Monday 26 July 1496 BCE (Julian); Monday 13 July 1496 BCE (Gregorian); 08 Abu / Av 2265 AM; Astronomical Julian Day JD_X: 1175215.5; RD_Y: 827219; RD_X: -546209; Sabbatical Cycle: 4; Metonic Cycle: 4;
1976 cf. Appendix Table X.127
1977 Tuesday 27 July 1496 BCE (Julian); Tuesday 14 July 1496 BCE (Gregorian); 09 Abu / Av 2265 AM; Astronomical Julian Day JD_X: 1175216.5; RD_Y: 827220; RD_X: -546208; Sabbatical Cycle: 4; Metonic Cycle: 4;
1978 cf. Appendix Table X.127
1979 cf. Appendix Table X.135
1980 cf. Numbers 20:1
1981 cf. Appendix Table X.136
1982 cf. Numbers 20:22-29; 33:38
1983 Monday 21 July 1458 BCE (Julian); Monday 08 July 1458 BCE (Gregorian); 01 Abu / Av 2303 AM; Astronomical Julian Day JD_X: 1189089.5; RD_Y: 841093; RD_X: -532335; Sabbatical Cycle: 7; Metonic Cycle: 4;
1984 cf. Appendix Table X.140

XX. The entire congregation saw that Aharon had died, and they cried (i.e. mourned) for him 30 days of the 5th month of Abu / Av until day one (1) of the following 6th month of Ululu / Elul.[1985] [1986]

XX. In YEAR 40 OF THE EXODUS, in the 11th month, on [day] one (1), Moses spoke to all of Israel his final address (i.e. the Book of Deuteronomy).[1987] This works out to be Wednesday 14 January 1457 BCE (Julian),[1988] [1989]

XX. Although not recorded in the Hebrew Bible, according to Jewish tradition, the death of Moses[1990] occurred on 07 Addaru / Adar. Using a chain of texts, this date is derived backwards from the known date of Israel's crossing the Jordan river into the Promised Land on 10 Nisannu / Nisan[1991] of YEAR 41 OF THE EXODUS == YEAR ONE (1) OF THE PROMISED LAND.[1992]

XX. GOD's command to cross was given "after the death of Moses", and Joshua told the people to prepare because they would cross in three (3) days.[1993] So the command was given on 07 Nisannu / Nisan of YEAR 41 OF THE EXODUS == YEAR ONE (1) OF THE PROMISED LAND.[1994]

XX. Israel mourned Moses for 30 days. The Rabbis assume the command in Joshua came right after the mourning ended. Therefore counting back 30 days of mourning from 07 Nisannu / Nisan arrives to 07 Addaru / Adar of YEAR 40 OF THE EXODUS – the day tradition holds that Moses died. This works out to be Thursday 19 February 1457 BCE (Julian).[1995] [1996] [1997]

XX. Although not recorded in the Hebrew Bible, according to Jewish tradition, the day/month of Moses' death was the same day/month as his birth.[1998] This was concluded by the ancient Jewish sages from Moses' statement in the Book of Deuteronomy, "120 years I [am] today."[1999] Therefore, the traditional date of Moses' birthday works out to be Sunday 25 February 1577 BCE (Julian).[2000]

1985 cf. Appendix Table X.140, X.141
1986 cf. Numbers 20:22-29; 33:38
1987 cf. Deuteronomy 1:1-3
1988 Wednesday 14 January 1457 BCE (Julian); Wednesday 01 January 1457 BCE (Gregorian); 01 Shabatu / Shvat 2304 AM; Astronomical Julian Day JD_X: 1189266.5; RD_Y: 841270; RD_X: -532158; Sabbatical Cycle: 1; Metonic Cycle: 5;
1989 cf. Appendix Table X.146
1990 cf. Deuteronomy 34:1-12
1991 cf. Joshua 4:19
1992 cf. Appendix Table X.148
1993 cf. Joshua 1:1-2, 11
1994 cf. Appendix Table X.149
1995 Thursday 19 February 1457 BCE (Julian); Thursday 06 February 1457 BCE (Gregorian); 07 Addaru / Adar 2304 AM; Astronomical Julian Day JD_X: 1189302.5; RD_Y: 841306; RD_X: -532122; Sabbatical Cycle: 1; Metonic Cycle: 5;
1996 cf. Appendix Table X.147
1997 cf. Deuteronomy 34:1-12; Babylonian Talmud, *Kiddushin* 38a
1998 cf. Babylonian Talmud: *Sotah* 12b-13a; *Megillah* 13b; *Kiddushin* 38a
1999 cf. Deuteronomy 31:2
2000 Sunday 25 February 1577 BCE (Julian); Sunday 11 February 1577 BCE (Gregorian); 07 Addaru / Adar 2184 AM; Astronomical Julian Day JD_X: 1145478.5; RD_Y: 797482; RD_X: -575946; Sabbatical Cycle: 7; Metonic Cycle: 18;

61.0. Entering the Promised Land

XX. After Moses had died in YEAR 40 OF THE EXODUS, in the following month, i.e. the 1st month Nisannu / Nisan, in YEAR 41 OF THE EXODUS == YEAR ONE (1) OF THE PROMISED LAND,[2001] Joshua led the congregation of Israel into the Promised Land. They crossed the Jordan River on 10 Nisannu / Nisan 2304 AM.[2002] This works out to be Monday 22 March 1457 BCE (Julian).[2003] [2004]

XX. Calculating back three (3) days, Joshua's command to prepare to cross the Jordan river in three (3) days was given on Friday 19 March 1457 BCE (Julian).[2005] [2006] [2007]

XX. The date of Israel's crossing the Jordan river into the Promised Land in the Hebrew year 2304 AM marks the beginning of exactly 860 Babylonian / Hebrew years (to the very day) until YEAR ONE (1) OF THE EXILE OF YEHOYACHIN (JEHOIACHIN) on 10 Nisannu / Nisan 3164 AM, i.e. Saturday 23 March 597 BCE (Julian).[2008] [2009] In other words, Israel sojourned exactly 860 Babylonian / Hebrew years (i.e. exactly 314116 days) in the land of Israel from 1497 BCE (2304 AM) until being exiled to Babylon in 597 BCE (3164 AM).

62.0. Recalibrating the Chronologies of Assyria, Babylon, and Egypt

XX. After confirming that this chronology appears evidently feasible and plausible as we apply it prolepticly to the events recorded in the Hebrew Scriptures, we can then use synchronisms that link the chronologies of the Bible to Egyptology, and Assyriology (i.e. Assyria and Babylon). These chronologies can then be calibrated to the mathematically precise biblical chronology in order to provide fresh insights into their histories.

XX. For example, we can calculate and recalibrate the precise years for synchronisms with Assyrian and Babylonian kings including the final year of the destruction of the ten (10) tribes of the northern kingdom of Israel in the 9th Regnal Year of Hoshea ben Elah to be definitively 719 – 718 BCE (Julian) under Assyrian king Shalmaneser. We can then determine the identity of the heretofore mysterious

2001 cf. Appendix Table X.148
2002 cf. Joshua 4:19
2003 Monday 22 March 1457 BCE (Julian); Monday 09 March 1457 BCE (Gregorian); 10 Nisannu / Nisan 2304 AM; Astronomical Julian Day JD_X: 1189334.5; RD_Y: 841338; RD_X: -532090; Sabbatical Cycle: 1; Metonic Cycle: 5;
2004 cf. Appendix Table X.149
2005 Friday 19 March 1457 BCE (Julian); Friday 06 March 1457 BCE (Gregorian); 07 Nisannu / Nisan 2304 AM; Astronomical Julian Day JD_X: 1189331.5; RD_Y: 841335; RD_X: -532093; Sabbatical Cycle: 1; Metonic Cycle: 5;
2006 cf. Appendix Table X.149
2007 cf. Joshua 1:1-2, 11
2008 Saturday 23 March 597 BCE (Julian); Saturday 17 March 597 BCE (Gregorian); 10 Nisannu / Nisan 3164 AM; Astronomical Julian Day JD_X: 1503450.5; RD_Y: 1155454; RD_X: -217974; Sabbatical Cycle: 7; Metonic Cycle: 10;
2009 cf. Appendix Table X.34

Pharaoh So of Egypt by cross-calibrating with Egyptian relative chronology using the precise, mathematical Biblical chronology as the Rosetta Stone that synchronizes everything.[2010] [2011]

XX. We can do this for all Assyrian, Babylonian, and Egyptian kings in order to synchronize and precisely recalibrate all of ancient Near East history.

63.0. Applying this Biblical Chronology forward in Time

XX. Not only can this precise biblical chronology be applied proleptically back in time, we can also use it to confirm (or correct) dates as we move forward in time applying it to the Seleucid Calendar, the Era of Contracts (also known as the Era of Alexander), and medieval Jewish history, e.g. Maimonides/Rambam that were recorded with this same calendar.

XX. For example, we can calculate a confirmed or corrected date (i.e. precise month / day) for the death of Alexander the Great.

XX. According to Depuydt, in no modern scholarly work (in fact not even in earlier discussions from the Renaissance onwards) has Alexander's death ever been dated to any other month than June of 323 BCE of the Julian calendar, even if – before the mid-1950s, when cuneiform evidence for the date came to light – this month date was only supported by the Egyptian date of 04 Pharmouthi (i.e. 8th Egyptian month) found in Pseudo-Callisthenes (3.35), which corresponds to Julian 13 June 323 BCE (Julian).[2012]

XX. According to Depuydt, the answers to precisely which day and which hour Alexander died have differed over time. However, according to Depuydt, it is safe to say that since a cuneiform date emerged in 1955 CE (Gregorian), the evening of 10 June 323 BCE (Julian)[2013] [2014] has become the generally accepted time for Alexander's death. According to him, since the cuneiform date became known, the earlier discussion on Alexander's death has become in large part obsolete.[2015]

XX. The evening of 10 June 323 BCE (Julian) is defended in two (2) important studies on Ptolemaic chronology.[2016] The first study was published in 1962 CE (Gregorian) by A.E. Samuel as a revised

2010 cf. Appendix Table X.2, X.194
2011 cf. 2 Kings 17:1-6; 18:7, 9-12
2012 Depuydt 1997: 117
2013 Sunday 10 June 323 BCE (Julian); Sunday 05 June 323 BCE (Gregorian); 29 Simannu / Sivan 3438 AM; Astronomical Julian Day JD_X: 1603607.5; RD_Y: 1255611; RD_X: -117817; Sabbatical Cycle: 1; Metonic Cycle: 18; 01 (Egyptian Month 08 Pharmouthi) 425 AN; 154970 days since beginning of Epoch;
2014 Monday 11 June 323 BCE (Julian); Monday 06 June 323 BCE (Gregorian); 30 Simannu / Sivan 3438 AM; Astronomical Julian Day JD_X: 1603608.5; RD_Y: 1255612; RD_X: -117816; Sabbatical Cycle: 1; Metonic Cycle: 18; 02 (Egyptian Month 08 Pharmouthi) 425 AN; 154971 days since beginning of Epoch;
2015 Depuydt 1997: 117
2016 Depuydt 1997: 117

version of a Yale doctoral dissertation.[2017] The second was published in 1990 CE (Gregorian) by E. Grzybek as a revised version of a University of Geneva doctoral dissertation. It should be noted, though, that the two authors obtain the evening of 10 June 323 BCE (Julian) by different arguments.[2018]

XX. Before the mid-1950s, it is 13 June 323 BCE (Julian) – the Julian equivalent of Pseudo-Callisthenes's Egyptian 04 Pharmouthi (i.e. 8th Egyptian month)[2019] – that is most commonly found in historical works. For example, students of Ptolemaic Egypt will readily encounter this date in the notes to the second edition of T. C. Skeat's tables for converting Ptolemaic dates into Julian dates. However, as Skeat observes in the introduction to this work, the notes were not updated from the first edition of 1954 CE (Gregorian). Dates other than 13 June 323 BCE (Julian) are only rarely proposed in earlier scholarship.

XX. Alexander's life falls in the first three (3) millennia of human history, when different nations as a rule used different calendars. Only after Julius Caesar's reform of 46 / 45 BCE (Julian) would it become normal for many nations to share a single calendar, i.e. the so-called Julian calendar. It is therefore not surprising that the four (4) dates of Alexander's death transmitted in ancient sources follow three different calendars.[2020]

XX. The three (3) major cultural hubs of Alexander's world were Babylonia, Egypt, and Greece. One of the four dates for his death follows the Babylonian calendar: 29 Ayyaru / Iyar (i.e. the 2nd Babylonian month). One follows the Egyptian calendar: 04 Pharmouthi (i.e. 8th Egyptian month). The two (2) others are Macedonian: One Macedonian date, 28 Daisios (i.e. 8th Macedonian month) is found in the Royal Diaries, as cited by Plutarch. The other, "30" Daisios (i.e. 8th Macedonian month),[2021] is given by Aristoboulos, a general of Alexander turned biographer, also as cited by Plutarch.[2022]

XX. The most important date is the Babylonian date, 29 Ayyaru / Iyar.[2023] According to Depuydt, the Babylonian date leaves no doubt that he died between morning and evening during the day on 11 June 323 BCE (Julian). According to him, the formulation of the cuneiform text in the tablet British Museum BM 49562 leaves no doubt that Alexander died between morning and evening of 11 June 323 BCE

2017 Depuydt 1997: 117-118
2018 Depuydt 1997: 118
2019 Wednesday 13 June 323 BCE (Julian); Wednesday 08 June 323 BCE (Gregorian); 02 Du'uzu / Tammuz 3438 AM; Astronomical Julian Day JD_X: 1603610.5; RD_Y: 1255614; RD_X: -117814; Sabbatical Cycle: 1; Metonic Cycle: 18; 04 (Egyptian Month 08 Pharmouthi) 425 AN; 154973 days since beginning of Epoch;
2020 Depuydt 1997: 119
2021 Depuydt notes the day of "30" with "sic", i.e. an ancient mistake.
2022 Depuydt 1997: 119
2023 Depuydt 1997: 119

(Julian) and could therefore not have died in the evening of 10 June 323 BCE (Julian).[2024] [2025] [2026]

XX. 04 Pharmouthi (i.e. 8th Egyptian month), i.e. the Egyptian date transmitted in the Greek Pseudo-Callisthenes, is without a doubt the 24-hour day lasting from the morning of 13 June to the morning of 14 June 323 BCE (Julian).[2027] [2028] [2029] According to Depuydt, this Egyptian date is immediately suspicious for two (2) reasons:[2030]

1.) the first reason is that the textual transmission is garbled, as the manuscripts do not agree;

2.) The second reason is that its Julian equivalent, 13 June 323 BCE (Julian), is somewhat removed from the New Moon or Conjunction of Babylon – that Depuydt dates to 11 June 323 BCE (Julian) 00:06 AM[2031] [which is plausible] – yet the text seems to be trying to say that Alexander died very close to New Moon.[2032]

XX. In the Babylonian cuneiform Astronomical Diaries, which have a sterling reputation for meticulousness,[2033] in Cuneiform Tablet BM 45962 for the year 323 BCE (Julian), the king is reported to have died on day 29 of the 2nd month Ayyaru / Iyar.[2034] [2035] According to Depuydt, it is now generally accepted that this cuneiform date that came to light a few decades ago limits the time of death of Alexander to the 24 hour period between the evening of 10 June 323 BCE (Julian) and the evening of

2024 Sunday 10 June 323 BCE (Julian); Sunday 05 June 323 BCE (Gregorian); 29 Simannu / Sivan 3438 AM; Astronomical Julian Day JD_X: 1603607.5; RD_Y: 1255611; RD_X: -117817; Sabbatical Cycle: 1; Metonic Cycle: 18; 01 (Egyptian Month 08 Pharmouthi) 425 AN; 154970 days since beginning of Epoch;

2025 Monday 11 June 323 BCE (Julian); Monday 06 June 323 BCE (Gregorian); 30 Simannu / Sivan 3438 AM; Astronomical Julian Day JD_X: 1603608.5; RD_Y: 1255612; RD_X: -117816; Sabbatical Cycle: 1; Metonic Cycle: 18; 02 (Egyptian Month 08 Pharmouthi) 425 AN; 154971 days since beginning of Epoch;

2026 Depuydt 1997: 120

2027 Wednesday 13 June 323 BCE (Julian); Wednesday 08 June 323 BCE (Gregorian); 02 Du'uzu / Tammuz 3438 AM; Astronomical Julian Day JD_X: 1603610.5; RD_Y: 1255614; RD_X: -117814; Sabbatical Cycle: 1; Metonic Cycle: 18; 04 (Egyptian Month 08 Pharmouthi) 425 AN; 154973 days since beginning of Epoch;

2028 Thursday 14 June 323 BCE (Julian); Thursday 09 June 323 BCE (Gregorian); 03 Du'uzu / Tammuz 3438 AM; Astronomical Julian Day JD_X: 1603611.5; RD_Y: 1255615; RD_X: -117813; Sabbatical Cycle: 1; Metonic Cycle: 18; 05 (Egyptian Month 08 Pharmouthi) 425 AN; 154974 days since beginning of Epoch;

2029 Depuydt 1997: 120

2030 Depuydt 1997: 121

2031 This proposed date of conjunction is according to H. H. Goldstine, "New and Full Moons 1001 B.C. to A.D. 1651", *Memoirs*, American Philosophical Society, 94 (Philadephia 1973) – cited by Depuydt 1997: 121

2032 Depuydt 1997: 121

2033 Depuydt 1997: 123-124

2034 Sachs and Hunger 1988: 205-207; Depuydt 1997: 117; https://Livius.org; https://Livius.org/sources/content/oriental-varia/a-contemporary-account-of-the-death-of-alexander;

2035 Depuydt 1997: 121

11 June 323 BCE (Julian).[2036] [2037] [2038]

XX. According to Depuydt, there is no doubt that this date can be further narrowed down to the period between morning and evening of 11 June 323 BCE (Julian), a period of about 14 hours because the date is close to the summer solstice. Importantly – according to Depuydt – this renders the now standard dating of Alexander's death to 10 June 323 BCE (Julian) impossible.[2039]

XX. According to Depuydt, there is no doubt about two (2) matters:[2040]

1.) 29 Ayyaru / Iyar is the day lasting from the evening of 10 June 323 BCE (Julian) to the evening of 11 June 323 BCE (Julian);

2.) Alexander is the king that died in that period.[2041]

XX. According to Depuydt, the Babylonian calendar is sufficiently well known that the end of the lunar month Ayyaru / Iyar and the beginning of the following lunar month Simannu / Sivan can be safely identified with the conjunction that occurred on 11 June 323 BCE (Julian) at 00:06 AM. According to him, all that is necessary to know is how soon after conjunction the first crescent was spotted in the evening so that 01 Simannu / Sivan could begin, and also whether Ayyaru / Iyar had 29 or 30 days. Although this is not clear from tablet BM 45962 itself, Depuydt notes it is certain from another tablet – BM 34075 published by Sachs as LBAT 1431 – that Ayyaru / Iyar had 29 days.[2042]

XX. It should be noted that in the proleptic Hebrew calendar the 2nd month Ayyaru / Iyar always has 29 days, whereas the Babylonian Ayyaru / Iyar had 29 or 30 days by observation.

XX. The Egyptian date 04 Pharmouthi (i.e. 8th Egyptian month) long served as a pillar in dating Alexander's death. According to Depuydt [correctly if the Egyptian days begin at Sunrise], there is no doubt that in 323 BCE (Julian) 04 Pharmouthi lasts from the morning of 13 June to the morning of 14

2036 Sunday 10 June 323 BCE (Julian); Sunday 05 June 323 BCE (Gregorian); 29 Simannu / Sivan 3438 AM; Astronomical Julian Day JD_X: 1603607.5; RD_Y: 1255611; RD_X: -117817; Sabbatical Cycle: 1; Metonic Cycle: 18; 01 (Egyptian Month 08 Pharmouthi) 425 AN; 154970 days since beginning of Epoch;

2037 Monday 11 June 323 BCE (Julian); Monday 06 June 323 BCE (Gregorian); 30 Simannu / Sivan 3438 AM; Astronomical Julian Day JD_X: 1603608.5; RD_Y: 1255612; RD_X: -117816; Sabbatical Cycle: 1; Metonic Cycle: 18; 02 (Egyptian Month 08 Pharmouthi) 425 AN; 154971 days since beginning of Epoch;

2038 Depuydt 1997: 121

2039 Depuydt 1997: 121

2040 Depuydt 1997: 123

2041 Depuydt 1997: 123

2042 Depuydt 1997: 123

June 323 BCE (Julian).[2043] [2044] However, there are two (2) main problems with 04 Pharmouthi.[2045]

XX. The first problem is that the textual transmission is garbled. The second problem is that 13 June 323 BCE (Julian) is simply too late to allow 04 Pharmouthi to overlap with the last day of a lunar month, the day date reported by Aristoboulos and BM 45962 as the date of Alexander's death, let alone with the penultimate day, Day 28, reported by the Royal Diaries.[2046]

XX. In 323 BCE (Julian) 01 Pharmouthi lasts from the morning of 10 June to the morning of 11 June and therefore conveniently overlaps with the evening of 10 June.[2047] [2048] However, according to Depuydt, the Babylonian evidence leaves no doubt that Alexander died during the day on 11 June 323 BCE (Julian), which is Egyptian 02 Pharmouthi. Depuydt concludes that neither 04 Pharmouthi nor its emendation 01 Pharmouthi, are therefore suitable. In fact, upon closer examination, it appears that the different versions do not unanimously and unambiguously exhibit 04 Pharmouthi. Depuydt concludes that this fact encourages an examination of the veracity of 04 Pharmouthi.[2049]

XX. According to Depuydt, conjunction occurred at 00:06 AM on 11 June 323 BCE (Julian). The first crescent must have been visible – according to Depuydt – by the evening of 12 June 323 BCE (Julian).[2050] In fact, we know from tablet BM 34075 that it was visible – according to Depuydt – in the evening of 11 June 323 BCE (Julian) in Babylon.[2051]

XX. In more recent times, the accepted date for Alexander's death has been the evening of 10 June 323 BCE (Julian).[2052] Depuydt concludes that it is certain that the Babylonian scribes assigned Alexander's death to daytime of 11 June 323 BCE (Julian).[2053]

2043 Wednesday 13 June 323 BCE (Julian); Wednesday 08 June 323 BCE (Gregorian); 02 Du'uzu / Tammuz 3438 AM; Astronomical Julian Day JD_X: 1603610.5; RD_Y: 1255614; RD_X: -117814; Sabbatical Cycle: 1; Metonic Cycle: 18; 04 (Egyptian Month 08 Pharmouthi) 425 AN; 154973 days since beginning of Epoch;

2044 Thursday 14 June 323 BCE (Julian); Thursday 09 June 323 BCE (Gregorian); 03 Du'uzu / Tammuz 3438 AM; Astronomical Julian Day JD_X: 1603611.5; RD_Y: 1255615; RD_X: -117813; Sabbatical Cycle: 1; Metonic Cycle: 18; 05 (Egyptian Month 08 Pharmouthi) 425 AN; 154974 days since beginning of Epoch;

2045 Depuydt 1997: 130

2046 Depuydt 1997: 130

2047 Sunday 10 June 323 BCE (Julian); Sunday 05 June 323 BCE (Gregorian); 29 Simannu / Sivan 3438 AM; Astronomical Julian Day JD_X: 1603607.5; RD_Y: 1255611; RD_X: -117817; Sabbatical Cycle: 1; Metonic Cycle: 18; 01 (Egyptian Month 08 Pharmouthi) 425 AN; 154970 days since beginning of Epoch;

2048 Monday 11 June 323 BCE (Julian); Monday 06 June 323 BCE (Gregorian); 30 Simannu / Sivan 3438 AM; Astronomical Julian Day JD_X: 1603608.5; RD_Y: 1255612; RD_X: -117816; Sabbatical Cycle: 1; Metonic Cycle: 18; 02 (Egyptian Month 08 Pharmouthi) 425 AN; 154971 days since beginning of Epoch;

2049 Depuydt 1997: 131

2050 Tuesday 12 June 323 BCE (Julian); Tuesday 07 June 323 BCE (Gregorian); 01 Du'uzu / Tammuz 3438 AM; Astronomical Julian Day JD_X: 1603609.5; RD_Y: 1255613; RD_X: -117815; Sabbatical Cycle: 1; Metonic Cycle: 18; 03 (Egyptian Month 08 Pharmouthi) 425 AN; 154972 days since beginning of Epoch;

2051 Depuydt 1997: 130

2052 Depuydt 1997: 130

2053 Depuydt 1997: 124

XX. Alexander the Great's Babylonian regnal count began right after he defeated Darius III, the last Achaemenid Persian king, at the Battle of Gaugamela in Autumn 331 BCE (Julian).

XX. Babylonian scribes then started counting his reign the same way they counted any king's reign with a caveat: Although they normally used a Babylonian Year 0 Accession Year system with the 1st Regnal Year beginning the next Babylonian New Year on 01 Nisannu / Nisan 330 BCE (Julian), Babylonian scribes evidently counted Alexander's reign as a continuation from his accession as king of Macedon prior to conquering Babylon.

XX. Scholars have recorded Alexander to have died in the year 323 BCE (Julian). In the Babylonian Astronomical Diaries for that year, the king is reported to have died on day 29 of the 2nd month Ayyaru / Iyar,[2054] traditionally dated to Monday 11 June 323 BCE (Julian). The proleptic Hebrew calendar confirms this date.

XX. In that year the Hebrew calendar ran one (1) month ahead of the Babylonian since the Babylonian calendar had intercalated its Addaru 2 / Adar 2 in March 323 BCE (Julian), a year before the Hebrew Addaru 2 / Adar 2 of 3439 AM. Babylonian 29 Ayyaru / Iyar is therefore Hebrew 30 Sivan 3438 AM, i.e. Monday 11 June 323 BCE (Julian). The Hebrew 29 Ayyaru / Iyar 3438 AM, Saturday 12 May 323 BCE (Julian), falls one (1) month earlier.[2055]

64.0. The Controversy with the Book of Daniel

XX. Montgomery (1927) and later Casey (1976) quote Pusey (1880) who asserts, "The book of Daniel is especially fitted to be a battle-ground between faith and unbelief. It admits no half-way measures. It is either Divine or an imposture."[2056]

XX. The Book of Daniel purports to describe events during the Neo-Babylonian and [Persian] Achaemenid periods, naming the rulers Nebuchadnezzar, Belshazzar, Darius "the Mede", and Cyrus – that is to say [historical figures and events from] the 6th and early 5th centuries BCE (Julian). However, many modern scholars [following the 3rd century CE (Julian) skeptic Porphyry and subsequent 18th century CE (Gregorian) "Higher Criticism"] believe the Book of Daniel to be a composition of the 2nd century BCE (Julian).[2057]

2054 Sachs and Hunger 1988: 205-207; Depuydt 1997: 117; https://Livius.org; https://Livius.org/sources/content/oriental-varia/a-contemporary-account-of-the-death-of-alexander;

2055 Saturday 12 May 323 BCE (Julian); Saturday 07 May 323 BCE (Gregorian); 29 Ayyaru / Iyar 3438 AM; Astronomical Julian Day JD_X: 1603578.5; RD_Y: 1255582; RD_X: -117846; Sabbatical Cycle: 1; Metonic Cycle: 18; 02 (Egyptian Month 07 Phamenoth) 425 AN; 154941 days since beginning of Epoch;

2056 Montgomery 1927: 57-58; Casey 1976: 15

2057 Mitchell 1997: 71

XX. The Book of Daniel has been the objective of "higher criticism" as well as "apology" to an unparalleled extent, especially since the revelations [from the discoveries] of Assyriology [learned since the 18th century CE (Gregorian)].[2058] Quoting Rowley (1935), Wiseman notes that "the references to Darius the Mede in the Book of Daniel have long been recognized as providing the most serious historical problem in the book."[2059]

XX. Therefore the question on the identity of Darius the Mede – including the possibility that a third individual (i.e. neither Cyrus nor Gubaru) was king of Babylon during the first few months of Cyrus' reign – must remain open.[2060] However, the mathematics and logical reasing will prove that Darius the Mede is none other than Darius the Persian (the Great).

XX. [According to the current majority consensus of scholars, the book of Daniel] acquired its final form in the Hebrew Bible sometime between 167 and 164 BCE (Julian) (during the persecution of Antiochus [IV] Epiphanes), [and as such, current scholarship believes that] Daniel is the latest [i.e. the least ancient] composition to eventually be incorporated into the Jewish Scriptures.[2061]

XX. Casey cites Cameron (1967) who notes that modern scholars have been sympathetic towards Porphyry [3rd century CE (Julian)] for his "brilliant and definitive demolition of the traditional date [6th – 5th century BCE (Julian)] of Daniel".[2062]

XX. However, Mitchell argues that the book [of Daniel] is more a product of the [Persian] Achaemenid period – perhaps the 5th century BCE (Julian) – than of the Hellenistic [Greek] period in the 2nd century BCE (Julian).[2063]

65.0. The Most Common View of the Date of Composition

XX. The most common view of the date of the Book of Daniel is that it was composed in the 2nd century BCE (Julian), around 165 BCE (Julian), in the period of persecution under Antiochus IV Epiphanes (175 – 164 BCE (Julian)). This conclusion depends largely on the apparent precision with which visions described in the 2nd half of the book [of Daniel] match actual events, leading up to the prophecy in Daniel 11:31[2064] which concerns the "King of the North", presumed to be one of the Seleucids.[2065]

2058 Montgomery 1927: p. vii
2059 Wiseman 1965: 9
2060 Mitchell 1997: 77
2061 Stuckenbruck 2006: 102
2062 Casey 1976: 15
2063 Mitchell 1997: 77
2064 cf. Daniel 9:27; 12:11
2065 Mitchell 1997: 74

XX. These prophecies are understood to refer to the desecration of the Temple in Jerusalem by Antiochus [IV] in about 167 BCE (Julian). The Book of Daniel however does not mention the death of Antiochus [IV], which took place in 164 BCE (Julian), so it is concluded [by skeptic critics] that it was completed [sometime] between 167 and 164 BCE (Julian).[2066]

XX. In general, the alleged precision of the match between the visionary language in Daniel and [actual historical] events culminating in about 167 BCE (Julian) is said to be an example of a type of prophecy (known from Roman documents of the Hellenistic period) in which the device known as *vaticinium ex eventu* (i.e. "a prophecy from an outcome") employs the description of a past event as if it were an event in the future.[2067]

XX. This type of composition is also known, however, in various forms in Akkadian texts.[2068] Daniel's account of sacred history – presented in the form of vaticinium ex eventu [according to skeptic critics] – is concerned with events following the Exile [of Judea to Babylon] until the time of Antiochus [IV] Epiphanes.[2069]

66.0. Porphyry

XX. Centuries before the advent of modern biblical criticism, Porphyry was a Neo-Platonic philosopher who already argued that the Book of Daniel was a Maccabean [2nd century BCE (Julian)] pseudoepigraph.[2070] If the Book of Daniel was both written in – as well as wrote about events in – the 2nd century BCE (Julian), then it has no prophetic value at all. [If such were the case, then both the Jews and] the Christians were mistaken [about the origin of the Book of Daniel], and if Jesus cited it, then Jesus was mistaken as well.[2071]

XX. The [Christian] view – dominant since the 2nd century CE (Julian) – was that Daniel contains prophecies [that are] essential to establish certain elements of Christian belief.[2072] In the [Christian] Church the 1st commentator was Hippolytus of Rome whose "On Daniel" was written circa 202 CE (Julian).[2073]

XX. Both Hippolytus of Rome, as well as – [earlier] – Justin the Martyr [c. 100 – 165 CE (Julian)], argued that the chronology of Daniel accurately predicted the birth of the Messiah, the destruction of the [2nd] temple in Jerusalem, and the 2nd coming [of Jesus]. Hoffmann cites Wilken (1984) who asserts

2066 Mitchell 1997: 74
2067 Mitchell 1997: 74
2068 Mitchell 1997: 74
2069 Stuckenbruck 2006: 114
2070 Casey 1976: 15
2071 Hoffmann 1994: 168
2072 Hoffmann 1994: 167
2073 Montgomery 1927: 107

that what Porphyry did was to undermine a whole system of Christian interpretation based on the prophetic value of the book.[2074] [However, it will be proven mathematically that Porphyry failed.]

XX. The idea that the Book of Daniel was not prophetic was profoundly disturbing, [thereby] provoking responses from Eusebius, Methodius, and Jerome – whose commentary on Daniel was a defense of the traditional Christian view.[2075] Jerome's work is primarily an apology against the Neo-Platonic attack of Porphyry upon the historicity of [the Book of] Daniel, who claimed that it was Maccabean [i.e. 2nd century BCE (Julian)].[2076]

XX. Jerome has done the service of preserving Porphyry's argument in very full form – often in citation – and the polemic has caused him to compose a very careful work. According to Montgomery, Jerome's commentary is intrinsically valuable for its constant dependence upon the tradition of the [Jewish] rabbis under whom he studied, and Montgomery asserts that Jerome's work is a monument to the earliest stages of Jewish exegesis as appears from its frequent agreement with the [later] Medieval [Jewish] Rabbis [/Commentators].[2077] The immense debt of the [later Christian] Protestant commentaries and vernacular Bibles to the Jewish commentaries is – according to Montgomery – evident at every step in the exegesis of [the Book of] Daniel.[2078]

XX. Jerome wrote his great commentary on the Book of Daniel to neutralize the philosopher's – [as the skeptic, Hoffmann, terms them] – "scathing insights" into the nature of Biblical prophecy.[2079] According to Hoffmann, the sentiments expressed [by Porphyry] were "devastating" because they came from someone who knew the sacred books of the Christians and their doctrine intimately.[2080]

XX. Moreover – according to Hoffmann – in his attack, Porphyry denied the Christian teachers their favorite refuge: allegory. Porphyry dealt with the plain sense of words.[2081] In what Hoffman terms a "devastating critique" which has not survived – but which has evoked plenty of reaction from Porphyry's critics – the [3rd century CE (Julian) Neo-Platonist philosopher] began his *Against the Christians* with an attack on the Christian view of prophecy: Porphyry condemned the use of allegory as a means of explaining away difficulties and contradictions in the biblical text.[2082]

XX. However, despite his contempt for allegory, the philosopher was more concerned with chronology than interpretation. [Among his other objections], Porphyry [rejected] the ascription of the Book of

2074 Hoffmann 1994: 167
2075 Hoffmann 1994: 167
2076 Montgomery 1927: 107
2077 Montgomery 1927: 107
2078 Montgomery 1927: 106
2079 Hoffmann 1994: 17
2080 Hoffmann 1994: 17
2081 Hoffmann 1994: 17
2082 Hoffmann 1994: 166

Daniel to the period before [sic: during] the Babylonian captivity in the 6th century BCE (Julian).[2083] On philological grounds, Porphyry seems to have argued that the book belonged to a later period, and described the events [as being] of the authors' [or author's] own time, i.e. *vaticinium ex eventu* ("a prophecy from an outcome") from the Maccabean period of the 2nd century BCE (Julian). [Porphyry rejected the idea that] events were still to unfold [after his time in the 3rd century CE (Julian)].[2084]

XX. Porphyry did not stop at the conclusion that the Book of Daniel was written [approximately] four (4) centuries later than [what] Jews and Christians taught.[2085] According to Jerome, Porphyry insisted that the prophetic interpretation was inconsistent even with the claims that Christians made for the book.[2086]

XX. The Christians seemed to say that the Book of Daniel was "doubly prophetic" in that it pointed forward to events of the 2nd century BCE (Julian), namely the Syrian [Greek Seleucid] violation of the temple precinct in 167 BCE (Julian), as well as pointing to events which took place under the Romans after the time of Jesus, [e.g.] the burning of the [2nd] temple. Christian teachers claimed that Jesus himself had invoked the prophecy in forecasting the destruction of the temple.[2087] [2088]

XX. [Unfortunately,] all of Porphyry's fifteen (15) books against the Christians are lost. The exact title [of Porphyry's work] is not known, and its popular [Greek] title – *Kata Christianōn* – can be dated securely only from the Middle Ages. The [Christian] Church was unusually successful in its efforts to eradicate all traces of *Against the Christians* from at least 448 CE (Julian).[2089] What we know of the book comes from fragments preserved in the context of refutation by Christian teachers such as Eusebius, Apollinarius, [Methodius, and Jerome].[2090]

XX. Not only were the books of Porphyry destroyed, but [also] many of the works of Christian[2091] writers who incorporated sections of Porphyry's polemic were burned in order to eliminate what one (1) critic (the bishop Apollinarius) called the "poison of his thought".[2092]

XX. Porphyry was vilified and his worked destroyed, and thus we are dependent upon fragments preserved by hostile critics for our knowledge.[2093] Of Porphyry's work we know nothing further, and his position as to the date of [the Book of] Daniel has been vindicated by most of modern scholarship.[2094]

2083 Hoffmann 1994: 166
2084 Hoffmann 1994: 167
2085 Hoffmann 1994: 167
2086 Hoffmann 1994: 167-168
2087 Hoffmann 1994: 168
2088 cf. Mark 13:2
2089 Hoffmann 1994: 164
2090 Hoffmann 1994: 17
2091 Hoffmann 1994: 164
2092 Hoffmann 1994: 165
2093 Casey 1976: 15
2094 Montgomery 1927: 107

XX. However, current scholarship is sometimes – periodically throughout history – mistaken, and sometimes it requires rethinking in order to recalibrate itself with scientifically established facts and absolute mathematical truth.

67.0. Messianic Expectation

XX. According to contemporary historians, the 1st century CE (Julian) was a time of great Messianic expectation among the Jews. The feeling was widespread that some prophecy regarding the time of his coming was now [i.e. then in the 1st century CE (Julian)] fulfilled.[2095]

XX. By the middle of the 3rd century CE (Julian), a mood of resignation seems to have set in as Rab admits (in Babylonian Talmud tractate *Sanhedrin* 97b) that "all the predestined dates have passed." He explains the delay by suggesting that the Messiah's coming now depends only on Israel's repentance and good works.[2096]

XX. But what Old Testament prophecy lies behind this expectation that the Messiah should have come in the 1st century of our [common] era CE (Julian)?[2097]

XX. Conservative Christians have long believed that Daniel 9:24-27 gives just such a prediction, although there has been considerable disagreement on how the details of this prophecy fit the ministry of Jesus.[2098]

XX. Perhaps the most popular interpretation of this passage has been given by Sir Robert Anderson. He pinpoints the end of the 69th week, the coming of "Messiah the Prince", as Sunday 06 April 32 CE (Julian) and claims that this was the very day of Jesus' triumphal entry into Jerusalem.[2099]

XX. Unfortunately, this view – as spectacular as it is – faces some serious problems.[2100]

68.0. Sir Robert Anderson

XX. Sir Robert Anderson (1841 – 1918 CE (Gregorian)) was an investigator with Scotland Yard who lived in the late 19th to early 20th centuries CE (Gregorian). He turned his investigative skills to Bible

2095 Newman 1973: 229
2096 Newman 1973: 230
2097 Newman 1973: 230
2098 Newman 1973: 230
2099 Newman 1973: 230
2100 Newman 1973: 230

prophecy, especially the book of Daniel, and the 70 weeks.[2101] He attempted to "unfold an elucidation" of the [prophetic] vision of the Seventy Weeks[2102] [of the 9th chapter of the Book of Daniel, i.e. Daniel 9:24-27].

XX. Among interpreters of the seventy (70) weeks, various arguments have been constructed to fix the proper starting point for the prophecy.[2103] Anderson's choice of 445 BCE (Julian), the 20th Regnal Year of Artaxerxes I [purportedly] mentioned in Nehemiah 2:1-6, uses the only suggested passage which specifically mentions building the city [of Jerusalem]. Chronological studies since Anderson's time have not changed this date.[2104]

XX. Even though the exact birth date of Jesus was nowhere recorded in the New Testament (NT) or elsewhere, and referring to Luke 3:1-2, Sir Robert Anderson makes the bold claim that no date in history – sacred or profane – is fixed with greater certainty than that of the year in which Jesus began his public ministry.[2105] He concludes that the 1st Passover of Jesus' ministry can be dated to Hebrew month Nisannu / Nisan year 29 CE (Julian), and that the date of Jesus' crucifixion can be fixed "with absolute certainty" to Nisannu / Nisan year 32 CE (Julian).[2106]

XX. Christian Commentator, John F. Walvoord, argues (according to him): Sir Robert Anderson has made a detailed study of a possible chronology for this period beginning with the well-established date of 445 BCE (Julian) when Nehemiah's decree was issued and culminating in 32 CE (Julian) on the very day of Christ's triumphal entry into Jerusalem shortly before His crucifixion. Sir Robert Anderson specifies that the seventy sevens began on the 01 Nisannu / Nisan, 14 March 445 BCE (Julian), and ended on 06 April 32 CE (Julian), i.e. 10 Nisannu / Nisan. Anderson's complicated computation is based upon "prophetic years" of 360 days totaling 173,880 days. This would be exactly 483 years according to biblical chronology.[2107]

XX. In order for the prophecy to come out right, Anderson assumes that the Jews used "[prophetic] years" of only 360 days. He seeks justification for this from Revelation 11:2-3, and he calls such a period a "prophetic year".[2108]

XX. However, the Old Testament connects the Passover festival (in the middle of the 1st month [of Nisannu / Nisan]) to the offering of the first-ripe grain.[2109] Therefore the Jewish calendar was locked into the seasons. Both the Talmud and archaeology indicate that this was accomplished by adding an

2101 Anderson 2014: 4
2102 Anderson 2014: 9
2103 Newman 1973: 232-233
2104 Newman 1973: 233
2105 Anderson 2014: 7-8
2106 Anderson 2014: 8
2107 Walvoord 1971: Chapter 9
2108 Newman 1973: 230
2109 cf. Leviticus 23:6-14

extra lunar month every few years so that in the long run the average length of the Jewish year matches our solar year of just under 365 ¼ days.[2110]

XX. Even granting the validity of "prophetic years" [of 360 days] and 32 CE (Julian) [for Anderson's proposed date of the crucifixion], it also appears – according to Newman – that Anderson may be off several days because he confused the Julian and Gregorian calendars.[2111]

XX. Indeed, as is evident, the actual date of 01 Nisannu / Nisan 445 BCE (Julian) is Thursday 13 March 445 BCE (Julian)[2112] [2113] – not Friday 14 March 445 BCE (Julian).[2114] [2115]

XX. Additionally, the actual date of 10 Nisannu / Nisan 32 CE (Julian) is Thursday 10 April 32 CE (Julian)[2116] [2117] – not Sunday 06 April 32 CE (Julian).[2118] [2119]

XX. Moreover, the number of days between Anderson's two (2) incorrect dates (i.e. from Friday 14 March 445 BCE (Julian)[2120] [2121] to Sunday 06 April 32 CE (Julian))[2122] [2123] is 173,882 days – not 173,880. This works out to be 483 "prophetic 360-day" years plus an extra two (2) days. Therefore, Anderson's calculations are imprecise to say the very least.

XX. Furthermore, even by taking the actual dates, the number of days between Thursday 13 March 445 BCE (Julian)[2124] [2125] to Thursday 10 April 32 CE (Julian)[2126] [2127] is 173,887 days. This works out to be

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2110 Newman 1973: 230
2111 Newman 1973: 231
2112 Thursday 13 March 445 BCE (Julian); Thursday 08 March 445 BCE (Gregorian); 01 Nisannu / Nisan 3316 AM; Astronomical Julian Day JD_X: 1558958.5; RD_Y: 1210962; RD_X: -162466; Sabbatical Cycle: 5; Metonic Cycle: 10;
2113 cf. Appendix Table X.104
2114 Friday 14 March 445 BCE (Julian); Friday 09 March 445 BCE (Gregorian); 02 Nisannu / Nisan 3316 AM; Astronomical Julian Day JD_X: 1558959.5; RD_Y: 1210963; RD_X: -162465; Sabbatical Cycle: 5; Metonic Cycle: 10;
2115 cf. Appendix Table X.104
2116 Thursday 10 April 32 CE (Julian); Thursday 08 April 32 BCE (Gregorian); 10 Nisannu / Nisan 3792 AM; Astronomical Julian Day JD_X: 1732845.5; RD_Y: 1384849; RD_X: 11421; Sabbatical Cycle: 5; Metonic Cycle: 11;
2117 cf. Appendix Table X.183
2118 Sunday 06 April 32 CE (Julian); Sunday 04 April 32 BCE (Gregorian); 06 Nisannu / Nisan 3792 AM; Astronomical Julian Day JD_X: 1732841.5; RD_Y: 1384845; RD_X: 11417; Sabbatical Cycle: 5; Metonic Cycle: 11;
2119 cf. Appendix Table X.183
2120 Friday 14 March 445 BCE (Julian); Friday 09 March 445 BCE (Gregorian); 02 Nisannu / Nisan 3316 AM; Astronomical Julian Day JD_X: 1558959.5; RD_Y: 1210963; RD_X: -162465; Sabbatical Cycle: 5; Metonic Cycle: 10;
2121 cf. Appendix Table X.104
2122 Sunday 06 April 32 CE (Julian); Sunday 04 April 32 BCE (Gregorian); 06 Nisannu / Nisan 3792 AM; Astronomical Julian Day JD_X: 1732841.5; RD_Y: 1384845; RD_X: 11417; Sabbatical Cycle: 5; Metonic Cycle: 11;
2123 cf. Appendix Table X.183
2124 Thursday 13 March 445 BCE (Julian); Thursday 08 March 445 BCE (Gregorian); 01 Nisannu / Nisan 3316 AM; Astronomical Julian Day JD_X: 1558958.5; RD_Y: 1210962; RD_X: -162466; Sabbatical Cycle: 5; Metonic Cycle: 10;
2125 cf. Appendix Table X.104
2126 Thursday 10 April 32 CE (Julian); Thursday 08 April 32 BCE (Gregorian); 10 Nisannu / Nisan 3792 AM; Astronomical Julian Day JD_X: 1732845.5; RD_Y: 1384849; RD_X: 11421; Sabbatical Cycle: 5; Metonic Cycle: 11;
2127 cf. Appendix Table X.183

483 "prophetic 360-day" years plus an extra seven (7) days – i.e. making Anderson's theory even more imprecise.

XX. In spite of all these objections, it appears that a good case can still be made for a definite fulfillment of this [Daniel 9:24-27] passage [that will be unfolded here] even though it is not as [has been calculated by Anderson or others].[2128]

XX. Spurred on by his faith in the Book of Daniel, as well as his hostility towards German "Higher Criticism", Sir Robert Anderson endeavored to inquire into Daniel's great prophecy of the "Seventy Weeks".[2129] Insodoing, Anderson makes the critical insight that the failure to distinguish between the several judgments of: 1.) the Servitude, 2.) the Captivity [i.e. Exile], and 3.) the Desolation is a source of error in the study of Daniel and the historical books of Scripture.[2130]

XX. Though Anderson's premise – of distinguishing three (3) distinct epochs with specific periods of 1.) Servitude, 2.) Captivity [i.e. Exile], and 3.) Desolation – is an insightful one, his proposed definition of these terms with their proposed epochs and their chronology is based upon dates that are now known to be mostly in error on his part. At the time of his writing, Anderson believed that most of the "historical errors in Daniel" that had made controversy in the century prior to his own day had been resolved by the [then-current] erudition and research of his own day in the late 19th – early 20th centuries CE (Gregorian).[2131]

69.0. Servitude, Captivity, and Desolation

XX. Anderson sets forth his premise as follows:

XX. Judah was brought under Servitude to Babylon for seventy years (70) because of national sin, and this was in the 3rd Regnal Year of King Jehoiakim in 606 BCE.[2132] [2133] It is now known, and has been demonstrated as well as mathematically proven, that Anderson's proposed dates are mostly incorrect. The correct date for the 3rd Regnal Year of Yehoyakim (Jehoiakim) is Spring 605 – 604 BCE (Julian) with this 3rd Regnal Year of Yehoyakim (Jehoiakim) having begun 01 Nisannu / Nisan (1st month).

XX. Furthmore, it will be argued here and has been mathematically proven that the Servitude, i.e. the servitude to another nation with the loss of national sovereignty of a truly independent Kingdom of

2128 Newman 1973: 231
2129 Anderson 2014: 5-6
2130 Anderson 2014: 6
2131 Anderson 2014: 6
2132 This year is now known to be incorrect. The correct year for the 3rd Regnal Year of Yehoyakim (Jehoiakim) is Spring 605 – 604 BCE (Julian).
2133 Anderson 2014: 6

Judah, officially began with the reign of Yehoyakim (Jehoiakim / Elyakim) who was installed as a vassal-puppet-slave-king by Egyptian Pharaoh Necho II in 608 BCE (Julian) and ended in 538 BCE (Julian) with the edict of Cyrus that Jews can return to the Land of Israel to rebuild their temple.

XX. Anderson reasons that Daniel was taken as captive "on the former capture of Jerusalem", i.e a 1st capture of Jerusalem that he reasons to be one of several such captures of Jerusalem. However, Anderson incorrectly dates this capture to 606 BCE (Julian).[2134] The actual date for this capture of Jerusale, and YEAR ONE (1) OF THE EXILE OF DANIEL is the 3rd Regnal Year of Yehoyakim (Jehoiakim) which is equivalent to the Year 0 Accession Year of Nebuchadezzar is 605 – 604 BCE (Julian).

XX. Anderson asserts that the far more severe judgment of the Captivity [i.e. Exile] fell upon them in 598 BCE.[2135] [2136] Again: Anderson's date is incorrect, but he correctly identifies the Captivity [i.e. Exile] as beginning with the EXILE OF YEHOYACHIN (JEHOIACHIN). Because of the discoveries of Assyriology since the 19th century CE (Gregorian), we now know the correct date for the 1st fall of (i.e. capture of) Jerusalem to be fixed to 02 Addaru / Adar (12th month) Late Winter 597 BCE (Julian) and the date of the beginning of YEAR ONE (1) OF THE EXILE OF YEHOYACHIN (JEHOIACHIN) to be 10 Nisannu / Nisan 597 BCE (Julian).

XX. Anderson continues that after a lapse of another nine (9) years [i.e. from the incorrect date of 598 BCE (Julian)], the terrible judgment of the Desolation was brought upon them, which was decreed to last for seventy (70) years.[2137] Anderson's recognition of the period of Desolation to be seventy (70) years is correct, but he incorrectly defines the epoch of those 70 years of Desolation as being on the day on which Jerusalem was invested (i.e. the day on which Judah was invaded by the Babylonians, and Jerusalem subsequently surrounded with Siege): day ten (10) of the 10th month (i.e. Tebetu / Tevet) in the 9th Regnal Year of Tzidkiyahu (Zedekiah) which Anderson dates to 589 BCE (Julian). The actual date of the beginning of the Siege of Jerusalem was Thursday 05 January 587 BCE (Julian),[2138] [2139] [2140] [2141]

XX. Although this date is very significant, it is not the starting point of the Epoch of Desolation since Jerusalem was not yet desolate when the Siege of Jerusalem began on 10 Tebetu / Tevet (i.e. 10th month) in the 9th Regnal Year of Tzidkiyahu (Zedekiah) in 588 – 587 BCE (Julian), and which

2134 Anderson 2014: 6
2135 This year is now known to be incorrect. The correct year for YEAR ONE (1) OF THE EXILE OF YEHOYACHIN (JEHOIACHIN) is Nisannu / Nisan 597 BCE (Julian).
2136 Anderson 2014: 6
2137 Anderson 2014: 6
2138 Thursday 05 January 587 BCE (Julian); Thursday 30 December 588 BCE (Gregorian); 10 Tebetu / Tevet 3174 AM; Astronomical Julian Day JD_X: 1507025.5; RD_Y: 1159029; RD_X: -214399; Sabbatical Cycle: 3; Metonic Cycle: 1;
2139 cf. Footnote #9
2140 cf. Appendix Table X.53
2141 Anderson 2014: 6

continued through the 11th Regnal Year of Tzidkiyahu (Zedekiah) in 587 – 586 BCE (Julian) until July 586 BCE (Julian) when the 1st Temple of Jerusalem was destroyed.

XX. Furthermore, since Anderson's initial dates were incorrect, he calculates an incorrect start date for the "Desolation" of Jerusalem to be 589 BCE (Julian) for when the city of Jerusalem was besieged, and later devastated and burned – according to Anderson in – 587 BCE (Julian).[2142] As we know, the actual date for the destruction and burning of Jerusalem began upon the arrival of Nebuzaradan to Jerusalem on Sunday 15 July 586 BCE (Julian),[2143] [2144] [2145] and his subsequent destroying of Solomon's First Temple on Wednesday 18 July 586 BCE (Julian).[2146] [2147] [2148]

70.0. Anderson's Prophetic Year of 360 Days

XX. [As mentioned, for his interpretation of Daniel's seventy-(70)-week prophecy, Anderson chooses the 20th Regnal Year of Artaxerxes I mentioned in Nehemiah 2:1-6, i.e. 445 – 444 BCE (Julian).] Anderson arbitrarily chooses the 1st day of the month Nisannu / Nisan 445 BCE (Julian) for his starting point, although the Biblical information[2149] gives only the month.[2150] However, for him to start even a week later would make it impossible to end the prophetic period before the crucifixion, even granting the validity of "prophetic years" and 32 CE (Julian).[2151]

XX. Moreover, the accuracy of Anderson's dates depends upon Jesus' crucifixion having occurred in the year 32 CE (Julian),[2152] which will be proven to be incorrect as Jesus was crucified in year 33 CE (Julian).

XX. Furthermore, Anderson incorrectly asserts that Daniel and Revelation definitely indicate that the prophetic year is one of 360 days. Such moreover – Anderson incorrectly asserts – was the sacred year of the Jewish calendar, and that as is well known, such was the ancient year of Eastern nations.[2153] Admittedly, an ancient Mesopotamian "ideal" calendar of 360-days did exist in antiquity, it seems it was an older, primitive version from which they ultimately developed the standard Mesopotamian

2142 Anderson 2014: 6
2143 Sunday 15 July 586 BCE (Julian); Sunday 09 July 586 BCE (Gregorian); 07 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507581.5; RD_Y: 1159585; RD_X: -213843; Sabbatical Cycle: 4; Metonic Cycle: 2;
2144 cf. Appendix Table X.59
2145 cf. 2 Kings 25:8-11
2146 Wednesday 18 July 586 BCE (Julian); Wednesday 12 July 586 BCE (Gregorian); 10 Abu / Av 3175 AM; Astronomical Julian Day JD_X: 1507584.5; RD_Y: 1159588; RD_X: -213840; Sabbatical Cycle: 4; Metonic Cycle: 2;
2147 cf. Appendix Table X.59
2148 cf. Jeremiah 52:12-14
2149 cf. Nehemiah 2:1
2150 Newman: 1973: 230-231
2151 Newman: 1973: 231
2152 Newman 1973: 230
2153 Anderson 2014: 7

calendar (through observation of the celestial bodies in the heavens with each season and refinement of the accuracy of the calendar), that we see still in existence today as the modern Hebrew calendar. However, most defintely a calendar of 360 days was not the sacred year of the Jewish calendar, and there is no evidence of its use in Jewish history.

XX. Anderson correctly calculates that seventy (70) years of 360 days contains exactly 25,200 days, and is correct that the Jewish New Year's day depended upon the equinoctial Moon [of the Vernal and Autumnal Equinoxes].

XX. However, notably, Anderson incorrectly assigns [Monday] 13 December 589 BCE (Julian)[2154] [2155] as the Julian date equivalent of 10 Tebetu / Tevet 589 BCE (Julian), yet correctly attributes [Sunday] 17 December 520 BCE (Julian)[2156] [2157] as the Julian date equivalent of day 24 of the 9th month of Kisilimu / Kislev in the 2nd Regnal Year of Darius of Persia, i.e. 520 – 521 BCE (Julian).[2158]

XX. Unfortunately, Anderson arrives to this correct Julian date of Sunday 17 December 520 BCE (Julian)[2159] [2160] by seeming accident because his incorrect calculations do not actually arrive to this correct date.[2161]

XX. Since he adds 25,200 days (i.e. seventy (70) years of 360-days each) counting from the incorrect starting date of Monday 13 December 589 BCE (Julian)[2162] [2163] (whose actual correct date calculates to Monday 11 December 520 BCE (Julian)),[2164] [2165] i.e. 18 Tebetu / Tevet 520 BCE (Julian), Anderson's proposed date is six (6) days short of the actual day for the correct Julian date of Sunday 24 Tebetu / Tevet 520 BCE (Julian), i.e. the 2nd Regnal Year of Darius.[2166]

XX. Following Anderson's logic, calculating the number of days between the correct dates for the beginning of the Siege of Jerusalem in the 9th Regnal Year of Tzidkiyahu (Zedekiah), 10 Tebetu / Tevet

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2154 Monday 13 December 589 BCE (Julian); Monday 07 December 589 BCE (Gregorian); 06 Tebetu / Tevet 3173 AM; Astronomical Julian Day JD_X: 1506637.5; RD_Y: 1158641; RD_X: -214787; Sabbatical Cycle: 2; Metonic Cycle: 19;
2155 cf. Appendix Table X.51
2156 Sunday 17 December 520 BCE (Julian); Sunday 11 December 520 BCE (Gregorian); 24 Kisilimu / Kislev 3242 AM; Astronomical Julian Day JD_X: 1531843.5; RD_Y: 1183847; RD_X: -189581; Sabbatical Cycle: 1; Metonic Cycle: 12;
2157 cf. Appendix Table X.81
2158 Anderson 2014: 6
2159 Sunday 17 December 520 BCE (Julian); Sunday 11 December 520 BCE (Gregorian); 24 Kisilimu / Kislev 3242 AM; Astronomical Julian Day JD_X: 1531843.5; RD_Y: 1183847; RD_X: -189581; Sabbatical Cycle: 1; Metonic Cycle: 12;
2160 cf. Appendix Table X.81
2161 Anderson 2014: 6
2162 Monday 13 December 589 BCE (Julian); Monday 07 December 589 BCE (Gregorian); 06 Tebetu / Tevet 3173 AM; Astronomical Julian Day JD_X: 1506637.5; RD_Y: 1158641; RD_X: -214787; Sabbatical Cycle: 2; Metonic Cycle: 19;
2163 cf. Appendix Table X.51
2164 Monday 11 December 520 BCE (Julian); Monday 05 December 520 BCE (Gregorian); 18 Kisilimu / Kislev 3242 AM; Astronomical Julian Day JD_X: 1531837.5; RD_Y: 1183841; RD_X: -189587; Sabbatical Cycle: 1; Metonic Cycle: 12;
2165 cf. Appendix Table X.81
2166 Anderson 2014: 6

587 BCE (Julian), i.e. Thursday 05 January 587 BCE (Julian),[2167] [2168] [2169] and through the completion of the rebuilding of the 2nd Temple in 24 Kisilimu / Kislev 520 BCE (Julian)[2170] [2171], i.e. Sunday 17 December 520 BCE (Julian) in the 2nd Regnal Year of Darius, this calculates to be 24,818 days, i.e. 182 extra days beyond the seventy (70) years of 360-day prophetic years.

XX. Current consensus in academic scholarship now recognizes that the Siege of Jerusalem began in 587 BCE (Julian), and that the city and temple were destroyed (i.e. the Seventy (70) Years of Desolation began) in the Babylonian / Hebrew 5th month of Abu / Av (i.e. July/August) 586 BCE (Julian).

XX. Anderson goes on to assert (incorrectly) that both the Servitude and the Captivity [i.e. Exile] ended [i.e. together] with the decree of Cyrus II in 536 BCE (Julian).[2172] [2173] Again: Anderson is here mistaken in his date of the decree of Cyrus II in his 1st Regnal Year over Babylon. We now know the decree of Cyrus II to be – in actuality – dated to 538 BCE (Julian).

XX. Later in this analysis we will address Anderson's claim that both the Servitude and the Captivity ended [together] with this decree of Cyrus II in 538 BCE (Julian). Specifically: we will take issue with the assertion that since the Servitude and the Captivity each began on separate dates, is it not both reasonable and logical to understand that they would each end on separate dates?

XX. It is worthwhile to apply Anderson's theory to 483 actual Babylonian / Hebrew (and/or Julian) years as the interpretation for the prophecy of seventy (70) weeks.

XX. If we use his proposed starting point of day one (1) of Nisannu / Nisanu of the 20th Regnal Year of Artaxerxes I, i.e. 445 BCE (Julian), and attempt to count forward 483 years, we arrive to the 477th year in year 32 CE (Julian) in which Jesus was not crucified, and the 478th year in year 33 CE (Julian) in which Jesus was actually crucified – this is imprecise to say the least with an imprecision of five (5) years.

XX. However, not only is this interpretation and application of 483 years imprecise, it is also incorrect because, as proven mathematically, Nehemiah's mention of the 20th Regnal Year of Artaxerxes I in Nehemiah 2 was a simple scribal error, and it is clear that the scribe meant to write the 21st Regnal Year.

2167 Thursday 05 January 587 BCE (Julian); Thursday 30 December 588 BCE (Gregorian); 10 Tebetu / Tevet 3174 AM; Astronomical Julian Day JD_X: 1507025.5; RD_Y: 1159029; RD_X: -214399; Sabbatical Cycle: 3; Metonic Cycle: 1;

2168 cf. Appendix Table X.53

2169 cf. Footnote #9

2170 Sunday 17 December 520 BCE (Julian); Sunday 11 December 520 BCE (Gregorian); 24 Kisilimu / Kislev 3242 AM; Astronomical Julian Day JD_X: 1531843.5; RD_Y: 1183847; RD_X: -189581; Sabbatical Cycle: 1; Metonic Cycle: 12;

2171 cf. Appendix Table X.81

2172 This year is now known to be incorrect. The correct year for the decree in the 1st Regnal Year of Cyrus (over Babylon) is 538 BCE (Julian).

2173 Anderson 2014: 6

XX. Even so, applying Anderson's theory of applying 483 years to the actual correct date of the command to rebuild the city walls of Jerusalem renders Anderson's thesis even more precise: beginning from Anderson's starting point of day one (1) of Nisannu / Nisanu in the 21st Regnal Year of Artaxerxes I in 444 BCE (Julian), and attempting to count forward 483 years, we arrive to the 476th year in year 32 CE (Julian), and the 477th year in year 33 CE (Julian) – this is even more imprecise with an imprecision of six (6) years.

XX. Therefore, we cannot consider 483 years that are applied to either the 20th or 21st Regnal Year of Artaxerxes I as being a correct interpretation of Daniel's prophecy of seventy (70) weeks, even if we separate the 1st sixty nine (69) weeks.

71.0. Daniel's Vision of the 70 Weeks

XX. In the first two (2) verses of Daniel chapter 9, we have the setting for Daniel's vision of the Seventy Weeks:[2174]

1 In year one of Daryavesh (Darius) son of Ahash-verosh (Ahasuerus), from the seed of Media, who was made king over the monarchy of the Chaldeans. 2 In year one of his reign, I Daniel, made insight into the books the number of years that was the Word of YHVH to Yermeyah (Jeremiah) the Prophet to fulfill for the ruins of Jerusalem: Seventy years.[2175]

XX. Daniel has just understood from "the books" (plural) that the desolation of Jerusalem would last seventy (70) years. Since Jeremiah is mentioned by name, his prophecy is obviously one of the books (i.e. the length of the captivity is predicted in Jeremiah 25:11-12 and 29:10)[2176] [that Daniel had read].

XX. The 2nd book of Chronicles [i.e. 2 Chronicles] also mentions the seventy (70) years,[2177] but it probably was not completed at the time of Daniel's vision. However, the Chronicler explains that the captivity was seventy (70) years long in order to compensate for seventy (70) sabbath years in which the Jews had disobeyed God's command for the land to lie fallow.[2178] [2179]

XX. In fact, Leviticus 26[2180] predicts that this specific punishment would come upon Israel if they violated the sabbatical-year regulation. Perhaps Exodus and Leviticus were the other books Daniel

2174 Newman 1973: 231
2175 cf. Daniel 9:1-2
2176 Newman 1973: 231
2177 cf. 2 Chronicles 36:21
2178 Newman 1973: 231
2179 cf. Exodus 23:10-11; Leviticus 25:3-7, 18-22
2180 cf. Leviticus 26:32-35, 43

consulted. Daniel at least had all the materials necessary to reach the conclusion found in 2 Chronicles even if he never saw that work.[2181]

XX. It is therefore not unreasonable to suppose that Daniel had been thinking about the [Sabbath Year] seven-(7)-year land-use cycle and the period of seventy (70) such cycles during which Israel had disobeyed this command. If so, the message which the angel brings, "Seventy weeks[2182] are determined[2183] upon they people..."[2184] suddenly seems much less obscure.[2185]

XX. It is interesting to note that a remark in the Talmud also associates the coming of the Messiah with a seven-(7)-year period in Babylonian Talmud tractate *Sanhedrin* 97a. Newman therefore suggests that this prophecy uses the term "week"[2186] for the Old Testament Sabbath-Year Cycle. Newman considers only the coming of the Messiah which involves only the first sixty-nine (69) weeks.[2187]

72.0. Newman's Methodology of the Sabbath-Year Cycle

XX. Although Newman's methodology has merit and will be redeemed mathematically, his calculations are ultimately flawed because he applies the sabbatical year cycle to the incorrect Regnal Year of Artaxerxes I, as well as assumes an incorrect date (i.e. 30 CE (Julian)) for the crucifixion of Jesus.

XX. Following Anderson, Newman takes the calculation forward from 445 BCE (Julian), i.e. the 20th Regnal Year of Artaxerxes I. However, unlike Anderson, Newman seeks to use the actual 69 sabbatical cycles rather than calculating (7 x 69) years, i.e. 483 years, since Newman suggests that this method of sabbatical cycles better fits the context. Newman locates these seven-(7)-year cycles in antiquity.[2188]

XX. In modern times the Jews in Israel still observe a seven-(7)-year sabbatical cycle. According to the Encyclopedia Judaica, the Hebrew civil year beginning in September 1951 – 1952 CE (Gregorian) was a sabbatical year.[2189] Although this date needs to be corrected to have begun Monday 01 October 1951 CE (Gregorian),[2190] [2191] indeed this Babylonian / Hebrew year that began 01 Tashritu / Tishri 5712 AM (i.e. October 1951 – 1952 CE (Gregorian)) can be confirmed mathematically as being a sabbatical 7th

2181 Newman 1973: 231
2182 Literally "sevens [-of-years]"
2183 Literally "cut"
2184 cf. Daniel 9:24-27
2185 Newman 1973: 231
2186 Literally "sevens [-of-years]"
2187 Newman 1973: 231
2188 Newman 1973: 233
2189 Newman 1973: 233
2190 Monday 18 September 1951 CE (Julian); Monday 01 October 1951 CE (Gregorian); 01 Tashritu / Tishri 5712 AM; Astronomical Julian Day JD_X: 2433920.5; RD_Y: 2085924; RD_X: 712496; Sabbatical Cycle: 7; Metonic Cycle: 12;
2191 cf. Appendix Table X.166

year.

XX. This Hebrew civil year [being a sabbatical year] is consistent with that of the Hebrew civil year beginning in September 54 – 55 CE (Julian)[2192] [2193] given by Sir William Ramsay (1966) and that of the Hebrew civil year beginning in September 40 – 41 CE (Julian)[2194] [2195] mentioned in a footnote of the Socino Talmud,[2196] but it is not clear where these got their information.[2197] Indeed, these can be confirmed mathematically as sabbatical 7th years.

XX. In the 1st book of the Maccabees, the primary historical source for the Maccabean period, we find that Jewish resistance to the Syrians was once weakened because of a sabbatical year.[2198] This sabbatical 7th year began Monday 21 September 164 BCE (Julian).[2199] [2200] Indeed, this can be confirmed mathematically as a sabbatical 7th year.

XX. A reference earlier in chapter 6 of the 1st book of the Maccabees[2201] indicates that this weakening of Jewish resistance to the Syrians occurred in the 150th year of the Seleucid era. According to Finegan, the 150th year could be either 163 – 162 BCE (Julian)[2202] [2203] or 162 – 161 BCE (Julian),[2204] [2205] depending upon whether the Macedonian [which began Autumn 312 BCE (Julian)] or Babylonian / Hebrew [which began Spring 311 BCE (Julian)] reckoning was in use.[2206]

XX. According to Newman, the former case fits the modern sabbatical year very well: the year 164 – 163 BCE (Julian)[2207] [2208] would have been the sabbath year so that famine conditions would have been most acute in the following year before crops could be harvested. Newman suggests then that we take

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2192 Saturday 21 September 54 CE (Julian); Saturday 19 September 54 CE (Gregorian); 01 Tashritu / Tishri 3815 AM; Astronomical Julian Day JD_X: 1741044.5; RD_Y: 1393048; RD_X: 19620; Sabbatical Cycle: 7; Metonic Cycle: 15;
2193 cf. Appendix Table X.164
2194 Monday 26 September 40 CE (Julian); Monday 24 September 40 CE (Gregorian); 01 Tashritu / Tishri 3801 AM; Astronomical Julian Day JD_X: 1735936.5; RD_Y: 1387940; RD_X: 14512; Sabbatical Cycle: 7; Metonic Cycle: 1;
2195 cf. Appendix Table X.162
2196 cf. Babylonian Talmud *Sotah* 41a (Socino ed.) XVI, 202n
2197 Newman 1973: 233
2198 cf. 1 Maccabees 6:49, 53-54
2199 Monday 21 September 164 BCE (Julian); Monday 18 September 164 BCE (Gregorian); 01 Tashritu / Tishri 3598 AM; Astronomical Julian Day JD_X: 1661785.5; RD_Y: 1313789; RD_X: -59639; Sabbatical Cycle: 7; Metonic Cycle: 7;
2200 cf. Appendix Table X.159
2201 cf. 1 Maccabees 6:20
2202 Thursday 09 September 163 BCE (Julian); Thursday 06 September 163 BCE (Gregorian); 01 Tashritu / Tishri 3599 AM; Astronomical Julian Day JD_X: 1662138.5; RD_Y: 1314142; RD_X: -59286; Sabbatical Cycle: 1; Metonic Cycle: 8;
2203 cf. Appendix Table X.160
2204 Thursday 29 September 162 BCE (Julian); Thursday 26 September 162 BCE (Gregorian); 01 Tashritu / Tishri 3600 AM; Astronomical Julian Day JD_X: 1662523.5; RD_Y: 1314527; RD_X: -58901; Sabbatical Cycle: 2; Metonic Cycle: 9;
2205 cf. Appendix Table X.161
2206 Newman 1973: 233
2207 Monday 21 September 164 BCE (Julian); Monday 18 September 164 BCE (Gregorian); 01 Tashritu / Tishri 3598 AM; Astronomical Julian Day JD_X: 1661785.5; RD_Y: 1313789; RD_X: -59639; Sabbatical Cycle: 7; Metonic Cycle: 7;
2208 cf. Appendix Table X.159

the modern cycle as valid in antiquity also.[2209] Indeed, this can be confirmed mathematically.

XX. Indeed, it can be confirmed that Newman's methodology does have some validity – in theory since his application of it is in error, not exact, and requires some loose interpretation as well as correction of his proposed dates.

XX. Newman reasons as follows: 1951 – 1952 CE (Gregorian) was a sabbatical year, i.e. the 7th year of a seven-(7)-year sabbatical cycle that began Monday 01 October 1951 CE (Gregorian)[2210] [2211] and lasted until the 1st year of the next sabbatical cycle (i.e. 1952 – 1953 CE (Gregorian)) on Saturday 20 September 1952 CE (Gregorian).[2212] [2213] Indeed, this can be confirmed mathematically.

XX. Newman continues that the corresponding seven-(7)-year cycle for this sabbatical year of 1951 – 1952 CE (Gregorian) would be [Autumn] 1945 – [Autumn] 1952 CE (Gregorian). Indeed, this can be confirmed mathematically.

XX. However, Newman states that if we calculate backwards [i.e. presumably to year 445 BCE (Julian)], and then forward again to the 69th sabbatical cycle, that this is 27 – 34 CE (Julian). However, this requires loose interpretation of application of the mathematics.

XX. The 20th Regnal Year of Artaxerxes, i.e. beginning 01 Nisannu / Nisan, Thursday 13 March 445 BCE (Julian), was the midpoint of the 5th year in a sabbatical cycle,[2214] [2215] as the Hebrew civil year 3316 AM (i.e. September 446 – 445 BCE (Julian)) count would have begun Tuesday 18 September 446 BCE (Julian).[2216] [2217]

XX. Therefore, following Newman, if we begin our count of 69 sabbatical cycles from this 5th year in this 1st sabbatical cycle (i.e. already in progress since September 446 BCE (Julian), and of course counting from the midpoint of 01 Nisannu / Nisan 445 BCE (Julian)),[2218] [2219] then the 5th year in the 69th

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2209 Newman 1973: 233

2210 Monday 18 September 1951 CE (Julian); Monday 01 October 1951 CE (Gregorian); 01 Tashritu / Tishri 5712 AM; Astronomical Julian Day JD_X: 2433920.5; RD_Y: 2085924; RD_X: 712496; Sabbatical Cycle: 7; Metonic Cycle: 12;

2211 cf. Appendix Table X.166

2212 Saturday 07 September 1952 CE (Julian); Saturday 20 September 1952 CE (Gregorian); 01 Tashritu / Tishri 5713 AM; Astronomical Julian Day JD_X: 2434275.5; RD_Y: 2086279; RD_X: 712851; Sabbatical Cycle: 1; Metonic Cycle: 13;

2213 cf. Appendix Table X.167

2214 Thursday 13 March 445 BCE (Julian); Thursday 08 March 445 BCE (Gregorian); 01 Nisannu / Nisan 3316 AM; Astronomical Julian Day JD_X: 1558958.5; RD_Y: 1210962; RD_X: -162466; Sabbatical Cycle: 5; Metonic Cycle: 10;

2215 cf. Appendix Table X.169

2216 Tuesday 18 September 446 BCE (Julian); Tuesday 13 September 446 BCE (Gregorian); 01 Tashritu / Tishri 3316 AM ; Astronomical Julian Day JD_X: 1558781.5; RD_Y: 1210785; RD_X: -162643; Sabbatical Cycle: 5; Metonic Cycle: 10;

2217 cf. Appendix Table X.168

2218 Thursday 13 March 445 BCE (Julian); Thursday 08 March 445 BCE (Gregorian); 01 Nisannu / Nisan 3316 AM; Astronomical Julian Day JD_X: 1558958.5; RD_Y: 1210962; RD_X: -162466; Sabbatical Cycle: 5; Metonic Cycle: 10;

2219 cf. Appendix Table X.169

sabbatical cycle is indeed Nisannu / Nisan 32 CE (Julian),[2220] [2221] i.e. within the 27 – 34 CE (Julian) sabbatical cycle – albeit from the 5th year of the 1st sabbatical cycle to the 5th year of the 69th sabbatical cycle.

XX. However, as mentioned, the 20th Regnal Year of Artaxerxes as recorded in Nehemiah 2:1, i.e. Nisannu / Nisan 445 – 444 BCE (Julian),[2222] was clearly (mathematically and logically) a scribal error, and Jesus was not – and could not have been – crucified in 32 CE (Julian).[2223] [2224]

XX. Thus if we apply Newman's methodology to the actual sabbatical 7th year of this 1st sabbatical cycle, i.e. September 444 – 443 BCE (Julian)[2225] [2226] – and of course counting from the midpoint of 01 Nisannu / Nisan 443 BCE (Julian)[2227] [2228] – then the actual sabbatical 7th year in the 69th sabbatical cycle works out to be 01 Nisannu / Nisan 34 CE (Julian),[2229] [2230] i.e. within the 27 – 34 CE (Julian) sabbatical cycle.

XX. However, this would work out to be the 22nd Regnal Year of Artaxerxes I, i.e. Nisannu / Nisan 443 – 442 BCE (Julian),[2231] and Jesus was not – and could not have been – crucified in 34 CE (Julian).[2232] [2233]

XX. Furthermore, if we apply Newman's methodology to the 1st year of the actual next sabbatical cycle, i.e. September 443 – 442 BCE (Julian)[2234] [2235] – and of course counting from the midpoint of 01

2220 Tuesday 01 April 32 CE (Julian); Tuesday 30 March 32 CE (Gregorian); 01 Nisannu / Nisan 3792 AM; Astronomical Julian Day JD_X: 1732836.5; RD_Y: 1384840; RD_X: 11412; Sabbatical Cycle: 5; Metonic Cycle: 11;
2221 cf. Appendix Table X.182
2222 cf. Appendix Table X.169
2223 Monday 14 April 32 CE (Julian); Monday 12 April 32 CE (Gregorian); 14 Nisannu / Nisan 3792 AM; Astronomical Julian Day JD_X: 1732849.5; RD_Y: 1384853; RD_X: 11425; Sabbatical Cycle: 5; Metonic Cycle: 11;
2224 cf. Appendix Table X.183
2225 Saturday 26 September 444 BCE (Julian); Saturday 21 September 444 BCE (Gregorian); 01 Tashritu / Tishri 3318 AM; Astronomical Julian Day JD_X: 1559520.5; RD_Y: 1211524; RD_X: -161904; Sabbatical Cycle: 7; Metonic Cycle: 12;
2226 cf. Appendix Table X.170
2227 Tuesday 23 March 443 BCE (Julian); Tuesday 18 March 443 BCE (Gregorian); 01 Nisannu / Nisan 3318 AM; Astronomical Julian Day JD_X: 1559698.5; RD_Y: 1211702; RD_X: -161726; Sabbatical Cycle: 7; Metonic Cycle: 12;
2228 cf. Appendix Table X.171
2229 Tuesday 09 March 34 CE (Julian); Tuesday 07 March 34 CE (Gregorian); 01 Nisannu / Nisan 3794 AM; Astronomical Julian Day JD_X: 1733543.5; RD_Y: 1385547; RD_X: 12119; Sabbatical Cycle: 7; Metonic Cycle: 13;
2230 cf. Appendix Table X.186
2231 cf. Appendix Table X.171
2232 Monday 22 March 34 CE (Julian); Monday 20 March 34 CE (Gregorian); 14 Nisannu / Nisan 3794 AM; Astronomical Julian Day JD_X: 1733556.5; RD_Y: 1385560; RD_X: 12132; Sabbatical Cycle: 7; Metonic Cycle: 13;
2233 cf. Appendix Table X.187
2234 Thursday 16 September 443 BCE (Julian); Thursday 11 September 443 BCE (Gregorian); 01 Tashritu / Tishri 3319 AM; Astronomical Julian Day JD_X: 1559875.5; RD_Y: 1211879; RD_X: -161549; Sabbatical Cycle: 1; Metonic Cycle: 13;
2235 cf. Appendix Table X.171

Nisannu / Nisan 442 BCE (Julian)[2236] [2237] – then the 1st year in the 69th sabbatical cycle works out to be 01 Nisannu / Nisan 35 CE (Julian),[2238] [2239] i.e. not within the 27 – 34 CE (Julian) sabbatical cycle.

XX. Furthermore, this would work out to be the 23rd Regnal Year of Artaxerxes I, i.e. Nisannu / Nisan 442 – 441 BCE (Julian),[2240] and of course Jesus was not – and could not have been – crucified in 35 CE (Julian).[2241] [2242]

XX. Recalling that the 20th Regnal Year of Artaxerxes I as recorded in Nehemiah 2:1, i.e. Nisannu / Nisan 445 – 444 BCE (Julian),[2243] was clearly (mathematically and logically) a scribal error, we will correctly apply Newman's methodology and begin our count from the 21st Regnal Year of Artaxerxes I, i.e. Nisannu / Nisan 444 – 443 BCE (Julian).[2244]

XX. Thus, if we apply Newman's methodology to the 6th year of that 1st of 69 sabbatical cycles that began in September 445– 444 BCE (Julian)[2245] [2246] – and of course counting from the midpoint of 01 Nisannu / Nisan 444 BCE (Julian)[2247] [2248] at the beginning of the 21st Regnal Year of Artaxerxes I – then the 6th year of the 69th of 69 sabbatical cycles works out to be 01 Nisannu / Nisan 33 CE (Julian),[2249] [2250] i.e. the exact year and month that Jesus was crucified – thus fulfilling Daniel 9:24-27 precisely to not only the exact 69th sabbatical cycle, and not only the exact year, but also the exact month.

XX. Moreover, although the exact day is not mentioned in Nehemiah 2:1, if the command to rebuild the walls of Jerusalem had been given by given by Artaxerxes I to Nehemiah on the 10th through the 14th day of Nisannu / Nisan at the climax of the Akitu festival, then indeed it is feasible that this prophecy could have been fulfilled on the precise day of either Jesus' arrest or his crucifixion.[2251]

2236 Saturday 12 March 442 BCE (Julian); Saturday 07 March 442 BCE (Gregorian); 01 Nisannu / Nisan 3319 AM; Astronomical Julian Day JD_X: 1560052.5; RD_Y: 1212056; RD_X: -161372; Sabbatical Cycle: 1; Metonic Cycle: 13;
2237 cf. Appendix Table X.172
2238 Tuesday 29 March 35 CE (Julian); Tuesday 27 March 35 CE (Gregorian); 01 Nisannu / Nisan 3795 AM; Astronomical Julian Day JD_X: 1733928.5; RD_Y: 1385932; RD_X: 12504; Sabbatical Cycle: 1; Metonic Cycle: 14;
2239 cf. Appendix Table X.189
2240 cf. Appendix Table X.172
2241 Monday 11 April 35 CE (Julian); Monday 09 April 35 CE (Gregorian); 14 Nisannu / Nisan 3795 AM; Astronomical Julian Day JD_X: 1733941.5; RD_Y: 1385945; RD_X: 12517; Sabbatical Cycle: 1; Metonic Cycle: 14;
2242 cf. Appendix Table X.190
2243 cf. Appendix Table X.169
2244 cf. Appendix Table X.170
2245 Saturday 06 September 445 BCE (Julian); Saturday 01 September 445 BCE (Gregorian); 01 Tashritu / Tishri 3317 AM; Astronomical Julian Day JD_X: 1559135.5; RD_Y: 1211139; RD_X: -162289; Sabbatical Cycle: 6; Metonic Cycle: 11;
2246 cf. Appendix Table X.169
2247 Thursday 02 April 444 BCE (Julian); Thursday 28 March 444 BCE (Gregorian); 01 Nisannu / Nisan 3317 AM; Astronomical Julian Day JD_X: 1559343.5; RD_Y: 1211347; RD_X: -162081; Sabbatical Cycle: 6; Metonic Cycle: 11;
2248 cf. Appendix Table X.170
2249 Saturday 21 March 33 CE (Julian); Saturday 19 March 33 CE (Gregorian); 01 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733190.5; RD_Y: 1385194; RD_X: 11766; Sabbatical Cycle: 6; Metonic Cycle: 12;
2250 cf. Appendix Table X.184, X.185, X.193
2251 cf. Appendix Table X.185

XX. Therefore, when applied correctly, Newman's methodology of counting these 69 seven-(7)-year sabbatical year cycles do indeed calculate correctly, and are completed in Hebrew Year 3793 AM, i.e. 33 CE (Julian), which is the the 6th year in the 69th of 69 sabbatical cycles, and the 12th year of the 19-year Metonic Cycle – i.e. fulfilling exactly 69 sabbatical year cycles exactly in the 1st month of Nisannu / Nisan 33 CE (Julian), the year and month of Jesus' Crucifixion.

73.0. An Alternate Literal (Cryptic) Reading of Daniel 9:24-27

XX. If that were not enough, it is possible to read, translate, and understand Daniel 9:24-27 in a secondary literal manner (i.e. in addition to the seventy (70) seven-(7)-year sabbatical cycles), and see this prophecy fulfilled mathematically a second time in a second (overlapping) way.

24 Seventy weeks[-of-years][2252] has been cut[2253] upon your people and upon the city of your holiness, to imprison the intentional-willful-transgressional-sin, and to seal[2254] sins, and to make atonement [for] iniquity, and to bring justice [of the] worlds,[2255] and to sign a vision and a prophet, and to anoint [the] Holiness of Holinesses.

25 And you will know, and you will contemplate: From the taking out of the word to return [the exiles] and to build Jerusalem until a Mashiach Nagid:[2256] seven weeks and sixty weeks and two [years].[2257] [She][2258] [2259] will return, and she [will be] built, street and moat, and in distress of the times.

26 And after the weeks [-of-years, there will be] sixty two [months].[2260] A Mashiach will be cut off, and he does not have, and the city and the holiness: the people of the coming Nagid[2261] will corrupt,[2262] and his end [is] in the flood, and until [his / the / an] end:[2263] a determined war [of] desolations.[2264]

2252 Alternate translation: "seventy sevens"
2253 Literally: "cut", i.e. "divided"; Figuratively: "determined"
2254 Literally: "to sign"
2255 Alternate translation: "and to bring eternal justice"
2256 i.e. "a Messiah, [one who is] spoken of"
2257 Alternate translation: "seven weeks[-of-years] and sixty-two weeks[-of-years]'
2258 i.e. She: the city of Jerusalem – feminine gender in Hebrew
2259 Alternate translation: "[You, Daniel,] will return [to Jerusalem]"
2260 Alternate translation: "sixty-two weeks[-of-years]"
2261 i.e. "[one who is] spoken of"
2262 Alternate translation: "will destroy"
2263 Alternate translation: "and until an end of a determined war: desolations."
2264 Alternate translation: "a determined war, desolations."

27 And [he] strengthened a covenant to many[2265] one week,[2266] and half of the week[2267] [he] will cause to cease sacrifice and offering, and by means of[2268] a wing of abominations [he] makes desolate and until [it] completes, and [she[2269] is] determined: [she] will melt upon a desolate.[2270]

XX. Recalling that the 20th Regnal Year of Artaxerxes as recorded in Nehemiah 2:1, i.e. Nisannu / Nisan 445 – 444 BCE (Julian),[2271] was clearly (mathematically and logically) a scribal error, we will begin our count from the 21st Regnal Year of Artaxerxes I, i.e. 444 – 443 BCE (Julian).[2272]

XX. It is literally written in Daniel:

From the taking out of the word to return [the exiles] and to build Jerusalem until a Mashiach Nagid:[2273] seven weeks and sixty weeks and two [years].

seven (7) weeks-of-years + sixty (60) weeks-of-years == 469 years.

+ two (2) years

== 471 years

XX. Counting 471 years from the 1st month of Nisannu / Nisan of 21st Regnal Year of Artaxerxes I, i.e. 444 – 443 BCE (Julian),[2274] [2275] brings us to the 1st month of Nisannu / Nisan 28 BCE (Julian).[2276] [2277] What event of significance could have happened in this year?

XX. To anoint [the] Holiness of Holinesses:[2278]

XX. According to the Gospel of Luke,[2279] John the Baptist commenced his ministry in the 15th Regnal

2265 cf. Babylonian Talmud *Yoma* 54
2266 Alternate translation: "one seven"
2267 Alternate translation: "half of the seven"; cf. Babylonian Talmud *Yoma* 54a; Rashi: "a Roman king"
2268 Literally: "on/upon"
2269 i.e. She: war – feminine gender in Hebrew
2270 A desolate what? Suggested: "desolation"
2271 cf. Appendix Table X.169
2272 cf. Appendix Table X.170
2273 i.e. "a Messiah, [one who is] spoken of"
2274 Thursday 02 April 444 BCE (Julian); Thursday 28 March 444 BCE (Gregorian); 01 Nisannu / Nisan 3317 AM; Astronomical Julian Day JD_X: 1559343.5; RD_Y: 1211347; RD_X: -162081; Sabbatical Cycle: 6; Metonic Cycle: 11;
2275 cf. Appendix Table X.170
2276 Tuesday 16 March 28 CE (Julian); Tuesday 14 March 28 CE (Gregorian); 01 Nisannu / Nisan 3788 AM; Astronomical Julian Day JD_X: 1731359.5; RD_Y: 1383363; RD_X: 9935; Sabbatical Cycle: 1; Metonic Cycle: 7;
2277 cf. Appendix Table X.174
2278 cf. Daniel 9:24
2279 cf. Luke 3:1-3

Year of Tiberius Caesar [r. 14 – 37 CE] with Jesus subsequently baptized by John [i.e. in that year].[2280] Since Tiberius was a Roman ruler, and since Luke was writing to Theophilus who appears to have been a Roman official, Luke would probably have reckoned the 15th Regnal Year of Tiberius according to a Roman [Julian] calendar.[2281]

XX. If Luke was using the [Roman] Julian calendar and reckoning according to the [Year 0] Accession Year system as used by various Roman historians such as Tacitus, then the 15th Regnal Year would have been from 01 January to 31 December 29 CE.[2282]

XX. Alternatively [and the one that will be demonstrated to be mathematically correct with Jesus' date of baptism by John in year 28 CE (Julian)], Luke may have used the normal Roman Regnal Year [i.e. Year 1 Non Accession Year] method of reckoning – in which case the 15th Regnal Year would have been from 19 August 28 CE (Julian) to 18 August 29 CE (Julian). If in fact Luke had used Jewish ecclesiastical reckoning [for dates of Roman kings: which is not the case], the 15th Regnal Year would have been Spring 29-30 CE.[2283]

XX. From the coins issued in Tiberius' reign, it is clear that Tiberius himself reckoned the 1st year of his reign from the death of Augustus in [August] 14 CE (Julian). Moreover, the New Testament scholar John Meier writes: "As a matter of fact, all the major Roman historians who calculate the years of Tiberius' rule – namely, Tacitus, Suetonius, and Cassius Dio – count from AD 14, the year of Augustus' death."[2284] Since Tacitus tells us that Jesus died in Tiberius' reign, Jesus must therefore have died in the period 14-37 CE.[2285]

XX. According to historians, the 1st Regnal Year of Tiberius began upon the death of Augustus on 19 August 14 CE (Julian), thus making August 28 – 29 CE (Julian) his 15th Regnal Year.

XX. It is not known the month when Jesus was baptized by John the Baptist. However, if John had begun his ministry after August 28 CE (Julian), then Jesus would have been baptized after this time.

XX. If one were to refer a time period of seven (7) years and two (2) months, then it is reasonable to refer to this period as whole by saying, "after these years" with it being understood that the smaller time units of two (2) months are included in the period of larger time units of seven (7) years and two (2) months.

XX. In like manner, it is reasonable (and can be understood) that referring to the 67 weeks-of-years as

2280 Humphreys and Waddington 1992: 337
2281 Humphreys and Waddington 1992: 337
2282 Humphreys and Waddington 1992: 337-338
2283 Humphreys and Waddington 1992: 338
2284 Humphreys 2011: 18
2285 Humphreys 2011: 18

"after these weeks[-of-years]" (i.e. 469 years) also includes the extra two (2) years, i.e. 471 years:

> *And after the weeks [-of-years, there will be] sixty two [months].*[2286]

XX. And after these weeks-of-years (including the two (2) single years) totaling 471 years from the 1st month of Nisannu / Nisan 444 BCE to the 1st month of Nisannu / Nisan 28 CE (Julian), one can count from the 1st month of Nisannu / Nisan 28 CE (Julian)[2287] [2288] exactly 62 Babylonian / Hebrew months, and arrive exactly to the 1st month Nisannu / Nisan 33 CE (Julian),[2289] [2290] i.e. the exact year and month of Jesus' crucifixion.

74.0. Darius the Mede

XX. As noted by H.H. Rowley (1935), the references to Darius the Mede in the Book of Daniel have long been recognized as providing the most serious historical problem in the book.[2291] Josephus and Jerome already grappled with attempting to reconcile Darius the Mede with known history centuries earlier (i.e. 1st century CE (Julian) and 4th century CE (Julian) respectively).

XX. The discovery of the Cyrus Cylinder (BM 90920) and Nabonidus Chronicle (BM 35382) in 1879 CE (Gregorian) sharpened the problem decades before Rowley's definitive landmark study since these cuneiform sources confirmed Herodotus' (ca. 484 – 425 BCE (Julian)) account and found no Median king between Nabonidus and Cyrus.

74.1. The Non-Chronological Structure of the Book of Daniel

XX. Upon careful reading of the Book of Daniel, it becomes clear that – like the Book of Ezekiel – it is not written in chronological order of events. The following table presents each chapter of the Book of Daniel alongside the ruler and regnal year cited in the text (if known), thus illustrating where the book's internal chronology departs from strict sequential order:

Table 74.1. Rulers and Regnal Years Cited by Chapter in the Book of Daniel

Chapter	Ruler	Year Cited	Verses
1	Nebuchadnezzar	3rd Regnal Year of Yehoyakim (Jehoiakim) king of	1:1, 1:21

2286 Alternate translation: "sixty-two weeks[-of-years]"
2287 Tuesday 16 March 28 CE (Julian); Tuesday 14 March 28 CE (Gregorian); 01 Nisannu / Nisan 3788 AM; Astronomical Julian Day JD_X: 1731359.5; RD_Y: 1383363; RD_X: 9935; Sabbatical Cycle: 1; Metonic Cycle: 7;
2288 cf. Appendix Table X.174
2289 Saturday 21 March 33 CE (Julian); Saturday 19 March 33 CE (Gregorian); 01 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733190.5; RD_Y: 1385194; RD_X: 11766; Sabbatical Cycle: 6; Metonic Cycle: 12;
2290 cf. Appendix Table X.184
2291 Rowley 1935: 8

		Judah; mentions Daniel was until the 1st Regnal Year of Cyrus	
2	Nebuchadnezzar	2nd Regnal Year of Nebuchadnezzar	2:1
3	Nebuchadnezzar	No Year Given	–
4	Nebuchadnezzar	No Year Given	–
5	Belshazzar	No Year Given	–
6	Darius of the Medes	No Year Given	–
7	Belshazzar	1st Regnal Year of Belshazzar	7:1
8	Belshazzar	3rd Regnal Year of Belshazzar	8:1
9	Darius the son of Ahasuerus from the seed of Media	1st Regnal Year of Darius	9:1
10	Cyrus	3rd Regnal Year of Cyrus	10:1
11	Cyrus	3rd Regnal Year of Cyrus	10:1
12	Cyrus	3rd Regnal Year of Cyrus	10:1

74.2. The Occurrences of Darius in the Text

XX. In the Book of Daniel – reading from the Hebrew / Aramaic Masoretic text whose chapter and verse divisions differ slightly from the King James Version – there are eight (8) mentions of Darius as a ruler of Babylon distributed across chapters, 6, 9, and 11:

1.) Daniel 6:1:[2292] "*And Daryavesh (Darius) of the Medes received the kingdom [at] about sixty two years [of age].*"

2.) Daniel 6:2:[2293] "*It was good before Daryavesh (Darius), and he established over the kingdom for [purpose of being] satraps – one hundred and twenty – that they would be throughout the entire kingdom.*"

3.) Daniel 6:7:[2294] "*Then these viziers and satraps assembled upon the king and thus said to him: "Daryavesh (Darius) the king, forever live!*"

4.) Daniel 6:10:[2295] "*Because of this, the king Daryavesh (Darius) inscribed the writing and prohibition.*"

5.) Daniel 6:26:[2296] "*Then Daryavesh (Darius) the king wrote to all the peoples, nations, and tongues that dwell in all the land of his kingdom: "Your peace will be great!*"[2297]

2292 King James Version: Daniel 5:31
2293 King James Version: Daniel 6:1
2294 King James Version: Daniel 6:6
2295 King James Version: Daniel 6:9
2296 King James Version: Daniel 6:25
2297 Literally: "Your peace will be high [up]!"

6.) Daniel 6:29:[2298] "*And this Daniel succeeded in the kingdom of Daryavesh (Darius) and in the kingdom of Coresh (Cyrus) the Persian.*"

7.) Daniel 9:1: "*In year one of Daryavesh (Darius) son of Ahash-verosh (Ahasuerus), from the seed of Media, who was made king over the monarchy of the Chaldeans.*"

8.) Daniel 11:1: "*And I, in year one of Daryavesh (Darius) the Mede, my position[2299] [will be] for maintaining and for a stronghold to him.*"

74.3. The Earliest Extra-Biblical References to Darius the Mede

XX. Outside of the Book of Daniel itself, the oldest known reference to Darius the Mede appears in the writings of the first-century Roman-Jewish historian, Flavius Josephus. In his *Antiquities of the Jews*, written around 93 – 94 CE (Julian). Josephus states that Babylon was taken by Darius who – together with his kinsman Cyrus – put an end to Babylonian rule, and that Darius was sixty-two (62) years old and the son of Astyages, and had another name among the Greeks.[2300]

XX. The next earliest extant reference appears roughly a half century later (i.e. ca. 140 – 160 CE (Julian)) in the rabbinic chronological work *Seder Olam Rabbah*,[2301] compiled in the mid-second century CE (Julian), which lists Darius the Mede as the first of four (4) Persian-period kings,[2302] tying his reign to Daniel's prophecy given in Darius's first year.

74.4. Reassessing the Assumed Succession from Belshazzar to Darius

XX. Scholars have long presumed, based on the placement of Darius's perceived accession at the close of chapter 5 in the King James Version, that Darius the Mede acceded to the throne immediately after Belshazzar's death that is recorded in this chapter. This assumption rests on the English versification rather than on the text itself, and warrants a closer examination before being accepted as established fact.

XX. In the Masoretic Hebrew / Aramaic text, this verse is not numbered 5:31 but rather 6:1, thus marking the beginning of a new chapter. Upon closer reading of the Aramaic text, it becomes clear that this verse begins a new paragraph (with new subject matter) which is distinct from the preceding

2298 King James Version: Daniel 6:28
2299 Most certainly it is clear from the Hebrew that this is not in the past tense.
2300 Flavius Josephus, *Antiquities of the Jews*, Book 10, Chapter 11, Section 248 – Whiston Translation
2301 Rabbi Yose ben Halafta, *Seder Olam Rabbah*, Chapter 28
2302 i.e. Darius the Mede; Cyrus; Achashverosh (i.e. Ahasuerus / Xerxes); Darius I (the Great)

narrative unit describing Belshazzar's death, and thus should not be read as a direct continuation of it.

XX. Nabonidus was the last king of the Neo-Babylonian empire, ruling officially for 17 years (r. 555 – 539 BCE (Julian)). As noted in section 30.0., from his 3rd Regnal Year to the beginning of his 13th Regnal Year, his son Belshazzar is recorded as co-regent. Neither king nor son is attested performing official duties after the start of the 13th Regnal Year with Belshazzar's last attestation in this capacity dated to the 3rd month of that 13th Regnal Year. However, as noted by Beaulieu, this may simply reflect a gap in surviving (extant) documentation since study of official letters indicates that Nabonidus had resumed rule by the middle of his 13th Regnal Year.

XX. Absence evidence requiring it, the simplest explanation requiring the fewest unnecessary assumptions is to be preferred – it becomes clear that the traditional reading rests on an assumption that is unnecessary and unsupported by anything beyond the order in which the chapters appear:

1.) It is unnecessary to conclude that Belshazzar was still king of Babylon on the night Cyrus' forces conquered the city given that his last attestation in the administrative record predates the conquest by roughtly four (4) years;

2.) It is unnecessary to assume that the night of Belshazzar's death in chapter 5 was the same night that Cyrus's forces conquered Babylon; the text does not state this.

3.) It is unnecessary to assume that Xenophon's account in *Cyropaedia* (ca. 370 BCE (Julian)) of a king killed the night of Cyrus's conquering Babylon refers to Belshazzar; Xenophon (ca. 430 – 354 BCE (Julian) does not name the king, and he was more likely referring to Nabonidus – the actual, official final king of the Neo-Babylonian empire.

4.) It is unnecessary to assume that Darius the Mede succeeded Belshazzar directly simply because the account of his accession in chapter 6 which follows the account of Belshazzar's death in chapter 5.

XX. The sequence of the biblical text cannot be treated as evidence of an historical, chronological sequence precisely because it has already been demonstrated that the Book of Daniel – just as with the Book of Ezekiel – is not written in chronological order. Therefore, there is no reason to assume that sequential chapters necessarily reflect sequential events.

74.5. Identifying Darius the Mede as Darius the Great

XX. Daniel 9:1 states specifically that Darius was "*made king*": "*In year one of Daryavesh (Darius) son of Ahash-verosh (Ahasuerus), from the seed of Media, who was made king over the monarchy of the*

Chaldeans."

74.5.1. Darius Was "Made King": Daniel 9:1 and the Manner of His Accession

XX. Herodotus provides the primary account of how Darius the Great actually became king: After the seven (7) conspirators who had overthrown the usurper Gaumata (Bardiya) agreed on monarchy as the form of government that Persia's government would take, they needed to determine who among them would rule.

XX. Herodotus records that seven (7) Persian nobles conspired to overthrow the usurper Gaumata (Bardiya),[2303] but Otanes withdrew his claim to the throne before the succession was decided,[2304] thus leaving six (6) to ride in the contest that determined who would become king:[2305]

"*Concerning the appointment of a king, the resolve to which they came was the following: They would ride out together next morning into the skirts of the city, and he whose steed first neighed after the sun was up should have the kingdom.*[2306]

"*It seemed as if the heavens conspired with Darius, and hereby inaugurated him king: so the five other nobles leaped with one accord from their steeds, and bowed down before him and owned him for their king.*"[2307]

74.5.2 The Age of Sixty-Two (62): An Internal Chronological Problem

XX. Daniel 6:1 (Masoretic verse numbering) states specifically "*And Daryavesh (Darius) of the Medes received the kingdom [at] about sixty two years [of age].*"

XX. Darius the Great reigned for thirty-six (36) years from 522 – 486 BCE (Julian). If Daniel 6:1 is correct that he received the kingdom at the age of sixty-two (62), then he would have died at approximately ninety-eight (98) years of age. Although this is possible, it is an implausible lifespan for the period especially for a ruling king, and is not attested in any source describing his life or death.

XX. This internal inconsistency suggests that the number sixty-two (62) is itself the more likely source of error. Two (2) distinct scribal-error explanations may account for how the number sixty-two (62) entered the text, and both are offered here for the reader to weigh rather than it being resolved in favor

2303 Herodotus, *Histories* 3.70
2304 Herodotus, *Histories* 3.83
2305 Herodotus, *Histories* 3.86
2306 Herodotus, *Histories* 3.84
2307 Herodotus, *Histories* 3.86

of one:

1.) A plausible explanation is that a Hebrew Jewish scribe, transmitting the number either from an earlier exemplar or from memory, reversed the two (2) digits in his head, thus rendering the words twenty-six (26) as sixty-two (62).

2.) A second possibility is that the scribe, who would have been thinking in Hebrew, confused the number with its Hebrew counterpart: the words for sixty (ששים) *shishim* and thirty (שלשים) *shloshim* are very similar with only one (1) letter (ל) *Lamed* differentiating them, and both share the letter (ש) *Shin*. Thus a scribe transmitting an Aramaic text – while thinking in Hebrew terms – may have rendered thirty-two (32) as sixty-two (62).

XX. A 3rd possibility is that Darius the Great was indeed 62 years old when he received the kingdom, and simply lived to be ninety-eight (98). However, external evidence weighs against this.

XX. Herodotus records that at the time of Cyrus' campaign against the Massagetae, roughly 530 BCE (Julian), Darius was "about twenty years old".[2308] This would place his birth around 550 BCE (Julian), making him approximately twenty-eight (28) at his accession in 522 BCE (Julian) – not sixty-two (62).

XX. While Herodotus's figure is itself an approximation rather than a precise date, it stands as the only external data point bearing on Darius' age, and it points away from the reading preserved in Daniel 6:1. Weighing all three (3) possibilities together, the reader will likely conclude that a scribal digit-reversal rendering the words for 26 as 62 is the most probable and likely explanation.

XX. A 4th – and least likely – possibility is that Darius the Mede was an actual historical king, distinct from Darius the Great, who briefly ruled over Babylon and instituted the system of 120 satraps sometime between the reigns of Cyrus and Darius the Great, but for whom no corroborating evidence has yet been discovered. Given the extensive administrative and chronological documentation surviving from this period including the Nabonidus Chronicle, the Cyrus Cylinder, and the combined administrative and astronomical chronology studied by Parker and Dubberstein (1956, 1971), as well as the dated celestial observations recorded in the Astronomical Diaries compiled by Sachs and Hunger (1988), the absence off any trace of such a figure makes this the weakest of the four (4) possibilities.

74.5.3. "Of the Seed of the Medes": The Question of Median Ancestry

XX. Daniel 6:1 and 9:1 both identify Darius as a Mede with Daniel 6:1 stating "*Daryavesh (Darius) of the Medes*" and Daniel 9:1 specifying "*from the seed of Media*". The standard objection is that Darius the Great was Persian – not Median – through his father Hystaspes. However, this assumes that Median

2308 Herodotus, *Histories* 1.209

and Persian identity remained sharply separated after Cyrus conquered Media in 550 BCE (Julian)[2309] – which is not the case as Medes continued to serve among the Persian ruling elite, and Cyrus himself was half-Median through his mother Mandane (daughter of the Median king Astyages).[2310]

XX. Given this blending at the highest levels of Achaemenid society, it is entirely possible that Darius the Great carried Median ancestry not recorded in surviving geneaologies. The assumption that he could not have been of Median descent is not established by the evidence; it is simply the default reading in the absence of information to the contrary.[2311]

XX. Darius the Great left several inscriptions describing his own ancestry and titles. Among these, the inscription at his tomb at Naqsh-e Rostam (known as DNa)[2312] and a parallel inscription as Susa (DSe)[2313] both open with the same formula: "*I am Darius, the great king, king of kings… son of Hystaspes, an Achaemenid, a Persian, son of a Persian, an Aryan, of Aryan seed.*"[2314]

XX. The term "Aryan" (Old Persian: *Ariya*) was not exclusive to the Persians, but served as a broader ethnolinguistic designation covering the Iran peoples generally including the Medes. Herodotus records that the Medes were anciently called by everyone "Arians",[2315] and the *Encyclopaedia Iranica* confirms this usage was not confined to Persians alone.[2316]

XX. Read in this light, Darius's formula "*a Persian, son of a Persian, an Aryan, of Aryan seed*" may carry a double-sense: the first pair affirming his father's Persian lineage, and the second being a broader term leaving room for a mother of Median descent since a Mede would equally have been considered "Aryan". The inscriptions are therefore silent on his mother's ancestry rather than excluding a Median line.

74.5.4. The Father's Name: Ahasuerus and/or Hystaspes

XX. These same inscriptions also name Darius's father directly: Hystaspes[2317] (Old Persian: *Vištāspa*), a

2309 The Nabonidus Chronicle records Astyages's defeat; Herodotus, *Histories* 1.127-130, gives the narrative account. See also Encyclopedia Iranica, "Cyrus ii. Cyrus I."

2310 Herodotus, *Histories* 1.107-108, 111, states directly that Cyrus's mother, Mandane, was the daughter of Astyages, king of the Medes, and his father, Cambyses I, was Persian.

2311 Herodotus repeatedly uses the terms (i.e. "Mede" and "Persian") interchangeably for the ruling nobility (e.g. calling Cyrus "king of the Medes" at points. For more information about Median-Persian integration in administration and nobility, see Pierre Briant, *From Cyrus to Alexander: A History of the Persian Empire* (Eisenbrauns, 2002).

2312 https://Livius.org/sources/content/achaemenid-royal-inscriptions/dna/

2313 https://Livius.org/sources/content/achaemenid-royal-inscriptions/dse/

2314 See also Roland G. Kent, *Old Persian: Grammar, Texts, Lexicon*, 2nd ed. (New Haven: American Oriental Society, 1953); https://ia600601.us.archive.org/19/items/OldPersian/Kent-OldPersian_text.pdf

2315 Herodotus, *Histories* 7.62

2316 https://IranicaOnline.org/articles/iranian-identity-ii-pre-islamic-period/

2317 Hystaspes is a Greek rendition used by Herodotus, and by convention: in English scholarship

Persian noble of the Achaemenid line, attested both in Darius's own inscriptions and in Herodotus. The name Hystaspes bears no phonetic resemblance to the Hebrew rendition (אחשורוש) *Ahashverosh* (which is the standard Hebrew rendering used elsewhere in the bible (i.e. Esther, Ezra) for the Persian king known as Xerxes I (Old Persian: *Xšayāršā*)). This Hebrew (אחשורוש) *Ahashverosh* is rendered to English as *Ahasuerus*, i.e. Darius's father's name given in Daniel 9:1.

XX. If Daniel's "Darius of the Medes" is indeed Darius the Great, then this discrepancy must be accounted for:

1.) Daniel 9:1 may be the redactors' own independent Hebrew rendering for *Vištāspa* – unconnected to its standard Hebrew rendering *Ahashverosh* or English rendering *Ahasuerus* elsewhere in scripture.

2.) The Men of the Great Assembly (אנשי כנסת הגדולה) *Anshei Knesset HaGdolah* (ca. 516 – 332 or until ca. 200 BCE (Julian))[2318] – who per Jewish tradition wrote the Book of Daniel[2319] many years later after the events – may simply have confused the direction of the relationship: naming Darius's son Xerxes I (i.e. *Ahasuerus* / *Ahashverosh* / *Xšayāršā*) as his father instead.

74.6. The Retrospective Naming of Kings in Daniel 6:28

XX. Daniel 6:29 (Masoretic numbering) states: "*And this Daniel succeeded in the kingdom of Daryavesh (Darius) and in the kingdom of Coresh (Cyrus) the Persian.*"

XX. Although this ordering may at first suggest that Darius preceded Cyrus, it should instead be read as the product of a scribe writing at a later period looking back in time, and naming the most recent king first before reaching back to Cyrus who conquered Babylon. Such retrospective ordering reflects the vantage point of the writers, i.e. the Men of the Great Assembly – not the actual sequence of the two (2) reigns.

74.7. Darius the Mede in Daniel 11:1

XX. As mentioned in section 34.0, the vision of Daniel 10 is dated to the 3rd Regnal Year of Cyrus, and

2318 Scholarly opinion is sharply divided about exact dates for the Great Assembly: The Orthodox Union describes it spanning approximately 320 years from ca. 520 – 200 BCE (Julian); Aish.com dates it more narrowly from 410 – 310 BCE (Julian). Traditional sources mark its conclusion with the death of Simeon the Just (its last member) but his identity and date are themselves disputed: rabbinic tradition (Babylonian Talmud, *Yoma* 69a) and Josephus identity him with Simon I, high priest 310 – 291 BCE (Julian); Chabad.org gives his death as 273 BCE (Julian), while other scholars identify him instead with Simon II, high priest 219 – 199 BCE (Julian) – (Encyclopedia.com; *Britannica*)

2319 Babylonian Talmud, *Bava Batra* 15a

continues through chapters 11 and 12 to the end of the book. In this vision, the "man dressed in fabrics", who addresses Daniel throughout the vision, refers to Darius the Mede in chapter 11:

20 And he said, "Do you know why I have come to you? And now I will return to battle with the Minister of Persia, and I am leaving, and behold the Minister of Greece is coming. 21 However, I will tell you the recorded [matter] in [the] Writing of Truth, and there is not [any] one who is becoming strong with me upon these [Ministers of Persia and Greece] except for Michael your[2320] Minister."[2321]

XX. The man dressed in fabrics continues speaking in chapter 11:

1 "And I, in year one of Daryavesh (Darius) the Mede,[2322] my position[2323] [will be] for maintaining and for a stronghold to him. 2 And now truth I will tell to you: behold another three kings[2324] stand for Persia, and the fourth[2325] will make wealthy, a great wealth from all [before him], and at his strength in his wealth, he will arouse all [his kingdom to war] with the kingdom of Greece. 3 And a mighty-warrior king[2326] stood and ruled a great dominion, and did as [according to] his will. 4 And at his standing his kingdom will break and will be divided to the four winds[2327] of the skies, and not for his end, and not as his dominion that he ruled because his kingdom will be uprooted, and to others besides these."[2328]

XX. This reference to Darius the Mede has traditionally been read as the speaker recalling a past event because of the mistaken assumption that Darius's reign had already occurred by the time of this vision in the 3rd Regnal Year of Cyrus. However, the underlying Hebrew word rendered as "my position" (עמדי) is an infinitive construct, i.e. a form that does not grammatically encode tense. Thus a past-tense reading adopted by most English translations is therefore an interpretive choice based upon the assumption that Darius the Mede ruled prior to Cyrus, and thus is not a grammatical requirement.

XX. Grammatically, the verse describes the speaker's future role in relation to Darius – not a past one – whose 1st Regnal Year had not yet arrived by the time of this vision. Such a reading requires no such false assumption that Darius the Mede had already reigned, and is fully consistent with the vision's own stated date in the 3rd Regnal Year of Cyrus (536 BCE (Julian)). Therefore, this entire vision – spanning chapters 10 through 12 – should be read as prophecy given to Daniel speaking of events still future to him at the time it was given, including the specific prophetic naming of Darius as the future

2320 The word "your" here is the plural tense.
2321 cf. Daniel 10:20-21
2322 i.e. Darius (r. 521 – 486 BCE (Julian))
2323 Most certainly it is clear from the Hebrew that this is not in the past tense.
2324 i.e. another three (3) kings counting from the time of speaking in the 3rd Regnal Year of Cyrus: 1.) Cyrus II (r. 538 – 530 BCE (Julian)); 2.) Cambyses II (r. 529 – 522 BCE (Julian)); 3.) Darius (r. 521 – 486 BCE (Julian));
2325 i.e. Xerxes I (r. 485 – 465 BCE (Julian))
2326 i.e. Alexander the Great (r. 336 – 323 BCE (Julian))
2327 Hebrew: "four spirits"
2328 cf. Daniel 11:1-4

king fourteen (14) years before he acceded to the throne in 522 BCE (Julian).

75.0. Crucifixion or Cruci-Fiction?

XX. No single matter of chronology has been examined so thoroughly as the date of death of Jesus of Nazareth. It is chronology's premier topic. All agree that only a handful of days in the late 20s and early 30s of the 1st century CE (Julian) qualify. These dates have all been defended at one time or another.[2329]

XX. The date of the Crucifixion [of Jesus] has been debated for many years, but there has been no agreement on the year nor the day. However astronomy can be used to reconstruct the Jewish calendar in the 1st century CE, and hence to rule out many possibilities.[2330]

XX. Sir Isaac Newton (1733) was the first to derive a date for the crucifixion of [Jesus] Christ by calculating when the crescent of the New Moon was first visible [i.e. via the geometrical method of arcus visionis] in order to correlate the Judean and Julian calendars.[2331] Researchers from Newton to modern researchers have essentially agreed on dates of [theoretical, geometrically calculated based values of arcus visionis] lunar visibility, but not on the resultant crucifixion date. The two (2) most accepted dates are Friday 07 April 30 CE (Julian)[2332] [2333] and Friday 03 April 33 CE (Julian).[2334] [2335] [2336] It must be noted that the former date, i.e. Friday 07 April 30 CE (Julian), occurred on 16 Nisannu / Nisan 3790 AM which eliminates it as a possibility for Jesus' crucifixion.

XX. It has been known that Newton preferred Friday 23 April 34 CE (Julian) [for the date of Jesus crucifixion].[2337] It must be noted that this date of Friday 23 April 34 CE (Julian) as calculated by Newton did not in actuality occur in the 1st month of Nisannu / Nisan, but in rather the 2nd month of Ayyaru / Iyar.[2338] [2339]

XX. Newton did not base his date for the crucifixion on any traditions about the birth of Christ or the

2329 Depuydt 2002: 466
2330 Humphreys and Waddington 1992: 331
2331 Pratt 1991: 301
2332 Friday 07 April 30 CE (Julian); Friday 05 April 30 CE (Gregorian); 16 Nisannu / Nisan 3790 AM; Astronomical Julian Day JD_X: 1732111.5; RD_Y: 1384115; RD_X: 10687; Sabbatical Cycle: 3; Metonic Cycle: 9;
2333 cf. Appendix Table X.179
2334 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2335 cf. Appendix Table X.185
2336 Pratt 1991: 301
2337 Pratt 1991: 301
2338 Friday 23 April 34 CE (Julian); Friday 21 April 34 CE (Gregorian); 16 Ayyaru / Iyar 3794 AM; Astronomical Julian Day JD_X: 1733588.5; RD_Y: 1385592; RD_X: 12164; Sabbatical Cycle: 7; Metonic Cycle: 13;
2339 cf. Appendix Table X.188

length of his life. After he reviewed several opinions about the date of Christ's birth, he concluded that "there is no tradition worth considering" and then went on to deduce the [incorrect] crucifixion date.[2340]

XX. Newton began by dating the baptism of [Jesus] Christ in 29 CE (Julian) during the [2nd half of the] 15th year of Tiberius Caesar [which began in August 28 CE (Julian)].[2341] He then cited evidence for a total of five (5) – rather than four (4) – Passovers during the ministry [i.e. from the date of Jesus' baptism in the 15th year of Tiberius] implying a 34 CE crucifixion.[2342]

XX. Then Newton did the calendrical analysis almost exactly as has been done since: determining in which years the crucifixion day – 14 Nisannu / Nisan on the Hebrew calendar – could have been a Friday.[2343] He stated that the Jews deemed the crescent visible "about 18h after the true conjunction" which is optimistic even near the Spring Equinox.[2344]

XX. Using that rule to determine the 1st day of Nisan, Newton [attempted to] calculate the day of the week for 14 Nisannu / Nisan for the years 31 – 36 CE (Julian). He thus excluded 31, 32, and 35 CE (Julian) because 14 Nisannu / Nisan could not have been a Friday which has been confirmed by all modern researchers.[2345] Indeed it can be confirmed that these three (3) dates for 14 Nisannu / Nisan in each of these years all fall on a Monday.[2346] [2347] [2348] [2349]

XX. After next excluding 35 and 36 CE (Julian) on historical grounds, Newton had narrowed the field to 03 April 33 CE (Julian)[2350] [2351] and 23 April 34 CE (Julian).[2352] [2353] He then found a final argument for 34 CE (Julian) in the ripeness of the corn at Passover based on Luke 6:1 referring to the last day of Passover two (2) years prior to the crucifixion.[2354]

XX. Thus Newton first narrowed the possible years down to 33 CE (Julian) and 34 CE (Julian). Had he

2340 Pratt 1991: 302
2341 cf. Luke 3:1, 21-23
2342 Pratt 1991: 301
2343 cf. John 19:14, 42
2344 Pratt 1991: 301
2345 Pratt 1991: 301
2346 Monday 26 March 31 CE (Julian); Monday 24 March 31 CE (Gregorian); 14 Nisannu / Nisan 3791 AM; Astronomical Julian Day JD_X: 1732464.5; RD_Y: 1384468; RD_X: 11040; Sabbatical Cycle: 4; Metonic Cycle: 10;
2347 Monday 14 April 32 CE (Julian); Monday 12 April 32 CE (Gregorian); 14 Nisannu / Nisan 3792 AM; Astronomical Julian Day JD_X: 1732849.5; RD_Y: 1384853; RD_X: 11425; Sabbatical Cycle: 5; Metonic Cycle: 11;
2348 Monday 11 April 35 CE (Julian); Monday 09 April 35 CE (Gregorian); 14 Nisannu / Nisan 3795 AM; Astronomical Julian Day JD_X: 1733941.5; RD_Y: 1385945; RD_X: 12517; Sabbatical Cycle: 1; Metonic Cycle: 14;
2349 cf. Appendix Table X.181, X.183, X.190
2350 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2351 cf. Appendix Table X.185
2352 Friday 23 April 34 CE (Julian); Friday 21 April 34 CE (Gregorian); 16 Ayyaru / Iyar 3794 AM; Astronomical Julian Day JD_X: 1733588.5; RD_Y: 1385592; RD_X: 12164; Sabbatical Cycle: 7; Metonic Cycle: 13;
2353 cf. Appendix Table X.188
2354 Pratt 1991: 301

preferred 33 CE (Julian), his reasoning would probably have been researched sooner because that date now appears to be correct.[2355] The 34 CE (Julian) choice has few supporters as it – according to Pratt – conflicts with the date of Paul's conversion which Newton had apparently not considered, and arguments for a 5th Passover and those based on the ripeness of the corn are not compelling.[2356]

XX. Moreover, Newton's choice of Friday 23 April 34 CE (Julian)[2357] [2358] rather than Thursday 22 April 34 CE (Julian) depended on invoking a postponement rule from the modern Hebrew calendar which Zeitlin (1966) has effectively argued was not used at that time. However, although Newton's arguments for 34 CE (Julian) have not passed the test of time, the basis for his second choice of 33 CE (Julian) is still sound.[2359]

XX. Since Newton, scholarly opinion has favored 30 CE (Julian)[2360] [2361] and is only recently returning to the 33 CE (Julian) date. The principal reason for having preferred 30 CE (Julian) was that [Jesus] Christ was supposedly born about 6 BCE (Julian) [according to tradition] before the death of Herod in 4 BCE (Julian) – which favors 30 CE (Julian) because [Jesus] Christ allegedly lived about 33 years.[2362]

XX. To support this theory, it was conjectured that Luke reckoned the 15th Regnal Year of Tiberius in the 26 – 28 CE (Julian) period rather than 29 CE (Julian). Currently that theory is being challenged for several reasons.[2363]

XX. According to Pratt, the strongest is what Newton implied: Roman history indicates 29 CE (Julian) was Tiberius' 15th year and no substantial evidence to the contrary has surfaced. Moreover, Maier (1968) has pointed out that Pilate's capitulation when accused of not being "Caesar's friend",[2364] and his desire to appease Herod Antipas[2365] require the crucifixion to have been after the death of Sejanus in 31 CE (Julian).[2366]

XX. Newton reasons as follows: Thus there remain only the years 33 CE (Julian) and 34 CE (Julian) to be considered, and the year 33 CE (Julian) Newton excludes by this argument: In the Passover two (2) years before the Passion when Christ went through the corn, and his disciples plucked the ears and

2355 Pratt 1991: 301
2356 Pratt 1991: 301-302
2357 Friday 23 April 34 CE (Julian); Friday 21 April 34 CE (Gregorian); 16 Ayyaru / Iyar 3794 AM; Astronomical Julian Day JD_X: 1733588.5; RD_Y: 1385592; RD_X: 12164; Sabbatical Cycle: 7; Metonic Cycle: 13;
2358 cf. Appendix Table X.188
2359 Pratt 1991: 302
2360 Friday 07 April 30 CE (Julian); Friday 05 April 30 CE (Gregorian); 16 Nisannu / Nisan 3790 AM; Astronomical Julian Day JD_X: 1732111.5; RD_Y: 1384115; RD_X: 10687; Sabbatical Cycle: 3; Metonic Cycle: 9;
2361 cf. Appendix Table X.179
2362 Pratt 1991: 302
2363 Pratt 1991: 302
2364 cf. John 19:12
2365 cf. Luke 23:12
2366 Pratt 1991: 302

rubbed them with their hands to eat, this ripeness of the corn shows that the Passover then fell late. And so did the Passover 14 April 32 CE (Julian)[2367] [2368] but the Passover 28 March 31 CE (Julian) fell very early.[2369] It should be noted that Newton's date of 28 March 31 CE (Julian) needs to be corrected to be 26 March 31 CE (Julian).[2370] [2371]

XX. Newton reckons that it was not therefore two (2) years after year 31 CE (Julian), but [rather] two (2) years after 32 CE (Julian) that Christ suffered.[2372] Thus – according to Newton (1733 CE (Julian)) – all the characters of the Passion agree to the year 34 CE (Julian), and that is the only year to which they all agree.[2373] As has been proven mathematically, Newton miscalculated the date of 14 Nisannu / Nisan for 34 CE.[2374] [2375] [2376]

XX. Humphreys and Waddington (1983) have suggested that reports that "the Moon appeared like blood" at the crucifixion probably referred to the lunar eclipse on Friday 03 April 33 CE (Julian),[2377] [2378] and Humphreys (1990) has refuted Schaefer's conclusion that the eclipse could not have been seen from Jerusalem. Finally, evidence is mounting that Herod died later than 4 BCE (Julian) – perhaps as late as 1 CE (Julian).[2379]

XX. Much of Newton's derivation of the crucifixion date is not only still sound, it was based on reconstructing the Hebrew calendar which is the modern technique. It was only in Newton's final step of preferring 34 CE (Julian) over 33 CE (Julian) that he no longer appears to be correct.[2380]

75.1. Jesus Died on the Eve of Passover (Friday 14 Nisannu / Nisan)

XX. [Various other sources apart from the Gospels such as] the apocryphal "Gospel of Peter" states explicitly that Jesus' death was on the eve of Passover (that is on 14 Nisannu / Nisan) and a Jewish

2367 Monday 14 April 32 CE (Julian); Monday 12 April 32 CE (Gregorian); 14 Nisannu / Nisan 3792 AM; Astronomical Julian Day JD_X: 1732849.5; RD_Y: 1384853; RD_X: 11425; Sabbatical Cycle: 5; Metonic Cycle: 11;
2368 cf. Appendix Table X.183
2369 Newton 1733: 167
2370 Monday 26 March 31 CE (Julian); Monday 24 March 31 CE (Gregorian); 14 Nisannu / Nisan 3791 AM; Astronomical Julian Day JD_X: 1732464.5; RD_Y: 1384468; RD_X: 11040; Sabbatical Cycle: 4; Metonic Cycle: 10;
2371 cf. Appendix Table X.181
2372 Newton 1733: 167-168
2373 Newton 1733: 168
2374 Monday 22 March 34 CE (Julian); Monday 20 March 34 CE (Gregorian); 14 Nisannu / Nisan 3794 AM; Astronomical Julian Day JD_X: 1733556.5; RD_Y: 1385560; RD_X: 12132; Sabbatical Cycle: 7; Metonic Cycle: 13;
2375 Friday 23 April 34 CE (Julian); Friday 21 April 34 CE (Gregorian); 16 Ayyaru / Iyar 3794 AM; Astronomical Julian Day JD_X: 1733588.5; RD_Y: 1385592; RD_X: 12164; Sabbatical Cycle: 7; Metonic Cycle: 13;
2376 cf. Appendix Table X.187, X.188
2377 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2378 cf. Appendix Table X.185
2379 Pratt 1991: 302
2380 Pratt 1991: 303

source, the Babylonian Talmud, records that "on the eve of Passover they hanged Yeshu" – earlier referred to as "Yeshu the Nazarene" (*b. Sanhedrin 43a*; "Jesus" is the Graeco-Roman equivalent of the Semitic "Yeshu").[2381]

XX. Another key piece of information is the day of the week on which he died.[2382] All four (4) gospels agree that Jesus died the day before the Jewish Sabbath. The gospels agree that Jesus died the day before the Sabbath. Hence Jesus died on a Friday.[2383]

XX. Although the Jews started their day at Sunset, they measured their hours of daylight from Sunrise. It is rather like using both a twenty-four-(24)-hour and a twelve-(12)-hour clock. At Passover time (our March or April), Sunrise in Jerusalem occurs at about 6 a.m. So Midday [i.e. approximately Noon (12:00)] is the 6th hour.[2384]

XX. Matthew's gospel tells us that Jesus died at "about the ninth hour".[2385] We can therefore deduce from the Bible that Jesus died at about 3 p.m. [i.e. 15:00.] We now have both the day (Friday) and the time (3 p.m.) [15:00] of the death of Jesus.[2386]

XX. [In fact,] there is a striking unanimity from all sources that the Crucifixion was on 14 Nisannu / Nisan and consequently – according to Humphreys and Waddington – the only two (2) plausible years for the Crucifixion are 30 CE (Julian)[2387] [2388] and 33 CE (Julian).[2389] [2390] [2391]

XX. Since the ministry of Jesus started after that of John the Baptist, the earliest possible date for the commencement of the ministry of Jesus is 28 CE (Julian), and John's Gospel records three (3) different Passovers occurring during his ministry (including the one at the Crucifixion).[2392] Many scholars believe that John omitted mention of a further Passover.[2393]

XX. [If Jesus were 28 (or thereabouts) when he was baptized in year 28 CE (i.e. the 15th Regnal Year of Tiberius)] when he commenced his ministry, then this is consistent with Luke 3:23 that he was "about

2381 Humphreys and Waddington 1992: 339
2382 Humphreys 2011: 20-21
2383 Humphreys 2011: 21
2384 Humphreys 2011: 22
2385 cf. Matthew 27:46
2386 Humphreys 2011: 22
2387 Friday 07 April 30 CE (Julian); Friday 05 April 30 CE (Gregorian); 16 Nisannu / Nisan 3790 AM; Astronomical Julian Day JD_X: 1732111.5; RD_Y: 1384115; RD_X: 10687; Sabbatical Cycle: 3; Metonic Cycle: 9;
2388 cf. Appendix Table X.179
2389 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2390 cf. Appendix Table X.185
2391 Humphreys and Waddington 1992: 339
2392 Humphreys and Waddington 1992: 339
2393 Humphreys and Waddington 1992: 340

thirty" at this time since – according to Humphreys and Waddington – the Greek [expression] translated "about thirty" is an imprecise expression meaning any age between 26 and 34. It should also be remembered that there is no Year 0 between 1 BCE (Julian) and 1 CE (Julian) in the Julian calendar.[2394]

XX. The year 34 CE (Julian) is almost certainly too late for the Crucifixion since – according to Humphreys and Waddington – it would conflict with the probable date of Paul's conversion. We can quite confidently date the later events in Paul's life. Working back from these, and using time intervals given by Paul himself (e.g., three (3) years and 14 years, see Galatians 1:18, 2:1) leads many scholars to infer that Paul's conversion was in 34 CE.[2395]

XX. In addition – according to Humphreys and Waddington – 34 CE (Julian) is only a possibility for the Crucifixion date if the weather was exceptionally severe. There is no positive evidence in favor of 34 CE (Julian) and Humphreys and Waddington therefore rule it out.[2396]

XX. Pilate was appointed procurator of Judea in 26 CE (Julian) by Sejanus, a noted anti-Semite. Tiberius executed Sejanus for sedition in October 31 CE (Julian), and subsequently ordered all governors not to mistreat the Jews. Hence the portrayal of Pilate in the Gospels is consistent with the Crucifixion occurring after October 31 CE (Julian), that is in 33 CE (Julian).[2397] [2398] [2399]

XX. Hence – as reasoned by Humphreys and Waddington – if this evidence is accepted, then 30 CE (Julian)[2400] [2401] cannot be the Crucifixion year, thus leaving 33 CE (Julian)[2402] [2403] as the only possibility.[2404]

75.2. The Date of Jesus' Crucifixion: 03 April 33 (Julian) CE // 01 April 33 CE (Gregorian)

XX. Humphreys and Waddington use two (2) approaches [i.e. methodologies] to dating the Crucifixion: 1.) a process of elimination showing that every year but one (1) is incompatible with the available evidence; 2.) lunar eclipse evidence which positively identifies a particular date. Both

2394 Humphreys and Waddington 1992: 350
2395 Humphreys and Waddington 1992: 338
2396 Humphreys and Waddington 1992: 338
2397 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2398 cf. Appendix Table X.185
2399 Humphreys and Waddington 1992: 340
2400 Friday 07 April 30 CE (Julian); Friday 05 April 30 CE (Gregorian); 16 Nisannu / Nisan 3790 AM; Astronomical Julian Day JD_X: 1732111.5; RD_Y: 1384115; RD_X: 10687; Sabbatical Cycle: 3; Metonic Cycle: 9;
2401 cf. Appendix Table X.179
2402 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2403 cf. Appendix Table X.185
2404 Humphreys and Waddington 1992: 339

approaches yield the same date of 03 April 33 CE (Julian).[2405] [2406] [2407]

XX. The date of the Crucifixion is the key date in the chronology of the life of Jesus since it is directly relevant to the date and nature of the Last Supper, and to the length of his ministry. It is also indirectly relevant to his date of birth. Humphreys and Waddington assert that the only certainty is that Jesus' death occurred during the 10 years that Pontius Pilate was procurator of Judea (26 – 36 CE (Julian)).[2408]

XX. There are advocates for nearly every year in this period. In addition to the unknown year, the day of the execution of Jesus is also uncertain since there appears to be – according to Humphreys and Waddington – a difference of one (1) day between the date given by the Gospel of John and that indicated by the Synoptics.[2409]

XX. There are three (3) main pieces of biblical evidence for dating the Crucifixion:[2410]

1.) Jesus was crucified when Pontius Pilate was procurator of Judea during 26 – 36 CE (Julian) (all four Gospels; also Tacitus, *Ann*. 15:44).[2411]

2.) All four (4) Gospels agree that Jesus died a few hours before the commencement of the Jewish Sabbath, that is, he died before nightfall on a Friday. In addition, the earliest writings that explicitly state the date of the Crucifixion all have it as a Friday.[2412]

3.) All four (4) Gospels agree to within about a day (see below) that the Crucifixion was at the time of Passover.[2413]

XX. The problem that then has to be solved is determining in which of the years 26 – 36 CE the 14th and 15th Nisannu / Nisan fell on a Friday [evening – Humphreys and Waddington group together the 14th and 15th of Nisannu / Nisan into one (1) Friday which is only possible if you mean the conjunction at evening that officially ends day 14 and begins day 15 of Nisannu / Nisan]. As is well known, various authors have attempted to use astronomy to provide a solution to this problem.[2414]

XX. However this is not straightforward since although astronomical calculations can accurately

2405 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2406 cf. Appendix Table X.185
2407 Humphreys and Waddington 1992: 331
2408 Humphreys and Waddington 1992: 332
2409 Humphreys and Waddington 1992: 332
2410 Humphreys and Waddington 1992: 332
2411 Humphreys and Waddington 1992: 332
2412 Humphreys and Waddington 1992: 332
2413 Humphreys and Waddington 1992: 332
2414 Humphreys and Waddington 1992: 334

specify the times of New Moons and Full Moons, we do not know with what skill the Jews of the 1st century CE (Julian) could detect the first faintly glowing lunar crescent following Conjunction with the Sun (i.e. the [actual Conjunction or "No Moon"] itself being invisible of course).[2415]

XX. Although in the 1st century CE (Julian) the beginning of the Jewish lunar month (in the official calendar) was fixed rigorously by astronomical observation, according to Humphreys and Waddington there remain two (2) uncertain calendrical factors: intercalary (or leap) months and the possibility of a cloudy sky rendering the Moon invisible.[2416]

XX. It is worth adding that the meteorology of a desert region – such as around Jerusalem – gives a high proportion of cloudless nights.[2417] Unfortunately we possess no historical reports as to the proclamation of leap-months in the years 26 – 36 CE.[2418] Thus – as noted by Humphreys and Waddington – a knowledge of the Jewish calendar for the time of [Jesus] Christ is a key piece of information for New Testament studies.[2419]

XX. In 1934 CE (Gregorian), J. K. Fotheringham published his influential paper, "The Evidence of Astronomy and Technical Chronology for the Date of the Crucifixion". Fotheringham used a more realistic [theoretical geometric visibility with values of arcus visionis] criterion for when the crescent of the New Moon would first be seen than the thirty-(30)-hour one that had been used previously, and his calculations – according to Humphreys – correctly predicted many observations of New Moons.[2420]

XX. Many biblical scholars were initially enthusiastic about these calculations until it was realized that several New Moons had been recently observed which should not have been seen according to Fotheringham's calculations. The enthusiasm of scholars then understandably waned because the calculations were clearly not totally reliable – yet according to Humphreys they were correct about 90 per cent of the time.[2421]

XX. Using a different approach, C.J. Humphreys (an Oxford materials scientist) and W.J. Waddington (an Oxford astrophysicist) attempted to improve upon the calculations made in 1934 CE (Gregorian) by Fotheringham, and together they submitted their paper to a leading scientific journal, *Nature*.[2422] According to Humphreys, leading experts who peer-reviewed and refereed the paper agreed that calculations of Humphreys and Waddington were significantly more accurate than those of Fotheringham, and hence their paper merited publication in *Nature*.[2423]

2415 Humphreys and Waddington 1992: 334
2416 Humphreys and Waddington 1992: 335
2417 Humphreys and Waddington 1992: 336
2418 Humphreys and Waddington 1992: 336
2419 Humphreys and Waddington 1992: 337
2420 Humphreys 2011: 48
2421 Humphreys 2011: 48
2422 Humphreys 2011: 49
2423 Humphreys 2011: 49-50

XX. Waddington checked the predictions of his calculations against 1,282 recorded observations of New Moons and Humphreys [and Waddington] assert that they gave the correct answer every time, including the occasions on which Fotheringham's [geometrical] calculations of [theoretical lunar crescent Moon visibility with values of arcus visionis] failed to give the right answer. Humphreys asserts that we can therefore have confidence in Waddington's calculations.

XX. The critical questions arise: How – and by what means – are they testing the accuracy of these calculations? On what basis does Humphreys (presumably based on other astronomers' expert opinions since he is a materials scientist) assert that Fotheringham's calculations are correct about 90 percent of the time? – and for that matter that Humphreys and Waddington's calculations are significantly more accurate than Fotheringham's to correctly predict many observations of New Moons?

XX. Waddington's calculated Julian dates for the date of 14 Nisannu / Nisan – based on [theoretical geometrical calculations with values of arcus visionis] earliest possible sighting of the crescent Moon (which gives the date of 01 Nisannu / Nisan) for the period 26 – 36 CE are given in the table.[2424] As is clearly evident, their geometric calculations of theorical lunar crescent Moon visibility according to theorical geometrical criteria with values of arcus visionis do not – apart from one exception – equate to actual dates of 14 Nisannu / Nisan. Indeed – despite Humphrey's assertion that all Waddington's calculations are correct, the only Julian date that Waddington accurately calculated to be 14 Nisannu / Nisan is for year 33 CE (Julian).[2425] [2426]

75.3. The Timeline of the Last Week of Jesus

XX. According to the Gospel of John, six (6) days before the Passover, Jesus came to Bethany where Lazarus was (who had been dead) whom he raised from the dead.[2427] In Bethany, Jesus was anointed with oil.[2428] This particular Sabbath immediately before the Passover (once per year) on which Jesus was anointed is known as the Great Sabbath (*HaShabbat HaGadol*) and is still celebrated today in Israel and among Jews globally. The date of this Great Sabbath in year 33 CE (Julian) works out to be Saturday 28 March 33 CE (Julian).[2429] [2430]

XX. On the next day, Sunday 09 Nisannu / Nisan 3793 AM, many people who had come for the

2424 Humphreys and Waddington 1992: 335; Humphreys 2011: 50
2425 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2426 cf. Appendix Table X.185
2427 cf. John 12: 1-8
2428 cf. Matthew 26:6-13; Mark 14:3-9; John 12: 1-8
2429 Saturday 28 March 33 CE (Julian); Saturday 26 March 33 CE (Gregorian); 08 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733197.5; RD_Y: 1385201; RD_X: 11773; Sabbatical Cycle: 6; Metonic Cycle: 12;
2430 cf. Appendix Table X.185

Passover in Jerusalem later that week (on Friday 14 Nisannu / Nisan 3793 AM that flows and continues into the immediately subsequent Feast of Matzot (Unleavened Bread) on Sabbath/Saturday 15 Nisannu / Nisan 3793 AM), when they heard that Jesus was coming to Jerusalem, they took branches of palm trees and went forth to meet him and cried, "Hosha'na: Blessed is the King of Israel that comes in the name of the LORD."[2431] This day, now known as Palm Sunday, works out to be Sunday 29 March 33 CE (Julian).[2432] [2433]

XX. The next day, Monday, would have been the 10 Nisannu / Nisan 3793 AM that is a very significant date in relation to the Passover in which the Passover sacrifice was selected and guarded for four (4) days from exactly 10 Nisannu / Nisan 3793 AM until its slaughter on 14 Nisannu / Nisan 3793 AM.[2434] This works out to be Monday 30 March 33 CE (Julian).[2435] [2436]

XX. If we put together all the events listed in the gospels as occurring between the last supper and the crucifixion, then we obtain the following list:[2437]

- Last supper

- Mount of Olives

- Gethsemane

- Jesus prayed

- Disciples fell asleep

- Jesus prayed again

- Disciples fell asleep again

- Jesus prayed a third time

- Disciples fell asleep a third time

2431 cf. John 12:12

2432 Sunday 29 March 33 CE (Julian); Sunday 27 March 33 CE (Gregorian); 09 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733198.5; RD_Y: 1385202; RD_X: 11774; Sabbatical Cycle: 6; Metonic Cycle: 12;

2433 cf. Appendix Table X.185

2434 cf. Exodus 12:1-6

2435 Monday 30 March 33 CE (Julian); Monday 28 March 33 CE (Gregorian); 10 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733199.5; RD_Y: 1385203; RD_X: 11775; Sabbatical Cycle: 6; Metonic Cycle: 12;

2436 cf. Appendix Table X.185

2437 Humphreys 2011: 9

- Jesus arrested

- Taken to Annas

- Peter's first denial and first cock crow

- Interrogation by Annas

- Taken to Caiaphas

- Peter's second and third denials

- Cock crows again

- Trial before the Sanhedrin

- Jesus handed over to the guards

- Jesus before the Sanhedrin again

- Trial of Jesus before Pilate

- Trial before Herod

- Trial before Pilate again

- Pilate releases Barabbas

- Jesus is flogged

- Soldiers mock Jesus

- Condemned by Pilate

- Simon carries Jesus' cross

- Crucifixion of Jesus[2438]

XX. Here comes the problem: Biblical scholars and Christians throughout the world [currently] believe

2438 Humphreys 2011: 9

that the last supper started after Sunset on Thursday [13 Nisannu / Nisan 3793 AM] night [which would then technically be the start of day 14 Nisannu / Nisan 3793 AM], and that the crucifixion of Jesus happened the following morning, Friday [14 Nisannu / Nisan 3793 AM], at about 9 a.m. (i.e. the 3rd hour[2439] counting from Sunrise).[2440]

XX. The trials of Jesus before the Sanhedrin, Pilate, and Herod all occurred in different parts of Jerusalem. Scholars have literally rushed around Jerusalem with a stop-watch to see how all the events recorded in the gospels could have fitted between Thursday night [14 Nisannu / Nisan 3793 AM since Hebrew day starts at Sunset] and Friday morning [i.e. still 14 Nisannu / Nisan 3793 AM]. Most conclude that it is impossible.[2441]

XX. According to Humphreys, the biblical scholar F. F. Bruce was not wrong when he called the date and nature of the last supper "the thorniest problem in the New Testament".[2442]

XX. According to the Gospel of Luke, "Then came the day of unleavened bread, when the Passover [sacrifice] must be killed. And he sent Peter and John, saying, 'Go and prepare us the Passover [sacrifice], that we may eat.'"[2443]

XX. It appears that the KJV mistranslated (into the English simple preterite tense) the Greek verb tense here in this verse for "to come", i.e. Greek *erchomai*, Strong's G2064. According to the word root etymology, *erchomai* is the middle voice of a primary verb used only in the present and imperfect tenses.[2444] That is to say, this verse should be translated either as "Then comes..." // "Then is coming..." OR "Then was coming..." Thus the date of Passover was approaching, and still in the future, but had not yet arrived, i.e. it was still before Passover.

XX. Therefore, a better, more accurate translation would be, "Then was coming the day of unleavened bread, when the Passover [sacrifice] must be killed. And he sent Peter and John, saying, 'Go and prepare us the Passover [sacrifice], that we may eat.'"[2445]

XX. Therefore, we should understand this date to be the 09 Nisannu / Nisan 3793 AM (i.e. Palm Sunday) during the day that would become the 10 Nisannu / Nisan 3793 AM that same evening, i.e. the date of the Last Supper – or earlier: perhaps as early as 01 Nisannu / Nisan 3793 AM. Jesus was sending Peter and John and instructing them as was written in the Torah for the Passover sacrifice to be acquired by (and held from) the 10 Nisannu / Nisan 3793 AM and held in guard until the 14 Nisannu /

2439 cf. Mark 15:25
2440 Humphreys 2011: 9
2441 Humphreys 2011: 10
2442 Humphreys 2011: 25
2443 cf. Luke 22:7-8
2444 https://www.blueletterbible.org/lang/lexicon/lexicon.cfm?Strongs=G2064&t=KJV
2445 cf. Luke 22:7-8

Nisan 3793 AM with all of the assembly of congregration of Israel slaughtering the Passover sacrifice at Sunset before evening of 14 Nisannu / Nisan 3793 AM.[2446]

XX. Thus we can understand the Greek of Luke 22:7-13 to mean that Passover was still fast approaching (i.e. coming yet still in the future), and Peter and John were sent to purchase and guard the Passover sacrifice on/by Palm Sunday 09 Nisannu / Nisan 3793 AM so that it would be with them by that evening at Sunset when 10 Nisannu / Nisan 3793 AM began as commanded in Exodus 12:1-6.

XX. Likewise Luke 22:14-20 should be understood as the Last Supper which occurred on Sunday evening 10 Nisannu / Nisan 3793 AM, and consisted of wine (of which Jesus notably does not drink), and bread – this was their humble Last Supper together for Jesus and his disciples on the evening at the end of the day of Palm Sunday, i.e. the 09 Nisannu / Nisan (before Sunset) that became 10 Nisannu / Nisan (commencing at Sunset).

XX. This accords with the explicit placement of the Last Supper in the Gospel of John before the Feast of Passover which began on Friday 14 Nisannu / Nisan 3793 AM at evening thus starting the new day of 15 Nisannu / Nisan 3793 AM (i.e. the Feast of Matzot (Unleavened Bread)).[2447]

XX. The Gospels of Matthew, Mark, and Luke evidently have been sources of confusion and whose meanings have been (up to this point) possibly lost in translation. In Matthew 26:17-25 this "first [day] of unleavened bread" should be understood as referring to the upcoming 1st day of Unleavened Bread, i.e. Passover 14/15 Nisannu / Nisan 3793 AM between Sunset and Evening;[2448] This is clear from the parallel passage of Mark 14:12-16.[2449] Moreover, the verses of Luke 22:7:13 confirm this understanding as referring to the upcoming Passover..[2450]

XX. Indeed in these exact same verses (i.e. Matthew 26:17, Mark 14:12, Luke 22:9) Jesus' disciples ask him, "Where do you want us to prepare for you to eat the Passover?" Logically this question would not be possible if the Feast of Passover (14/15 Nisannu / Nisan 3793 AM) and Feast of Matzot (Unleavened Bread: 15 – 21 Nisannu / Nisan 3793 AM) had already taken place. Furthermore, the reference to killing the Passover in Mark 14:12 should be understood as being an abstract desription of what happens on the Feast of Matzot (Unleavened Bread), and not to be confused as the killing of the Passover sacrifice having already taken place.

XX. Therefore, this "first [day] of unleavened bread" in Matthew 26:17 and Mark 14:12 should be understood just as it is Luke 22:7-13, i.e. during the day of Palm Sunday 09 Nisannu / Nisan 3793 AM, immediately after the Great Sabbath 08 Nisannu / Nisan 3793 AM, speaking of the approaching 10

2446 cf. Exodus 12:1-6
2447 cf. John 13:1-2
2448 cf. Matthew 26:17-25
2449 cf. Mark 14:12-16
2450 cf. Exodus 12:1-6; Luke 22:7-13

Nisannu / Nisan 3793 AM (which began that same day Sunday 09 Nisannu / Nisan 3793 AM at evening) to acquire and guard the Passover sacrifice from 10 Nisannu / Nisan 3793 AM (i.e. that Sunday evening beginning at Sunset) until Friday the 14 Nisannu / Nisan 3793 AM when all of Israel are to slaughter this Passover sacrifice between Sunset and Evening as commanded to do in the Torah of Moses.[2451]

XX. According to the Gospels of Matthew, Mark, and Luke, the disciples did as Jesus had directed them, and they prepared the Passover sacrifice.[2452] Now when Sunday evening of Palm Sunday had come [i.e. 9/10 Nisannu / Nisan 3793 AM], Jesus sat down with the twelve disciples to eat their last supper together[2453] before his arrest later that night (or early morning of) Monday 10 Nisannu / Nisan 3793 AM.

XX. According to the Gospels, after the Last Supper Jesus and his disciples went to the Olive Grove Garden of Gethsemane on the Mount of Olives.[2454] Later that night Jesus was arrested (i.e. late Sunday night after Midnight (00:00) through early Monday morning), i.e. the night and early morning of Monday 10 Nisannu / Nisan 3793 AM.[2455] [2456] [2457]
XX. According to the Gospel of Luke, as soon as it was day (i.e. early morning of Monday 10 Nisannu / Nisan 3793 AM) the elders of the people and the chief priests and the scribes came together to try Jesus.[2458] This should be understood as taking place in the morning of Monday the 10 Nisannu / Nisan 3793 AM[2459] [2460] whose Hebrew day began the previous evening at Sunset on Palm Sunday 09 Nisannu / Nisan 3793 AM.[2461] [2462]

XX. Therefore, the date of the Last Supper should be understood as having taken place the Sunday evening after Sunset, i.e. 10 Nisannu / Nisan 3793 AM that began at Sunset.

XX. According to the Gospels, when morning/day (of Monday 10 Nisannu / Nisan 3793 AM) came,[2463] all the chief priests and elders of the people took counsel against Jesus to put him to death.[2464]

2451 cf. Exodus 12:1-6
2452 cf. Matthew 26:19; Mark 14:16; Luke 22:13
2453 cf. Matthew 26:20-29; Mark 14:17-25; Luke 22:14-38; John 13:1-17:26
2454 cf. Matthew 26:30; Mark 14:26; Luke 22:39; John 18:1
2455 Monday 30 March 33 CE (Julian); Monday 28 March 33 CE (Gregorian); 10 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733199.5; RD_Y: 1385203; RD_X: 11775; Sabbatical Cycle: 6; Metonic Cycle: 12;
2456 cf. Appendix Table X.185
2457 cf. Matthew 26:47-56; Mark 14:43-53; Luke 22:47-54; John 18:1-13
2458 cf. Luke 22:66
2459 Monday 30 March 33 CE (Julian); Monday 28 March 33 CE (Gregorian); 10 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733199.5; RD_Y: 1385203; RD_X: 11775; Sabbatical Cycle: 6; Metonic Cycle: 12;
2460 cf. Appendix Table X.185
2461 Sunday 29 March 33 CE (Julian); Sunday 27 March 33 CE (Gregorian); 09 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733198.5; RD_Y: 1385202; RD_X: 11774; Sabbatical Cycle: 6; Metonic Cycle: 12;
2462 cf. Appendix Table X.185
2463 cf. Matthew 27:1; Mark 15:1; Luke 22:66
2464 cf. Matthew 27:1; Mark 14:55;

XX. According to the Gospel of Mark, immediately in the morning the chief priests held a consultation with the elders and scribes and the whole council; and they bound Jesus, led him away, and delivered him to Pilate.[2465]

XX. According to the Gospel of Matthew, after they had bound Jesus, they delivered him to Pontius Pilate, the governor.[2466]

XX. According to the Gospel of Luke, then [in the morning] the whole multitude of them arose, and led him unto Pilate.[2467]

XX. According to the Gospel of John, then they led Jesus from Caiaphas to the Praetorium, and it was early morning. But they themselves did not go into the Praetorium lest they should be defiled, but that they might eat the Passover.[2468] Then Pilate entered the Praetorium and called Jesus and asked him, "Are you the King of the Jews?"[2469]

XX. Thus all of these gospel accounts of Jesus' first meeting with Pilate should be understood as taking place during the early morning of Monday 10 Nisannu / Nisan 3793 AM.[2470] [2471] The Gospels of Matthew and Mark both only account for this first meeting of Pilate when Jesus remained mostly silent before his accusers.

XX. Only the Gospel of Luke details the sequence of events from:

1.) Pilate's 1st trial of Jesus;[2472]

2.) Herod's trial of Jesus;[2473]

3.) Pilate's 2nd trial of Jesus;[2474]

XX. The Gospel of John seems to combine the two (2) trials of Pilate into one (1) account with the first trial being from John 18:28-33 (i.e. when Jesus was mostly silent), and the second trial from John

2465 cf. Mark 15:1
2466 cf. Matthew 27:2
2467 cf. Luke 23:1
2468 cf. John 18:28
2469 cf. John 18:33
2470 Monday 30 March 33 CE (Julian); Monday 28 March 33 CE (Gregorian); 10 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733199.5; RD_Y: 1385203; RD_X: 11775; Sabbatical Cycle: 6; Metonic Cycle: 12;
2471 cf. Appendix Table X.185
2472 cf. Luke 23:1-7
2473 cf. Luke 23:8-12
2474 cf. Luke 23:13-25

18:34-38 (i.e. when Jesus answered more of Pilate's questions).

XX. At Pilate's first trial of Jesus, he hears accusations against Jesus, and questions him.[2475] This should be understood as taking place from approximately 09:00 a.m. [i.e. the 3rd hour of the day] until approximately Noon (12:00) [i.e. the 6th hour] or even possibly until as late as approximately 15:00 p.m. [i.e. the 9th hour] of Monday 10 Nisannu / Nisan 3793 AM. Pilate then sends Jesus to Herod[2476]

XX. Organizing, and leading, and moving a multitude (i.e. an angry, accusatory mob) of people to and from Pilate's Praetorium, and then conducting this 1st trial of Jesus by Pilate (and then to Herod and back to Pilate) was no quick task, so it likely must have taken some time to take Jesus to Herod – especially if the Pilate's 1st trial of Jesus didn't end until afternoon. Therefore it is reasonable and logical that the Jewish priests and scribes would have held Jesus in custody overnight Monday 10/11 Nisannu / Nisan 3793 AM until the next early morning (Tuesday the 11 Nisannu / Nisan 3793 AM) before delivering Jesus to Herod for trial early the next morning of Tuesday 11 Nisannu / Nisan 3793 AM.[2477] [2478]

XX. According to the Gospel of Luke, Herod received and questioned Jesus. Herod was enthusastic to finally meet Jesus because he had heard so many things about him and hoped to see a miracle done by him[2479] Herod then questioned Jesus with many words.[2480] Then Herod heard the vehement accusations of the chief priests and scribes[2481] against Jesus. Then Herod – with his men of war – mistreated with contempt and mocked Jesus, arrayed him in a gorgeous robe [i.e. likely purple and/or scarlet] and sent him back to Pilate.[2482]

XX. All the events described here of Herod's trial of Jesus logically would take a significant amount of time of no less than a couple of (or several or perhaps many) hours. It is probably not likely that everything described at Herod's trial could have been accomplished in the few hours from afternoon to evening (i.e. after Pilate's 1st trial of Jesus) on Monday 10 Nisannu / Nisan 3793 AM[2483] [2484] – that is of course if the multitude was able to bring Jesus in time before Sunset with enough time for the events described above.

XX. Therefore it is logically more reasonable that Herod received Jesus and the multitude on the early

2475 cf. Luke 23:3
2476 cf. Luke 23:7
2477 Tuesday 31 March 33 CE (Julian); Tuesday 29 March 33 CE (Gregorian); 11 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733200.5; RD_Y: 1385204; RD_X: 11776; Sabbatical Cycle: 6; Metonic Cycle: 12;
2478 cf. Appendix Table X.185
2479 cf. Luke 23:8
2480 cf. Luke 23:9
2481 cf. Luke 23:10
2482 cf. Luke 23:11
2483 Monday 30 March 33 CE (Julian); Monday 28 March 33 CE (Gregorian); 10 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733199.5; RD_Y: 1385203; RD_X: 11775; Sabbatical Cycle: 6; Metonic Cycle: 12;
2484 cf. Appendix Table X.185

morning approximately 09:00 [i.e. 3rd hour] Tuesday 11 Nisannu / Nisan 3793 AM,[2485] [2486] and that by Midday Noon (12:00) [i.e. 6th hour] or afternoon (15:00) [i.e. 9th hour] Herod was done with Jesus, and sent him back on his way to Pilate – i.e Jesus likely left Herod's custody that same day of Tuesday the 11 Nisannu / Nisan 3793 AM.[2487] [2488]

XX. According to the Gospel of Luke, Jesus was sent back to Pilate, and he (Pilate) and Herod that same day were made friends together, for before they were at enmity between themselves.[2489] It is conceivable that Herod's trial of Jesus took the entire day of daylight business hours of Tuesday 11 Nisannu / Nisan 3793 AM,[2490] [2491] and that although Herod sent Jesus back that day, it is possible and plausible (and logically likely) that Jesus again was held overnight in custody of the Jewish leadership and only delivered to Pilate the next day on the early morning of Wednesday 12 Nisannu / Nisan 3793 AM.[2492] [2493]

XX. However, it is possible that Jesus was both sent back to – and received by – Pilate that same day on Tuesday 11 Nisannu / Nisan 3793 AM. Luke 23:11 should be understood (upon first glance) as the uncertain day that Jesus was received back by Pilate (i.e. either Tuesday 11 Nisannu / Nisan 3793 AM[2494] [2495] or Wednesday 12 Nisannu / Nisan 3793 AM).[2496] [2497]

XX. Luke 23:12 should be understood as the definitive day that Jesus arrived back to Pilate in which both he and Herod were made friends. However, Luke 23:12 does not necessarily imply that Jesus arrived back to Pilate's custody that evening of Tuesday 11 Nisannu / Nisan 3793 AM[2498] [2499] (i.e.

2485 Tuesday 31 March 33 CE (Julian); Tuesday 29 March 33 CE (Gregorian); 11 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733200.5; RD_Y: 1385204; RD_X: 11776; Sabbatical Cycle: 6; Metonic Cycle: 12;
2486 cf. Appendix Table X.185
2487 Tuesday 31 March 33 CE (Julian); Tuesday 29 March 33 CE (Gregorian); 11 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733200.5; RD_Y: 1385204; RD_X: 11776; Sabbatical Cycle: 6; Metonic Cycle: 12;
2488 cf. Appendix Table X.185
2489 cf. Luke 23:12
2490 Tuesday 31 March 33 CE (Julian); Tuesday 29 March 33 CE (Gregorian); 11 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733200.5; RD_Y: 1385204; RD_X: 11776; Sabbatical Cycle: 6; Metonic Cycle: 12;
2491 cf. Appendix Table X.185
2492 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2493 cf. Appendix Table X.185
2494 Tuesday 31 March 33 CE (Julian); Tuesday 29 March 33 CE (Gregorian); 11 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733200.5; RD_Y: 1385204; RD_X: 11776; Sabbatical Cycle: 6; Metonic Cycle: 12;
2495 cf. Appendix Table X.185
2496 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2497 cf. Appendix Table X.185
2498 Tuesday 31 March 33 CE (Julian); Tuesday 29 March 33 CE (Gregorian); 11 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733200.5; RD_Y: 1385204; RD_X: 11776; Sabbatical Cycle: 6; Metonic Cycle: 12;
2499 cf. Appendix Table X.185

beginning 12 Nisannu / Nisan 3793 AM at Sunset)[2500] [2501] when Herod sent Jesus back to Pilate – although it is possible since it is not explicitly clear. Therefore it will necessary to deal with both possibilities.

XX. According to the Gospel of Luke, after having received Jesus back from Herod, the same day (i.e. possibly Tuesday 11 Nisannu / Nisan 3793 AM afternoon/evening,[2502] [2503] but also possible Wednesday 12 Nisannu / Nisan 3793 AM early-morning)[2504] [2505] they became friends (i.e. likely Herod sent a letter back to Pilate with Jesus that was well-received by Pilate enough to stir up his personal endearment for Herod).

XX. Upon receipt of Jesus back to his custody, Pilate had to call together the chief priests and the rulers of the people:[2506] Something that would have taken several hours at the very least to organize and send messengers (with either written or oral messages) across the city and perhaps even into the local surrounding local rural regions of Judea to notify the Jewish officials of Pilate's summoning them, and then for these individuals to each organize himself, come, gather, and organize together as a group before Pilate – all of this takes time.

XX. Therefore we can reason that:

EITHER 1.) if Pilate received back Jesus late afternoon to early evening approximately 15:00 – 18:00 [i.e. the 9th hour – 12th hour] of Tuesday 11 Nisannu / Nisan 3793 AM,[2507] [2508] then subsequently Pilate sent out messages that afternoon or evening to the Jewish leadership to come and gather during daylight hours the following day (i.e. Wednesday 12 Nisannu / Nisan 3793 AM).[2509] [2510]

OR alternatively, 2.) if Pilate received Jesus back on early morning 06:00 – 09:00 [i.e. the 1st – 3rd hour] of Wednesday 12 Nisannu / Nisan 3793 AM,[2511] [2512] then he subsequently sent out messages

2500 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2501 cf. Appendix Table X.185
2502 Tuesday 31 March 33 CE (Julian); Tuesday 29 March 33 CE (Gregorian); 11 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733200.5; RD_Y: 1385204; RD_X: 11776; Sabbatical Cycle: 6; Metonic Cycle: 12;
2503 cf. Appendix Table X.185
2504 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2505 cf. Appendix Table X.185
2506 cf. Luke 23:13
2507 Tuesday 31 March 33 CE (Julian); Tuesday 29 March 33 CE (Gregorian); 11 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733200.5; RD_Y: 1385204; RD_X: 11776; Sabbatical Cycle: 6; Metonic Cycle: 12;
2508 cf. Appendix Table X.185
2509 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2510 cf. Appendix Table X.185
2511 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2512 cf. Appendix Table X.185

later that morning or early afternoon that same day to the Jewish leadership to come and gather during daylight hours the following day (i.e. Thursday 13 Nisannu / Nisan 3793 AM).[2513] [2514]

XX. If Jesus arrived back to Pilate on late afternoon or evening of Tuesday 11 Nisannu / Nisan 3793 AM,[2515] [2516] then his actual trial before the summoned Jewish leadership began as early as early morning Wednesday 12 Nisannu / Nisan 3793 AM.[2517] [2518] If this is the case, then Jesus' actual trial before the summoned Jewish leadership could have taken place as early as early morning 06:00 – 09:00 [i.e. 1st – 3rd hour], but also any time as late as later that same day in the afternoon 14:00 – 15:00 [i.e. 8th – 9th hour] on Wednesday 12 Nisannu / Nisan 3793 AM.[2519] [2520]

XX. Alternatively, if Jesus arrived back to Pilate on early morning 06:00 – 09:00 [i.e. 1st hour – 3rd hour] of Wednesday 12 Nisannu / Nisan 3793 AM,[2521] [2522] then Pilate likely only sent out messages to call the chief priests and scribes sometime after Jesus' arrival back from Herod that same day. If this is the case, then Jesus' actual trial before the summoned Jewish leadership could have begun as late as early morning (earlier than about the 6th hour [i.e. approximately Noon (12:00)]) of Thursday 13 Nisannu / Nisan 3793 AM[2523] [2524] since Jesus was presented to the multitude in a crown of thorns and robe after his first round of torture by the Roman soldiers at about the 6th hour [i.e. approximately Noon (12:00)].[2525]

XX. At Pilate's 2nd trial of Jesus (which is only recorded in the Gospels of Luke and John since the Gospels of Matthew and Mark only write of Pilate's 1st trial of Jesus), when Pilate had called together the chief priests and the rulers of the people,[2526] he proclaimed that neither he nor Herod has found any fault in Jesus [worthy of death], and offers to only chastise [i.e. torture] him and release him.[2527]

2513 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2514 cf. Appendix Table X.185
2515 Tuesday 31 March 33 CE (Julian); Tuesday 29 March 33 CE (Gregorian); 11 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733200.5; RD_Y: 1385204; RD_X: 11776; Sabbatical Cycle: 6; Metonic Cycle: 12;
2516 cf. Appendix Table X.185
2517 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2518 cf. Appendix Table X.185
2519 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2520 cf. Appendix Table X.185
2521 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2522 cf. Appendix Table X.185
2523 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2524 cf. Appendix Table X.185
2525 cf. John 19:5, 14
2526 cf. Luke 23:13
2527 cf. Luke 23:14-17

XX. This (i.e. Pilate's 2nd trial of Jesus) should be understood as taking place as early as afternoon on Wednesday 12 Nisannu / Nisan 3793 AM[2528] [2529] or as late as early morning of Thursday 13 Nisannu / Nisan 3793 AM[2530] [2531] (i.e. depending which day Jesus arrived back to Pilate from Herod).

XX. However, the religious establishment with its multitude wanted Jesus dead.[2532] Pilate tried again a 3rd time to save Jesus' life: He asked what evil had Jesus done and proclaimed that he has found no cause worthy of death, and again tried to let it suffice to just chastise and let Jesus go[2533] because he found no fault in Jesus,[2534] and then Jesus came out dressed in the robe and crown of thorns[2535] at about the 6th hour [i.e. approximately Noon (12:00)] on Thursday 13 Nisannu / Nisan 3793 AM.[2536] [2537]

XX. During this time Pilate offered to release Jesus as per custom at the [i.e. before] Passover[2538] Instead, the multitude demanded that Pilate release Barabbas.[2539] After speaking again with Jesus, Pilate again tried to release him,[2540] but after the multitude insisted, Pilate decreed the death sentence for Jesus that they demanded.[2541]

XX. Between verse 25 and the next verse in the Gospel of Luke (23:26) which recounts the episode of Simon's carrying Jesus' cross, a significant sequence of events is missing. The Gospels of Matthew, Mark, and John complete the detail of this sequence of events (that is missing from the Gospel of Luke) completely from this point of 2nd trial before Pilate where the death sentence is pronounced to the time of Simon carrying Jesus' cross in Luke 23:26. Therefore we can understand the events immediately after Luke 23:22-25 as follows:

XX. According to the Gospels of Matthew, Mark, and John, after giving the death sentence that was demanded,[2542] then Pilate therefore took Jesus and scourged [i.e. tortured] him.[2543] That is to say: Then the soldiers of the governor Pilate led him away into the hall called the Praetorium, and they called

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2528 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2529 cf. Appendix Table X.185
2530 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2531 cf. Appendix Table X.185
2532 cf. Luke 23:18-21
2533 cf. Luke 23:22-25
2534 cf. John 19:4
2535 cf. John 19:5
2536 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2537 cf. Appendix Table X.185
2538 cf. Luke 23:20; John 18:39
2539 cf. Luke 23:18; John 18:40
2540 cf. John 19:6-12
2541 cf. Luke 23:22-25; John 19:13-16
2542 cf. Luke 23:22-25
2543 cf. Mark 15:15; John 19:1

together the whole garrison around him.[2544] This should be understood as having taken place either late afternoon Wednesday 12 Nisannu / Nisan 3793 AM[2545] [2546] or early-morning Thursday 13 Nisannu / Nisan 3793 AM.[2547] [2548]

XX. Then they stripped Jesus [naked], and they clothed him with with a scarlet robe[2549] and with purple;[2550] they twisted a crown of thorns, and put it on his head.[2551] They put a reed in his right hand, and they bowed the knee before him and mocked him,[2552] and began to salute him saying, "Hail, King of the Jews!"[2553]

XX. Then they spat on him, and took the reed and struck him on the head.[2554] They struck him on the head with a reed and spat on him; and bowing the knee, they worshipped him.[2555] And they struck him with their hands.[2556]

XX. The Gospel of John tell us that at about the 6th hour [i.e. approximately Noon (12:00)], while still in the midst of being tortured by the garrison of Pilate's Roman soldiers (and still wearing a crown of thorns and purple robe), Jesus is presented by Pilate to the multitude.[2557] Although Pilate tries to release Jesus, the multitude demands that Jesus be crucified.[2558]

XX. This presentation of Jesus to the multitude at the 6th hour [i.e. approximately Noon (12:00)] could not have occurred on Sunday 09 Nisannu / Nisan [i.e. Palm Sunday], Monday 10 Nisannu / Nisan [Trials by Sanhedrin and 1st trial by Pilate], Tuesday 11 Nisannu / Nisan [Trial by Herod], or Friday 14 Nisannu / Nisan [Crucifixion Day eve of Passover]. The only days possible for this 2nd trial by Pilate and condemnation are Wednesday 12 Nisannu / Nisan 3793 AM[2559] [2560] and/or Thursday 13 Nisannu /

2544 cf. Matthew 27:27; Mark 15:16
2545 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2546 cf. Appendix Table X.185
2547 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2548 cf. Appendix Table X.185
2549 cf. Matthew 27:28
2550 cf. Mark 15:17; John 19:2
2551 cf. Matthew 27:29; Mark 15:17; John 19:2
2552 cf. Matthew 27:29
2553 cf. Matthew 27:29; Mark 15:18; John 19:3
2554 cf. Matthew 27:30
2555 cf. Mark 15:19
2556 cf. John 19:3
2557 cf. John 19:5, 14
2558 cf. John 19:6-15
2559 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2560 cf. Appendix Table X.185

Nisan 3793 AM.[2561] [2562]

XX. Reasoning that if Jesus' 2nd trial by Pilate was as early as beginning in the afternoon 14:00 – 15:00 [i.e. 8th – 9th hour] on Wednesday 12 Nisannu / Nisan 3793 AM[2563] [2564] (after having sent out urgent notice and summons to the Jewish leadership earlier that morning upon receipt of Jesus back from Herod after [Tuesday/Wednesday] overnight custody with Jewish leadership), it is simply not possible for Jesus to have been tortured in time to be presented at about the 6th hour [i.e. approximately Noon (12:00)] the same day.[2565]

XX. If Jesus' 2nd trial by Pilate was as early as later in the afternoon 14:00 – 15:00 [i.e. 8th – 9th hour] on Wednesday 12 Nisannu / Nisan 3793 AM,[2566] [2567] then it is more likely that Jesus endured more than 24 hours of torture by the Roman garrison and was presented – as it is written in the Gospel of John – at about the 6th hour [i.e. approximately Noon (12:00)] only the next day on Thursday 13 Nisannu / Nisan.[2568] [2569] [2570]

XX. Using similar reasoning mentioned above for Wednesday 12 Nisannu / Nisan 3793 AM,[2571] [2572] if Jesus' 2nd trial by Pilate began on early morning Thursday 13 Nisannu / Nisan 3793 AM,[2573] [2574] then it is also not likely or plausible for Jesus to have been both tried and tortured in time to be presented at about the 6th hour [i.e. approximately Noon (12:00)] on the same day.[2575]

XX. Thus it is logical to conclude that Pilate's 2nd trial of Jesus began the previous day on Wednesday 12 Nisannu / Nisan 3793 AM,[2576] [2577] and that Jesus was delivered to Pilate's garrison of soldiers the

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2561 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2562 cf. Appendix Table X.185
2563 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2564 cf. Appendix Table X.185
2565 cf. John 19:5, 14
2566 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2567 cf. Appendix Table X.185
2568 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2569 cf. Appendix Table X.185
2570 cf. John 19:5, 14
2571 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2572 cf. Appendix Table X.185
2573 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2574 cf. Appendix Table X.185
2575 cf. John 19:5, 14
2576 Wednesday 01 April 33 CE (Julian); Wednesday 30 March 33 CE (Gregorian); 12 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733201.5; RD_Y: 1385205; RD_X: 11777; Sabbatical Cycle: 6; Metonic Cycle: 12;
2577 cf. Appendix Table X.185

same day to be scourged and otherwise tortured. Then – the next day (i.e. Thursday 13 Nisannu / Nisan 3793 AM)[2578] [2579] – Jesus was presented to the multitude in his crown of thorns and robe at about the 6th hour [i.e. approximately Noon (12:00)] on Thursday 13 Nisannu / Nisan 3793 AM.[2580] [2581] [2582]

XX. Therefore then on Thursday 13 Nisannu / Nisan 3793 AM[2583] [2584] after the 6th hour [i.e. approximately Noon (12:00)], Pilate delivers Jesus back to his garrison of soldiers to be crucified[2585] – and it is logical to reason that Jesus was delivered back to the soldiers to be further tortured, and that some of the garrison would have continued to torture Jesus for another 18 hours through the day, evening, and night while the others made preparations for Jesus' crucifixion and the accompanying military security required to implement the death penalty for Jesus (as well as others) the next day Friday 14 Nisannu / Nisan 3793 AM,[2586] [2587] the eve of Passover in Jerusalem, 33 CE (Julian).

XX. And then when they had mocked him,[2588] they took the [scarlet] robe off him,[2589] they took the purple off him,[2590] put his own clothes on him, and led him out and away to crucify him.[2591]

XX. Most certainly the torture of Jesus (most especially after the pronouncement of the death sentence) by the entire garrison of the governor Pilate's soldiers was not a rushed or quick affair. It would most certainly would have been drawn out to prolong the suffering of Jesus and thereby also prolong the sadistic entertainment of the Roman soldiers.

XX. Therefore this second round of torture (after Pilate's decree of death by crucifixion) should be understood as taking place from afternoon of Thursday the 13 Nisannu / Nisan 3793 AM[2592] [2593] after about the 6th hour [i.e. approximately Noon (12:00)],[2594] during which time Jesus is returned to the

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2578 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2579 cf. Appendix Table X.185
2580 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2581 cf. Appendix Table X.185
2582 cf. John 19:5, 14
2583 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2584 cf. Appendix Table X.185
2585 cf. Matthew 27:26; Mark 15:15; John 19:16
2586 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2587 cf. Appendix Table X.185
2588 cf. Matthew 27:31; Mark 15:20
2589 cf. Matthew 27:31
2590 cf. Mark 15:20
2591 cf. Matthew 27:31; Mark 15:20
2592 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2593 cf. Appendix Table X.185
2594 cf. John 19:5, 14

garrison to be crucified, i.e. another day of pre-crucifixion torture that afternoon, evening, and night of the same day (i.e. Thursday 13 Nisannu / Nisan 3793 AM)[2595] [2596] until the early morning hours of Friday the 14 Nisannu / Nisan 3793 AM[2597] [2598] (which began the previous Thursday evening at Sunset), the eve of Passover.

XX. Therefore we can understand that Jesus was held in prison in poignant symbolism just as it is written in the Torah where the Passover sacrifice is held in guarding [i.e. prison] from 10 Nisannu / Nisan 3793 AM through 14 Nisannu / Nisan 3793 AM with all of Israel slaughtering this Passover sacrifice before Sunset (the end of day 14) and observing the Feast of Unleavened Bread at Sunset/Evening (the beginning of day 15) Nisannu / Nisan 3793 AM.[2599]

XX. This whole series of events during this week before Passover gives new meaning and understanding to the term, "The Crucifixion Week".

75.4. The Timeline of Jesus' Day of Crucifixion and the Moon Turning to Blood

XX. According to the Gospel of Mark, Jesus was crucified beginning in the 3rd hour (i.e. 09:00) [of the day].[2600] The Gospels of Matthew, Mark, and Luke report that from the 6th hour [i.e. Noon (12:00)] there was darkness over all the land unto the 9th hour [i.e. 15:00]. And about the 9th hour [i.e. 15:00], Jesus cried with a loud voice saying, "Eli, Eli, lama sabachthani?" That is to say, "My God, my God, why hast thou forsaken me?"[2601] Therefore we can see that it took approximately six (6) hours for Jesus to die on the cross that day Friday 14 Nisannu / Nisan 33 CE (Julian).[2602] [2603]

XX. Since the occurrence of a solar eclipse at Passover time has long been known to be astronomically impossible [because solar eclipses can only occur at Conjunction, i.e. No Moon, but not possible at Opposition, i.e. Full Moon], but that of a lunar eclipse may well occur,[2604] as is well known, the mechanism by which the Sun was darkened may [very likely] have been – according to Humphreys and Waddington – a khamsin dust storm.[2605]

2595 Thursday 02 April 33 CE (Julian); Thursday 31 March 33 CE (Gregorian); 13 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733202.5; RD_Y: 1385206; RD_X: 11778; Sabbatical Cycle: 6; Metonic Cycle: 12;
2596 cf. Appendix Table X.185
2597 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2598 cf. Appendix Table X.185
2599 cf. Exodus 12:1-6
2600 cf. Mark 15:25
2601 cf. Matt 27:45-46; Mark 15:33-34; Luke 23:44-45
2602 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2603 cf. Appendix Table X.185
2604 Humphreys and Waddington 1992: 348
2605 Humphreys and Waddington 1992: 341

XX. Since the darkened Sun occurred at the Crucifixion,[2606] it is reasonable to suppose that the Moon turned to blood that same evening, "before that great and glorious day" – [i.e. the resurrection]. This interpretation of Acts 2:20 is supported by the New Testament scholar, F.F. Bruce (1952).[2607]

XX. There is some other evidence that on the evening of the day of the Crucifixion the Moon appeared like blood. The so-called "Report of Pilate", a New Testament apocryphal fragment, states "Jesus was delivered to him by Herod, Archelaus, Philip, Annas, Caiphas, and all the people. At his Crucifixion the Sun was darkened; the stars appeared and in all the world people lighted lamps from the sixth hour [i.e. Noon (12:00)] till evening; the Moon appeared like blood".[2608]

XX. The Moon turning to blood is a graphic description of a lunar eclipse. The reason an eclipsed Moon appears blood-red is well known and the effect has been well documented.[2609]

XX. Even though during an eclipse the Moon is geometrically in the Earth's shadow, some sunlight still reaches it by the refraction of light passing through the Earth's atmosphere. The light reaching the Moon is red since scattering by air molecules and very small particles along its long path through the atmosphere preferentially removes the blue end of the spectrum.[2610]

XX. The phrase "Moon turned to blood" has been used by writers and historians to describe lunar eclipses for many centuries, and the expression dates back to at least 300 BCE (Julian). Descriptions of some well-documented ancient eclipses have been compiled by F. K. Ginzel (1899) and [purportedly] matched with his calculated eclipse dates.[2611]

XX. In the medieval European annals compiled by G. H. Pertz there are so many lunar eclipses described by "the Moon turned to blood" that the phrase appears to be used as a standard description.[2612] F. R. Stephenson considers that the prophesy of Joel [i.e. Joel 2:28-32 in Acts 2:14-21] clearly alludes to a lunar eclipse.[2613]

XX. There is therefore strong evidence that when Peter [in Acts 2:14-21], the "Report of Pilate", and Cyril of Alexandria (the orthodox Patriarch of Alexandria in 412 CE (Julian)) refer to the Moon turning to blood on the evening of the Crucifixion, that they were describing a lunar eclipse. It is surprising that this deduction does not appear to have been made before, although F. F. Bruce almost reaches this conclusion. Presumably Bruce and other commentators have not been aware that a blood-red Moon is a

2606 cf. Luke 23:44-45
2607 Humphreys and Waddington 1992: 341
2608 Humphreys and Waddington 1992: 342
2609 Humphreys and Waddington 1992: 343
2610 Humphreys and Waddington 1992: 343
2611 Humphreys and Waddington 1992: 343
2612 Humphreys and Waddington 1992: 343-344
2613 Humphreys and Waddington 1992: 344

well-documented description of a lunar eclipse.[2614]

XX. Humphreys and Waddington have determined the eclipses relevant to their work by the use of J. Meeus and H. Mucke's Canon of Lunar Eclipses (which is – according to Humphreys and Waddington – the most comprehensive data available) as corrected by F. R. Stephenson and L. V. Morrison who used ancient Babylonian eclipse records to refine the calculations to take into account more accurately the effects of long-term changes in the Earth's rate of rotation.[2615]

XX. According to Humphreys and Waddington these calculations agree very well with records of Babylonian and Chinese eclipse observations (the probable error in an eclipse time 2000 years ago is – according to Humphreys and Waddington – typically five (5) minutes).[2616]

XX. [Data from] all lunar eclipses (total and partial) visible from Jerusalem between 26 CE (Julian) and 36 CE (Julian) show that in the period 26 – 36 CE (Julian) there was one (1) – and only one (1) – lunar eclipse at Passover time visible from Jerusalem, namely that of Friday 03 April 33 CE (Julian).[2617]

XX. Humphreys and Waddington state with confidence that this lunar eclipse would have been visible from Jerusalem.[2618] This date is the most probable date for the Crucifixion [not only because of astronomical reasons, but also] deduced independently by Humphreys and Waddington using other data.[2619]

XX. The interpretation of Peter's words [citing Joel 2:28-32 in Acts 2:14-21] in terms of a lunar eclipse is therefore not only astronomically and calendrically possible, but it also allows us with reasonable certainty to specify Friday 03 April 33 CE (Julian)[2620] [2621] as being the date of the Crucifixion. Of course the random probability of a lunar eclipse occurring at Moonrise on a particular date is small.[2622]

XX. According to Humphreys and Waddington, calculations show that the eclipse on 03 April 33 CE (Julian)[2623] [2624] was visible from Jerusalem at Moonrise (all times quoted below are local Jerusalem times as measured by a sundial.) Being below the horizon, the start of the eclipse was invisible from Jerusalem. The Moon rose above the Jerusalem horizon at about 6:20 p.m. ([i.e. 18:20] the start of the

2614 Humphreys and Waddington 1992: 342, 344
2615 Humphreys and Waddington 1992: 344, 346
2616 Humphreys and Waddington 1992: 344
2617 Humphreys and Waddington 1992: 344-345
2618 Humphreys and Waddington 1992: 346
2619 Humphreys and Waddington 1992: 345
2620 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2621 cf. Appendix Table X.185
2622 Humphreys and Waddington 1992: 345
2623 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2624 cf. Appendix Table X.185

Jewish Sabbath and also the start of Passover day in 33 CE (Julian)) with about 20% of its disc in the umbra of the earth's shadow and the remainder in the penumbra.[2625]

XX. However – according to Humphreys and Waddington – the ancients made no distinction between the umbral and penumbral shadows and to a casual observer about 60 per cent of the Moon's disc would have been perceived as being "in eclipse" at Moonrise. The 60 per cent "bite" out of the Moon was positioned close to the top (that is the leading edge) of the Moon [i.e. giving the Moon horns].[2626]

XX. Thus – according to Humphreys and Waddington – most of the visible area of the rising Moon would initially have been seen as fully eclipsed. The perceived eclipse ended at about 7:10 p.m. [i.e. 19:10].[2627]

XX. For the case of the eclipse of 03 April 33 CE (Julian)[2628] [2629] the Moon was just above the horizon and was seen against a relatively bright sky background (the Sun having just set). The most probable color of the rising Moon would be red in the umbral shadow and yellow-orange elsewhere.[2630]

XX. At Moonrise, the initially small yellow-orange region would indicate that the Moon had risen, but with most of its visible area "turned to blood". If in fact a massive dust storm was responsible for darkening the Sun, and this storm subsided at about 3 p.m. [i.e. 15:00] (the darkness is recorded in the Gospels to have lasted from [the 6th hour to the 9th hour, i.e.] Noon [12:00] to 3 p.m. [i.e. 15:00]), then any dust still suspended in the atmosphere would tend to modify the above colors. The nature of such a modification would depend upon the size distribution of the particles.[2631]

XX. The majority of lunar eclipses pass unnoticed, occurring when people are asleep or indoors [either during day or night]. However, this eclipse would probably have been seen by most of the population of Israel since the Jews on Passover Day [i.e. 14/15 Nisannu / Nisan 3793 AM] would be looking for both Sunset and Moonrise in order to commence their Passover meal.[2632]

XX. Instead of seeing the expected Full Paschal Moon rising, they would have initially seen a Moon with a red "bite" removed [i.e. with horns]. The effect would be dramatic: The Moon would appear to grow to Full Moon in the next hour. The crowd on the day of Pentecost[2633] would undoubtedly have understood Peter's words [citing Joel 2:28-32] about the Moon turning to blood as referring to this

2625 Humphreys and Waddington 1992: 346
2626 Humphreys and Waddington 1992: 346
2627 Humphreys and Waddington 1992: 346
2628 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2629 cf. Appendix Table X.185
2630 Humphreys and Waddington 1992: 347
2631 Humphreys and Waddington 1992: 347
2632 Humphreys and Waddington 1992: 347
2633 cf. Acts 2:14-21

eclipse that they had seen [on the evening of Passover 14/15 Nisannu / Nisan] 33 CE (Julian).[2634] [2635] [2636]

XX. If this date was indeed the date of the Crucifixion we would expect some textual reference to this lunar eclipse. Humphreys and Waddington present three (3) such textual references: the Acts of the Apostles [Acts 2:14-21 citing Joel 2:28-32], the "Report of Pilate", and the writings of Cyril of Alexandria. There are therefore strong convergent arguments pointing to 03 April 33 CE (Julian)[2637] [2638] as being the date of the Crucifixion.[2639]

XX. Given the above, the Resurrection of Jesus Christ took place on Sunday 05 April 33 CE (Julian));[2640] [2641]

76.0. The Problems of Chronology of the Ancient Near East

XX. The chronologies of Mesopotamia and Egypt are often cited as a basis for substantiating a long history and prehistory for the development of civilization in the Near East.[2642] The multiple sources available for the reconstruction of the chronological background of ancient Near Eastern history are a major factor distinguishing these cultures from Egypt, where ultimately there is but one single historical thread.[2643]

XX. By contrast, Mesopotamia and the neighboring regions offer a series of at least partially independent sources. There are admittedly important differences in the distribution of these sources over time and space such that phases with abundant material contrast with others which are less satisfying.[2644]

XX. In the ancient Near East, three methods for distinguishing one year from another were used:[2645]

1.) From the start of the Early Dynastic Period in Egypt, c. 3000 BCE (Julian) – and from about

2634 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2635 cf. Appendix Table X.185
2636 Humphreys and Waddington 1992: 347
2637 Friday 03 April 33 CE (Julian); Friday 01 April 33 CE (Gregorian); 14 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733203.5; RD_Y: 1385207; RD_X: 11779; Sabbatical Cycle: 6; Metonic Cycle: 12;
2638 cf. Appendix Table X.185
2639 Humphreys and Waddington 1992: 350
2640 Sunday 05 April 33 CE (Julian); Sunday 03 April 33 CE (Gregorian); 16 Nisannu / Nisan 3793 AM; Astronomical Julian Day JD_X: 1733205.5; RD_Y: 1385209; RD_X: 11781; Sabbatical Cycle: 6; Metonic Cycle: 12;
2641 cf. Appendix Table X.185
2642 Hasel 2004: 6
2643 Klinger 2006: 304
2644 Klinger 2006: 304
2645 Millard 1994: 1

2400 BCE (Julian) in Babylonia – come examples of years named after an important event. This system, or a variation, continued until the end of the Old Kingdom in Egypt, c. 2150 BCE (Julian), and until the end of the First Dynasty of Babylon in the east c. 1595 BCE (Julian).[2646]

2.) Then, in Babylonia, it gave way to a simple numbering of years of each king's reign, and in Egypt, to other cyclical reckonings. When a king died, the remaining months of the current year were usually termed the "accession year" of the next ruler [with] his first year commencing the next new calendrical year [in the Spring].[2647]

3.) The third way of dating was by eponyms. Each year was named after a high officer of state, termed *limu* (or *limmu*) in Akkadian. How and when this method began is uncertain, for although dating by officials is found in some Sumerian texts of the mid-3rd millennium BCE (Julian), it is otherwise confined to Assyria from the 19th to the 7th centuries BCE (Julian). From Assyria, this system of naming years is believed to have passed to Greece in the archonship, and to Rome in the consular dating.[2648]

XX. Applications of all these systems are known from ancient Mesopotamia in dating records of royal campaigns, prestigious building projects, or diplomatic exchanges, but most widely on legal, administrative and business documents that required a date by their very nature. However, neither ancient secretaries [i.e. scribes] nor modern scholars could set documents dated by year names or by eponyms in order without knowledge of the names in correct sequence.[2649]

XX. Accordingly, the scribes drew up lists and some of them (reaching into the 3rd millennium BCE (Julian)) have survived, though incompletely. With year-names, distinction of one from another was relatively easy, while under the eponym system an official might hold the office more than once, or there might be two eponyms bearing the same name, so the men's titles could be added to distinguish between them.[2650]

XX. For certain periods there is simply insufficient material to draw any conclusions, regardless of the number of different threads available. The middle of the 2nd millennium BCE (Julian) is such a period, and this has led to the conundrum of three parallel chronological systems,[2651] each based on the Venus tablet recording year six (6) of Ammisaduqa of Babylon.[2652]

XX. Central in Babylonian astronomical observing practice is the recording of the dates of first

2646 Millard 1994: 1
2647 Millard 1994: 1
2648 Millard 1994: 1
2649 Millard 1994: 1
2650 Millard 1994: 1
2651 i.e. Low, Middle, High (aka Short, Middle, Long) Chronologies;
2652 Klinger 2006: 304

appearance and/or disappearance of stars and planets in the morning or evening twilight sky close to the horizon. The first appearance of a star or planet is sometimes also referred to as first visibility, heliacal rising, or morning first (MF). The oldest known Babylonian planetary observations known so far are those of Venus during the reign of the Old Babylonian king Ammisaduqa (~1650 BCE (Julian)). The oldest known stellar observations are the ones listed in MUL.APIN.[2653]

XX. Collections of observations of first appearance and disappearance of the planets were probably routinely done already early in the 2nd millennium BCE (Julian) and were systematically recorded from about 750 BCE (Julian) onwards in the *Astronomical Diaries* (oldest extant copy dating from 652 BCE (Julian)).[2654] On the basis of this kind of observations planetary periods were derived by the Babylonian astronomers in the 6th and 5th centuries BCE (Julian), and these periods were used in the so-called "Goal-Year" texts to compute future planetary positions.[2655] They also form the basis for the development of the ingenious arithmetical schemes to compute planetary ephemeredes known from the Seleucid era.[2656]

XX. This fix[ed] point [based on the observations of Venus] became a pivot upon which the various chronologies turned with the later periods firmly anchored, and the earlier periods floating freely, the date for the fall of [the First Dynasty of] Babylon being 1595 BCE (Julian) according to the Middle chronology, or 1531 BCE (Julian) according to the Low chronology.[2657] The conquest of Babylon by Murshili I is an important event in the Hittite versions, but cannot be used chronologically. The Babylonian records indicate that it took place at the "time of Samsuditana".[2658]

XX. A standard table with dates for Mesopotamian rulers following the Middle Chronology was prepared by J. A. Brinkman for A. L. Oppenheim and E. Reiner, *Ancient Mesopotamia: Portrait of a Dead Civilization* (Chicago, 1977), [pp.] 335-348.[2659]

XX. The preservation of the very different strands from the beginning of the Ur III period (2111 – 2003 BCE (Julian), Middle Chronology) – with its rich harvest of chronological data – to the end of the Old Babylonian period are so tightly woven that the relative chronology for the history of the first half of the 2nd millennium – according to Klinger – can be viewed as certain. The prospective of either larger hiati or a significant reduction – according to Klinger – can be excluded.[2660]

XX. Even with this Venus tablet there seems to be some confusion about the year since Pannekoek (1961), Klinger (2006), and others note year six (6) of Ammisaduqa while Kugler (1912) and others

2653 De Jong 2007: 108
2654 De Jong 2007: 108
2655 See Hunger and Pingree 1999 – cited by De Jong 2007: 108
2656 De Jong 2007: 108
2657 Klinger 2006: 304-305
2658 Klinger 2006: 305
2659 Klinger 2006: 304
2660 Klinger 2006: 305

specify year eight (8). If chronology has been based on this systemic uncertainty, then how can we trust such a chronological system to be reliable?

XX. When studying the ancient world, scholars use two (2) principal approaches to asses the passage of time. One is relative chronology which aims to establish the order of events as they occurred without any concern for the length of time involved. The other is absolute chronology which assigns events in the past to specific calendar years.[2661]

XX. The relative chronology of the 2nd millennium BCE (Julian) in western Asia is generally straightforward. When it works best it is based on sequences of changing artifact styles anchored to stratigraphic contexts.[2662] It is often on the basis of the archaeological evidence that the arguments are made for the reinterpretation of the biblical chronological data.[2663]

XX. A problem with this type of periodization is that ceramic, architectural, and artistic styles do not necessarily change at the same time as ruling dynasties or predominant metals do. Even so, archaeologists can usually identify artifact types peculiar to the chronological subdivisions in general use, and the system works reasonably well.[2664]

XX. Archaeologists in the Levant and Anatolia usually subdivide the millennium into the Middle Bronze Age (early 2nd millennium BCE (Julian)) [MBA], Late Bronze Age (mid- to late 2nd millennium BCE (Julian)) [LBA], and early Iron Age (ca. 1200 – 1000 BCE (Julian) [IA], with each phase characterized by diagnostic pottery and other material culture types.[2665]

XX. In Mesopotamia where written records are more profuse than in the Levant or Anatolia, archaeologists employ historically defined periods such as Isin-Larsa, Old Babylonian, and Kassite, whose names refer to ruling dynasties discussed in ancient texts.[2666]

76.1. Assyriology and Mesopotamian (Assyrian and Babylonian Chronologies)

XX. Mesopotamian chronology is [a relative chronology that is] conventionally based on texts: king lists, eponym lists, dated documents, synchronisms, royal inscriptions, etc.[2667] At present the relative chronologies of the Old Assyrian period and of the Hammurabi dynasty – together covering about three

2661 Schwartz 2008: 450
2662 Schwartz 2008: 450
2663 Hasel 2004: 6
2664 Schwartz 2008: 450
2665 Schwartz 2008: 450
2666 Schwartz 2008: 450
2667 Hunger 2009 – cited by De Jong 2013

(3) centuries – are fairly well established.[2668] [2669]

XX. Putting this relative chronological framework on an absolute footing has so far turned out to be quite difficult[2670] [because the absolute] chronology of the ancient Near East (2000 – 1500 BCE (Julian)) is not well established.[2671]

XX. Present proposals for the beginning of the reign of Hammurabi range from 1848 BCE (Julian) (Long [High] Chronology),[2672] to 1696 BCE (Julian) (Ultra Short [Low] Chronology),[2673] [thus] corresponding to an uncertainty margin of about 150 years.[2674]

XX. To avoid the confusion created by this uncertainty (and in the absence of anything better) the Assyriological community has adopted the so-called (High) Middle Chronology (Hammurabi [Regnal Year] 1 == 1792 BCE (Julian)) as a working hypothesis. The Middle Chronology has served as a general historical reference frame for the Old Babylonian period over the past half century.[2675]

76.2. Absolute Chronology for Mesopotamia

XX. The absolute dating of relative chronological sequences has traditionally been based on ancient records of astronomical observations like solar and lunar eclipses, observations of heliacal risings and settings of the planets and on calendar information. Until recently astronomical chronology was the only "hard" scientific method available to provide absolute calibrations of historical records. During the last decades dendrochronology and radiocarbon dating have become increasingly important as additional chronological tools for the Ancient Near East.[2676] [2677]

XX. The chronology of Mesopotamia is often spoken of as being "fixed" by astronomical observations. That is when a scribe recorded a king's name in conjunction with a datable natural occurrence such as an eclipse, that date may be referred to as absolute or "exact". Hasel cites Glass and notes that it is important to observe that when attempting to go back further in time than the 1st millennium BCE (Julian), however, one encounters problems for it is [currently] not possible to simply extend the absolute dates of the 1st millennium BCE (Julian) back into the 2nd millennium BCE (Julian).[2678] [2679]

2668 Pruzsinszky 2009 – cited by De Jong 2013
2669 De Jong 2013: 147
2670 De Jong 2013: 147
2671 De Jong and Foertmeyer 2010: 141
2672 Huber et al. 1982 – cited by De Jong 2013
2673 Gasche et al. 1998 – cited by De Jong 2013
2674 De Jong 2013: 147
2675 De Jong 2013: 147
2676 e.g. Manning et al. 2001 – cited by De Jong 2013
2677 De Jong 2013: 147
2678 Glass 1984: 92 – cited by Hasel 2004
2679 Hasel 2004: 7

XX. The 2nd millennium BCE (Julian) dates are decidedly relative. Furthermore, two (2) historical gaps of unknown duration occur between the 24th and 16th centuries BCE (Julian) the former at about 2000 BCE (Julian), the latter about 1600 BCE (Julian) so that specific incidents and entire eras remain unanchored in absolute time.[2680] Before that only approximate dates are possible: time estimations are derived from purely archaeological evidence, from paleographic data... and by all too few radiocarbon dates.[2681] [2682]

76.3. The Venus Tablet from Ashurbanipal's Library

XX. The Assyrians – as rough and warlike conquerors often do – adopted the culture of the conquered and carried on its forms and traditions. The Assyrian pantheon was identical with that of Babylon except that now Ashur was the new god of gods.[2683]

XX. The first hardy [Assyrian] warriors were succeeded by princes who loved and protected the arts and sciences.[2684] It was then that [the last Assyrian king] Ashurbanipal (668 – 626 BCE (Julian)) installed a library in his palace and ordered the old texts from all the ancient sites and temples of Babylonia [that had been conquered by the Assyrians] to be assembled and copied.[2685]

XX. The thousands of clay tablets were deposited in orderly rows, with their titles at the sides, supplemented and explained by catalogs, dictionaries and commentaries, continuously augmented by new archive items, the reports to the king and his correspondence with officials. The more than 13,000 fragments dug up from this site alone and preserved in the British Museum (and the many thousands from other sites) give us a good picture of the customs and ideas, business life and culture, religion and economy of that society, and of its astronomy also.[2686]

XX. [Among the texts found at Ashurbanipal's library] there exists a truly astronomical document which we can say with certainty derives from the first Babylonian dynasty [from the 2nd millennium BCE (Julian)]. It is a text from Ashurbanipal's library and now preserved in the British Museum, with data concerning the planet Venus, and is a copy of earlier texts.[2687]

XX. The discovery of the Venus tablet by Henry Austin Layard, and translated by Rawlinson and Smith

2680 Glass 1984: 92 – cited by Hasel 2004
2681 Knapp 1992: 716 – cited by Hasel 2004
2682 Hasel 2004: 7
2683 Pannekoek 1961: 36
2684 Pannekoek 1961: 36-37
2685 Pannekoek 1961: 37
2686 Pannekoek 1961: 37
2687 Pannekoek 1961: 33

(1870) received considerable attention[2688] for mentioning the year formula for year eight (8) of Ammisaduqa.[2689] In 1911, Kugler succeeded in deciphering its contents.[2690]

XX. The Venus Tablet of Ammisaduqa is part of Enuma Anu Enlil and contains 59 omens that deal with phenomena of the planet Venus and with predictions corresponding with the observed phenomena. Most of the observed phenomena are pairs of last and first visibilities of Venus and most of the omens' apodoses refer to events of a turbulent period (floods, food supply, activities of kings, and wars).[2691] The importance of the VT for Mesopotamian chronology is the possibly dated Venus observation of omen 10, which forms the basis for dating of the tablet's Venus cycles.[2692]

XX. This omen records the disappearance of Venus on the 25th day of the 12th month (Addaru) of the "Year of the golden throne", which is usually identified as the year-name of the 8th year of Ammisaduqa.[2693] In 1975 the VT, now known in eleven (11) exemplars, was reedited by Reiner and Pingree, who warned that due to the long tradition of the text, its data are highly corrupted and therefore highly unreliable for chronological purposes.[2694]

XX. An overview of proposed dates (in chronological order of publication) for Ammisaduqa year 1 according to the VT are as follows:[2695]

Table 76.1. Proposed Dates for Ammisaduqa Year 1 according to the Venus Tablet – from Pruzsinszky 2009: 72

Authors	**Dates**
KUGLER (1912)	between -2060 and -1800: -1976/75
WEIDNER (1914)	-2000
WEIDNER (1917)	-1808/1807
FOTHERINGHAM – LANGDON (1923)	-1920/1919
KUGLER (1924)	-1800/1799
SCHNABEL (1925)	-1920/1919
LANGDON – FOTHERINGHAM –SCHOCH (1928)	-1920/1919
NEUGEBAUER (1929)	rejected the use of the VT for dating the fall of the Babylon I dynasty
SEWELL – SMITH (1940)	-1645/1644 + -1920/1919
SIDERSKY (1940)	-1701/1700
UNGNAD (1940)	-1659/1658

2688 The history is documented by Langdon, Fotheringham and Schoch 1928: 28-44; cf. Reiner and Pingree 1975 – cited by Hasel 2004: 7
2689 Hasel 2004: 7
2690 Pannekoek 1961: 33
2691 Pruzsinszky 2009: 70
2692 Pruzsinszky 2009: 70-71
2693 Pruzsinszky 2009: 71
2694 Pruzsinszky 2009: 71
2695 Pruzsinszky 2009: 72

NEUGEBAUER (1941)	rejected the use of the VT for dating the fall of the Babylon I dynasty
CORNELIUS (1942)	-1581/1580
VAN DER WAERDEN (1943–1965)	-1581/1580
WEIR (1972)	-1645/1644
PINGREE – REINER (1975)	only 8-year cycle can be determined based on omens 1-10; the VT is insufficient to establish dates
HUBER (1982)	-1701/1700 (in combination with attested 30-day months and attested intercalations, statistical analyses)
GURZADYAN, Dating...	-1551/1550 (8-year cycle)

XX. The original of EAE 63 is generally believed to have been written during the reign of Ammisaduqa since the end of the first eight-year cycle has the same year-name of the 8th year of Ammisaduqa. By the time its text had reached the Nineveh library (7th century BCE (Julian)) the VT must have undergone considerable expansion and corruption through extensive recopying in addition to any errors made by the original scribe. The main difficulty in the evaluation of the Venus observation is the number of inconsistencies of disappearance and (re)appearance dates, the duration of invisibility, and the discrepancies within duplicates.[2696]

XX. One part consists of many groups of lines describing the phenomena of the planet (here called *Nin-dar-anna*, 'mistress of the heavens') as astrological omens followed by corresponding predictions. They cannot be observations or astronomical computations because the same time-intervals always recur and the imaginary dates of beginning (regularly alternating between morning and evening phenomena) are always of one (1) month and one (1) day.[2697]

XX. The interesting point with this famous Venus Tablet is that the dates, months, and days are not random, but agree pretty well with dates expected from the natural sequence of the Venus phenomena. It is therefore very likely that the dates are based on observations made at some point, and predictions were added to them. For us, the dates – that is the presumed observations – are the interesting part.[2698]

XX. Apart from two (2) easy-to-correct errors, these dates are what one expects from celestial mechanics. The last date is not followed by an omen, but by a year-name referring to a "golden throne". The year was identified by Kugler with the year eight (8) of Ammisaduqa. This set of dates, after correction, can be considered to reflect observations from the first eight (8) years of Ammisaduqa's reign. Unfortunately, they do not suffice to uniquely identify these years in our calendar.[2699]

XX. Even more unfortunately, however, the remaining dates in the text are far more corrupt. About 1/3

2696 Pruzsinszky 2009: 71
2697 Pannekoek 1961: 33
2698 Hunger 2009: 150
2699 Hunger 2009: 150

of them are astronomically impossible, so one has to emend them. Pingree stated that the first 10 omens, corresponding to an 8-year cycle of Venus phenomena, are a valid negative argument in the chronology debate. In other words, he thought that these data were reliable enough that one could expect the correct chronology to agree with them. However, he considered the rest of the data as hopelessly corrupt, so much so that he doubted their relation to Ammisaduqa's reign.[2700]

XX. The dates of the first appearances and the last appearances of Venus in the intercalated lunar calendar of the Babylonians virtually repeat every 56 (+/- 8) years. In this way then, although we cannot [yet] give unique dates for Ammisaduqa, [scholars believe that] we can exclude all dates other than the triplet cluster of years every 56 (+/- 8) years as utterly impossible. The date to be assigned to this First Dynasty of Babylon is still a matter concerning which scholars have not been able to agree.[2701]

XX. At the moment (1970), scholarly opinion is unwilling to consider more than three (3) such triplets, leading to dates 1650 BCE, 1594 BCE, 1586 BCE, and 1530 BCE (Julian) for the end of the First Dynasty of Babylon. It is no accident that these proposed dates are spaced 56 (+/- 8) years apart. The arguments for these possibilities are rather complicated and involved, and are particularly sensitive to events in the middle of the 2nd millennium BCE (Julian) in the peripheral regions of Northern Syria and Anatolia.[2702]

XX. From the time of king Ammisaduqa, who began his reign 52 years before the end of the First Dynasty of Babylon, the dates of the first and last visibility of Venus as morning star and as evening star are recorded for the 21 years of his rule. Unfortunately, we have these reports in a somewhat corrupted form many hundreds of years after the events.[2703]

XX. The intervals between eastern appearance and disappearance [of Venus] are always eight (8) months and five (5) days; then come three (3) months until the western appearance; again eight (8) months and five (5) days until disappearance; then seven (7) days until appearance in the East. The total is 19 months and 17 days – i.e. the correct Venus period.[2704]

XX. These intervals and dates of appearance and disappearance have been deduced from observational data contained in the remaining part of the text, but among the dates there are many erroneous figures wrongly copied, or incorrect phenomena. It is a curious fact that intervals from erroneously copied values have been used to deduce the mean values applied in the schematic predictions.[2705]

XX. A fault in copying put the long period of invisibility (after the eastern disappearance, which is

2700 Hunger 2009: 150
2701 Sachs 1970: 21
2702 Sachs 1970: 21
2703 Sachs 1970: 21
2704 Pannekoek 1961: 33
2705 Pannekoek 1961: 33

normally about two (2) months) as five (5) months and 16 days and the average became three (3) months. This shows that the man who later handled the materials to insert the omens had so little astronomical knowledge or paid so little attention that he included the grossly deviating value in the average. Since we know the meaning of the numbers, some of the errors in these observational data which comprise 21 years can be corrected.[2706]

XX. Among the misplaced phenomena of Venus is found, in the 8th year, a line reading "year of the golden throne". The same designation is found in the texts of civil contracts of the 8th year of King Ammi-zaduga, the penultimate king of the dynasty [and great-great-grandson of Hammurabi], who reigned for 21 years.[2707]

XX. So the obvious conclusion [that Assyriologists have made] was that the 21 years of the Venus text corresponded exactly to this king's 21 years' reign. This was confirmed by the fact that the years of 13 months in the Venus text corresponded to the civil texts. Since the Venus phenomena for that time can be accurately computed from modern data [using values for Arcus Visionus], this afforded a means of ascertaining the precise dates for that king's reign, and the entire chronology of about 2000 BCE (Julian) was put on a firm basis.[2708]

XX. The method followed by Kugler was to use the statement that in the 6th year,[2709] on the 26th day of the month Arachsamna / Marheshvan.[2710] Venus had disappeared in the West and, on the 3rd day of the next month Kisilimu / Kislev,[2711] [Venus] had reappeared in the East. Hence its conjunction with the Sun nearly coincided with the conjunction of the Moon with the Sun in a season roughly corresponding to December or January [of the Julian / Gregorian calendars].[2712]

XX. Kugler found that these conditions were best fulfilled by 23 January 1971 BCE (Julian). From this it would follow that the first Babylonian dynasty reigned from 2225 to 1926 BCE (Julian) and that Hammurabi reigned from 2123 to 2081 BCE (Julian). The mean date of 01 Nisannu / Nisan[2713] thus corresponded to 26 April of our [Julian / Gregorian] calendar.[2714] [2715]

XX. Such [so-called] precise identification of such ancient dates was generally recognized as an admirable example of astronomical-chronological research, but it is interesting that the first result was

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2706 Pannekoek 1961: 33
2707 Pannekoek 1961: 34
2708 Pannekoek 1961: 34
2709 Does Pannekoek incorrectly write here 6th year when he intends 8th year?
2710 i.e. the 8th month, Hebrew: Heshvan/Marheshvan
2711 i.e. the 9th month
2712 Pannekoek 1961: 34
2713 i.e. the 1st month
2714 What does he mean: the "mean date" corresponds to…? The "average" date of 01 Nisannu / Nisan? An average date is not a precise accurate date!
2715 Pannekoek 1961: 34

entirely erroneous. There are several dates in accordance with the data.[2716]

XX. The [Inferior and Superior] Conjunctions of Venus with the Sun return every eight (8) years, but 2.4 days earlier. Those of the Moon with the Sun return after eight (8) years, but 1.6 days late [i.e. a total difference of 4.0 days every eight (8) years].[2717]

XX. Hence two (2) dates eight (8) years apart may sufficiently conform to the phenomena, after which the Conjunctions move increasingly apart, the Venus [Inferior and Superior] Conjunctions coming [2.4 days] earlier, the Moon Conjunctions [1.6 days] later [i.e. a total difference of 4.0 days every eight (8) years]. After seven (7) periods of eight (8) years, the Venus Conjunctions come 17 days earlier, and the Moon Conjunctions eleven (11) days later, so that a coincidence of the Conjunctions again occurs [i.e. a total difference of 28.0 days after 56 years], but now 29½ days earlier[2718] [which equates to a time of ~1.5 days when Old Moon would become No Moon in Lunar Conjunction sometime during this ~1.5-day period before becoming New Moon at day 30 (for 29-day months) or day 31 (for 30-day months)].

XX. Hence we find a series of dates (or pairs of dates [approximately] eight (8) years apart) [for coinciding Conjunctions of both Venus and the Moon relative to the Earth] following one another at intervals of 56 [years minus 28.0 days] or 64 years [minus 32.0 days], and the astronomer depends entirely on the historian to indicate the right century.[2719]

XX. When Kugler made his investigations, historians agreed to fix the First Babylonian Dynasty at about 2000 BCE (Julian). Other investigators who tried to correct Kugler's work did not dare to put the events more than 120 years later. However there existed an old chronicle of kings by Berossus,[2720] a Babylonian priest teaching in Greece, which placed the [First] Dynasty [of Babylon] four (4) centuries later [i.e. ca. 1600 BCE (Julian)].[2721]

XX. This had always been rejected by historians, but recently they reversed their opinion.[2722] So now – according to Pannekoek – it is deemed most probable that the date of coinciding Conjunctions [of both Venus and the Moon relative to the Earth] was 25 December 1641 BCE (Julian), that Hammurabi reigned from 1792 to 1750 BCE (Julian), and that the entire dynasty ruled from 1894 to 1595 BCE (Julian) or perhaps 64 years later, for there are still uncertainties in the historical documents.[2723]

XX. However this may be, it appears that the planet Venus was already being observed with special attention during the first centuries of Babylonian power – perhaps even earlier. Had other planets been

2716 Pannekoek 1961: 34
2717 Pannekoek 1961: 34
2718 Pannekoek 1961: 34
2719 Pannekoek 1961: 34
2720 c. 3rd century BCE (Julian)
2721 Pannekoek 1961: 34
2722 Pannekoek 1961: 34
2723 Pannekoek 1961: 34-35

observed with the same care, references to such observations would certainly have been preserved in Assyrian copies [at the Library of Ashurbanipal], but Venus alone is mentioned.[2724]

XX. It is quite clear that the priest[-astronomers], watching for the Moon's crescent, were struck by this most brilliant of stars [i.e. Venus] and that it must have appeared to them as an exceptional luminary.[2725] It does not seem impossible that in the clear atmosphere of these lands the horns of the Venus crescent were perceived [since] modern observers also have mentioned such instances.[2726] In later texts [the] Sun, Moon, and Venus are often named together as a triad of related deities, distinguished from the other four (4) planets.[2727]

XX. Thus it can be still better understood that in these ancient times the Babylonian priest-astronomers devoted special attention to Ishtar [i.e. Venus] as a sister-star of the Moon. Not only did they watch its appearances and disappearances with religious zest, but they were struck by its regularities and tried – though in a primitive manner – to find the periods and use them for prediction. However this may have been the work of later centuries of political depression, e.g. between 1500 and 1000 BCE (Julian).[2728]

XX. That [Venus] tablet has become one of the key pieces of evidence to fix the chronology of the early 2nd millennium BCE (Julian) by astronomical means.[2729] [2730] One reason that Mesopotamian historical chronology during the 2nd and 3rd millennium BCE (Julian) has faced increasing complexity concerns the cycle in which astronomical movement occurs.[2731] Since the planetary movements of Venus occur in cycles of 56 or 64 years,[2732] they offer a series of possible dates rather than an agreed upon single date.[2733] [2734]

XX. Kugler (1912) concluded that the following years were candidates for Ammisaduqa Year 1: -2040, -1976, -1856. He settled for a middle chronology date of -1976, but later on preferred an -1800 solution.[2735] The next year five (5) dates (-1976, -1920, -1856, -1808, -1800) were considered by Langdon, Fotheringham and Schoch before they settled on -1920.[2736] [2737]

XX. Hence, experts differentiate between three (3) and up to five (5) distinctive chronologies for the 2nd

2724 Pannekoek 1961:35
2725 Pannekoek 1961:35
2726 Pannekoek 1961: 35
2727 Pannekoek 1961:35
2728 Pannekoek 1961:35
2729 Poebel 1942; Rowton 1960; Huber 1987: 5 – cited by Hasel 2004
2730 Hasel 2004: 7
2731 Hasel 2004: 7
2732 Ungnad 1940 – cited by Hasel 2004
2733 Knapp 1992: 716 – cited by Hasel 2004
2734 Hasel 2004: 7-8
2735 Kugler 1924: 622-627; Huber 1987: 5 – cited by Hasel 2004
2736 Langdon, Fotheringham, and Schoch 1928: 61-62; Huber 1987: 6 – cited by Hasel 2004
2737 Hasel 2004: 8

millennium BCE (Julian): the ultra-high,[2738] the high,[2739] the middle,[2740] the low,[2741] and the ultra-low.[2742] These competing chronologies allow for up to 226 years of flexibility.[2743]

XX. [Therefore, understandably] the credibility of the Venus text and its observations has been seriously questioned in recent years.[2744]

Table 76.2. Proposed Mesopotamian Chronologies for Hammurabi's Reign – adapted from Hasel 2004: 8

Mesopotamian Chronologies	**Hammurabi's Reign**
Ultra-High Chronology	1930 – 1888 BCE (Julian)
High Chronology	1848 – 1806 BCE (Julian)
Middle Chronology	1792 – 1750 BCE (Julian)
Low Chronology	1728 – 1686 BCE (Julian)
Ultra-Low Chronology	1704 – 1662 BCE (Julian)

XX. In the case of absolute chronology, [current scholarship is] relatively confident in addressing the last four (4) centuries of the 2nd millenium BCE (Julian). This period is well dated because the record of a solar eclipse in 763 BCE (Julian) is anchored to a list of Assyrian eponyms, i.e. officials whose names were used to designate individual years,.[2745] Specifically, the inscription for this solar eclipse is dated to the eponym year of Būr-saggilê which in turn is reckoned to be equivalent with the 9th or 10th year of the Assyrian king Aššur-Dan III. Precise astronomical calculations have determined this solar eclipse to have occurred – according to Hasel – on 14 / 15 June 763 BCE (Julian).[2746] [2747] [2748] [2749]

XX. The lists of eponyms and kings kept by the Assyrians and Babylonians allow us to date events back from the 1st millennium BCE (Julian) to the 14th century BCE (Julian).[2750] Because of the existence of eponym lists which provide a complete sequence of events in the reigns of various Assyrian kings, it

2738 Landsberger 1954 – cited by Hasel 2004
2739 Goetze 1951; 1957; Thureau-Dangin 1942; Huber 1982; 1987 – cited by Hasel 2004
2740 Kugler 1912; S. Smith 1940; 1945; Rowton 1962 – cited by Hasel 2004
2741 Albright 1956; Cornelius 1956; cf. Aström 1987 – cited by Hasel 2004
2742 Weidner 1945-51 – cited by Hasel 2004
2743 Hasel 2004: 8
2744 Schwartz 2008: 450-451
2745 Schwartz 2008: 450
2746 Monday 15 June 763 BCE (Julian); Monday 07 June 763 BCE (Gregorian); 30 Simannu / Sivan 2998 AM; Astronomical Julian Day JD_X: 1442902.5; RD_Y: 1094906; RD_X: -278522; Sabbatical Cycle: 2; Metonic Cycle: 15;
2747 Tuesday 16 June 763 BCE (Julian); Tuesday 08 June 763 BCE (Gregorian); 01 Du'uzu / Tammuz 2998 AM; Astronomical Julian Day JD_X: 1442903.5; RD_Y: 1094907; RD_X: -278521; Sabbatical Cycle: 2; Metonic Cycle: 15;
2748 cf. Appendix Table X.2, X.194
2749 Hasel 2004: 7
2750 Schwartz 2008: 450

is [theoretically] possible to derive from this one (1) fixed date [of solar eclipse of 14 / 15 June 763 BCE (Julian) – if it is the correct 9th or 10th year of the Assyrian king Aššur-Dan III which may need to be revised] [2751] [2752] [2753] an absolute chronology extending back to 910 BCE (Julian).[2754]

XX. Mesopotamian chronology in the 14th century BCE (Julian) can also be linked to adjacent regions based on the diplomatic archives discovered at Tell el-Amarna, the capital of the "heretic" [Egyptian] pharaoh Akhenaten, which reveal that Akhenaten was a contemporary of Tushratta, king of Mitanni in Upper Mesopotamia, as well as Suppiluliuma I, king of the Hittites in Anatolia, and the Kassite king Burnaburiash II of Babylon.[2755]

XX. By contrast, establishing an absolute chronology for the period before the 14th century BCE [for Mesopotamia] has been a frustrating process marked by controversy and disagreement. Most problematic is the paucity of written records and archaeological evidence for the epoch between the downfall of the First Dynasty of Babylon and the appearance of the subsequent Kassite dynasty.[2756]

XX. For historians of ancient Mesopotamia, the dating of the First Dynasty of Babylon is a very important problem. Not only is the well-documented relative chronology of the First Dynasty of Babylon – which is 300 years long – unhinged over a period of at least 120 years, but an additional relative chronology of more than 500 years preceding the First Dynasty of Babylon could be translated into absolute chronology if one were able to date the First Dynasty of Babylon.[2757]

XX. The absence of documentation for this so-called Dark Age (whose length is uncertain), prevents us from connecting the well-dated events of the 14th century BCE (Julian) to the chronology of the earlier 2nd millennium BCE (Julian).[2758] The historical, chronological reckoning of sequences in the period prior to 1450 BCE (Julian) is based primarily on the Assyrian King Lists[2759] and that the Mesopotamian chronology hinges on Assyrian records for its chronological framework.[2760] Furthermore, the Akkadian King List remains the essential source of historical reconstruction for the 3rd millennium BCE (Julian).[2761]

XX. Although Hittite records in Anatolia span this poorly understood era, they add little to the

2751 Monday 15 June 763 BCE (Julian); Monday 07 June 763 BCE (Gregorian); 30 Simannu / Sivan 2998 AM; Astronomical Julian Day JD_X: 1442902.5; RD_Y: 1094906; RD_X: -278522; Sabbatical Cycle: 2; Metonic Cycle: 15;
2752 Tuesday 16 June 763 BCE (Julian); Tuesday 08 June 763 BCE (Gregorian); 01 Du'uzu / Tammuz 2998 AM; Astronomical Julian Day JD_X: 1442903.5; RD_Y: 1094907; RD_X: -278521; Sabbatical Cycle: 2; Metonic Cycle: 15;
2753 cf. Appendix Table X.2, X.194
2754 Hasel 2004: 7
2755 Schwartz 2008: 450
2756 Schwartz 2008: 450
2757 Sachs 1970: 21
2758 Schwartz 2008: 450
2759 Poebel 1942; Weidner 1926; 1945-51; Gelb 1954 – cited by Hasel 2004
2760 Knapp 1992; Grayson 1992a; 1992b – cited by Hasel 2004
2761 Hasel 2004: 11

chronological discussion because the Hittite scribes failed to use a system of chronological reckoning such as eponyms or year names. Complicating matters, Hittite kings often adopted the names of their predecessors, and the scribes neglected to distinguish one from the other.[2762]

76.4. Synchronisms in Mesopotamia

XX. Partly ameliorating the problem are texts from periods of increased international contact which demonstrate that a ruler in one area lived at the same time as a ruler in another area. For example, we know that in the earlier part of the [2nd] millennium BCE (Julian), Zimri-Lim, king of Mari on the Syrian Euphrates, was a contemporary of Hammurabi in the First Dynasty of Babylon, while a different Hammurabi ruled the powerful Yamhad kingdom in northern Syria. In the middle of the [2nd] millennium BCE (Julian) Mursili I, king of the Hittites, conquered both Yamhad and the First Dynasty of Babylon.[2763]

XX. The challenge is to relate these synchronisms and their associated regional chronologies with absolute dates. To do this, earlier generations of scholarship relied on observations of the planet Venus recorded in a group of Mesopotamian omen tablets called the Enuma Anu Enlil series, which dated to the reign of Ammi-saduqa of the First Dynasty of Babylon.[2764]

XX. These observations were related to a celestial cycle that repeats every sixty-four (64) years, resulting in the proposal of three (3) possible absolute chronologies, each associated with a different turn of the cycle (“high”, “middle”, “low”). The majority of scholars adopted the Middle Chronology, dating Hammurabi of Babylon’s reign to 1792 – 1750 BCE (Julian) and the fall of his dynasty to 1595 BCE (Julian). This remains the chronological scheme in common use.[2765]

76.5. The Venus Tablet of Ammī-Ṣaduqa within Enūma Anu Enlil

XX. Venus is one of the two (2) inner planets in the solar system. Inner planets have orbits around the Sun interior to that of the Earth. As a consequence the elongation of Venus from the Sun never exceeds 47° and Venus may appear on both sides of the Sun as evening and as morning star.[2766]

XX. A much debated anchor point for this chronology is provided by a series of observed dates of the first appearance and the disappearance of the planet Venus during the reign of the Babylonian king Ammisaduqa. The Venus observations allow four (4) different chronologies: Ammisaduqa Year 1 ==

2762 Schwartz 2008: 450
2763 Schwartz 2008: 450
2764 Schwartz 2008: 450
2765 Schwartz 2008: 450
2766 De Jong and Foertmeyer 2010: 143

1702 BCE (Julian), the Long Chronology; 1646 BCE (Julian), the High Chronology; 1638 BCE (Julian), the Low Middle Chronology; and 1582 BCE (Julian), the Short Chronology.[2767] Other independent historical and/or archaeological evidence is required to choose among these chronologies.[2768]

XX. All previous studies of the Venus observations of Ammisaduqa have been based on the use of the well-known concept of the arcus visionis, the distance in degrees between Venus and the Sun measured perpendicular to the horizon on the date of first and last appearance of Venus. At the present day the most comprehensive study of this kind is still the one by Huber et al. (1982). It is based on the authoritative text edition of the Venus tablet by Reiner and Pingree (1975) and it makes use of robust statistical techniques to analyse the data.[2769]

XX. In 1912 CE (Gregorian), the German Jesuit Franz Xaver Kugler published his discovery that the year formula of year eight (8) of the Old Babylonian king Ammī-Ṣaduqa appeared in a list of observations of the planet Venus in the omen series Enūma Anu Enlil. Ever since that time the Venus observations of Ammī-Ṣaduqa have taken a central position in all attempts to put the Old Babylonian chronology on an absolute footing.[2770]

XX. Since then many authors have tried to use these observations to determine a chronology for the ancient Near East by fitting them to computed dates of first appearances and disappearances of Venus.[2771] In spite of all efforts the matter remains undecided and controversial.[2772]

XX. Over the past century there has been a continuous debate among specialists to try to settle this chronological question. Important landmarks in the discussion were provided by the new edition of tablet 63 of Enuma Anu Enlil by Reiner and Pingree (1975) and the thorough reanalysis of Huber et al. (1982) of the Venus observations. Huber et al. confirmed and considerably strengthened the original results of Van der Waerden, and based on an extensive statistical analysis of the Venus data and of Old Babylonian month lengths they made a strong case for the Long Chronology.[2773]

XX. The omen series Enūma Anu Enlil consists of 70 tablets, copied and in use for celestial divination during two (2) millennia of Mesopotamian cultural activity. Tablet 63 contains dates in the Babylonian lunar calendar of observations of Venus and associated omina.[2774] In omen no. 10 reference is made to "The year of the Golden Throne", known to be the 8th year of the reign of king Ammī-Ṣaduqa, great-great grandson of the famous Babylonian king Hammurabi. Venus is named in the text as Nin-si$_4$-an-na,

2767 De Jong and Foertmeyer 2010: 141
2768 De Jong and Foertmeyer 2010: 142
2769 De Jong and Foertmeyer 2010: 144
2770 De Jong 2013: 148
2771 De Jong and Foertmeyer 2010: 142
2772 De Jong and Foertmeyer 2010: 142
2773 De Jong and Foertmeyer 2010: 142
2774 i.e. omens

the name current in the late Sumerian period and in the time of the Old Babylonian dynasty (2000 – 1600 BCE (Julian)).[2775]

XX. Over the past century there have been many attempts to astronomically date the reign of Ammī-Ṣaduqa by selecting a "best data set" from the text and fitting that to computed Venus dates. The most authoritative study is that by Huber et al. (1982) who find that there are four (4) possible Venus chronologies: Ammī-Ṣaduqa [Year] 1 == A.) 1702 BCE (Julian), "Long Chronology"; B./C.) 1646/1638 BCE (Julian), "Middle Chronologies"; or D.) 1582 BCE (Julian), "Short Chronology".[2776]

XX. Already more than 70 years ago Neugebauer (1941) had pointed out that the Venus observations – while providing a number of possible candidate chronologies – are by themselves not sufficiently discriminating to make a definitive choice. Other historical, archeological and/or astronomical constraints are required to choose between the different possible chronologies.[2777]

XX. Huber et al. (1982) expressed preference for the Long Chronology (1702 BCE (Julian)) on the basis of a best fit to (independent) Old Babylonian month length data. Using archeological evidence largely based on pottery sequences Gasche et al. (1998) have argued that the Old Babylonian chronology has to be shortened considerably compared to the traditional Middle Chronology. They proposed a date for Ammī-Ṣaduqa year 1 of 1550 BCE (Julian), implying a date of 1696 BCE (Julian) for the beginning of the reign of Hammurabi (i.e. New Chronology, also referred to as Ultra Short Chronology).[2778]

76.6. The Venus Tablet Data is Problematic for Chronology

XX. Reiner and Pingree rightly warn about the risks involved in the use of certain parts of the Venus tablet for dating.[2779] As noted by Huber, even if the Venus Tablet almost certainly has a genuine observational kernel, it may or may not record observations from the [actual] time of Ammisaduqa.[2780]

XX. However, Huber asserts that if anybody is qualified to judge such matters, then he is [qualified] since he has made his career as a statistician by dealing with contaminated and otherwise corrupted data, and with data analysis.[2781] Huber has come to a much more favorable assessment of the situation than Reiner and Pingree.[2782]

2775 De Jong 2013: 148
2776 De Jong 2013: 149
2777 De Jong 2013: 149
2778 De Jong 2013: 149
2779 Reiner and Pingree 1975: 23 – cited by Huber 2000
2780 Huber 2000: 9
2781 cf. Huber (1981) – cited by Huber 2000
2782 Huber 2000: 8

XX. To follow an orderly judicial procedure, we need at least a second, independent source of evidence, confirming the attribution (which was based on the name of Ammisaduqa's year eight (8), "Year of the Golden Throne", occurring in the corrupt passage between years eight (8) and nine (9) of the tablet). According to Huber, the Old Babylonian (OB) month-lengths permit such an independent confirmation[2783] as he asserts that the month-lengths now furnish a second, independent source of evidence, reinforcing that conclusion by allotting very low probabilities to these [lower] chronologies.[2784]

XX. However, Huber admits that clearly the Venus tablet data are badly contaminated, and one must be careful. Fortunately, a fairly large fraction (about 20%) of the data can be recognized as being corrupted without prejudging the chronology, and robust methods should be fully able to handle the rest. In particular, there can be hardly any doubt – Huber asserts – that the omens traditionally attributed to the Years 1-17 are consecutive.[2785]

XX. Those of the years 1-8 and 9-17 show an exactly parallel statistical behavior across different chronologies – that is they show the behavior to be expected from consecutive data,[2786] and it would be exceedingly difficult to find a different home for the second half of them without destroying this parallelism.[2787]

XX. From the number of internal inconsistencies (between dates of disappearance and reappearance and the stated duration of invisibility), of discrepancies between duplicates and of astronomical impossibilities, Huber once guessed that at least 20% – maybe 40% – of the numbers contained in that text are grossly wrong, or as he prefers to express it: that between 60% and 80% appear to be in order. They must have been corrupted by copying errors during the long period between the creation of the original text and the writing of the surviving manuscripts [e.g. that were found in the library of Ashurbanibal which was a library of copies of more ancient manuscripts].[2788]

XX. Many of the corrupted passages [in the Venus Tablet] are signaled by impossible durations of invisibility. Since several (but not all) of the impossible durations are consistent with the dates of disappearance and reappearence given in the text, Huber concludes that the Venus tablet goes back to an original text containing dates, including year names, but no durations of invisibility/visibility, and that the latter were added only after the text already had considerably deteriorated.[2789]

XX. The preliminary screening [of Venus dates] is primarily based on astronomically impossible

2783 Huber 2000: 9
2784 Huber 2000: 10
2785 Huber 2000: 8
2786 cf. Table 4.3a of Huber et al. (1982)
2787 Huber 2000: 8
2788 Huber 2000: 3
2789 Huber 2000: 3

durations of invisibility: at Inferior Conjunction (Venus between Sun and Earth) the expected duration ranges from about 1 to 19 days; at Superior Conjunction (Venus behind Sun), from about 55 to 75 days.[2790] [2791]

XX. To summarize the process of purging and emending, Huber notes that the text originally covered 13 synodic periods of Venus, or 52 individual dates of disappearance and appearance. Huber disregards the conjectural restoration of year 18; this leaves us with 50 dates. Among them, Huber has marked twelve (12) as probably corrupt (without making any chronological assumptions). Thus Huber concludes that 38 usable dates remain (of them, 4 are fairly secure conjectural emendations – again made without making any assumptions about the chronology)[2792]

XX. According to Huber, the data shows that for Ammisaduqa year 1 == -1549 (the Gasche-Gurzadyan chronology) [i.e. 1550 BCE (Julian)] there is a systematic offset of about half a lunar month between calculations and observations. For example, Venus events that are calculated to occur just after New Moon, in reality are – according to Huber – observed just after Full Moon, and so on (this argument clearly is independent of the intercalations).[2793]

XX. Huber reasons that since Venus events shift about four (4) days in the lunar calendar for each eight-(8)-year Venus period, one would have to move by four (4) Venus periods (or 32 years [and 16 days]), either forward to the Supershort chronology -1517 [i.e. 1518 BCE (Julian)] or backward to the Short Chronology -1581 [i.e. 1582 BCE (Julian)], in order to get the events in step with the lunar month.[2794]

76.7. Huber's Long (High) Chronology

XX. Huber gave a renewed evaluation of chronologies, partly in response to Gasche et al. (1998)[2795] He considered five (5) types of astronomical data: According to Huber, the currently available astronomical evidence relevant for the absolute chronology of the late 3rd and early 2nd millennium BCE (Julian) has five (5) independent parts:[2796] [2797]

A.) Old Babylonian (OB) Venus data; The Venus Tablet;

B.) Old Babylonian (OB) month-lengths;

2790 For more details, see Huber et al. (1982) – cited by Huber 2000
2791 Huber 2000: 3
2792 Huber 2000: 4
2793 Huber 2000: 5
2794 Huber 2000: 5
2795 Hunger 2009: 150
2796 Huber 2000: 1
2797 Hunger 2009: 151

C.) Ur III month-lengths;

D.) Ur III eclipses; Ur III lunar eclipses;

E.) Akkad eclipses; Lunar eclipses of the Akkadian dynasty.

XX. Both A.) the OB Venus data, and B.) the OB month-lengths are linked to Ammiṣaduqa Year 1; Both C.) Ur III month-lengths and D.) Ur III eclipses are linked to Amar-Sin Year 1; And E.) Akkad eclipses are linked to Sargon Year 1.[2798]

XX. According to Huber, there are approximately 400 years between Ammiṣaduqa Year 1 and Amar-Sin Year 1; Furthermore, Huber also estimates approximately 287 years between Amar-Sin Year 1 and Sargon Year 1.[2799]

XX. As noted by Huber, none of them [i.e. this astronomical evidence] is fully trustworthy. The Venus data (A) are corrupted through a long scribal tradition and it is not *a priori* obvious that they represent genuine observations from the time of Ammiṣaduqa. The month-length data (B) and (C) are noisy. The eclipses (D) and (E) are from the omen series Enuma Anu Enlil, and it is not clear to what extent such omens contain genuine eclipse observations, or merely learned systematization. Also the identification of the apodoses with historical events is uncertain (e.g. whether the omen EAE 20-III pertains to the death of Šulgi or of some other king).[2800]

XX. Huber reckons that if they have the same statistical structure as their Neo-Babylonian counterparts (which is not proven), a correct chronology induces an agreement rate of merely 67% between calculated and observed 30-day months. This is uncomfortably close to the agreement rate 53% pertaining to a randomly chosen wrong chronology.[2801]

XX. According to Huber, for establishing absolute dates on the basis of month-lengths alone one would need between 250 and 300 precisely spaced month-lengths (i.e. several decades of data, which may have lacunas, but with perfectly known intercalations).[2802]

XX. According to Huber, around 2000 BCE (Julian), eclipse times are subject to clocktime errors of give or take about one (1) hour, caused by the irregular rotation of the earth.[2803] [2804] Moreover, the year

2798 Huber 2000: 1
2799 Huber 2000: 1
2800 Huber 2000: 1
2801 Huber 2000: 1
2802 Huber 2000: 1
2803 i.e. ΔT (Delta T)
2804 Huber 2000: 1

counts establishing the distances between the First Dynasty of Babylon [i.e. Ammiṣaduqa Year 1], the Third Dynasty of Ur [i.e. Amar-Sin Year 1], and the Akkad dynasty [i.e. Sargon Year 1], may have unsuspectedly large uncertainties.[2805]

XX. Huber argues that the only way to check astronomical consistency of the evidence is by trying – that is: by attempting to fit chronologies to the data. The best fitting chronology is not necessarily the correct one, but if some chronologies can be fitted coherently, then this supports validity of the data.[2806]

XX. Huber admits that certain aspects of the Venus data are chronologically insensitive. For example, the durations of invisibility depend mainly on the common longitude of the Sun and the planet at the time of conjunction – in other words: [durations of invisibility depend] on the season, and one gets a good agreement once every eight (8) years [and four (4) days]. According to Huber, it seems that Gurzadyan had zeroed in on such aspects.[2807]

XX. Huber notes that what is chronologically sensitive [about the Venus data] are the dates of [Venus] events relative to the lunar month: they shift four (4) days every eight (8) years, and thus Venus events are in step with the months only about once every 56 or 64 years. However, it is not possible to argue in a simple-minded fashion in terms of such Venus periods (8, 56, or 64 years). According to Huber, some astronomically feasible Venus chronologies are spaced 35 years (e.g. -1517 [i.e. 1518 BCE (Julian)] and -1482 [i.e. 1483 BCE (Julian)]), and it is necessary to check compatibility by explicit computation.[2808]

XX. On the evaluation of the dates, P. J. Huber (2000) still maintains that a statistical analysis of the possibly faulty dates is possible, and that the ensuing results he considers to be compatible with a date for Ammisaduqa 1 are 1582 BCE (Julian) or 1516 BCE (Julian).[2809]

76.8. Gasche's Ultra-Low (Ultra-Short) Chronology

XX. It was just a question of time until a new attempt would be made to revise the entire chronology of the period before the 1st millennium BCE (Julian). The model proposed by Gasche et al. in 1998 thus gave momentum to a new and intense debate which endures up to the time of writing [by Klinger (2006)].[2810]

XX. In 1998 CE (Gregorian), a new approach by a team led by Hermann Gasche, was published. Based

2805 Huber 2000: 1-2
2806 Huber 2000: 2
2807 Huber 2000: 2
2808 Huber 2000: 2
2809 Klinger 2006: 305
2810 Klinger 2006: 306

on Gasche's evaluation of pottery development, they considered a lower chronology (i.e., lower than the "Middle chronology") to be necessary. They then searched for a possible re-interpretation of the Venus Tablet, of eclipse omens referring to the destruction of Ur, and of other evidence adduced in the past, and presented such a new, lower chronology. It gives Hammurabi a reign from 1696 – 1654 BCE (Julian), and places the end of the dynasty in 1499 BCE (Julian).[2811]

XX. This was greeted both with agreement and with dissent from colleagues. It is therefore evident that the problem of Mesopotamian chronology of the early 2nd millennium has not been solved; at least, there are several people who disagree among each other.[2812]

XX. Whether a new chronological framework similar to that prevailing with the Middle Chronology will emerge remains to be seen. At present, the earlier chronological proposals are maintained as conventions.[2813]

XX. Largely rejecting the use of the Venus tablets, a team of scholars led by Hermann Gasche has recently made a formidable attack on the Middle Chronology [i.e. the compromise chronology that is currently accepted by the majority of mainstream scholars of Assyriology and the academia establishment], advocating an "ultra-low" system that would place the fall of the First Dynasty of Babylon in 1499 BCE (Julian), roughly a century later than the date established by the Middle Chronology.[2814]

XX. This proposal – based on a consideration of pottery, stratigraphy, texts, and astronomical data – would eliminate the Dark Age traditionally assigned to the 16th century BCE (Julian) and place the fall of Babylon just prior to the well-dated Kassite rulers of the 14th century BCE (Julian). Its proponents argue that the styles of pottery and other material culture from the era before the fall of Babylon are too similar to the styles of the 14th century BCE (Julian) to have been separated by such a long period of time.[2815]

XX. While this conclusion may well prove to be correct, it is by no means clear that modern scholars are able to assess the likelihood of different rates of material culture change in antiquity. For example, one might note the extraordinary conservatism of the texts from Terqa on the Middle Euphrates in which tablets from the early 2nd millennium BCE (Julian) were so similar to those from late in the [same 2nd] millennium BCE (Julian) that scholars had first assumed they were written within a few decades of each other.[2816]

2811 Hunger 2009: 149
2812 Hunger 2009: 150
2813 Klinger 2006: 306
2814 Schwartz 2008: 451
2815 Schwartz 2008: 451
2816 Schwartz 2008: 451

XX. If the Ultra-Low Chronology is correct and the Dark Age century eliminated, a century must be added to one of the periods prior to the 2nd millennium BCE (Julian). Champions of the Ultra-Low Chronology have not yet indicated which earlier period they would propose to lengthen.[2817]

XX. Possible support for the Ultra-Low Chronology has been offered from the results of the excavations at Hazor in the southern Levant, a site mentioned in the Mari texts as an important city and a trading partner of the Mari king. According to its excavators, Hazor did not achieve an urban size and prosperity until the Middle Bronze IIB period, which must therefore correspond to the period of the Mari texts.[2818]

XX. At Tell el-Dab'a (Avaris) in northern Egypt, Levantine-style Middle Bronze IIB pottery was found in contexts associated with Dynasty 13 and assigned to the early 17th century BCE (Julian). As a result, excavators date the Middle Bronze IIB city of Hazor and, by extension, the Mari texts to the early 17th century BCE (Julian).[2819]

XX. This conclusion is based on the assumption that Egyptian absolute chronology is relatively secure compared with that of western Asia.[2820]

XX. An early 17th-century BCE (Julian) date for the Mari archives diverges from the early- to mid-18th-century BCE (Julian) date assigned by the Middle Chronology and conforms to the Ultra-Low Chronology. It remains to be seen whether the various strands of this argument (including the archaeological interpretation of Hazor's urban status, the ceramic parallels at Tell el-Dab'a, and the absolute dating of the strata at Tell el-Dab'a) are upheld.[2821]

XX. Manfred Bietak, the excavator of Tell el-Dab'a, has proposed that apart from the dating of Hazor, the Egyptian dating of the Levantine materials from his site has wide repercussions for 2nd-millennium BCE (Julian) Levantine chronology. He has proposed a lowering of the traditional dates for the Levantine Middle Bronze Age, placing its beginning at about 1900 BCE (Julian) as opposed to 2000 BCE (Julian) and the transition from Middle Bronze IIA to Middle Bronze IIB at about 1700 BCE (Julian) rather than 1800 BCE (Julian).[2822]

XX. While some archaeologists working in the southern Levant have been amenable to this proposition, others – chief among them William G. Dever – have vehemently opposed it, rejecting Bietak's interpretation of both the Egyptian and Canaanite evidence.[2823]

2817 Schwartz 2008: 451
2818 Schwartz 2008: 451
2819 Schwartz 2008: 451
2820 Schwartz 2008: 451
2821 Schwartz 2008: 451
2822 Schwartz 2008: 451
2823 Schwartz 2008: 451

76.9. Gurzadyan's and Gasche et al.'s Ultra-Low (Ultra-Short) Chronology

XX. As mentioned by Huber and others, the data from the Venus Tablet has been thrown into doubt due to the numerous errors which have corrupted the text.[2824] The Old Babylonian calendar was not continually modified [i.e. not intercalated according to set calculated rule at this time], but was altered only when the [observed] discrepancy between the seasons and the calendar forced the issue.[2825] [2826]

XX. In Old Babylonian times the pattern of intercalation of lunar months was governed by the intention to keep the lunar calendar lined up to the agricultural year, characterized by the date harvest in the fall and the barley harvest in the spring. Fotheringham (1928) has shown that on average the date harvest fell in Ululu / Elul (month VI) and the barley harvest in Addaru / Adar (month XII).[2827]

XX. In the Old Babylonian kingdom the intercalated month was [usually] a 2nd Addaru / Adar [i.e. 13th month after the 12th month of 1st Addaru / Adar] at the end of the year. The lack of regularity in these intercalations (as exhibited by the civil contract dates) shows that they took place empirically according to the [observed] ripeness of the crops, or when it otherwise appeared necessary.[2828]

XX. Even two (2) consecutive years [(or more) is known to have] occurred with 13 months when the calendar was too far out. A few times, when apparently a deviation had to be corrected rapidly, the 6th month [i.e. Ululu / Elul] was duplicated.[2829]

XX. Occasionally the Old Babylonian calendar could be quite a bit out of step with the agricultural year. During Hammurabi's reign the beginning of the year at one time moved forward by about 2.5 months due to four (4) intercalations in four (4) consecutive years, and during the reign of Ammisaduqa the beginning of the year at one time slipped backwards by about seven (7) weeks in five (5) consecutive years without being corrected by intercalation.[2830]

XX. [Gurzadyan challenges Huber's results, and offers his own Methodology which upon close examination is realized to be flawed, and significantly prone to the metaphorical 'Lost in Translation' of the many "moving parts"/variables errors of his Methodology]. The date for Ammisaduqa Year 1 == 1550 BCE (Julian) that Gurzadyan calculates is thus – according to his own words – 'anchored by the lunar eclipses but not by the lunar calendar'.[2831]

2824 Gurzadyan 2000: 179
2825 Neugebauer 1975 – cited by Gurzadyan 2000: 179
2826 Gurzadyan 2000: 179
2827 De Jong and Foertmeyer 2010: 152
2828 Pannekoek 1961: 30
2829 Pannekoek 1961: 30
2830 De Jong and Foertmeyer 2010: 152
2831 Gurzadyan 2000: 181

XX. However, further study reveals Gurzadyan's methodology of isolating the eclipse parameters of E[numa] A[nu] E[nlil] 20 and 21 and extracting the limited data from the Venus Tablet. [Despite that Gurzadyan asserts that these admittedly are very limited data, and nonetheless, according to him they] allow a high degree of confidence in employing these as the astronomical anchor for the Ultra-Low Chronology proposed in Gasche et al. (1998).[2832]

XX. The relative sequence of the Inferior and Superior Conjunctions and not the lunar calendar lie at the base of Gurzadyan's approach [i.e. methodology] to the 8-year Venus cycles. This sequence [8-year Venus cycles] was – according to Gurzadyan – unique and its signature would have been – again according to Gurzadyan – the last to have been effaced and lost as a result of careless copying and inattentive observation. The date for Ammisaduqa Year 1 == 1550 BCE (Julian) is thus – according to Gurzadyan – anchored by the lunar eclipses but not by the [actual] lunar calendar.[2833]

XX. To illustrate the significance and value of his method, Gurzadyan calculates the Venus data for Ammisaduqa Year 1 as 1702, 1646, 1582 and 1550 BCE (Julian), corresponding to the traditional High, Middle, Low Chronologies, and Ultra-Low Chronology.[2834] In order to avoid the possible impact of local visibility conditions and other uncertainties, the elongation alone is presented. The data thus relies on the angular separation of Venus from the Sun,[2835] by means of which Venus becomes invisible from a specific critical angle Θ_{cr}.[2836] The glare critical angle for average vision is given by the formula (where m_v is the visual stellar magnitude):[2837] [2838]

$$\log \Theta_{cr} == 0.2(9.28 + m_v)$$

XX. [Thus calculating] dates of Venus' passage according to the [specific critical] angle ($\Theta_{cr} == 11°$).[2839]

XX. The relative magnitudes for each chronology are calculated [with the flawed Methodology of not only the data set being extremely limited, but this methodology also relies on using a theoretical (calculated) formula for possible geometric values of *arc visionis* of Venus relative to the Sun], and Universal Time [UT] is set at 00:41[2840] [i.e. 00:41 local Greenwich Mean Time (GMT) and then equated with time differences to e.g. Babylon local time: This methodology has too many variables // moving parts that yield a potential large margin of error for precise measurements of positions and dates.]

2832 Gurzadyan 2000: 183
2833 Gurzadyan 2000: 181
2834 Gasche et al. 1998 – cited by Gurzadyan 2000: 181
2835 i.e. the *arcus visionis*
2836 i.e. the *arcus visionis*
2837 Schaefer 1991 – cited by Gurzadyan 2000: 181
2838 Gurzadyan 2000: 181
2839 Gurzadyan 2000: 181
2840 Gurzadyan 2000: 181

XX. Gurzadyan's calculations were performed with the Ephemeris - Tool 4.1 Version 4.1003, 2000 software, elaborated by M. Dings based on Newcomb's theories, which is – according to Gurzadyan – quite sufficient for this purpose. The sequence of the conjunctions [of Venus] can be clearly followed for each chronology. The data in the Venus Tablet can only be linked to any given chronology if anchored by an alternative link [e.g. the precise Babylonian / Hebrew calendar]. Without any reliable data (such as the Ur III eclipses) – as Gurzadyan admits – the signal cannot be anchored with confidence.[2841]

XX. [In support of his position, Gurzadyan cites support from the following academic disciplines:] Scholars not associated with the development of his hypothesis [in other fields of scholarship of such as] studying seals (e.g., Gualandi 1998), Egyptian materials,[2842] and Anatolian dendrochronological evidence[2843] have all provided additional interpretations of evidence – that according to Gurzadyan is – compatible with the Ultra-Low Chronology. Other contributors present at the Ghent Colloquium (e.g. the contributions of Gates, Van Soldt and Warburton) likewise presented interpretations – that Gurzadyan claims are – compatible with the Ultra-Low Chronology.[2844]

XX. Furthermore, while no compelling arguments have been brought against any elements of the astronomical or archaeological evidence, Gurzadyan states that Hunger (2000) has likewise come to share the opinion of the Venus data, suggesting that only the 8-year cycles can be taken as reliable markers.[2845]

XX. However, as noted by De Jong, although this proposal has led to a lively debate in the literature,[2846] it has not resulted in finally resolving the question. The fact that the Ultra Short Chronology is not supported by the Venus observations as pointed out by Huber (2000) has cast severe doubt on its credibility.[2847] De Jong and Foertmeyer state more emphatically that Huber (2000) has shown unambiguously that this [Ultra-Low/Ultra-Short] Chronology is inconsistent with – and in fact is refuted by – the Venus observations.[2848]

76.10. The Planet Venus and Its Four (4) Chronologies

XX. Planetary observations were carried out in Mesopotamia as early as the reign of King Ammisaduqa. Venus observations are recorded in the 63rd tablet of an extensive series entitled *Enuma*

2841 Gurzadyan 2000: 181
2842 Krauss 1998 and personal communication – cited by Gurzadyan 2000: 184
2843 Kuniholm et al. 1996 – cited by Gurzadyan 2000: 184
2844 Gurzadyan 2000: 184
2845 Gurzadyan 2000: 184
2846 See Pruzsinszky 2009; Roaf 2012 – cited by De Jong 2013
2847 De Jong 2013: 149
2848 De Jong and Foertmeyer 2010: 143

Anu Enlil, a list of omens, which Neugebauer (1969: 101) likens to papal bulls of the Middle Ages. A more detailed list of observations of the Moon, planets, and stars from Babylonia from ~700 BCE (Julian) – the ([mul]APIN) is discussed in Kelley and Milone 2011, section §7.1.[2849]

XX. Central in Babylonian astronomical practice is the recording of the dates of first appearance and disappearance of the planets. Because the inner planet Venus always stays within an angular distance of 48° from the Sun as seen from the Earth, it experiences a repeating cycle of (4) four consecutive appearances and disappearances: two (2) as Morning Star (first appearance and disappearance in the East), and two (2) as Evening Star (first appearance and disappearance in the West).[2850]

XX. Several passages in the Babylonian astronomical compendium MUL.APIN [the [mul]APIN] show that already in the 2nd millennium BCE (Julian) Babylonian astronomers were well aware of the intricacies of the motion of the planet Venus, its appearances and disappearances and its periods of invisibility. The Venus tablet of Ammī-Ṣaduqa contains the earliest Babylonian collection of such observations.[2851]

XX. It takes Venus 584 days to go through the cycle of two (2) first appearances and two (2) disappearances. After five (5) cycles (i.e. ten (10) first appearances and ten (10) disappearances) Venus reappears at approximately the same position in the sky (shifted backwards by about 2.5° in longitude). These five (5) cycles correspond to a little less than eight (8) years, the well-known Babylonian 8-year period of Venus (more precisely 99 lunar months minus (-) 4 days) in the Babylonian lunar calendar.[2852] [2853]

XX. The minimum angular distance from the Sun for Venus to become visible near the horizon (the so-called *arcus visionis*) can be estimated from Neo-Babylonian observations of Venus during the last six (6) centuries BCE (Julian). This has most recently been done in two (2) independent studies by Mebert (2010) and by De Jong (2012) with quite similar results, based on about 100 usable observational records from the Astronomical Diaries.[2854]

XX. De Jong (2013) makes six (6) conclusions, the first of which is the estimation of the traditional time window of possible dates in the Mesopotamian Chronology:

1.) According to De Jong, there exists in the Old Babylonian (OB) chronology the traditional uncertainty margin [among current Assyriology] of all possible Venus chronologies where Ammī-

2849 Kelley and Milone 2011: 75
2850 De Jong 2013: 150
2851 De Jong 2013: 150
2852 See Hunger and Pingree 1999, p. 203ff – cited by De Jong 2013
2853 De Jong 2013: 150
2854 De Jong 2013: 150

Ṣaduqa Year 1 varies from 1710 to 1550 BCE (Julian)[2855] – that in the traditional historical window [of uncertainty from] 1710 – 1550 BCE (Julian) the Venus observations allow – according to De Jong – three (3) pairs of candidate chronologies: Ammī-Ṣaduqa Year 1 == A.) 1702/1694 BCE (Julian); B.) 1646/1638 BCE (Julian); and C.) 1581/1574 BCE (Julian).[2856]

2.) According to De Jong, of these [possible dates in the traditional window of chronological, historical uncertainty] the 1646 BCE (Julian), 1638 BCE (Julian), the 1581 BCE (Julian), and the 1574 BCE (Julian) chronologies provide the best fit to the Venus data (standard deviations [of error] ≤ 3.9 days).[2857]

3.) According to De Jong, the canonical High Middle Chronology (1646 BCE (Julian)) is less probable because the lunar calendar shift exceeds the standard deviation [of error].[2858]

4.) According to De Jong, the [Low] 1574 BCE (Julian) Chronology proposed by Mebert (2010) is a valid candidate chronology.[2859]

5.) According to De Jong, the [Ultra-Low] 1550 BCE (Julian) Chronology proposed by Gasche et al. (1998) is "not supported" by the Venus observations.[2860]

6.) According to De Jong, the Venus observations by themselves are insufficient to choose between the six (6) candidate chronologies so that additional constraints are required to make a final choice,[2861] [e.g. the precise Babylonian / Hebrew calendar].

XX. [Within the Venus Tablet of Ammī-Ṣaduqa within] observation no. 2 [records] the first appearance in the East (Morning First (MF)) of Venus on the 18th day of the month Šabatu / Shvat (month XI) in Year 1 of Ammī-Ṣaduqa,[2862] Since the Babylonian date of the Venus observation is 18 Šabatu / Shvat (month XI), this implies that the first day 01 Nisannu / Nisan of Year 2 of Ammī-Ṣaduqa falls 42 days after the Venus observation;[2863] We do not know how well the Old Babylonian calendar was lined up with the seasons and since large deviations (up to several months) are known to have occurred during the reigns of Hammurabi and Ammī-Ṣaduqa.[2864]

2855 De Jong 2013: 150
2856 De Jong 2013: 152
2857 De Jong 2013: 152
2858 De Jong 2013: 152
2859 De Jong 2013: 152
2860 De Jong 2013: 153
2861 De Jong 2013: 153
2862 De Jong 2013: 150
2863 By calculating 42 days, De Jong is assuming that this was not a leap year with an intercalated 13th month of Addaru II / Adar II
2864 De Jong 2013: 151

XX. The astronomical data that have been used in the past to try to make a choice between candidate Venus chronologies are:[2865]

1.) the distribution of 30-day months in the Old Babylonian calendar;[2866]

2.) a lunar eclipse reported by the diviner Asqūdum to the king of Mari;[2867]

3.) and the record of a solar eclipse in the Mari Eponym Chronicle in the year after the birth of Šamši-Adad.[2868]

XX. However, so far all attempts to establish absolute dates for the Old Assyrian and Old Babylonian chronology have been been inconclusive.[2869] [2870]

XX. According to Reiner and Pingree (1975) the first 20 observations, covering the first eight (8) years of the reign of Ammī-Ṣaduqa, constitute the most reliable set.[2871] Leaving aside the [Ultra-Low] 1550 BCE (Julian) Chronology which – according to De Jong – provides a poor fit to almost all observations,[2872] [De Jong's Methodology which attempts to calculate] the predicted dates are computed using the [i.e. minimum angular distance] arcus visionis,[2873] [i.e. a geometrical methodology which is not only flawed and seriously prone to error because of aforementioned too many variables ("moving parts"), but whose data set of Venus observations is severely limited.]

XX. Notice that on average the differences between observed and computed dates are significantly smaller for Evening Last (EL) and Morning First (MF) observations than for Morning Last (ML) and Evening First (EF) observations. This was first noted by Van der Waerden (1945/1948) and is discussed in detail by de Jong (2012). It is due to differences in the relative motion of Venus with respect to the Sun near inner [i.e. Inferior] and outer [i.e. Superior] Conjunction. It also shows up in the analysis of the arcus visionis [i.e. minimum angular distance] values derived from Neo- and Late Babylonian observation by de Jong (2012).[2874]

XX. De Jong and Foertmeyer's analysis (2010) of the Venus observations does not allow them to choose between the four (4) Venus chronologies. Therefore they admit that it seems that we are still at the stage first quite clearly formulated by Neugebauer (1941) that additional independent historical

2865 De Jong 2013: 153
2866 Huber et al. 1982; Mebert 2010 – cited by De Jong 2013
2867 Banjevic 2006; Mebert 2010 – cited by De Jong 2013
2868 Michel and Rocher 1997/2000; Mebert 2010 – cited by De Jong 2013
2869 See the review by Pruzsinszky 2009 – cited by De Jong 2013
2870 De Jong 2013: 153
2871 De Jong 2013: 151
2872 De Jong 2013: 152
2873 De Jong 2013: 151
2874 De Jong 2013: 152

evidence is required to choose between the different possible Venus chronologies.[2875]

XX. Two (2) studies presenting such independent evidence (although with contradictory results) have been published: One study, by Huber et al. (1982), uses a list of textually attested Old Babylonian month lengths to demonstrate that the Long [i.e. High] Chronology is by far to be preferred. The other study by Manning et al. (2001) shows that radiocarbon dating of tree rings in Anatolian archaeological monuments supports the (Low) Middle Chronology. A third suggestion by Gurzadyan, published as part of a general reappraisal of the archaeological evidence, was convincingly shown by Huber (2000) to be invalid.[2876]

XX. These Venus observations seem to have offered the basis for the astronomical calculations, since this observational event repeats itself every 56 / 64 years, and thus the date means that the year six (6) of Ammisaduqa[2877] can be set on a spectrum ranging from 1641 to 1577 BCE (Julian). This point has now been lost since the observations are no longer viewed as reliable, with the result that the long debate about the relative merits of the Middle or Low chronologies must now be abandoned, having become pointless as the various alternatives for an absolute date are no longer separated by intervals of 64 years.[2878]

XX. Explicitly: the use of a date 1531 or 1595 BCE (Julian) [i.e. a difference of 64 years] for the Hittite conquest of Babylon – which is certainly relevant for the beginning of Hittite history – is today a mere convention. According to Klinger, limiting it [i.e. the conquest of Babylon] to a period of 10 – 20 years is possible. Instead [of the various obsolete alternatives for an absolute date separated by intervals of 64 years], we now have a moving link separating a block of 500 years from the later periods where the chronology is relatively reliable.[2879]

76.11. Astronomical Computer Programs

XX. As noted by Huber, we now have a substantial body of completely different astronomical programs and tables, partly based on analytical expansions, partly on brute force numerical integration. In addition, we have a large body of ancient naked-eye observations. The results of various comparisons can be summarized as follows [below]:[2880]

XX. In order to avoid too much repetition, Huber plays some variations on the themes treated in Huber et al. (1982); for a change, Huber argues with discrepancies expressed in days (which is slightly more

2875 De Jong and Foertmeyer 2010: 153
2876 De Jong and Foertmeyer 2010: 153
2877 Does Klinger mean Year Eight (8)?
2878 Klinger 2006: 305
2879 Klinger 2006: 305
2880 Huber 2000: 18

intuitive) rather than in *arcus visionis* values [which is the degree angle of Venus relative to the Earth and the Sun].[2881]

XX. According to Huber, there are no problems with the celestial mechanics of modern programs (Bretagnon, Chapront, Meeus, ...): For the historical periods in question, they are – according to Huber – more accurate than ancient visual observations.[2882]

XX. According to Huber, there are minor problems with empirical visual parameters [of ancient visual observations that uses *arcus visionis*]: systematic errors of about two (2) minutes in rising and setting times, five (5) minutes in beginning and end of lunar eclipses and of totality, and random effects about twice as large.[2883]

XX. According to Huber, there are some problems with lunar orbital acceleration: now resolved, thanks to lunar laser ranging. The correct value is in the range of -25.8" to -26"/cy^2 (the value -23.8946 used in the commercially available 1991 CE (Gregorian) version of the Chapront program is – according to Huber – much too low).[2884]

XX. As observed by Huber, there are major problems with the irregular rotation of the earth. The problems are in the difference ([ΔT, Delta] T == ET – UT) between Ephemeris Time (ET) (the [theoretical] uniform time scale used in celestial mechanics) and Universal Time (UT) (based on the [actual observed] rotation of of the earth [which as noted above is irregular], i.e. roughly equivalent to Greenwich Mean Time (GMT)) and its extrapolation beyond [Astronomical Year] -500 [i.e. -501 BCE (Julian)]. At present, most approaches are directly or indirectly based on Stephenson and Morrison, 1982 and 1984.[2885]

XX. According to Huber, a statististical analysis and stochastic modeling of the Length-of-Day (LOD) process shows that in the range of one (1) year to 2700 years it can be excellently modeled by a Brownian motion with drift.[2886] For the uncertainties in the extrapolation of [ΔT, Delta] T, caused by irregular fluctuations in the length of day (LOD), one obtains – according to Huber – the following standard [deviations of] errors:[2887]

by [Astronomical Year] -1000 [i.e. 1001 BCE (Julian)]: 10 minutes

by [Astronomical Year] -1500 [i.e. 1501 BCE (Julian)]: 30 minutes

2881 Huber 2000: 2
2882 Huber 2000: 18
2883 Huber 2000: 18
2884 Huber 2000: 19
2885 Huber 2000: 19
2886 Huber 2000 – cited by Huber 2000
2887 Huber 2000: 19

by [Astronomical Year] -2000 [i.e. 2001 BCE (Julian)]: 60 minutes

76.12. Huber's Conclusions about the Venus Tablet and Mesopotamian Chronology

XX. Huber's conclusions can be summarized as:[2888]

1.) There is hardly any doubt that the Venus Tablet contains a hard core of genuine, consecutive observations. They must have extended at least over the Years 1 to 17 (with a corrupted passage in Years 8 – 9). Whether the physically separate (and rather corrupt) Section III is a continuation for Years 19 to 21 (skipping Year 18) is undecidable. All this holds independently of the attribution of the observations to Ammisaduqa's reign.[2889]

2.) The Venus chronology proposed by Gasche-Gurzadyan (Ammisaduqa year 1 == -1549 [i.e. 1550 BCE (Julian)]) has no basis in the Venus Tablet; In fact, it is flatly contradicted by it. The nearest compatible chronologies are 32 years earlier (-1581 [i.e. 1582 BCE (Julian)]) or 32 years later (-1517 [1518 BCE (Julian)]), respectively.[2890]

XX. According to Huber, the precise version of the Gasche chronology (Ammisaduqa Year 1 == -1549 [i.e. 1550 BCE (Julian)], fall of Babylon 51 years later) is not supported by the astronomical evidence, contrary to the claims made by Gasche et al. (1998). Huber argues that this Ultra Low Chronology is flatly contradicted by the Venus text, [with] the nearest compatible Venus dates being -1581 [i.e. 1582 BCE (Julian)] and -1517 [i.e. 1518 BCE (Julian)]. Also the eclipses do not support it. Therefore, according to Huber, a set of long [high] chronologies gives a coherent, near-perfect fit for all five (5) components (A) to (E) of the astrononomical evidence:[2891]

Sargon Year 1 == -2380 or -2326 [i.e. -2381 BCE (Julian) or -2327 BCE (Julian)]

Amar-Sin Year 1 == -2093 [i.e. -2094 BCE (Julian)]

Ammisaduqa Year 1 == -1701 [i.e. -1702 BCE (Julian)]

Fall of Babylon == -1650 [i.e. -1651 BCE (Julian)]

XX. According to Huber, the overall, coherent fit is so good that it must be taken seriously. It cannot be

2888 Huber 2000: 8
2889 Huber 2000: 8
2890 Huber 2000: 8
2891 Huber 2000: 17

disregarded simply because it disagrees with the (by necessity chronologically softer) archaeological evidence.[2892]

XX. In other words, according to Huber: the best "short" [or "low" chronology] fit is unconvincingly poor, and therefore the combined astronomical evidence – far from supporting it – argues against a short [low] chronology. In Huber's opinion, tree-ring data from a single site[2893] are not strong enough to overcome this, and Huber does not think that the conflict can be resolved without additional evidence.[2894]

XX. According to Hunger, leaving aside the lunar eclipses, a combination of the Venus Tablet data and Old Babylonian month lengths alone supports only the so-called High chronology. According to Huber, the data has been misunderstood by Gasche and his team; their astronomical calculations are also marred by errors. In particular, the insistence by Gurzadyan that the Venus Tablet can only be used to establish an 8-year cycle of Venus phenomena beginning with Ammisaduqa year 1 does not seem to reflect an understanding the Babylonian lunar calendar. While Gasche's chronology is based on the archaeological material, it cannot be considered verified by astronomy. [2895]

XX. Hunger sums up the situation as follows: if the Venus Tablet contains reliable records from the time of Ammisaduqa, and if the Old Babylonian month lengths are usable data, then the Huber's "High Chronology" is the most likely one. If we think that we are forced by other evidence, ceramics for instance, to arrive at a different (lower) chronology, we have to abandon the use of these astronomical data.[2896]

76.13. The Historical Chronology of Assyria

XX. Assyrian chronology is based on the Assyrian King List (AKL).[2897] Any chronological chart of Assyrian kings is ultimately based on the AKL, which has been called the "backbone" of Assyrian chronology.[2898] The AKL has some gaps and omissions, and these may be explained as a consequence of political confusion during those periods.[2899]

XX. Chronological problems have centered on conflicts or lacunae within the Assyrian tradition. Despite some gaps, omissions and conflicting numbers, the AKL is generally considered the "backbone" of Assyrian chronology. The AKL together with the dates for the Babylon I dynasty based

2892 Huber 2000: 17
2893 See Kuniholm et al. (1996) – cited by Huber 2000
2894 Huber 2000: 18
2895 Hunger 2009: 151
2896 Hunger 2009: 151
2897 Hunger 2009: 145
2898 Pruzsinszky 2009: 48
2899 Pruzsinszky 2009: 53

on the astronomical data of the Venus Tablet (VT) attributed to the reign of Ammisaduqa, can securely establish Mesopotamian chronology for the 2nd millennium BCE (Julian). Some problems still remain as to the type of a calendar in use in Assyria during the 2nd millennium [whether it was a solar calendar like in Egypt, or a strictly lunar calendar that was not intercalated].[2900]

XX. The Assyrian king-lists [after the Venus Tablet of Ammisaduqa] are the second major source for the establishment of an Ancient Near Eastern chronology.[2901] The Assyrian King List's five (5) main exemplars represent one (1) text with minor variants due to scribal mistakes.[2902] The AKL covers the entire 2nd millennium BCE (Julian) and continues to the reign of Šalmaneser V (726 – 722 BCE (Julian)).[2903]

XX. The AKL contains information on Assyrian rulers going back to the time before 2000 BCE (Julian). The compilers of the lists aimed for completeness. The reign lengths of some of the earliest kings seem to have been unknown at the time of the composition of the AKL, as may be concluded from the beginning part, which is divided into three (3) sections or groups.[2904]

XX. Besides the Assyrian King List (AKL), there are also lists of eponyms (i.e. Assyrian Eponym Canon (AEC). Eponyms are high officials whose names were used to give a name to a year. Instead of a number, the year is identified by the eponym's name, similar to the dating by consuls used in ancient Rome. Lists were necessary to keep a record of the sequence of the eponyms. Unfortunately, the lists that are preserved do not provide a continuous sequence.[2905]

XX. The greatest lack in the cuneiform sources continues to be the lack of adequate data combined with the absence of an overarching system of year names. In Assyria, the traditional principle of naming years and counting according to the holder of a certain office (*limu / limmu*) was maintained, but the sequence of officials in the books of eponyms [i.e. the Assyrian Eponym Canon (AEC)] eliminated the need for naming the years.[2906]

XX. This system was already established in the first quarter of the 2nd millennium BCE (Julian), and became a standard dating method as can be seen in the eponym dates in the thousands of Old Assyrian texts from the karum-period in Anatolia, and as is confirmed by the recent discovery of two (2) tablets with *limu* lists.[2907] With the overlap between these lists and the eponym chronicle from Mari, a period of almost exactly 200 years is not absolutely precisely dated in relative terms.[2908]

2900 Pruzsinszky 2009: 52
2901 Klinger 2006: 307
2902 Hunger 2009: 145
2903 Pruzsinszky 2009: 52
2904 Pruzsinszky 2009: 48
2905 Hunger 2009: 146
2906 Klinger 2006: 306
2907 Klinger 2006: 306
2908 Klinger 2006: 306-307

XX. Yet here too [with the Assyrian king-lists] it must be conceded that the later periods are better documented and more reliable than both the earlier era and the neighboring regions. These lists are the spine of all dates stretching back to the 12th century BCE (Julian), complemented by eponym-lists [i.e. the Assyrian Eponym Canon (AEC)] (generally complete for the first half of the 1st millennium BCE (Julian)), and themselves anchored [seemingly] absolutely with a solar eclipse dated to 15 June 763 BCE (Julian), recorded in the eponym-lists [i.e. AEC]. The general compatibility of the versions of the Assyrian king-lists (AKL) thus leaves only small margins in dating the reigns of the Assyrian kings back to the 12th century BCE (Julian).[2909]

XX. The eponym list(s) for the first half of the 1st millennium BCE (Julian) are usually dated by means of a solar eclipse mentioned in one of the exemplars.[2910]

"*In the eponymy of Bur-sagale, (governor) of Guzana: revolt in the city of Assur; in the month Simanu, the sun made an eclipse.*"[2911]

XX. According to Hunger, in order to use the eclipse, one has to make additional assumptions about the possible date of the list or any item mentioned in it.[2912] One needs the number of years between a secure date and some event in the eponym list. Secure dates can be found in astronomical events, provided they can be identified with certainty.[2913]

XX. A tablet collecting lunar eclipses mentions an eclipse in year 1 of Mukin-zeri, a Babylonian king who fought against the Assyrian Tiglath-pilesar III. Since it is known from the so-called Babylonian Chronicle that this year corresponded to Tiglath-pilesar's 15th Regnal Year, the reign of the Assyrian is thereby astronomically dated. A similar eclipse in the same month would at best occur 18 years earlier or later.[2914]

XX. Hunger concludes that we are therefore entitled to trust the eponym list. Disagreeing with Bloch (2012a) and Jeffers (2017) who believe that the Assyrians used a strictly lunar [non-intercalary] calendar during this 2nd millennium BCE (Julian) period, Hunger believes that at this time the AEC uses solar years (whatever their exact definition), and not lunar ones.[2915]

XX. Since the Assyrian King List (AKL) is most likely derived from eponym lists [of the Assyrian Eponym Canon] (AEC), Hunger believes that it too should be based on solar years. It seems very

2909 Klinger 2006: 307
2910 Hunger 2009: 146
2911 Millard 1994: 41 – cited by Hunger 2009: 146
2912 Hunger 2009: 146
2913 Hunger 2009: 146
2914 Hunger 2009: 146
2915 Hunger 2009: 146

unlikely to Hunger that this relation would have been abandoned in the Middle Assyrian period, and then taken up again by Tiglath-pileser I.[2916]

XX. For the time before 910 BCE (Julian), where the preserved exemplars of the eponym list [of the Assyrian Eponym Canon] (AEC) break off, we have to rely on the Assyrian King List (AKL) to reconstruct the chronology. In the case of variant readings or missing lengths of reign we can only rarely invoke the eponym list; its fragments for the second half of the second millennium do not provide enough information.[2917]

XX. Fundamental for the Neo-Assyrian eponyms is A. Millard, *The Eponyms of the Assyrian Empire 910-612 BC* (Helsinki, 1994). For the Middle Assyrian period, H. Freydank, *Beiträge zur mittelassyrischen Chronologie und Geschichte* (Berlin, 1991) is indispensable. For the specifications of this solar eclipse of 15 June 763 BCE (Julian), cf. also Stephenson (1997), *Historical Eclipses and Earth's Rotation*, pp. 126-127.[2918] [2919]

76.14. The Historical Chronology of Babylonia

XX. In comparison with the Assyrian sources, the Babylonian material is far poorer. The Babylonians did indeed move relatively rapidly to a more elegant and relatively more accessible means of dating, namely simply counting the years sequentially according to the king's reign which became the norm as early as the first half of the 14th century BCE (Julian).[2920]

XX. However there is a comparative dearth of written sources so that the Babylonian chronology must be established by synchronisms with Assyrian history where works such as the synchronic history are important – if partial – witnesses. This covers the period from the middle of the 2nd millennium BCE (Julian) through Adad-Nerari III in the 8th century BCE (Julian).[2921]

XX. The general situation is more or less comparable in all of the other areas as a consistent sequential system of dates was not used anywhere, or is at least not preserved. This means that any attempt to establish a precise chronology for the history of the ancient Near East must rely initially on the Assyrian data, and only then on that of their neighbors, and their predecessors.[2922]

XX. Once these relative chronologies have been established, the second step is to identify the best

2916 Hunger 2009: 148
2917 Hunger 2009: 146
2918 Klinger 2006: 307
2919 Klinger 2006: 307
2920 Klinger 2006: 308
2921 Klinger 2006: 308
2922 Klinger 2006: 308

possible synchronisms with the Assyrian chronology. Due to the multitude of sources – and especially the diplomatic correspondence of certain periods – it is possible to establish not merely one-to-one links, but actually to weave a fabric of relations which can actually be established without any internal contradictions.[2923]

XX. Arriving at absolute dates depends upon a very different and diverse set of conditions. For the 1st millennium BCE (Julian), people and events can be dated with near precision as the uncertainty is highly constrained. For the earlier periods, some closed blocks can be isolated in which the relative chronology is certain, but anchoring these to the absolute dating of the 1st millennium BCE (Julian) leaves a considerable margin of error.[2924]

XX. There is a significant contrast with respect to the precise chronological synchronizations between Mesopotamia and Egypt for the 1st and 2nd millennia BCE (Julian). For the 1st millennium BCE (Julian) where the Assyrian data is reliable and precise, the correlation difficulties lie with Egypt. Whereas before this the reverse is true.[2925]

XX. In the middle of the 2nd millennium BCE (Julian), there is a gap in our information on the Babylonian side. Prior to this gap, we know the dynasties of Ur, Isin, Larsa, and Babylon in the first half of the 2nd millennium BCE (Julian) as their relative chronology is well established. We have lists of year names covering altogether about 400 years. Although one might try to bridge this gap by different hypotheses, it seems better to look for an independent way to date the earlier dynasties. The main problem is how to anchor them to a reliably dated event.[2926]

XX. It was therefore a sensation when, at the beginning of the 20th century, F. X. Kugler (1910) found that cuneiform omen texts, using successive appearances and disappearances of Venus as ominous signs, could be connected with the Old Babylonian king Ammisaduqa. These omens are preserved in first-millennium copies; so they are separated from the presumed time of observation by almost a millennium. Unfortunately, the calendar dates of these phenomena had been very much corrupted in transmission so that in order to use the text for chronology, emendations were required, not all of which had the same probability. As research went on, three main possible datings of Ammisaduqa emerged, labeled as the “High”, “Middle” and “Low” Chronologies.[2927]

XX. Depending on what other evidence scholars adduced, they arrived at different choices among these three. Applying the three (3) chronologies to the well-known king Hammurabi of Babylon, his reign is dated as follows:[2928]

2923 Klinger 2006: 308
2924 Klinger 2006: 308
2925 Klinger 2006: 308
2926 Hunger 2009: 149
2927 Hunger 2009: 149
2928 Hunger 2009: 149

High Chronology: 1848 – 1806 BCE (Julian)

Middle Chronology: 1792 – 1750 BCE (Julian)

Low Chronology: 1728 – 1686 BCE (Julian)

XX. The end of Hammurabi's dynasty, which occurred 155 years after his death, is therefore placed in either 1651, 1595, or 1531 BCE (Julian), respectively.[2929]

XX. Šamši-Adad I is known from the Mari correspondence to have died in year 18 of Hammurabi. Then we can remember the solar eclipse reported for the year after the birth of Šamši-Adad I in the so-called Mari Eponym Chronicle.[2930]

XX. This part of Mesopotamian chronology is more or less contemporaneous with the Minoan eruption of Santorini (although, as far as Hunger knows, no Mesopotamian record – archaeological or textual – of the eruption exists).[2931] Since the Minoan eruption of Santorini took place around the middle of the 2nd millennium BCE (Julian), we are concerned either with Old Babylonian times or with the period immediately thereafter (depending upon when the eruption took place and whether the First Dynasty of Babylon ended c. 1595, 1531 or 1499 BCE (Julian)). Thus, according to Hunger, we can disregard the earlier periods (which would take us well into the 3rd millennium BCE (Julian) and more than half a millennium away from the Minoan eruption, without any useful input).[2932]

XX. Despite the millennia during which these civilizations existed, documents confirming direct contacts and exchange between Mesopotamia and it neighbors on the one hand, and Egypt on the other, are rather rare and restricted to certain historical phases. Although one can assume that at all times there will have been people who were traveling back and forth or trading between these lands, this has left virtually no trace in the sources.[2933]

XX. We only have access to major state activities – diplomatic contacts or military engagements – which were generally exceptional. The lucky find of the Amarna archive appears to be a period of particularly intensive contacts, but may actually have been repeated at very different times. The seemingly comprehensive documentation in the Mari archives does not reveal one single direct link to Egypt.[2934]

2929 Hunger 2009: 149
2930 Hunger 2009: 151
2931 Hunger 2009: 150
2932 Hunger 2009: 151
2933 Klinger 2006: 309
2934 Klinger 2006: 309

XX. Despite such obstacles, there is a general consensus among scholars on the basic framework of the chronological framework which can be established using the basis provided by the Assyrian records. This includes also the reduction of circa one (1) decade in the Assyrian chronology proposed some time ago – fundamental is the contribution by Boese & Wilhelm.[2935]

XX. Another constraint is the revised model proposed by H. Gasche et al. which would result in a significant reduction even with respect to the Low Chronology. Affecting the ancient Near East up to the middle of the 2nd millennium BCE (Julian), it touches a period for which the Assyrian sources do not provide complete cover.[2936]

XX. [Fundamentally disagreeing with Huber's High Chronology, Klinger asserts that] the arguments by Gasche et al. are supported not only by philological historical sources, but fundamentally based on archaeological criteria.[2937] For the period from the end of the 14th century BCE (Julian), the dates of this new proposal do not differ significantly from those used hitherto, as a comparison between the Boese/Wilhelm shortened Assyrian chronology and the "ultra-low chronology" produced by Gasche et al. shows.[2938]

XX. The further back in Assyrian history one goes, the greater the divergences and the greater the room for differing chronological assumptions so that Gasche et al. favor bringing the end of the First Dynasty of Babylon down to 1499 BCE (Julian), rather than 1531 or 1595 BCE (Julian). At present, from the philological standpoint, only the statements of the periodic intervals (*Distanzangaben*) can serve as control mechanisms.[2939] For example, an inscription of Shalmaneser I concerning his reconstruction of the Ekursagkurkurra specifies a period of 580 years separated his reign from that of Shamshi-Adad I.[2940]

XX. Unfortunately there is no consensus on their value and chronological relevance. Gasche et al. argue that the temporal intervals preserved as *Distanzangaben* are either incorrect or they try to provide a specific interpretation for the relevant interval.[2941] They argue, "Our analysis of the *Distanzangaben* makes it apparent that no firm chronological conclusions can be drawn from them."[2942]

XX. Klinger asserts that this statement [by Gasche et al.] only applies to the model they propose.[2943] Simply rejecting the *Distanzangaben* completely is apparently not appropriate as can be surmised from

2935 Klinger 2006: 309
2936 Klinger 2006: 309
2937 Klinger 2006: 309
2938 Klinger 2006: 310
2939 Klinger 2006: 310
2940 Klinger 2006: 310-311
2941 Klinger 2006: 310
2942 Klinger 2006: 310
2943 Klinger 2006: 310

the analysis of the dates by Boese and Wilhelm.[2944]

XX. Veenhof dates the accession of Shamshi-Adad to the year 1833 BCE (Julian), on the basis of the Middle Chronology. Important is also his observation that the *Distanzangaben* of the later Assyrian tradition relate to Irishum I and Shamshi-Adad I, meaning that these are therefore probably realistic, effectively negating the hypothesis that the *Distanzangaben* are not chronologically relevant. Following the traditional Middle Chronology, Veenhof specifies the following Old Assyrian dates: the accession of Irishum I in 1974 BCE (Julian) and the death of Shamshi-Adad I in 1776 BCE (Julian).[2945]

XX. It must be stated that this significant reduction not only entails suggesting a reduction of the length of the Assyrian period of almost 200 years (whereby roughly a quarter of a century will have passed since the end of the Third Dynasty of Ur), but also that a close link between the dynasty ruling in Yamkhad probably allows a link between the Old Hittite Period and the earlier periods, and that this simply cannot be reduced at will.[2946]

XX. The precise year of the death of Shamshi-Adad I is still discussed, as it may have been in year 12, 13 or 17 of the reign of Hammurabi.[2947] In addition there is a solar eclipse listed in the Mari chronicle for the year after the birth of Shamshi-Adad I.[2948]

XX. The links in the chronological system of the first half of the 2nd millennium BCE (Julian) between Assyria and Babylon (with the death of the Assyrian king Shamshi-Adad I during the 2nd decade of the reign of the Babylonian king Hammurabi) ultimately allows the synchronization of Hittite history with Mesopotamian history through the campaign of Murshili I leading to the end of the First Dynasty of Babylon in year 31 of Samsuditana.[2949] Dating this event to the year 1499 BCE (Julian) creates major problems for Hittite history.[2950]

XX. Gasche et al. did not attempt to integrate the difficulties of Hittite history between Murshili I and Tutkhaliya I. G. Beckman ("Hittite Chronology", *Akkadica* 119-120, (2000)), discussed precisely these issues at length and came to the conclusion "that the Anatolian evidence does not favor the suggested downward revision of the date of the fall of Babylon".[2951]

76.15. The Assyrian Calendar

2944 Klinger 2006: 311
2945 Klinger 2006: 307
2946 Klinger 2006: 311
2947 Klinger 2006: 312
2948 Klinger 2006: 311
2949 Klinger 2006: 311-312
2950 Klinger 2006: 312
2951 Klinger 2006: 312

XX. The question of the form of the Assyrian calendar is – according to Klinger – less serious than the differing evaluation of the *Distanzangaben*, but nevertheless important. The issue is whether it depended upon [purely] lunar months and whether or not intercalary months were used to match the solar year, and thus whether or not calculations lead to a difference in the number of years in the chronology with respect to solar years. Fortunately, there is – according to Klinger – evidence favoring the use of solar years in Assyria during the 2nd millennium BCE (Julian) and not just later.[2952]

XX.An additional problem [i.e. uncertainty in absolute chronology] is caused by the use of a lunar calendar in Middle Assyrian times. Despite their awareness of the practice, most scholars have hitherto assumed that in Middle Assyrian times regnal years were counted according to a solar calendar, and the dates provided by Frahm (2017) are based on this supposition as well.[2953]

XX. However since there is no evidence for intercalation in Middle Assyrian documents from before the 11th century BCE (Julian) – and due to a number of other reasons – it may be more likely (as recently argued in studies by Yigal Bloch (2010, 2012a, 2012b) and Joshua Jeffers (2017)) that the regnal years of Middle Assyrian kings preceding Assur-bel-kala (no. 89) were all lunar years ca. 11 days shorter than the solar year with its 365.24 days. If so, the dates of [Assyrian] rulers [on which we base our chronology of the Ancient Near East] would have to be modified.[2954]

XX. The reign of Tiglath-pileser I (1114 –1076 BCE (Julian))[2955] was a turning point in the history of the calendar in the ancient Assyrian kingdom. During his reign, Assyria adopted a calendar, which had already existed by that time for several centuries in southern Mesopotamia – the so-called Standard Mesopotamian calendar [2956] which had emerged between ca. 1750 BCE (Julian) and ca. 1500 BCE (Julian).[2957] [2958]

XX. In the reign of Tiglath-pileser I, Assyria adopted a southern Mesopotamian calendar, in which the year consisted of twelve (12) lunar months:[2959] [2960]

1.) Nisannu / Nisan;

2.) Ayyāru / Iyar;

3.) Simānu / Sivan;

2952 Klinger 2006: 312
2953 Frahm 2017: 616
2954 Frahm 2017: 616
2955 Bloch 2012b: 25
2956 Bloch 2012a: 20
2957 Cohen 1993: 297-305 – cited by Bloch 2012a: 49
2958 Bloch 2012a: 49
2959 Cohen 1993: 297-301 – cited by Bloch 2012b: 25
2960 Bloch 2012b: 25

4.) Duʾūzu / Tammuz;

5.) Abu / Av;

6.) Ulūlu / Elul;

7.) Tašrītu / Tishri (Tishrei);

8.) Araḫšamnu / Marheshvan (Heshvan);

9.) Kissilīmu / Kislev;

10.) Ṭebētu (Kanūnu) / Tevet;

11.) Šabāṭu / Shvat;

12.) Addaru / Adar;

XX. The same [linguistically/dialectically-similar] month names were adopted in the Jewish calendar from the 6th century BCE (Julian) onwards.[2961] [2962]

XX. Twelve (12) lunar months amount on the average to ca. 354.36 days, which is 10.89 days less than the [close to] tropical solar [Julian] year (ca. 365.25 days). Consequently, in order to keep the calendar in pace with the solar year, the years in the [Standard Mesopotamian] Babylonian calendar were occasionally intercalated.[2963] Since the main polity using this calendar at the time of its adoption by Tiglath-pileser I was Babylonia, it is justified to speak of an Assyrian adoption of the [Standard Mesopotamian] Babylonian calendar in his reign.[2964]

XX. Once in a while, an intercalary 13th month – a second Ulūlu/Elul (recorded as ^{ITU}KIN.^{d}INANNA.2.KAM) or Addaru / Adar (recorded as ^{ITU}DIRI.ŠE.KIN.KUD, "the additional Addaru", or occasionally simply as ^{ITU}DIRI, "the additional month") – was added to the year in order to keep the calendar in pace with the solar year cycle.[2965] [2966] In other words, the [Standard Mesopotamian] Babylonian calendar was not purely lunar but [rather it was] luni-solar. Since in the long run intercalation was intended to keep the lunar calendar dates in pace with the solar year, one can assume

2961 Cohen 1993: 297-301 – cited by Bloch 2012b: 25
2962 Bloch 2012b: 25
2963 Bloch 2012a: 20
2964 Bloch 2012b: 25
2965 Cohen 1993: 297-301 – cited by Bloch 2012b: 25
2966 Bloch 2012a: 20; Bloch 2012b: 25

that on the average, the 13th month would be added once in (29.53 / 10.89 == ca. 2.7 years).[2967]

XX. Given that the mechanism of intercalation should have rendered the average length of the Babylonian year equal to the Julian solar year, and given that the [*Assyrian King List*] AKL allows a precise reconstruction of the chronology of the kings of Assyria from Tiglath-pileser I to the kings of the 9th – 8th centuries BCE (Julian), then according to Bloch the reign of Tiglath-pileser I can be dated with near complete certainty to 1114 – 1076 BCE (Julian). Tiglath-pileser I has been, until now, the earliest Assyrian king for whose reign such measure of certainty in absolute dating is possible.[2968]

XX. Before the reign of Tiglath-pileser I, Assyria used a different calendar, with totally different month names. The order of the twelve (12) months of that calendar was established by Hans Ehelolf and Benno Landsberger.[2969] [2970]

XX. In the same study, Ehelolf and Landsberger also observed that there is no evidence for specifically designated intercalary months in the traditional Assyrian calendar. [This is] in contrast to the Babylonian calendar, in which intercalary months were specifically designated as such.[2971] [2972] [Thus] documents indicating parallels between specific Assyrian and Babylonian months imply that different Assyrian months could be contemporaneous with Nisannu, the [consistent] 1st month of the [Standard Mesopotamian] Babylonian year.[2973]

XX. The [Assyrian] months were lunar, as is clear from the fact that they lasted either 29 or 30 days.[2974] The question is whether the Middle Assyrian calendar, practiced before the adoption of the Babylonian calendar by Tiglath-pileser I, employed any mechanism for intercalation.[2975]

XX. According to Jeffers (2017), the months of the Assyrian year were in fact not permanently affixed to particular seasons of the solar year, [thereby] revealing that its year was not intercalated and was thus perpetually rotating [or wandering/migrating] slowly through the solar year.[2976] Bloch also concludes that during the 13th – 12th centuries BCE (Julian), the Assyrian calendar must have been purely lunar without intercalation (similar to the modern Islamic calendar).[2977] If so, this would have serious implications for the dating of Assyrian regnal years. The generally accepted regnal dates that are currently assigned to the reigns of the Middle Assyrian (MA) kings would have to be lowered

2967 Bloch 2012a: 20
2968 Bloch 2012b: 25
2969 Ehelolf and Landsberger 1920 – cited by Bloch 2012b: 25
2970 Bloch 2012b: 25
2971 see Brinkman 1976: 400-401 cited by Bloch 2012b: 25
2972 Bloch 2012b: 25
2973 Bloch 2012b: 25-26
2974 Bloch 2012a: 20-21
2975 Bloch 2012a: 21
2976 Jeffers 2017: 151
2977 Bloch 2012a: 19

because lunar years are shorter than their solar counterparts.[2978]

XX. During the 1st millennium BCE (Julian), Assyria used the so-called Standard Mesopotamian calendar, which emerged in southern Mesopotamia sometime between ca. 1750 and ca. 1500 BCE.[2979] Since during the second half of the 2nd millennium BCE (Julian), the most important polity using the Standard Mesopotamian calendar was Babylonia (under the rule of the Kassite – and then the Second Isin – dynasties),[2980] it is justified to call it the [Standard Mesopotamian] Babylonian calendar.[2981]

XX. Therefore, it is justified to speak of the adoption of the [Standard Mesopotamian] Babylonian calendar in Assyria in the reign of Tiglath-pileser I.[2982] The adoption of the [Standard Mesopotamian] Babylonian calendar took place probably in the final two (2) decades of the reign of Tiglath-pileser I, following his conquest of Babylonia.[2983]

XX. Among ca. 2,000 dated Assyrian documents from the second half of the 2nd millennium BCE (Julian), published until today, none mentions an intercalary Assyrian month (or contains any other indication of a 13th month being added to the year).[2984] Hence, it appears that intercalation by the addition of an explicitly indicated 13th month to a year, as known from the Babylonian calendar, was not practiced in Assyria in the relevant period.[2985]

XX. This situation was clear already by the 1920s, and scholars working on reconstruction of the Middle Assyrian calendar had to propose other mechanisms of intercalation if they thought that intercalation was practiced in Assyria in the second half of the 2nd millennium BCE (Julian).[2986]

XX. In contradistinction to such intercalation mechanisms proposed by Weidner (1920s) and Koch (1989), several scholars maintained that intercalation was not practiced at all in the Middle Assyrian calendar.[2987] [Among these,] Bloch argues [– as well as Jeffers –] that the Middle Assyrian calendar was purely lunar and did not employ any intercalation mechanism (similar to the modern Islamic calendar).[2988]

XX. According to Bloch, while in the 13th – 12th centuries BCE (Julian) the calendar practiced in Assyria was purely lunar, it was apparently not the original calendar of Assyria. For the period

2978 Jeffers 2017: 153
2979 Cohen 1993: 297-305 – cited by Bloch 2012b: 334
2980 Bloch 2012b: 334; Bloch 2012a: 20
2981 Bloch 2012a: 20
2982 Cohen 1993: 300-301 – cited by Bloch 2012b: 334
2983 Bloch 2012b: 334
2984 Bloch 2012a: 21; Bloch 2012b: 337
2985 Bloch 2012a: 21; Bloch 2012b: 337-338
2986 Bloch 2012a: 21
2987 Bloch 2012a: 21-22
2988 Bloch 2012a: 22; Bloch 2012b: 340

spanning the 17th – 14th centuries BCE (Julian), very few Assyrian documents are known, and they do not allow any clear conclusion concerning the structure of the calendar used in Assyria in that period. On the other hand, for the Old Assyrian period [during the] 20th – 18th centuries BCE (Julian) documentary evidence is abundant.[2989]

XX. The practice [of intercalation of lunar months] is known since the Ur III period, and letters from the Old Babylonian period record an intercalary month based on royal decrees, but the continuous usage of some rules cannot be documented – at least partially because of the lack of sources. In Assyria, the regular use of calibrating by using the lunar calendar is only known from the 1st millennium BCE (Julian).[2990]

XX. For issues of absolute chronology, this question of whether the Assyrian calendar depended solely upon lunar months, and whether or not intercalary months were used to match the solar year, is a critical issue for precise absolute chronology. Thus whether or not calculations lead to a difference in the number of years in the chronology with respect to solar years is a critical issue.

76.16. Synchronisms across the Ancient Near East

XX. There are no direct synchronisms before the Amarna period.[2991] We can thus use the dates for the Assyrian kings as proposed by Boese / Wilhelm as the basis for the presentation of the synchronisms with Egyptian history.[2992]

XX. The earliest contacts are recorded in the Amarna letters which refer to earlier events, and specifically to the relations between Egypt and the Hurrian Mittani state, the most important major power in Western Asia in the 15th century BCE (Julian).[2993]

XX. After intense diplomatic contacts under Amenhotep II, a dynastic marriage was arranged during the reigns of Thutmose IV and Artatama (I) which should be dated to the first decades of the 14th century BCE (Julian) according to the Egyptian chronology. The tradition was maintained by Shuttarna II and Amenhotep III who married a Mittani princess in his year 10, and later a daughter of Tushratta, who had since become ruler. In Shuppiluliuma I of Hatti, Tushratta met a dangerous foe in the first decade of the second half of the 14th century BCE (Julian).[2994]

XX. In light of the complete absence of any original Hurrian or Mittani state sources, not much more

2989 Bloch 2012a: 40
2990 Klinger 2006: 312
2991 Klinger 2006: 312-313
2992 Klinger 2006: 313
2993 Klinger 2006: 313
2994 Klinger 2006: 313

chronological information can be gleaned from the information of the Amarna letters.[2995]

XX. It is only with the 14th century BCE (Julian) letters of the Assyrian king Aššur-Uballit I to Amenhotep IV (EA 15 and 16) that a real synchronism can be won from the Amarna letters that mentions an exchange of letters between Aššur-nadin-aḫḫe II and Amenhotep III.[2996] There would appear to be considerable room for debate since the Assyrian ruled for more than 30 years, but the contents of the first letter place it relatively near the start of his reign (1353 BCE (Julian))[2997]

XX. In the absence of detailed information about the temporal relationship between the predecessors of Eriba-Adad I and Amenhotep III and his successor Amenhotep IV, scholars can only deduce that the exchange must have taken place around 1381 BCE (Julian).[2998] Beckerath, *Chronologie*, 61 places this "shortly after his accession to the throne" which he dates to 1355 BCE (Julian).[2999]

XX. Therefore according to Beckerath's chronology he dates the accession of Amenhotep IV to the end of 1351 BCE (Julian), thus allowing an indirect synchronism as EA 15 does not name the pharaoh. Quite apart from this, there is sufficient time in the reign of Amenhotep IV for the second letter to have been addressed to him, despite the uncertainties with respect to the reading of the name.[3000]

XX. Thereafter follows a gap of several centuries in the historical record of direct contacts between the two (2) powers [of Egypt and Mittani].[3001]

XX. For further links between the cuneiform-using states and Egypt, we can turn to Babylonian and Hittite sources. As preserved, the Amarna correspondence documents contacts over several generations – as in letters between the Babylonian king Kadashman-Enlil I and [Egyptian Pharaoh] Amenhotep III, and their successors Burnaburiash II and [Pharaoh] Amenhotep IV where the Egyptian must have come to the throne during the reign of the second Babylonian king (as the latter is in touch with both of these pharaohs). The preserved letters (above all from Burnaburiash II to [Pharaoh] Amenhotep IV) contain indications that – of the earlier Kassite kings – Kara-indash had contact with [Pharaoh] Thutmose IV, and Kurigalzu I with [Pharaoh] Amenhotep III.[3002]

XX. Since the reigns of the Kassite Kings before Kadashman-Enlil are unknown these vague hints appear unpromising. Yet one can actually create a chronological framework since we know that Kara-indash reigned at the same time as the Assyrian king Aššurbel-nishu (1407 – 1399 BCE (Julian)),

2995 Klinger 2006: 313
2996 Klinger 2006: 313
2997 Klinger 2006: 313-314
2998 Klinger 2006: 314
2999 Klinger 2006: 314
3000 Klinger 2006: 314
3001 Klinger 2006: 314
3002 Klinger 2006: 314

which puts some limits for the not particularly long reign of [Pharaoh] Thutmose IV.[3003]

XX. The sources are more precise for Kadashman-Enlil I and Burnaburiash II as the reign of the latter is known, and the synchronic history links the end of his reign with the Assyrian chronology.[3004] Although the exchange of letters only began after his 30th year, [Burnaburiash II] actually corresponded with [Pharaoh] Amenhotep III, [and therefore] the entire reign of [Pharaoh] Amenhotep IV would thus parallel that of Burnaburiash II who must have reigned until the accession to the throne of the successor of this pharaoh, and thus to the year 1323 BCE (Julian) according to the shortened chronology.[3005]

XX. His [i.e. Burnaburiash II's] accession to office must then have taken place in 1349 BCE (Julian), when [Pharaoh] Amenhotep III would still have been reigning according to the synchronism of the Amarna letters. However some of the numbers remain uncertain due to factors which have not yet been eliminated in the reigns of the Kassite rulers, and thus the figures cannot be taken as absolute limits.[3006]

XX. According to Egyptologist Jörg Klinger, discrepancies with Egyptian dates can easily be explained as lying in the Babylonian sources.[3007] The reduction in the Babylonian chronology in this case would stand in contradiction to the assumed accession of [Pharaoh] Amenhotep IV ca. 1351/50 BCE (Julian), as Burnaburiash would only have ascended to the throne in the following year, and thus a synchronism with the older of the two (2) pharaohs would be excluded, although precisely – according to Klinger – this is fact reliably attested. According to Klinger, the margin of uncertainty in this phase remains the +/- 5 years mentioned.[3008]

XX. We must therefore confirm that – according to Egyptologist Jörg Klinger – the known synchronisms between Egypt and Babylon cannot provide and exact chronological fix[ed] points due to the imprecision inherent in the frame of reference, or in the uncertainties in the chronological framework itself – although they do place some further limits on the range of variation.[3009]

76.17. Egyptology and Egyptian Chronology

XX. In the early 20th century CE (Gregorian), W. F. Petrie, the recognized founder of Egyptology and the archaeological principles of stratigraphy and superimposition, published the beginning of the historic/dynastic period (Dynasty I) at 5510 BCE (Julian).[3010] [3011] His conclusions were based primarily

3003 Klinger 2006: 314
3004 Klinger 2006: 314-315
3005 Klinger 2006: 315
3006 Klinger 2006: 315
3007 Klinger 2006: 315
3008 Klinger 2006: 315
3009 Klinger 2006: 315
3010 Petrie 1906:175 – cited by Hasel 2004
3011 Hasel 2004: 11

on "the authority of the Turin papyrus and of Manetho," for "it is only reasonable to accept those lists as substantially accurate".[3012] [3013]

XX. Other noted Egyptologists later placed Dynasties I and II at 3425 – 3000 BCE (Julian)[3014] or 3400 – 2980 BCE (Julian).[3015] It is significant that the most extensive adjustments have occurred in the Old Kingdom where relative dates allow for more change due to the length of the First Intermediate Period.[3016]

XX. However, the last century of Egyptological research (expanded by incredible amounts of new data from excavations and textual studies) has raised serious questions not only for the length of the historic periods, but also for the absolute nature of early dates. Several serious questions have been raised which necessitate revisions for the traditional absolute chronology prior to the Egyptian New Kingdom.[3017] [3018]

XX. Since the founding of the international research programme SCIEM 2000 ("The Synchronization of Civilizations in the Eastern Mediterranean in the Second Millennium BC"), scholars have been working toward establishing a general framework of interregional chronology of the 2nd millennium BCE (Julian).[3019]

XX. It became clear that this aim could be achieved in reasonable time – according to Bietak and Höflmayer – only within the relative chronology, which means that the periodization of the different regions in the Eastern Mediterranean could be shown through their relationships to each other. In order to apply such a scheme with absolute dates, we still have the problem of two (2) chronologies:[3020]

1.) the historical chronology, based mainly on [relative] Egyptian chronologies [that relies mostly on dead-reckoning of minimal(!) years of reign] and Assyrian chronologies and their interrelationship;

2.) radiocarbon chronology;

XX. Despite all attempts to discuss these differences away or at least to minimize them, one has to realize that there are periods with a considerable difference between radiocarbon chronology and historical chronology which cannot be denied nor be reconciled at the moment. The high chronology

3012 Petrie 1906:175 – cited by Hasel 2004
3013 Hasel 2004: 11-12
3014 Weigall 1910: xvii – cited by Hasel 2004
3015 Breasted 1916: 419 – cited by Hasel 2004
3016 Hasel 2004: 12
3017 Ward 1992:55 – cited by Hasel 2004
3018 Hasel 2004: 12
3019 Bietak and Höflmayer 2007: 13
3020 Bietak and Höflmayer 2007: 13

(HC) rests now on the strength and weakness of the radiocarbon dating alone.[3021]

XX. The alternative to Radiocarbon dating, the [relative] Egyptian chronology, is based on a combination of astrochronology (Sothis- and lunar dates), incomplete or corrupted king lists, incomplete regnal data, genealogies of officials and time estimates based on them, and even such records as the stelae of the holy Apis bulls, recording their lifespan and the kings under which they were born or have passed away. Such dead reckoning [of minimal dates] from undisputed dates of the 1st millennium BCE (Julian) backwards, such as the conquest of Egypt by Cambyses at 525 BCE (Julian), is today the preferred method of arriving at a historical chronology of the New Kingdom.[3022] Historical Egyptian chronology also relies on the interrelationship with other chronologies such as the Assyrian one, which offers with its eponyms' lists a framework which is considered within minor margins of error fairly accurate.[3023]

XX. The chronological experts in Egyptology agree on a date for the beginning of the New Kingdom from c. 1550 to 1540 BCE (Julian). There is also an understanding that the margin of error may be within – according to Bietak and Höflmayer – 20 years both ways, but this would put pressure on the genealogies and on some specific lengths of reign (e.g. Tuthmosis II, Tuthmosis IV, Horemheb).[3024]

XX. The two (2) systems – the [hard-science] radiocarbon method and the [relative dating] historical chronology – have periods of agreement such as the 14th and 13th centuries BCE (Julian). However, it is wrong to claim the time from the 18th century BCE (Julian) backwards as a period of agreement again, thus limiting the disagreement to two (2) to three (3) centuries.[3025]

XX. For the time before the New Kingdom we don't have such a close control over the historical chronology as we do for the New Kingdom, especially not for the time of the Old Kingdom. The radiocarbon dates obtained from this period are also not consistent. For the Middle Kingdom, we have a disagreement within the historical chronology between a high chronology (HC) and a low chronology (LC), which are – according to Bietak and Höflmayer – about 42 years apart. Therefore we are not in the position to say if the radiocarbon dates are in agreement with the historical chronology or not.[3026]

XX. On the contrary, the recent investigation of radiocarbon dates from short-lived samples throughout the stratigraphy of Tell el-Dab'a by the VERA laboratory, covering successive strata from the 20th to the 15th centuries BCE (Julian), show a series of time consistent dates with an offset – according to Bietak and Höflmayer – between 100 and 150 years higher than the historical chronology when using the high

3021 Bietak and Höflmayer 2007: 13
3022 Bietak and Höflmayer 2007: 13
3023 Bietak and Höflmayer 2007: 13-14
3024 Bietak and Höflmayer 2007: 14
3025 Bietak and Höflmayer 2007: 14
3026 Bietak and Höflmayer 2007: 14

chronology of the Middle Kingdom to cover the first part of this stratigraphy.[3027] In addition, the seriation of pottery types at Tell el-Dab'a helps to establish cross relationships to other sites with great precision.[3028]

XX. The dates of the successive strata are anchored by two (2) historical datum lines to the year 5 of Sesostris III (1868 BCE (Julian) High Chronology (HC)) and the conquest and abandonment of Avaris c. 1530 BCE.[3029] Of course Tell el-Dab'a alone cannot establish interregional chronology, but its local chronology should show that the offset between the series of radiocarbon dates from Tell el-Dab'a and the Egyptian historical chronology is indeed real as well as significant.[3030]

XX. This time difference of – according to Bietak and Höflmayer – ca. 100 years or more repeats the offset between traditional (low) chronology (Thera eruption around 1500 BCE (Julian)) and the new radiocarbon-based high Aegean chronology (middle to second half of 17th century BCE (Julian)). Therefore – according to Bietak and Höflmayer – it would not make sense to try to remedy this situation by unilaterally raising the Aegean chronology by 100 to 150 years, claiming that a new proportion of the relationship between Egypt and the Aegean has been found.[3031]

XX. Such a unilateral rise – according to Bietak and Höflmayer – is also not necessary, as it seems that for the 17th to the 15th centuries BCE (Julian) the offsets of radiocarbon versus traditional chronology are the same in Egypt and the Aegean. Therefore – according to Bietak and Höflmayer – one can come only to the conclusion that either the radiocarbon chronology or the historic chronology is wrong, or both have a defect. In such a case, the mutual control would not be possible without the help of an independent absolute dating method such as dendrochronology[3032] [or precise mathematical astronomy].

XX. Working our way backwards, we can observe an increasingly higher date by the radiocarbon technology versus the historical chronology.[3033]

XX. According to Bietak and Höflmayer, in order to harmonize the two (2) chronologies, one would be obliged to inflate the regnal years of kings of the 15th and early 14th century. This would be possible with the kings Tuthmosis II and Tuthmosis IV reaching a practically unsupportable maximum of 20 years and creating among the high officials of this time unusually high ages. However, to reach the age of 100 or even more is completely out of the question. This shows that the major reason (or fault) for this offset – according to Bietak and Höflmayer – cannot be blamed on historical chronology.[3034]

3027 Bietak and Höflmayer 2007: 14
3028 Bietak and Höflmayer 2007: 16
3029 Bietak and Höflmayer 2007: 14
3030 Bietak and Höflmayer 2007: 16
3031 Bietak and Höflmayer 2007: 16
3032 Bietak and Höflmayer 2007: 19
3033 Bietak and Höflmayer 2007: 20
3034 Bietak and Höflmayer 2007: 20

XX. That there is an offset and not a false understanding in the relationship of the Egyptian and the Aegean periodization (so the thesis of Sturt Manning) is – according to Bietak and Höflmayer – shown by the Tell el-Dab'a-series of Walter Kutschera et al. and by the late first appearance of Thera pumice in the archaeological contexts of Egypt and the Levant, i.e. not before the Late Bronze Age in the Levant and not before the Tuthmoside Period, i.e. 15th century in Egypt.[3035]

XX. In summation, the agreement between carbon 14 (C14) dating and historical chronology in the 14th century BCE (Julian) and the sharp rise of an offset a century earlier of up to 100 to 150 years as well as in the preceding centuries only shows that – according to Bietak and Höflmayer – the calibrated radiocarbon dates presented by Manning, Bronk Ramsey et al. cannot be considered as a series of chronometric precision, but as a series where the precision seems to deviate considerably from the 15th century backwards. According to Bietak and Höflmayer, this conclusion is the more cogent one as – within the historical chronology of the 18th Dynasty with its dense network of regnal and genealogical data – nobody could claim that a mistake of more than 100 years could have mounted up from the Amarna period to the early Tuthmosides (within a century).[3036]

XX. Under such auspices, one has to ask if it would not be worthwhile to investigate if a systemic failure in the Mediterranean C14 evaluation could be discovered, or if the absorption of C14 was, for environmental reasons, different from the 15th century BCE (Julian) backwards. Probably, we do not know enough about what may affect radiocarbon and its evaluation process. For this reason it, would be – according to Bietak and Höflmayer – very important to close the gaps in Anatolian dendrochronology and to do the same with the cedar tree from Lebanon. Such new standards could be used to build up regional calibration.[3037]

XX. In Egyptian chronology there are also problems in the first half of the 1st and the whole 2nd millennium BCE (Julian) as well as the time before which also have to be worked out in respect to maximal margins of errors [because of the minimal years of reign calculated by dead-reckoning].[3038]

XX. According to Bietak and Höflmayer, one should do this without being biased by absolute chronologies. One may expect that at least achieving the relative interregional timetables could solve some problems in absolute chronology. Above all, it seems to be most important that scientists should take the difference between Radiocarbon and historical chronology as seriously as Bietak and Höflmayer do.[3039]

3035 Bietak and Höflmayer 2007: 20
3036 Bietak and Höflmayer 2007: 20
3037 Bietak and Höflmayer 2007: 20
3038 Bietak and Höflmayer 2007: 21
3039 Bietak and Höflmayer 2007: 21

76.18. Manetho as an Authority for Egyptian Chronology

XX. During the 3rd century BCE (Julian) a "history of Egypt" was written in Greek by a[n Egyptian] priest named Manetho.[3040] It seemed almost certain that Manetho had compiled his history from some of the same lists that modern historians use for the reconstruction of Egyptian chronology.[3041] In spite of early reliance on Manetho,[3042] and attempts to legitimize it as dependable,[3043] an increasing number of errors and inaccuracies have been discovered.[3044] This has caused some historians to abandon this tradition altogether[3045] or to show that dependence on Manetho is not necessary, "at least not for the chronology of the new kingdom and later".[3046] Because of these problems "his work ceased to be the basis of Egyptian chronology many decades ago".[3047] New evidence on the length of reigns and the existence of co-regencies have contributed to the abandonment of Manetho as a credible source.[3048]

76.19. New Evidence on the Length of Reigns

XX. New data about the length of certain reigns have caused revisions. For example, it has been established that Sesostris III reigned for 19 years rather than 39 as previously thought,[3049] and Amenemhet IV for 13 rather than nine (9) years.[3050] During the 19th Dynasty, Merenptah reigned no more than ten (10) years rather than Manetho's 19.[3051] The general tendency of new data that has come to light supports a shortening [i.e. a bringing lower] rather than lengthening [i.e. raising higher] of the chronology.[3052]

76.20. The Question of Coregencies

XX. The matter of coregencies remains unresolved.[3053] One example will suffice. A short coregency between Thutmose III and Amenhotep II is often assumed.[3054] Redford[3055] allows one (1) and a third

3040 Waddell 1940; Helck 1956 – cited by Hasel 2004
3041 Knapp 1992: 716 – cited by Hasel 2004
3042 Petrie 1906; Weigall 1910; Breasted 1916 and others – cited by Hasel 2004
3043 Helck 1956 – cited by Hasel 2004
3044 Redford 1986: 231-332 – cited by Hasel 2004
3045 Wente and van Siclen 1976: 217-218; Ward 1984: 155-156 – cited by Hasel 2004
3046 Hornung 1987 – cited by Hasel 2004
3047 Kitchen 2000: 39 – cited by Hasel 2004
3048 Hasel 2004: 12
3049 Simpson 1972; 1984; cf. Kitchen 1992:329; 2000 – cited by Hasel 2004: 12
3050 Beckerath 1976:50 – cited by Hasel 2004: 12
3051 Bierbrier 1975:118 note 2; on Merenptah's reign and see Hasel 1994; 1998; 2003; 2004 – cited by Hasel 2004: 12
3052 Hasel 2004: 12-13
3053 Ward 1992: 54 – cited by Hasel 2004
3054 See Casperson 1986 – cited by Hasel 2004
3055 Redford 1966: 120 – cited by Hasel 2004

(1/3) years, Wente and van Siclen[3056] two (2) and a third (1/3) years as does Murnane[3057] who insists that this coregency is "remarkably well attested". Krauss,[3058] on the other hand, allows for no such coregency. The existence of coregencies would also shorten [i.e. lower] Egyptian chronology [and/or mathematically / logically suggest possible interregna between rulers hitherto and heretofore not considered] which has tended to place rulers in consecutive rather than overlapping succession.[3059]

76.21. The Egyptian 365.0-day Civil Calendar

XX. Though when exactly the Egyptian 365.0-day civil calendar originated, from the 3rd millennium BCE (Julian), the year of the Egyptian civil calendar consisted of 365.0 days with no leap year. Dates were expressed in terms of the year (of the king's reign), month (of each of the three (3) seasons), and the day (in terms of the 30-day months, and by the individual five (5) days for the epagomenai). Thus, in the Egyptian system, within a year, a date is expressed according to day, season and month [that we reckon today as] days numbered from 1 to 365.[3060]

XX. [For Egyptian historical chronology] three (3) calendrical structures are relevant:[3061]

1.) the Egyptian [365.0-day] civil calendar;

2.) Egyptian lunar time-reckoning;

3.) the Babylonian luni-solar calendar.

XX. Throughout Egyptian history, the [365.0-day] civil calendar was the dominant calendar of daily life.[3062]

XX. Probably no calendaric institution has continued over a longer period than the Egyptian [365.0-day year civil] calendar. After its uninterrupted use during all of Egyptian history, the Hellenistic astronomers adopted the Egyptian [365.0-day civil] year for their calculations. Ptolemy based all his tables in the *Almagest* on Egyptian [365.0-day civil] years. Even as late as 1543 CE (Julian), Copernicus (in *De revolutionibus orbium caelestium*) used Egyptian years.[3063]

XX. Dates within a year were expressed in terms of the three (3) seasons of four (4) months each:

3056 Wente and van Siclen 1976: 227-228 – cited by Hasel 2004
3057 Murnane 1977:44-57 – cited by Hasel 2004
3058 Krauss 1978:174-175 – cited by Hasel 2004
3059 Hasel 2004: 13
3060 Krauss and Warburton 2009: 126
3061 Depuydt 2006a: 459
3062 Depuydt 2006a: 459
3063 Neugebauer 1942/1983: 196

Akhet "inundation"; *Peret* "winter" (presumably the season of the "emergence" of the fields from the flood); *Shemu* "summer".[3064] [The Egyptian 365.0-day civil year consisted of] 12 months of 30 days plus five (5) epagomenal days, and its three (3) seasons of four (4) months [each numbered I-IV]: *Akhet*, *Peret*, and *Shemu*.[3065] [3066]

XX. In what follows, they will be called [Egyptian] Months 1 through 12. Month names have also been preserved.[3067] [3068] Originally months were numbered (not named), but a few month names are known from the Middle Kingdom (MK).[3069]

XX. In the period at hand, the names appear almost exclusively in 5th century BCE (Julian) Aramaic papyri from Egypt. The Greek variants are now conventionally used in modern historiography.[3070]

Table 76.2. Ancient Egyptian Months (Greek variants) – Adapted from Depuydt 2006a: 459 and Dershowitz and Reingold 2018: 30

3064 Hornung, Krauss, and Warburton 2006: 47
3065 cf. Hornung, Krauss, and Warburton (2006), Part I, Chapter 5 – cited by Depuydt 2006a: 459
3066 Depuydt 2006a: 459
3067 Depuydt 2006a: 459
3068 Dershowitz and Reingold 2018: 30
3069 Hornung, Krauss, and Warburton 2006: 47
3070 Depuydt 2006a: 459

#	**Greek**	**Egyptian**	**Days**	#	**Greek**	**Egyptian**	**Days**
1	Thoth	I Akhet	30	7	Phamenoth	III Peret	30
2	Phaophi	II Akhet	30	8	Pharmouthi	IV Peret	30
3	Hathyr/Athyr	III Akhet	30	9	Pachon	I Shemu	30
4	Choiak	IV Akhet	30	10	Payni	II Shemu	30
5	Tybi	I Peret	30	11	Epiphi	III Shemu	30
6	Mecheir/Mechir	II Peret	30	12	Mesore/Mesori	IV Shemu	30

				13	Epagomenae	(Unnamed)	5

XX. According to Neugebauer, the Egyptian calendar is an ideal tool: twelve (12) months of thirty (30) days each, five (5) additional days at the end, and no intercalation whatsoever. It is no wonder that the Hellenistic astronomers preferred this system to the Babylonian lunar calendar with its very irregularly changing months of twenty-nine (29) and thirty (30) days combined with a complicated [so-called Metonic 19-year] cyclic intercalation – not to mention the chaos of Greek and Roman calendars.[3071]

XX. Neugebauer remarks that for the same reason [of preferred simplicity] modern astronomy does not use the [more complicated] Gregorian calendar for computations but [rather the less complicated] Julian years instead.[3072] [However, since we no longer use the Julian calendar and instead use the Gregorian upgrade of the Julian, we (in modern times) think and date things in terms of the perspective relative to our current Gregorian dating system.]

XX. However, the ideal simplicity of the Egyptian calendar raises serious problems for the historian. Should we assume that [ancient Egyptian] astronomers – for the sake of their own calculations – imposed on the rest of the population a calendar with no respect for Sun and Moon? No scholar will accept this viewpoint.[3073]

XX. Only one other solution seems to remain: the simplicity of the Egyptian calendar is a sign of its primitivity.[3074] It is the remainder of prehistoric crudeness, preserved without change by the Egyptians, who – according to Neugebauer – are considered to be the most conservative race known in human history.[3075]

XX. What Neugebauer means is the fact that there is no astronomical phenomenon which possibly could impress on the mind of a primitive observer that a lunar month lasts [always exactly] 30 days and a solar year contains [exactly] 365 days. Observation during one (1) year is sufficient to convince anybody that in about six (6) cases out of twelve (12) the Moon repeats all its phases in only 29 days and never in more than 30 [days].[3076]

XX. [Furthermore] forty (40) years' observation of the Sun (e.g. [observation] of the dates of the equinoxes) must make it obvious that the [Egyptian] years fell short by 10 days [after this 40-year period, i.e. (365.25 – 365.0 days) == (0.25 days) x (40 years) == (10 days) in relation to the Julian solar year]. The inevitable consequence of these facts is – it seems to Neugebauer (1942) – that every theory

3071 Neugebauer 1942/1983: 196
3072 Neugebauer 1942/1983: 196
3073 Neugebauer 1942/1983: 196
3074 Neugebauer 1942/1983: 196
3075 Neugebauer 1942/1983: 196-197
3076 Neugebauer 1942/1983: 197

of the origin of the Egyptian calendar which assumes an astronomical foundation is doomed to failure.[3077]

XX. The Egyptian 365.0-day year is almost a quarter (¼) day shorter than the solar [Gregorian / Tropical] year – or cycle of seasons – of about 365.2422 days. The Egyptian [365.0-day civil] year therefore wanders or shifts [similar to the way the Islamic Pure Lunar calendar wanders, shifts, or migrates approximately 11 days every year] in relation to the solar [Gregorian / Tropical] year at a rate of about [i.e. approximately] one (1) day in four (4) years, and of exactly one (1) day in four (4) years in relation to the Julian calendar year of 365.25 days (that is three (3) years of 365 days followed by one (1) [year] of 366).[3078]

XX. Because the [Egyptian 365.0-day civil] calendar did not provide for an extra day every four (4) years, the date of the rising of Sirius [as well as the 1st day of the Egyptian month Thoth, i.e. Egyptian Sothic New Year's Day] in the [Egyptian 365.0-day] civil calendar shifted accordingly. The concept of a "Sothic cycle" for a complete shift of 1460 [Julian] years (i.e. 365.0 days × 4) is first documented in the Hellenistic Period. The earliest examples for dating in the [Egyptian 365.0-day] civil calendar of 365 days come from Djoser's reign [i.e. 3rd Dynasty], or ca. 150 years after its introduction.[3079]

XX. It is indeed correct that Pharaonic sources do not attest an intercalated day. It is well known that in 238 BCE (Julian) Ptolemy III attempted to introduce an improvement in the Egyptian calendar by adding a 366th day every four (4) years, anticipating the later Julian calendar reform, but his edict went unheeded.[3080]

XX. A second attempt under Augustus succeeded in imposing a form of the Julian calendar on Egypt. This so-called Alexandrian calendar remained in use in Egypt until the Arab conquest, and it still survives today as the liturgical calendar of the Coptic and Ethiopian churches.[3081]

XX. The Egyptian [365.0-day Civil] New Year [i.e. 1st day of Egyptian month Thoth, Egyptian Sothic New Year's Day that wanders] returns to the same day in the Julian calendar in exactly 1460 (365.0 × 4) [Julian year vs. 1461 Egyptian] years. In 664 BCE (Julian), [Egyptian] New Year [i.e. the 1st day of month Thoth] fell – according to Depuydt – on 05 February 664 BCE (Julian) for the first of four (4) successive years. By 332 BCE (Julian), it had receded – according to Depuydt – to 14 November.[3082]

XX. According to Depuydt, two (2) [Egyptian 365.0-day Civil] New Years fall in 1.) 521 BCE (Julian), which is a leap year of 366 days, like all years BCE (Julian) divisible by 4 after subtracting 1: the first

3077 Neugebauer 1942/1983: 197
3078 Depuydt 2006a: 459
3079 Hornung, Krauss, and Warburton 2006: 47
3080 Hornung, Krauss, and Warburton 2006: 48
3081 Hornung, Krauss, and Warburton 2006: 48
3082 Depuydt 2006a: 459

on 01 January 521 BCE (Julian), and the second 2.) 365 days later on 31 Dec 521 BCE (Julian).[3083] According to Depuydt, [Egyptian 365.0-day Civil] New Year then shifts to a day earlier in the Julian calendar – to 30 December – already in 517 BCE (Julian), not in 516 BCE (Julian).[3084]

XX. Alongside the [Egyptian] 365.0-day calendar, a lunar calendar was also used [in Egypt]. Specialists [in Egyptology and Astronomy] are not in accord on how it functioned in detail.[3085]

XX. There are no reports of solar and lunar eclipses from pharaonic Egypt. Thus ancient Egyptian data are absent in modern astronomical analysis of recorded eclipses.[3086] Eclipses that were recorded in the Near East such as the solar eclipse of 15 June 763 BCE (Julian)[3087] [3088] [3089] [3090] are of indirect importance to Egyptian chronology. The solar omen of year 10 of Murshili II would be important for the chronology of the Amarna period and its aftermath, if it were indeed an eclipse.[3091]

76.22. The Annual Flooding of the Nile

XX. The rising and falling of the Nile and the harvesting of grain and fruit occur regularly at certain times within the solar year. The Nile flood results principally from monsoon rains that fall over the Ethiopian plateau between mid-May and September. The Nile sunk to its lowest level in April/May,[3092] [and] towards the end of May and the beginning of June it began to rise.[3093]

XX. In modern times the dams constructed at Aswan beginning around 1900 CE (Gregorian) have prevented the annual flooding of the Nile Valley. Data recorded in the 19th century CE (Gregorian) and the Middle Ages provide information about the onset and duration of the flood which are crucial for evaluating pharaonic dates.[3094] [Alas such fresh, new annual Nile flood data is not possible because of the Aswan Dam.]

76.23. The Star of Sirius (Sothis / Sepdet (Spdt))

3083 Depuydt 2006a: 459-460
3084 Depuydt 2006a: 460
3085 Hornung, Krauss, and Warburton 2006: 48
3086 Krauss 2006a: 377
3087 Monday 15 June 763 BCE (Julian); Monday 07 June 763 BCE (Gregorian); 30 Simannu / Sivan 2998 AM; Astronomical Julian Day JD_X: 1442902.5; RD_Y: 1094906; RD_X: -278522; Sabbatical Cycle: 2; Metonic Cycle: 15;
3088 Tuesday 16 June 763 BCE (Julian); Tuesday 08 June 763 BCE (Gregorian); 01 Du'uzu / Tammuz 2998 AM; Astronomical Julian Day JD_X: 1442903.5; RD_Y: 1094907; RD_X: -278521; Sabbatical Cycle: 2; Metonic Cycle: 15;
3089 cf. Appendix Table X.2, X.194
3090 See Klinger 2006: 308 – cited by Krauss 2006a: 378
3091 Krauss 2006a: 378
3092 Krauss 2006a: 369
3093 Krauss 2006a: 369-370
3094 Krauss 2006a: 369

XX. Many stars and other celestial bodies undergo periods of invisibility when the Sun is nearby. These periods of invisibility are bounded by the dates of the star's heliacal rising and [heliacal] setting. The star is first glimpsed during morning twilight on the date of heliacal rising. The apparition (or time of observability) of the star ends on the date of heliacal setting when the Sun approaches too close to the object.[3095]

XX. As is well known, the ancient Egyptians based their calendar on the heliacal rising of Syrius (Sirius)[3096] and even built temples for its observations. The Maya divided their calendar into four (4) intervals based on heliacal phenomena of Venus,[3097] and the first day of an Islamic month is defined by the heliacal rising of the Moon[3098] [in the west at the beginning of the month exactly as the Babylonian / Hebrew month, and then rises all month before it sets in the east at the end of the month].

XX. A more exact marker of the new solar year [for the ancient Egyptians] was the annual reappearance of Sirius, the brightest star in the eastern sky just before Sunrise, signaling the solar induced climatic seasons of the agricultural year.[3099] The heliacal rising of the fixed star Sirius[3100] was the only astronomical event which occurred on a certain day in the [Egyptian] 365.0-day civil calendar [i.e. approximately every four (4) years the heliacal rising of Sirius (Sothis) moved one (1) day on the civil calendar].[3101]

XX. [For the Ancient Egyptians] the helical rising of the star Sothis [i.e. Sirius] provided an assured signal every year of the beginning of the new solar year. It kept to the strict solar timetable of 365.25 days, but its appearance was recorded on a calendar composed of only 365.0 days. As a result, the heliacal rising of Sothis would appear on the same day for four (4) years then on the next day of the civil calendar for the next four (4) years, and so on. It would take approximately 1460 [365.25-day Julian] years for the Sothic cycle to once again be synchronized with the [Egyptian] civil calendar [of 365.0 days].[3102]

XX. Sirius, the Dogstar in the constellation of Canis Major, was known to Egyptians as *Spdt* [*Sepdet*] after their goddess Sopdet, and as Sothis by the Greeks. As the Earth orbited around the Sun, Sirius could be observed for all but the 70 days of the year when it was obliterated from view by the Sun's light. Its reappearance came predictably every 365¼ days, known as its heliacal rising, and could be seen by the naked eye in Egypt's cloudless summer sky, but its observation depended on the arc of vision (arcus visionis). It was a reliable indicator of the beginning of the solar year, and that the

3095 Schaefer 1987: S19
3096 Lockyer: 1894 – cited by Schaefer 1987: S19
3097 Lounsbury 1978 – cited by Schaefer 1987: S19
3098 Schaefer 1987: S19
3099 Tetley 2014: 41
3100 Greek: Sothis
3101 Hornung, Krauss, and Warburton 2006: 47
3102 Tetley 2014: 40

anticipated inundation beginning the agricultural cycle was near.[3103]

XX. The striking reappearance of Sothis after 70 days was an expected event because the ancient Egyptians scrupulously observed the stars that were seen above the horizon throughout the year. Sirius was preceded by the constellation of Orion. R.A. Wells (1994) writes: "The red giant at the left shoulder of the figure of Orion, Betelgeuse (α Ori) [i.e. Alpha Ori], and the slightly fainter, bluer star in the right leg, Rigel (β Ori) [i.e. Beta Ori], rise close together in time. When they are high enough in the sky so that Sirius can just be seen rising, the 3 stars together form a very distinctive triangle pointing downwards." Together with other attendant stars the rising of Sirius was eagerly awaited and celebrated by the ancient Egyptians.[3104]

XX. In the course of 4000 years the date of the heliacal rising of Sirius moves forward with respect to the summer solstice by one (1) day in about 120 years. This is due to precession but tempered by the fact that Sirius lies far (~40°) south of the ecliptic. For a star in the ecliptic this forward motion would be one (1) day in about 75 years, as expected for a rate of precession of 1° in 72 years. The rather large proper motion of Sirius, which causes a displacement in the sky of about 1.5° in 4000 years, only causes a minor shift in the date of heliacal rising of about one (1) day over that period.[3105]

XX. The heliacal rise of Sirius was an important marker for the ancient Egyptian calendar. As such, a variety of studies have used calculated dates as a basis for deriving an absolute astronomical chronology.[3106] The ancient Egyptians made observations of the heliacal rising of Sirius as the basis for a sidereal calendar.[3107] [3108]

XX. In recent years, the exact date of the heliacal rise from Egypt has become critical in several areas of inquiry. Gurshtein has made a key point in his broad reconstruction of the history of the zodiac based on the dates on which the heliacal rising of Sirius starts its Great Cycle [of 1460 (365.25-day) Julian years as the basis but also taking into account the effects of precession, as well as its position on the ecliptic, and its forwardd proper motion mentioned above]. At the Oxford VI conference in La Laguna, several questioners tried to refute Gurshtein's thesis by arguing that his dates of the heliacal rising of Sirius were wrong since there should also be the effects of precession.[3109]

XX. Also at the same conference, Krauss offered an analysis of the heliacal rise of Sirius based on the Illahun papyri in an effort to improve the absolute chronology in the early 2nd millennium BCE (Julian). In addition, various older papers attempt to establish an absolute chronology with the heliacal rise of

3103 Tetley 2014: 41
3104 Tetley 2014: 41
3105 De Jong 2006: 438
3106 Schaefer 2000: 149
3107 Lockyer 1894 – cited by Schaefer 1987: S29
3108 Schaefer 1987: S29
3109 Schaefer 2000: 149

Sirius as the starting point: Read (1970, 1995); Parker (1970); Krauss (1985); van Oosterhout (1992).[3110]

XX. Currently, Egyptologists use dates for the heliacal rise of Sirius derived by the "Arcus Visionis" method going back to Ptolemy through the work of Schoch (1928). The latest works (Ingham (1969); van Oosterhout (1992)) have no improvements upon the older work.[3111]

XX. Astronomy has made great advances – both observationally and theoretically – since the time of Schoch. In particular, the old rule-of-thumb based on the Arcus Visionis is now known frequently to lead to large errors that invalidate many old conclusions. With these advances [in astronomy], modern astronomers can calculate the dates of Sirius' heliacal rise to a much greater accuracy than was possible by the old rules-of-thumb.[3112]

XX. The position of Sirius changes with time due to proper motion and precession. The proper motion of Sirius was taken from Hoffleit (1997).[3113] For exact precession, Schaefer has adopted the rigorous method of Meeus (1991).[3114]

XX. Any change in the observer's latitude in 30° north will result in close to one (1) day change for every degree of difference in latitude (i.e. the heliacal rise date becomes later as the observer moves north).[3115]

XX. What is the altitude of both Sirius and the Sun at the time of best visibility on the date of the heliacal rising? From Ptolemy to Schoch to Ingham, the situation is idealized with Sirius exactly on the horizon while the Sun is below the horizon by an angle equal to the Arcus Visionis.[3116]

76.24. Arcus Visionis

XX. For those few researchers who consider the question (such as Lockyer (1894) and Aveni (1972)), Sirius is always taken to be visible to the horizon at heliacal rise. But this is easily disproven since the extinction (0.35 magnitudes per air mass) [i.e. variable] at the horizon (with the equivalent of roughly 40 air masses) [i.e. variable] will dim Sirius by 14 magnitudes. Even one (1) degree above the horizon (with 26 air masses) and a good extinction (0.26 magnitudes per air mass), Sirius will still only be barely visible even under a dark night sky.[3117]

3110 Schaefer 2000: 149
3111 Schaefer 2000: 149
3112 Schaefer 2000: 149
3113 Schaefer 2000: 150
3114 Schaefer 2000: 150-151
3115 Schaefer 2000: 151
3116 Schaefer 2000: 151
3117 Schaefer 2000: 151

XX. In general, the altitude of best visibility [i.e. variable] will be a trade-off between sky brightness [i.e. variable] and extinction [i.e. variable]. For the particular case of Sirius, the altitude of Sirius at the heliacal rise will be around 6° while the altitude of the Sun will be around -5°. Although the Arcus Visionis formulation is flawed (it is nevertheless convenient for historical discussions), the Arcus Visionis for the heliacal rise of Sirius from Egypt is around 11°.[3118]

XX. The natural variations [i.e. variables] in the extinction coefficient [i.e. variable], clouds [i.e. variable], and the phase of the Earth's rotation [i.e. variable] all create an apparently random scatter in the observed date of the heliacal rise of a few days and in the threshold Arcus Visionis of a few degrees.[3119]

XX. This uncertainty must be carried along in all evaluations. Frequently, in the past, differences in Arcus Visionis or date have formed the basis for significant conclusions, whereas we now know those differences to be insignificant. The moral is that strict mathematical models with no attention to error bars can easily result in wrong answers when the model is pushed too far.[3120]

XX. In the Almagest and Planetary hypotheses, Ptolemy gives the Arcus Visionis for Jupiter (magnitude -2.0) of 10° and 9°, and for Mercury (magnitude -1.7) of 10° and 12°. No value is quoted for Sirius.[3121]

XX. To compare these historical reports with the modern calculations of Sirius, a correction of roughly 1° must be subtracted to allow for the larger typical azimuth of Sirius as compared to sources on the ecliptic, and a correction of roughly 1° in the other direction is required to correct for the magnitudes. So based on Ptolemy's Arcus Visionis values, Sirius would have an Arcus Visionis of 9° – 10° or 10° – 12°.[3122]

XX. The understanding of heliacal phenomena is important to the growing studies of archaeoastronomy and the history of science. However, the latest theoretical work on this problem for stars and planets is by Ptolemy. Ptolemy's method uses the Arcus Visionis as a function of magnitude and is discussed and tabulated by Lockyer (1894), Aveni (1972), Schoch (1924), and Neugebauer (1929). To make matters worse, the most recent sets of actual observations in the literature are those of the Babylonians and Egyptians.[3123]

XX. Schoch (1924), Neugebauer (1929), and Huber (1977) have analyzed Babylonian cuneiform texts

3118 Schaefer 2000: 151
3119 Schaefer 2000: 151-152
3120 Schaefer 2000: 152
3121 Schaefer 2000: 152
3122 Schaefer 2000: 152
3123 Schaefer 1987: S19

and derived the Arcus Visionis for over a hundred observations. Typical values are within a degree or two (2) of 6°.[3124]

76.25. Ptolemy's Almagest vs. Babylonian Astronomers

XX. Ptolemy's great astronomical treatise, usually known as the *Almagest*, has probably aroused more interest among historians of astronomy than any other work from before the Renaissance. According to Steele, it was written by Claudius Ptolemaeus (c. 100 – 175 CE (Julian)) in Alexandria, and according to Toomer, it was published sometime between 150 CE (Julian) and 161 CE (Julian),[3125] the *Mathematike Syntaxis*, to give it its Greek name, was a comprehensive treatise of mathematical astronomy.[3126]

XX. The *Almagest* is primarily a work of astronomical theory which includes observations of lunar eclipses, occultations, equinoxes, and over one thousand stars. The reliability of Ptolemy's observations has been questioned since the 19th century CE (Gregorian).[3127]

XX. Around the turn of the century a number of astronomers made investigations into the secular [lunar orbital tidal] accelerations of the Moon and Sun, which are related to long-term changes [i.e. ΔT (Delta T)] in the Earth's rate of rotation [i.e. Length of Day (LOD)]. If the rate of slowing of the Earth's rotation were constant, then the difference between a clock measuring time by the Earth's rotation and a uniformly running clock, known as the Earth's Rotational Clock Error or ΔT (Delta T), would increase quadratically over time.[3128]

XX. In fact there are non-tidal effects which produce a deviation from uniform deceleration, but to a first approximation this quadratic effect dominates. By analyzing the circumstances of eclipses in the past it is possible to determine the change in the Earth's rotational clock error [i.e. ΔT (Delta T)] over time.[3129]

XX. Clearly, reports from antiquity are of great value in such determinations as the clock error is – according to Steele – large at this period and so can be detected in relatively – according to Steele – imprecise observations. Thus the eclipse observations in the *Almagest*, which date back to 721 BCE (Julian), were viewed by Newcomb (1878) and others with great interest.[3130]

3124 Schaefer 1987: S30
3125 Toomer 1984 – cited by Steele 2000b: 89
3126 Steele 2000b: 89
3127 Steele 2000b: 89
3128 Steele 2000b: 90
3129 Steele 2000b: 90
3130 Steele 2000b: 90

XX. The only other eclipse observations from antiquity available to these astronomers [apart from ancient Babylonian and Assyrian records] were those recorded in the Greek and Latin classics. However, there are problems of interpretation and dating for many of these records.[3131]

XX. With the development of lunar laser ranging techniques to measure the lunar acceleration directly (Dickey et al. 1994) and the discovery of a large number of historical eclipse observations from Babylon and China, the eclipse observations in the *Almagest* are – according to Steele – not now so important in determinations of the Earth's rotational clock error. These recent improvements in our knowledge of the changes in the rate of rotation of the Earth allow accurate determinations of the circumstances of eclipses in the past to be made.[3132]

XX. Older eclipse canons, such as the famous *Canon der Finsternisse* of von Oppolzer (1887), which have been widely used to evaluate historical eclipse observations, do not make sufficient allowance for the Earth's variable rate of rotation and can be in error – according to Steele – by as much as an hour for the times of eclipses in early antiquity.[3133]

XX. In addition, the recovery of the large, although far from complete, archive of eclipse records from Babylon allows us to compare the Babylonian eclipses reported by Ptolemy with contemporary examples from the original source material.[3134]

XX. The many astronomical observations reported by Ptolemy in his *Almagest* fall into three (3) distinct groups: 1.) those said to have been observed in Babylon; 2.) those observed by early Greek astronomers; 3.) those observed by Ptolemy and his contemporaries. It would appear that Hipparchus was the source of most of the observations in the first two (2) categories.[3135]

XX. It is known that Hipparchus made a compilation of eclipse records for the 600 years before his time, and as Ptolemy must have had available to him a more or less complete record of lunar eclipses so that he could select those that fulfilled the criteria for his calculations [i.e. variables], it would seem that this compilation was his source.[3136] [3137]

XX. In deducing the basic parameters of his lunar model, Ptolemy chose to use observations of lunar eclipses for "these are the only observations which allow one to determine the lunar position precisely; all others, whether they are taken from passages [of the Moon] near fixed stars, or from [sightings with] instruments, or from solar eclipses, can contain a considerable error due to parallax"[3138] Ptolemy

3131 Steele 2000b: 90
3132 Steele 2000b: 90
3133 Steele 2000b: 90-91
3134 Steele 2000b: 91
3135 Steele 2000b: 91
3136 Toomer 1988 – cited by Steele 2000b: 91
3137 Steele 2000b: 91
3138 Almagest iv, 1; trans. Toomer (1984: 173) – cited by Steele 2000b: 91

describes no observations of solar eclipses, only 18 lunar obscurations.[3139]

XX. Ptolemy used his own chronology throughout the *Almagest*. This commenced with the beginning of the reign of Nabonassar (equivalent to 26 February 747 BCE (Julian)) for, as he says, "that is the era beginning from which the ancient observations are, on the whole, preserved down to our own time."[3140] He also uses the Egyptian year of twelve (12) 30-day months followed by 5 extra days, making a total of 365 days. This system, which allows accurate day counts to be made with ease, is probably due to Hipparchus.[3141] In addition, when discussing observations made during the night, Ptolemy often gives a double date to avoid confusion.[3142]

XX. Most of the times quoted by Ptolemy are in seasonal hours.[3143] A seasonal hour is defined as being one-twelfth (1/12) of the length of the day or night [i.e. variable], and is thus dependent on the time of year. Ptolemy's first step [i.e. variable] in using these times is to convert them to equinoctial hours, a constant unit of time defined as one twenty-fourth (1/24) of the length of time between two (2) consecutive midnights.[3144]

XX. Ptolemy uses Alexandria as his meridian [i.e. not Babylon] and so in his analysis of the lunar eclipse observations, reduces all of his local times [i.e. variable] to this meridian [i.e. variable].[3145]

XX. The earliest ten (10) lunar eclipses reported in the *Almagest* are all said to have been observed in Babylon. According to Ptolemy, Babylon is 5/6 of an hour – or 12.5 degrees – to the east of Alexandria. The true difference in longitude is 14.5 degrees. This discrepancy reflects one of the greatest problems faced by the ancient Greek astronomers: the determination of geographical longitudes.[3146] Therefore, this imprecise value [i.e. variable] may have affected Ptolemy's calculations.

XX. The earliest known Neo-Babylonian [astronomical] observational record apparently dates from the [Year 0] Accession Year of Nabonassar, the same year as Ptolemy claims that the earliest Babylonian records were available to him. Whether this is because Nabonassar either initiated the program of regular astronomical observation or destroyed all of the earlier historical records as claimed by Berossus,[3147] or is simply a curious coincidence may never be known unless earlier records come to light in the future.[3148]

3139 Steele 2000b: 91
3140 Almagest iii, 7; trans. Toomer (1984: 166)) – cited by Steele 2000b: 93
3141 Toomer 1996 CE – cited by Steele 2000b: 93
3142 Steele 2000b: 93
3143 i.e. temporal hours, *horae temporales*
3144 Steele 2000b: 93
3145 Steele 2000b: 94
3146 Steele 2000b: 94
3147 Brinkman 1968: 227
3148 Steele 2000b: 94

XX. Around the same time the astronomers in Babylon were making systematic observations of eclipses. These were recorded in the astronomical diaries: detailed records of a variety of astronomical observations and predictions including solstices and equinoxes, planetary positions and visibilities, and lunar phenomena.[3149]

XX. The other striking difference between the two (2) records is the way they state the time of the eclipse. The cuneiform records give the times in units called *bēru* and UŠ, where there are 30 UŠ [i.e. 120 minutes] in one (1) *bēru* [i.e. 2 hours]. One (1) UŠ is equal to four (4) minutes of time, and so there are 360 UŠ [i.e. 1440 minutes] in one (1) day. It has therefore become customary to render UŠ as "degree" in translations. The UŠ [i.e. 4 minutes] is an equinoctial unit of time as it does not vary in length with the seasons.[3150]

XX. However, Ptolemy generally gives the time of the Babylonian observations in seasonal hours [i.e. variable and thus room for "lost-in-translation" time-conversion errors]. Whilst there is some evidence that the Babylonians used seasonal hours in their horoscopes, they did not use them in their astronomical texts. The only seasonal time unit that is ever used in a description of an eclipse is the period of length one-third of the night, known as a "watch". However, the use of watches generally only occurs in the reports of the astrologers and scholars.[3151]

XX. Customarily the Babylonians reported the time of the eclipse, the duration of the phases of the eclipse, and an estimate of the magnitude of the eclipse. The time of the eclipse relates to the moment that the eclipse began. This is measured relative to either Sunset or Sunrise, whichever is nearer.[3152]

XX. If we are to assume, as seems unavoidable, that the astronomical diaries or the eclipse lists were the source of the Babylonian observations in the *Almagest*, we must ask how the times came to be converted into seasonal hours.[3153]

XX. There is only one (1) record of an eclipse in the *Almagest* that is also preserved on an extant Babylonian tablet dated to 16 July 523 BCE (Julian).[3154] [3155] [3156]Furthermore, the time [i.e. variable] of the eclipse given in the cuneiform record is for the start of the eclipse, whereas Ptolemy has the time of its middle [which begs the question if Ptolemy's calculation with a different time for the eclipse

3149 Steele 2000b: 95
3150 Steele 2000b: 96
3151 Steele 2000b: 96
3152 Steele 2000b: 96
3153 Steele 2000b: 97
3154 Wednesday 16 July 523 BCE (Julian); Wednesday 10 July 523 BCE (Gregorian); 14 Du'uzu / Tammuz 3238 AM; Astronomical Julian Day JD_X: 1530593.5; RD_Y: 1182597; RD_X: -190831; Sabbatical Cycle: 4; Metonic Cycle: 8; 17 (Egyptian Month 07 Phamenoth) 225 AN; 81956 days since beginning of Epoch;
3155 Thursday 17 July 523 BCE (Julian); Thursday 11 July 523 BCE (Gregorian); 15 Du'uzu / Tammuz 3238 AM; Astronomical Julian Day JD_X: 1530594.5; RD_Y: 1182598; RD_X: -190830; Sabbatical Cycle: 4; Metonic Cycle: 8; 18 (Egyptian Month 07 Phamenoth) 225 AN; 81957 days since beginning of Epoch;
3156 Steele 2000b: 97

affected the results and most certainly would affect the variable ΔT (Delta T)].[3157]

XX. It is found on the Babylonian tablet BM 33066 as the total eclipse of Regnal Year 7 of Cambyses, month IV, night 14, 1 2/3 *bēru* after Sunset. In the *Almagest* v, 14 it is recorded as having occurred in the 7th year of Cambyses, which is the 225th year from Nabonassar (AN), Phamenoth 17/18 in the Egyptian calendar:[3158]

(5) BC 523 July 16 / 17 (mag. = 0.53)

... Again in the seventh year of Kambyses, which is the 225th year from Nabonassar, Phamenoth [month VII] 17/18 in the Egyptian calendar [-522 July 16/17], 1 [equinoctial] hour before midnight at Babylon, the Moon was eclipsed half its diameter from the north. Thus the eclipse occurred about 1 5/6 equinoctial hours before midnight at Alexandria.[3159]

XX. According to Steele, with the exception of the eclipse in 523 BCE (Julian), all of the recorded magnitudes of the partial lunar eclipses [by the Babylonian astronomers] are overestimated. Interestingly, the 523 BCE (Julian) eclipse is the only case in which more than half of the Moon was eclipsed. Thus – according to Steele – these reports suggest that the Babylonian astronomers tended to overestimate the magnitude of small eclipses.[3160] According to him, this is probably due to the difficulty in distinguishing between the penumbral and umbral shadow.[3161] However, Steele concludes that most Babylonian estimates of magnitude are fairly accurate.[3162]

XX. The three (3) groups of eclipse observations recorded in the *Almagest* show distinct differences in their variations from modern computations. Unsurprisingly, given their age – according to Steele – the Babylonian observations appear to be the least accurate of the three (3) groups. However [as if of seeming no import to Steele], unlike both the early and later Greek observations, they show negligible systematic error in their timing. Indeed the systematic error in the timing of the early Greek observations is very serious and must lead us to question how representative these records are of the observations being made at this period.[3163]

XX. According to Steele, the Babylonian astronomers appear to have been greatly inferior to their Greek counterparts in estimating the magnitude of a lunar eclipse. Perhaps the Greek astronomers developed techniques for directly measuring the fraction of the diameter of the Moon covered by shadow using instruments [i.e. a human-made variable], whereas the Babylonians may have simply

3157 Steele 2000b: 98
3158 Steele 2000b: 97-98
3159 *Almagest*, V, 14; trans. Toomer (1984: 253) – cited by Stephenson 1997: 101
3160 Steele 2000b: 100
3161 Steele 2000b: 100
3162 Steele 2000b: 98
3163 Steele 2000b: 105

estimated the degree of obscuration by eye. The trend of the Babylonians to – according to Steele – overestimate the extent of small eclipses and to – according to Steele – underestimate the extent of large eclipses – according to Steele – supports this hypothesis.[3164]

76.26. Dendrochronology and Carbon-14 Dating

XX. Turning to the natural sciences, two (2) absolute dating methods are particularly relevant, although neither is without its problems. Dendrochronology dates samples of ancient wood by counting annual growth rings, a satisfying precise method. However, dendrochronology can inform only as to the year a tree was cut down, while wood components found in excavated buildings may have been used and reused for centuries after the tree was felled.[3165]

XX. Peter Kuniholm and his colleagues have been instrumental in providing a sequence of tree-ring dates for Anatolia spanning a millennium and a half, from the late 3rd to the early 1st millennium BCE (Julian). Unfortunately, this sequence is "floating", that is [to say]: not yet anchored to a deposit with a known calendar-year date.[3166]

XX. A date range for this floating sequence has been proposed through the use of high-precision carbon-14 dates of wood samples from within the sequence. The uncertainties of these methods are illustrated by the fact that the date range proposed in 1996 was revised upward in 2003 by twenty-two (22) years.[3167]

XX. Carbon-14 (radiocarbon) analysis allows for the dating of organic material through estimation of the number of years since organism died, calculated through the assessment of the amount of decay of carbon-14 absorbed during the organism's lifetime. Contrary to the assumptions of the method's originators, however, the levels of atmospheric carbon-14 were not consistent [i.e. variable] through time, and thus carbon-14 dates must be calibrated by comparison to tree-ring data of known age (note how carbon-14 and dendrochronology are used as checks on one another).[3168]

XX. Adding a further note of uncertainty is the steady modification and revision of calibration methods and results over the past few decades. Once a carbon-14 date is calibrated, it will often have a standard deviation [of error] that allows a confident assignment only within a century or two (2) [i.e. inaccuracy of possibly 100 – 200 years].[3169]

3164 Steele 2000b: 105
3165 Schwartz 2008: 451
3166 Schwartz 2008: 451
3167 Schwartz 2008: 451
3168 Schwartz 2008: 451
3169 Schwartz 2008: 451

XX. Because the uncertainties in absolute chronology during the 2nd millennium BCE (Julian) rarely involve much more than a century, the usefulness of the carbon-14 method has been limited. Radiocarbon dating, however, has grown increasingly precise in recent years, and the technique may provide much-needed assistance in the near future.[3170]

XX. According to Schwartz (2008), the most significant use of natural-science methods in addressing the problems of 2nd-millennium BCE (Julian) chronology has involved a combination of dendrochronological and historical evidence.[3171] Excavations of the Sarıkaya palace at Acemhöyük in central Anatolia [i.e. Turkey] revealed clay seal impressions associated with the northern Mesopotamian king Shamshi-Adad I, presumably removed from deliveries of goods sent from his realm.[3172]

XX. Dendrochronological analysis of wood beams from the palace yielded a date of 1774 +4/-7 BCE (Julian), suggesting that Shamshi-Adad died sometime after that date – unless the sealings had been moved from an earlier building.[3173]

76.27. Dendrochronology, Solar Eclipses, and Shamshi-Adad

XX. This result is important because Shamshi-Adad is known to have died midway through the reign of Hammurabi of Babylon. Dating Shamshi-Adad's reign from the late 19th to the early 18th century BCE (Julian) coincides with the traditional Middle Chronology dates.[3174]

XX. Related data has recently been provided by analysis of the so-called Mari Eponym Chronicle which records a solar eclipse observed in the year after Shamshi-Adad's birth. Bearing the Acemhöyük results in mind, Cécile Michel reviewed the years during which total solar eclipses occurred and proposed a date of 1834 BCE for the birth of Shamshi-Adad.[3175]

XX. In the Mari Eponym Chronicle a "darkening of the Sun" is mentioned around the birth of Šamši-Adad. The text has been generally interpreted as referring to the occurrence of a solar eclipse. There have been several attempts to date the birth of Šamši-Adad based on this solar eclipse.[3176] Since Šamši-Adad was not born in Assur but most probabably near Eshnunna, where his family resided at the time, it is quite well possible that the solar eclipse around his birth was observed near Eshnunna, about 200 km South East of Assur.[3177]

3170 Schwartz 2008: 451
3171 Schwartz 2008: 451
3172 Schwartz 2008: 451-452
3173 Schwartz 2008: 452
3174 Schwartz 2008: 452
3175 Schwartz 2008: 452
3176 De Jong 2013: 156
3177 De Jong 2013: 157

XX. According to the text of the Mari Eponym Chronicle the eclipse took place in the year after Šamši-Adad was born.[3178] Partial solar eclipses will pass unnoticed for a naked-eye observer unless the Sun is more than about 95% eclipsed[3179] or the eclipse happens very close to the horizon (within a few degrees) so that the Sun can be looked at without blinding the observer.[3180]

XX. Because Shamshi-Adad is said to have died seventy-three (73) years after the eclipse [i.e. at approximately age 74-75], his death would be assigned to 1760 BCE (Julian) [and thus his birth to 1834 BCE (Julian)], requiring a "modified" Middle Chronology that lowers the dates used for him in this volume by sixteen (16) years. It must be noted, however, that the authenticity of the recorded event could be doubted, given the likely propagandistic intentions of the document from which it originated. Even if the eclipse were historical, it is uncertain where it was observed and whether it was total or partial.[3181]

XX. The eclipse data [that De Jong reckons for this solar eclipse] have been computed with a program that uses state-of-the-art astronomical ephemerides of the Sun[3182] and the Moon[3183] with modern values [i.e. variable] for the secular lunar [orbital tidal] acceleration [i.e. variable] and the clock-time correction [i.e. variable] implemented. The eclipse predictions of this program have been checked against the *Five Millennium Catalog of Solar Eclipses: -1999 to +3000 by Espenak and Meeus (2009)*.[3184]

XX. According to De Jong, the program has two (2) unique features: 1.) proper corrections for atmospheric refraction [i.e. variable] are applied near the horizon [i.e. variable], and 2.) extrapolation errors [i.e. variable] in the clock-time correction [i.e. variable] can be specified as a free parameter [i.e. variable].[3185]

XX. Extrapolation errors in the clock-time correction [i.e. ΔT (Delta T)] represent the uncertainty in the gradual slowing down of the rate of rotation of the Earth which affects the exact location on Earth of the zone of totality of a solar eclipse and, consequently, also the magnitude of partial solar eclipses at geographical locations outside the zone of totality. The magnitude [i.e. variable] of the standard error [i.e. variable] in the clock-time correction [i.e. variable] during the 2nd millennium BCE (Julian) has recently been estimated by Huber (2006, Table 3) based on an extrapolation of the errors in the clock-time corrections of well-documented historical eclipses during the 1st millennium BCE (Julian) taken

3178 De Jong 2013: 156
3179 Muller and Stephenson 1975: 467 – cited by De Jong 2013
3180 De Jong 2013: 157
3181 Schwartz 2008: 452
3182 Bretagnon and Simon 1986 – cited by De Jong 2013
3183 Chapront-Touzé and Chapront 1991 – cited by De Jong 2013
3184 De Jong 2013: 156
3185 De Jong 2013: 156

from Morrison and Stephenson (1982).[3186]

XX. De Jong's first step in the fine-tuning of the Old Babylonian chronology is to make use of the well-established synchronism that Šamši-Adad died in the last month of the Assyrian year with eponym Ṭāb-Ṣilli-Aššur (REL 197) during the 18th year of the Old Babylonian king Hammurabi.[3187] According to Barjamovic et al., Šamši-Adad was born in REL 126 so that he died at the advanced age of 71 years in REL 197 which corresponds to absolute time 1776 BCE (Julian) ± 10 years (at the 95% confidence level).[3188]

XX. Based on this synchronism the date of the first year of the reign of Hammurabi is then constrained to 1793 BCE (Julian) ± 10 years. Using the well-established Old Babylonian relative chronology of the Hammurabi dynasty (e.g. Hunger and Pingree 1999, p. ix) this implies that the first year of the reign of king Ammī-ṣaduqa is constrained to 1647 BCE (Julian) ± 10 years, and the fall of Babylon to 1596 BCE (Julian) ± 10 years.[3189]

XX. However, according to De Jong, these much stricter limits to the absolute dating of the Old Babylonian chronology imply that of all the candidate chronologies allowed by the Venus observations, only the two (2) Middle Chronologies remain as viable candidates.[3190]

76.28. Egyptian Historical Chronology and Radiocarbon Dating

XX. While state-of-the-art application of radiocarbon dating and Bayesian modeling are able to provide essential contributions to (and refinements of) Egyptian historical chronology, it was in fact the Egyptian historical chronology that stood at the beginning of the development of the method of radiocarbon dating.[3191]

XX. In the seminal work on Ancient Egyptian Chronology edited by Erik Hornung, Rolf Krauss, and David A. Warburton and published in 2006,[3192] despite the skeptical view of the editors (see Hornung's earlier statement), a special chapter was devoted to radiocarbon that was authored by Sturt W. Manning.[3193] Still the method stood on the fringes of the discipline of Egyptology, even in a volume that was especially devoted to chronology.[3194]

3186 De Jong 2013: 156
3187 Charpin and Ziegler 2003: p.170ff – cited by De Jong 2013: 158
3188 De Jong 2013: 158
3189 De Jong 2013: 158
3190 De Jong 2013: 158
3191 Höflmayer 2016: 3
3192 Hornung, Krauss, and Warburton 2006 – cited by Höflmayer 2016: 8
3193 Manning 2006 – cited by Höflmayer 2016: 8
3194 Höflmayer 2016: 8

XX. Manning noted that far too few good examples of modern research programs existed that provided transparent, good, and useful data that actually could have helped to refine the Egyptian historical chronology. And thus the author concluded: "High quality radiocarbon dating offers an important but as yet not fully exploited resource for Egyptology."[3195] [3196]

XX. Although up to the mid-2000s radiocarbon dating for Dynastic Egypt was almost never carried out in a systematic way with clear research questions, sampling strategy, and interpretation, since then results of two independent projects set the radiocarbon record for Egypt on new grounds.[3197] The first project was entitled "Radiocarbon Dating and the Egyptian Historical Chronology" and was carried out by Christopher Bronk Ramsey, Michael W. Dee, and colleagues at the University of Oxford[3198] In general, this project provided results that are in agreement with the Egyptian historical chronology.[3199]

XX. The second project was SCIEM 2000 ("The Synchronization of Civilizations in the Eastern Mediterranean in the Second Millennium BC"), directed by Manfred Bietak and carried out at the Austrian Academy of Sciences. Although this project primarily focused on establishing chronological synchronisms between Egypt, the ancient Near East, Cyprus, and the Aegean based on archaeological data, especially the first appearances of key-pottery wares (see Bietak 2013), it also included a subproject on radiocarbon dating directed by Walter Kutschera.[3200]

XX. This subproject produced the most substantial sequence of radiocarbon dates that is currently available from a single site in Egypt, from Tell el-Dabʿa, ancient Avaris, in the eastern Nile Delta.[3201]. Although the Oxford project provided dates that were in agreement with historical estimates, radiocarbon dates for Tell el-Dabʿa were in gross conflict with the archaeological/historical date estimations of the excavator. Although at this time the question of what leads to this divergence cannot be conclusively answered, several lines of evidence seem to suggest that a major redating of archaeological phases and synchronisms within Egypt and the Near East might be in order.[3202] [3203]

XX. The radiocarbon sequence for the site of Tell el-Dabʿa (ancient Avaris) in the eastern Nile Delta provided results that are in gross conflict with calendar dates proposed by the excavator (Kutschera et al. 2012). This difference of about 120 years is of considerable significance not only for the site itself but also for our understanding of the history and development of Egyptian history and cultural development during the Second Intermediate Period and early New Kingdom.[3204]

3195 Manning 2006: 352
3196 Höflmayer 2016: 8
3197 Höflmayer 2016: 8
3198 Bronk Ramsey et al. 2010; Shortland and Bronk Ramsey 2013 – cited by Höflmayer 2016: 8-9
3199 Höflmayer 2016: 9
3200 Höflmayer 2016: 9
3201 Kutschera et al. 2012 – cited by Höflmayer 2016: 9
3202 See Manning et al. 2014; Höflmayer 2015; Höflmayer forthcoming – cited by Höflmayer 2016: 9
3203 Höflmayer 2016: 9
3204 Höflmayer 2016: 12

XX. Indeed, many open questions regarding chronology and absolute dates center on this site and its links with the historical chronology. Because the site is of utmost relevance, not only for the history of the Egyptian Second Intermediate Period but also for linking the relative chronological sequences of the Levant, Cyprus, and (partly) of the Aegean to Egypt.[3205]

XX. Overall, the site spans a stratigraphy of about 600 years (start of the Middle Kingdom to Amenhotep II [of the New Kingdom]. A substantial sequence of radiocarbon evidence from the site of Tell el-Dabʿa, however, challenges the absolute calendar dates for the site's stratigraphy as put forward by the excavator.[3206]

XX. On average, radiocarbon dates are about 120 years higher than dates for the site's stratigraphy based on the assumed datum lines with the Egyptian historical chronology. These results challenge the validity of the proposed datum lines and require a rigorous reanalysis of the site's stratigraphy and dating before reaching a final verdict.[3207]

XX. If the high radiocarbon dates for Tell el-Dabʿa were to be accepted, much of the site's stratigraphy would have to be shifted upward, while at the same time the Egyptian historical chronology only has limited flexibility in the amount of some three (3) to four (4) decades, for example, for the beginning of the New Kingdom. If Tell el-Dabʿa had to be redated, also the absolute calendar dates for the relative sequences of the Middle and early Late Bronze Ages of the Levant would have to be adjusted, because much depends on the archaeological synchronisms with Tell el-Dabʿa. Indeed, preliminary analysis of available radiocarbon dates from the Levant (Tell el-Burak, Lebanon, and Tel Ifshar, Israel) point to similar high dates.[3208] [3209]

XX. The great benefit of radiocarbon dating is that organic remains can be dated independently in different regions using the same method. We are no longer dependent on archaeological synchronisms between, for example, the Nile Valley and the Levant, Cyprus, or the Aegean to create a coherent framework of, for example, the eastern Mediterranean.[3210]

XX. So far, absolute calendar dates for the Egyptian Old, Middle, and New Kingdoms can be based on radiocarbon dates. There are also a few sequences for the Early and Middle Bronze Ages of the Levant available that allow chronological synchronization based on radiocarbon dating.[3211] [3212]

3205 Höflmayer 2016: 12
3206 Höflmayer 2016: 13
3207 Höflmayer 2016: 14
3208 Höflmayer 2015 – cited by Höflmayer 2016: 14
3209 Höflmayer 2016: 14
3210 Höflmayer 2016: 15
3211 See e.g., Regev et al. 2012; Marcus 2013; Höflmayer et al. 2014; Höflmayer 2015; Höflmayer et al. 2016a; Höflmayer et al. 2016b – cited by Höflmayer 2016: 15
3212 Höflmayer 2016: 15

XX. The special case of Tell el-Dabʿa with radiocarbon results that are about 120 years higher than dates proposed by the excavator will undoubtedly raise discussions for the years to come. If they were to be accepted, much of the Middle Bronze Age chronological framework for the Levant would have to be shifted upward. So far, radiocarbon results from sites in the Levant support the high dates from Tell el-Dabʿa.[3213]

XX. Future work will fill current gaps and continue to refine the Egyptian chronology. It is hoped that the appeal once expressed by Sturt Manning will indeed be heard in our discipline: "radiocarbon dating should become the friend of Egyptologists."[3214] [3215]

76.29. Problems of Chronology: Egypt and the Aegean

XX. According to Cline, relative dating is essentially settled for the 2nd millennium BCE (Julian) in both the Aegean and Egypt. For example, we know the order in which the New Kingdom pharaohs ruled: Amenhotep III ruled earlier than did Tutankhamun who in turn ruled earlier than did Ramesses II. We also know that the Late Bronze Age followed the Middle Bronze Age in both the Aegean and the ancient Near East.[3216]

XX. It is the absolute dating that poses the problems in both the Aegean and Egypt during the 2nd millennium BCE (Julian). Attempting to determine the years during which a pharaoh ruled, or when an event such as the eruption of the Thera volcano in the Aegean took place, is not easy when dealing with people and events that occurred nearly four (4) thousand years ago.[3217]

XX. For example, until fairly recently it was accepted by most Aegean prehistorians that the eruption of the volcanic island of Thera took place in approximately 1480+/- BCE (ca. 1525 BCE (Julian) when modified) and that the resulting volcanic ash, tephra, and other debris may have rained down upon the nearby island of Crete, dramatically affecting the crops and perhaps even paving the way for an invasion of Crete by Mycenaeans from the Greek mainland.[3218]

XX. In the late 1980s and early 1990s, however, Philip Betancourt and Sturt Manning published a series of initially independent articles suggesting that the eruption had taken place nearly 150 years earlier in about 1628 BCE (Julian). Their evidence, which eventually included radiocarbon dates from short-lived plants on the island of Thera itself as well as data from ice cores taken from as far away as

3213 Höflmayer 2016: 15
3214 Manning 2006: 354 – cited by Höflmayer 2016: 15
3215 Höflmayer 2016: 15
3216 Cline 2008: 453
3217 Cline 2008: 453
3218 Cline 2008: 453

Greenland, provoked a chronological debate that has to date lasted more than two (2) decades and shows no indication of being resolved.[3219]

XX. As a result, two (2) sets of chronologies of the Aegean area are currently in use: the so-called Traditional Chronology which begins the Late Helladic and Late Minoan periods at about 1625 BCE (Julian), and the High Chronology which begins those periods at about 1700 BCE (Julian).[3220]

XX. There are similar problems in Egypt where one finds High, Middle, Low, and Ultra-Low Chronologies in use by various archaeologists and Egyptologists. While convention usually dictates the use of the Middle Chronology as a compromise, some scholars adhere vigorously to alternative schemes. As long as a scholar makes clear which chronology is being used and provides absolute dates (e.g. 1390 – 1352 BCE (Julian) for the reign of Amenhotep III), others can accordingly adjust their own dating schemes.[3221]

XX. Of all the Egyptian paintings depicting Aegean peoples, places, or goods, the most recently discovered, and perhaps the most interesting, are those found by Manfred Bietak at Tell el-Dabʿa in the Nile Delta region of Egypt. Now identified as the site of ancient Avaris, capital city of the Hyksos who ruled Lower Egypt from about 1640 to 1550 BCE (Julian), Tell el-Dabʿa was also rebuilt and reoccupied by the Egyptian rulers of [New Kingdom] Dynasty 18.[3222]

XX. The fragmentary paintings that adorned the walls of one or more palaces at the site were originally dated by Bietak and his team to the Hyksos period (just prior to the New Kingdom), but are now thought by the excavators to date to the 15th century BCE (Julian), the time of Thutmose III and a period of known – and relatively intense – contacts between Egypt and the Aegean.[3223]

76.30. Current State of Scholarship of Absolute Chronology

XX. [In regards to Mesopotamia, Anatolia, and the Syro-Levantine Region] at present, there is no conclusive evidence to favor any of the competing approaches to absolute chronology for the early- to mid-2nd millennium BCE (Julian). Scientific methods and historical sources all involve problems and uncertainties. Nevertheless, continuing refinement of natural-science methods, new or reanalyzed texts, bold reappraisals such as that of Gasche and his colleagues, and international chronological projects such as that organized by Bietak bode well for future progress.[3224]

3219 Cline 2008: 453
3220 Cline 2008: 453
3221 Cline 2008: 453
3222 Cline 2008: 454
3223 Cline 2008: 454
3224 Schwartz 2008: 452

XX. Debate over the absolute dating of both the Aegean region and Egypt during the 2nd millennium BCE (Julian) will undoubtedly continue for the foreseeable future. Egyptologists feel confident that changes in absolute dates will probably be relatively minor. They feel that we can rest assured that – at the very least – we have a fairly good grasp of the relative chronology of these two (2) regions.[3225]

XX. Because debate continues and problems of chronology persist, correlations between the Bronze Age Aegean and Egypt during the 2nd millennium BCE (Julian) will vary depending on whether the Traditional [Middle] or the High Chronology is used.[3226]

XX. The overall goal of the Special Research Programme, SCIEM 2000,[3227] is to establish an absolute chronology of the Eastern Mediterranean and the adjacent regions for the 2nd millennium BCE (Julian). Mesopotamia is of special importance here because it has a tradition of king lists and similar historiographic texts which would provide such a reliable chronology – were they only completely preserved.[3228]

XX. Frequent synchronisms link Mesopotamia to the Levant and Egypt. According to Hermann Hunger, from Pruzsinsky's investigations it became clear that a solution for Mesopotamian chronology could not yet be achieved.[3229] The main problem remains the correlation of Aegean with Near Eastern chronologies.[3230]

XX. Mesopotamian history and chronology has been mainly reconstructed on the basis of various king lists (KLs) supplemented by royal inscriptions and chronicles. The Assyrian King List (AKL) still remains the "backbone" of Mesopotamian chronology as it represents the only stable and relatively fixed scheme with which all the data is to be compared, and thus is the starting point for any chronology-related issue. Other historiographical texts, such as Babylonian King List (BKL), the Synchronistic History, as well as other chronicles, are to be included and possibly linked to the data of the AKL.[3231]

XX. The Mesopotamian chronology systems – particularly the High Chronology (HC), Middle Chronology (MC), and Low Chronology (LC) – solely depend on calculations based upon the astronomical data in the Venus Tablet (VT) and not on the interpretation of king list data. However, within the past few years more emphasis has been given to sources other than the VT and the eclipse data in the omen tablets of Enuma Anu Elish (EAE), which naturally has resulted in different

3225 Cline 2008: 454
3226 Cline 2008: 454
3227 "The Synchronisation of Civilisations in the Eastern Mediterranean in the Second Millennium B.C." of the Austrian Academy of Sciences
3228 Pruzsinszky 2009: 13
3229 Pruzsinszky 2009: 13
3230 Pruzsinszky 2009: 20
3231 Pruzsinszky 2009: 22

chronologies.[3232]

XX. Tadmor (1970) stressed the chronological importance of the Venus Tablet (VT), the Assyrian King List (AKL), and the synchronism between Hammurabi and Šamšī-Adad I documented in the Mari texts. Since then major improvements have been made in Babylonian history and chronology.[3233]

XX. Absolute Mesopotamian chronology is securely established for most of 1st millennium BCE (Julian) Babylonia and Assyria. Lengths of reigns, chronologically fixed by astronomical observations, are known from Nabopolassar (626 – 605 BCE (Julian)) onwards. Before the 8th century BCE (Julian) no continuous sequence of Babylonian kings can be securely established. However, Ptolemy lists some Neo-Assyrian kings in parallel with Babylonian kings, and because the Assyrian King List (AKL) and the Assyrian Eponym List (EL) [i.e. the Assyrian Eponym Canon (AEC)] enable us to set an absolute chronology from 910 to 649 BCE (Julian), the absolute dates of contemporary Babylonian kings can be fixed via synchronisms.[3234]

XX. Prior to 910 BCE (Julian) there are some gaps in our knowledge of eponyms. Assyrian royal chronology can go back as far as ca. 1420 / 1430 BCE (Julian), the reign of Enlil-naṣir II, with an uncertainty of ten (10) years. Beyond this, reign lengths are poorly known, so that dates cannot be as certain. For instance, the reign lengths of Enlil-naṣir's predecessors have been lost in all versions of the AKL.[3235]

XX. In Babylonia, most kings' reign lengths are known for the second half of the 2nd millennium BCE (Julian). However, these data are only sufficient for a relative chronology. For absolute dates one has to rely on synchronisms with Assyria.[3236]

XX. The term "Old Babylonian period" is closely connected with the Babylon I dynasty and its famous ruler Hammurabi. The time span between the end of the Ur III period and the end of the Babylon I dynasty is generally termed "Old Babylonian" – the linguistic designation for the Akkadian [language] used in Babylonia during this time.[3237]

XX. The absolute dating of the early Old Babylonian period is mainly based on astronomical events (lunar eclipses in the Akkad and Ur III period) mentioned in the Enuma Anu Elish (EAE) combined with the month-length data of the Old Babylonian period and calculations derived from the risings and settings of the Venus recorded in the Venus Tablet (VT) dated to the 8th year of Ammisaduqa of the Babylon I dynasty. More information on the early 2nd millennium can be drawn from various King Lists

3232 Pruzsinszky 2009: 23
3233 Pruzsinszky 2009: 24
3234 Pruzsinszky 2009: 17
3235 Pruzsinszky 2009: 17
3236 Pruzsinszky 2009: 17
3237 Pruzsinszky 2009: 40

(KLs), Babylonian King List (BKL), Sumerian King List (SKL), Synchronistic KL, date-lists containing year-names, and chronicles.[3238]

XX. Around the middle of the 2nd millennium BCE (Julian), after the end of the Babylon I dynasty and during the early Kassite period, there is a chronological gap in all of our information from Mesopotamia.[3239] The fall of Babylon marks the beginning of the Dark Age, a period of uncertain length characterized by a paucity of texts and inscriptions.[3240]

XX. This Dark Age makes it impossible to establish absolute chronology of the earlier half of the 2nd millennium BCE (Julian) and earlier. Since we lack absolute dates, which could be anchored within the first half of the 2nd – or the end of the 3rd – millennia, we simply do not know the actual length of the Dark Age. Only relative dates can be provided prior to ca. 1430 / 1420.[3241]

XX. This Dark Age lasted according to the New (i.e. Ultra Low) Chronology (NC) zero (0) years, the Low Chronology (LC) ca. 20 years, the Middle Chronology (MC) ca. 85 years, and the High Chronology (HC) ca. 140 years. According to the Ultra High Chronology (UHC), the Dark Age may have lasted ca. 200 – 230 years. This period covers the interval between the end of the Babylon I dynasty and the beginning of the Middle Assyrian period (the Middle Asyrian Eponym List [i.e. in the Assyrian Eponym Canon (AEC)] starts with Aššur-nīrārī II) when textual evidence increases again.[3242]

XX. Few documents are known for the beginning of the Kassite period: only with Kara-indaš, the 15th king of this dynasty dated to the late 15th centurty BCE (Julian), does our knowledge of historical events improve. Basically, we need to date the Babylon I dynasty to establish some fixed points of chronological reference for the first half of the 2nd millennium BCE (Julian).[3243]

XX. Chronological relations and synchronisms are sufficiently known for the time before the onset of the Dark Age: Hence one of the primary tasks is the coordination of the chronological data of early 2nd millennium BCE (Julian) Assyria, Babylonia, Egypt, and Anatolia (Hatti).[3244]

XX. The central problem of Mesopotamian chronology is the dating of the Babylon I dynasty. Attempts have been made to compute an absolute date for this dynasty by means of the "Venus Tablet" (VT) written – it would seem – during the reign of the Old Babylonian king Ammisaduqa. However, the data in the table is difficult to interpret and has resulted in three (3) chronologies: the High (HC), the Middle (MC), and the Low (LC). In most publications, the Middle Chronology (MC) is used. However, this is

3238 Pruzsinszky 2009: 41
3239 Pruzsinszky 2009: 17
3240 Pruzsinszky 2009: 40
3241 Pruzsinszky 2009: 17
3242 Pruzsinszky 2009: 29
3243 Pruzsinszky 2009: 29
3244 Pruzsinszky 2009: 17

for reasons of "convenience", not because the Middle Chronology has been "proven".[3245]

XX. Currently the difference between the chronologies is 230 years. The central question of the chronology discussions remains the duration of the Dark Age.[3246] The solution to the problem of Mesopotamian absolute chronology during the 2nd millennium BCE (Julian) hinges on the length of the Dark Age which is conventionally set in the 16th century BCE (Julian).[3247]

XX. In the past, the approach of this problem was through computations based on astronomical texts (the Venus Tablet (VT), omens, etc.).[3248] Neugebauer (1941) had serious doubts about the value of astronomical data, particularly the Venus data for Mesopotamian absolute chronology.[3249]

XX. In fact today the Middle Chronology (MC) is considered the least likely chronology.[3250] During recent years writers specifically dealing with chronological matters have preferred the Low Chronology (LC), Ultra Low Chronology (i.e. New Chronology (NC)), or a lowered Middle Chronology (MC).[3251] The High Chronology (HC) is now accepted only by Huber and a few philologists such as Dalley (1984), Frayne (1990), Hunger-Pingree (1999), and Selz (2002).[3252]

XX. In the past few years, the Low Chronology (LC) has been the most frequently used chronology in the scholarly community. However, lately the very low Ultra Low Chronology (i.e. New Chronology (NC)) by Gasche et al. has provoked renewed debate on chronologies. They based their study on ceramic sequences, which could not be accommodated within the Middle Chronology (MC), and "demanded" a drastic lowering of the Middle Chronology (MC) by ca. 100 years. The New Chronology (NC) was eagerly accepted by archaeologists. However, few historians or philologists have accepted the New Chronology (NC) so far.[3253]

XX. The High Chronology (HC) is still supported as "the most likely solution" by Huber on the basis of his calculations with the astronomical data and month-lengths. The Middle Chronology (MC) and Low Chronology (LC) are also based on the Venus Tablet, but differ by one Venus period / cycle of 64 years. Thus according to the main chronological systems of the present, the dates of the reign of Hammurabi are:[3254]

1848 – 1806 BCE (Julian) High Chronology;

3245 Pruzsinszky 2009: 17
3246 Pruzsinszky 2009: 24
3247 Pruzsinszky 2009: 30
3248 Pruzsinszky 2009: 29
3249 Pruzsinszky 2009: 27
3250 Pruzsinszky 2009: 19
3251 Pruzsinszky 2009: 20
3252 Pruzsinszky 2009: 20
3253 Pruzsinszky 2009: 21
3254 Pruzsinszky 2009: 25

1792 – 1750 BCE (Julian) Middle Chronology;

1728 – 1686 BCE (Julian) Low Chronology;

XX. As valuable as data from dendrochronological or astronomical research may be, its direct connection with specific historical events is not easy to establish. For the time being, we are still missing this crucial connection between the sciences and archaeological-historical studies.[3255]

XX. Since the sources for the 3rd millennium BCE (Julian) are too unreliable and insufficient to provide an absolute chronology, the absolute dates of the 3rd millennium BCE (Julian) will have to be based on those of the 2nd millennium BCE (Julian).[3256]

XX. In any case, this variety of views and opinions indicates that the discussion of the absolute dates of the 2nd millennium BCE (Julian) is still very much alive.[3257] Advances in historical research in the middle of the 2nd millennium will certainly clarify chronological issues.[3258]

XX. The basis for chronology after the middle of the 15th century BCE (Julian) is the Assyrian King List (AKL) and Babylonian King List (BKL) in combination with dated tablets and the Assyrian eponym lists (ELs) [i.e. in the Assyrian Eponym Canon (AEC)]. For the second half of the 2nd millennium BCE (Julian) – according to Pruzsinszky – an uncertainty of 10 years is to be reckoned with.[3259]

XX. It is still uncertain how much time separated the middle of the 15th century BCE (Julian) from the end of the Babylon I dynasty (1651 == HC, 1595 == MC, 1531 == LC, 1499 == NC sometimes also referred to as the Ultra Low Chronology (ULC)).[3260]

XX. After the discovery of the synchronism between Hammurabi and Šamšī-Adad I, it became evident from a historical point of view that the High Chronology (HC) – which like other chronologies is primarily based on the evaluation of astronomical data (the Venus Tablet (VT) combined with lunar eclipses) – according to Pruzsinszky cannot be correct.[3261]

XX. The absolute chronology of the second half of the 2nd millennium BCE – the Late Bronze Age (LBA) – is dependent on synchronisms among the great powers of the era (Egypt, Assyria, Hatti and its

3255 Pruzsinszky 2009: 30
3256 Pruzsinszky 2009: 18
3257 Pruzsinszky 2009: 21
3258 Pruzsinszky 2009: 19
3259 Pruzsinszky 2009: 19
3260 Pruzsinszky 2009: 19
3261 Pruzsinszky 2009: 19-20

Syrian dependencies) and shows an uncertainty of only 5 – 10 years according to Pruzsinszky.[3262] As of today, most scholars accept the Low Chronology (LC) for Egypt (1479 BCE (Julian) accession of Tutmosis III and 1279 BCE (Julian) accession of Ramses II), the Middle Chronology (MC) for Mesopotamia in the Old Babylonian period (Hammurabi's accession in 1792 BCE (Julian)) and the shortened chronology (ten (10) years) proposed by Boese – Wilhelm (1979) for the Middle Assyrian kings which is connected with Hittite chronology based on the drastic shortening of Šuppiluliuma's I reign from 40 to 20 years: 1343 – 1322 / 1318.[3263]

XX. Moreover, further evidence from various sources can be expected that will give greater insight in to the Eponym Lists (ELs) and date-lists. New material has been added from astrochronology (solar and lunar eclipses). Thus the time is ripe for a new and updated general study on the chronology of 2nd millennium BCE (Julian) Mesopotamia.[3264]

XX. Since the exact dates for the 2nd and 3rd millennium BCE (Julian) Mesopotamia are still conjectural, we largely depend on king lists and chronicles as well local chronographers. Especially helpful to establish synchronisms between Mesopotamia and peripheral regions – as well as Egypt – are synchronistic accounts (the Synchronistic History, Synchronistic KL, Chronicle P, etc.), diplomatic correspondences and other documents. Relative and absolute chronology may be established by interlocking synchronisms (called the "comparative chronology" by Rowton in his 1959 and 1966 articles), which aims to collect all known synchronisms in order to provide an understanding of studies that have applied this method to gain absolute dates.[3265]

XX. Babylonian chronology of the second half of the 2nd millennium BCE (Julian) is basically tied to Assyrian chronology, which in turn is based primarily on the Assyrian King List (AKL), the eponym list [in the Assyrian Eponym Canon (AEC)] and eclipse dates for the 1st millennium BCE (Julian).[3266] A solar eclipse mentioned in the Eponym Chronicle provides a reliable absolute date for the year of eponym Būr-Saggile 763 BCE (Julian).[3267]

XX. Egyptian chronology is tied to Mesopotamian chronology, on both of which are Hittite and Elamite chronology dependent.[3268]

XX. Hittite chronology is also dependent on Assyrian chronology. Therefore, a variety of absolute dates – depending on which Mesopotamian chronology is adopted – can be observed in the literature. Because the dates of the 2nd millennium BCE (Julian) rulers of Syria and Elam ultimately depend on the

3262 Pruzsinszky 2009: 21
3263 Pruzsinszky 2009: 21-22
3264 Pruzsinszky 2009: 29
3265 Pruzsinszky 2009: 30
3266 Pruzsinszky 2009: 30
3267 Pruzsinszky 2009: 17
3268 Pruzsinszky 2009: 30

Assyrian and Babylonian chronology, they can be only approximate.[3269]

XX. However, major improvements have recently been made in the chronology of peripheral areas, and the study of Mesopotamian chronology can benefit from this, especially with respect to the synchronisms. Synchronisms constitute the most secure textual evidence for the dating of most rulers of minor dynasties. Even indirect synchronisms, when handled with care, have value for absolute chronology, but it must be kept in mind that in most cases we are dealing with synchronisms that cover time spans and are not tied to specific dates.[3270]

XX. By establishing the absolute chronological dates for Hammurabi of the Babylon I Dynasty, one can in turn establish the approximate regnal dates of the contemporary kings Neferhotep I of the 13th Dynasty of Egypt, Iantin-ʿAmmu of Byblos, and Zimri-Līm of Mari – thus connecting Egypt, the Levant, and Syria with Mesopotamia. The important synchronism between Hammurabi and Šamšī-Adad I of Aššur given by *Vorderasiatische Bibliothek* (5, 284, 11f.), which dates Šamšī-Adad I to the 10th year of Hammurabi, and the Mari correspondence (which notes the synchronism between Šamšī-Adad's death and year 18 of Hammurabi) correlates Babylonian chronology with Assyrian chronology based on the AKL.[3271]

XX. Since the Hittite ruler Muršili I is understood to have been responsible for the fall of Babylon, he must be a contemporary of the last ruler of the Babylon I Dynasty, Samsuditana. The preceding ruler Ammiṣaduqa is known to have ruled at the same time as Kuk-Našur II of the Elamite Sukkalmaḫ dynasty. For a later period we possess another synchronism between Babylonia and Elam: the Kassite Kadašman-Ḫarbe I and Tepti-ahar of the Kidinuids. From about 1450 BCE (Julian) onwards the chronology of the Near East is based on interlocking information from King Lists and synchronisms that prevent the possibility of gross errors.[3272]

XX. Basically, the Assyrian dates rely on the analysis of the AKL (with "minor" variants: uncertainties due to the differences in interpretation of the expression DUB-*pi-šu*, variants in the manuscripts of the AKL, the assumed length of first Assyrian Dark Age succeeding the reign of Išme-Dagān I, etc.) plus the assumption on whether the Assyrians used the lunar or the solar calendar before Tiglath-pileser I.[3273]

77.0. The Measurement of Time, ΔT (Delta T), and Eclipses

XX. Throughout the history of time measurement – from sundials to atomic clocks – time scales have

3269 Pruzsinszky 2009: 32
3270 Pruzsinszky 2009: 31
3271 Pruzsinszky 2009: 31
3272 Pruzsinszky 2009: 31
3273 Pruzsinszky 2009: 31-32

always been established by taking into account prevailing technology and needs.[3274] The earliest attempts to measure time were probably based on observations of celestial and meteorological phenomena: the seasonal motion of the Sun along the horizon, the progression of Moon phases, the shifting of winds, and so on.[3275]

XX. The necessity to divide and regulate time has in various ways led different primitive peoples and tribes to observe the celestial bodies, and hence to a beginning of astronomical knowledge.[3276] Thus a higher development of astronomy proceeded directly from the demands of time-reckoning, and more especially from the problem of adapting the lunar calendar to the solar year.[3277] However, here a difficulty arises because the [Julian / Gregorian] solar-year dates of New Moon and Full Moon move up and down [i.e. fluctuate], so that the phases of the Moon cannot indicate a determinate seasonal date.[3278]

XX. Then the nightly stars, already known through movement and orientation, offer a better solution. Attentive observation reveals that the position of the stars at the same hour of the night regularly changes with the season. Gradually the same position comes earlier in the night: the most westerly stars disappear in the evening twilight [of the western sky as the celestial sphere – from the perspective of Earth – apparently spins to the West], and at Dawn new stars emerge on the eastern horizon, appearing ever earlier in each following month.[3279]

XX. This morning appearance and evening disappearance – the so-called "heliacal rising and setting" – determined by the yearly course of the Sun through the ecliptic, repeats itself every year at the same [tropical Gregorian solar-year] date. The same happens with the moment at which a star rises in the evening twilight (the end of the observable risings) or sets just before dawn, the so-called "acronychian" [i.e. acronychal] rising or setting.[3280]

XX. Stellar phenomena are not the only means of fixing the dates of the solar year. The Sun itself can also be used.[3281]

XX. Time-keeping (which we take to mean the measurement of subdivisions of the day) has long involved observations of both celestial and terrestrial phenomena. On the celestial side is the measurement of shadows cast by a vertical gnomon which led to the development of sundials.[3282]

3274 Nelson et al. 2001: 526
3275 Seidelmann 1992: 73
3276 Pannekoek 1961: 22
3277 Pannekoek 1961: 25
3278 Pannekoek 1961: 22
3279 Pannekoek 1961: 22
3280 Pannekoek 1961: 22
3281 Pannekoek 1961: 22
3282 Seidelmann 1992: 73

Sundials were among the first instruments used to measure the time of day.[3283]

XX. On the terrestrial side is the development of water clocks (*clepsydrae*).[3284] When the first reliable water clocks were constructed, great care was taken to reflect the behavior of a sundial instead of the apparent motion of the heavens.[3285]

XX. The origins of both approaches predate historical records. Historical references to water clocks are found in Babylonian and Egyptian records of the 16th century BCE (Julian).[3286] In each stage of the evolution of timekeeping, there has been an incremental step away from the Sun as the measure of time in favor of a more uniform, accessible, or convenient standard.[3287]

XX. The system of hours developed first in Egypt where day and night were each divided into 12 parts. This results in a system in which the lengths of daylight and nighttime hours varied with the seasons.[3288]

XX. In summer, an hour of daylight was longer than an hour of darkness; in winter the situation was reversed. Only at equinoxes were hours of day and night of equal length.[3289]

XX. When this system of temporal hours [of varying length] spread throughout Europe, it became a function of latitude [i.e. variable] as well as season [i.e. variable]. It continued in civil use until the spread of mechanical clocks in the 15th and 16th centuries CE (Julian through 04 October 1582 CE) encouraged the gradual acceptance of equal hours.[3290]

XX. Our familiar system of 24 equal hours comes from Greek astronomy. At least as early as Hipparchus, the Egyptian system of 24 temporal hours was merged with the Babylonian division of the day into 360 equal units. The result was a system of 24 equal hours with each hour divided into 60 minutes. This established a tradition for astronomical tables that continues to this day.[3291]

XX. Although their origin is a mystery, mechanical clocks were in use in Europe near the beginning of the 14th century CE (Julian).[3292] It was not until the 14th century CE (Julian) that an hour of uniform length became customary due to the invention of mechanical clocks. These clocks were significant not only because they were masterpieces of mechanical ingenuity, but also because they altered the public's

3283 Nelson et al. 2001: 510
3284 Seidelmann 1992: 73
3285 Nelson et al. 2001: 510
3286 Seidelmann 1992: 73
3287 Nelson et al. 2001: 526
3288 Seidelmann 1992: 74
3289 Seidelmann 1992: 74
3290 Seidelmann 1992: 74
3291 Seidelmann 1992: 74
3292 Seidelmann 1992: 74

perception of time.[3293]

XX. In the era of telescopic observations, pendulum clocks served as the standard means of keeping time until the introduction of modern electronics. Quartz-crystal clocks were developed as an outgrowth of radio technology in the 1920s and 1930s CE (Gregorian). In 1948 CE (Gregorian), Harold Lyons at the National Bureau of Standards in Washington D.C. (now the National Institute of Standards and Technology, Gaithersburg, MD) constructed the first atomic clock.[3294]

XX. Once practical atomic clocks became operational, the *Bureau International de l'Heure* (BIH) and several national laboratories began to establish atomic time scales. The responsibility for the maintenance of the international standard is now given to the *Bureau International des Poids et Mesures* (BIPM).[3295]

XX. Some form of atomic time has been maintained continuously since 1955 CE (Gregorian).[3296] The methods and accuracy of timekeeping have changed dramatically since 1960 CE (Gregorian).[3297]

77.1. Establishing a System of Time

XX. As noted by Nelson et al., two (2) elements are needed to measure the passage of time:[3298]

1.) a time "reckoner", i.e. a repeatable phenomenon [e.g. the Sun or the Moon] whose motion (or change-of-state) is observable as well as [that] obeys a definite law;

2.) a time reference, i.e. with respect to which [e.g. coordinate position on the Earth] the position (or state) of the time reckoner [e.g. the Sun or the Moon] can be determined;[3299]

XX. To establish a system of time one must define two (2) quantities:[3300]

1.) the unit of duration (e.g. the second or the day);

2.) the epoch (or the zero (0) [moment starting point]) of the chosen time;[3301]

3293 Nelson et al. 2001: 510
3294 Nelson et al. 2001: 510
3295 Nelson et al. 2001: 510
3296 Nelson et al. 2001: 510
3297 Seidelmann 1992: 39
3298 Nelson et al. 2001: 509-510
3299 Nelson et al. 2001: 509-510
3300 Seidelmann 1992: 39
3301 Seidelmann 1992: 39

XX. These elements correspond to the two (2) properties of time measurement:[3302]

1.) interval;

2.) epoch;

XX. The epoch is its starting date. Every calendar has an epoch. This date is virtually never the [same] date that the calendar was adopted, but rather a hypothetical starting point for the first day [of that calendar system].[3303]

XX. Together the A.) time reckoner [e.g. the Sun or the Moon] and the B.) time reference [e.g. a position on the Earth] constitute – what Nelson et al. term – a clock.[3304]

XX. In other words, as Einstein famously proved: time is relative.

XX. Nelson et al. note three (3) primary methods of measuring time that have been in common use for modern applications in astronomy, physics, and engineering. These [three (3)] methods have evolved as the design and construction of clocks have advanced in precision and sophistication. Each of these [three (3)] measures of time has had a variety of refinements and modifications for particular applications.[3305]

XX. Dershowitz and Reingold also note three (3) distinct methods of measuring time that are in use today:[3306] [and they will be correlated (below) with Nelson et al.'s classification of the three (3) primary methods of measuring time, as well as with Seidelmann's four (4) types of time-systems in use in Physics and Astronomy]:

1.) Universal Time (UT), based on the rotation of earth around its own axis;

2.) Ephemeris Time (ET), based on the revolution of the Earth in its orbit around the Sun;

3.) Atomic Time (AT, or *Temps atomique international* (TAI)) based on the quantum mechanics of the atom;

XX. Each of these bases for measure of time has its history of evolution.[3307]

3302 Nelson et al. 2001: 510
3303 Dershowitz and Reingold 1997: 13; Dershowitz and Reingold 2008: 14
3304 Nelson et al. 2001: 510
3305 Nelson et al. 2001: 510
3306 Dershowitz and Reingold 2008: 174
3307 Nelson et al. 2001: 510; Sana, Muhammad, and Shahid 2008b: 363

1.) The 1st is Universal Time (UT) which is the time scale based on the rotation of the Earth and its [own] axis.[3308] There are several closely related types of Universal Time (UT).[3309] In Universal Time (UT), the unit of duration represents the solar day [which is] defined to be as uniform as possible despite variations in the rotation of the Earth.[3310] [This corresponds with Dershowitz and Reingold's *Solar Time*.] Solar Time is based on the solar day which measures the time between successive transits of the Sun across the meridian (i.e. the north-south line, through the zenith [directly] overhead in the sky). This period varies because of the non-uniform motion of the Earth.[3311]

2.) The 2nd is Ephemeris Time (ET) [– equivalent with Seidelmann's Dynamical Time –] which is the time scale based on the revolution of the Earth in its orbit around the Sun[3312] in which the unit of duration is based on the orbital motion of the Earth, Moon, and planets.[3313] [This corresponds to Dershowitz and Reingold's *Dynamical Time*.] Dershowitz and Reingold's Dynamical Time is a uniform measure taking the frequency of oscillation of certain atoms as the basic building block. Various forms of Dynamical Time use different frames of references which make a difference in a universe governed by relativity.[3314] Dynamical Time represents [a time scale that includes] the independent variable of the equations of motion of the bodies in the solar system. According to the theory of relativity, this independent variable depends on the coordinate system being used as the system of reference.[3315]

3.) The 3rd is Atomic Time (AT, or *Temps Atomique International* (TAI)) which is the time scale based on the quantum mechanics of the atom.[3316] in which the unit of duration corresponds to a defined number of wavelengths of radiation of a specified atomic transition of a chosen isotope.[3317] The fundamental unit of atomic time is the *Système International* (SI) second.[3318] International Atomic Time (*Temps Atomique International* (TAI)) is a practical time standard that conforms as closely as possible to the definition of the SI second. It fulfills the needs of accuracy, long-term stability, and reliability for astronomy.[3319] The SI Second and International Atomic Time [TAI] are used as bases for interpolation and prediction of the other timescales.[3320]

4.) A 4th method of measuring time, Sidereal Time, is enumerated by Seidelmann (1992) in which the unit of duration is the period of the Earth's rotation with respect to a point nearly fixed with

3308 Nelson et al. 2001: 510
3309 Dershowitz and Reingold 2008: 175
3310 Seidelmann 1992: 39
3311 Dershowitz and Reingold 2008: 174
3312 Nelson et al. 2001: 510
3313 Seidelmann 1992: 39
3314 Dershowitz and Reingold 2008: 174
3315 Seidelmann 1992: 41
3316 Nelson et al. 2001: 510
3317 Seidelmann 1992: 39
3318 Seidelmann 1992: 40
3319 Seidelmann 1992: 40
3320 Seidelmann 1992: 40

respect to the stars.[3321] [This corresponds with Dershowitz and Reingold's Sidereal Time.] Sidereal Time varies less than solar time. It is measured as the right ascension of a point in the sky crossing the meridian at any given moment. Thus local [at any given point on Earth, i.e. variable] sidereal time depends on terrestrial longitude [i.e. variable] and differs from observatory to observatory [i.e. variable].[3322]

XX. Each of these measures of time has had a variety of refinements and modifications for particular applications.[3323] In each of these systems various definitions of the unit of duration and of the epoch have prevailed in different periods of recent astronomical history.[3324] Astronomical calculations are typically done using Dynamical Time [i.e. Ephemeris Time (ET)] with its unchanging time units. There are various forms of Dynamical Time.[3325]

XX. On the other hand, Solar Time [i.e. Universal Time (UT)] units are not constant mainly because of the retarding effects of tides and the atmosphere which cause a relatively steady lengthening of the day [Length of Day (LOD)] and contribute what is called a "secular" (i.e. steadily changing) term to its length. This slowdown causes the mean solar day to increase in length (LOD) by about 1.7 milliseconds per century.[3326]

XX. Because Universal Time (UT) is based on the Earth's speed of rotation – which is slowly decreasing – the discrepancy [or difference, i.e. ΔT (Delta T)] between Universal Time (UT) and Dynamical Time (DT) is growing. It now stands at about 65 seconds and is currently increasing at about an average of one (1) second per year.[3327]

XX. To account for the vagaries in the length (LOD) of a UT day, every now and then a leap second is inserted (usually between 31 December and 01 January), thereby keeping our clocks – which show Universal Time (UT) – in tune with the gradually slowing rotation of the Earth. Dershowitz and Reingold note that this accumulated discrepancy is not entirely predictable.[3328]

77.2. The Vernal Equinox is a Fundamental Reference Point

XX. The [Sun's] ecliptic is the apparent annual path of the Sun against the background of stars. The intersection of the ecliptic with the celestial equator provides a fundamental reference point called the

3321 Seidelmann 1992: 39
3322 Dershowitz and Reingold 2008: 174
3323 Nelson et al 2001: 510
3324 Seidelmann 1992: 40
3325 Dershowitz and Reingold 2008: 177
3326 Dershowitz and Reingold 2008: 177
3327 Dershowitz and Reingold 2008: 177
3328 Dershowitz and Reingold 2008: 177

vernal equinox.[3329]

XX. The vernal equinox is the moment when the Sun's longitudinal position crosses the *true celestial equator* (i.e. the line in the sky above [extended out from] the Earth's [terrestrial] equator) from south to north on approximately 20 March (Gregorian) each year. At that time the day and night are 12 hours each all over the world, and the Earth's axis of rotation is perpendicular to the line connecting the centers of the Earth and Sun.[3330]

XX. In practice, Universal Time (UT) is determined not by the meridian transit of the mean Sun, but by the diurnal motion of the vernal equinox in accordance with a conventional formula specifying UT1 in terms of Greenwich Mean Sidereal Time (GMST).[3331]

XX. Initially the period of revolution of the Earth was understood to be the sidereal year. However, it was subsequently pointed out by André Danjon (1955 / 1957) that the tropical year is more fundamental than the sidereal year. The length of the tropical year (equinox to equinox) is derived directly from Newcomb's formula (1895) for the geometric mean longitude of the Sun for an epoch of 00 January 1900 CE (Gregorian), 12^h (UT), whereas the length of the sidereal year (fixed star to fixed star) depends upon the adopted value [i.e. variable] of the [Earth's] precession [of the Equinoxes].[3332]

77.3. Universal Time (UT)

XX. Universal Time (UT) is the measure of time used as the basis for all civil time-keeping. It conforms closely to the mean diurnal motion of the Sun. The apparent diurnal motion of the Sun involves both the nonuniform diurnal rotation of the Earth and the motion of the Earth in its orbit around the Sun.[3333]

XX. Although it would be possible to define a system of time measurement in terms of the hour angle of the Sun, such a system could never be related precisely to sidereal time and could not therefore be determined by observations of star transits. As a result, Universal Time (UT) is directly related to sidereal time by means of a numerical formula.[3334] It does not refer to the motion of the Earth and is not precisely related to the hour angle of the Sun.[3335]

XX. Universal Time (UT) at any instant can be derived from observations of the diurnal motion of the

3329 Nelson et al. 2001: 510
3330 Dershowitz and Reingold 2008: 185
3331 Nelson et al. 2001: 510
3332 Nelson et al. 2001: 512
3333 Seidelmann 1992: 50
3334 See Seidelmann 1992: 50, equation 2.24-1
3335 Seidelmann 1992: 50

stars or radio sources. The uncorrected observed rotational timescale (which is dependent on the place of observation) [i.e. variable] is designated UT0.[3336] The variation of UT1 with respect to uniform time truly [i.e. UT0] represents the irregularities of the rotation of the Earth.[3337]

XX. Correcting this timescale for the shift in longitude of the observing station caused by polar motion produces the UT1 timescale (which is independent of observing location), but is influenced by the slightly variable rotation of the Earth.[3338]

XX. 01 January 1984 CE (Gregorian) Greenwich Mean Sidereal Time (GMST) has been related to UT1 [through a complicated equation] as follows:[3339]

(GMST1 of 0^hUT1) == (24110.54841 seconds) + (8640184.812866 T_u seconds) + (0.093104 T_u^2 seconds) – (6.2 x 10^{-6} T_u^3)

where (T_u == d_u / 36525) and (d_u) is the number of days of Universal Time (UT) elapsed since (JD 2451545.0 UT1), i.e. 01 January 2000 CE (Gregorian) 12^h UT1, taking on values of (+/- 0.5), (+/- 1.5), (+/- 2.5), (+/- 3.5), [etc.].[3340]

XX. Since 01 January 1984 CE (Gregorian) Greenwich Mean Sidereal Time (GMST) has been related to UT1 [as per equation above], this relation conforms to the position and motion of the equinox specified in the IAU 1976 System of Astronomical Constants, the 1980 IAU Theory of Nutation, and the positions and proper motions of stars in the FK5 catalog.[3341]

XX. This equation is often considered to be the definition of UT1 because when it was established the observation of transits of the FK5 stars (or of other stars with their positions and proper motions in the FK5) was the best means to obtain UT1. When new techniques for measuring UT1 became operational, the arbitrary constants involved in their implementation were set to align the UT1 values with those obtained by observations of FK5 stars and to be consistent with this equation.[3342]

XX. The constant term and the numerical coefficient of (T_u) [i.e. .variables] in this equation were chosen so that UT1 would be continuous in value and rate through the change from the old to the new system of constant (01 January 1985 CE (Gregorian)). The small differences due to the change in the theory of nutation [i.e. variable] were neglected.[3343]

3336 Seidelmann 1992: 50
3337 Seidelmann 1992: 51
3338 Seidelmann 1992: 50
3339 Seidelmann 1992: 50
3340 Seidelmann 1992: 50
3341 Seidelmann 1992: 50
3342 Seidelmann 1992: 50
3343 Seidelmann 1992: 51

XX. For the dynamics of the solar system, the study of the rotation of the Earth, and the study of the Galaxy, one needs the relationship between the position and motion of the equinox and those of a point on the equator, i.e. the hour angle of which truly represents the rotation of the Earth in space (sidereal rotation).[3344]

XX. There are several ways [i.e. variables] to establish this relationship including using statistics on the proper motions of stars, the dynamical method, and observations of quasars. These lead to slightly different [i.e. variable] realizations of the equinox, and therefore of sidereal time. Therefore an agreement has been made to take as a reference the equinox and its motion as realized by the IAU 1976 System of Astronomical Constants, the 1980 IAU Theory of Nutation, and the positions and proper motions of stars of the FK5 catalog.[3345]

XX. Although the lengths of a day of UT1 and a day of Mean Sidereal Time vary slightly with variations in the Earth's rotation, the ratio of UT1 to Mean Sidereal Time is given by an equation[3346] that is the established method of determining Universal Time (UT) by multiplying an interval of sidereal time elapsed from (0^h UT) by a fixed conversion factor [i.e. variable]. According to Seidelmann, this maintains the constancy of the ratio of Universal Time (UT) to sidereal time irrespective of variations in the rate of rotation of the Earth.[3347]

(Mean Sidereal Day) == ($23^h56^m04.090524^s$ of Universal Time (UT))

(Day of Universal Time (UT) == ($24^h03^m56.5553678^s$ of Mean Sidereal Time)

XX. The rotational period of the Earth is (86164.09890369732) seconds of UT1 or ($23^h56^m04.09890369732^s$). This gives a ratio of a UT1-day to the period of rotation of (1.002737811906). The rate of rotation is (15.04106717866910) seconds of arc per second of time.[3348]

77.4. Coordinated Universal Time (UTC)

XX. The basis of [our modern] civil time is Coordinated Universal Time (UTC),[3349] an atomic time scale that agrees in rate with International Atomic Time (TAI, from French: *Temps atomique international*), but differs by an integral number of seconds, and is the basis of [modern] civil time.[3350] TAI is the fundamental atomic time scale "in the background" from which other scales of uniform time

3344 Seidelmann 1992: 51
3345 Seidelmann 1992: 51
3346 See Seidelman 1992: 52 equation 2.241-4
3347 Seidelmann 1992: 52
3348 Seidelmann 1992: 52
3349 Nelson et al. 2001: 510
3350 Nelson et al. 2001: 509

are derived.[3351] The advantage of TAI is that it is a continuous, atomic time scale without steps.[3352] TAI is a practical realization of a uniform time scale.[3353]

XX. Civil time keeping uses Coordinated Universal Time (UTC) which since 1972 CE (Gregorian) has been atomic time adjusted periodically by leap seconds to keep it close to the prime meridian's mean solar time.[3354] [UTC is an] atomic time scale that corresponds exactly in rate with TAI but is kept within 0.9 second of UT1 by the occasional insertion – or deletion – of a one-(1)-second step (leap seconds) when necessary, usually at the end of June or December.[3355]

XX. The present UTC system was established in 1972 CE (Gregorian),[3356] and since then – at Nelson et al.'s time of writing in 2001 CE (Gregorian) – there have been twenty-two (22) leap seconds, all of which have been positive.[3357]

XX. Coordinated Universal Time (UTC) differs from TAI by an integer number of seconds.[3358] TAI is related to UTC by the relation:[3359]

(TAI) == (UTC) + (ΔAT)[3360]

XX. When it was introduced, UTC was still the most readily available worldwide system for independent determination of position.[3361]

XX. International Atomic Time (TAI) represents the most precise achievable determination of time. This timescale is completely determined by applying the methods of physics to the determination of time. However we live – and make astronomical observations on – the surface of the rotating Earth. Thus it is necessary to relate TAI to the less consistent (UT) timescale determined from the Earth's rotation.[3362]

XX. Each longitudinal degree (°) of separation gives rise to a four (4) minute difference in local time:[3363]

3351 Nelson et al. 2001: 525
3352 Nelson et al. 2001: 525
3353 Nelson et al.2001: 510
3354 Dershowitz and Reingold 2008: 175
3355 Nelson et al. 2001: 510; Seidelmann 1992: 53-54
3356 Nelson et al. 2001: 518
3357 Nelson et al 2001: 510
3358 Seidelmann 1992: 53
3359 Nelson et al. 2001: 525
3360 or (ΔTAI), i.e. Delta (or Difference in / Change in) of Atomic Time
3361 Nelson et al. 2001: 520
3362 Seidelmann 1992: 58
3363 Dershowitz and Reingold 2008: 175

(four (4) minutes per degree (°)) x (360 celestial degrees (°)) == (1440 minutes per day)

(1440 minutes per day) / (60 minutes per hour) == (24 hours per day)

XX. This 4-minute unit per degree also happens to be the Ancient Babylonian unit USH which is equivalent to 4 minutes.

XX. The local mean time zone changes every 15°.[3364] For example, Jerusalem is 35.2° east of Greenwich [which is 0.0° longitude]. Its time zone is (UT + 2^{h}). To obtain standard time in Jerusalem from local mean time, a net offset of $20^{m}48^{s}$ is added.[3365]

XX. Nelson et al. posit that the most basic issue in the future of UTC is the nature of the social requirement to adjust an extremely precise, uniform time scale to the time determined using the variable rotation of the Earth. As they note, common practice today has already compromised this requirement to the point that we are content with conventional [man-made] constructions such as mean solar time, zone time, and Daylight Saving Time.[3366]

77.5. Greenwich Mean Time (GMT)

XX. The advent of the railroads in the 2nd quarter of the 19th century CE (Gregorian) introduced an era of high-speed transport and mobility. Efforts to coordinate schedules culminated in the adoption of regional zones of Standard Time and the choice of Greenwich as the international reference for the prime meridian.[3367]

XX. Originally Greenwich Mean Time (GMT) was defined as Mean Solar Time on the meridian of Greenwich reckoned from Mean Noon (12:00). However, beginning in 1925 CE (Gregorian0, the British Nautical Almanac and many other national ephemerides reckoned GMT from Midnight (00:00) to coincide with the civil [Gregorian] day, rather than Noon (12:00) as had been the traditional astronomical practice [previously that coincided with Julian Astronomical Days].[3368]

XX. The Mean Solar Day beginning at Noon (12:00), twelve (12) hours after Midnight (00:00) at the beginning of the same civil [Julian / Gregorian] date is known as the Astronomical Day [i.e. Julian Day (JD or JD_X)]. Local Mean Time (LMT) was reckoned from Mean Noon (12:00) on a local meridian.[3369]

3364 Dershowitz and Reingold 2008: 175
3365 Dershowitz and Reingold 2008: 177
3366 Nelson et al. 2001: 526
3367 Nelson et al. 2001: 523
3368 Nelson et al. 2001: 523
3369 Seidelmann 1992: 76

XX. Beginning with the national ephemerides for 1925 CE (Gregorian), a discontinuity of 12^{h} [i.e. 0.5 day] was introduced in the tabular arguments so – for example – the instant designated December 31.5 in the volumes for 1924 CE (Gregorian) was designated 1.0 January in volumes for 1925 CE (Gregorian).[3370]

XX. A confusion in terminology between Greenwich Mean Time (GMT) and Greenwich Civil Time (GCT) [lead to eventual] dropping of both designations and substituting Universal Time (UT). Care is necessary to avoid confusion: The designation UT always refers to time reckoned from Greenwich Midnight (00:00) even for epochs before 1925 CE (Gregorian).[3371]

XX. For navigation, Greenwich Mean Time (GMT) has meant UT1. Thus GMT has two (2) meanings that differ by as much as 0.9^{s}. For these reasons GMT should not be used for precise purposes.[3372]

XX. In 1928 CE (Gregorian), the redefined GMT was designated Universal Time (UT) by the IAU. However, the term GMT persisted in almanacs and navigation publications, and the ambiguity in its intended meaning [of GMT originally being equivalent with Mean Noon (12:00)] was the cause of some confusion. [Today] the terms "Mean Solar Time" and "GMT" have come to be recognized as being synonymous with UTC in ordinary language.[3373]

77.6. Universal Time 1 (UT1) is Nominally Equivalent with Mean Solar Time (and GMT since 1925 CE)

XX. In contrast to Coordinated Universal Time (UTC), Universal Time (UT1) is an astronomical time scale defined by the Earth's rotation and is used in celestial navigation.[3374] UT1 is the measure of astronomical time defined by the rotation of the Earth on its [own] axis with respect to the Sun. It is nominally equivalent to Mean Solar Time (MST) [that is] referred to [by] the meridian of Greenwich, United Kingdom [i.e. GMT] and reckoned from Midnight (00:00).[3375]

XX. The ordinary method of measuring time is called Universal Time (UT). It is Local Mean Solar Time (LMST), reckoned from Midnight (00:00) at the observatory in Greenwich, England, U.K., the 0° [longitude] meridian. The formal recognition of Greenwich as the "prime meridian" dates from the International Meridian Conference of 1884 CE (Gregorian), but it had been informal practice from 1767 CE (Gregorian).[3376] Most astronomical predictions are usually presented in terms of Universal

3370 Seidelmann 1992: 76
3371 Seidelmann 1992: 76
3372 Seidelmann 1992: 76
3373 Nelson et al. 2001: 523
3374 Nelson et al. 2001: 509
3375 Nelson et al. 2001: 510
3376 Dershowitz and Reingold 2008: p. xv, 174

Time (UT).[3377]

XX. The equivalent designation "Greenwich Mean Time", abbreviated "GMT" has fallen into disfavor with astronomers because of confusion as to whether days begin at Midnight (00:00) or Noon (12:00). Before 1925 CE (Gregorian), 0^h GMT meant Noon (12:00); from 1925 CE (Gregorian) onwards it has meant Midnight (00:00).[3378]

77.7. UT1 is the True Measure of Earth's Rotation

XX. The true measure of the Earth's rotation is UT1 which is the form of Universal Time (UT) [i.e. UT0 that is] corrected for polar motion.[3379] This time measured by the rotation of Earth is denoted by [the abbreviation] UT.[3380]

XX. However, because of irregularities in the Earth's rotation, UT1 is not uniform.[3381] As Universal Time (UT) is based on the variable rotation of the Earth, the quantity ΔT (Delta T) reflects the irregularities of that rotation,[3382] so it varies slowly but rather irregularly with time.[3383]

XX. For most practical purposes today, the time system known as Universal Time (UT) – formerly known as GMT – is adopted which is defined by the (variable) rotation of the Earth. However, UT is not suitable for computation of the positions of the Moon, Sun, and planets using gravitational theories of their motion. Such [gravitational] theories do not allow for variations in the rate of rotation of the Earth on its [own] axis.[3384]

XX. Universal Time (UT), or Greenwich Civil Time [i.e. Greenwich Mean Time (GMT)], which is needed for [modern] civil life, is based on the rotation of the Earth. However, the Earth's rotation is slowing down and – moreover – this occurs with unpredictable irregularities.[3385] For this reason UT is not a uniform time-scale.[3386]

XX. Our clocks – locked on Universal Time (UT) – are gradually slowing down with respect to the uniform [Terrestrial Time] TT [or Ephemeris Time (ET), see below]. In fact – as noted by Meeus – by

3377 Espenak: "Eclipse Predictions by Fred Espenak"; EclipseWise.com; https://www.eclipsewise.com/help/timezone.html
3378 Dershowitz and Reingold 2008: 174-175
3379 Nelson et al. 2001: 510
3380 Tanikawa, Yamamoto, and Sôma 2010: 797
3381 Nelson et al. 2001: 510
3382 i.e. ΔT itself is variable in its values
3383 Meeus and Simons 2000: 323
3384 Morrison and Stephenson 2004: 327
3385 Meeus 1998b: 154
3386 Meeus 1998b: 154

using UT, astronomical events seem to occur earlier than predicted in a uniform time-scale.[3387]

77.8. Ephemeris Time (ET)

XX. Ephemeris Time (ET) is a theoretically uniform time scale defined by the Newtonian dynamical laws of motion of the Earth, Moon, and planets.[3388] Although ET was a uniform time scale, it was not easily realized or disseminated.[3389] This measure of time has been succeeded by several new time scales that are consistent with [Einstein's] general theory of relativity.[3390]

XX. Ephemeris Time (ET) is not affected by the irregularities of the axial spin of the Earth, as it is evaluated from the revolution of the Earth in its orbit around the Sun in view of the principles of celestial mechanics. In this context, it is clear that ET and UT are not perfectly in agreement[3391] [because] due to irregularities in the axial rotation of the Earth – i.e. the Universal Time (UT) is not a uniform time.[3392]

XX. From evidence compiled by de Sitter (1927) of fluctuations in the Earth's rotation, Danjon (1929) recognized that Mean Solar Time (MST) – which was observationally tied to the rotation of the Earth – did not satisfy the need for a uniform timescale. He suggested using the timescale of the Newtonian laws of planetary motion.[3393]

XX. His call for reform which appeared in a nontechnical journal was perhaps unnoticed by specialists – certainly it was unheeded. Two (2) decades later Clemence (1948) published a more specific proposal apparently without being aware of Danjon's paper (Clemence 1971).[3394]

XX. From the results of Spencer Jones (1939), Clemence derived the factor ΔT[3395] that would reduce Universal Time (UT) to the measure defined by Newcomb's tables. In addition, Clemence determined the correction to Brown's lunar theory that would make the independent variable of the lunar theory consistent with that of Newcomb's Sun. Thus a comparison of lunar observations with Brown's (corrected) theory – according to Seidelmann – could be used to determine ΔT.[3396]

XX. In 1950 CE (Gregorian) a proposal to establish a more uniform fundamental standard of time was

3387 Meeus 1998a: 77; Sana, Muhammad, and Shahid 2008a: 1
3388 Nelson et al. 2001: 510
3389 Nelson et al. 2001: 514
3390 Nelson et al. 2001: 510
3391 Seidelmann 1985 – cited by Sana, Muhammad, and Shahid 2008b: 363
3392 Husfeld and Kronberg 1996 – cited by Sana, Muhammad, and Shahid 2008b: 363
3393 Seidelmann 1992: 79
3394 Seidelmann 1992: 79
3395 Can we derive and determine a more accurate factor for the value(s) of ΔT?
3396 Seidelmann 1992: 79

considered at the Conference on the Fundamental Constants of Astronomy held in Paris. At this conference Clemence proposed using the measure of time defined by Newcomb's Tables of the Sun.[3397]

XX. The Conference adopted a resolution recommending that this measure of time be adopted, expressed in units of the sidereal year at 1900.0 CE (Gregorian), and be designated by the name Ephemeris Time (ET) as suggested by Brouwer. This recommendation was adopted in 1952 CE (Gregorian) by the International Astronomical Union (IAU).[3398]

XX. Upon further consideration, the tropical[3399] [solar] year, which was considered more fundamental than the sidereal year, was chosen as the unit of time. Accordingly, the fundamental unit of time – the second – was redefined as (1 / 31556925.975) of the length of the tropical year for 1900.0 CE (Gregorian).[3400]

XX. In 1955 CE (Gregorian) the IAU approved this proposed definition.[3401] However the tropical year was understood to be the mean tropical year defined by Newcomb's expression for the geometric mean longitude [i.e. variable] of the Sun.[3402] The value of the second required for exact agreement with Newcomb's tables is (1 / 31556925.97474) of the tropical year.[3403]

XX. The primary unit of Ephemeris Time (ET) was the tropical year at the fundamental epoch of 0 January 1900 CE (Gregorian), 12^{h} (12:00) ET. The tropical year was defined as the interval during which the Sun's longitude (referred to the mean equinox of the date) increased by 360°.[3404]

XX. With its basic unit and initial epoch defined in this way, Ephemeris Time (ET) was equivalent to the system of time of Newcomb's solar tables. Newcomb originally identified this time scale with Mean Solar Time (MST) which he considered to be uniform.[3405]

XX. When Mean Solar Time (as determined from observations) was shown to be nonuniform, Newcomb's timescale was identified directly with Ephemeris Time (ET). The first two (2) terms of the Sun's geometric mean longitude [i.e. variable] were defined as absolute constants, and the corresponding values [i.e. variables] for the Moon and other planets were subject to possible revision to bring them into accord with observation.[3406]

3397 Seidelmann 1992: 79
3398 Seidelmann 1992: 79
3399 i.e. Gregorian.
3400 Seidelmann 1992: 79
3401 Seidelmann 1992: 79
3402 Seidelmann 1992: 79-80
3403 Seidelmann 1992: 80
3404 Seidelmann 1992: 80
3405 Seidelmann 1992: 80
3406 Seidelmann 1992: 80

XX. The symbol (*T*) in these expressions was therefore intended to represent Ephemeris Time (ET) not only in the theory of the Earth's motion around the Sun, but also in the heliocentric theories of the other planets.[3407] Ephemerides derived from Newcomb's tables of the Sun and planets could therefore be regarded as having Ephemeris Time (ET) as the independent time argument.[3408]

XX. The mean longitude of any other planets – or even of the Moon – could have been used to define the origin and rate of a uniform time, and Ephemerides of the Sun, Moon, and planets could have been constructed with that time system as the independent argument.[3409]

XX. In principle, the Ephemeris Time (ET) at any moment can be determined by comparing the observed positions of the Sun, Moon, and planets with their corresponding ephemerides. The tabular time for which the observations agreed with the ephemerides would be the Ephemeris Time (ET).[3410]

XX. In practice, Universal Time (UT) (which can be determined very accurately and with little delay from observations of the diurnal motions of the stars) was used as an intermediary measure of time. The difference between the two (2) measures of time, ($\Delta T == ET - UT$), was then obtained by comparing observations with ephemerides.[3411]

XX. Observations of the Moon (whose geocentric motion is much greater than those of other bodies) provided the most accurate determination of ΔT. However, direct comparison with the lunar ephemeris calculated from Brown's lunar tables did not give ΔT immediately since Brown's theory was not strictly gravitational and his tables were not in complete accord with Newcomb's solar tables. This was rectified by eliminating the empirical term from the mean longitude of the Moon [i.e. variable] and by applying a [mathematical, equational] correction [i.e. variable] to the tabular mean longitude.[3412]

XX. Additional corrections [i.e. variables] were needed for some of the periodic terms in longitude, latitude, and parallax. Beginning in 1960 CE (Gregorian), the lunar ephemeris was calculated from this amended theory – directly from the theoretical expressions for the longitude, latitude, and parallax – instead of from Brown's tables.[3413]

XX. A distinction had to be drawn between ($UT + \Delta T$) and (ET) when (ΔT) was determined using the previous equations and observations of the Moon. ($UT + \Delta T$) differed from (ET) in two (2) main respects:[3414]

3407 Seidelmann 1992: 80-81
3408 Seidelmann 1992: 81
3409 Seidelmann 1992: 81
3410 Seidelmann 1992: 81
3411 Seidelmann 1992: 81
3412 Seidelmann 1992: 81
3413 Seidelmann 1992: 81
3414 Seidelmann 1992: 81

1.) by a quadratic expression in (T) of the form ($a + bT + cT^2$), the coefficients of which had been observationally determined to be zero (0), but which almost certainly differed from zero (0) by significant amounts (it should be noted that the term (cT^2) is more of a fundamental physical character than ($a + bT$)).[3415]

2.) by any deficiencies that might have been present in Brown's lunar theory including any revision of any constants [i.e. variables] involved; in particular Brown used (1 / 294) for the flattening of the Earth.[3416]

XX. In the period from 1960 to 1984 CE (Gregorian) improvements were introduced in the lunar theory, and the different versions of the lunar theory were designated by (j = 0), (j = 1), and (j = 2). The Ephemeris Times (ET) determined from these lunar theories were correspondingly labeled ET0, ET1, and ET2.[3417]

XX. Thus [it is critical to note and understand how] the value of ΔT could differ systematically depending upon which method (or body [e.g. the Sun, the Moon, Venus, etc.]) was used for its determination. This will continue to be true when ΔT is extrapolated back in time prior to the availability of Atomic Time (TAI).[3418]

77.9. Difficulties in the Definition of Ephemeris Time (ET)

XX. In hindsight, the definition of Ephemeris Time (ET) was subject to a number of difficulties both in concept and in implementation. It was based on Newcomb's theory of the Sun and an associated set of astronomical constants [i.e. variables]. Both the theory and the constants were replaced in 1984 CE (Gregorian).[3419]

XX. However, at a deeper level ET suffered from its prerelativistic origins. Since its definition ignored general relativity, ET cannot be categorized as a geocentric or barycentric timescale, nor as a proper time or coordinate time.[3420]

XX. Both the fundamental unit and the epoch of ET contained technical problems. The tropical year was chosen as the unit of time under the assumption that it was independent of astronomical constants [i.e. variables] – particularly the constant of precession [i.e. variable].[3421]

3415 Seidelmann 1992: 81
3416 Seidelmann 1992: 81
3417 Seidelmann 1992: 82
3418 Seidelmann 1992: 82
3419 Seidelmann 1992: 82
3420 Seidelmann 1992: 82
3421 Seidelmann 1992: 82

XX. In reality, the determinations of both the tropical year and the sidereal year depend on the adopted system of astronomical constants [i.e. variables]. Unfortunately, the possibility of a time-dependent variation in the equinox (commonly called equinox motion) was not recognized. As a result, Ephemeris Time (ET) was subject to corrections introduced in the system of constants [i.e. variables] in 1984 CE (Gregorian).[3422]

XX. Although the fundamental epoch (designated 0 January 1900 CE (Gregorian), 12^h ET) was definitive – that the instant at which the geometric mean longitude of the Sun referred to the mean equinox of [the above] date was (279° 41' 48".04) [i.e. 279 degrees, 41 minutes, 48.04 seconds] – its determination depended on a comparison of observations with an apparent ephemeris of the Sun.[3423]

XX. The observations were themselves definitive, but the apparent ephemeris as deduced from the geometric mean longitude depended on the value adopted for the constant of aberration [i.e. variable]. All relevant observations and determinations had been made using 20".47 [i.e. 20.47 seconds] for the constant of aberration.[3424]

XX. A change in this value would lead to a change in the determination of the instant of the fundamental epoch, and thus to a corresponding change in the measures of Ephemeris Time (ET) [calculated and] assigned to all other instants.[3425]

XX. Thus Ephemeris Time (ET) was defined by the motion of a body [i.e. the Sun] that is difficult to observe. Determination of ET [i.e. a theoretical uniform timescale] was better accomplished by observations of rapidly moving bodies – e.g. the Moon, Mercury, and Venus.[3426]

XX. However, there was no specified relationship between ET as defined, and ET as determined from observations of the Moon, Mercury, or Venus.[3427] In practice, the determination of ET from the motion of the Moon was based on accepted values of various constants [i.e. variables].[3428] Thus changes in the constants resulted in different measures of ET.[3429]

XX. Ephemeris Time (ET), which takes the orbital motions in the solar system as the basic building block, is an outdated time scale as of 1984 CE (Gregorian).[3430]

3422 Seidelmann 1992: 82
3423 Seidelmann 1992: 82
3424 Seidelmann 1992: 82
3425 Seidelmann 1992: 82
3426 Seidelmann 1992: 82
3427 Seidelmann 1992: 82
3428 Seidelmann 1992: 82-83
3429 Seidelmann 1992: 83
3430 Dershowitz and Reingold 2008: 174

77.10. Terrestrial Dynamical Time (TDT) or Terrestrial Time (TT)

XX. Terrestrial Dynamical Time (TDT) – or Terrestrial Time is (TT) – is the theoretical timescale of apparent geocentric ephemerides of bodies in the solar system. TDT is specified with respect to TAI in order to take advantage of the direct availability of Coordinated Universal Time (UTC) which is based on the SI second. TDT is currently set equal to (TAI + 32.184 seconds).[3431]

XX. The relationship between TDT and TAI provides continuity with Ephemeris Time (ET). The chosen offset between TDT and TAI is equal to the estimate of the difference between ET and TAI at the time that TDT was introduced.[3432]

XX. Since TAI is a statistical timescale based on a large number of clocks operating on the Earth, and TDT is an idealized uniform timescale, the two (2) timescales cannot be made identical. TAI is subject to systematic errors in the length of the TAI second and in the method of forming TAI.[3433]

XX. For years prior to 1955 CE (Gregorian) when atomic time (TAI) was not available, TDT must be extrapolated backward using the theory of a rapidly moving celestial body that has been fit to observations made since 1955 CE (Gregorian). The Moon and Mercury are the most suitable bodies for such purposes.[3434]

XX. The accuracy of the backward extrapolation will depend on the length of the comparison after 1955 CE (Gregorian) and will deteriorate as a function of the length of time prior to 1955 CE (Gregorian). Use of the Moon for this extrapolation is complicated by the Moon's [secular lunar orbital] tidal [friction] acceleration. At this time there are uncertainties concerning the value of the [secular lunar orbital] tidal [friction] acceleration and the systematic variations between the theoretically calculated position of the Moon and the observations.[3435]

XX. The IAU (1976) definition of Dynamical Time (TDT) relates the dynamical time of solar-system ephemerides to the timescale of terrestrial atomic physics. However, distinctions must be made between an ideal dynamical time which is the independent variable in the equations of motion for the solar system, and both TDT and TDB which are currently defined as ideal forms of atomic time.[3436]

XX. In principle, the time in the equations of motion of the Sun, Moon, and planets could diverge from the time determined from observations of phenomena of terrestrial physics. An ideal Dynamical Time

3431 Seidelmann 1992: 42, 47
3432 Seidelmann 1992: 42
3433 Seidelmann 1992: 42
3434 Seidelmann 1992: 42
3435 Seidelmann 1992: 42
3436 Seidelmann 1992: 63

[TDT] can be determined only by analysis of observations over an extended period of time.[3437]

XX. Occultation observations – specifically observations of the time when the Moon passes in front of a star – provide the most accurate current method for investigating dynamical time (TDT). The observations require a recording of the time [i.e. variable] when the occultation took place as well as the location [i.e. variable] of the observer.[3438]

77.11. Sidereal Time is the Measure of Earth's Actual Rotation

XX. In general terms, sidereal time is the hour angle of the vernal equinox which is also known as the *First Point of Aries*. Apart from the inherent motion of the equinox, due to precession [i.e. variable] and nutation [i.e. variable], Sidereal Time is a direct measure of the diurnal rotation of the Earth. Equal intervals of angular motion correspond to equal intervals of Mean Sidereal Time (MST) so that [Apparent] Sidereal Time (AST) reflects the actual rotation of the Earth, and can be determined by observations of celestial objects.[3439]

XX. [That is to say it would appear evident that conventional methodology of calculating arcus visionis would be calculating for Apparent and/or Mean Sidereal Time (i.e. the measure of the actual, variable, non-uniform rotation of the Earth), and would be slightly different than calculations that attempt to solve for Universal Time (UT) or (TDT) / (TT).]

77.12. Local Apparent Sidereal Time (LAST or LST)

XX. Local [Apparent] Sidereal Time, (LAST) or (LST), is the measure of astronomical time defined by the rotation of the Earth with respect to the stars. The sidereal day is the time interval between successive transits of the vernal equinox which represents the Earth's period of rotation relative to the stars, and is approximately (86164.0905) mean solar seconds.[3440] Local [Apparent] Sidereal Time, (LAST) or (LST), is not a uniform measure of astronomical time.[3441]

XX. Apparent Sidereal Time (AST) [observed and measured locally] is the Sidereal Time [i.e. relative to the stars] measured by the hour angle of the true equinox, i.e. the intersection of the true equator of date with the ecliptic of date. The position of the true equinox is affected by the nutation [i.e. variable, sub-wobble of the axis of precession (i.e. variable)] of the axis of the Earth which consequently introduces periodic inequalities [i.e. variables] into the Apparent Sidereal Time (AST) [which would

3437 Seidelmann 1992: 63
3438 Seidelmann 1992: 63
3439 Seidelmann 1992: 48
3440 Nelson et al. 2001: 511
3441 Nelson et al. 2001: 511

affect the accuracy of any astronomical calculations that include these variables].[3442]

XX. Owing to the precession [i.e. variable] of the Earth's axis with respect to the celestial reference system, the sidereal day is about (0.0084) seconds shorter than the actual period of rotation [of the Earth] in inertial space.[3443] The apparent sidereal day – nominally of 24 hours of Apparent Sidereal Time (AST) – differs from the period of rotation [of the Earth] by a variable amount depending on the nutation [and precession, i.e. variables] .[3444]

XX. Thus the true rotational period of the Earth is approximately (86164.0989) mean solar seconds. However, Nelson notes that the mean solar day currently exceeds a [standard reference] day of exactly 86400 SI (from French: *Le Système International d'Unités*) seconds by about 2.5 milliseconds. Therefore, the Earth's period of rotation is currently about (86164.1014) SI seconds.[3445]

77.13. Mean Sidereal Time (MST)

XX. In the early 20th century CE (Gregorian), free-pendulum clocks revealed the periodic effects of nutation [i.e. variable]. The principal term consists of an [approximately] 18-year [and more specifically 18.6-year] oscillation with an amplitude of about one (1) second.[3446]

XX. These effects cannot be neglected, and [thus] it became necessary to introduce the concept of Mean Sidereal Time (MST) which is affected only by [the Earth's] precession [i.e. variable].[3447] Mean Sidereal Time (MST) is the time measured by the diurnal motion of the mean equinox of date which is affected by only the secular inequalities due to the precession of the axis.[3448]

XX. [Local] Apparent Sidereal Time, (LAST) / (LST) / (AST), minus Mean Sidereal Time (MST) is the equation of the equinoxes. In ephemerides immediately preceding 1960 CE (Gregorian) the equation of the equinoxes was called the nutation in right ascension.[3449]

XX. Greenwich Mean Sidereal Time (GMST) is Mean Sidereal Time (MST) with respect to the Greenwich meridian from which Universal Time (UT1) is derived.[3450] To each local meridian on the Earth there corresponds on a Local [Apparent or Mean] Sidereal Time [(LAST) / (LST) / (AST) or (LMST)] connected with the sidereal time of the Greenwich meridian by means of the [Global Time

3442 Seidelmann 1992: 48
3443 Nelson et al. 2001: 511
3444 Seidelmann 1992: 49
3445 Nelson et al. 2001: 511
3446 Nelson et al. 2001: 511
3447 Nelson et al. 2001: 511
3448 Seidelmann 1992: 48
3449 Seidelmann 1992: 48
3450 Nelson et al. 2001: 511

Zone] relation:[3451]

XX. Local [Apparent or Mean] Sidereal Time [(LAST) / (LST) / (AST) or (LMST)] == Greenwich [Apparent or Mean] Sidereal Time [(GAST) or (GMST)] + (East Longitude).[3452]

XX. Sidereal time is conventionally measured in hours, minutes, and seconds so that longitude is measured (positively to the east) in time at the rate of one (1) hour to 15°. An object transits over the local meridian when the Local Sidereal Time (LST) is equal to its right ascension.[3453]

XX. Owing to precession, the mean sidereal day of 24 hours of Mean Sidereal Time (MST) is shorter than the actual period of rotation [UT1] of the Earth by about (0.0084^{S}), i.e. the amount of precession in right ascension in one (1) day.[3454] These numbers are not rigorously constant [i.e. variable] because the sidereal motion of the equinox due to precession is proportional to the length of the UT day, i.e. [proportional] to the period of the rotation of the Earth, whereas the angular measure of the complete rotation is of course constant.[3455]

XX. Because of its variable rate, Apparent Sidereal Time (AST) is used only as a measure of epoch – it is not used as a measure of time interval. Observations of the diurnal motions of the stars provide a direct measure of Apparent Sidereal Time (AST) as their right ascensions are measured from the true equinox.[3456]

XX. However, in many practical methods of determining time the right ascensions are diminished by the equation of the equinoxes [i.e. known as nutation in right ascension in ephemerides in the period immediately preceding 1960 CE (Gregorian)] so that Mean Sidereal Time (MST) is deduced directly from the observations. Today, Sidereal Time is determined most accurately by interferometer observations of radio sources, as well as from lunar and satellite laser-ranging.[3457]

XX. Since the rotation of the Earth is subject to irregular forces [i.e. variables], Sidereal Time is [also] irregular [since it is the measure of Earth's actual non-uniform rotation] with respect to atomic time (TAI). The practical determination of Sidereal Time requires that short-period irregularities and polar motion [i.e. variables] be evaluated at the same time.[3458]

77.14. The SI Second

3451 Seidelmann 1992: 49
3452 Seidelmann 1992: 49
3453 Seidelmann 1992: 49
3454 Seidelmann 1992: 48
3455 Seidelmann 1992: 48-49
3456 Seidelmann 1992: 49
3457 Seidelmann 1992: 49
3458 Seidelmann 1992: 49

XX. The definition of the second – the most precisely reproducible unit in the Système International (SI) – was once based on the rotation of the Earth. Now the second is based on natural frequencies of the atoms of selected elements.[3459]

XX. Since 1967 CE (Gregorian) the Système International (SI) second has been the standard unit of time in all timescales. The length of the SI second was defined by Markowitz et al. (1958) in terms of the observationally determined value of the second of Ephemeris Time (ET). Thus the SI second closely matches an Ephemeris second and provides continuity with Ephemeris Time (ET).[3460]()

XX. The SI second is equivalent to the second of Ephemeris Time (ET), which in turn is equal to the mean solar second of the early 19th century CE (Gregorian). The length of the day (LOD) was exactly (86400) SI seconds in about 1820 CE (Gregorian). Before then, the mean solar day was less than (86400) seconds, and since then it has been greater than (86400) seconds.[3461]

XX. At the rate of about (1.4) milliseconds per century over the past 180 years, the length of the day (LOD) has increased by roughly (2.5) milliseconds, so that today the length of the day (LOD) is about (86400.0025) SI seconds. The difference of (2.5) milliseconds per day accumulates to nearly one (1) second over an entire year. The present length of the mean solar day is about (2.5) milliseconds longer than a day of precisely (86400) SI seconds.[3462]

XX. It is this accumulated difference that is compensated [for] by the occasional insertion of a leap second to make the length of the year one (1) second longer. As noted by Nelson et al., a change in the frequency of occurrence of leap seconds is an indication of the slowing down or acceleration of the Earth's rotation. As a consequence of this long-term trend, the Earth's rotation runs slow with respect to atomic time (TAI).[3463]

77.15. Earth is Decelerating

XX. Although [the concept that] the possibility [of] the Earth's rate of rotation can vary was first suggested as long ago as the mid-18th century CE (Gregorian), it was not until 1939 CE (Gregorian) that this inference was fully substantiated.[3464] As Universal Time (UT) is based on the variable rotation of the Earth, the quantity ΔT (Delta T) reflects the irregularities of that rotation, so it varies slowly but

3459 Seidelmann 1992: 39
3460 Seidelmann 1992: 84
3461 Nelson et al. 2001: 518
3462 Nelson et al. 2001: 518
3463 Nelson et al. 2001: 518
3464 Morrison and Stephenson 2004: 328

rather irregularly with time.[3465]

XX. The cause of the long-term deceleration of the Earth's rotation is due to a number of circumstances, but principally due to ([secular] lunar [orbital]) tidal friction [acceleration].[3466] The cumulative slowing of the Earth's rotation means that events in the far [distant, ancient] past actually occurred later than we would calculate them to have occurred on the basis of uniform time [i.e. Terrestial Time, TT].[3467]

XX. Astronomical observations have shown that the rotation rate of the Earth around its axis very slowly decreases and that the Moon is accelerated[3468] in its orbit around the Earth. Both effects are very small, corresponding to an increase in the length of day (LOD) by about two (2) milliseconds per century and an increase in the length of the sidereal month (i.e. the time required for the Moon to complete one (1) orbital period [with respect to the background stars]) of about 20 milliseconds per century.[3469]

XX. However small these so-called "secular accelerations" may be, their effect accumulates over centuries to quite noticeable differences in time and in the position of the Moon. For instance, using modern standard values (*Explanatory Supplement of the Astronomical Ephemeris (ESAE)* 1961) De Jong and Van Soldt compute that in 1000 BCE (Julian) the differences in time and in lunar longitude have accumulated to about seven (7) hours and to about 2.5° respectively.[3470]

XX. However, Meeus notes that in the last few years prior to his writing (1998), the International Astronomical Union had introduced subtle changes in the reference frame used for the coordinates of celestial objects (both within as well as far beyond our solar system). Meeus stresses that so sweeping were these revisions that a highly respected work for professional astronomers, the *Explanatory Supplement to the Astronomical Ephemeris (ESAE)* – published in 1961 – is now [i.e. 1998] seriously out of date.[3471]

XX. [Nonetheless, regardless of the accuracy of current calculations heretofore – as well as the current disagreement among scholars heretofore about which exact numbers (i.e. variables) to use for critical calculations of other variables such as ΔT –] ancient astronomical observations could be of great help in determining the magnitude of the secular [lunar orbital tidal friction] accelerations [of the Moon].[3472]

XX. Among the astronomical data available from antiquity, solar [as well as lunar] eclipse records are

3465 Meeus and Simons 2000: 323
3466 Kelley and Milone 2011: 104
3467 Kelley and Milone 2011: 105
3468 Secular, lunar, orbital, tidal friction acceleration.
3469 De Jong and Van Soldt 1989: 65
3470 De Jong and Van Soldt 1989: 65
3471 Meeus 1998a: p. ii
3472 De Jong and Van Soldt 1989: 65

most useful for this purpose because the time of an eclipse depends upon the position of the Moon, and the geographical location of the zone of totality of an eclipse depends upon the attitude of the earth at the time of the eclipse.[3473]

XX. Meeus explains that without too much calculation, it is possible to obtain with good accuracy the principal characteristics of an eclipse of the Sun or the Moon. For a solar eclipse, the situation is complicated by the fact that the phases of the event are different for different observers [at different locations] at the Earth's surface. In the case of a lunar eclipse, all observers see the same phase at the same instant.[3474]

XX. Unlike solar eclipses, the duration of each phase of a lunar obscuration is independent of the observer's location.[3475] Hence – according to Stephenson and Fatoohi – computation need take no account of the errors in local time arising from irregularities in the Earth's rate of rotation [i.e. ΔT (Delta T)].[3476] It is thus possible to compare these measured intervals directly with their computed equivalents.[3477]

XX. Moreover, the zone of totality of a solar eclipse traces a very narrow path (typically about 100 kilometers wide) on the Earth's surface.[3478] As a consequence, total solar eclipses are very rare events (on average: approximately one (1) per century at any location on Earth), so that an eclipse can usually be identified when only the place of observation [i.e. variable] and the approximate year of observation [i.e. variable] are known.[3479]

XX. In 1975 CE (Gregorian) Muller and Stephenson attempted to determine the deceleration of the Earth and the acceleration of the Moon by fitting data for about 25 carefully selected solar eclipses covering about 25 centuries from 1375 BCE (Julian) to 1267 CE (Julian). Their analysis yields values for the secular accelerations that are substantially larger than the standard values obtained from modern ([from] approximately ~ 1700 CE (Gregorian) to the present time) astronomical observations.[3480]

XX. The [secular] lunar [orbital tidal friction] acceleration is exclusively due to the transfer of angular momentum from the Earth to the Moon by tidal interaction, and there is no reason to assume large fluctuations in the tidal torques over historical times.[3481] Thus, the difference between the modern standard value and that found by Muller and Stephenson (1975) is puzzling.[3482]

3473 De Jong and Van Soldt 1989: 65
3474 Meeus 1998a: 379
3475 Stephenson and Fatoohi 1993: 259
3476 Stephenson and Fatoohi 1993: 259-260
3477 Stephenson and Fatoohi 1993: 260
3478 De Jong and Van Soldt 1989: 65
3479 De Jong and Van Soldt 1989: 65-66
3480 De Jong and Van Soldt 1989: 66
3481 Why do they assume that it remained steady, constant, and invariable?
3482 De Jong and Van Soldt 1989: 66

XX. However, it is not clear why De Jong and Van Soldt assume that there is no reason to assume that there may have been large fluctuations in the tidal torques over historical times.

77.16. Earth is Decelerating at a Non-Uniform Rate

XX. Unfortunately, Earth's rotation is not slowing down at a uniform rate. Non-tidal effects of climate change,[3483] polar ice caps, ocean depths, and the dynamics of Earth's molten core make it impossible to predict the exact value of ΔT (Delta T) in the remote past or distant future.[3484]

XX. The Earth's rotation slows with respect to atomic time (TAI).[3485] Because the Earth's rotation is gradually slowing down, as well as having both random and periodic fluctuations [in the speed and/or movements of this rotation of the Earth], it is not a uniform measure of time.[3486] However, astronomers need a uniform time-scale for their accurate calculations (e.g. celestial mechanics, orbits, ephemerides).[3487]

XX. The exact value of the difference (ΔT == TD – UT) cannot be predicted and hence can only be deduced from observations.[3488] Citing Robert, it is noted by Sana, Muhammad, and Shahid that the knowledge of the exact value of ΔT is essential for predicting the correct time of an astronomical event such as eclipses,[3489] occultation, transits, etc. If ΔT is not considered [correctly], the predicted value may contain an error of several seconds. In some cases this error could be in minutes.[3490]

XX. In comparison, the secular change in the rotation rate of Earth currently increases the length of day (LOD) by 2.3 milliseconds per century. While this amount may seem astonishingly small, its accumulated effects have important consequences. In one (1) century, Earth loses about 40 seconds, while in one (1) millennium, the planet is over one (1) hour "behind schedule". Astronomers use the quantity ΔT to describe this time difference.[3491]

XX. Morrison and Stephenson note that the analysis of historical eclipses shows that the length of the day (LOD) is increasing on the average by 1.8 milliseconds per century, as distinct from the value of

3483 Espenak says literally "global warming", though this could be debated, and thus the more inclusive term of "climate change" has been used here.
3484 Espenak: "Eclipse Predictions by Fred Espenak"; EclipseWise.com; https://www.eclipsewise.com/help/rotation.html
3485 Nelson et al. 2001: 518
3486 Nelson et al. 2001: 517
3487 Meeus 1998b: 154
3488 Meeus 1998a: 77; Sana, Muhammad, and Shahid 2008a: 1
3489 Sana, Muhammad, and Shahid 2008a: 1
3490 Sana, Muhammad, and Shahid 2008a: 2
3491 Espenak, "Eclipse Predictions by Fred Espenak": EclipseWise.com; https://www.eclipsewise.com/help/rotation.html

2.3 milliseconds per century expected from tidal friction.[3492] From analyses of the discordance between observations and calculations of eclipses, the average measured increase in the length of the day (LOD) is 1.8 milliseconds per century, from which it is concluded that besides the tidal contribution there is another long-term component [i.e. variable] acting to decrease the length of the day (LOD) by 0.5 milliseconds per century.[3493]

XX. The tidal torque [i.e. variable] exerted by the Moon (and to a lesser extent, the Sun) is the predominant mechanism in reducing the Earth's rate of spin. It is known that by this mechanism, the length of the day (LOD) is increasing by 2.3 millseconds per century.[3494]

XX. This component, which is thought to result from the decrease of the Earth's oblateness following the last ice age, is consistent with recent measurements made by artificial satellites. Besides these secular trends, there is clear evidence of fluctuations [i.e. variables] of the length of the day (LOD) on a timescale of centuries, with an amplitude of 4 milliseconds which is similar to that of the decade fluctuations [i.e. variables].[3495]

XX. Both lunar and solar tides raised in the oceans and – to a lesser extent – in the solid body of the Earth gradually decrease the Earth's rate of spin, and thus increase the LOD. Global sea-level studies suggest that tidal friction has not changed significantly over the last three (3) millennia or so.[3496]

XX. The gravitational attraction of the Moon is the principal cause of the ocean tides on Earth.[3497] The ocean tides are caused by the gravitational pull of the Moon and – to a lesser extent – the Sun.[3498]

XX. However, as the tides are attracted to the Moon, the oceans appear to rise and fall while Earth rotates beneath them. This tidal friction gradually transfers angular momentum from Earth to the Moon. Earth loses energy and slows down while the Moon gains the energy and consequently its orbital period and distance from Earth increase.[3499]

77.17. The Moon is Spinning Away from the Earth

XX. The slower spin of the Earth causes the Moon to edge farther [away] from our planet.[3500] The Moon's average distance from the Earth is increasing by 3.8 centimeters per year. Such a precise value

3492 Morrison and Stephenson 2002: 338
3493 Morrison and Stephenson 2001: 247
3494 Morrison and Stephenson 2001: 247
3495 Morrison and Stephenson 2001: 247
3496 Morrison and Stephenson 2004: 328
3497 Littman, Willcox, and Espenak 1999: MrEclipse.com; https://www.mreclipse.com/Totality2/TotalityExtinct.html
3498 Espenak, "Eclipse Predictions by Fred Espenak": EclipseWise.com; https://www.eclipsewise.com/help/rotation.html
3499 Espenak, "Eclipse Predictions by Fred Espenak": EclipseWise.com; https://www.eclipsewise.com/help/rotation.html
3500 Littman, Willcox, and Espenak 1999: MrEclipse.com; https://www.mreclipse.com/Totality2/TotalityExtinct.html

is possible because of the Apollo laser reflectors which the astronauts left behind during the lunar landing missions.[3501]

XX. From 1969 to 1972 CE (Gregorian), the Apollo astronauts left a series of laser reflectors on the Moon's surface. Since then, scientists on Earth have been bouncing powerful lasers off these reflectors.[3502]

XX. Laser-pulse ranging between Earth-based telescopes and retroreflectors installed on the surface of the Moon is currently done to a precision of approximately [+/-] two (2) cm. A capability of determining Universal Time (UT) to a fraction of a millisecond was demonstrated by Silverberg (1974) and is effectively shown by recent results.[3503]

XX. Because of the precision of the measurements, these observations are very sensitive to the dynamics of the Earth-Moon system. They provide very accurate corrections to the theory of the Moon's motion and librations, and the theory of general relativity.[3504]

XX. Other significant parameters [i.e. variables] are those that affect the positions of the Moon-based retroreflectors and the Earth-based telescope – including Earth tides, continental drift, plate motion, and polar motion. The observations are also sensitive to the constant of precession [i.e. variable], the theory of nutation [i.e. variable], and the Earth's orbit [i.e. variable]. All of these parameters are correlated in complex ways.[3505]

XX. By timing the round trip of each laser pulse, the Moon's distance can be measured to an accuracy of several inches: the Moon is receding from the Earth at the rate of about 1.5 inches (3.8 centimeters) per year.[3506]

XX. Eventually the Moon's distance will increase so much that it will be too far away to produce total eclipses of the Sun.[3507] Total eclipses are moving toward extinction. As the Moon recedes from the Earth, its apparent disk becomes smaller. Thus total eclipses become more rare, and annular eclipses [become] more frequent.[3508]

XX. When the Moon's mean distance from the Earth increases by 14550 miles (23410 kilometers), the Moon's apparent disk will be too small to cover the entire sun [as seen from Earth], even when the

3501 Espenak, "Eclipse Predictions by Fred Espenak": EclipseWise.com; https://www.eclipsewise.com/help/rotation.html
3502 Littman, Willcox, and Espenak 1999: MrEclipse.com; https://www.mreclipse.com/Totality2/TotalityExtinct.html
3503 Seidelmann 1992: 62
3504 Seidelmann 1992: 62
3505 Seidelmann 1992: 62
3506 Littman, Willcox, and Espenak 1999: MrEclipse.com; https://www.mreclipse.com/Totality2/TotalityExtinct.html
3507 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/help/rotation.html
3508 Littman, Willcox, and Espenak 1999: MrEclipse.com; https://www.mreclipse.com/Totality2/TotalityExtinct.html

Moon's elliptical orbit carries it closest to Earth. [Eventually] total eclipses will no longer be possible.[3509] With the Moon receding at 1.5 inches (3.8 centimeters) per year, the last total solar eclipse visible from the surface of the Earth will take place – according to Littman, Willcox, and Espenak – approximately 620 million years from now.[3510]

XX. The reciprocal effect of the lunar tides produces a significant acceleration (actually negative) of the lunar motion. Because of the complexity of tidal friction, an accurate value for this [secular] lunar orbital [tidal friction] acceleration ($\dot{n}$) can be derived only empirically.[3511]

XX. Although it materially affects all computations of the Moon's position in the past, this [secular] lunar [orbital tidal friction] acceleration ($\dot{n}$) is especially important in the computation of Eclipses of the Sun and Moon. Reliable measurements of the [secular] lunar [orbital] tidal [friction] acceleration ($\dot{n}$) have been available only during the last twenty (20) years or so [prior to 2004 CE (Gregorian)].[3512]

XX. Recent (2001) global weather conditions have contributed to a short-term change in the length of day (LOD). Decade fluctuations due to the interaction between the Earth's core, mantle, and global ocean circulation may also contribute. At present [at time of Netson et al.'s writing in 2001] the day is actually closer to (86400) SI seconds, and thus leap seconds have not been required. However, this condition cannot persist and the long-term trend – according to Nelson et al. – will be eventually restored.[3513]

77.18. Determination of the Past Variations in Length of Day (LOD) and the Position of the Pole

XX. The determination of historical variations in the rotational rate of the Earth has been a topic of discussion since the latter part of the 19th century CE (Gregorian) when such variations were first suspected. Most of the research in this area has been on the use of ancient astronomical observations of eclipses and telescopic observations of lunar occultations in order to determine ΔT (Delta T, i.e. the difference between Earth rotational time (UT1) and Terrestrial Dynamical Time (TDT) or formerly Ephemeris Time (ET)). The time derivative of this value is the Length of Day (LOD).[3514]

77.19. Historical Variations in UT1 and Length of Day (LOD)

3509 Littman, Willcox, and Espenak 1999: MrEclipse.com; https://www.mreclipse.com/Totality2/TotalityExtinct.html
3510 Littman, Willcox, and Espenak 1999: MrEclipse.com; https://www.mreclipse.com/Totality2/TotalityExtinct.html
3511 Morrison and Stephenson 2004: 328
3512 Morrison and Stephenson 2004: 328
3513 Nelson et al. 2001: 519
3514 Seidelmann 1992: 265

XX. Since studies by Newcomb[3515] indicated that measurements of the transits of Mercury, the Moon, Jupiter's satellites all showed similar variations possibly related to the Earth's rotation (i.e. LOD), many studies and papers have been undertaken to determine accurately these variations over time.[3516]

XX. However, these studies have been complicated by their dependence on the position of the Moon (during eclipses or occultations), and hence on whether there has ever been any change in the acceleration of the Moon's mean motion [i.e. secular lunar orbital tidal friction acceleration] ($\dot{n}$) (and effectively then any change in the rate of ET determinations based on the Moon's motion).[3517]

XX. Many studies of this problem that have been undertaken up until the 1970s CE (Gregorian) attempted to solve simultaneously for ($\dot{n}$) and ΔT. An example of this approach is that by Newton (1970).[3518]

XX. However, ($\dot{n}$) – according to Seidelmann – can be assumed to be constant, an assumption that seems to be supported by modern astronomical observations (including radar and LLR measurements) of the Moon. Many studies during the 1970s and 1980s CE (Gregorian) have been undertaken with this assumption as their basis.[3519]

XX. [However, it may be mistaken to assume that the rate of change of the Moon's mean motion (i.e. secular lunar orbital tidal friction acceleration) ($\dot{n}$) has been constant extending into the distant, ancient past under discussion.]

XX. The most complete current references on historical UT1 and LOD variations (with ($\dot{n}$) assumed constant) are by Stephenson and Morrison (1984), and McCarthy and Babcock (1986) from which much of the material presented in Seidelmann (1992) is derived.[3520] The results presented in both of these references agree well with differences due only to slightly different data sources and different degrees of smoothing before 1861 CE (Gregorian).[3521] Some new results are also available in Stephenson and Lieske (1987).[3522]

XX. The results can – according to Seidelmann – conveniently be considered by examining various historical periods separately based upon the type of observations being observed. The specific periods to be mentioned in Seidelmann (1992) will include (working backward): 1980 CE (Gregorian) to present, 1955 – 1980 CE (Gregorian), 1620 – 1954 CE (Gregorian), 700 – 1600 CE (Julian –

3515 Newcomb 1882: 465 – cited by Seidelmann 1992: 265
3516 Seidelmann 1992: 265
3517 Seidelmann 1992: 265
3518 Seidelmann 1992: 265
3519 Seidelmann 1992: 265
3520 Seidelmann 1992: 265
3521 Seidelmann 1992: 265-266
3522 Seidelmann 1992: 266

Gregorian), and 700 BCE – 200 CE (Julian).[3523]

XX. Observations also exist for the period 200 – 700 CE (Julian), but have not yet [1992 CE (Gregorian)] become available (in published form). For example Hilton et al.[3524] report ΔT values determined from 58 occultations of planets by the Moon observed by the Chinese in the period 68 BCE to 575 CE (Julian).[3525]

XX. For further information on the methods used when considering lunar occultation data, Morrison[3526] may be consulted. Descriptions of how eclipse data were used are presented in Stephenson and Morrison[3527] along with many additional references on data sources.[3528]

XX. Observations of UT1 from 1955 to 1980 CE (Gregorian) are unique in that they can be referenced to TDT due to the advent of the first atomic time standards in the 1950s CE (Gregorian). However since most optical techniques (and then-developing space-based techniques) were in use, the accuracy levels were an order of magnitude lower than at the present time. Determinations by the BIH provide the best results from 1963 – 1980 CE (Gregorian) (originally in the form of (TAI - UT1) with TAI different from TDT only by an additive constant to TAI of (32.184) seconds).[3529]

XX. The period of 1620 to 1955 CE (Gregorian) covers substantially the period where telescopic observations may be used in determinations of UT1. Lunar occultations alone provide fairly precise values from 1861 to 1954 CE (Gregorian) as described by Morrison.[3530] The sparseness of such observations before that time results in some difficulties in recovering UT1.[3531]

XX. Results from occultations and now also eclipses are fitted with cubic splines in order to obtain a continuous series of values. Beyond Stephenson and Morrison's 1984 [CE] treatment, part of this period (1672 to 1806 [CE]) has been reconsidered with newly available solar eclipse timings by Stephenson and Lieske.[3532] [3533]

XX. The “medieval period” of 700 to 1600 CE (Julian – Gregorian) refers to the general period from timed Arabian eclipse observations and other untimed observations until the invention of the telescope. During this period a single “curve” of values of ΔT may no longer be estimated.[3534]

3523 Seidelmann 1992: 266
3524 Hilton et al. 1989 – cited by Seidelmann 1992: 266
3525 Seidelmann 1992: 266
3526 Morrison 1979b – cited by Seidelmann 1992: 266
3527 Morrison 1984 – cited by Seidelmann 1992: 266
3528 Seidelmann 1992: 266
3529 Seidelmann 1992: 266
3530 Morrison 1979a – cited by Seidelmann 1992: 266
3531 Seidelmann 1992: 266
3532 Stephenson and Lieske 1987 – cited by Seidelmann 1992: 266
3533 Seidelmann 1992: 266
3534 Seidelmann 1992: 266

XX. One may only set limits on the values at sporadic epochs during the period.[3535] The timed Arabian observations actually ran from approximately 800 to 1000 CE (Julian) with the timings made by measuring the altitude of the Moon or a clock star.[3536] The untimed observations were made largely from Europe usually in nonastronomical works.[3537]

XX. Ancient observations during the period circa 700 BCE to 200 CE (Julian) are of a rather limited number, but in some cases are quite useful. Stephenson and Morrison[3538] have been able to use:

1.) five (5) Chinese sightings of total and near total eclipses between 198 BCE and 120 CE (Julian);

2.) a Babylonian observation of a total solar eclipse in 136 BCE (Julian);

3.) "timed" (possibly with water clocks) Babylonian observations of lunar eclipses;

4.) twenty (20) Babylonian records indicating that the Moon rose or set while eclipsed.[3539]

XX. The advent of the International Latitude Service (ILS) in the 1890s CE (Gregorian) brought about monitoring of polar motion that has continued to this day. However except for sporadic periods during the 19th century CE (Gregorian), little data on polar motion exists before that time. Extremely high-precision polar-motion data (as with UT1 data) exists only from 1980 CE (Gregorian) on with the introduction of space-based geodetic techniques.[3540]

XX. The stations of the ILS provided monitoring of polar motion from September 1899 until 1987 CE (Gregorian). Before 1900 CE (Gregorian) data that can be used for polar-motion determination are quite sporadic.[3541]

77.20. The Length of Day (LOD)

XX. The variations in the Earth's rate of rotation are at present unpredictable and are due to several causes: There is a long-term slowing down of the rate of rotation that appears to be due largely to the effect of tidal friction [with the Earth's oceans]. There are irregular changes with characteristic periods

3535 Seidelmann 1992: 266
3536 Seidelmann 1992: 266-267
3537 Seidelmann 1992: 267
3538 Stephenson and Morrison 1984 – cited by Seidelmann 1992: 267
3539 Seidelmann 1992: 267
3540 Seidelmann 1992: 269
3541 Seidelmann 1992: 270

of a few years that appear to be due to an interchange of angular momentum between the mantle and the core of the Earth.[3542]

XX. There are also seasonal and short-period irregular changes that appear to be due largely to exchanges of angular momentum between the solid Earth and the atmosphere [of air and other gases]. The total effect on the orientation of the Earth in space is determined from observations by an international monitoring service that provides future predictions based on the observations.[3543]

XX. These variations are usually represented by the corresponding changes in the length of the (universal) day (LOD) – with respect to a day of (86400) SI seconds – and by the accumulated difference between UT and TAI (or ET before 1955 CE (Gregorian)):[3544]

(UT – TAI) == (ΔT)

(UT – ET) == (ΔT)

(UT – TT) == (ΔT)

XX. The latter difference in time (i.e. ΔT, Delta T) may be expressed as a corresponding difference in angular orientation [i.e. longitudinal / latitudinal position] of the Earth.[3545]

XX. The accumulated effect of the changes (i.e. ΔT, Delta T) in the rate of rotation can be seen most clearly in the geographic longitudes and times at which certain phenomena (such as eclipses of the Sun [and Moon]) were observed in early history.[3546]

XX. According to Seidelmann, this was a source of much confusion[3547] when the rotation of the Earth provided the standard timescale for all purposes.[3548] It is now recognized that Sidereal Time and Universal Time (UT) are best regarded as measures of angles that serve to specify the orientation of terrestrial coordinate systems with respect to celestial coordinate systems.[3549] The angle of rotation varies rapidly as the Earth rotates and is known as the Greenwich Apparent Sidereal Time (GAST).[3550]

XX. The [measurements of Sidereal Time and Universal Time (UT)] must be determined from

3542 Seidelmann 1992: 19
3543 Seidelmann 1992: 19
3544 Seidelmann 1992: 19
3545 Seidelmann 1992: 19
3546 Seidelmann 1992: 19
3547 i.e. confusion for modern scholars who study the matters of LOD and ΔT who evidently were confused when comparing modern calculations with ancient practices, understandings, and records.
3548 Seidelmann 1992: 19
3549 Seidelmann 1992: 19-20
3550 Seidelmann 1992: 20

observations, although man-made clocks [and computer programs] can be used for interpolation and extrapolation.[3551]

XX. Citing Spencer (1954), Nelson et al. note three (3) types of variation in the Earth's rotation that have been identified:[3552]

1.) a steady deceleration;

2.) random fluctuations;

3.) periodic [seasonal] changes;

77.21. Earth's Steady, Long-Term Deceleration

XX. As early as 1695 CE (Gregorian) Sir Edmond Halley was led to suspect an acceleration in the mean motion of the Moon [i.e. secular lunar orbital tidal friction acceleration (ṅ)] from a study of ancient eclipses of the Sun recorded by Claudius Ptolemy and the medieval Arabian astronomer Muhammad al-Battānī. By the mid-18th century CE (Gregorian) the [concept of secular] lunar [orbital tidal friction] acceleration [(ṅ)] was fully established.[3553]

XX. In 1754 CE (Gregorian) Immanuel Kant suggested that this acceleration might be an apparent phenomenon caused by a steady deceleration in the Earth's rotation due to tidal friction. Part of the effect was later attributed to the variation in the solar perturbation on the Moon's orbit.[3554]

XX. As shown by Pierre-Simon Laplace (1787)[3555] and John Couch Adams (1853),[3556] the planetary perturbations cause the Earth's orbital eccentricity to diminish and – as a consequence – the Sun's mean action on the Moon also diminishes. In addition, the observed [secular] lunar [orbital tidal friction] acceleration [(ṅ)] is affected by the recession of the Moon [away] from the Earth in order to compensate [for] the decrease in the Earth's rate of spin [i.e. thus creating a longer LOD] by conservation of angular momentum. It was not until the 20th century CE (Gregorian) that an apparent acceleration of the Sun was also identified.[3557]

XX. Long-term variations of the length of day (LOD) are mainly produced by lunar and solar tides, but

3551 Seidelmann 1992: 20
3552 Nelson et al. 2001: 511
3553 Nelson et al. 2001: 511
3554 Nelson et al. 2001: 511
3555 Van Flandern 1970: 657; Krajnović 2020: 11
3556 Krajnović 2020: 11
3557 Nelson et al. 2001: 511

other causes – such as the continuing rise of land that was glaciated during the last ice-age – are also significant.[3558]

XX. Citing studies of eclipses by F.R. Stephenson and L.V. Morrison (1984; 1995), Nelson et al. reiterate their suggestion that the long-term average rate of increase in the length of the day (LOD) is about 1.7 milliseconds per century (-4.5×10^{-22} rad/s^2).[3559] Although the increase in the length of day (LOD) seems miniscule, it has a cumulative effect on a time scale based on the Earth's rotation.[3560] In the past 2000 years the Earth – acting as a clock – has lost over 3 hours.[3561]

XX. It was Sir Harold Jeffreys (1920) who made the first quantitative estimate of global tidal friction.[3562] The rate of energy dissipation by tidal friction is now considered to correspond to a rate of increase in the length of day (LOD) of 2.3 milliseconds per century (-6.1×10^{-22} rad/s^2). To account for the observed deceleration [of the Earth's rate of rotation] there must also be a component in the opposite direction of about 0.6 milliseconds per century ($+1.6 \times 10^{-22}$ rad/s^2). Nelson cites several scholars who propose that this [observed deceleration rate of the Earth] is possibly associated with changes in the Earth's oblateness parameter (J_2) caused by post-glacial rebound or with deep ocean dissipation.[3563]

XX. Evidence for long-term deceleration in the Earth's rotation – extending over millions of years – also exists in coral fossils that exhibit both daily and annual growth rings. For example, several corals dating from the middle of the Devonian Period (approximately 370 million years ago) indicate that the number of days in the year was between 385 and 410. Citing Runcorn (1966), Nelson et al. assert that the rate of deceleration was substantially the same then as it is now.[3564]

77.22. Random Fluctuations in Earth's Deceleration

XX. Besides a steady decrease, the Earth's rotation is subject to frequent small changes that are random and cumulative. This variation was inferred from studies of statistical irregularities in the displacements of the Moon, Sun, Mercury, and Venus in proportion to their mean motions. [Such] random fluctuations were first observed directly by atomic clocks in the mid-1950s CE (Gregorian).[3565]

77.23. Periodic Seasonal Variation in Earth's Deceleration

3558 Morrison and Stephenson 2004: 327
3559 Nelson et al. 2001: 511
3560 Nelson et al. 2001: 511-512
3561 Nelson et al. 2001: 512
3562 Nelson et al. 2001: 512
3563 Nelson et al. 2001: 512
3564 Nelson et al. 2001: 512
3565 Nelson et al. 2001: 512

XX. [Furthermore,] there is also a periodic seasonal variation in the rotation of the Earth caused principally by meteorological effects. This seasonal variation was first detected by means of excellent crystal clocks and first reported in 1936 CE (Gregorian) by A. Scheibe and U. Adelsberger.[3566] Previously the research on long-term variations in the rotation of the Earth was correlated with the motion of the Moon which was the only clock available with sufficient precision.[3567]

XX. In 1937 CE (Gregorian) N. Stoyko at the BIH found that the length of day (LOD) in January exceeded the LOD in July by two (2) milliseconds. The rotation of the Earth runs slow by about 30 milliseconds in May and runs fast by a similar amount in November. The seasonal variation in the length of day (LOD) is now known to be of the order 0.5 milliseconds about the mean LOD.[3568]

XX. By international agreement, an empirical correction for the seasonal variation has been applied since 01 January 1956 CE (Gregorian) resulting in the time scale UT2.[3569] [That is to say] UT2 is UT1 corrected for the seasonal variation.[3570]

XX. From 1960 to 1983 CE (Gregorian) the great astronomical almanacs such as the *Astronomical Ephemeris* used a [theoretical] uniform time that was called Ephemeris Time (ET), which was based on planetary motions.[3571] In 1960 CE (Gregorian) ET replaced UT1 as the independent variable of astronomical ephemerides. However, ET did not include relativistic effects [of Einstein's theory of relativity] and did not distinguish between proper time and coordinate time.[3572]

XX. In 1976 CE (Gregorian) the IAU defined time-like arguments that distinguish coordinate systems, [the 1st with] origin at the center of the Earth, and [the 2nd with origin at] the center of the solar system, which are consistent with [Einstein's] general theory of relativity. In 1979 CE (Gregorian) these time scales received the names Terrestrial Dynamical Time (TDT) and Barycentric Dynamical Time (TDB).[3573] TDB varies from TDT (or TT) by only periodic variations, with amplitudes [of] less than (0.002) seconds.[3574]

XX. In 1984 CE (Gregorian) TDT replaced ET as the tabular argument of the fundamental geocentric ephemerides. TDT has an origin of 01 January 1977 CE (Gregorian) zero (0) hour TAI with a unit interval equal to the SI second, and maintains continuity with ET. In 1991 CE (Gregorian) the IAU

3566 Nelson et al. 2001: 512; Seidelmann 1992: 85
3567 Seidelmann 1992: 85
3568 Nelson et al. 2001: 512
3569 Nelson et al. 2001: 512
3570 Nelson et al. 2001: 510
3571 Meeus 1998b: 154
3572 Nelson et al. 2001: 513
3573 Nelson et al. 2001: 513
3574 Nelson et al. 2001: 513

renamed TDT to simply Terrestrial Time (TT).[3575] [Terrestrial] Dynamical Time (TDT) or Terrestrial Time (TT)] – which is defined by atomic clocks[3576] – is homogeneously flowing time,[3577] and is in practice a continuation of Ephemeris Time (ET).[3578]

XX. [Therefore] the difference between TT and UT is generally called ΔT (Delta T).[3579] [The following equations are all virtually equivalent to one another:]

(UT – TAI) == (ΔT)

(UT – ET) == (ΔT)

(UT – TT) == (ΔT)

XX. The relationship between TT (Terrestrial Time) and TAI [from French: *Temps atomique international*] is not strictly rigorous for two (2) fundamental reasons:[3580]

1.) While TT is theoretically uniform, TAI is a statistically-formed [i.e. averaged] time scale based on contributions from major time centers;

2.) A scale of time based on the laws of gravitation may not be philosophically equivalent to one based on the quantum mechanics of an atom;

XX. A prevalent opinion among astronomers in the mid-1960s CE (Gregorian) had been that the atomic standards could provide the unit of time, but not the continuous scale of time that they needed. However, to the contrary, the BIH was convinced that an atomic standard was the best reference for time and devoted its resources to the establishment of a practical international scale of atomic time.[3581]

XX. TAI was formally adopted in 1971 CE (Gregorian) and was an extension of the BIH atomic time scale that had been continuous back to 1955 CE (Gregorian). A distribution of approximately 200 clocks maintained in 50 laboratories contributes to TAI using an optimized weighting algorithm.[3582]

XX. [However, the interests of] two (2) communities of users [were at play]. Some – such as astronomers, geodesists, and navigators – wanted a broadcast time connected with the angle of the Earth's rotation in space. Others – such as physicists and engineers at time and frequency laboratories –

3575 Nelson et al. 2001: 513
3576 Meeus 1998b: 154
3577 Tanikawa, Yamamoto, and Sôma 2010: 797
3578 Meeus 1998b: 154
3579 Meeus 1998b: 154
3580 Nelson et al. 2001: 513
3581 Nelson et al. 2001: 514
3582 Nelson et al. 2001: 515

wanted it to be perfectly uniform to agree with the best clocks. Attempts to meet the needs of both communities led to the creation of Coordinated Universal Time (UTC) [in 1972 CE (Gregorian)].[3583]

77.24. Isaac Newton and Edmund Halley

XX. When Isaac Newton first published his revolutionary theory of gravitation in the *Principia* (1687 CE (Gregorian)), it laid the ground work for the prediction of planetary motion throughout the solar system. Edmund Halley played a pivotal role in motivating Newton to develop this mathematical description of gravity. In fact, Halley even financed much of the *Principia*'s publication costs.[3584]

XX. Although not as well-known [as his other achievements], Halley also made important scientific contributions in his studies of eclipses. Among them, he rediscovered the [ancient Babylonian] Saros cycle of 18 years plus 10 or 11 days used by Chaldeans and Babylonians (and later the Greeks) for simple lunar eclipse predictions – as it had been [previously] unknown in Halley's day.[3585]

XX. Using Newton's Theory of Moon's Motion (TMM) and the Saros cycle, Halley made a series of calculations to identify ancient eclipses in the literature. However, Halley soon encountered a problem: the eclipse paths that he predicted were shifted with respect to the historical records.[3586]

XX. In 1695 CE (Gregorian), Edmund Halley discovered that eclipses recorded in ancient history did not match calculations for the times or places of those eclipses. Starting with records of eclipses in his day and the observed motion of the Moon and Sun, he used Isaac Newton's new theory of universal gravitation (1687) to calculate when and where ancient [eclipses] should have occurred, and compared them with eclipses actually observed more than 2,000 years earlier: they did not match.[3587]

XX. [Halley had reasoned –] Either:[3588]

1.) the Moon was accelerating in its orbit;

2.) the Earth's rotation rate was slowing down, i.e. the length of the day (LOD) was increasing.

XX. Halley had great confidence in [Newton's] theory of gravitation and resisted the temptation to conclude that the force of gravity changed as time passed. Instead, he proposed that the length of a day

3583 Nelson et al. 2001: 515
3584 Espenak, "Eclipse Predictions by Fred Espenak": EclipseWise.com; https://www.eclipsewise.com/help/rotation.html
3585 Espenak, "Eclipse Predictions by Fred Espenak": EclipseWise.com; https://www.eclipsewise.com/help/rotation.html
3586 Espenak, "Eclipse Predictions by Fred Espenak": EclipseWise.com; https://www.eclipsewise.com/help/rotation.html
3587 Littman, Willcox, and Espenak 1999: MrEclipse.com; https://www.mreclipse.com/Totality2/TotalityExtinct.html
3588 Espenak, "Eclipse Predictions by Fred Espenak": EclipseWise.com; https://www.eclipsewise.com/help/rotation.html

(LOD) on Earth must have increased by a small amount.[3589]

XX. Although both are actually true; Halley correctly identified the increasing length of the day (LOD) as the primary culprit. It took another 300 years to understand why.[3590]

XX. It had been discovered by Laplace as early as 1786 CE (Gregorian) that if the Earth's orbital eccentricity slowly decreased, the Moon would accelerate.[3591] This slowing of the Earth's rotation and the magnitude of the [secular lunar orbital] tidal friction acceleration (ṅ) [i.e. the rate of change of the mean motion of the Moon] were not fully realized until the 20th century CE (Gregorian).[3592]

XX. Although the computation of Earth's deceleration is determined from timed observations of lunar and planetary phenomena, observation of an event with a recorded local solar time [including times recorded at night for observations of lunar eclipses] provides an invaluable and necessary check.[3593]

XX. The slowing of the Earth's rotation means that events in the distance past occurred later than calculated, and eclipses are seen at more easterly sites, under the assumption of uniform time [i.e. Terrestrial Time (TT) where] ($\Delta T == 0$).[3594]

77.25. Eclipses of the Moon and Sun

XX. Eclipses can only occur when the Sun and Moon are near the nodes on the lunar orbit so that eclipses begin to occur when the Sun is within the eclipse limits, and the Moon is New [or No] Moon (for a solar eclipse) or Full Moon (for a lunar eclipse).[3595] The Earth and the Moon throw conical shadows into space. The Earth's shadow is large enough to immerse the Moon completely (i.e. to eclipse it) and lies on the ecliptic opposite the Sun.[3596]

XX. Citing Toomer (1984), Kelley and Milone note that [a lunar eclipse] appears to be eclipsed for all parts of the Earth alike, both in size of the eclipse and the length of the intervals of the various phases.[3597] Because the observation of a solar eclipse is so dependent on the location of the observer – in longitude as well as in latitude – all records have implicit time information. However, all historical records are subject to distortion and to copyist error especially if they represent recollection long after

3589 Littman, Willcox, and Espenak 1999: MrEclipse.com; https://www.mreclipse.com/Totality2/TotalityExtinct.html
3590 Espenak, "Eclipse Predictions by Fred Espenak": EclipseWise.com; https://www.eclipsewise.com/help/rotation.html
3591 Kelley and Milone 2011: 126
3592 Kelley and Milone 2011: 127
3593 Kelley and Milone 2011: 105
3594 Kelley and Milone 2011: 106
3595 Kelley and Milone 2011: 123
3596 Kaler 2003: 34
3597 Kelley and Milone 2011: 122

the event.[3598]

XX. The global frequency of lunar eclipses is less than solar eclipses, but because of their greater visibility, lunar eclipses can be seen more often at any single location on Earth. There can be no more than a total of seven (7) eclipses – lunar and solar – in any one (1) year. Total solar eclipses at a particular spot on Earth may not repeat for 300 years, although partial eclipses seen from that spot are more frequent.[3599]

XX. Eclipses are possible in all months. However, since the Babylonian calendar has strictly lunar months, lunar eclipses are possible only around the middle of the month on days 12 to 15.[3600] Nevertheless, the 14th day seems to have been the "normal" day for a lunar eclipse.[3601]

XX. Ptolemy (ca. 150 CE (Julian)) himself made use of ancient eclipse observations. The earliest lunar eclipse entry in Ptolemy's list was that seen in Alexandria in the year 546 of the Nabonassar Era [i.e. Anno Nabonassar (AN)] on the 27th / 28th day of the 7th Egyptian month Phamenoth. Ptolemy provides the records of four (4) lunar eclipses, and these are useful in providing information on the local dates that he used (i.e. year of the reign of Hadrian and months and days in the Egyptian calendar).[3602]

XX. For study of the motion of the Moon, Ptolemy recognized the importance of lunar over solar eclipse records because the latter phenomenon is strongly dependent on the position of the observer, but the former is visible wherever the eclipsed Moon is above the horizon.[3603]

XX. At Full Moon, when the Moon is also opposite the Sun [i.e. Opposition], it can be eclipsed. Because of the five-(5)-degree tilt of the lunar orbit [relative to the ecliptic of the Sun], the Full Moon usually passes above or below the shadow [of the Earth], and we see no eclipse.[3604]

XX. However, not quite twice a year, the Moon passes full phase while sufficiently close to the ecliptic, and we get a lunar eclipse. If all of the Moon goes into the shadow, the eclipse is "total". If only part of it is immersed, it is "partial".[3605]

XX. During totality, the Moon does not disappear, but [rather] takes on a deep reddish color.[3606] The color of the totally eclipsed Moon is a dark dull red because of the refraction and scattering of sunlight

3598 Kelley and Milone 2011: 123
3599 Kelley and Milone 2011: 124
3600 Hunger and Pingree 1999: 15
3601 Hunger and Pingree 1999: 16
3602 Kelley and Milone 2011: 122
3603 Kelley and Milone 2011: 122
3604 Kaler 2003: 34
3605 Kaler 2003: 34
3606 Kaler 2003: 34

by the Earth's atmosphere into its shadow.[3607]

XX. [This is] the same reason that the Sun is reddened at Sunset when its light shines through our thick blanket of air.[3608] In some cases other colors may actually appear on the Moon because of clouds and other atmospheric phenomena.[3609]

XX. A central total [lunar] eclipse takes about four (4) hours from start to finish. Half (½) the [lunar] eclipses occur during daylight when we cannot see them, reducing the number visible to somewhat less than one (1) per year.[3610]

XX. The Moon is only a quarter (¼) the size of Earth, which makes the lunar shadow similarly shorter [than the shadow of the Earth]. At and near perigee, the shadow [of the Moon] can just barely reach us, while near apogee, it stops short. That is, at apogee [i.e. farthest from the Earth], the lunar disk is slightly smaller than the Sun's half-(½)-degree angular diameter, while at perigee [i.e. closest to the Earth] it is slightly larger.[3611]

XX. The Moon's angular size of half (½) a degree is much smaller than it appears. That the Moon seems huge when rising over the horizon is an optical illusion.[3612]

XX. A "total eclipse" of the Sun, in which the shadow [of the Moon] hits the Earth and the Moon fully covers the Sun, can take place only near perigee. If the Moon is near apogee, we see a ring – an annulus – surrounding the lunar disk, and therefore an "annular eclipse".[3613]

XX. At best, the shadow spot [of the Moon during a total eclipse of the Sun that is] on the Earth is very small: less than 300 km across. As the Moon orbits and the Earth rotates, the shadow zips along at some 2,000 km/hr, leading to a maximum duration of not quite eight (8) minutes. For any given location, total [solar] eclipses are therefore rare, recurring on the order of centuries. Partial [solar] eclipses, in which the Moon is seen to cover only a portion of the Sun, are much more common.[3614]

XX. Globally, solar eclipses (of both kinds) are more common than lunar eclipses. The geometry of the solar eclipse dictates at least two (2) per year, and there can be as many as five (5)![3615]

3607 Hunger and Pingree 1999: 16; Kaler 2003: 34
3608 Kaler 2003: 34-35
3609 Hunger and Pingree 1999: 16
3610 Kaler 2003: 34
3611 Kaler 2003: 35
3612 Kaler 2003: 32
3613 Kaler 2003: 35
3614 Kaler 2003: 35
3615 Kaler 2003: 35

77.26. Ancient Eclipses as Tools to Check Earth's Deceleration Rate

XX. Ancient eclipses have been – and still are – valuable to study for many reasons.[3616] Ancient eclipses are among the most useful events for checking the [Earth's] deceleration rate. The ancient lunar and solar eclipses used by several authors include Babylonian, Chinese, Greek, and Arabic sources.[3617]

XX. This work has important consequences for historical studies: chronologies are sometimes based on eclipse records (at least in part) so it is important to sort out what can be determined convincingly without using the very events that are to be calibrated(!).[3618] When archaeoastronomy can be used in conjunction with other techniques, it may often substantially increase the precision of a date.[3619]

XX. Only the convergence of philological, historical, and astronomical data gives a high probability of a certain dating[3620] [by] proposing to the humanist several possibilities from which he will choose according to his specific knowledge of text, philology, history.[3621]

XX. Many atlases of eclipses are available with which ancient observations can be compared:[3622]

XX. For solar and lunar eclipses of the entire ancient world, only Theodor von Oppolzer's *Canon der Finsternisse* (Canon of Eclipses (1887 / 1962)) has a nearly complete record,[3623] and [it] stands as one of the greatest accomplishments in computational astronomy of the 19th century CE (Gregorian).[3624] He lists 8000 solar eclipses and 5200 lunar eclipses.[3625] [However,] the 19th century CE (Gregorian) ephemerides for the Sun and Moon, which are critical to eclipse predictions, cannot compare to the accuracy and precision of modern ephemerides.[3626]

XX. Oppolzer included a correction [i.e. a variable] for the secular variation in the lunar longitude.[3627] [However,] the 1887 CE (Gregorian) canon did not take into account the shifts in latitude and timing of ancient eclipses due to Earth's variable rotation rate and the secular acceleration of the Moon. Nevertheless, von Oppolzer's canon remained the seminal reference on predictions of eclipses until well into the 20th century CE (Gregorian).[3628]

3616 Kelley and Milone 2011: 121
3617 Kelley and Milone 2011: 105
3618 Kelley and Milone 2011: 105
3619 Kelley and Milone 2011: 104
3620 De Meis 2004: 119
3621 De Meis 2004: 122
3622 Kelley and Milone 2011: 126
3623 Kelley and Milone 2011: 126
3624 Espenak and Meeus 2009: p. v
3625 Kelley and Milone 2011: 126
3626 Espenak and Meeus 2009: p. v
3627 Kelley and Milone 2011: 126
3628 Espenak and Meeus 2009: p. v

XX. Schove (1984) states that the beginning and end points of von Oppolzer's eclipses could have errors as large as 100 km on the polar plots provided.[3629] P.V. Neugebauer (1931a, 1931b) suggested that the error in timings amounted to only approximately ~20 minutes in 700 BCE (Julian).[3630]

XX. Other eclipses tables [relevant] for selective purposes are:[3631]

1.) Ginzel (1899) have solar eclipses for the Mediterranean from 900 BCE to 600 CE (Julian).[3632]

2.) P.V. Neugebauer and Hiller (1931) for Near East sites for solar eclipses from the interval 4200 BCE to 900 BCE (Julian), and for lunar eclipses for 3450 BCE to 900 BCE (Julian).[3633]

3.) Link (1969) delineates the geometric details of lunar eclipses, both ancient and modern.[3634]

4.) Kudlek and Mickler (1971) for Near East sites [solar and lunar eclipses] back to 3000 BCE.[3635]

5.) Meeus and Mucke (1983) provide lunar eclipse tables for the interval 2003 BCE (Julian) to 2526 CE (Gregorian).[3636] In 1979 CE (Gregorian), Meeus and Mucke published *Canon of Lunar Eclipses: -2002 to +2526* containing the Besselian elements of 10,936 lunar eclipses. It was intended mainly for historical research and served as the modern day successor of von Oppolzer's great canon. The Meeus-Mucke work also contains data on penumbral eclipses that are not included in von Oppolzer's canon. Neither of these publications offers diagrams or maps to illustrate the geometry or visibility of each eclipse.[3637]

6.) Espenak's (1989) *Fifty Year Canon of Lunar Eclipses* includes individual Moon-shadow geometry diagrams and eclipse visibility maps of all lunar eclipses, but it covers a relatively short period from +1986 to +2035 CE (Gregorian).[3638] Both of these recent lunar eclipse canons [i.e. that of Meeus-Mucke and that of Espenak] are based on Newcomb's *Tables of the Sun* (1895) and Brown's lunar theory (1905), subject to later modifications in the *Improved Lunar Ephemeris* (1954 CE

3629 Kelley and Milone 2011: 126
3630 Kelley and Milone 2011: 127
3631 Kelley and Milone 2011: 127
3632 Kelley and Milone 2011: 127
3633 Kelley and Milone 2011: 127
3634 Kelley and Milone 2011: 121
3635 Kudlek, M., and Mickler, E.H. 1971; *Solar and Lunar Eclipses of the Ancient Near East 3000 B.C. to 0 with Maps* – cited by Kelley and Milone 2011: 127; 548; Kelley and Milone have typo of "Hudek and Mickler" on page 127.
3636 Kelley and Milone 2011: 127
3637 Espenak and Meeus 2009: p. v
3638 Espenak and Meeus 2009: p. v

(Gregorian)). These were the best ephemerides of their day but they have since been superseded.[3639]

7.) Stephenson and Houlden (1986) have solar eclipse maps for the Far East – with modeled values of ΔT – for the interval 1500 BCE (Julian) to 1900 CE (Gregorian).[3640]

8.) Liu and Fiala's (1992) *The Canon of Lunar Eclipses 1500 B.C.- A.D. 3000* uses modern theories of the Sun and the Moon prepared by the Bureau des Longitudes of Paris.[3641] [It] is a modern and comprehensive canon of world lunar eclipses covering the interval -1500 BCE (Julian) to +3000 CE (Gregorian) – with computer software to recreate the circumstances of the appearance and the visibility of eclipses on images of anywhere on the Earth – with ΔT corrections. Their Table 3.3 contains corrections [i.e. variables] corresponding to the formulae of Stephenson and Morrison (1984) for the interval -1500 BCE (Julian) to +1500 CE (Gregorian) but with a slightly different [secular] lunar [orbital] tidal [friction acceleration ($\dot{n}$), or the rate of change of the Moon's mean motion] term (-23.895" /cy^2 from the value -26" /cy^2) [i.e. variable]. This enables the user to subtract out the Stephenson and Morrison correction [i.e. variable] and to add other corrections [i.e. variables] as improved values become available.[3642]

9.) Espenak and Meeus (2006, 2009a, 2009b, 2009c) published a massive set of lunar and solar catalogs and canons covering a 5000-year span.[3643] [It] is the first to offer eclipse geometry diagrams and visibility maps for every lunar eclipse (12,064 eclipses) over a period covering five thousand years from -1999 (Julian / Gregorian Astronomical) to +3000 CE (Gregorian). The following points highlight the features and characteristics of this work:[3644]

1.) [It is] based on modern theories of the Sun and the Moon constructed at the Bureau des Longitudes of Paris rather than the older Newcomb and Brown ephemerides;

2.) [It contains] ephemerides and eclipse predictions performed in Terrestrial Dynamical Time (TDT);

3.) [It] covers historical period of eclipses as well as one (1) millennium into the future;

4.) [It contains] diagrams for each eclipse [that] depict the Moon's path through Earth's penumbral and umbral shadows;

5.) [It contains] world maps [that] identify geographic regions of visibility for each phase of

3639 Espenak and Meeus 2009: p. v
3640 Kelley and Milone 2011: 127
3641 Espenak and Meeus 2009: p. v
3642 Kelley and Milone 2011: 121, 127
3643 Kelley and Milone 2011: 121
3644 Espenak and Meeus 2009: p. v-vi

every eclipse;

6.) [Its] visibility maps are based on the most current determination of the historical values of ΔT;

7.) [Its] estimates of eclipse visibility map accuracy based on the uncertainty in the value of ΔT (i.e. standard error in ΔT);

XX. A primary goal of Espenak and Meeus' work is to assist historians and archeologists in the identification and dating of eclipses found in references and records from antiquity.[3645]

XX. In a review of the degree of reliability of ancient eclipse observations, Muller and Stephenson (1989) conclude that only four (4) early solar eclipse records can be considered of the superior class "A" reliability:[3646]

1.) Total solar eclipse of 07 July 709 BCE (Julian)at Chu Fu, China

2.) Total solar eclipse of 26 September 322 BCE (Julian) at Babylon

3.) Total solar eclipse of 04 March 181 BCE (Julian) at Ch'ang An, China

4.) Total solar eclipse of 15 April 136 BCE (Julian) at Babylon

The records of the eclipse of 15 April 136 BCE (Gregorian) were evaluated by Stephenson and Clark (1978) as the most accurate before the invention of the telescope.[3647]

The question of total [solar] eclipses is limited for Babylon: from -750 [751 BCE (Julian)] to the [astronomical] year zero [0 == -1 BCE (Julian)] there were only four (4) [total solar] eclipses visible from there as such on:[3648]

1.) -435 May 31

2.) -401 January 18

3.) -135 April 15

3645 Espenak and Meeus 2009: p. v-vi
3646 Kelley and Milone 2011: 123
3647 Kelley and Milone 2011: 123
3648 De Meis 2004: 120

4.) -9 June 19

XX. The other [solar eclipses] were annular (two (2) only) or partial.[3649]

XX. It is notable that of the total solar eclipses studied by scholars that were supposedly visible at Babylon, only the total solar eclipse of 15 April -135 (136 BCE (Julian)) is mentioned by both Muller and Stephenson (1989) as well as De Meis (2004). Muller and Stephenson also mention a total solar eclipse of 26 September -321 (322 BCE (Julian)) which De Meis does not include.

77.27. ΔT (Delta T)

XX. The variations in the rotation of the Earth can be determined from observations, though early observations are – according to Seidelmann – much less accurate than modern observations. The rate of rotation does not decrease uniformly as it would if tidal friction were the only mechanism [i.e. variable] affecting the Earth's rotation.[3650]

XX. To determine the geographic visibility of an eclipse, TD[T] [or TT or ET] is converted to Universal Time (UT) using the parameter ΔT.[3651] ΔT [is] the clock error in Universal Time (UT) due to the variation of the Earth's rotation. Any trend and fluctuation in the rotation may be expected to have a counterpart in the time series of ΔT.[3652]

XX. For a uniform time system [e.g. TDT / TT or ET], the value of ΔT is obtained when the actual location of the totality zone of a solar [or lunar] eclipse is known. Of course, the actual location will be defined [only] when ΔT is known.[3653] [3654]

XX. Accurate knowledge of the value of ΔT is often crucial in assessing the local circumstances of solar [and lunar] eclipses. Neglect of variations in the Earth's spin rate would materially affect the calculated positions of where these phenomena could be seen on the Earth's surface.[3655] [Indeed, such uncertainty in ancient values for ΔT could also significantly affect the accuracy of calculations of dates and times of first or last lunar crescent visibility in antiquity when using geometric arcus visionis that is based on lunar ephemerides, and hence the calculation and dating of ancient lunar calendar dates may contain errors.]

3649 De Meis 2004: 120
3650 Seidelmann 1992: 83
3651 Espenak and Meeus 2009: 4
3652 Han and Qiao 2003: 569
3653 "Grasshopper: when you can take the pebble from my hand, it will be time for you to leave."
3654 Han and Qiao 2003: 572
3655 Morrison and Stephenson 2004: 327

XX. Where is the actual totality zone when the eclipse occurred? The value of ΔT leads to a relative displacement between the actual and calculated totality zone. So we define the totality zone by changing the [variable] ΔT – there is an interesting phenomenon.[3656] [3657]

XX. [This leads to a situation in which] it is almost impossible to avoid circular arguments: Because one must calculate the time difference (ΔT) between TT and UT (i.e. ΔT == TT – UT), the transformation between [i.e. conversion to and from] TT and UT requires a set of useful timed records made by ancient observers. [However, this is precisely] the disadvantage in that the UT must be reconstructed from calculations of ancient observations[!!].[3658] [That is to say:] Its exact value can be deduced only *a posteriori* from [ancient] observations.[3659]

XX. Therefore, ΔT can only be deduced from [ancient] observations. The knowledge of the exact value of ΔT is essential for predicting the correct time of an astronomical event or to confirm the time of historical events. If ΔT is not considered [correctly] the result may contain an error of several seconds [3660] [or more(!!)].

XX. The difference between uniform Dynamical Time (TD[T] [or Terrestrial Time (TT)]) and Universal Time (UT) – known as ΔT (Delta T) – is needed in several astronomical calculations such as eclipses and occultations, or to reduce [i.e. to convert] measured positions of minor planets to a uniform time scale for the aim of orbital determinations.[3661] This difference between TT [or ET] and UT is generally called ΔT (Delta T).[3662]

XX. The time difference ΔT == (ET – UT1) == (TT – UT1) represents the difference between Ephemeris Time (ET)] (or Terrestrial Time (TT))[3663] – i.e. a theoretically uniform time scale defined by the Newtonian dynamical laws of motion of the Earth, Moon, and planets.[3664] – and the variable scale of Universal Time (UT1) [which is the true measure of Earth's rotation and based upon the variable, non-uniform rotation of the Earth on its own axis], i.e. ΔT == (ET – UT1); or ΔT == (TT – UT1).[3665]

XX. There is an uncertainty in Earth's rotational period expressed in the parameter [i.e. variable] ΔT and its impact on the geographic visibility of eclipses in the past and future.[3666] The uncertainty in the value of ΔT is of particular interest in the calculation of eclipse paths in the distant past and future.[3667]

3656 "Aaaaaaah, so!"
3657 Han and Qiao 2003: 572
3658 Henriksson 2009: 167
3659 Meeus 1998b: 154
3660 Sana, Muhammad, and Shahid 2008a: 1
3661 Meeus and Simons 2000: 323
3662 Meeus 1998b: 154
3663 Nelson et al. 2001: 517
3664 Nelson et al. 2001: 510
3665 Nelson et al. 2001: 517; Seidelmann 1992: 78
3666 Espenak and Meeus 2009: https://EclipseWise.com; https://www.eclipsewise.com/lunar/Lepubs/5MCLE.html;
3667 Espenak and Meeus 2006: https://EclipseWise.com; https://www.eclipsewise.com/help/uncertainty.html;

XX. [However,] correlating historical records with specific eclipses [has heretofore been] no easy task since there are usually several possible candidates [when attempting to calculate using ΔT with its inherent uncertainty in both its possible varying values as well as its standard error]. Accurate visibility maps using the best available values of ΔT coupled with estimates in the standard error of ΔT are critical in discriminating among potential eclipses candidates. Ultimately, Espenak and Meeus note that historical eclipse identification can lead to improved chronologies in the time line of a particular culture.[3668]

77.28. Values of ΔT

XX. Good values of ΔT only exist sometime after the invention of the telescope (1610 CE (Gregorian)). Careful analysis of telescopic timings of stellar occultations by the Moon permits the direct measurement of ΔT during this time period. Prior to the 1600s CE (Gregorian), values of ΔT must rely on historical records of the naked eye observations of eclipses and occultations. Such observations are rare in the literature and – according to Espenak – of coarse precision.[3669]

XX. Unfortunately, estimating [not only ΔT itself, but also] the standard error in ΔT prior to 1600 CE (Gregorian) is a difficult problem. It depends upon a number of factors which include the accuracy of determining ΔT from historical eclipse records as well as modeling the physical processes producing changes in Earth's rotation.[3670]

XX. Values of ΔT over most of the telescopic period are usually determined from a comparison of the observed and computed positions of the Moon. Most of these observations – numbering many thousands – are in the form of lunar occultations of stars.[3671] Prior to the telescopic era (between about 1620 – 1955[CE (Gregorian)), eclipses provide the most effective data for determining results for ΔT. The long-term parabolic trend is deduced from an extensive series of observations of ancient and medieval eclipses.[3672]

XX. As noted by Henriksson (2009 CE (Gregorian)), the most extensive study of ΔT has been performed by Richard Stephenson (1997 CE (Gregorian)). His analysis is based on the value for the secular [lunar orbital tidal friction] acceleration (ṅ) [i.e. the rate of change of the mean motion] of the Moon to be (26 +/-2" /cy^2) determined from telescopic observations of transits of Mercury 1677 – 1973 CE (Gregorian), analyzed by Morrison and Ward (1975). However, as noted by Henriksson, the oldest

3668 Espenak and Meeus 2009: p. vi
3669 Espenak, "Eclipse Predictions by Fred Espenak", https://EclipseWise.com; https://www.eclipsewise.com/help/rotation.html;
3670 Espenak and Meeus 2006: https://EclipseWise.com; https://www.eclipsewise.com/help/uncertainty.html;
3671 Morrison and Stephenson 2004: 328
3672 Morrison and Stephenson 2004: 329

of these observations have low quality and there is no motivation [i.e. no logical reason] for an extrapolation of this 300-years interval by a factor of seven (7) back to the ancient observations [due to the uncertainty of accuracy when these values are extrapolated into the distant ancient past].[3673]

XX. Stephenson and collaborators have made a number of important contributions concerning Earth's rotation during the past several millennia. In particular, they have identified hundreds of eclipse and occultation observations in early European, Middle Eastern, and Chinese annals, manuscripts, canons, and records. In spite of their relatively low precision, these data represent our only record of the value of ΔT during the past several millennia.[3674]

XX. In particular, Stephenson and Morrison (1984) have fit hundreds of records with simple polynomials to achieve an [attempted] best fit for describing the value of ΔT from 700 BCE (Julian) to 1600 CE (Gregorian). In more recent work Stephenson (1997) has made improvements in his analysis of ΔT using additional historical records.[3675]

XX. [Another] such technique was developed by Jean Meeus and Larry Simons in 2000 CE (Gregorian) that gives a set of polynomial approximation for ΔT for the range of 1620 to 2000 CE (Gregorian).[3676] Meeus stresses that it should be noted that for the very [distant] past and future, his calculated values of ΔT may be in error by many minutes, as fluctuations due to the variable rotation of the Earth are unknown for those remote epochs.[3677] For any instant during a [specific] year one needs some approximation technique to deduce the value of ΔT.[3678]

77.29. The Uncertainty of ΔT

XX. The uncertainty in ΔT means that reliable eclipse paths prior to about 1500 BCE (Julian) are – according to Espenak – not possible. Similarly, all future values of ΔT are simple extrapolations of current values and trends. Such estimates are prone to growing uncertainty as one extrapolates further and further into the future. By the year 3000 CE (Gregorian), the value of ΔT could be on the order of one (1) hour with an extrapolated uncertainty of about ten (10) minutes or several degrees (°) in longitude.[3679]

XX. According to De Jong, Huber has shown that this clock time correction ΔT (Delta T) is accurate within a standard deviation [of error] of ±5 minutes back to 500 BCE (Julian).[3680] Due to the complete

3673 Henriksson 2009: 167
3674 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/help/rotation.html;
3675 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/help/rotation.html;
3676 Meeus and Simons 2000: 323-324 – cited by Sana, Muhammad, and Shahid 2008b: 364
3677 Meeus 2002: 8
3678 Meeus and Simons 2000: 323-324 – cited by Sana, Muhammad, and Shahid 2008b: 364
3679 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/help/rotation.html;
3680 De Jong 2012: 395

lack of systematic records of astronomical observations in ancient Egypt, no visibility lines based on Egyptian data alone can be obtained for Egyptian lunar observations.[3681]

XX. If a computer program does not incorporate these values into its calculations, then the resultant eclipse paths will contain longitude shifts inconsistent with the best estimates for ΔT.[3682] It is therefore a matter of concern that at present there appears to be a degree of confusion and misapprehension among historians of astronomy over the choice of values of ΔT that should be used in making retrospective computations of lunar and solar positions.[3683]

XX. A correct identification of ancient solar [and lunar] eclipses is not only important for historical reasons, but also gives the possibility to determine the [secular lunar orbital tidal friction] acceleration (ṅ) [or the rate of change of the mean motion] of the longitude of the Moon to a high precision.[3684] Eclipses – especially solar ones according to De Meis – are most useful to date an historical document. In the near past [prior to the days of our computer age], to avoid cumbersome and difficult calculations one used to consult eclipse tables or "Canons".[3685]

XX. However, [ephemerides] tables gave rise to equivocal [i.e. uncertain, ambiguous] results: for example, one would find a day in which an eclipse occurred, but that eclipse actually was visible elsewhere [because of the phenomenon of ΔT], and the table was silent on this.[3686] Old catalogs and Canons either ignored ΔT or gave wrong values, as the measurement and extrapolation of this variable is quite recent, especially by Meeus (1991) and Huber (2000).[3687]

XX. This quantity ΔT is the difference between Dynamical Time (TD[T] or TT) and Universal Time (UT), [i.e. ΔT == TT – UT] that is between a constantly running [uniform] time (TD[T] or TT) and the actual perturbed time (UT) (due to the variations in the rotation of the Earth).[3688]

XX. [According to current scholarship] this quantity ΔT amounts to about six (6) hours in -700 [-701 BCE (Julian)], and this means that the actual visibility of a phenomenon occurs at a longitude about 90° east of the point calculated in Dynamical Time (TD[T] or TT).[3689]

XX. Exact values of ΔT are subject to various considerations [i.e. variables], however De Meis believes that good mean values can be computed.[3690] However, as De Meis correctly notes: to ignore ΔT [or to

3681 Gautschy 2014: 80
3682 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/help/rotation.html;
3683 Morrison and Stephenson 2004: 327
3684 Henriksson 2009: 166
3685 De Meis 2004: 120
3686 De Meis 2004: 120
3687 De Meis 2004: 121
3688 De Meis 2004: 121
3689 De Meis 2004: 121
3690 De Meis 2004: 121

use an incorrect value for ΔT in calculating dates of ancient eclipses] is clearly a source of errors.[3691]

77.30. The Rate of Change of the Moon's Mean Motion: Secular Lunar Orbital Tidal Friction Acceleration (ṅ)

XX. Stephenson (1997) published his monumental work on the investigation of changes in the Earth's rotation using pre-telescopic observations of eclipses between 763 BCE (Julian) and 1772 CE (Gregorian). According to him, he determined the Earth's clock error ΔT for each of the eclipses.[3692]

XX. For the interval 700 BCE (Julian) to the present, prior work by Stephenson and Morrison (1995 CE (Gregorian)) suggests that the slowing of the Earth's rotation has measurable variation. They have revised estimates of the underlying deceleration. In particular they (and others) have argued that action of tidal friction [on the Earth] due to the Moon is (at present (2011 CE (Gregorian)) partly offset by a negative (deceleration) term which implies a speedup.[3693]

XX. With an adopted value for the [secular] lunar [orbital tidal friction] acceleration [or the rate of change of the mean motion of the Moon], (ṅ) == (-26 ± 0.5" /cy^2) [i.e. -26 ± 0.5 arcseconds per century squared], Stephenson and Morrison (1995) found a slowing of the Earth due to tidal friction alone of (2.3 ± 0.1) milliseconds per century (ms /cy).[3694] A [secular] lunar [orbital tidal friction] acceleration (ṅ) of -26 arcseconds /cy^2 results in a recession of the Moon from the Earth of 3.86 meters /century [m /cy], and an increase in the length of the month of (0.038) seconds /century [s/cy].[3695]

XX. From Babylonian and other data, Stephenson and Morrison (1995) revise their preferred value for the observed ΔT to be:[3696]

$$\Delta T == +31.0\ (\pm\ 0.9)\ t^2\ s$$

XX. This implies a rate of change of the length of the day (LOD) of (+1.70 ± 0.05 ms/cy). Because +2.3 ms /cy is the expected rate from tidal friction, the difference of -0.6 ms /cy suggests a source of speedup.[3697]

XX. Further residuals in their fittings of Babylonian, Chinese, Arab, and European data lead Stephenson and Morrison (1995) to conclude that there are periodic fluctuations of ± 4 milliseconds

3691 De Meis 2004: 121
3692 Sôma and Tanikawa 2015: 11
3693 Kelley and Milone 2011: 105
3694 Kelley and Milone 2011: 105-106
3695 Kelley and Milone 2011: 105
3696 Kelley and Milone 2011: 106
3697 Kelley and Milone 2011: 106

over a period of [approximately] ~1500 years.[3698] These fluctuations provide a limitation on the usefulness of ancient records.[3699]

XX. Sôma and Tanikawa have been trying to determine more precise values of ΔT using contemporaneous solar eclipses. Additionally, they have been trying to investigate whether or not the Moon's [secular lunar orbital] tidal [friction] acceleration (ṅ) has been constant since ancient times.[3700]

XX. According to Sôma and Tanikawa, the Moon's [secular lunar orbital] tidal [friction] acceleration (ṅ) has been almost constant since the 17th century CE (Gregorian), but this is no guarantee that the [secular] lunar [orbital] tidal [friction] acceleration (ṅ) has been constant since ancient times[!]. Therefore it is important to investigate ancient records of astronomical phenomena and determine whether this value has been constant [or not] from ancient times.[3701]

XX. In all modern calculations, the value for the secular lunar [orbital] tidal [friction] acceleration (ṅ) – -26 arcseconds /century^2 ([denoted as] -26" /cy^2) – has been used from the transits of Mercury 1677 – 1973 CE (Gregorian), determined by Morrison and Ward (1975).[3702]

XX. [Secular lunar orbital tidal friction acceleration (ṅ) is a critical variable whose value can significantly affect the accuracy of the geometric arcus visionis methodology based on calculations of lunar ephemerides, but as noted above: the calculate value for this variable (ṅ) are determined from the observed / measured / calculated transits of Mercury 1677 – 1973 CE (Gregorian).] Published values of (ṅ) have varied considerably between 1931 and 2002 derived by different methods.[3703] [The specific value a study adopts – and whether it is consistent with the ΔT value that is subsequently derived using (ṅ) – affects the computed lunar positions on which the geometric arcus visionis depends.] If another value of (n) [i.e. written (ṅ) elsewhere] is used [calculated from a different methodology, e.g. the motions of the Moon], then according to Seidelmann, the values of (TDT – UT1) should be corrected by the addition of [the following equation]:[3704]

$$(\Delta T)\ (TDT - UT1) == (-1.821s)\ (\Delta n / 2)\ (T - 19.55)$$

where (T) is measured in Julian centuries from 1900.0 CE (Gregorian).[3705]

Table 77.1. Principal values of [secular] lunar [orbital] tidal [friction] acceleration (ṅ) expressed in arcseconds /century^2 [i.e. arcseconds per century per century OR arcseconds per century

3698 Kelley and Milone 2011: 106-107
3699 Kelley and Milone 2011: 107
3700 Sôma and Tanikawa 2015: 11
3701 Sôma and Tanikawa 2015: 13
3702 Henriksson 2009: 166
3703 McCarthy and Babcock 1986 – cited by Seidelmann 1992: 84
3704 Seidelmann 1992: 84
3705 McCarthy and Babcock 1986 – cited by Seidelmann 1992: 84

squared, denoted as a numeral x" /cy^2] obtained so far[3706]

Author(s)	**Year**	**Secular Lunar Orbital Tidal Friction Acceleration (ṅ) (" /cy^2)**
Schoch	1931	-29.68 arcseconds /century^2
Spencer Jones	1939	-22.44
Van Flandern	1970	-56.0
Oesterwinter and Cohen	1972	-38
Morrison	1973	-42.0
Van Flandern	1975	-65.0
Morrison and Ward	1975	-26.0
Muller	1976	-30
Calame and Mulholland	1978	-24.6
Williams et al.	1978	-23.8
Ferrari et al.	1980	-23.8
Dickey et al.	1982	-23.8
Dickey and Williams	1982	-25.12
Newhall et al.	1988	-24.90
Dickey et al.	1994	-25.88
Chapront and Chapront-Touzé	1997	-25.64
Chapront et al.	1999	-25.78
Chapront et al.	2000	-25.836
Chapront et al.	2002	-25.858

XX. These values of [secular] lunar [orbital] tidal [friction] acceleration (ṅ) are twice the coefficient of the (T^2) tidal term (where (T) is the time in [Julian] centuries). The value listed for Spencer Jones (1939) was actually derived by Clemence (1948), and based on Spencer Jones' results [of 1939].[3707] [The value listed for Carl Schoch (1931) has been added to this table from external data cited in Henriksson (2009)]. The values obtained from the lunar laser ranging observations since the 1970s CE (Gregorian) converge well at about -25.9" cy^{-2}.[3708]

XX. At the time of Morrison and Stephenson's writing (2004), reliable measurements of the [secular]

3706 Sôma and Tanikawa 2015: 12
3707 Sôma and Tanikawa 2015: 12
3708 Sôma and Tanikawa 2015: 13

lunar [orbital] tidal [friction] acceleration (ṅ) had only been available during the previous 20 years or so. The accurate results for [secular lunar orbital tidal friction acceleration] (ṅ) are obtained from lunar laser ranging observations, and these indicate a value very close to -26 arcseconds per century per century [or: -26 arcseconds per century squared] ([denoted as] -26" /cy^2).[3709]

XX. Morrison and Stephenson's own results for ΔT are all derived using this figure [of -26" /cy^2]. [As mentioned above,] they note that global sea-level studies suggest that tidal friction has not changed significantly over the last three (3) millennia or so.[3710]

XX. This is essentially the value used in major Jet Propulsion Laboratory (JPL) Development Ephemerides such as LE51 and the LE400 series. However, for several decades the most commonly used value for [secular lunar orbital tidal friction acceleration] (ṅ) had been -22.44" /cy^2 (i.e. negative 22.44 arcseconds per century per century [or: negative 22.44 arcseconds per century squared]), which was derived from the work of Jones (1939). This value was used in the *Astronomical Almanac* until 1984 CE (Gregorian).[3711]

XX. However, according to Morrison and Stephenson (2004), use of such a low figure for [secular lunar orbital tidal friction acceleration] (ṅ) leads to errors in the lunar position of as much as 0.5° (roughly the Moon's apparent diameter) around 1000 BCE (Julian). For Morrison and Stephenson's own results for ΔT, in all their computations of lunar co-ordinates they both use and recommend using [secular lunar orbital tidal friction acceleration] (ṅ) == -26"/cy^2 (i.e. negative 26 arcseconds per century per century [or: negative 26 arcseconds per century squared]).[3712] [Let us recall again Schloch's value for secular lunar orbital tidal friction acceleration (ṅ) == -29.68 arcseconds /century^2 advocated by Henriksson (2009).]

XX. Thus it is possible to see that there is disagreement among the leading scholars who study the Earth's ΔT as well as the rate of change of the Moon's mean motion, i.e. secular lunar orbital tidal friction acceleration (ṅ). Preceding efforts to attempt to determine exact values of ΔT remain uncertain, questionable, unresolved, and unsettled as exact values of ΔT are influenced by exact values of (ṅ) which have been supposedly determined from the transits of Mercury 1677 – 1973 CE (Gregorian), analyzed by Morrison and Ward (1975).

XX. However, let us recall Henriksson who notes the oldest of these observations have low quality and there is no motivation [i.e. no logical reason] for an extrapolation of this 300-years interval by a factor of seven (7) back to the ancient observations [due to the uncertainty of accuracy when these values are extrapolated into the distant ancient past].[3713]

3709 Morrison and Stephenson 2004: 328
3710 Morrison and Stephenson 2004: 328
3711 Morrison and Stephenson 2004: 328
3712 Morrison and Stephenson 2004: 328
3713 Henriksson 2009: 167

77.31. The Lunar Laser Range (LLR)

XX. The Lunar Laser Range (LLR) data consist of time-of-flight measurements from McDonald Observatory to any one (1) of four (4) retroreflectors on the Moon and back again.[3714] The retroreflectors are at the Apollo 11, 14, and 15 landing sites and on the Lunakhod 2 vehicle.[3715]

XX. These 2954 range points are distributed from August 1969 to June 1980 CE (Gregorian). During the least-squares fit the ranges have been weighted according to the instrumental errors that accompany each point. The general trend is toward improving accuracy with time.[3716]

XX. For the Moon, the mean orbital elements that affect its geocentric distance will be strongly determined by the laser-ranging data. The eccentricity is implicitly determined to a few parts in 10^9, and the mean anomaly to a few milliarcseconds.[3717]

XX. There is a strong determination of differential geocentric lunar and solar ecliptic longitudes resulting from strong solar perturbations on the lunar distance [from the Earth] (amplitudes of about 3000 km and 4000 km for the two (2) leading terms). During the span of the observations the differential longitudes are known to a few milliarcseconds.[3718]

XX. Outside the data span [provided by the Lunar Laser Range (LLR)] the error in the lunar longitude is dominated by the uncertainty in the [secular] lunar [orbital] tidal [friction] acceleration ($\dot{n}$).[3719]

77.32. Calculations of ΔT

XX. An ideally [i.e. theoretical] uniform time-scale [i.e. not Universal Time (UT) since Universal Time (UT) is not a uniform time-scale] (known as Terrestrial Time (TT) [which maintains continuity with Ephemeris Time (ET)]) is implicit in these theories.[3720] By definition: ΔT is equal to the difference [of] (TT – UT):[3721]

$$(\Delta T) == (TT - UT)$$

XX. In defining TT, astronomers have adopted the standard length of day (LOD) as defined by the

3714 Seidelmann 1992: 299
3715 Seidelmann 1992: 299-300
3716 Seidelmann 1992: 300
3717 Seidelmann 1992: 300
3718 Seidelmann 1992: 300
3719 See Williams and Standish 1989 – cited by Seidelmann 1992: 300
3720 Morrison and Stephenson 2004: 327-328
3721 Morrison and Stephenson 2004: 328

average rate of rotation of the Earth over the period from 1750 – 1892 CE (Gregorian) [with] the effective mean epoch thus [being] near 1820 CE (Gregorian). Unless adequate allowance is made for changes [ΔT] in the Earth's spin rate [i.e. rate of rotation], the computed positions of the Moon and Sun (and also the planets) for some particular time in the past will be correspondingly in error.[3722]

XX. According to the calculations of Espenak and Meeus: during the 5000-year period from -1999 to +3000 (2000 BCE (Julian) to 3000 CE (Gregorian)), Earth will experience 12064 eclipses of the Moon. The statistical distribution of eclipse types for this interval is – according to Espenak and Meeus – as follows: 4378 penumbral eclipses, 4207 partial eclipses, and 3479 total eclipses.[3723]

XX. The largest uncertainty in the eclipse predictions is caused by fluctuations in Earth's rotation due primarily to tidal friction of [the Earth's oceans caused by] the Moon. The resultant drift in apparent clock time [i.e. TT – UT] is expressed as ΔT and is determined as follows:[3724]

1.) pre-1950s CE (Gregorian): ΔT is calculated from empirical fits to historical records derived by Morrison and Stephenson (2004);

2.) 1955 – 2006 CE (Gregorian): ΔT is obtained from published observations;

3.) Post-2006 CE (Gregorian): ΔT is extrapolated from current values weighted by the long-term trend from tidal effects;

XX. Before 1955 CE (Gregorian) the values given by (ΔT == (ET – UT1)) are based upon observations of the Moon. After 1955 CE (Gregorian) values given by (ΔT == (TT – UT1)) == (TAI + 32.184 seconds – UT1) are from measurements by atomic clocks as published by the BIH and the BIPM.[3725]

XX. Citing Stephenson and Morrison's (1995) mathematical formula, Nelson asserts that over the past 2700 years ΔT can be represented by a parabola of approximately the [following] form, where ΔT is expressed in seconds, and [the variable] (*T*) is the year:[3726]

$$(\Delta T) == (31s /cy^2) ((T - 1820)^2 / (100)^2) - (20s)$$

XX. However, as Nelson et al. admit, actual values of ΔT based on astronomical data may differ somewhat from this [proposed equation by Stephenson and Morrison for ΔT] as no single parabola can satisfactorily represent all modern and historical data.[3727]

3722 Morrison and Stephenson 2004: 328
3723 Espenak and Meeus 2009: 1; EclipseWise.com; https://www.eclipsewise.com/lunar/Lepubs/5MCLE.html;
3724 Espenak and Meeus 2009: ; EclipseWise.com; https://www.eclipsewise.com/lunar/Lepubs/5MCLE.html;
3725 Nelson et al. 2001: 517
3726 Nelson et al. 2001: 517
3727 Nelson et al. 2001: 517

XX. In the mainstream theory used today, the formulae [i.e. formulas] are expressed in the so-called Ephemeris Time (ET) which is a time flow proportional to:[3728]

1.) the motion of the planets in their orbits;

2.) after 1955 CE (Gregorian) [these formulas are expressed in] Atomic Time (TAI). This combined time scale is called Terrestrial Time (TT);

XX. In June 1985 CE (Gregorian), Henriksson completed a computer program for the calculation of ancient solar eclipses. He argues that it has since been successfully tested against all well-defined observations back to 2500 BCE (Julian). Henriksson's program is mainly based on the theory by Carl Schoch (1931) with Henriksson's own improvements included concerning modern astro-physical parameters [i.e. variables].[3729]

XX. In comparing his program to Stephenson's program, Henriksson maintains that he has always used Carl Schoch's (1931) values [i.e. variables] for the sidereal lunar secular [tidal friction] acceleration [i.e. rate of change of the mean motion of the Moon], ($\dot{n}$) == (-29.68 arcseconds /century^2), and for the braking of the rotation of the earth [he uses Schoch's] (36.28 seconds /century^2).[3730]

XX. All the formulas used by Henriksson in his program are expressed in Universal Time (UT) as used by the ancient observers. This means that the slowing down of the rotation of the Earth can be calibrated by a direct comparison with ancient observations.[3731]

XX. In comparison, Steele (2000) tries to identify ancient eclipses by a computer program designed by F.R. Stephenson. His values for ΔT are taken from Stephenson and Morrison (1995). As cited by Henriksson, Steele notes that, "These programs are based upon the solar ephemeris of Newcomb (1895) and a corrected version of the lunar ephemeris designated (*j==2*) (IAU 1968), incorporating a [secular] lunar [orbital tidal friction] acceleration ($\dot{n}$) of (-26" /cy^2) as determined by Morrison & Ward (1975) from analysis of the transits of Mercury".[3732]

XX. At the time of Henriksson's writing (2009), he notes that during the previous 25 years [prior to his writing] that the use of Terrestrial Time (TT) has dominated as the time basis in all professional or commercially available computer programs for the calculation of eclipses. Unfortunately, however, it [i.e. Terrestrial Time (TT)] has difficulties before 500 CE (Julian), and is – according to Henriksson – completely useless for observations made before 700 BCE (Julian). Henriksson argues that his

3728 Henriksson 2009: 166
3729 Henriksson 2009: 166
3730 Henriksson 2009: 167
3731 Henriksson 2009: 166
3732 Henriksson 2009: 167

computer program is useful at least back to 3000 BCE (Julian) with errors of – according to him – just a few minutes [which he asserts are] caused by quasi-periodic non-tidal effects.[3733]

XX. Although actual changes in the LOD amount to only a few (1 / 100ths) of a second over several thousand years, the cumulative effect (known as ΔT) of these minute changes can be very large. During this interval, the Moon can change position by nearly 4°.[3734]

XX. Furthermore, the derivative of ΔT is [the change (Δ, Delta) in the length of day (LOD)]:[3735]

$$\Delta L_{day} == (0.0017 \text{ s / d / cy}) ((T - 1820) / 100)$$

XX. This equation represents the change Δ [Delta, or difference] in the length of day (LOD) (i.e. ΔL_{day}) in SI seconds relative to the standard reference day of exactly 86400 SI seconds. According to this long-term trend – as calculated by Stephenson and Morrison – the rate of increase in the length of the day (LOD) is about 1.7 milliseconds per century.[3736]

XX. Citing Morrison and Stephenson (1986 CE (Gregorian)), Nelson notes that during the era of telescopic observations from the year 1620 CE (Gregorian) onwards, observations of changes in the length of day (LOD) illustrate that the LOD has been increasing at about 1.4 milliseconds per century. That is to say that a day [at the time of Nelson et al.'s writing] in 2001 CE (Gregorian) is approximately 1.4 milliseconds longer than a day one (1) century prior.[3737]

XX. However, while during the years 1968 to 1980 CE (Gregorian) the increase [in ΔT and its derivative ΔL_{day}] was almost exactly one (1) second per year, from 01 June 1997 CE to 01 June 1998 CE (Gregorian) it was only 0.65 second, and during the next 12 months it was only 0.38 second, which means that during the last [few] years [preceding Meeus and Simon's time of writing in 2000 CE (Gregorian)] the rotation of the Earth was temporarily speeding up.[3738]

XX. As noted by Nelson et al., other studies (McCarthy and Babcock (1986); McCarthy and Carter (1990) imply slightly different values. Furthermore, Nelson et al. note that the actual value of the LOD will depart from any long-term trend because of short-term fluctuations of between -3 milliseconds and +4 milliseconds on a time scale of decades.[3739]

XX. In the long run, ΔT should increase more and more rapidly the farther we go into the future since

3733 Henriksson 2009: 167
3734 Morrison and Stephenson 2004: 327
3735 Nelson et al. 2001: 517
3736 Nelson et al. 2001: 518
3737 Nelson et al. 2001: 518
3738 Meeus and Simons 2000: 324
3739 Nelson et al. 2001: 518

the rotation of the Earth is generally slowing down.[3740]

77.33. International Atomic Time (TAI)

XX. Until 1960 CE (Gregorian), the unit of time – the second – was defined as a specified fraction (1 / 86400) of the Mean Solar Day. With the adoption of Ephemeris Time (ET) the second was defined to be a specified fraction of the [solar] year. This definition was more precise than the definition based on the Mean Solar Day, but it was difficult to measure from observational data and to implement with operational clocks.[3741]

XX. Beginning in July 1955 CE (Gregorian) atomic timescales were available from organizations in several countries. In 1971 CE (Gregorian) the experimental atomic timescale was adopted as the worldwide time reference system under the name International Atomic Time (TAI – from [French:] *Temps Atomique International*). The origin [i.e epoch] of TAI was arbitrarily chosen so that the TAI and UT1 readings at 01 January 1958 CE (Gregorian), 0^h [i.e. Midnight (00:00)] were the same.[3742]

77.34. The Types of Months

XX. To appreciate the full complexity of the Moon's behavior in the sky and the roles that these [aspects of behavior] play in calendar problems – and in eclipse prediction – the Moon's orbit must be examined.[3743] There are actually several kinds of months which help to highlight aspects of the Moon's complex motions. With one (1) exception [i.e. the civil Gregorian / Julian month], these types of months are the periods of the Moon in its orbit with respect to particular reference points or directions:[3744]

1.) The *New Moon* [i.e. technically precisely: *No Moon* since *New Moon* is actually the first sighting of the *New Crescent Moon* after the conjunction of *No Moon*] occurs when the Sun and Moon have the same celestial longitude. The time of New Moon (i.e. conjunction of Sun and Moon) to [next] New Moon – a *lunation* – is called the *Synodic Month*.[3745] Its value today ranges from approximately (29.27) to (29.84) days with a mean [i.e. average] currently of about (29.530588853) Mean Solar Days in days of 86,400 atomic [TAI equivalent to SI] seconds [per day]. The mean and true times of the New Moon can differ by up to about 14 hours.[3746]

3740 Meeus and Simons 2000: 324
3741 Seidelmann 1992: 84
3742 Seidelmann 1992: 84
3743 Kelley and Milone 2011: 30
3744 Kelley and Milone 2011: 34
3745 Dershowitz and Reingold 2008: 193
3746 Dershowitz and Reingold 2008: 193

Because the Synodic Month is not an integral number of days, this has had important consequences for calendars involving the Moon.[3747] The Synodic Month is not constant, but [rather] is decreasing in mean length by about (3.6×10^{-7}) solar days per century (though it is increasing in length by about (0.021) atomic [TAI equivalent to SI] seconds per century). The net effect of the decreases in [the mean lengths of] Synodic Month and Tropical Year is to increase the number of months from its current value of about (12.3682670) per year by (0.3×10^{-6}) months per century.[3748]

When the word lunation is mentioned, it is usually referring to the Synodic Month.[3749] The Synodic Month is the month of phases,[3750] i.e. the interval from New Moon to New Moon. It is the period with respect to a line between the Earth and the Sun. The Earth's motion around the Sun shows up in the eastward shift of the Sun moving[3751] [relative to its position against the background of the constellations of stars – as seen from Earth] East among the stars by [approximately] ~30° in the course of a month,[3752] i.e. in the [same] direction of the Moon's orbital motion. The Moon's catching up to the Sun causes the Synodic Month to be longer than the Sidereal Month.[3753]

2.) The *Sidereal Month* is the time it takes the Moon to make one (1) revolution around the Earth,[3754] i.e. it is the period of [orbital] revolution of the Moon around the Earth with respect to a line to a distant star.[3755] Its mean value is (27.32166) days. In the interim, the Earth has moved in its orbit around the Sun. Thus the difference in longitude between the Sun and Moon has increased, which is why the Synodic Month is longer [than the Sidereal Month].[3756]

3.) The *Anomalistic Month* is the time between consecutive perigees (i.e. points at which the Moon is closest to Earth).[3757] The *argument [i.e. variable] of perigee*, i.e. the angle between ascending node and the point of perigee in the orbit, is moving eastward in the direction of the Moon's orbital motion, and so the Anomalistic Month has a longer length than the Sidereal Month.[3758] The Anomalistic Month averages (27.55455) days.[3759]

4.) The *Draconitic*, *Draconic*, or *Nodal month* is the interval between successive lunar passages through the ascending node. Because the node is regressing at a relatively high rate, the Moon meets it

3747 Kelley and Milone 2011: 34-35
3748 Dershowitz and Reingold 2008: 195
3749 Kelley and Milone 2011: 35
3750 Kelley and Milone 2011: 30, 34
3751 Kelley and Milone 2011: 34
3752 Kelley and Milone 2011: 30
3753 Kelley and Milone 2011: 34
3754 Kelley and Milone 2011: 34
3755 Dershowitz and Reingold 2008: 195
3756 Kelley and Milone 2011: 34
3757 Dershowitz and Reingold 2008: 195; Kelley and Milone 2011: 34
3758 Kelley and Milone 2011: 34
3759 Dershowitz and Reingold 2008: 195

much sooner than it would a line to a distant star. The Draconic Month is therefore much shorter than the Sidereal Month.[3760]

5.) The *Tropical Month* is the interval between successive passages of the Moon through the vernal equinox. Due to precession (from the long-term wobbling of the Earth, as the Moon and Sun act to pull the equatorial bulge into the ecliptic plane;[3761] the vernal equinox is slowly moving westward in the sky at a rate of about 50" /year [i.e. approximately 50 arcseconds per year].[3762]

6.) The *Civil Month* is the unit of month in use within a certain political jurisdiction. The modern Civil [Gregorian] Month has 28, 29, 30, or 31 integral days, depending on the particular month and year. It evolved from the Synodic Month.[3763] [3764]

XX. We also use a notion of a *Solar Month*, i.e. the time for the Sun's position in the sky to traverse one (1) sign of the zodiac (30° of longitude). Its mean [i.e. average] value is one twelfth (1 / 12) of a Solar Year and ranges from (29.44) days in Northern Hemisphere winter (to traverse Capricorn) to (31.43) days in Northern Hemisphere summer.[3765]

Table 77.2. Lengths of Lunar Months where the units for all months are in Mean Solar Days (MSDs)[3766]

Type of Month	Length of Month
P_{syn}	29.530589 days == 29^{d} 12^{h} 44^{m} 02.9^{s}
P_{sid}	27.321662 days == 27^{d} 07^{h} 43^{m} 11.6^{s}
P_{anom}	27.554550 days == 27^{d} 13^{h} 18^{m} 33.1^{s}
P_{drac}	27.212221 days == 27^{d} 05^{h} 05^{m} 35.9^{s}
P_{trop}	27.321582 days == 27^{d} 07^{h} 43^{m} 04.7^{s}

XX. The Moon's motions are difficult to follow.[3767] Isaac Newton used to say that his head ached when he thought about the Moon.[3768] It is to the great credit of the ancient observers that they made as much sense of these motions as they did. In addition, they were able to find several examples of regularity in

3760 Kelley and Milone 2011: 34
3761 See Kelley and Milone 2011: §3.1.3
3762 Kelley and Milone 2011: 34
3763 See Kelley and Milone 2011: §4.2
3764 Kelley and Milone 2011: 35
3765 Dershowitz and Reingold 2008: 195
3766 Kelley and Milone 2011: 35
3767 Kelley and Milone 2011: 35
3768 De Meis 2004: 121; Kelley and Milone 2011: 33

lunar motion.[3769]

XX. The Moon orbits the Earth on a path close to – but not on – the ecliptic, changing phase as it does so and basically replicating the [eastward] motion of the Sun [relative to the stars] but at a much faster rate, as well as at a more variable celestial latitude. The Sun travels its path in a year, and the Moon [travels its path] in a month.[3770]

XX. In the course of a month, the Sun moves East among the stars by [approximately] ~30°.[3771] This means that the synodic month must be longer than the time it takes for the Moon to encircle the Earth with respect to a line to the distant stars. This affects lunar and solar calendars, as well as the occurrence of eclipses.[3772]

XX. The Full Moon, because it is opposite the Sun, rides high across the (Northern Hemisphere) midwinter sky, and low across the midsummer sky.[3773] Although the Sun's rise and set points on the horizon vary slowly from day to day, those of the Moon change much more rapidly from day to day.[3774] For obscure reasons, when four (4) Full Moons occur within one (1) solar season (from equinox to solstice or solstice from equinox), the 3rd is termed a *Blue Moon*.[3775]

XX. The Moon's declination is sometimes less and sometimes more than is the Sun's extreme values [of declination] (±23.5° at present). This means that the amplitude of its azimuth variation over the month varies from month to month in an 18.6-year cycle. This fact is of importance in studying alignments to the Moon.[3776]

XX. The perturbations on the Moon are particularly great because it moves nearly on the ecliptic, but not exactly on it. That the orbit should be near the ecliptic is curious because as a satellite of the Earth, we could expect it to move near the plane of the Earth's equator which is the case for most of the other major satellites of the planets.[3777] These perturbations must be taken into account in lunar orbit calculations; without them, the results could be wrong by several degrees.[3778]

XX. However, our [lunar] satellite is far enough from the Earth at present that its motion is effectively dominated by the Sun so that the Earth and Moon form a kind of double-planet system. [However, to be more precise] the Earth-Moon [system] is really an Earth-Moon-Sun system, i.e. a three-(3)-body

3769 Kelley and Milone 2011: 35
3770 Kelley and Milone 2011: 29
3771 Kelley and Milone 2011: 30
3772 Kelley and Milone 2011: 30
3773 Kelley and Milone 2011: 30
3774 Kelley and Milone 2011: 30
3775 Dershowitz and Reingold 2008: 195; Dershowitz and Reingold 2018: 227;
3776 Kelley and Milone 2011: 30
3777 Kelley and Milone 2011: 33
3778 Kelley and Milone 2011: 34

system. [Within a wider context] the solar system is an *n*-body system, and each object is accelerated by [and thus affects] all the other objects.[3779] All bodies are affected by all other bodies. No planet can have a perfect elliptical orbit because it will be pulled on by the other planets.[3780]

XX. Even so, the Moon is close enough to the Earth to undergo – as well as cause – tidal effects that have slowed the Moon's rotation to equal its orbital period [i.e. thus causing the same near side of the Moon to always face the Earth, known as tidal lock], and to result in an increasing distance from the Earth, and an increasing length of month [and increasing length of day (LOD) for the Moon since the Moon's LOM == LOD]. Tidal effects on the Earth are resulting in a slowing down of Earth's rotation[3781] [i.e. increasing length of day (LOD) for the Earth], which affects the timing of ancient phenomena, such as eclipses.[3782] [3783]

XX. It is sometimes said that the Moon "comes back to the same place" after not one (1), but [rather] three (3) sidereal months. After one (1) sidereal month the Moon is at the same position among the stars. So if the phrase has meaning, it must refer to the location in the sky of the observer.[3784]

XX. After one (1) sidereal month, the Moon will not be at the same hour angle because the sidereal month interval is not an integral number of days. However, three (3) sidereal months amount to nearly an integral number of days:[3785]

(3) (P_{sid}) == (81.964986 days) == (81^{d} 23^{h} 9^{m} 35^{s})

XX. Thus the Moon will be about an hour east of the meridian after three (3) sidereal months at the same time of night – and in the same constellation – but it will be at a different lunation phase. An interval of three (3) synodic months covers (88.59177) days that at the end of which the lunar phase is repeated so that the phase after three (3) sidereal months will be earlier by an angle of roughly:[3786]

(ΔΦ [Delta Phi]) == (88.59177 days – 81.69499 days) / (29.530589 days) == (0.23355 lunation)

XX. Thus if the Moon was initially Full [Moon], three (3) sidereal months later, it would be in a waxing gibbous phase, i.e. just after 1st quarter, having slipped back not quite a quarter (1/4) phase. For every subsequent three-(3)-month interval, the Moon's phase will slip by an additional (0.234) [i.e. 0.23355] lunation on average. In 12 sidereal months, (4 x (0.23355) == 0.9342), the phase repeats more

3779 Kelley and Milone 2011: 33
3780 Kaler 2003: 47
3781 See Kelley and Milone 2011: §4.5
3782 See Kelley and Milone 2011: §5.2, especially §5.2.1.3
3783 Kelley and Milone 2011: 33
3784 Kelley and Milone 2011: 35
3785 Kelley and Milone 2011: 35
3786 Kelley and Milone 2011: 35

closely but results in a slight phase shift for each such 12-sidereal-month interval.[3787]

XX. The movement of the Moon among the stars requires that the Moon traverse a different region each day for the approximately 27 days of its sidereal period. In an anthropomorphic sense, it spends each night in a different "house".[3788]

XX. The regression of the nodes of the Moon's orbit [i.e. a variable position] has a major consequence for the behavior of the Moon in our sky, which is spectacular at high latitude locations on Earth. Over an 18.6-year period, the shifting node alters the range in declination achieved by the Moon during the month: from ~±18½° to ~±28½°.[3789] [This 18.6-year period is also the same period of the Earth's nutation (or sub-wobble) of the Earth's precession. Could this regression of the nodes be a direct or indirect result / consequence of the nutation (sub-wobble) of the Earth that is compounded with the effects of the precession of the Earth?]

XX. This variation changes the azimuthal amplitude, resulting in a striking weaving movement of the rise and set points of the Moon on the horizon over an 18.6-year interval. The phenomenon bears strongly on the question of megalithic lunar alignments.[3790] [3791]

XX. That there is a difference between the anomalistic and sidereal periods implies the rotation of the orbit of the Moon such that the line of apsides (i.e. the major axis) moves forward (i.e. [apsidial motion moves] eastward in the direction of the Moon's motion in its orbit). This motion was apparently also known in ancient China.[3792]

77.35. The Saros Cycles

XX. Throughout most – perhaps all – of the Late Babylonian period, the basic method of predicting eclipses of the Sun and Moon in Babylonia was based upon the so-called Saros Cycle (SC) of 223 synodic months. This method was not restricted to simply identifying dates [that are] one (1) or more Saroi [i.e. multiple Saros cycles] after a visible eclipse, but could [also] be used to identify the dates of all eclipse possibilities (EPs).[3793]

XX. As has often been noted, the term "Saros" is modern. To the Babylonians this period was simply called 18 MU.MEŠ: "18 years."[3794] It almost certainly originated with lunar eclipses and was then

3787 Kelley and Milone 2011: 35
3788 Kelley and Milone 2011: 35
3789 Kelley and Milone 2011: 35
3790 Discussed in Kelley and Milone 2011: §3.2.1 and at length, in applications, in Kelley and Milone 2011: §6.
3791 Kelley and Milone 2011: 35-36
3792 Kelley and Milone 2011: 36
3793 Brack-Bernsen and Steele 2005: 181
3794 Steele 2000a: 424

applied to solar eclipses. Indeed it is most likely that all Babylonian eclipse theories – for want of a better term – were developed from lunar eclipse observations and used for predicting lunar eclipses and then applied by analogy to solar eclipses.[3795] This period, which is equal to about 6585 1/3 days – or slightly more[3796] (i.e. about 11 days longer)[3797] than 18 years – has become known as the "Saros."[3798]

XX. Following Steele (2000a CE (Gregorian)), when discussing the Saros, the following definitions of terms will be used: "Saros Cycle" is the [one (1)] period of 223 synodic months containing 38 eclipse possibilities. "Saros Series" refers to a collection of eclipse possibilities each separated by one (1) Saros of 223 synodic months from the preceding eclipse possibility. "Saros Scheme" will be taken to mean the particular distribution of eclipse possibilities within a[ny one] Saros Cycle at a given time.[3799]

XX. It is not known exactly when this scheme for predicting eclipses was developed, but it was almost certainly before about 575 BCE (Julian) [– i.e. the exact time that Daniel was purported to be in the service of Nebuchadnezzar in Babylon]. The tablet LBAT 1420 [BM 38462], published as Text No. 6 in Hunger (2001 CE (Gregorian)), contains a collection of eclipse observations for the period from at least 604 BCE to 576 BCE (Julian), and there is nothing to suggest that the text was compiled significantly after the date of its contents. Interspersed between the observed eclipses, the months of EPs are also noted. The distribution of EPs is exactly what we would expect from the Saros Scheme described above and thus implies that scheme was in use by at least 575 BCE (Julian).[3800]

XX. It may well have been used earlier. The earliest eclipse records we possess apparently begin in 747 BCE (Julian), and already they include predicted eclipses. Furthermore, we have three (3) tablets that originally formed part of a large compilation of lunar eclipse observations and predictions covering 24 Saroi. The earliest preserved eclipse on these tablets was in 731 BCE (Julian), but it is very probable from the structure of the compilation that – when complete – it covered the period from 747 BCE to 315 BCE (Julian).[3801]

XX. The compilation is set out in exactly the form of the Saros Scheme matrix described above, and predictions are given of the times of all EPs that were not observed. The structure of the large compilation as reconstructed by Steele (2000a CE (Gregorian)) from other extant eclipse records is discussed by Brack-Bersen and Steele (2005 CE (Gregorian)).[3802]

XX. Of course, this compilation does not tell us anything about when the Saros Scheme was brought into use since it is quite possible that some of the "predicted" eclipses were calculated backwards when

3795 Brack-Bernsen and Steele 2005: 181
3796 Steele 2000a: 424
3797 Brack-Bernsen and Steele 2005: 201
3798 Steele 2000a: 424
3799 Steele 2000a: 424
3800 Brack-Bernsen and Steele 2005: 182
3801 Brack-Bernsen and Steele 2005: 182
3802 Brack-Bernsen and Steele 2005: 182

the text was compiled to fill it out. Nevertheless, we have the evidence from LBAT 1420 that the Saros Scheme was being used by at least circa 575 BCE (Julian) [– i.e. the exact time that Daniel was purported to be in the service of Nebuchadnezzar in Babylon].[3803]

XX. However, determining which months are EPs is not the end of the matter. Many of the extant records of eclipse predictions also include the time of day or night when the eclipse was expected to occur. According to Brack-Bernsen and Steele (2005 CE (Gregorian)), as has been shown by Steele and Stephenson (1997 CE (Gregorian)), this time relates to the moment when the eclipse was expected to begin. This in itself indicates that the times were probably calculated by means of the Saros.[3804]

XX. If mathematical astronomy had been used to determine the time of the eclipse, it would naturally result in the time of syzygy. To obtain the moment when the eclipse began, a correction to this time would have to be made based upon the expected duration of the eclipse, but with the possible exception of one (1) column of numbers on a single text (i.e. the so-called Text S – see Brack-Bernsen and Steele (2005 CE (Gregorian))), no functions for the duration of an eclipse are known from the mathematical astronomical texts.[3805]

XX. However, when an eclipse was observed, the time when it began was recorded. Adding on the Saros length to this time, the time of the predicted eclipse would also be for when it was expected to begin.[3806] The length of the Saros depends upon the positions of the Sun and Moon in their cycles of anomalistic motion and varies between about 1,35 and 2,15 UŠ in excess of an integer number of days.[3807]

XX. In the mathematical lunar theory known as ACT System A (Neugebauer 1955 CE (Gregorian)) the length of the Saros assuming a constant solar velocity of 30° per month is given by the so-called column Φ [Greek: *Phi*]. However, to obtain the true length of the Saros, a correction [i.e. variable] for solar anomaly would have to be added to Φ [Greek: *Phi*], but such a function is so far unattested in the preserved texts.[3808]

XX. Thus it would not be possible to use Φ [Greek: *Phi*] to determine the time of predicted eclipses from earlier observations.[3809] In any case, eclipse times were being calculated well before it is believed Φ [Greek: *Phi*] was probably constructed.[3810] Therefore the length of the Saros that was used to predict eclipse times must be sought elsewhere.[3811]

3803 Brack-Bernsen and Steele 2005: 182
3804 Brack-Bernsen and Steele 2005: 182
3805 Brack-Bernsen and Steele 2005: 182
3806 Brack-Bernsen and Steele 2005: 182
3807 Brack-Bernsen and Steele 2005: 182-183
3808 Brack-Bernsen and Steele 2005: 183
3809 Brack-Bernsen and Steele 2005: 183-184
3810 Brack-Bernsen and Steele 2005: 183-184
3811 Brack-Bernsen and Steele 2005: 184

XX. The method for predicting lunar eclipses itself arises by combining the Saros with the realization that EPs generally occur every 6 months but occasionally after only 5 months. Therefore if within a Saros of 223 months there are (*a*) eclipses at a 6-month interval and (*b*) eclipses at a 5-month interval, simple mathematics shows from [the following algebraic formula] that (*a* == 33) and (*b* == 5):[3812]

(6a) + (5b) == (223)

XX. Within a Saros of 223 months, there are therefore 38 EPs, of which 33 are at a 6-month interval and the remaining 5 are at a 5-month interval: (*a* == 33) and (*b* == 5).[3813]

XX. Distributing the [five (5)] 5-month intervals as evenly as possible, we obtain five (5) groups of eclipses in which the first eclipse of each group follows a 5-month interval and the remainder are at 6-month intervals.[3814] The groups contain respectively: eight (8), seven (7), eight (8), seven (7), and eight (8) eclipses.[3815]

XX. Since EPs repeat after [one] (1) Saros, this simple distribution results in a matrix-like scheme by which all EPs can be predicted. All that needs to be done is to decide which EP corresponds to the first eclipse in the first column of the matrix.[3816]

XX. With respect to the fixed background of stars, the Moon moves around the Earth in an approximately circular orbit with an average period of (27.321662 days), known as the Sidereal Month.[3817] However, from the Earth the Sun also appears to circle us returning to the same location relative to the fixed stars in a period of (365.2564 days) known as the Sidereal Year.[3818]

XX. Therefore over the course of a Sidereal Month (27.321662 days) the Sun has moved slightly ahead of the fixed stars, and so it takes a little more than another two (2) days [i.e. specifically an additional (2.208926999999999 days)] for the Moon and Sun to reach conjunction. The average time interval between two (2) conjunctions or oppositions of the Moon and Sun is equal to (29.530589 days) and is known as the Synodic Month.[3819]

XX. The Saros [cycles] arise from a harmonic between three (3) of the Moon's orbital cycles. All three (3) periods are subject to slow variations over long time scales, but their values as of 2000 CE

3812 Brack-Bernsen and Steele 2005: 181
3813 Brack-Bernsen and Steele 2005: 181
3814 Brack-Bernsen and Steele 2005: 181-182
3815 Brack-Bernsen and Steele 2005: 182
3816 Brack-Bernsen and Steele 2005: 182
3817 Note this is close to – but not the same as – both the Anomalistic Month (perigee to perigee) == 27.554550 days, as well as the Draconic Month (node to node) == 27.212221 days. cf. also Steele 2000a.
3818 Steele 2000a: 422
3819 Steele 2000a: 422

(Gregorian) – according to Espenak – are:[3820]

Synodic Month (*New Moon* to *New Moon*) == (29.530589 days) == (29^{d} 12^{h} 44^{m} 03^{s})

Anomalistic Month (perigee to perigee) == (27.554550 days) == (27^{d} 13^{h} 18^{m} 33^{s})

Draconic Month (node to node) == (27.212221 days) == (27^{d} 05^{h} 05^{m} 36^{s})

XX. One (1) Saros is equal to 223 Synodic Months. However, 239 Anomalistic Months and 242 Draconic Months are also equal (within a few hours) to this same period:[3821]

223 Synodic Months == (6585.3223 days) == (6585^{d} 07^{h} 43^{m})

239 Anomalistic Months == (6585.5375 days) == (6585^{d} 12^{h} 54^{m})

242 Draconic Months == (6585.3575 days) == (6585^{d} 08^{h} 35^{m})

XX. With a period of approximately 6585.32 days (approximately ~18 years, 11 days, 8 hours), the Saros is a valuable tool in investigating the periodicity and recurrence of eclipses. It was first known to the Chaldeans[3822] as an interval when lunar eclipses repeat, but the Saros is applicable to solar eclipses as well.[3823]

XX. Any two (2) eclipses separated by one (1) Saros cycle [i.e. 223 Synodic (Lunar) Months == 6585.3223 days == 6585^{d} 07^{h} 43^{m} == approximately ~18 (Lunar) Years, 11 Days, 8 Hours] share similar characteristics. They occur at the same node with the Moon at nearly the same distance from Earth and the same time of year. Because the Saros period is not equal to a whole number of days, its biggest drawback as an eclipse predictor is that subsequent eclipses are visible from different parts of the globe [i.e. an example of the critical significance of Delta T to determine precise times and locations].[3824]

XX. [The 24 hours of a day on Earth == 360°.] The extra 1/3 day displacement [in the period of the Saros cycle] means that Earth must rotate an additional [approximately] ~8 hours or ~120° with each [Saros] cycle [i.e. 223 Synodic [Lunar] Months == (6585.3223 days) == (6585^{d} 07^{h} 43^{m}) ==

3820 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEperiodicity.html#section104;
3821 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEperiodicity.html#section104;
3822 Does he mean ethnic Chaldeans literally specifically, or as general term for ancient Babylonians?
3823 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEperiodicity.html#section104;
3824 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEperiodicity.html#section104;

approximately ~18 [Lunar] Years, 11 Days, 8 Hours]. For lunar eclipses, this results in a shift of [approximately] ~120° west in the visibility zones of each succeeding eclipse. Thus a Saros series returns to approximately the same geographic region every three (3) Saros periods ([i.e. approximately] ~54 years and 34 days). This triple Saros cycle is known as the *Exeligmos*.[3825]

XX. The westward migration in the zone of eclipse visibility illustrates the effect of the extra 1/3-day in the Saros period. Although the Moon's path through Earth's shadows is similar from one (1) eclipse to the next, it is not exact. The Moon-shadow geometry changes slowly as do the characteristics of each lunar eclipse in a Saros series.[3826]

XX. Saros series do not last indefinitely because the Synodic, Anomalistic, and Draconic Months are not perfectly commensurate with one another. In particular, the Moon's node [i.e. the two (2) points at which the orbit of the Moon intersects the ecliptic] shifts eastward by about 0.48° with each eclipse in a series.[3827]

XX. Any two (2) eclipses separated by one (1) Saros cycle share very similar geometries. They occur at the same node with the Moon at nearly the same distance from Earth and at the same time of year. Because the Saros period is not equal to a whole number of days, its biggest drawback is that subsequent eclipses are visible from different parts of the globe. The extra 1/3 day displacement means that Earth must rotate an additional ~8 hours or ~120º with each cycle. For solar eclipses, this results in the shifting of each successive eclipse path by ~120º westward. Thus a Saros series returns to about the same geographic region every three (3) Saroses [i.e. every three (3) Saros cycles which together total] (54 years and 34 days).[3828]

XX. Since there are two (2) to five (5) solar eclipses every year, there are approximately forty (40) different Saros series in progress at any one (1) time. For instance, during the later half of the 20th century CE (Gregorian), there are 41 individual [Saros] series and 26 of them are producing central eclipses. As old series terminate, new ones are beginning and take their places.[3829]

XX. Because of the ellipticity of the orbits of Earth and the Moon, the exact duration and number of eclipses in a complete Saros series is not constant. A series may last 1226 to 1587 years and is composed of 69 to 89 eclipses. A series can begin with 6 to 25 penumbral eclipses, followed by 6 to 24

3825 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEperiodicity.html#section104;
3826 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEperiodicity.html#section104;
3827 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEperiodicity.html#section104;
3828 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEsaros.html;
3829 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEsaros.html;

partial eclipses. During the mid-life of a Saros series, there are 11 to 29 total eclipses. Finally, a series ends with 6 to 24 partial eclipses followed by 7 to 25 penumbral eclipses. At present (2008 CE (Gregorian)), there are 40 active Saros series numbered 110 to 149. The number of eclipses in each of these series ranges from 70 to 83, however, the majority of them (80%) are composed of 70 to 73 eclipses.[3830]

XX. Historically speaking, the word Saros derives from the Babylonian [Akkadian] term "sar" which is an interval of 3600 years.[3831] According to Neugebauer (1969), there is no clear evidence that the Saros eclipse cycle (of 18^{y} 11^{d}) was used in ancient Mesopotamia.[3832] [In fact] it was never used as an eclipse period until English astronomer Edmund Halley adopted it in 1691 CE (Gregorian).[3833]

XX. However, Neugebauer (1969) suggests that some evidence points to a crude 18-year cycle as the means for a lunar eclipse repetition cycle prior to the Seleucid Era.[3834] According to R. H. van Gent, Halley "...extracted [the Saros eclipse cycle] from the lexicon of the 11th-century [CE (Julian)] Byzantine scholar Suidas who in turn erroneously linked it to an (unnamed) 223-month Babylonian eclipse period mentioned by Pliny the Elder (*Naturalis Historia* II.10[56])."[3835]

XX. The periodicity and recurrence of lunar eclipses is governed by the Saros cycle, a period of approximately 6585.3 days (18 years, 11 days, 8 hours). When two (2) eclipses are separated by a period of one (1) Saros, they share a very similar geometry. These two (2) eclipses occur at the same node with the Moon at nearly the same distance from the Earth and the same time of year.[3836]

XX. Each [Saros] series typically lasts 12 to 13 centuries and contains 70 or more eclipses. Every Saros series begins with a number of penumbral lunar eclipses. The series will then produce several dozen partial eclipses, followed by several dozen total eclipses. The later portion of the series produces another set of partial eclipses before ending with a final group of penumbral eclipses. Thus, the Saros is useful for organizing eclipses into families or series.[3837]

3830 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEperiodicity.html#section104;
3831 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEsaros.html;
3832 Kelley and Milone 2011: 125
3833 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEsaros.html;
3834 Kelley and Milone 2011: 125
3835 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEsaros.html; – Square brackets in *Naturalis Historia* II.10[56] are by Espenak
3836 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEsaros/LEsaroscat.html;
3837 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEsaros/LEsaroscat.html;

XX. The time interval between any two (2) successive lunar eclipses can be either 1, 5, or 6 lunations (Synodic Months). However, lunar eclipses separated by 1, 5, or 6 lunations are usually quite dissimilar. They are frequently of unlike types (i.e., penumbral, partial, or total) with diverse Sun-Moon-Earth alignment geometries, and with different lunar orbital characteristics (i.e., longitude of perigee and longitude of ascending node). More importantly, these short periods are of no value as predictors of future eclipses because they do not repeat in a recognizable pattern.[3838]

XX. Although lists of eclipses by the Saros cycles are known (the so-called Saros canon),[3839] we do not know exactly how these calculations were made.[3840]

77.36. Gamma (Greek: γ)

XX. [For a lunar eclipse] NASA defines the variable Gamma (γ) as the distance from the center of the shadow cone axis [of the Earth] to the center of the Moon (in units of Earth's equatorial radii) at the instant of greatest eclipse.[3841] [For a solar eclipse] NASA defines Gamma (γ) as the distance of the Moon's shadow axis from Earth's center in units of equatorial Earth radii. It is defined at the instant of greatest eclipse when its absolute value is at a minimum.[3842]

XX. In the case of a lunar eclipse, Meeus defines Gamma (γ) as the least distance from the center of the Moon to the axis of the Earth's shadow, in units of the Earth's equatorial radius. The quantity γ [Gamma] is positive or negative depending upon the Moon's center passing north or south of the axis of the shadow.[3843]

XX. Gamma (γ) changes monotonically throughout any single Saros series. The change in Gamma (γ) is larger when Earth is near its aphelion (June to July [i.e. possible Hebrew months Iyar / Sivan / Tammuz / Av]) than when it is near perihelion (December to January [i.e. possible Hebrew months Kislev / Tevet / Shvat / Adar]). For odd numbered series (descending node), Gamma (γ) increases, while for even numbered series (ascending node), Gamma (γ) decreases. This simple rule describes the current behavior of Gamma (γ), but this has not always been the case.[3844]

XX. The eccentricity of Earth's orbit is presently (0.0167), and is slowly decreasing. It was (0.0181) in the year -2000 [i.e. 2001 BCE (Julian)] and will be (0.0163) in +3000 CE (Gregorian). In the past,

3838 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/lunar/LEhelp/LEperiodicity.html;
3839 De Meis 2004: 123-124
3840 De Meis 2004: 123
3841 https://eclipse.gsfc.nasa.gov/LEsaros/LEsaroscatkey.html;
3842 https://eclipse.gsfc.nasa.gov/SEhelp/SEglossary.html;
3843 Meeus 1998a: 382
3844 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://eclipsewise.com/lunar/LEhelp/LEperiodicity.html#section105;

when the eccentricity was larger, there were Saros series in which the trend in Gamma (γ) reversed for one (1) or more Saros cycles before resuming its original direction. These instances occur near perihelion when the Sun's apparent motion is highest and may, in fact, overtake the eastward shift of the node. The resulting effect is a relative shift west of the node after one (1) Saros cycle instead of the usual eastward shift. Consequently, Gamma (γ) reverses direction.[3845]

XX. Among 204 Saros series examined (-20 to 183), there are many other examples of temporary shifts in the monotonic nature of Gamma (γ), although none as unusual as Saros 13. The reversals are short and rarely last for more than four (4) or five (5) eclipses in a series. Some series even have two (2) separate reversals in Gamma (γ). The most recent eclipse with a Gamma (γ) reversal – according to Espenak – was on 10 January 1648 CE (Gregorian) (Saros 138). In past millennia, the Gamma (γ) reversals were more frequent because Earth's orbital eccentricity was larger.[3846]

78.0. Conclusion (PRE-PRINT WORK-IN-PROGRESS)

XX. Mathematics never lies. It is absolute truth, and it is the one standard by which every chronology of the ancient world must ultimately be measured. The only absolute chronology dating system for ancient history is the precision of astronomy.

XX. If we are to scientifically study the dates and numbers recorded in the Hebrew Scriptures, the Babylonian Chronicles, the Astronomical Diaries, as well as other texts, then we must be willing to follow mathematics, logic, and reason wherever they lead. We must draw our conclusions from what can be proven with – and inferred from – precise and exact mathematics which is the only foundation that is sufficient enough to bear the weight of absolute, precise historical truth that can be objectively (scientifically) confirmed by others.

XX. When an accepted conclusion (or academic consensus) conflicts with the mathematics, it is the conclusion (and the academic consensus) that must yield. Our task is then to reconcile our understanding to the mathematics and to accept the more precise picture of reality that results – no matter what that new reality may suggest, imply, reveal, require, or prove.

XX. This is the scientific method applied to history: every mathematical count can be repeated, every number or calculation can be verified, and every synchronism can be tested: Every result and conclusion can be confirmed or disproved by anyone willing to do the arithmetic and apply logic, as well as sound reasoning.

3845 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://eclipsewise.com/lunar/LEhelp/LEperiodicity.html#section105;

3846 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://eclipsewise.com/lunar/LEhelp/LEperiodicity.html#section105;

XX. Applied consistently, precise mathematics allows us to reconstruct and calibrate a single, accurate chronology of ancient history. That chronology synchronizes the interlocking chronologies of every culture whose records intersect with it, and it is accurate not only to the year, and not only to the month, but even to the exact day. The Hebrew biblical chronology, the Mesopotamian Akkadian (Babylonian and Assyrian) chronologies, the Egyptian chronology, the Aegean chronology, the Hittite chronology, the Elamite chronology, the Greek Chronology, the Seleucid Chronology are not separate and competing systems. Properly calibrated, they form one record of one history.

XX. Jeremiah 25:1 is the key that unlocks the mathematically and astronomically precise biblical chronology. This verse is our "Rosetta Stone" that allows us to unfold the chronology recorded in the entire Hebrew Bible as well as synchronize and recalibrate all other ancient chronologies that intersected with this astronomically precise, correct biblical chronology: Assyrian, Babylonian, Egyptian, Aegean, Hittite, Elamite, Greek, Seleucid, and continuing through our modern era via the Jewish Anno Mundi calendar used by Talmudic Sages and Rabbis including Maimonidies (Moses ben Maimon) in the Middle Ages.

XX. The discoveries, insights, and ramifications are far-reaching that extend across many academic disciplines: Assyriology, Egyptology, Bible Research, Jewish Studies, Ancient History, Archaeology, and Astronomy. The implications across multiple disciplines of academia are so enormous that scholars in these disciplines will need to rethink and reevaluate their current knowledge base, and entire volumes can be written on the revolutionary implications of this research.

XX. As a direct result of observing the mathematical precision of this ancient biblical chronology, this corrected chronology can be used to unlock and mathematically determine the precise dates of many other events across all of recorded ancient history, as well as to come to correct and precise conclusions (with implications) that were not previously visible or possible to surmise without the absolute truth of mathematics.

XX. The same method of applying the mathematical precision of an absolute astronomical chronology applies to every period and every people whose records survive. Of course it is not possible to detail all of the conclusions or implications across the many academic disciplines that are unlocked with the precise chronology of the Hebrew Bible. However, it is clearly obvious that the Bible is the correct and absolutely precise chronology of ancient history, and indeed it is the correct "backbone" of ancient chronology and world history upon which all other chronologies will need to synchronize and recalibrate.

XX. The examples presented in this study as well as in the conclusions that follow only illustrate the method; they do not exhaust it. We must leave it to other scholars and researchers (each in his or her specialty academic field and discipline) to help clarify an updated, correct understanding of their fields.

78.1. Ancient Calendar Systems

XX. When pondering the ancient past, what can we surmise?

XX. It is very likely that the ancient Sumerians began noticing astronomical patterns in the heavens as the annual four (4) seasons progressed and periodically repeated: Spring, Summer, Autumn, Winter. As the seasons changed, they also began to count the repeating patterns of approximately twelve (12) cycles of the Moon that circled the Earth going eastward every month throughout the year, as well as the background constellations of stars that rotated westward annually, and returned each year one (1) year later.

XX. At some point, they noticed that the Moon repeats its patterns approximately every 30 days, and thus established a primitive "ideal" calendar of twelve (12) months of 30 days, i.e. 360 days, in order to attempt to approximate and measure (and predict) future expected patterns of 29- and 30-day months. This would give a semi-precise mathematical approximation of what they observed.

XX. However, they would also have observed 29-day months, often (but not always) alternating with the 30-day months. Then they would have begun their attempts at measuring, refining, and calibrating the actual lunisolar year with their "ideal" (or theorized) year of 360 days relative to the actual observed year of approximately ~354 1/3 days per lunar year of twelve (12) months.

XX. Eventually, some brilliant mind noticed that approximately every two (2) or three (3) years, the ~354 1/3 days of the twelve-(12)-month lunar year could be recalibrated with the 365-day solar year with an intercalary extra month of 30 days. Thus the first attempts at periodic intercalation were enacted by observation (and decree) as necessary in order to synchronize the months with their expected annual agricultural harvest seasons throughout the solar year of approximately ~365 days.

XX. Brack-Bernsen shows that two (2) calendars were used side by side in Mesopotamia from about 2600 BCE (Julian) to about 300 BCE (Julian):[3847]

1.) The cultic or civil calendar was lunisolar. Each month began on the evening when the new crescent was first visible after conjunction. Each month therefore had 29 or 30 days.[3848]

2.) The administrative calendar counted every month as 30 days and every year as 360 days. It was used for calculation and never for dating.[3849] Its use in accounting goes back to the protoliterate

3847 Brack-Bernsen 2007: 83, 98
3848 Brack-Bernsen 2007: 83
3849 Brack-Bernsen 2007: 83

texts of Uruk (c. 3200 BCE (Julian)).[3850]

XX. The administrative year was kept in step with the lunisolar year. In an intercalary (diri) year of the Ur III period the extra month was also counted as 30 days, which gave 390 days.[3851] A text of Shulgi states: "It is (a period of) 16 years, including 6 extra months".[3852][5] This is one (1) intercalation about every three (3) years.

XX. Later astronomers kept the 360-day year as an "ideal" or "schematic" year of 12 × 30 days in Enūma Anu Enlil Tablet XIV and MUL.APIN.[3853] A new month that began on day one (1), after a 30-day month, was a good omen. A new month that began on day 30, after a 29-day month, was a bad omen. In Brack-Bernsen's words, "it was taken as a good sign when nature behaved as the 'ideal' 360-day year".[3854]

XX. In MUL.APIN the ideal year served as "a grid imposed upon and adjusted to nature". The text also calculates the effect of an intercalation every three (3) years.[3855] Brack-Bernsen concludes: "The astronomical lunisolar calendar regulated life, while the artificial accounting or ideal calendar (which I see as a practical and regular approximation to nature) could be used for calculation."[3856][8]

XX. In Egypt (at approximately the 3rd millennium BCE (Julian) or prior), the ancient Egyptians may have (but not necessarily) had some interaction with the ancient Sumerians, and may have theorized that a more accurate solar calendar of 365 days could be used instead of one containing an "ideal" 360 days (or the observed lunar ~354 1/3 days), and thus the competing calendar system of the solar year was utilized that approximated the 365-day solar year more accurately than the lunar year.

XX. Over time, the Egyptians may have noticed that even their 365-day calendar was slipping relative to the actual solar year of ~365.25 days per year (with the heliacal annual rising of the "Dog Star" Sirius), and yet chose to remain with the 365-day solar calendar without intercalation that best approximated both the ~365.25 solar year, the annual rising of Sirius, as well as the annual flooding of the Nile River.

XX. The Hebrew lunisolar calendar that we have today is identical to the ancient Babylonian Standard Mesopotamian Calendar that was in widespread use throughout the ancient Near East by around 1500 BCE (Julian): the same months and the same structure. The two (2) differ only in the application of the intercalary months. This proleptic Hebrew (Babylonian) calendar can be viewed as the culmination of

3850 Brack-Bernsen 2007: 83-84
3851 Brack-Bernsen 2007: 88
3852 Brack-Bernsen 2007: 89
3853 Brack-Bernsen 2007: 93
3854 Brack-Bernsen 2007: 93
3855 Brack-Bernsen 2007: 97
3856 Brack-Bernsen 2007: 98

the correct observation of the numerical cycles observed in celestial bodies (i.e. the Moon, the Sun, the constellations) with the refinement of the calendar through correct mathematical measurement, intercalation, and prediction to account for these astronomical cycles.

XX. Over the course of its 19-year cycles, the Babylonian / Hebrew lunisolar calendar is arguably the most accurate and synchronized calendar of its kind. It takes into account the movements of the Moon, the Sun and the stars. Thus the mathematically precise proleptic Hebrew calendar can be viewed as the ultimate refinement of this long process of measurement. It is the goal the ancient astronomers were working toward from their first "ideal" year of 360 days.

78.2. Attempting to Establish and Synchronize Ancient Chronologies

XX. Mesopotamian history and chronology has been mainly reconstructed on the basis of various king lists (KLs) supplemented by royal inscriptions and chronicles. The Assyrian King List (AKL) still remains the "backbone" of Mesopotamian chronology as it represents the only stable and relatively fixed scheme with which all the data is to be compared, and thus is the starting point for any chronology-related issue. Other historiographical texts, such as Babylonian King List (BKL), the Synchronistic History, as well as other chronicles, are to be included and possibly linked to the data of the AKL.[3857]

XX. Absolute Mesopotamian chronology is securely established for most of 1st millennium BCE (Julian) Babylonia and Assyria. Lengths of reigns, chronologically fixed by astronomical observations, are known from Nabopolassar (626 – 605 BCE (Julian)) onwards. Before the 8th century BCE (Julian) no continuous sequence of Babylonian kings can be securely established. However, Ptolemy lists some Neo-Assyrian kings in parallel with Babylonian kings, and because the Assyrian King List (AKL) and the Assyrian Eponym List (EL) [i.e. the Assyrian Eponym Canon (AEC)] enable us to set an absolute chronology from 910 to 649 BCE (Julian), the absolute dates of contemporary Babylonian kings can be fixed via synchronisms.[3858]

XX. Prior to 910 BCE (Julian) there are some gaps in our knowledge of eponyms. Assyrian royal chronology can go back as far as ca. 1420 / 1430 BCE (Julian), the reign of Enlil-naṣir II, with an uncertainty of – according to Pruzsinszky – ten (10) years. Beyond this, reign lengths are poorly known, so that dates cannot be as certain. For instance, the reign lengths of Enlil-naṣir's predecessors have been lost in all versions of the AKL.[3859]

XX. The Mesopotamian chronology systems – particularly the High Chronology (HC), Middle

3857 Pruzsinszky 2009: 22
3858 Pruzsinszky 2009: 17
3859 Pruzsinszky 2009: 17

Chronology (MC), and Low Chronology (LC) – solely depend on calculations based upon the astronomical data in the Venus Tablet (VT) and not on the interpretation of king list data. However, within the past few years more emphasis has been given to sources other than the VT and the eclipse data in the omen tablets of Enuma Anu Elish (EAE), which naturally has resulted in different chronologies.[3860]

XX. In Babylonia, most kings' reign lengths are known for the second half of the 2nd millennium BCE (Julian). However, these data are only sufficient for a relative chronology. For absolute dates one has to rely on synchronisms with Assyria.[3861]

XX. Babylonian chronology of the second half of the 2nd millennium BCE (Julian) is basically tied to Assyrian chronology, which in turn is based primarily on the Assyrian King List (AKL), the eponym list [in the Assyrian Eponym Canon (AEC)] and eclipse dates for the 1st millennium BCE (Julian).[3862] A solar eclipse mentioned in the Eponym Chronicle provides a reliable absolute date for the year of eponym Būr-Saggile 763 BCE (Julian).[3863]

XX. The basis for chronology after the middle of the 15th century BCE (Julian) is the Assyrian King List (AKL) and Babylonian King List (BKL) in combination with dated tablets and the Assyrian eponym lists (ELs) [i.e. in the Assyrian Eponym Canon (AEC)]. For the second half of the 2nd millennium BCE (Julian) – according to Pruzsinszky – an uncertainty of 10 years is to be reckoned with.[3864] From about 1450 BCE (Julian) onwards the chronology of the Near East is based on interlocking information from King Lists and synchronisms that prevent the possibility of gross errors.[3865]

XX. Chronological relations and synchronisms are sufficiently known for the time before the onset of the Dark Age: Hence one of the primary tasks is the coordination of the chronological data of early 2nd millennium BCE (Julian) Assyria, Babylonia, Egypt, and Anatolia (Hatti).[3866]

XX. The central problem of Mesopotamian chronology is the dating of the Babylon I dynasty. Attempts have been made to compute an absolute date for this dynasty by means of the "Venus Tablet" (VT) written – it would seem – during the reign of the Old Babylonian king Ammisaduqa. However, the data in the table is difficult to interpret and has resulted in three (3) chronologies: the High (HC), the Middle (MC), and the Low (LC). In most publications, the Middle Chronology (MC) is used. However, this is for reasons of "convenience", not because the Middle Chronology has been "proven".[3867]

3860 Pruzsinszky 2009: 23
3861 Pruzsinszky 2009: 17
3862 Pruzsinszky 2009: 30
3863 Pruzsinszky 2009: 17
3864 Pruzsinszky 2009: 19
3865 Pruzsinszky 2009: 31
3866 Pruzsinszky 2009: 17
3867 Pruzsinszky 2009: 17

XX. Currently the difference between the chronologies is 230 years. The central question of the chronology discussions remains the duration of the Dark Age.[3868]

XX. Few documents are known for the beginning of the Kassite period: only with Kara-indaš, the 15th king of this dynasty dated to the late 15th century BCE (Julian), does our knowledge of historical events improve. Basically, we need to date the Babylon I dynasty to establish some fixed points of chronological reference for the first half of the 2nd millennium BCE (Julian).[3869]

XX. The solution to the problem of Mesopotamian absolute chronology during the 2nd millennium BCE (Julian) hinges on the length of the Dark Age which is conventionally set in the 16th century BCE (Julian).[3870]

XX. After the discovery of the synchronism between Hammurabi and Šamšī-Adad I, it became evident from a historical point of view that the High Chronology (HC) – which like other chronologies is primarily based on the evaluation of astronomical data (the Venus Tablet (VT) combined with lunar eclipses) – according to Pruzsinszky cannot be correct.[3871]

XX. The absolute chronology of the second half of the 2nd millennium BCE – the Late Bronze Age (LBA) – is dependent on synchronisms among the great powers of the era (Egypt, Assyria, Hatti and its Syrian dependencies) and shows an uncertainty of – according to Pruzsinszky – only 5 to 10 years.[3872] As of today, most scholars accept the Low Chronology (LC) for Egypt (1479 BCE (Julian) accession of Tutmosis III and 1279 BCE (Julian) accession of Ramses II), the Middle Chronology (MC) for Mesopotamia in the Old Babylonian period (Hammurabi's accession in 1792 BCE (Julian)) and the shortened chronology (ten (10) years) proposed by Boese – Wilhelm (1979) for the Middle Assyrian kings which is connected with Hittite chronology based on the drastic shortening of Šuppiluliuma's I reign from 40 to 20 years: 1343 – 1322 / 1318.[3873]

XX. Since the sources for the 3rd millennium BCE (Julian) are too unreliable and insufficient to provide an absolute chronology, the absolute dates of the 3rd millennium BCE (Julian) will have to be based on those of the 2nd millennium BCE (Julian).[3874]

XX. Hittite chronology is also dependent on Assyrian chronology. Therefore, a variety of absolute dates – depending on which Mesopotamian chronology is adopted – can be observed in the literature.

3868 Pruzsinszky 2009: 24
3869 Pruzsinszky 2009: 29
3870 Pruzsinszky 2009: 30
3871 Pruzsinszky 2009: 19-20
3872 Pruzsinszky 2009: 21
3873 Pruzsinszky 2009: 21-22
3874 Pruzsinszky 2009: 18

Because the dates of the 2nd millennium BCE (Julian) rulers of Syria and Elam ultimately depend on the Assyrian and Babylonian chronology, they can be only approximate.[3875]

XX. Frequent synchronisms link Mesopotamia to the Levant and Egypt. According to Hermann Hunger, from Pruzsinsky's investigations it became clear that a solution for Mesopotamian chronology could not yet be achieved.[3876] The main problem remains the correlation of Aegean with Near Eastern chronologies.[3877]

XX. Within the past few years the amount of chronologically-useful astronomical data from Mesopotamia has not increased substantially, but the discussion on the known data has not diminished. A focus of the debate is how the astronomical data fits together with the historical and archaeological data.[3878]

XX. According to Pruzsinszky, it seems that there has been and always will be a discrepancy in the results. As long as we lack a sufficient number of astronomical dates which can be linked to archaeological or historical facts, the debate cannot be resolved.[3879]

XX. As has been mentioned, astronomical data, specifically the 56 / 64-year Venus cycles, form the basis of the Mesopotamian chronological systems, the High Chronology (HC), Middle Chronology (MC), and the Low Chronology (LC). Any reference to one of those chronologies therefore touches on issues connected with astronomical observations.[3880]

XX. Absolute chronology could not be established without knowledge of the calendars in use and astronomical events observed. Egyptian chronology is based on Sothis dates and several lunar dates, and 1st millennium BCE (Julian) Mesopotamian absolute chronology is firmly established on the basis of the solar eclipse date of 763 BCE (Julian).[3881]

XX. Although the 763 BCE (Julian) solar eclipse helped to establish the chronology of the eponyms of the 1st and the latter part of the 2nd millennia BCE (Julian), major difficulties are still present in the evaluation of the astronomical data for the early 2nd and later 3rd millennia BCE (Julian).[3882] The solar eclipse of 763 BCE (Julian) securely fixes absolute chronology of 1st millennium Mesopotamia. We still lack a similar astronomical sign-post for the 2nd millennium.[3883]

3875 Pruzsinszky 2009: 32
3876 Pruzsinszky 2009: 13
3877 Pruzsinszky 2009: 20
3878 Pruzsinszky 2009: 77
3879 Pruzsinszky 2009: 77
3880 Pruzsinszky 2009: 77
3881 Pruzsinszky 2009: 77
3882 Pruzsinszky 2009: 77
3883 Pruzsinszky 2009: 81

XX. Irregularities in observed astronomical data could have been caused by volcanic eruptions, such as that of Thera saturating the atmosphere with volcanic dust, which would have particularly interfered with observations near the horizon and thus affected the dates of Venus' first and last visibility. The date of the Thera eruption is still uncertain: 1645, 1628 or 1520 BCE (Julian) but perhaps may eventually be determined by dendrochronological data, the ice cores of Greenland, and dated archaeological strata with traces of ash and pumice deriving from this eruption.[3884]

XX. The lower chronologies generally are opposed only by those who rely on the astronomical data for chronology. However, Hunger (2000: 155-158) pointed out, that the connection between eclipses of Enuma Anu Elish (EAE) and historical events is not secured at all, and presented a quite pessimistic view of the chronological usefulness of the omens EAE 20 and 21.[3885]

XX. The Venus Tablet (VT), which deals with the risings and settings of Venus during the reign of Ammisaduqa (probably), is chronologically anchored within the Babylon I dynasty, but its date is so vague that it leaves us with three (3) possible chronologies (if not more), the High Chronology (HC), Middle Chronology (MC), and the Low Chronology (LC) reflected in EAE 63.[3886]

XX. Omens listed in EAE 63 relate to Venus' synodic period, which lasts ca. 584 days or ca. 20 months. Babylonian calendar dates of first and last visibilities of Venus repeat themselves after five (5) synodic periods or ca. eight (8) solar years – more precisely 99 synodic months minus 4 days. After seven (7) or eight (8) such 8-year periods the four-(4)-day deficits accumulate to a full month, which means that after 56 or 64 years Venus phenomena are again more or less in step with the lunar calendar.[3887]

XX. Unfortunately the Old Babylonian observational data contains many errors. It is also difficult to find out according to which convention the astronomers would report on their observations (see Huber et al., OPNE 12-13). Despite all this, modern scholars have "aimed to squeeze objective negative information out of the Venus data by using statistical methods to fix the absolute chronology of the 2nd millennium BC." (Huber, High...1, 6).[3888]

XX. Hunger concludes that he regrets to say that there is conflicting evidence for Mesopotamian chronology: pottery development suggests a relatively Low Chronology, tree rings (assuming they are correctly interpreted) a somewhat higher chronology, and astronomy (if P. Huber is correct) a very high one. At the moment, a decision to Hunger seems impossible, but he hopes for better data.[3889]

3884 Pruzsinszky 2009: 78
3885 Pruzsinszky 2009: 78
3886 Pruzsinszky 2009: 79
3887 Pruzsinszky 2009: 79
3888 Pruzsinszky 2009: 79
3889 Hunger 2009: 152

78.3. Assyrian Chronologies

XX. Basically, the Assyrian dates rely on the analysis of the AKL (with "minor" variants: uncertainties due to the differences in interpretation of the expression DUB-*pi-šu*, variants in the manuscripts of the AKL, the assumed length of first Assyrian Dark Age succeeding the reign of Išme-Dagān I, etc.) plus the assumption on whether the Assyrians used the lunar or the solar calendar before Tiglath-pileser I.[3890]

XX. An additional problem [i.e. uncertainty in absolute chronology] is caused by the use of a lunar calendar in Middle Assyrian times. Despite their awareness of the practice, most scholars – e.g. Hunger (2009), Klinger (2006) have hitherto assumed that in Middle Assyrian times regnal years were counted according to a solar calendar, and the dates provided by Frahm (2017) are based on this supposition as well.[3891]

XX. However since there is no evidence for intercalation in Middle Assyrian documents from before the 11th century BCE (Julian) – and due to a number of other reasons – it may be more likely (as recently argued in studies by Yigal Bloch (2010, 2012a, 2012b) and Joshua Jeffers (2017)) that the regnal years of Middle Assyrian kings preceding Aššur-bēl-kala (no. 89) [who ruled for 18 years (1076 – 1057 BCE (Julian) and overlapped the reign of David ben Yishai king of Israel 1061 – 1022 BCE (Julian)] were all lunar years, ca. 11 days shorter than the solar year with its 365.24 days. If so, the dates of [Assyrian] rulers [on which we base our chronology of the Ancient Near East] would have to be modified.[3892]

XX. According to Jeffers (2017), the months of the Assyrian year were in fact not permanently affixed to particular seasons of the solar year, [thereby] revealing that its year was not intercalated and was thus perpetually rotating [or wandering/migrating] slowly through the solar year.[3893] Bloch also concludes that during the 13th – 12th centuries BCE (Julian), the Assyrian calendar must have been purely lunar without intercalation (similar to the modern Islamic calendar).[3894] If so, this would have serious implications for the dating of Assyrian regnal years. The generally accepted regnal dates that are currently assigned to the reigns of the Middle Assyrian (MA) kings would have to be lowered because lunar years are shorter than their solar counterparts.[3895]

XX. According to Bloch, while in the 13th – 12th centuries BCE (Julian) the calendar practiced in Assyria was purely lunar, it was apparently not the original calendar of Assyria. For the period spanning the 17th – 14th centuries BCE (Julian), very few Assyrian documents are known, and they do

3890 Pruzsinszky 2009: 31-32
3891 Frahm 2017: 616
3892 Frahm 2017: 616
3893 Jeffers 2017: 151
3894 Bloch 2012a: 19
3895 Jeffers 2017: 153

not allow any clear conclusion concerning the structure of the calendar used in Assyria in that period. On the other hand, for the Old Assyrian period [during the] 20th – 18th centuries BCE (Julian) documentary evidence is abundant.[3896]

XX. The practice [of intercalation of lunar months] is known since the Ur III period, and letters from the Old Babylonian period record an intercalary month based on royal decrees, but the continuous usage of some rules cannot be documented – at least partially because of the lack of sources. In Assyria, the regular use of calibrating by using the lunar calendar is only known from the 1st millennium BCE (Julian).[3897]

XX. The reign of Tiglath-pileser I (1114 – 1076 BCE (Julian))[3898] was a turning point in the history of the calendar in the ancient Assyrian kingdom. During his reign, Assyria adopted a calendar, which had already existed by that time for several centuries in southern Mesopotamia – the so-called Standard Mesopotamian calendar [3899] which had emerged between ca. 1750 BCE (Julian) and ca. 1500 BCE (Julian).[3900] [3901]

XX. During the 1st millennium BCE (Julian), Assyria used the so-called Standard Mesopotamian calendar, which emerged in southern Mesopotamia sometime between ca. 1750 and ca. 1500 BCE.[3902] Since during the second half of the 2nd millennium BCE (Julian), the most important polity using the Standard Mesopotamian calendar was Babylonia (under the rule of the Kassite – and then the Second Isin – dynasties),[3903] it is justified to call it the [Standard Mesopotamian] Babylonian calendar.[3904]

XX. Therefore, it is justified to speak of the adoption of the [Standard Mesopotamian] Babylonian calendar in Assyria in the reign of Tiglath-pileser I.[3905] The adoption of the [Standard Mesopotamian] Babylonian calendar took place probably in the final two (2) decades of the reign of Tiglath-pileser I, following his conquest of Babylonia.[3906]

XX. For issues of absolute chronology, this question of whether the Assyrian calendar depended solely upon lunar months, and whether or not intercalary months were used to match the solar year, is a critical issue for precise absolute chronology. Thus whether or not calculations lead to a difference in the number of years in the chronology with respect to solar years is a critical issue.

3896 Bloch 2012a: 40
3897 Klinger 2006: 312
3898 Bloch 2012b: 25
3899 Bloch 2012a: 20
3900 Cohen 1993: 297-305 – cited by Bloch 2012a: 49
3901 Bloch 2012a: 49
3902 Cohen 1993: 297-305 – cited by Bloch 2012b: 334
3903 Bloch 2012b: 334; Bloch 2012a: 20
3904 Bloch 2012a: 20
3905 Cohen 1993: 300-301 – cited by Bloch 2012b: 334
3906 Bloch 2012b: 334

78.4. Egyptian Chronology

XX. Egyptian chronology is tied to Mesopotamian chronology, on both of which are Hittite and Elamite chronology dependent.[3907]

XX. According to Cline, relative dating is essentially settled for the 2nd millennium BCE (Julian) in both the Aegean and Egypt.[3908] It is the absolute dating that poses the problems in both the Aegean and Egypt during the 2nd millennium BCE (Julian). Attempting to determine the years during which a pharaoh ruled, or when an event such as the eruption of the Thera volcano in the Aegean took place, is not easy when dealing with people and events that occurred nearly four (4) thousand years ago.[3909]

XX. Debate over the absolute dating of both the Aegean region and Egypt during the 2nd millennium BCE (Julian) will undoubtedly continue for the foreseeable future. Egyptologists feel confident that changes in absolute dates will probably be relatively minor. They feel that we can rest assured that – at the very least – we have a fairly good grasp of the relative chronology of these two (2) regions.[3910]

XX. However, as discussed, Egyptian chronology is very problematic, especially with its dead-reckoning of minimal years of reign and its preference of the relative chronologies over the absolute chronologies provided the hard sciences of carbon 14 dating, dendrochronology, and astronomical dating.

78.5. Unlocking, Unsealing, and Unfolding the Precise Biblical Chronology

XX. As has been presented here, by using Jeremiah 25:1 as our key and "Rosetta Stone", we are able to synchronize and recalibrate the biblical chronology with the Babylonian Chronology and subsequently the Assyrian Chronology – specifically, we were able to determine that the 4th Regnal Year of Yehoyakim (Jehoiakim) king of Judah (r. 608 – 597 BCE (Julian)) is equivalent to the 1st Regnal Year of Nebuchadezzar II king of Babylon (r. 604 – 562 BCE (Julian)).

XX. Indeed, Tadmor (1956) offers an elegant explanation:[3911]

XX. The only way to retain the synchronism for the date of the Battle of Carchemish in the last year of Nabopolassar between the [1st month] Nisannu / Nisan and [the 5th month] Abu / Av in 605 – 604 BCE

3907 Pruzsinszky 2009: 30
3908 Cline 2008: 453
3909 Cline 2008: 453
3910 Cline 2008: 454
3911 Tadmor 1956: 226-227

(Julian) to the 4th Regnal Year of Yehoyakim (Jehoiakim) is to make two (2) assumptions:

1.) Regnal Years of the Judean kings in that period were reckoned according to the Mesopotamian method in which Year 1 started only with the Nisannu / Nisan following the accession of the king (i.e. Post-Dating: Year 0 Accession Years). This assumption is generally accepted on the basis of the Judeo-Babylonian synchronisms for that period.

2.) Regnal Years began in Nisannu / Nisan.

XX. Tadmor admits that this last assumption is much debated as postulating a Nisan-to-Nisan year in Judah requires one to consequently equate the 4th Regnal Year of Yehoyakim (Jehoiakim) with the Year 0 Accession Year of Nebuchadnezzar rather than with the Babylonian king's 1st full Regnal Year.[3912]

XX. As demonstrated, according to the Talmud, the New Year for kings is Nisannu / Nisan and therefore all those – e.g. Thiele and those who follow him – who attribute a Tishri-to-Tishri year for the Hebrew kings are simply mistaken.

XX. From here, we were able to synchronize each year forward in time to confirm dates such as the fall of Jerusalem in March 597 BCE (Julian), the destruction of Jerusalem in July – not August – 586 BCE (Julian), as well as all dates recorded in the Hebrew Bible that contain precise dates of years, months, and/or days, e.g. the chronology of the Torah, Joshua, Judges (not covered in this opus), Kings, Chronicles, Isaiah, Jeremiah, Ezekiel, Daniel, Haggai, Zechariah, Esther, Ezra, and Nehemiah.

XX. For example, precise mathematics alone allows us to reconstruct in exact detail the final days of Yehoyakim (Jehoiakim / Elyakim), king of Judah (608 – 597 BCE (Julian)). The absolute precision of the mathematics includes the revelation of the co-regency of his final months with his son Yehoyachin (Jehoiachin), who reigned exactly three (3) months and ten (10) days.[3913]

XX. It also proves that the king captured on the 2nd day of the 12th month, Addaru / Adar, in Nebuchadnezzar's 7th regnal year (598 – 597 BCE (Julian)) – as recorded also in the Babylonian Chronicle – was Yehoyakim (Jehoiakim) himself.

XX. This Babylonian date fell on the proleptic Hebrew date 03 Nisan 3164 AM (Saturday 16 March 597 BCE (Julian)), just after day one (1) of the 1st month (Nisannu / Nisan) of the 11th Regnal Year of Yehoyakim (Jehoiakim). His son Yehoyachin (Jehoiachin) was called out to Nebuchadnezzar only a few days later, before the march into exile began on 10 Nisannu / Nisan 3164 AM (Saturday 23 March 597 BCE (Julian)).

3912 cf. Jeremiah 25:1; 46:2
3913 cf. 2 Kings 24:8-9; 2 Chronicles 36:9-10

XX. In so doing, the dates of the ancient lunar eclipses have been precisely confirmed, while dates previously assigned to biblical events by way of the Babylonian calendar (which are not recorded in the Babylonian record) have been corrected – such as the destruction of the Temple which fell in July, not August, 586 BCE (Julian). Furthermore, Parker and Dubberstein's own dates prove at times incorrect – where later evidence has overturned their reconstructed intercalations – and at other times precise to the very day. This has profound implications that are far-reaching across ancient history as well as both ancient and modern astronomy.

XX. Using the correctly synchronized biblical chronology, we were able to determine that the 31st (FINAL) Regnal Year of Yoshiyahu (Josiah) king of Judah (and death at the hands of Pharaoh Necho II of Egypt) was in 608 BCE (Julian) in the 18th Regnal Year of Nabopolassar king of Babylon, and not 609 BCE (Julian) in his 17th Regnal Year as per current academic consensus.

78.5.1. The United and Divided Hebrew Monarchies

XX. Taking the Hebrew Bible literally, we were able to use the interlocking synchronisms of the Books of Kings, Chronicles, Jeremiah, Isaiah, Daniel, Ezekiel, Haggai, Zechariah, Esther, Ezra, and Nehemiah to come to a correct understanding of the chronologies of the United Monarchy under king David (r. 1061 – 1022 BCE (Julian)) and his son Solomon (r. 1021 – 982 BCE (Julian)), as well as the 395 years of the Divided Monarchy from 981 – 586 BCE (Julian).

XX. By unfolding the mathematics of the chronologies of the kings of the United Monarchy of Israel, as well as a Divided Monarchy of the southern kingdom of Judah and the northern kingdom of Israel, it becomes clear with absolute mathematical precision which kings ruled alone, and which kings had coregencies. It also allows us to confirm that both kingdoms used the same Nisannu / Nisan (spring to spring) dating system of kings that was used throughout the ancient Near East.

XX. However, only the southern (legitimate line of David) kingdom of Judah used a Year 0 Accession Year followed by the king's 1st Regnal Year beginning the next Spring. Remarkably, the mathematics reveal that for all illegitimate rulers including all kings of Israel (as well as the Usurper Queen Atalyah of Judah), an immediate count of Year 1 from the 1st day of rule was used, and not the Year 0 Accession Year of legitimate kings.

XX. Furthermore, it is exclusively the mathematics of these interlocking chronologies of these two (2) kingdoms alone that allow us to determine not only the existence – but also the precise years – of the interregna in which each kingdom experienced interregna with no king to rule. Were it not for this interlocking system of chronology between the two (2) kingdoms of Judah and Israel, it would not be possible to determine these interregna.

XX. The northern kingdom of Israel had four (4) interregna:[3914]

1.) (934 – 931 BCE (Julian));

2.) (864 – 863 BCE (Julian));

3.) (792 – 769 / 768 BCE (Julian));

4.) (758 – 757 BCE (Julian));

XX. The southern kingdom of Judah had only one (1) interregnum:[3915]

1.) (818 – 807 / 806 BCE (Julian))

XX. As the Assyrian chronology is considered by Assyriologists as certain from the years 910 – 649 BCE (Julian), the correct biblical chronology will also make possible for us to determine synchronisms between cultures that were not possible previously not only within these years, but also before 910 BCE (Julian) and after 649 BCE (Julian) when the AKL and AEC are considered less precise and not for certain without error, and thus we can rely on the mathematical precision of the biblical chronology.

XX. For example, assuming the AKL and AEC are accurate back to the 1st Regnal Year of Tiglat-pileser I who ruled for 39 years (1115 – 1077 BCE (Julian)), then we can determine that king David acceded to the throne in the 13th Regnal Year of Assyrian king Aššur-bēl-kala (1062 – 1061 BCE (Julian)) who ruled for 18 years (1074 – 1057 BCE (Julian)). From this synchronized chronology, we can determine that king Solomon acceded to the throne in the 10th Regnal Year of Šalmaneser II (1022 – 1021 BCE (Julian)) who ruled for 12 years (1031 – 1020 BCE (Julian)).

XX. Therefore, in like manner we can synchronize all Hebrew kings with all Assyrian (as well as Babylonian and Egyptian) kings of whom we have a correct record during this period. Thus, we can correct Thiele who synchronizes the biblical chronology to the AEC by arguing that king Ahab was the Israelite king at the Battle of Qarqar in the 6th Year of Šalmaneser III (853 – 852 BCE (Julian)) who ruled for 35 years (858 – 824 BCE (Julian)).[3916] In fact, unless the Julian date associated with the Assyrian record is incorrect, this year of 853 – 852 BCE (Julian) was the 11th Regnal Year of Yeho'ahaz (Jeho-ahaz) ben Yehu (Jehu) king of Israel, and the 33rd Regnal Year of Yeho'ash (Yo'ash / Jo'ash) ben Ahazyah king of Judah.[3917]

3914 cf. Appendix Table X.2, X.194
3915 cf. Appendix Table X.2, X.194
3916 Kurkh Monolith; Oracc text (Shalmaneser III no. 2): https://oracc.museum.upenn.edu/riao/ria5/Q004607/
3917 cf. Appendix Table X.2, X.194

XX. However, if indeed the Assyrian record is correct that it was the Israelite king Ahab, who ruled from 924/923 through 903/902 BCE (Julian), recorded for the Battle of Qarqar in the 6th Year of Šalmaneser III (853 – 852 BCE (Julian)), then it would logically mean that the Assyrian record must be corrected and recalibrated with the mathematically precise biblical chronology of the Hebrew kings.

XX. Moreover, the Assyrian Annals record the 18th Regnal Year of Šalmaneser III in 841 – 840 BCE (Julian) in which he records receiving tribute from the Israelite king (in Akkadian): "Iāūa" of the "House of Omri". Thiele attributed this to Yehu (Jehu) ben Yehoshafat ben Nimshi who ruled from 892 – 865 BCE (Julian), and was not from the House of Omri.[3918]

XX. Neither could this be the actual son of Omri, Yehoram (Jehoram) ben Ahav ben Omri who ruled from 903 – 892 BCE (Julian). However, the 9th Regnal Year of Yo'ash (Jo'ash) ben Yeho'ahaz king of Israel in 841 – 840 BCE (Julian) synchronizes perfectly with the 18th Regnal Year of Šalmaneser III. Thus this Assyrian king must have used the term "House of Omri" as a generic term for the northern kingdom of Israel (because of the previous Omride dynasty that ruled from 931 – 930 BCE to 892 – 891 BCE (Julian)). Therefore, scholars must have attributed this to the incorrect "Yehu (Jehu) // Yeho (Jeho)" as the Akkadian term "Iāūa" seems to be referring to "Yo'ash ben Yeho'a[haz]".[3919]

XX. Furthermore, through the mathematics we can determine the precise year of the 9th (FINAL) Regnal Year of Hoshea ben Elah, the last king of Israel – and thus also correct the date of the fall of Samaria – to be (719 – 718 BCE (Julian)), and thus confirm that it was indeed Sargon II who destroyed Samaria and exiled the ten (10) northern tribes of Israel.

XX. Moreover, we can determine the year range in which Ahaz ben Yotam king of Judah asked for help (against Pekah ben Remalyahu king of Israel and Retzin king of Aram) from Tiglath-pileser III who ruled for 18 years (744 – 727 BCE (Julian)). This only could have happened when the years of reign of all three (or four with Retzin king of Aram) overlapped between 739 – 735 BCE (Julian).[3920] [3921]

XX. It is only through the mathematics that several scribal errors can be revealed such as Ba'asha ben Ahiyah king of Israel builds the "Rama" in the 26th Regnal Year of Asa ben Aviyah (Aviyam) king of Judah, and not the 36th Regnal Year. Another example: There was no 20th Regnal Year of Yotam ben Uziyah king of Judah.[3922] The scribe likely intends the 20th Regnal Year of Pekah ben Remalyahu king of Israel.[3923] [3924]

3918 Assyrian Annals, year 18; Oracc text (Shalmaneser III no. 12, lines 29-30): https://oracc.museum.upenn.edu/riao/ria5/Q004617/
3919 cf. Appendix Table X.2, X.194
3920 cf. 2 Kings 16: 5-10; 2 Chronicles 28: 16-25
3921 cf. Appendix Table X.2, X.194
3922 cf. 2 Kings 15:30
3923 cf. 2 Kings 15:19, 27, 29-30
3924 cf. Appendix Table X.2, X.194

XX. Another more serious scribal error (perhaps – see discussion below) that is revealed only through the mathematics is the account of Sennacherib's invasion in the 14th Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz king of Judah. If the Assyrian record is correct, this could not have happened during this year of 711 – 710 BCE (Julian) because this was equivalent to the 11th Regnal Year of Sargon II who ruled for 17 years (721 – 705 BCE (Julian)).[3925] However, Sennacherib who ruled for 24 years (704 – 681 BCE (Julian)), conducted his 3rd campaign – according to the Assyrian and Babylonian records – in which he invaded Judah in his 4th Regnal Year, 701 – 700 BCE (Julian), does align and accord perfectly with the 24th Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz king of Judah.

78.5.2. The Identity of Assyrian King Pul

XX. The mathematics of the precise astronomical Hebrew luni-solar calendar alone allow us to definitively determine the identity of the Assyrian King Pul mentioned in the Hebrew Bible to be Aššur-dān III who ruled for 18 years (772 – 755 BCE (Julian)) overlapping the reigns of:[3926]

1.) Azaryah (Uziyahu) ben Amatzyah king of Judah (807 – 755 BCE (Julian));

2.) Menahem ben Gadi king of Israel (768 – 758 BCE (Julian));

3.) Pekahyah ben Menahem king of Israel (757 – 755 BCE (Julian));

4.) Pekah ben Remalyahu king of Israel (755 – 736 BCE (Julian));

XX. Pul king of Assyria comes and Menahem ben Gadi king of Israel gives him tribute.[3927] Separate from the mathematics, from the Hebrew Bible it is clear that Pul and Tiglat-pilneser III (i.e. Tiglat-pileser III) are two (2) different individuals.[3928]

78.5.3. Synchronizing Kings of Assyria with Kings of Judah and Israel

XX. Tiglat-pileser III ruled for 18 years (744 – 727 BCE (Julian)) overlapping the reigns of:[3929]

1.) Yotam ben Uziyahu king of Judah (753 – 738 BCE (Julian)) – ruled 16 years;

2.) Ahaz ben Yotam king of Judah (738 – 723 BCE (Julian)) – ruled 16 years;

3925 cf. Appendix Table X.2, X.194
3926 cf. Appendix Table X.2, X.194
3927 cf. 2 Kings 15:19-20
3928 cf. 1 Chronicles 5:26
3929 cf. Appendix Table X.2, X.194

3.) Pekah ben Remalyahu king of Israel (755 – 736 BCE (Julian)) – ruled 20 years;

4.) Hoshea ben Elah king of Israel (727 – 719 BCE (Julian)) – ruled 9 years;

XX. Šalmaneser V ruled for 5 years (726 – 722 BCE (Julian)) overlapping the reigns of:

1.) Ahaz ben Yotam king of Judah (738 – 723 BCE (Julian)) – ruled 16 years;

2.) Hizkiyah ben Ahaz king of Judah (725 – 696 BCE (Julian)) – ruled 29 years;

3.) Hoshea ben Elah king of Israel (727 – 719 BCE (Julian)) – ruled 9 years;

XX. Sargon II ruled for 17 years (721 – 705 BCE (Julian)) overlapping the reigns of:

1.) Hizkiyah ben Ahaz king of Judah (725 – 696 BCE (Julian)) – ruled 29 years;

2.) Hoshea ben Elah king of Israel (727 – 719 BCE (Julian)) – ruled 9 years;

XX. Sennacherib ruled for 24 years (704 – 681 BCE (Julian)) overlapping the reigns of:

1.) Hizkiyah ben Ahaz king of Judah (725 – 696 BCE (Julian)) – ruled 29 years;

2.) Menasheh ben Hizkiyah king of Judah (696 – 641 BCE (Julian)) – ruled 55 years;

XX. In the 4th Regnal Year of Hizkiyah (Hezekiah) ben Ahaz king of Judah (which was equivalent to the 7th Regnal Year of Hoshea ben Elah king of Israel), i.e. 721 – 720 BCE (Julian), the Hebrew Bible records that Šalmaneser V king of Assyria came to the then-capital city Samaria (Shomron) in the kingdom of Israel, and began what would become a three-(3)-year siege.[3930]

XX. However, Šalmaneser V's 5th (FINAL) year was in the 3rd Regnal Year of Hizkiyah (Hezekiah) ben Ahaz king of Judah (which was equivalent to the 6th Regnal Year of Hoshea ben Elah king of Israel), i.e. 722 – 721 BCE (Julian), and therefore is evidently a scribal error mistaking the Assyrian king, perhaps not aware that Šalmaneser V had died, and that Sargon II had replaced him.[3931]

XX. The Assyrian and Babylonian Chronicle (ABC 1) records Šalmaneser V's death in his 5th Regnal Year, in the 10th month Ṭebētu / Tevet (December 722 – January 721 BCE (Julian)): "The fifth year: Šalmaneser went to his destiny in the month Tebêtu. For five years Šalmaneser ruled Akkad and

3930 cf. 2 Kings 17:1-6; 18:9-11
3931 cf. Appendix Table X.2, X.194

Assyria".[3932] The chronicle gives no day.

XX. The next line dates Sargon's accession to the same month: "On the twelfth day of the month Tebêtu, Sargon ascended the throne in Assyria".[3933] Sargon's Year 0 Accession Year therefore ran from 12 Ṭebētu / Tevet[3934] [3935] to the end of that year, and his 1st Regnal Year began the following Nisannu / Nisan.[3936] [3937]

XX. In this next year of his 1st Regnal Year, i.e. Spring 721 – 720 BCE (Julian), Sargon II became official king of Assyria, and continued the siege until it was captured in 9th Regnal Year of Hoshea ben Elah, the last king of Israel (which was equivalent to the 6th Regnal Year of Hizkiyah ben Ahaz king of Judah), i.e. 719 – 718 BCE (Julian), and exiled its inhabitants.[3938] [3939]

78.5.4. Identification of Merodach-(Berodach)-baladan

XX. The Babylonian king *Marduk-apla-iddina* II of the Babylonian Chronicle[3940] [3941] is the "Berodach-baladan son of Baladan, king of Babylon" of 2 Kings 20:12,[3942] who is also called "Merodach-baladan son of Baladan, king of Babylon" in the parallel Isaiah 39:1.[3943] The Hebrew *Merodach-baladan* (מְרֹדַךְ בַּלְאֲדָן) renders the Akkadian *Marduk-apla-iddina* ("Marduk has given an heir").

XX. The Book of Kings writes the first element with *b* (בְּרֹאדַךְ), while Isaiah writes with *m* (מְרֹדַךְ). The two (2) texts agree in his title, "king of Babylon", and in making him Hizkiyahu (Hezekiah) ben Ahaz king of Judah's contemporary.

3932 ABC 1 i 29-30: https://www.livius.org/sources/content/mesopotamian-chronicles-content/abc-1-from-nabu-nasir-to-samas-suma-ukin/
3933 ABC 1 i 31: https://www.livius.org/sources/content/mesopotamian-chronicles-content/abc-1-from-nabu-nasir-to-samas-suma-ukin/
3934 Sunday 18 December 722 BCE (Julian); Sunday 10 December 722 BCE (Gregorian); 12 Tebetu / Tevet 3040 AM; Astronomical Julian Day JD_X: 1458063.5; RD_Y: 1110067; RD_X: -263361; Sabbatical Cycle: 2; Metonic Cycle: 19;
3935 Proleptic Hebrew date. If the two (2) calendars were in step that year, the Babylonian 12 Ṭebētu fell about one (1) to two (2) days later. If a Babylonian intercalation had put them one (1) month apart, it fell about one (1) month earlier or later. This cannot yet be determined since Parker and Dubberstein's tables begin in 626 BCE (Julian).
3936 Tuesday 03 April 721 BCE (Julian); Tuesday 26 March 721 BCE (Gregorian); 01 Nisannu / Nisan 3040 AM; Astronomical Julian Day JD_X: 1458170.5; RD_Y: 1110174; RD_X: -263254; Sabbatical Cycle: 2; Metonic Cycle: 19;
3937 Proleptic Hebrew date. If the two (2) calendars were in step that year, the Babylonian 01 Nisannu fell about one (1) to two (2) days later. If a Babylonian intercalation had put them one (1) month apart, it fell about one (1) month earlier or later. This cannot yet be determined since Parker and Dubberstein's tables begin in 626 BCE (Julian).
3938 cf. 2 Kings 17:5-6; 18:9-11
3939 cf. Appendix Table X.2, X.194
3940 ABC 1 i 32 – ii 23: https://www.livius.org/sources/content/mesopotamian-chronicles-content/abc-1-from-nabu-nasir-to-samas-suma-ukin/
3941 ABC 1 i 32 – ii 23: https://www.livius.org/sources/content/mesopotamian-chronicles-content/abc-1-from-nabu-nasir-to-samas-suma-ukin/abc-1.ii
3942 cf. 2 Kings 20:12
3943 cf. Isaiah 39:1

78.5.5. The Accession of Merodach-(Berodach)-baladan II King of Babylon

XX. The Assyrian and Babylonian Chronicle (ABC) records: "In the month Nisannu, Marduk-apla-iddina ascended the throne in Babylon".[3944] His Year 0 Accession Year therefore falls in Sargon's 1st Regnal Year. The chronicle itself refers to "the accession year of Marduk-apla-iddina".[3945] That year is his Year 0 Accession Year, and his 1st Regnal Year begins the following Nisannu / Nisan being equivalent with the 2nd Regnal Year of Sargon II.[3946]

78.5.6. The Years of Merodach-(Berodach)-baladan's First Reign

XX. The Assyrian and Babylonian Chronicle (ABC) numbers his years:

"The second year of Marduk-apla-iddina: Humban-Nikaš, king of Elam, did battle against Sargon, king of Assyria, in the district of Der".[3947]

"The fifth year of Marduk-apla-iddina: Humban-Nikaš, king of Elam, went to his destiny".[3948]

"From the accession year of Marduk-apla-iddina until the tenth year Assyria was belligerent towards Marduk-apla-iddina".[3949]

"The twelfth year of Marduk-apla-iddina: Sargon went down to [Akkad] and did battle against Marduk-apla-iddina. Marduk-apla-iddina retreated before him and fled to Elam. For twelve years Marduk-apla-iddina ruled [Babylon]. Sargon ascended the throne in Babylon".[3950]

78.5.7. The Fall of Samaria in 719 BCE: Synchronizing Hizkiyahu (Hezekiah), Hoshea, Šalmaneser V, and Sargon II

3944 ABC 1 i 32: https://www.livius.org/sources/content/mesopotamian-chronicles-content/abc-1-from-nabu-nasir-to-samas-suma-ukin/
3945 ABC 1 i 41: https://www.livius.org/sources/content/mesopotamian-chronicles-content/abc-1-from-nabu-nasir-to-samas-suma-ukin/
3946 cf. Appendix Table X.2, X.194
3947 ABC 1 i 33-34: https://www.livius.org/sources/content/mesopotamian-chronicles-content/abc-1-from-nabu-nasir-to-samas-suma-ukin/
3948 ABC 1 i 38: https://www.livius.org/sources/content/mesopotamian-chronicles-content/abc-1-from-nabu-nasir-to-samas-suma-ukin/
3949 ABC 1 i 41-42: https://www.livius.org/sources/content/mesopotamian-chronicles-content/abc-1-from-nabu-nasir-to-samas-suma-ukin/
3950 ABC 1 ii 1-5: https://www.livius.org/sources/content/mesopotamian-chronicles-content/abc-1-from-nabu-nasir-to-samas-suma-ukin/abc-1.ii

XX. The Year 0 Accession Year of Hizkiyahu (Hezekiah) ben Ahaz king of Judah is equivalent to the 3rd Regnal Year of Hoshea ben Elah king of Israel,[3951] i.e. 725 – 724 BCE (Julian).[3952]

XX. The 3rd Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz is equivalent to the 6th Regnal Year of Hoshea ben Elah,[3953] i.e. 722 – 721 BCE (Julian). This is the 5th (FINAL) year of Šalmaneser V in which sometime during his reign – according to the Assyrian and Babylonian Chronicle (ABC) – "he ravaged Samaria". Šalmaneser V died in the 10th month, Tebetu / Tevet, and Sargon II ascended the throne on day twelve (12) of the same month, Tebetu / Tevet.[3954] This is the Year 0 Accession Year of Sargon II.[3955]

XX. The 4th Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz is equivalent to the 7th Regnal Year of Hoshea ben Elah,[3956] i.e. 721 – 720 BCE (Julian). This is the 1st Regnal Year of Sargon II.[3957]

XX. Samaria (Shomron) was taken by the king of Assyria (Sargon II) in the 9th (FINAL) Regnal Year of Hoshea ben Elah, which is equivalent to the 6th Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz,[3958] i.e. 719 – 718 BCE (Julian). In this year, which is the 3rd Regnal Year of Sargon II, Sargon completed the three-(3)-year siege of Samaria. Sargon destroyed Samaria and exiled its inhabitants.[3959]

XX. Therefore, we can conclude that the definitive date for the fall of Samaria (Shomron), and the subsequent destruction and exile of those who had remained (from the first exile by Tiglat-pileser III)[3960] in the northern kingdom of Israel was in 719 – 718 BCE (Julian), and not 722 – 721 BCE (Julian) as held by current academic consensus.

78.5.8. Hizkiyahu (Hezekiah) ben Ahaz's 14th Regnal Year

XX. Hizkiyahu (Hezekiah) ben Ahaz king of Judah reigned 29 years,[3961] and at his illness 15 years were added to his life.[3962] His illness therefore fell in his 14th Regnal Year, i.e. 711 – 710 BCE (Julian). This is equivalent to the 11th Regnal Year of Sargon II the king of Assyria, as well as to the 10th Regnal Year

3951 cf. 2 Kings 18:1
3952 cf. Appendix Table X.2, X.194
3953 cf. derived from 2 Kings 18:1, 9
3954 ABC 1 i 28-31: https://www.livius.org/sources/content/mesopotamian-chronicles-content/abc-1-from-nabu-nasir-to-samas-suma-ukin/
3955 cf. Appendix Table X.2, X.194
3956 cf. 2 Kings 18:9
3957 cf. Appendix Table X.2, X.194
3958 cf. 2 Kings 17:6; 18:10
3959 cf. Appendix Table X.2, X.194
3960 cf. 2 Kings 15:29; 1 Chronicles 5:6, 26
3961 cf. 2 Kings 18:2
3962 cf. 2 Kings 20:6; Isaiah 38:5

of Merodach-(Berodach)-baladan ben Baladan, king of Babylon.[3963]

XX. After hearing of his illness and of the wonder (i.e. the miracle, the sign) of the sundial shadow moving backwards being performed, Merodach-(Berodach)-baladan ben Baladan, king of Babylon, sends his envoys to Hizkiyahu (Hezekiah) ben Ahaz sometime after his illness in his 14th Regnal Year.[3964]

XX. A related verse, 2 Chronicles 32:31, mentions "the scoffers (mockers) of the Ministers of Babylon" without naming the king:[3965]

31 And thus in [the matter of] the scoffers of the Ministers of Babylon who sent [them] upon him to demand [explanation] of the wonder that was in the land; the Elohim left him to test him to know everything in his heart.

XX. The wonder mentioned is the wonder (i.e. the miracle, the sign) of the sundial shadow moving backward 10 degrees that was given to Hizkiyah (Hezekiah) ben Ahaz king of Judah (725 – 696 BCE (Julian)) when he was sick approaching death, and prayed to GOD, and subsequently received a reprieve of an additional 15 years of life prophesied to him by Isaiah the prophet.[3966] [3967]

XX. The Hebrew Bible dates the Assyrian king Sennacherib's invasion to the 14th Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz:[3968] [3969]

13 And in the year fourteen (14) of the king Hizkiyah, Sanheriv king of Assyria ascended upon all the fortified cities of Judah, and [he] seized them.

XX. Isaiah 36:1 contains nearly exactly the same wording as in 2 Kings 18:13:[3970] [3971]

1 And [it] was in the year fourteen (14) of the king Hizkiyahu, Sanheriv king of Assyria ascended upon all the fortified cities of Judah, and [he] seized them.

XX. However, according to the Assyrian and Babylonian Chronicle (ABC) in the 14th Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz, 711 – 710 BCE (Julian), Sennacherib was not yet king of Assyria as it was the 11th Regnal Year of Sargon II who ruled 17 years from 721 – 705 BCE (Julian).

3963 cf. Appendix Table X.2, X.194
3964 cf. Appendix Table X.2, X.194
3965 cf. 2 Chronicles 32:31
3966 cf. Appendix Table X.2, X.194
3967 cf. 2 Kings 20:1-19; Isaiah 38:1; 39:1-8; 2 Chronicles 32:34
3968 cf. Appendix Table X.2, X.194
3969 cf. 2 Kings 18:13
3970 cf. Appendix Table X.2, X.194
3971 cf. Isaiah 36:1

XX. According to the Assyrian and Babylonian Chronicle (ABC), the 1st Regnal Year of Sennacherib was not until 704 – 703 BCE (Julian)[3972] which is equivalent to the 21st Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz.

XX. Sennacherib's own inscriptions fix the date of his campaign against Judah to his third campaign. The Bellino Cylinder, dated to the second half of the eponymy of Nabû-lēʾi (702 BCE (Julian)), records only his first two (2) campaigns.[3973] The Rassam Cylinder, dated to the beginning of the eponymy of Mitūnu (700 BCE (Julian)), is the first to record the third campaign.[3974] The third campaign therefore took place in 701 BCE (Julian), the 4th Regnal Year of Sennacherib.[3975] This is equivalent to the 24th Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz (701 – 700 BCE (Julian)).[3976]

XX. The mathematics thus requires that the original text be read as "and in the year twenty-four" (וּבְאַרְבַּע וְעֶשְׂרִים שָׁנָה), or "and in the year four-and-twenty" and not "and in the year fourteen" (וּבְאַרְבַּע עֶשְׂרֵה שָׁנָה). The two (2) readings differ by only two (2) letters: עֶשְׂרֵה ("ten") for וְעֶשְׂרִים ("and twenty"). The order "four and twenty" is itself attested in the Hebrew Bible.[3977]

XX. The original may have been written in either of two (2) ways, both attested in the Hebrew Bible:

1.) (וּבְאַרְבַּע וְעֶשְׂרִים שָׁנָה) – "and in the year four-and-twenty" with the unit first. This differs from the received text only in one word: *'esreh* (עֶשְׂרֵה) "ten" for *'esrim* (וְעֶשְׂרִים) "and twenty" The unit-first order is attested for "four and twenty" in Numbers 25:9 and 1 Chronicles 24:18; 25:31, and for a regnal year in Jeremiah 52:30 (בִּשְׁנַת שָׁלֹשׁ וְעֶשְׂרִים לִנְבוּכַדְרֶאצַּר), "in the year three-and-twenty of Nebuchadrezzar".

2.) (וּבְעֶשְׂרִים וְאַרְבַּע שָׁנָה) – "and in the year twenty-and-four" with the tens first. This is the usual order for year counts in the Book of Kings, including the number twenty-four itself (עֶשְׂרִים וְאַרְבַּע שָׁנָה) in 1 Kings 15:33, and Hizkiyahu (Hezekiah) ben Ahaz's own reign of twenty-nine (29) years (וְעֶשְׂרִים וָתֵשַׁע שָׁנָה) in 2 Kings 18:2.[3978]

XX. That the order of such numbers could be reversed in transmission is shown by the parallel accounts of the age Yeho-ahaz (Jeho-ahaz) at his accession. 2 Kings 23:31 reads (בֶּן־עֶשְׂרִים וְשָׁלֹשׁ שָׁנָה),

3972 ABC 1; Grayson 1975; translation: Livius.org; ABC 1; https://www.livius.org/sources/content/mesopotamian-chronicles-content/abc-1-from-nabu-nasir-to-samas-suma-ukin/
3973 RINAP 3, Sennacherib 3 (Bellino Cylinder): https://oracc.museum.upenn.edu/rinap/rinap3/Q003477/
3974 RINAP 3, Sennacherib 4 (Rassam Cylinder): https://oracc.museum.upenn.edu/rinap/rinap3/Q003478/
3975 Dates of both editions: ORACC, Sennacherib Scores; Grayson and Novotny 2012 (RINAP 3/1)
3976 cf. Appendix Table X.194
3977 Numbers 25:9; Jeremiah 52:30; 1 Chronicles 24:18; 25:31
3978 cf. 1 Kings 15:33; 2 Kings 18:2

"twenty-and-three years [old]" while 2 Chronicles 36:2 reads (בֶּן שָׁלוֹשׁ וְעֶשְׂרִים שָׁנָה), "three-and-twenty years [old]".[3979]

XX. Therefore, one can reasonably conclude that this is a simple scribal error, most likely arising from the scribe's confusion with the 14th Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz's illness. The text dates that illness only "in those days",[3980] but it is mathematically fixed to his 14th year by the fifteen (15) years added in order to total a reign of twenty-nine (29) years.[3981]

XX. The error is readily explained: Hizkiyahu (Hezekiah) ben Ahaz's illness fell in his 14th Regnal Year (29 − 15 == 14).[3982] A scribe with the illness in mind in year fourteen (14) may simply have confused the numbers, or carelessly written "fourteen" in place of "twenty-four". Once the error stood in one text, it was carried into the parallel text.[3983]

XX. We can conclude that the error is ancient since it is already present in every surviving witness: the Masoretic Text (MT), the Septuagint (τῷ τεσσαρεσκαιδεκάτῳ ἔτει) (LXX),[3984] the Great Isaiah Scroll,[3985] and Josephus.[3986] Corrected to the 24th Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz, 701 – 700 BCE (Julian), the text agrees with the Assyrian and Babylonian records correctly to the exact year.

XX. Therefore, Sennacherib's third campaign against Judah that is dated by his own inscriptions to 701 BCE (Julian), falls in the 24th Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz (701 – 700 BCE (Julian)). In this way the biblical chronology, the Assyrian Eponym Canon (AEC), and the Assyrian and Babylonian Chronicle (ABC) stand together without contradiction.

XX. The emendation of "fourteenth" to "twenty-fourth" has been proposed before.[3987] However, those proposals reasoned that Hizkiyahu (Hezekiah) ben Ahaz's reign began in 727 BCE (Julian), so his 24th year would fall in 703 BCE (Julian) – not 701 BCE (Julian). The actual chronology based upon precise mathematics corrects all errors and reveals the correct biblical timeline that synchronizes perfectly with both Assyrian and Babylonian accounts: counted exactly, the 24th Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz is 701 – 700 BCE (Julian), the precise year of Sennacherib's third campaign.

XX. The only logical alternative explanation is that the biblical dates are all correct recording that Sennacherib's 3rd campaign that included Judah in the 14th Regnal Year of Hizkiyahu (Hezekiah) ben

3979 cf. 2 Kings 23:31; 2 Chronicles 36:2
3980 cf. 2 Kings 20:1; Isaiah 38:1
3981 cf. 2 Kings 18:2; 20:6
3982 cf. 2 Kings 18:2; 20:6
3983 cf. 2 Kings 18-20; Isaiah 36-39
3984 Rahlfs 1935, 4 Kingdoms 18:13; Isaiah 36:1: https://github.com/eliranwong/LXX-Rahlfs-1935
3985 1QIsa[a], col. 28, lines 28-29 (Abegg, Bowley and Cook transcription): https://github.com/ETCBC/dss
3986 Josephus, Antiquities 10.1.1: https://penelope.uchicago.edu/Josephus/ant-10.html
3987 Montgomery 1951: 483; Rowley 1962: 411-413; Young 1969: 2:541-542; Archer 1970: 207-208 – cited by Franz; https://www.lifeandland.org/2011/02/the-hezekiah-sennacherib-chronology-problem-reconsidered-chapter-3/

Ahaz, and that somehow – somewhere – the Assyrian and Babylonian records (or our current understanding of them) are incorrect.

XX. Time will tell, and mathematics will help us to uncover the absolute truth.

78.5.9. The Date of the Completion of the First Temple and the Exodus from Egypt

XX. After determining the precise dates of the reign of king Solomon, we were able to determine the date of his 4th Regnal Year which is critical to the calculation of dates back through the Exodus and those prior. Therefore, counting forward four (4) years from his Year 0 Accession Year of 1022 – 1021 BCE (Julian), we arrive to the 4th Regnal Year of Shlomo (Solomon) commencing 01 Nisannu / Nisan in Spring 1018 – 1017 BCE (Julian):[3988]

XX. The construction of the 1st Temple took 7.5 years from the 2nd month of the 4th Regnal Year (1018 – 1017 BCE (Julian)) through the 8th month of the 11th Regnal Year (1011 – 1010 BCE (Julian)) of Shlomo (Solomon). It stood for 424 years from the Babylonian / Hebrew year 2751 AM through 3175 AM. As mentioned, this number 424 is the gematria numerical value for the Hebrew term for Messiah son of David (*Mashiach ben David*, משיח בן דוד).

XX. We then were able to calculate back 480 years from king Solomon's 4th Regnal Year, and thus determine the exact year, month, and day of the Exodus in April 1497 BCE (Julian). This will not only make possible for Egyptologists to recalibrate their chronology to synchronize with the precise biblical chronology, but also to determine the heretofore elusive and mysterious identity of the Pharaoh of the Exodus: i.e. to determine who was Pharaoh in April 1497 BCE (Julian), and this will reveal who is the correct Pharaoh of the Exodus of Israel from Egypt.

XX. From here, we determined the exact year, month, and day of the receiving of the Torah on Mount Sinai in June 1497 BCE (Julian), as well as several dates of events in the Torah during their wandering in the desert for 40 years (1497 – 1457 BCE (Julian)) including – but not limited to – the deaths of Miriam (March – April 1458 BCE (Julian)) and Aharon (July 1458 BCE (Julian)), as well as the final address of Moses to the Nation of Israel (January 1457 BCE (Julian)).

XX. Subsequently, the date of the Entering of the Promised Land was determined to be in March 1457 BCE (Julian). Consequently, the mathematics reveal that the children of Israel sojourned in the land for exactly 860 luni-solar years of 314116 days from 10 Nisannu / Nisan 1457 BCE (Julian) through their exile from the land on 10 Nisannu / Nisan 597 BCE (Julian).

3988 cf. 1 Kings 6:1; 37-38

78.5.10. The Date of the Flood of Noaḥ

XX. If we are to believe in the literal accuracy of the details in the Hebrew Bible, we must consider the curious case of the account of the Flood of Noaḥ:[3989]

1 And YHVH said to Noaḥ, "Come – you and your entire house – to the ark because you I [have] seen righteous before Me in this generation. 2 From all the pure beast[s you] will take for you[rself] seven [pairs] (seven [pairs]): [each] man and his woman – and from the beast[s] that [is] not pure [you will take for yourself] two: [each] man and his woman. 3 And also from the flying [bird] of the skies seven [pairs] (seven [pairs]) – male and female – to make seed live upon the face of all the land. 4 Because in another seven days I [will] cause [it] to rain on the land forty days and forty nights, and I will erase the entire existence that I made [that is] upon the face of the Earth."[3990]

8 From the pure beast[s], and from the beast[s] that [is] not pure, and from the flying [birds], and all that tramples on the Earth. 9 Two [by] two [they] came to Noaḥ to the ark – male and female – when Elohim commanded Noaḥ. 10 And [it] was [that] in seven days, and the waters of the Flood were upon the land.[3991]

11 In the year six hundred years to Noaḥ's life – in the second month, on day seventeen of the month (on this day) – all the fountains of the great deep were split [open], and the lattices of the skies were opened. 12 And the rain was on the land forty days and forty nights.[3992]

XX. According to simple arithmetic, adding the ages of the ancestors of Noaḥ up to and including his age of 600 years at the Flood equals 1656. This number is also confirmed in *Seder Olam Rabbah*.[3993] Therefore, the date that the Flood began in the Hebrew year 1656 AM works out to be Monday 22 May 2105 BCE (Julian).[3994]

XX. It rained for forty (40) days and forty (40) nights, thus it rained until the morning of 01 July 2105 BCE (Julian).[3995]

17 And the Flood was forty days upon the land, and the waters [became] very great, and [they]

3989 Translation: Azariah 2022: https://DanielAzariah.com; https://danielazariah.substack.com;
3990 cf. Genesis 7:1-4
3991 cf. Genesis 8:8-10
3992 cf. Genesis 7:11-12
3993 Rabbi Yose ben Halafta, *Seder Olam Rabbah*, Chapter 1
3994 Monday 22 May 2105 BCE (Julian); Monday 04 May 2105 BCE (Gregorian); 17 Ayyaru / Iyar 1656 AM; Astronomical Julian Day JD_X: 952713.5; RD_Y: 604717; RD_X: -768711; Sabbatical Cycle: 4; Metonic Cycle: 3;
3995 Saturday 01 July 2105 BCE (Julian); Saturday 13 June 2105 BCE (Gregorian); 28 Simannu / Sivan 1656 AM; Astronomical Julian Day JD_X: 952753.5; RD_Y: 604757; RD_X: -768671; Sabbatical Cycle: 4; Metonic Cycle: 3;

carried the ark, and [it] rose [up high] from above the land. 18 And the waters overwhelmed [the land], and [became] very great upon the land, and the ark went [floating] upon the face of the waters.[3996]

XX. It would seem that after the forty (40) days and forty (40) nights of rain and flooding, there was another forty (40) days when the Flood was at its full force.

23 And [He] erased the entire existence that [was] upon the face of the Earth (from Adam to beast, to trampling [creature], and to the flying [bird] of the skies) – and [they] were erased from the land, and [there] remained only Noaḥ and [those] that [were] with him in the ark. 24 And the waters overwhelmed over the land one hundred fifty days.[3997]

XX. It would seem that including the forty (40) days and forty (40) nights of rain and flooding, as well as the subsequent forty (40) days of overwhelming the land at full force (i.e. 80 days), the Flood continued to overwhelm the land until a total of 150 days had passed.

1 And Elohim remembered Noaḥ and all the living [creatures], and all the beast[s] that [were] with him in the ark, and Elohim caused a Spirit to pass upon the land, and the waters abated. 2 And [some] fountains of [the] deep and the lattices of the skies were closed [up], and the rain was imprisoned from the skies. 3 And the waters returned from upon the land back and forth, and the waters receded from the end of one hundred fifty days.[3998]

XX. It is evident that after 150 days, that the waters slowly began receding.

4 And the ark rested in the seventh month on day seventeen of the month upon the mountains of Ararat. 5 And the waters were going and receding until the tenth month – on the tenth [month] on [day] one of the month the heads of the mountains were seen.[3999]

XX. Counting from the 1st day of the rains and flooding reaching full force from day (and night) forty (40) through day eighty (80), the ark rested upon the mountains of Ararat exactly 147 days later on Monday 16 October 2105 BCE (Julian).[4000] The waters began receding on day 150, and the heads of the mountains were seen on Wednesday 27 December 2105 BCE (Julian),[4001] i.e. 72 days after coming to rest on the mountains of Ararat.

3996 cf. Genesis 7:17-18
3997 cf. Genesis 7:23-24
3998 cf. Genesis 8:1-3
3999 cf. Genesis 8:4-5
4000 Monday 16 October 2105 BCE (Julian); Monday 28 September 2105 BCE (Gregorian); 17 Tashritu / Tishri 1657 AM; Astronomical Julian Day JD_X: 952860.5; RD_Y: 604864; RD_X: -768564; Sabbatical Cycle: 5; Metonic Cycle: 4;
4001 Wednesday 27 December 2105 BCE (Julian); Wednesday 09 December 2105 BCE (Gregorian); 01 Tebetu / Tevet 1657 AM; Astronomical Julian Day JD_X: 952932.5; RD_Y: 604936; RD_X: -768492; Sabbatical Cycle: 5; Metonic Cycle: 4;

6 And [it] was from the end of forty days, and Noaḥ opened the window of the ark that [he] made. 7 And [he] sent the raven, and [it most assuredly] left and [then] returned until the drying of the waters from upon the land.[4002]

XX. After the ark had come to rest, it is not clear if Noaḥ waited forty (40) days from that date, or from the day that the heads of the mountains were seen. If counting from the day that the ark came to rest, then at the end of waiting forty (40) days, Noaḥ opened the window of the ark on Saturday 25 November 2105 BCE (Julian),[4003] and sent the raven.

XX. If counting from the day that the heads of the mountains were seen, then at the end of waiting forty (40) days, Noaḥ opened the window of the ark on Monday 05 February 2104 BCE (Julian),[4004] and sent the raven.

XX. It is not clear how long the raven flew, or when it returned.

8 And [he] sent the pigeon from [being] with him to see if the waters [had] eased from upon the face of the Earth. 9 And the pigeon did not find a resting-place for the sole of her foot, and [she] returned to him – to the ark – because waters [were] upon the face of all of the land, and [he] sent [forth] his hand, and [he] took her, and [he] brought her to him[self] to the ark.[4005]

XX. It is not clear when Noaḥ sent the pigeon, how long it flew, or when it returned.

10 And [he] yearned again [for dry land] another seven days, and again [he] sent the pigeon from the ark. 11 And the pigeon came to him at the time of evening, and behold: an olive[-tree] leaf [that she had] devoured in her mouth, and Noaḥ knew that the waters [had] eased from upon the land.[4006]

XX. It is not clear when Noaḥ sent the pigeon the second time.

12 And [he] yearned again [for dry land] another seven days, and [he] sent the pigeon and [she] did not repeat to return to him again.[4007]

4002 cf. Genesis 8:6-7

4003 Saturday 25 November 2105 BCE (Julian); Saturday 07 November 2105 BCE (Gregorian); 27 Arahsamna / Marheshvan 1657 AM; Astronomical Julian Day JD_X: 952900.5; RD_Y: 604904; RD_X: -768524; Sabbatical Cycle: 5; Metonic Cycle: 4;

4004 Monday 05 February 2104 BCE (Julian); Monday 18 January 2104 BCE (Gregorian); 12 Shebatu / Shevat 1657 AM; Astronomical Julian Day JD_X: 952972.5; RD_Y: 604976; RD_X: -768452; Sabbatical Cycle: 5; Metonic Cycle: 4;

4005 cf. Genesis 8:8-9

4006 cf. Genesis 8:10-11

4007 cf. Genesis 8:12

13 And [it] was in year six hundred one in the first [month] on [day] one of the month the waters [had begun to] dry from upon the land, and Noaḥ removed the cover of the ark, and [he] saw, and behold: the face of the Earth [had begun to] dry. 14 And in the second month on day twenty seven of the month the land [had] dried [enough to leave the ark].[4008]

XX. In the year 1657 AM, the waters had begun to dry from upon the land, and Noaḥ removed the cover of the ark on Sunday 25 March 2104 BCE (Julian).[4009] The land had dried enough to leave the ark on Sunday 20 May 2104 BCE (Julian).[4010]

XX. Therefore, it was 363 days from the time that the rains and the Flood began to when the land had dried enough to leave the ark.[4011] How this date of the Flood (2105 – 2104 BCE (Julian)) reconciles with ancient Mesopotamian and Egyptian chronologies will be left for researchers in the fields of Assyriology and Egyptology to discuss and debate.

78.5.11. Synchronizing Kings of Babylon, Persia, and Greece

XX. Moving forward in time, the mathematical Hebrew luni-solar calendar demonstrates precise synchronization of the biblical chronology with both the Neo-Babylonian and Persian Achaemenid kings, and thus reveals and confirms the exact dates of the kings during that period including (but not limited to) Darius I (521 – 486 BCE (Julian)) in the Books of Daniel, Haggai, and Zechariah; Xerxes I (485 – 465 BCE (Julian)) in the Book of Esther; and Artaxerxes I (464 – 424 BCE (Julian)) in the Books of Ezra and Nehemiah. Furthermore, this calendar also confirms the date for the death of Alexander the Great in 323 BCE (Julian).

XX. The mathematics (and a correct, non-chronological reading of the Book of Daniel) reveal that Darius the Mede is none other than Darius I (521 – 486 BCE (Julian)).

XX. The mathematics also reveal the scribal error of Nehemiah 2:1 that should be read as the 21st Regnal Year of Artaxerxes I (444 – 443 BCE (Julian)) and not the 20th Regnal Year (445 – 444 BCE (Julian)).

4008 cf. Genesis 8:13-14

4009 Sunday 25 March 2104 BCE (Julian); Sunday 07 March 2104 BCE (Gregorian); 01 Nisannu / Nisan 1657 AM; Astronomical Julian Day JD_X: 953020.5; RD_Y: 605024; RD_X: -768404; Sabbatical Cycle: 5; Metonic Cycle: 4;

4010 Sunday 20 May 2104 BCE (Julian); Sunday 02 May 2104 BCE (Gregorian); 27 Ayyaru / Iyar 1657 AM; Astronomical Julian Day JD_X: 953076.5; RD_Y: 605080; RD_X: -768348; Sabbatical Cycle: 5; Metonic Cycle: 4;

4011 A friend of the author (who wishes to remain anonymous) is a researcher of Gematria in the Torah. He has the following discovery and insight to share about this number 363: This number 363 is the Gematria numerical value of the following Hebrew term: The Messiah (המשיח) HaMashiach.

78.5.12. The Prophecy of Seventy (70) Weeks in Daniel 9:24-27

XX. Having established precise dates for the reign of Artaxerxes I (464 – 424 BCE (Julian)), the mathematics of the Hebrew calendar makes it possible to correctly evaluate and analyze Sir Robert Anderson's interpretation of the messianic prophecy of seventy (70) weeks in the Book of Daniel 9:24-27. Consequently, we can confirm that Sir Robert Anderson was mistaken in his proposed dates and calculations, but his methodology has some merit when the text and actual dates are understood correctly.[4012]

XX. Newman attempted to apply a different methodology of counting literally sixty-nine (69) cycles of seven (7) years, but he made some mistakes with his proposed dates and assumptions of when to begin and end the count. However, when applied correctly – counted from the 21st Regnal Year of Artaxerxes I (444 – 443 BCE (Julian)) – Newman's methodology of counting sixty-nine (69) cycles of seven (7) years does indeed work out precisely to the exact year of Jesus' crucifixion in 33 CE (Julian).[4013]

XX. Furthermore, with an alternate, literal, cryptic (yet nonetheless correct) reading of the Hebrew Masoretic text of Daniel 9:24-27, applying Anderson's methodology to correctly count 471 (i.e. not 483) individual years from Nisannu / Nisan of the 21st Regnal Year of Artaxerxes I (i.e. 444 BCE (Julian)) calculates precisely to arrive at the year of Jesus' baptism by John the Baptist in 28 CE (Julian), i.e. "to anoint the Holiness of Holinesses".[4014] This same prophecy precisely counts the 62 months from that Nisannu / Nisan 28 CE (Julian) to Jesus' crucifixion in Nisannu / Nisan 33 CE (Julian).[4015] [4016]

XX. Even if Daniel was written in the Maccabean period circa 165 BCE (Julian) – as critics have always claimed since the time of Porphyry – how is it possible to explain the precise mathematical accuracy of the messianic prophecy of seventy (70) weeks of Daniel 9:24-27 that predicts the exact year of Jesus Christ's crucifixion by not only one (1) methodology (Newman 1973), but also a 2nd methodology (Azaryah 2026) that both work out exactly with perfect mathematical precision?

78.5.13. The Last (Crucifixion) Week of Jesus

XX. Applying logic and reasoning to evaluating the chronology of the last (crucifixion) week of Jesus, the mathematics reveal that the Last Supper of Jesus happened on day (i.e. beginning at night) Sunday / Monday 10 Nisannu / Nisan as foreshadowed in the Torah with the commandment to take the passover

4012 cf. Appendix Table X.192
4013 cf. Appendix Table X.193
4014 cf. Daniel 9:24
4015 cf. Daniel 9:25-26
4016 cf. Appendix Table X.192

lamb on 10 Nisannu / Nisan and guard him until (through) day 14 Nisannu / Nisan.[4017]

XX. Jesus was arrested that night of 10 Nisannu / Nisan 33 CE (Julian), and kept in custody the entire week, tried several times, tortured all week, and never released after his arrest. He was crucified on Friday 14 Nisannu / Nisan 33 CE (Julian).

78.6. Evaluating the Astronomical Anchors of Ancient Chronology

XX. The chronological framework underlying the study of the ancient Near East rests on a small number of eclipse-based anchor points, drawn from Babylonian and Assyrian astronomical records. Modern scholars date these records by calculating lunar conjunctions and oppositions, combined with crescent-visibility models such as Arcus Visionis. A systematic re-examination of five (5) of the most significant of these anchor points against the proleptic Hebrew calendar – applied to the same primary cuneiform and Ptolemaic sources – confirms the conventional dates.

XX. This thereby demonstrates that the mathematically precise proleptic Hebrew calendar can serve as a fixed reference standard (i.e. a grid) against which the ancient Babylonian calendar can be measured year by year. Since both are essentially the same Standard Mesopotamian Calendar, the differences between them are not errors. It is precisely these differences that reveal how (and when) the ancient Babylonians intercalated, and how (and when) they began their months back through ancient history.

XX. The modern calculations against which these ancient records have been measured rest, in turn, on a specific lineage of astronomical theory: Newcomb's tables of the Sun (1895), Brown's lunar theory (1919) and its later refinements, and – in more recent scholarship – numerical ephemerides such as those of the Jet Propulsion Laboratory (the DE series) and ELP-2000/82. These theories are calibrated and validated against modern and recent-historical observations, and their extrapolation into antiquity is standard practice.

XX. These methods are extrapolated into antiquity. They rely on the conjunction calculation combined with a geometrical crescent-visibility model (Arcus Visionis or an equivalent threshold-based method) to establish the beginning of each Babylonian month. The proleptic Hebrew calendar offers an independent, purely arithmetical check on these results. Where the two (2) agree, the conventional date is confirmed. Where they differ, the difference shows the state of the Babylonian calendar in that year.

XX. The earliest of these, the solar eclipse of the eponym year of Būr-saggilê (previously dated to 14 / 15 / 16 June 763 BCE (Julian)), illustrates a previously unresolved scholarly ambiguity that the proleptic Hebrew calendar resolves rather than an outright error: both 14 June 763 BCE (Julian) and 16 June 763 BCE (Julian) can be excluded as in the proleptic Hebrew calendar the former corresponds to

4017 cf. Exodus 12:3-6

day 29 of the 3rd month, Simannu / Sivan, and the latter corresponds to day one (1) of the following 4th month, Du'uzu / Tammuz.

XX. Both of these dates are impossible because a solar eclipse can only fall at the conjunction, which here falls on the last day of the Hebrew month, i.e. either day 29 or day 30 (depending upon whether it is a 29-day or a 30-day month). In this case, the last day is day 30 of the 3rd month (15 June 763 BCE (Julian)). The proleptic Hebrew calendar therefore places this eclipse in the same month as the Assyrian record, Simannu / Sivan, showing that the two (2) calendars were in step in 763 BCE (Julian).

XX. This eclipse is dated to the eponym year of Būr-saggilê, which in turn has been conventionally dated to the 9th or 10th Regnal Year of Ashur-dan III. If this date of 763 BCE (Julian) is correct – and if the Assyrian record is accurate, precisely certain, and error free between the years 910 – 649 BCE (Julian) – then we can definitively date this solar eclipse to the 10th Regnal Year of Ashur-dan III.[4018]

XX. The lunar eclipse in the Year 0 Accession Year of Nabonassar, conventionally dated to 06 February 747 BCE (Julian) – by Beaulieu and Britton (1999), Huber and De Meis (2004), and confirmed independently by Espenak and Meeus (2009) in the NASA F*ive Millennium Canon of Lunar Eclipses* – converts to day 16 of the 11th month, Shabatu / Shvat 3014 AM in the proleptic Hebrew calendar. In comparison, the Babylonian record places it on day 14 of the 12th month Addaru / Adar. Both are correct for their own calendar. The proleptic Hebrew month is calculated from the molad, whereas the Babylonian month began with the observed first visible crescent, about two (2) days later.

XX. This indicates that the Babylonian month count in that year ran one (1) month ahead of the proleptic Hebrew calendar, with the Babylonian month beginning about two (2) days later – a divergence in intercalation and month-beginning which merits further investigation.

XX. Likewise, the lunar eclipse dated to the 5th Regnal Year of Nabopolassar – 21 / 22 April 621 BCE (Julian) as recorded independently in Ptolemy's Almagest, and confirmed by Huber and De Meis (2004) as well as by the NASA lunar eclipse canon (2009) – is affirmed by this proposed methodology (of using the proleptic Hebrew lunisolar calendar to make calendrical calculations of ancient dates) as correct. In the proleptic Hebrew calendar it falls on 15 Ayyaru / Iyar 3140 AM, and in Parker and Dubberstein's tables on 14 Ayyaru / Iyar in the same month. The two (2) calendars were in step that year, with the usual one (1)-day offset. This case serves as a validation control for the proposed methodology's reliability.

XX. The lunar eclipse recorded in Astronomical Diary VAT 4956, dated to day 15 of the 3rd month, Simannu / Sivan of the 37th Regnal Year of Nebuchadnezzar II, is conventionally placed on 04 July 568 BCE (Julian). In Parker and Dubberstein's tables 01 Simannu / Sivan fell on 21 June, so 15 Simannu began on the evening of 04 July 568 BCE (Julian), in agreement with the conventional date.

4018 cf. Appendix Table X.194

XX. However, in the proleptic Hebrew calendar 04 July is 13 Tammuz 3193 AM, and 15 Simannu / Sivan falls one (1) month earlier on 06 June 568 BCE (Julian). The difference arises because the Hebrew calendar's leap month falls in 569 BCE (3192 AM). In comparison, tablets attest a Babylonian Addaru 2 / Adar 2 at the end of Nebuchadnezzar's 36th Regnal Year in 568 BCE.[4019] Here the proleptic Hebrew calendar identifies the precise year in which Babylonian intercalation diverged from it.

XX. A further, corroborating pattern emerges from Parker and Dubberstein's foundational lunar tables (1956, 1971) of Babylonian vs. Julian calendar dates. These remain the standard reference for Babylonian chronology, refined only in particular years by later scholarship.[4020] The principal problem involved is to determine the years in which a 13th lunar month – the "intercalary" month – was added to bring the lunar year back into line with the solar year, and which intercalary month was used: Ululu 2 / Elul 2 (month 6b), or Addaru 2 / Adar 2 (month 12b).[4021]

XX. Comparison with the proleptic Hebrew calendar shows that from 626 to 424 BCE (Julian) the two calendars began each year either in the same month or exactly one (1) month apart. In 104 of the 105 years in which they were in step, the Babylonian month began on the same day as the Hebrew or one (1) to two (2) days later, as expected of a month begun by the visible crescent.

XX. The one-(1)-month differences arise from the different years in which the two (2) calendars intercalated. Therefore, these differences are not errors. The proleptic Hebrew calendar makes them visible as a fixed reference standard. It reveals year by year when the ancient Babylonians intercalated and how they began their months.

XX. Admittedly, scholars have long debated when exactly the Babylonians adopted the 19-year Metonic cycle as a standard calculation rather than observation and decree. Their opinions range from the mid-6th century BCE (Julian) to the reign of Darius I (the Great) and later.

XX. Measured against the proleptic Hebrew calendar, the fixed pattern of Babylonian intercalation holds without exception from the 10th Regnal Year of Xerxes I (476 BCE (Julian)). According to Ossendrijver, the cycle became permanent only in the 24th Regnal Year of Artaxerxes II (381 BCE (Julian)).

XX. The Babylonian calendar was the product of centuries – even millennia – of refinement. Its aim was to measure the heavens precisely since the time of the Sumerians who were considered ancient not only by the Neo-Babylonians, but even by the more ancient Old Babylonians.

4019 Parker and Dubberstein, 1971: 5
4020 Walker 1997; Ossendrijver 2018a
4021 Walker 1997: 22

XX. As the ancient Sumerians and Babylonians observed and refined their measurements, counting and calculations, intercalation remained a matter of trial and error. Sometimes they inserted a second Addaru / Adar, sometimes a second Ululu / Elul, and sometimes they missed a month. The ancient record does not always preserve which intercalary month was used in every year.

XX. However, such errors were relatively brief. They were corrected once they were noticed. Each intercalation realigned the calendar with the solar year and restored two (2) or three (3) years of accurate dating. Unlike the Egyptian 365-day calendar that drifted steadily against the heliacal rising of Sirius and the solar year, any Babylonian discrepancy was temporary and never cumulative long-term.

XX. Therefore, we can view the Babylonian / Hebrew lunisolar calendar as it has been perfected today as the ultimate refinement of the precise mathematics that the ancient Sumerians and Old Babylonians attempted to understand. These mathematics define the exact movements of the Moon relative to the Sun and so determine the days of the month and the solar year. This precise Babylonian / Hebrew calendar is the ancient "backbone" by which to date ancient observations of the planets (e.g. Venus) and of certain stars (e.g. Sirius). We can use this calendar not only to calculate modern dates but also to calculate precise dates proleptically into the ancient past.

XX. The death of Alexander the Great presents a further case. The conventionally accepted date, 11 June 323 BCE (Julian), rests on the cuneiform record (BM 45962) which states day 29 of the 2nd month Ayyaru / Iyar, and on the calculated conjunction of 00:06 AM on 11 June 323 BCE (Julian) combined with crescent-visibility reasoning.

XX. By this period, the Babylonian 19-year cycle had already been standardized. However, the Babylonian and Hebrew cycles intercalate in different years. The Babylonian calendar inserted its Addaru 2 / Adar 2 in March 323 BCE (Julian), one (1) year before the Hebrew Addaru 2 / Adar 2 of 3439 AM. Therefore, in that year the Hebrew calendar ran one (1) month ahead, and Babylonian 29 Ayyaru / Iyar is Hebrew 30 Sivan 3438 AM, Monday 11 June 323 BCE (Julian). The proleptic Hebrew calendar thus confirms the conventional date.

XX. These five (5) cases involve independent scholarly treatments: Parker and Dubberstein (1956, 1971), Depuydt (1997), Beaulieu and Britton (1999), Huber and De Meis (2004), and the NASA *Five Millennium Canons of Lunar and Solar Eclipses* by Espenak and Meeus (2009, 2006). These are the standard references for ancient Babylonian dates and for eclipse chronology.

XX. In every case the proleptic Hebrew calendar confirms their dates and accounts for each apparent difference. Therefore, the proleptic Hebrew calendar can serve as an independent check on ancient dates across multiple academic disciplines.

XX. Measured against the proleptic Hebrew calendar, the ancient records and the modern eclipse

canons and lunar ephemerides – Newcomb's, Brown's, and the JPL / ELP-based systems – are in full agreement. Therefore, the proleptic Hebrew calendar stands as the fixed mathematical standard – the grid – against which ancient dates can be measured to the very day. It gives direct insight into the Babylonian and Assyrian calendars and, by consequence, into their chronologies and the Egyptian chronologies that interlink with them.

XX. Used in conjunction with the modern eclipse canons and lunar ephemerides, the proleptic Hebrew calendar can fix the precise date of each ancient observation. In turn, these precisely dated observations can refine our knowledge of the Earth's rotation and of ΔT in the distant, ancient past.

XX. Indeed, Stephenson (1997) details certain observational requirements for determining ΔT in the pre-telescopic period. Any early astronomical observation which is of value in studying changes in the Length of Day (LOD) in the past must satisfy a wide variety of criteria. These may be listed as follows:[4022]

1.) The observation must involve at least one (1) of the brighter and more rapidly moving objects in the solar system (i.e. the Moon, Sun, or one of the inner planets Mercury, Venus, and Mars). [The Moon is the central time "reckoner" of the Babylonian / Hebrew lunisolar calendar (i.e. a repeatable phenomenon whose motion or change of state is observable and obeys a definite law).[4023] It provides a yearly, monthly, and daily time-indicator.]

2.) The exact Julian or Gregorian date of the observation must either be specified directly or be able to be determined unambiguously. [The proleptic Hebrew lunisolar calendar provides exact proleptic Julian and Gregorian dates for the ancient past. It also measures the offset of each recorded Babylonian date year by year.]

3.) A reasonably accurate value of the Universal Time (UT) must be reducible from the reported circumstances, or – as in the case of total solar eclipses – the rotational phase of the Earth must be able to be derived. [The proleptic Hebrew lunisolar calendar makes possible just this sort of reasonably accurate value of UT to be measured because it not only provides the time reckoner, but also provides a time reference (i.e. with respect to which [e.g. coordinate position on the Earth] the position (or state) of the time reckoner [e.g. the Sun or the Moon] can be determined)[4024] – each day, each month, and each year resets its synchronicity with the actual rotation of the Earth, i.e. Universal Time (UT)]

4.) The corresponding Terrestrial Time (TT) must be able to be precisely calculated from the dynamical equations of motion of the celestial body or bodies concerned. [The proleptic Hebrew lunisolar calendar allows the precise calculation of TT from the dynamical equations of motion of the

4022 Stephenson 1997: 43-44
4023 Nelson et al. 2001: 509-510
4024 Nelson et al. 2001: 509-510

Moon by fixing the correct day of each observation.]

5.) In most cases the geographical position of the observer must be known with some precision – preferably to within a small fraction of a degree in both latitude and longitude. [The Babylonian / Hebrew lunisolar calendar as recorded from the perspective of Babylon, Assyria, or Jerusalem provides just this necessary certainty since the coordinates of these places are known precisely.]

XX. Since careful measurements of the co-ordinates of the Sun, Moon, and planets are fairly rare throughout the pre-telescopic period, studies of the Earth's past rotation have mainly concentrated on the observation of specific events, such as conjunctions between two (2) celestial bodies. Numerous early records of both solar and lunar eclipses satisfy all of the above conditions and form an impressive body of data. Other types of observation are of much more limited utility.[4025]

XX. Babylonian astronomers systematically observed eclipses along with many other celestial phenomena. Only about ten per cent of the original material is extant, and nearly all surviving tablets are badly damaged. Nevertheless, nearly 150 important observations of eclipses – mainly of the Moon – are available.[4026]

XX. From the point of view of Earth rotation studies, solar and lunar eclipses have a number of mutually distinct features. In the case of a solar obscuration, a shadow is cast by the Moon on the rotating Earth. As a result, the visibility of the various phases at any particular site depends very much on the value of the accumulated clock error ΔT at that date.[4027]

XX. A lunar eclipse occurs when the Moon enters the terrestrial shadow so that only the local time of each individual stage is a function of ΔT; parameters such as magnitude and duration are not usually affected by Earth's rotation. Nevertheless, early solar and lunar observations are sufficiently diverse as to complement one another in the investigation of long-term changes in the length of the day (LOD).[4028]

XX. However, not only do we have recorded dates for solar and lunar eclipses, but in the Assyrian and Babylonian records, we also have recorded dates for daily motions and observed positions of the Moon (often with specific timings of e.g. Moonrise and Moonset, i.e. the Lunar Six) as well as other celestial bodies, i.e. the planets. Therefore, we have even more ancient data with which to determine more precise ancient values for ΔT.

XX. These findings are presented here as testable, falsifiable claims, offered for independent peer review and replication. The underlying algorithms and data are available for that purpose. Further

4025 Stephenson 1997: 44
4026 Stephenson 1997: 59
4027 Stephenson 1997: 48
4028 Stephenson 1997: 48

research will be required to confirm, refine, or disprove them.

XX. In particular, it will be required to extend the year-by-year comparison of the Babylonian and proleptic Hebrew calendars before 626 BCE (Julian) and after 424 BCE (Julian). It will also be required to determine those Babylonian intercalations that no surviving record preserves such as whether an intercalary Addaru 2 / Adar 2 followed the lunar eclipse of 747 BCE (Julian).

XX. Stephenson (1997) asserts that Parker and Dubberstein's tables (1956) have proved very valuable in his investigation of historical eclipses, the Earth's rotation, ΔT, and the secular lunar orbital tidal friction acceleration of the Moon (ṅ). However, he is incorrect when he states that these tables are highly reliable, and that errors seldom exceed a single day.[4029]

XX. First, where Parker and Dubberstein reconstructed an intercalation that later evidence has ruled out – as in the 29th Regnal Year of Darius I and the 7th Regnal Year of Xerxes I[4030] – their Julian dates are wrong by an entire lunar month for every month affected.

XX. Second, their day-by-day dates rest on computed crescent visibility (Arcus Visionis) rather than on observation, and Parker and Dubberstein themselves state that their dates may be wrong by one (1) day.[4031]

XX. Measured against the fixed proleptic Hebrew calendar, these errors can be traced year by year. Supplementary data is provided as CSV files to accompany this work in order to allow scientific, independent, 3rd party confirmation.[4032]

XX. The conventional dates for the lunar eclipses of 747 BCE (Julian) and 568 BCE (Julian) correspond exactly to entries in the standard modern eclipse canons. The discrepancies identified here between the observed Babylonian calendar and the proleptic Hebrew calendar indicate how far the scribes' recorded month labels had drifted, as measured against the proleptic Babylonian / Hebrew calendar, and where later intercalation would have been required to restore alignment.

XX. These cases show where the Babylonian calendar diverged from the proleptic Hebrew calendar, which can be used to identify the years in which Babylonian intercalation differed.

XX. Where a recorded Babylonian date has been shown to differ from its proleptic Babylonian / Hebrew-calendar equivalent – as demonstrated for the eclipses of 747 BCE and 568 BCE – this difference reflects the state of the Babylonian calendar in those years. Therefore, any ΔT reconstruction

4029 Stephenson 1997: 116
4030 Walker 1997; Ossendrijver 2018a
4031 Parker and Dubberstein 1971: 25
4032 https://github.com/DanielAzaryah

relying on such records must rest on the astronomically fixed date of the event as well as its proleptic Hebrew date rather than on a conversion of the recorded calendar date.

XX. The eclipse of 747 BCE (Julian) illustrates the difference: a direct conversion reads the recorded Babylonian date, 14 Addaru / Adar, as if it were the Hebrew 14 Addaru / Adar 3014 AM. This yields 06 March 747 BCE (Julian), a day on which no eclipse occurred.

XX. However, the fixed proleptic Hebrew date of the eclipse is 16 Shevat 3014 AM, i.e. 06 February 747 BCE (Julian), a difference of one (1) month less two (2) days. The month reflects the Babylonian intercalation offset, and the two (2) days reflect the Babylonian crescent offset.

XX. Only the proleptic Hebrew calendar makes this difference visible. It applies mathematical precision to the ancient past, whereas the Babylonian calendar rested on human observation and decree. Therefore the proleptic Hebrew calendar is necessary to measure the Julian, Babylonian, and Hebrew dates against one another, thus ensuring that ΔT will be derived from 06 February 747 BCE (Julian), and not from the converted date of 06 March 747 BCE (Julian).

XX. Furthermore, this proposed new methodology of using the modern (ancient) Babylonian / Hebrew calendar to proleptically calculate ancient dates can be applied to all ancient dates in order to synchronize and recalibrate precise ancient chronologies of the entire ancient Near East.

78.7. ΔT and the Assumed Constancy of ṅ: A Proposed Methodology

XX. The precise dates established in the preceding section bear directly on a second, methodologically distinct problem in the study of ancient astronomy: the determination of (ΔT) which is the accumulated difference between a uniform dynamical time-scale (Terrestrial Time (TT)) and the actual, irregular, observed rotation of the Earth (Universal Time (UT)). (ΔT) is conventionally derived from ancient eclipse records by a method that itself depends on an unverified assumption, one which the present findings help to illuminate.

XX. In order to determine the value of ΔT from an eclipse observation, it is necessary to be able to calculate accurately the positions of the Sun and Moon at any selected epoch. By definition, the Sun has negligible acceleration on Terrestrial Time (TT). However, the longitude of the Moon contains an appreciable quadratic term – part of which (owing to the reciprocal action of the tides) can only be determined empirically.[4033]

XX. Since many of the eclipses recorded in history are remote from the present-day, the effect of the lunar accelerative term on their calculated visibility is substantial. Furthermore, knowledge of the tidal

4033 Stephenson 1997: 33

component of this acceleration (usually denoted by ($\dot{n}$)), leads directly to a determination of the effect of the tides on the Earth's spin. Consequently, it is important to investigate both the numerical value of ($\dot{n}$) and its constancy during the historical period.[4034]

XX. The standard method proceeds by fixing a value for ($\dot{n}$) in advance – which is the secular (long-term) rate of change of the Moon's mean motion, driven principally by tidal friction between the Earth and Moon. Stephenson (1997) adopts a value of ($\dot{n}$) ≈ -26.0 arcsec /century², derived exclusively from modern data: transits of Mercury, satellite geodesy, and lunar laser ranging. A later, independently derived value, ($\dot{n}$) ≈ -25.858 arcsec /century² (Chapront, Chapront-Touzé and Francou, 2002), likewise rests on lunar laser ranging.

XX. In neither case does any ancient eclipse or occultation record contribute to the determination of ($\dot{n}$). Having fixed ($\dot{n}$), the lunar ephemeris is then used to compute the expected circumstances of an ancient eclipse, and (ΔT) – not ($\dot{n}$) – is the sole parameter adjusted until the computed circumstances match the historical record.

XX. It follows from this acknowledged correlation that any error in the assumed value of ($\dot{n}$) is not merely left unaddressed by the standard method, but is absorbed into the resulting (ΔT) values, thus indistinguishable from genuine variation in the Earth's rotation; this inference is drawn from Stephenson's stated correlation between the two (2) parameters of ($\dot{n}$) and the Earth's rotational deceleration ($\dot{\omega} / \omega$), rather than being a claim made explicitly in those terms in the source itself.

XX. This procedure is not merely a simplifying convenience; it reflects a genuine mathematical limitation of the underlying data, acknowledged directly by Stephenson.[4035] Commencing in 1970, several investigators attempted to utilize ancient eclipse observations to solve simultaneously for both ($\dot{n}$) and the mean acceleration of the Earth's spin, ($\dot{\omega} / \omega$).[4036] The most comprehensive study of the medieval measurements is by Newton (1972a), who expressed his results in terms of Greek lowercase letter Omega (ω), the rate of spin of the Earth.[4037]

XX. In the analysis of ancient eclipses, ($\dot{n}$) and the Earth's rotational deceleration ($\dot{\omega} / \omega$) are highly correlated: only a linear combination of the two (2) quantities can be determined from eclipse observations alone. Earlier attempts by other investigators – Newton (1970, 1972b), Muller and Stephenson (1975), and Muller (1976) – to solve for both parameters simultaneously (with results for ($\dot{n}$) ranging from -42 to -30 arcsec /century²) proved to be of limited usefulness for this reason.[4038]

XX. All eclipse computations in Stephenson's book have been made using specially designed programs

4034 Stephenson 1997: 33
4035 Stephenson 1997: 34
4036 Stephenson 1997: 34
4037 Stephenson 1997: 45
4038 Stephenson 1997: 34

– which are adaptations of standard ephemerides for the Sun and Moon. In computing solar positions, Stephenson has used Newcomb's analytical ephemeris (Newcomb 1895a). All Fourier terms of amplitude greater than 0.025 arcsec in longitude and 0.005 arcsec in latitude have been included. For lunar positions a modification of the analytical ephemeris (j == 2) (IAU 1968), in which ((ṅ) == -26 arcsec /cy^2), has been adopted. In this case, all Fourier terms in longitude greater than 0.05 arcsec, and in latitude greater than 0.01 arcsec have been included.[4039]

XX. In recent years, investigators of early eclipse observations have assumed an independent value for (ṅ) in order to solve exclusively for ΔT. Fortunately for Earth rotation studies, several precise determinations of (ṅ) using a variety of other means have been made in recent years. Techniques include the following: transits of the planet Mercury across the solar disk, artificial satellite data, and lunar laser ranging.[4040]

XX. In general, the average depth of water on the continental shelves is about 100 m. A change in the global sea-level by 3 per cent (i.e. by 3 m) would probably alter the area of the shelves by a similar ratio.[4041] No model is available which can predict just how tidal dissipation would respond to variations in both area and water depth on the continental margins.[4042]

XX. However, according to Stephenson, the very small magnitude of these variations suggests that the effect on global dissipation will be negligible. Stephenson concludes that the assumption of a sensibly constant value for (ṅ) over the historical period seems justified.[4043]

XX. However, the degree of constancy of (ṅ) the [secular] lunar [orbital] tidal [friction] acceleration during the interval of approximately 2700 years covered by reliable historical observations is clearly a matter of some concern as it may not have been constant, and indeed may have varied at times [– and it is even more uncertain for the period before these observations, especially regarding the global Flood of Noah (2105 BCE (Julian)) that would have affected global sea level and the tides, so that the modern assumed value of (ṅ) may not apply at all]. This acceleration is directly proportional to the lunar tidal torque and hence to the rate of dissipation of the Earth's rotational energy.[4044]

XX. Analysis of ancient and medieval eclipse observations leads to a result for the total rate of change in the LOD owing to both tides and other causes. Hence any residual non-tidal lengthening of the day (for example produced by post-glacial uplift, sea-level variations, etc.) may be obtained by subtracting the tidal component which is produced by the action of both the Moon and Sun.[4045]

4039 Stephenson 1997: 40
4040 Stephenson 1997: 34
4041 Stephenson 1997: 36
4042 Stephenson 1997: 36-37
4043 Stephenson 1997: 37
4044 Stephenson 1997: 36
4045 Stephenson 1997: 37

XX. Despite their relatively low precision, pre-telescopic observations cover a sufficient time-span for long-term trends in the length of day (LOD) to become apparent. These trends cannot be discerned from modern measurements. Here we have the main reason why archaic observations are so important in the study of the Earth's past rotation.[4046]

XX. Among ancient and medieval civilizations, only the Babylonians and the inhabitants of East Asia (Chinese, Koreans and Japanese) recorded occultations on a regular basis. It is indeed fortunate that a promising array of both timed and untimed observations of eclipses is available. Reliable records of these events commence around 700 BCE (Julian) and continue throughout most of the pre-telescopic period.[4047]

XX. Eclipses of both Sun and Moon are frequently noted in historical writings, but in such works – according to Stephenson – only solar obscurations are of much value in the study of changes in the LOD.[4048] According to Stephenson, only a small proportion of these early data proves to be of value in the study of the Earth's past rotation. Most useful observations, notably timings of individual phases [of an eclipse] are the work of astronomers. Nevertheless, many useful accounts of eclipses – especially total obscurations of the Sun – were written by chroniclers and others who had no special interest in astronomy.[4049]

XX. The analysis of ancient and medieval eclipse records is just one of several techniques which have been utilised in recent years to investigate variations in the LOD over the historical past.[4050]

XX. The Babylonian Astronomical Diaries also record non-eclipse lunar observations – the Moon's position, occultations of Normal Stars, positions of the planets (e.g. Venus and Mars) relative to the Normal Stars, and the Lunar Six intervals – which may likewise be applicable.

XX. The precise dates established in Section 78.6 do not resolve the specific mathematical degeneracy between the constant-rate secular contribution of (ṅ) and the constant-rate secular contribution of Earth's non-tidal rotational change: because both produce an identical functional form in the data, no quantity of additional eclipse-type observations, however accurately dated, can by itself divide that combined secular trend into its two components.

XX. This differs from – and does not preclude – the detection of non-constant or periodic departures from that trend such as the roughly 1,500-year undulation previously identified in the literature which more accurate and more extensive data may indeed help to resolve. What the present findings do

4046 Stephenson 1997: 43
4047 Stephenson 1997: 47
4048 Stephenson 1997: 48
4049 Stephenson 1997: 48
4050 Stephenson 1997: 43

provide is a more reliable and precise placement in time of the observations underlying any such reconstruction.

XX. Where a recorded Babylonian date has been synchronized precisely with the proleptic Hebrew calendar – showing whether it falls on the exact same day, or stands one (1) day, two (2) days, or one (1) month apart, as demonstrated for the eclipses of 747 BCE and 568 BCE (Julian) and for the death of Alexander the Great in 323 BCE (Julian) – the Babylonians' own local time and location measurements (e.g. the Lunar Six) can be anchored to the correct day.

XX. The control case of the lunar eclipse of 621 BCE (Julian), recorded in Ptolemy's Almagest, confirms the precision of this method to the very day. This is independent of (and prior to) any question of (ṅ). Therefore, synchronizing such dates is a necessary component for a more accurate reconstruction of (ΔT) across the historical record.

XX. However, this improved (ΔT) still does not by itself allow (ṅ) to be separated out from the Earth's non-tidal rotational change within it. According to Stephenson, the two (2) quantities are highly correlated, and only a linear combination of them can be determined from eclipse observations alone.[4051]

XX. It should be noted that the differences identified in Section 78.6 and the (ṅ) / (ΔT) circularity described above may involve discrepancies of substantially different orders of magnitude – a question that merits further quantitative investigation rather than assumption. The differences identified in Section 78.6 originate in the calendars themselves – i.e. in Babylonian intercalation and crescent-based month-beginning – rather than in the astrophysical quantities (ṅ) or (ΔT).

XX. The extent to which these differences have affected any published values of (ΔT) for the ancient past – and the assumed constancy of (ṅ) on which those values rest – remains to be established rather than presumed. Parker and Dubberstein's conversions, and every conversion of a Babylonian date, depend on the variables of crescent visibility: Arcus Visionis, location, elevation, and weather. By contrast, the proleptic Hebrew calendar is purely arithmetic. It counts days – one rotation of the Earth each – free of all these variables. Therefore, it provides the fixed standard against which each Babylonian date can be measured.

XX. A further, narrower point concerns the Babylonian Lunar Six measurements – precise time-interval observations of the Moon near full and new moon, recorded in time-degrees (UŠ) probably via water clock. Given accurate dating of the day in question, these measurements permit a direct calculation of the Moon's motion for that specific ancient day. This is offered here only as a narrow, specific observation about what accurately-dated Lunar Six data can establish for a single period.

4051 Stephenson 1997: 34

XX. Additional assumptions must be made for retro-calculation (aside from the recent advancements in the measurement of planetary motion) because deceleration of the Earth's rate of rotation, which is dependent on tidal friction and numerous non-tidal mechanisms (changes in the moment of inertia of the earth, electromagnetic coupling between core and mantle, solar-induced atmospheric pressure variations), occurs at an irregular rate. These mechanics have an impact on the motion of the moon and consequently for calculation of past eclipses.[4052]

XX. It must be said though that the choice of solar eclipses for such early [ancient] times is a delicate procedure. We only know the year in which it is supposed to have happened, no month, season or time of day. For identification, additional assumptions have to be made which reflect evaluation of the changes in the Earth's rotation [i.e. ΔT]. Depending on such assumptions, eclipses which appear plausible to one scholar may seem unlikely to another.[4053]

XX. The proposed methodology therefore proceeds in three (3) steps:

1.) It gathers the ancient observations recorded in the Babylonian Astronomical Diaries: eclipses, the Lunar Six, occultations of Normal Stars by the Moon, and planetary observations;

2.) It fixes the exact day of each record by the proleptic Hebrew calendar. This synchronizes the records with one another and with the Julian calendar. It also shows whether each Babylonian date falls on the same day as the Hebrew, or stands one (1) day, two (2) days, or one (1) month apart;

3.) It applies the Babylonians' own timed measurements to that exact day. In this way the ancient record – dated precisely by the proleptic Hebrew calendar – becomes the principal means of measuring the Earth's rotation in antiquity;

XX. In summary, the present findings support three (3) related but distinct conclusions:

1.) The proleptic Hebrew calendar confirms the key eclipse-based anchor points in ancient Near Eastern history;

2.) The proleptic Hebrew calendar measures year by year the state of the Babylonian calendar, with consequences most importantly for the historical ancient chronologies built upon these anchor points – specifically the ancient chronologies of Assyria, Babylonia, Egypt, and the broader ancient Near East;

3.) The standard method for deriving (ΔT) from ancient eclipse records rests on an *a priori* assumption regarding (ṅ) that cannot be tested by the very data used to derive (ΔT) – a circularity

4052 Pruzsinszky 2009: 73; Footnote #277
4053 Huber 2000: 57 – cited by Hunger 2009: 152

acknowledged, though not resolved, in the foundational literature itself;

XX. These conclusions are offered here as testable claims, for confirmation, refinement, or refutation through independent scholarly and scientific peer review.

78.8. The Babylonian Intercalations Measured Against the Proleptic Hebrew Calendar

XX. Parker and Dubberstein's tables were compared year by year against the proleptic Hebrew calendar as a fixed reference standard. The comparison covers the 203 years from the Year 0 Accession Year of Nabopolassar to the 41st Regnal Year of Artaxerxes I (626 – 424 BCE (Julian)).[4054] [4055] In every one of those years the two (2) calendars began 01 Nisannu / Nisan either in the same month or exactly one (1) month apart, and in no year did the difference exceed one (1) month.

XX. In 104 of the 105 years in which the two (2) calendars began in the same month, the Babylonian 01 Nisannu / Nisan fell on the same day as the proleptic Hebrew 01 Nisannu / Nisan or one (1) to two (2) days later. Both calendars originally began the month with the first visible crescent after conjunction, in Babylon and in Jerusalem respectively.[4056] [4057] However, the proleptic Hebrew calendar used here is computed from the molad, the mean conjunction, which precedes first visibility by about one (1) to two (2) days.[4058]

XX. Parker and Dubberstein's dates are themselves computed and, by their own admission, may be wrong by one (1) day.[4059] Since Jerusalem (31.78° N, 35.23° E) lies about nine (9) degrees of longitude west of Babylon (32.54° N, 44.42° E),[4060] the crescent could also be seen there on the same evening or one (1) evening earlier.

XX. In 97 years the proleptic Hebrew calendar began the year one (1) month ahead of the Babylonian calendar, and in one (1) year it began one (1) month behind. The relation between the two (2) calendars changes only when one (1) of them intercalates a month and the other does not. When the Babylonians intercalated a month that the proleptic Hebrew calendar did not, the proleptic Hebrew calendar became one (1) month ahead; when the proleptic Hebrew calendar then intercalated its Addaru 2 / Adar 2, the Babylonian calendar caught up and the two (2) were again in step.

4054 Parker and Dubberstein 1971: 27-33

4055 Ossendrijver (2018a: 147) states that the 9th Regnal Year of Nabopolassar and the 27th Regnal Year of Nebuchadnezzar contained an Addaru 2 / Adar 2, which Parker and Dubberstein's tables do not show; this cannot be verified here.

4056 Brack-Bernsen 2007: 83

4057 cf. Mishnah *Rosh Hashanah* 2:5-7; Maimonides / Rambam, *Mishneh Torah Sefer Z'manim Kiddush HaChodesh*, Chapter 1-2

4058 Maimonides, Sanctification of the New Month 6:2–3

4059 Parker and Dubberstein 1971: 25

4060 UNESCO World Heritage List: no. 148, Old City of Jerusalem and its Walls (https://whc.unesco.org/en/list/148/maps/); no. 278, Babylon (https://whc.unesco.org/en/list/278/maps/)

Table 78.1. How an Intercalation Changes the Relation between the Two (2) Calendars

Event	Result
Neither calendar intercalates	The relation stays the same
The calendars are in step and the Babylonians intercalate	The proleptic Hebrew calendar becomes one (1) month ahead
The proleptic Hebrew calendar is one (1) month ahead and intercalates, and the Babylonians do not	The proleptic Hebrew calendar moves back one (1) month: the two (2) calendars are in step
The proleptic Hebrew calendar is one (1) month ahead and intercalates, and the Babylonians also intercalate about one (1) month later	In step for one (1) month only, then the proleptic Hebrew calendar is again one (1) month ahead
The calendars are in step, the Babylonians intercalate first, and the proleptic Hebrew calendar intercalates later in the same year	One (1) month ahead for part of the year, then in step again
The calendars are in step and only the proleptic Hebrew calendar intercalates	The proleptic Hebrew calendar becomes one (1) month behind. This occurs only in the 19th Regnal Year of Nabopolassar (607 – 606 BCE (Julian)), where Parker and Dubberstein themselves state that "there remains some question as to whether or not this tablet is correctly placed" and that it "results in an exceptionally low date of March 6 for Nisanu 1" (Parker and Dubberstein 1971: 4 n. 9) – see explanation below.

XX. When both calendars intercalated a month in the same year, each added one (1) month, so that the relation at the end of that year was the same as at its beginning. This occurred in eleven (11) years.

XX. In seven (7) of them the two (2) calendars began the year in step. In five (5) of these the Babylonian intercalation, an Ululu 2 / Elul 2, came first and put the proleptic Hebrew calendar one (1) month ahead, and the proleptic Hebrew Addaru 2 / Adar 2 later in the year brought them back into step.[4061]

XX. For example, in the 2nd Regnal Year of Cyrus II the Babylonian Ululu 2 / Elul 2 began 06 September 537 BCE (Julian), and the proleptic Hebrew Addaru 2 / Adar 2 of 3225 AM began 28 February 536 BCE (Julian).[4062] In the other two (2) both calendars began their Addaru 2 / Adar 2 one

4061 The five (5) years with a Babylonian Ululu 2 / Elul 2 are the 5th and 10th Regnal Years of Nabopolassar; the 5th and 41st Regnal Years of Nebuchadnezzar; and the 2nd Regnal Year of Cyrus II. (Parker and Dubberstein 1971: 4-5, 27-29)

4062 Parker and Dubberstein 1971: 29

(1) day apart.[4063] [4064]

Table 78.2. The 2nd Regnal Year of Cyrus II (537 – 536 BCE (Julian))

Date (Julian)	Hebrew 19-year Metonic cycle year	Event	Relation
11 / 12 March 537 BCE	13 (3224 AM)	Year begins (proleptic Hebrew / Babylonian)	In step
06 September 537 BCE	14 (3225 AM)	Babylonian Ululu 2 / Elul 2	Hebrew one (1) month ahead
28 February 536 BCE	14 (3225 AM)	Proleptic Hebrew Addaru 2 / Adar 2 of 3225 AM	In step

XX. In the other four (4) years the proleptic Hebrew calendar began the year already one (1) month ahead: the 11th, 14th, and 33rd Regnal Years of Nebuchadnezzar, as well as the 3rd Regnal Year of Nabonidus. Here the proleptic Hebrew Addaru 2 / Adar 2 came first and brought the two (2) calendars into step, but about one (1) month later the Babylonian Addaru 2 / Adar 2 put the proleptic Hebrew calendar one (1) month ahead again.[4065] For example, in the 11th Regnal Year of Nebuchadnezzar the proleptic Hebrew Addaru 2 / Adar 2 of 3168 AM began 01 March 593 BCE (Julian), and the Babylonian Addaru 2 / Adar 2 began 31 March 593 BCE (Julian).

Table 78.3. Years in which Both Calendars Intercalated an Addaru 2 / Adar 2 while the Proleptic Hebrew Calendar was One (1) Month ahead

Regnal Year	**Proleptic Hebrew Addaru 2 / Adar 2 began**	**Babylonian Addaru 2 / Adar 2 began**
Nebuchadnezzar 11	01 March 593 BCE (Julian)	31 March 593 BCE (Julian)
Nebuchadnezzar 14	26 February 590 BCE (Julian)	28 March 590 BCE (Julian)
Nebuchadnezzar 33	26 February 571 BCE (Julian)	28 March 571 BCE (Julian)
Nabonidus 3	25 February 552 BCE (Julian)	28 March 552 BCE (Julian)

XX. In these four (4) years the proleptic Hebrew intercalation that would normally have closed the difference was canceled out one (1) month later by the Babylonian intercalation, so that the difference continued into the following year. For this reason most periods in which the proleptic Hebrew calendar was one (1) month ahead lasted only one (1) or two (2) years. The three (3) longer periods each include one (1) or two (2) of these four (4) years:

4063 The two (2) years with both Addaru 2 / Adar 2 one (1) day apart are the 2nd and 7th Regnal Years of Nabopolassar (proleptic Hebrew / Babylonian: 02 / 03 March 623 BCE (Julian) and 07 / 08 March 618 BCE (Julian) respectively).
4064 Parker and Dubberstein 1971: 4-5, 27-29
4065 Parker and Dubberstein 1971: 5, 7, 28, 29

1.) the 10th – 16th Regnal Years of Nebuchadnezzar (595 – 588 BCE (Julian)), seven (7) years;

2.) the 32nd – 35th Regnal Years of Nebuchadnezzar (573 – 569 BCE (Julian)), four (4) years;

3.) the 2nd – 5th Regnal Years of Nabonidus (554 – 550 BCE (Julian)), four (4) years;

Table 78.4. Periods in which the Proleptic Hebrew Calendar began the Year One (1) Month ahead

Length	**Number of periods**	
1 year		38
2 years		22
4 years	2: Nebuchadnezzar 32 – 35 (573 – 569 BCE (Julian)); Nabonidus 2 – 5 (554 – 550 BCE (Julian))	
7 years	1: Nebuchadnezzar 10 – 16 (595 – 588 BCE (Julian))	
Total	63 periods covering 97 years	

XX. In only one (1) year was the Babylonian calendar one (1) month ahead of the proleptic Hebrew calendar: the 19th Regnal Year of Nabopolassar (607 – 606 BCE (Julian); Hebrew 19-year Metonic cycle year 1). Parker and Dubberstein themselves note that the tablet behind that year results in "an exceptionally low date" for 01 Nisannu / Nisan and that its placement remains in question.[4066]

XX. Measured against the mathematically precise proleptic Hebrew calendar, this single exception strongly suggests that the intercalation was misplaced. Had it been an Addaru 2 / Adar 2 at the end of the 18th Regnal Year instead of an Ululu 2 / Elul 2 in the 19th, both calendars would have intercalated together on 05 and 06 March 607 BCE (Julian). The 19th Regnal Year would then have begun on a normal date, 05 April 607 BCE (Julian), and no year in the whole period would show the Babylonian calendar ahead of the proleptic Hebrew calendar. However, confirmation would require new textual evidence.

XX. Over the whole period Parker and Dubberstein's tables contain 75 Babylonian intercalations: 55 of Addaru 2 / Adar 2 and 20 of Ululu 2 / Elul 2. The proleptic Hebrew calendar has 74 intercalations of Addaru 2 / Adar 2 in the same period. Of the 75 Babylonian intercalations, 70 are attested by ancient texts and only five (5) are reconstructions, which Parker and Dubberstein mark in italics.[4067]

4066 Parker and Dubberstein 1971: 4 n. 9
4067 Parker and Dubberstein 1971: 25, 27-33

XX. The one-(1)-month differences found here therefore rest largely on recorded intercalations in tables that Walker (1997) notes have never been seriously questioned. The principal problem involved is to determine the years in which a 13th lunar month – the "intercalary" month – was added to bring the lunar year back into line with the solar year, and which intercalary month was used: Ululu 2 / Elul 2 (month 6b), or Addaru 2 / Adar 2 (month 12b).[4068]

XX. For the years from 495 BCE (Julian) onward, later evidence corrects Parker and Dubberstein in three (3) places:[4069] [4070]

1.) The reconstructed Addaru 2 / Adar 2 of the 29th Regnal Year of Darius I is replaced by an attested Ululu 2 / Elul 2 in his 30th year (492 – 491 BCE (Julian); Hebrew 19-year Metonic cycle year 2);

2.) The reconstructed Addaru 2 / Adar 2 of the 7th Regnal Year of Xerxes I by an Ululu 2 / Elul 2 in his 8th year (478 – 477 BCE (Julian); Hebrew 19-year Metonic cycle year 16);

3.) Ossendrijver also provisionally replaces the Addaru 2 / Adar 2 of Xerxes I's 18th year with an Ululu 2 / Elul 2;

XX. With these corrections the result is unchanged: the two (2) calendars are never more than one (1) month apart – in step in 107 years, and one (1) month apart in 96. For the years before Darius I, the comparison rests on Parker and Dubberstein's tables as published.

XX. Parker and Dubberstein number the 19-year cycle from the Babylonian year that began in Nisannu / Nisan 747 BCE (Julian) as their cycle year one (1) – a starting point they chose for convenience.[4071] In this study each Babylonian year is numbered instead by the proleptic Hebrew 19-year Metonic cycle in two (2) ways:

1.) Counted from Nisannu / Nisan, the number gives the Babylonian year in which the intercalation was made.

2.) Counted from Tashritu / Tishri, it gives the Hebrew year in which the intercalary month fell.

XX. Because the Hebrew leap years are counted from Tashritu / Tishri, only the second numbering places the intercalations of both calendars within the same Hebrew year.

4068 Walker 1997: 22
4069 Walker 1997: 23; Ossendrijver 2018a: 141-142
4070 Ossendrijver 2018a: 142-143
4071 Parker and Dubberstein 1971: 1 n. 1, 6

XX. On that basis and with the corrections applied, every Babylonian intercalation from the 10th Regnal Year of Xerxes I (476 – 475 BCE (Julian); Hebrew 19-year Metonic cycle year 18) to the end of the period falls in Hebrew cycle years 1, 4, 7, 10, 12, 15 and 18 without exception, while in Parker and Dubberstein's numbering from 747 BCE (Julian) the same intercalations fall in their cycle years 3, 6, 8, 11, 14, 17 and 19.

XX. Each lies one (1) or two (2) years before a Hebrew leap year (3, 6, 8, 11, 14, 17 and 19),[4072] so that the two (2) calendars follow the same seven-(7)-step pattern offset by a fixed interval. The last Babylonian intercalation outside this pattern is the Ululu 2 / Elul 2 of the 8th Regnal Year of Xerxes I (478 – 477 BCE (Julian); Hebrew 19-year Metonic cycle year 16).

Table 78.5. One (1) Full Hebrew 19-year Metonic Cycle, 474 – 455 BCE (Julian) with the Corrections Applied

Year BCE (Julian)	Regnal Year	Hebrew cycle year (Tashritu / Tishri)	Hebrew cycle year (Nisannu / Nisan)	P&D cycle year	Babylonian intercalation began	Proleptic Hebrew Addaru 2 / Adar 2 began
474 – 473	Xerxes I 12	1	19	8	Addaru 2, 24 March 473	–
473 – 472	Xerxes I 13	2	1	9	–	–
472 – 471	Xerxes I 14	3	2	10	–	03 March 471
471 – 470	Xerxes I 15	4	3	11	Addaru 2, 22 March 470	–
470 – 469	Xerxes I 16	5	4	12	–	–
469 – 468	Xerxes I 17	6	5	13	–	27 February 468
468 – 467	Xerxes I 18	7	6	14	Ululu 2, 22 September 468 (provisional)	–
467 – 466	Xerxes I 19	8	7	15	–	07 March 466
466 – 465	Xerxes I 20	9	8	16	–	–
465 – 464	Xerxes I 21	10	9	17	Ululu 2, 20 September 465	–
464 – 463	Artaxerxes I 1	11	10	18	–	05 March 463
463 – 462	Artaxerxes I 2	12	11	19	Addaru 2, 24 March 462	–
462 – 461	Artaxerxes I 3	13	12	1	–	–
461 – 460	Artaxerxes I 4	14	13	2	–	01 March 460
460 – 459	Artaxerxes I 5	15	14	3	Addaru 2, 20 March 459	–

4072 Maimonides, Sanctification of the New Month 6:10-11

459 – 458	Artaxerxes I 6	16	15	4	–	–
458 – 457	Artaxerxes I 7	17	16	5	–	26 February 457
457 – 456	Artaxerxes I 8	18	17	6	Addaru 2, 16 March 456 (reconstructed)	–
456 – 455	Artaxerxes I 9	19	18	7	–	06 March 455

XX. Walker reports that historians had concluded that the cycle was known to the Babylonians "by about 500 BC."[4073] Applied to the corrected data, the proleptic Hebrew cycle places the onset exactly by precise mathematical calculation, where Ossendrijver places it from the cuneiform evidence: the earliest identifiable 19-year cycle begins in the 10th Regnal Year of Xerxes I (Hebrew 19-year Metonic cycle year 18), and the preceding intercalation in his 8th Regnal Year (Hebrew 19-year Metonic cycle year 16) "probably deviated from that pattern."[4074]

XX. The intercalation of the 10th Regnal Year of Xerxes I rests on a late astronomical text of questionable reliability, so the first attested intercalation within the pattern is that of his 12th Regnal Year (Hebrew 19-year Metonic cycle year 1), attested in a single document from Persepolis.[4075]

XX. Within the pattern, the kind of month inserted was not yet fixed: the 18th Regnal Year of Xerxes I (Hebrew 19-year Metonic cycle year 7) has an Ululu 2 / Elul 2 where the later cycle has an Addaru 2 / Adar 2; the 19th and 38th Regnal Years of Artaxerxes I (both Hebrew 19-year Metonic cycle year 10) have an Addaru 2 / Adar 2 where it has an Ululu 2 / Elul 2.[4076]

XX. According to Ossendrijver, the cycle became fully permanent only in the 24th Regnal Year of Artaxerxes II (381 – 380 BCE (Julian); Hebrew 19-year Metonic cycle year 18).[4077]

XX. In Parker and Dubberstein's uncorrected tables, the same pattern appears to begin earlier: in the 24th Regnal Year of Darius I (498 – 497 BCE (Julian); Hebrew 19-year Metonic cycle year 15). That earlier onset depends on the two (2) reconstructions later discarded.[4078] Placed on the proleptic Hebrew cycle, the dependence of the pattern on each reconstructed or corrected intercalation becomes directly visible.

XX. The proleptic Hebrew calendar used as a fixed standard thus makes the Babylonian pattern of

4073 Walker 1997: 23
4074 Ossendrijver 2018a: 148
4075 Ossendrijver 2018a: 142
4076 Ossendrijver 2018a: 148
4077 Ossendrijver 2018a: 148
4078 Parker and Dubberstein 1971: 30-31; Walker 1997: 23; Ossendrijver 2018a: 141-142

intercalation visible year by year: which month was inserted, in which year, and from which year the pattern holds without exception. However, it does not show when the Hebrew calendar in its present form came into use. Nonetheless, it does allow any Babylonian date to be converted into the proleptic Hebrew calendar with a determined – not an estimated – difference.

XX. Claude (Anthropic) was used in preparing this work. It was used to peruse, analyze, and verify the table data comparing Parker and Dubberstein's intercalary months with the proleptic Hebrew calendar; to confirm the corresponding Hebrew, Julian, and Gregorian dates which the author had originally computed with the algorithms of Dershowitz and Reingold (1997, 2008); to compile the comparison tables; to check calculations, dates, and citations throughout the concluding sections and the abstract; and to assist in drafting and revising text in those sections. The research, chronology, arguments, and conclusions are the author's own. Readers would therefore be prudent to independently check the accuracy of the work produced with Claude.

78.9. The Sign of the Sundial's Shadow Given to the Hizkiyahu (Hezekiah) ben Ahaz and the Long Day of Joshua

XX. As discussed in Section 77, the Earth's non-uniform rate of rotation is overall decelerating yet fluctuates with periodic accelerations. This phenomenon could possibly explain the Earth's reversing its spin for 10 degrees so that the sundial's shadow reversed its direction for Hizkiyahu (Hezekiah) ben Ahaz perhaps because of an internal geodynamic aberration of interaction within the Earth's core and/or mantle, or some other phenomenon within the Earth.

XX. This could also explain the Long Day of Joshua where the Earth stopped spinning for several hours. Although unusual, and not what we would expect as normal behavior for the Earth, this observed phenomenon of fluctuations in the Earth's rate of spin where the Earth could actually stop spinning (or spin backwards for 10 degrees) under a "perfect storm of circumstances" could explain these miracles that are recorded in the Bible.

XX. The irregular rotation of the Earth, which ΔT (Delta T) measures, is a possible explanation for the wonder/sign of the sundial's shadow for Hizkiyahu (Hezekiah) ben Ahaz, as well as the Long Day of Joshua. As we cannot yet prove this, we will simply leave it as a theory that has yet to be explained (or disproved) with scientific proof.

XX. It may be possible for this theory to be tested: if the whole Earth turned back ten (10) degrees in the 14th Regnal Year of Hizkiyahu (Hezekiah) ben Ahaz (711 – 710 BCE (Julian)), and never made up the lost time, then every eclipse before that year would be off by at least forty (40) minutes compared with every eclipse after it.

XX. We have eclipse records from both before and after that year: the solar eclipse of Būr-saggilê (763 BCE (Julian)), the lunar eclipse of 747 BCE (Julian), and the lunar eclipses of 721 – 720 BCE (Julian) in Ptolemy's Almagest from before it, and the long series of Babylonian records from after it. Dated precisely with the proleptic Hebrew calendar, these records may show whether such a jump happened, although the earliest records are few and their timings are not exact.

XX. However, if the Earth made up the lost time – for example, during the following night – or if the sign was local, "the wonder that was in the land",[4079] then there would be no lasting difference, and the eclipse records would show nothing.

XX. The Long Day of Joshua happened before any surviving eclipse records, so it cannot be tested this way.

78.10. Final Words (PRE-PRINT WORK-IN-PROGRESS)

XX. As demonstrated and proven, we can use the modern (yet ancient) Hebrew calendar to proleptically calculate accurate dates for the ancient past by synchronizing it with the ancient Babylonian "Standard Mesopotamian Calendar" as the Hebrew Calendar is itself also the Standard Mesopotamian Calendar. The dates in the Venus Tablet (VT) are not reliable or accurate, but could be recalibrated and either confirmed or disproved with this precise Babylonian / Hebrew lunisolar calendar.

XX. The Hebrew Bible is absolute truth as demonstrated by the precise mathematical and astronomical accuracy of the Hebrew lunisolar calendar through which we have been able to unlock the biblical chronology as recorded in the Hebrew Scriptures to reveal it as the most precise of ancient chronologies. Therefore, it is the biblical chronology that has proven itself as the new "backbone" with which all other ancient chronologies must now synchronize and recalibrate themselves. For example, the Egyptian dead-reckoning only offers a minimum framework, and thus may be missing years, i.e. necessitating the raising of years to a higher chronology.

XX. Consequently, this synchronization of ancient historical chronologies will allow scholars of Jewish studies to recalibrate the events recorded throughout Jewish history in order to find and account for the so-called "missing ~165 years", and thus determine and update precise dates.

XX. Moreover, this mathematical, lunisolar Babylonian / Hebrew calendar can then be used to fix precisely the day of every recorded ancient lunar and solar eclipse catalogued by NASA via Espenak

4079 cf. 2 Chronicles 32:31

and Meeus,[4080] [4081] (and others before them)[4082] as well as to systematically determine more precise ancient values for the critical astronomical variable, ΔT (Delta T), which reflects the irregularities of the Earth's rate of rotation and its measurement throughout history. It should be stressed that this variable, ΔT (Delta T), has particular relevance, significance, and critical importance for the study and measurement of times and positions of ancient eclipses.

XX. The critical variable ΔT (Delta T) is how astronomers calculate positions (where observations would be possible) for upcoming (and therefore also ancient) eclipses. This is especially relevant for solar eclipses. A correct identification of ancient solar [and lunar] eclipses is not only important for historical reasons, but also gives the possibility to determine the [secular lunar orbital tidal friction] acceleration (ṅ) [or the rate of change of the mean motion] of the longitude of the Moon to a high precision,[4083] [although as shown in Section 78.7, only in combination with the Earth's rotational change since eclipse observations alone cannot separate the two]

XX. If a computer program does not incorporate these values into its calculations, then the resultant eclipse paths will contain longitude shifts inconsistent with the best estimates for ΔT.[4084] It is therefore a matter of concern that at present there appears to be a degree of confusion and misapprehension among historians of astronomy over the choice of values of ΔT that should be used in making retrospective computations of lunar and solar positions.[4085]

XX. Just as ancient Babylonian astronomers used Goal-Year periods to predict phenomena, we can propose that the same method can be used to proleptically "predict" these phenomena into the ancient past by means of the 18-year Saros cycle. Used in conjunction with modern astronomical computation, it can refine the precision of measurement.

XX. Thus for recalibration of ancient values of ΔT (Delta T) because the Hebrew calendar ensures that each day the rotation of the Earth is reset for that day with a precise, predictable monthly position of the Moon, this may reduce the dependence on a geometric arcus visionis when calculating for ancient periods where the accumulation of error(s) from any (or many) variables can (and demonstrably does) produce inaccurate dates for the first visible crescent (New Moons).

XX. This recalibration will make it possible to date ancient lunar observations precisely – and also to refine ΔT (Delta T) for the ancient past – by means of the precise Hebrew calendar that is identical in its structure, its months, and its means of synchronizing the solar year with the ancient Babylonian

4080 NASA/TP–2009–214172: Espenak and Meeus (2009): *Five Millennium Canon of Lunar Eclipses: –1999 to +3000 (2000 BCE to 3000 CE)*
4081 NASA/TP–2006–214141: Espenak and Meeus (2006): *Five Millennium Canon of Solar Eclipses: –1999 to +3000 (2000 BCE to 3000 CE)*
4082 e.g. Huber and De Meis (2004); Beaulieu and Britton (1994); as well as others previously.
4083 Henriksson 2009: 166
4084 Espenak, "Eclipse Predictions by Fred Espenak", EclipseWise.com; https://www.eclipsewise.com/help/rotation.html;
4085 Morrison and Stephenson 2004: 327

"Standard Mesopotamian Calendar". This allows for seamless integration of calculated dates into the ancient past with this modern Hebrew calendar that is the "perfected refinement" of the correct observation and measurement of the Moon's motions, phases, intercalations, and its resulting calendar.

XX. Furthermore, this same Babylonian / Hebrew calendar system is the key to unsealing the Book of Daniel, and specifically chapter 9 with its Messianic prophecy[4086] that Sir Robert Anderson[4087] (as well as both ancient Jewish and Christian scholars before him) attempted to interpret and decipher as a sealed prediction of the exact day of the coming of the Messiah. Applying this same Hebrew calendar (together with a correct literal reading of the cryptic Hebrew text), this passage in the Book of Daniel can be unsealed to correctly interpret, reveal, and confirm the precise mathematical prediction and fulfillment of the exact year of Jesus' baptism by John the Baptist (28 CE (Julian)),[4088] [4089] as well as both the exact year and month of Jesus' Crucifixion (33 CE (Julian)).[4090] [4091] [4092]

XX. As a direct result of observation of the mathematics of this ancient chronology, it becomes clearly evident without a doubt that the mysterious, elusive identity of the heretofore unknown historical figure of "Darius the Mede" can now be confirmed as none other than Darius the Great (r. 521 – 486 BCE (Julian)) himself (i.e. hidden in plain sight). Thus the Book of Daniel must be read and correctly understood in this context.

XX. Therefore, for the first time in history, all of this correctly, thoroughly, and precisely proves both mathematically and scientifically not only that the Hebrew Bible is the most accurate historical book of ancient history (including the Book of Daniel), and not only that Jesus (Yeshua) was the Promised Messiah, but also by direct implication: It literally, undeniably, thoroughly, precisely, and irrevocably proves the existence of the Creator, GOD, for all to see with their own eyes, and scientifically (and mathematically) verify for themselves.

4086 cf. Daniel 9:24-27
4087 Anderson 1894: *The Coming Prince*
4088 cf. Appendix Table X.174
4089 cf. Daniel 9:24-25 – "to anoint the Holy of Holies".
4090 cf. Appendix Table X.184, X.185
4091 cf. Daniel 9:24-26 – "a Messiah will be cut off".
4092 Crucifixion or Cruci-Fiction?